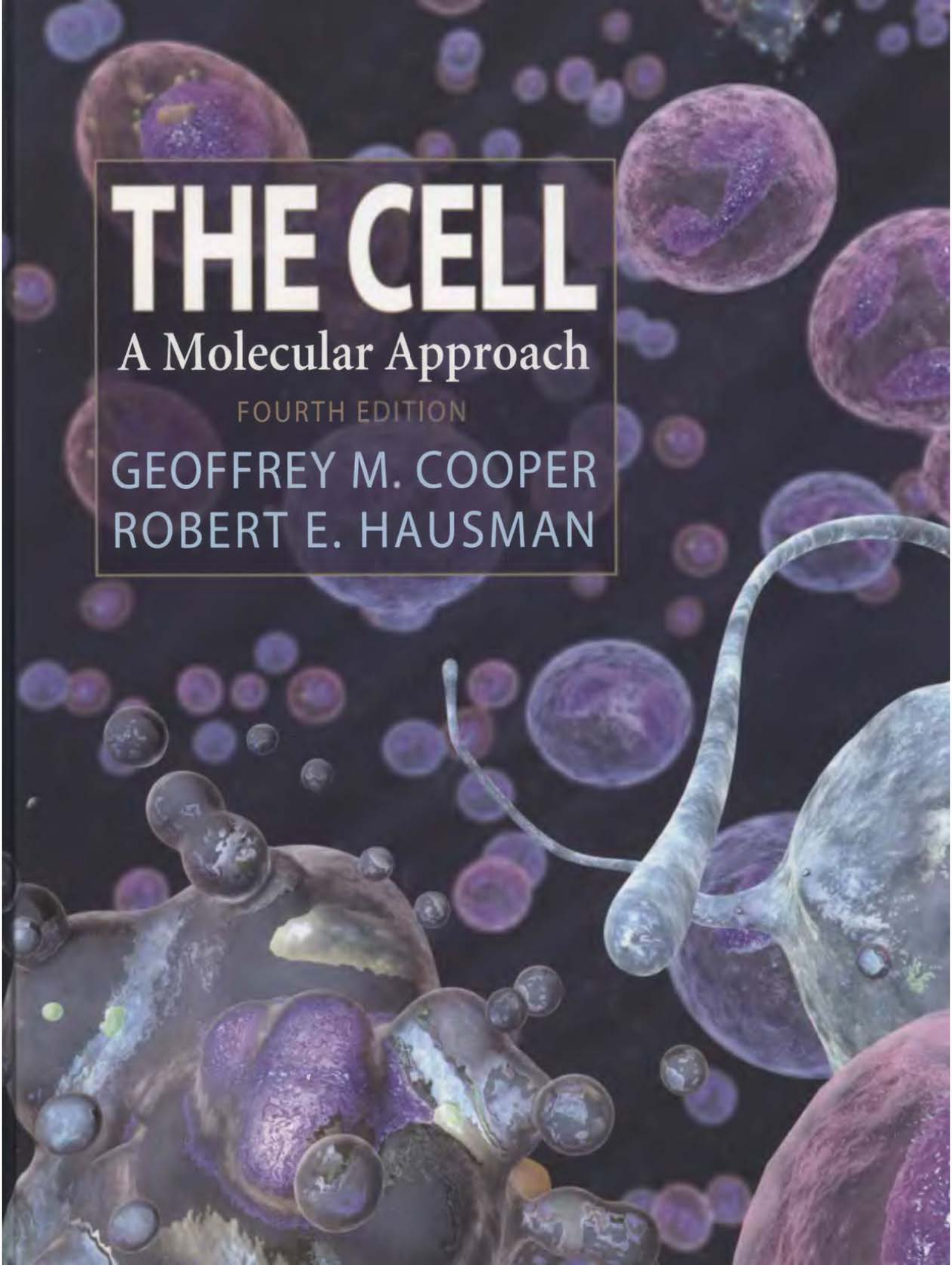
THE CELL
A Molecular Approach
FOURTH EDITION
GEOFFREY M. COOPER
ROBERT E. HAUSMAN

THE CELL

A Molecular Approach, Fourth Edition

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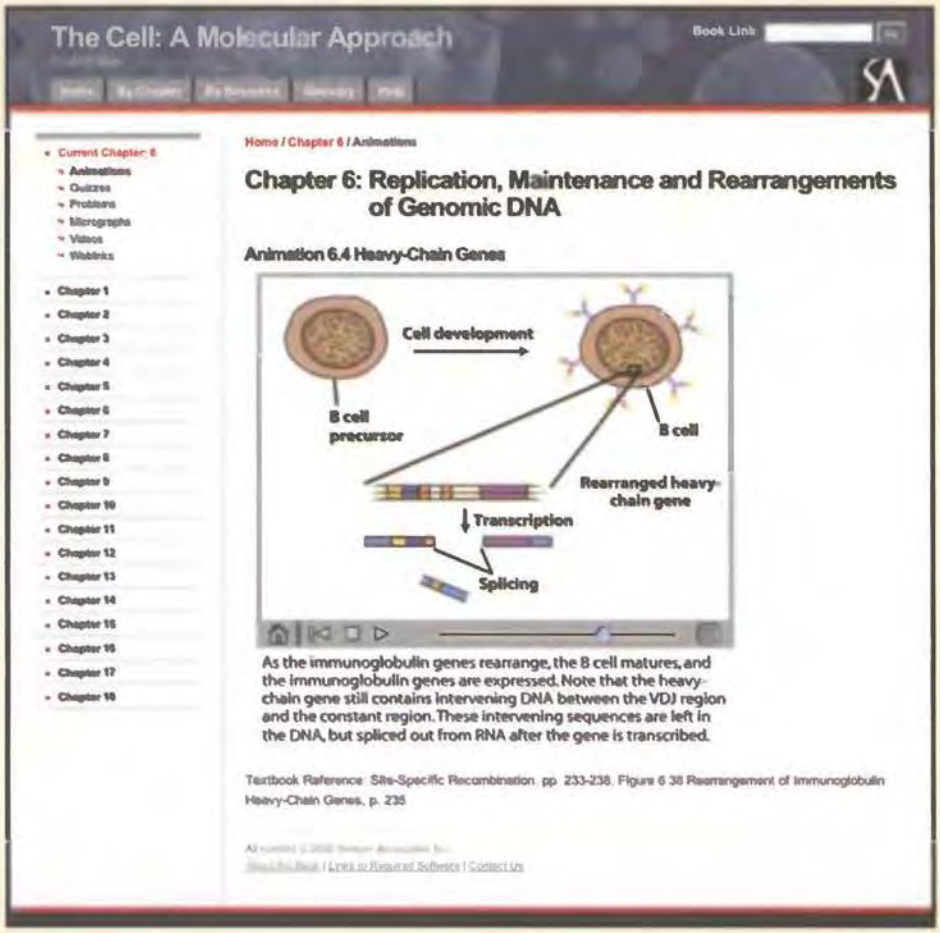

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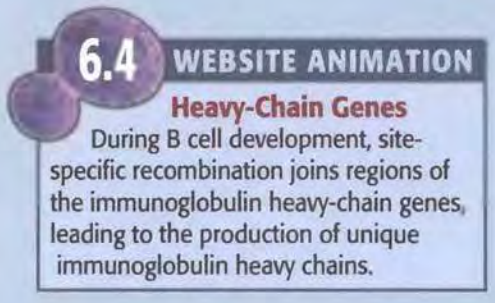

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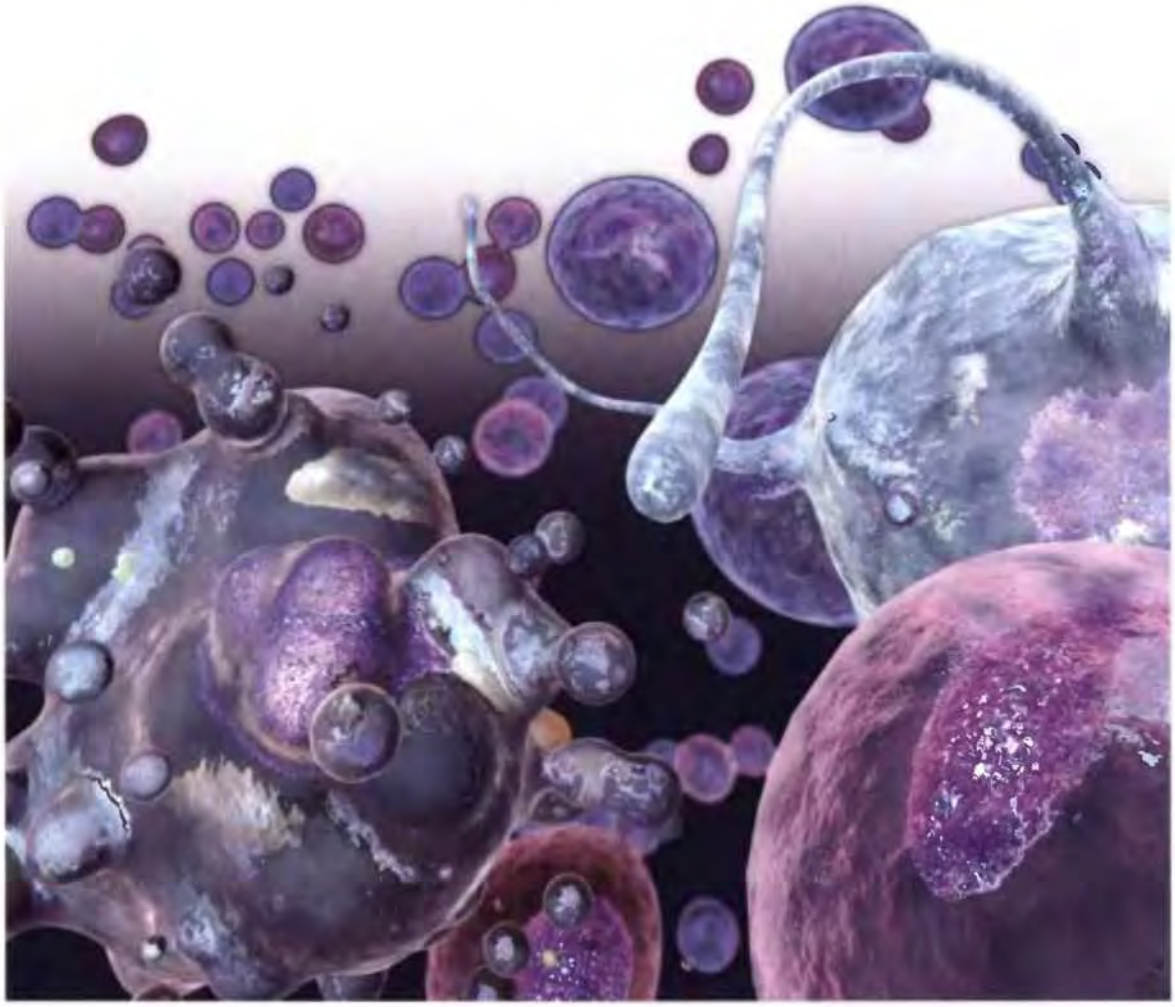

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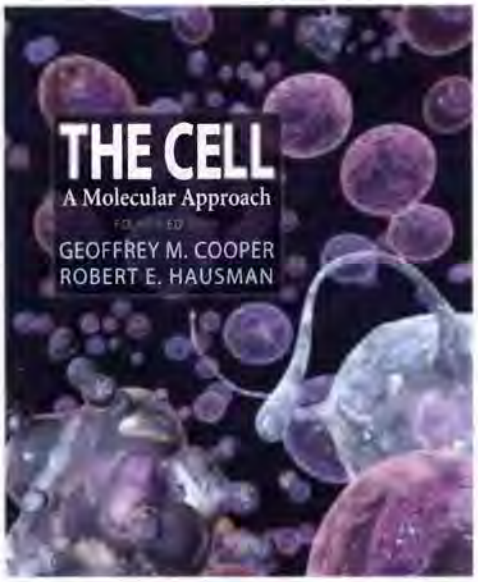

The cover

The cover illustrates the growth and maturation of blood cells. A cell extending processes is shown in the lower right and a cell undergoing apoptosis is in the lower left.

The artist

Chris Bickel is a professional medical illustrator residing near Washington, D.C. He specializes in visualizing and animating molecular and cellular biology.

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This book is dedicated to
Florence Haseltine, with thanks
for having taught me how
to do my first experiment.

G.M.C.

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Preface

The three years since publication of the Third Edition of *The Cell* in 2003 have been marked not only by rapid progress in many areas of cell biology, but also by revolutionary changes in some of the ways that molecular and cellular biologists are approaching their science. The human genome project introduced large scale experimentation to cell and molecular biology, together with the need for sophisticated computational approaches to analyze the vast amounts of data generated by genome sequencing. The field of bioinformatics developed in response to this need and has now become an integral part of our efforts to understand the molecular basis of cell behavior. The success of genomics has also spawned global experimental approaches in other areas and given rise to the new field of systems biology, which combines large-scale experimentation with quantitative analysis and modeling.

These changes in experimental approaches have been coupled with striking advances in several areas. Perhaps foremost among these, it has become apparent that microRNAs play widespread roles in regulating gene expression, the full biological significance of which remains to be discovered. Our understanding of stem cells has progressed enormously, together with the potential for using these cells to treat a variety of devastating diseases. Likewise, the last few years have seen major progress in the development of anticancer drugs that specifically target the abnormal proteins responsible for the uncontrolled growth of cancer cells.

The Fourth Edition of *The Cell* has been revised and updated to incorporate these advances and to highlight the exciting opportunities that mark the future of molecular and cellular biology. This edition not only provides an up-to-date review of genomics, but also includes major new sections dealing with the current challenges of bioinformatics, proteomics, and systems biology. The chapter on Cell Signaling has similarly been expanded to include a new section on signaling networks and quantitative analysis of signaling pathways within the cell. A new chapter on Cell Death and Cell Renewal provides expanded treatment of both programmed cell death and stem cells, including their medical applications. And the chapter on Cancer has been updated to cover the most recent developments in designer drugs.

While the text has been updated, the goals and approaches of *The Cell* remain the same as in previous editions. *The Cell* continues to be an accessible and readable text that can be appreciated and mastered by undergraduate students who are taking a first course in cell and molecular biology. Without sacrificing accessibility, we have also sought to ensure that *The Cell* is an intellectually satisfying and rigorous text that will engage students and enable them to appreciate the outstanding unanswered questions in our field. With this goal in mind, we have emphasized experimental approaches and the frontiers of current knowledge throughout the book. Each chapter contains one or two Key Experiment essays that describe a seminal paper and its background in detail, with the intent of giving students a sense of "doing science." Most chapters also contain Molecu-

lar Medicine essays that relate basic science to clinical applications. The Fourth Edition of *The Cell* also includes a new feature: each chapter contains several short "sidebars" designed to stimulate students by highlighting additional areas of interest and medical relevance. Finally, we have provided an updated and expanded set of questions at the end of each chapter, with answers to all questions at the back of the book. These questions are designed not only to facilitate review of the chapter, but also to stimulate students to think about the design of experiments and interpretation of experimental results.

The overarching theme of this and previous editions of *The Cell* has been to convey not only the facts, but also a sense of the excitement and challenges of research in contemporary molecular and cellular biology. We hope that *The Cell* stimulates our students to meet these challenges and contribute to the research upon which future texts will be based.

Acknowledgments

The Fourth Edition of *The Cell* has benefited, as always, from the comments of instructors and students who used the previous edition. Many readers have offered helpful suggestions that have been incorporated into the current edition, and we are grateful for their advice.

We are also grateful to the following colleagues who reviewed the new edition and whose comments and criticisms have improved the text:

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We are also grateful to our publishers and editors. Andy Sinauer and Dean Scudder at Sinauer Associates and Jeff Holtmeier at ASM Press have been continuing sources of support and help. Christopher Small, Jefferson Johnson, and Janice Holabird did an exceptional job crafting the book. Suzanne Lain did an outstanding job of copyediting the manuscript, and Chelsea Holabird once again coordinated the production of *The Cell* with skill, patience, and good humor.

GEOFFREY M. COOPER • ROBERT E. HAUSMAN • APRIL 2006

Organization and Features of *The Cell* A Molecular Approach

The Cell has been designed to be an approachable and teachable text that can be covered in a single semester while allowing students to master the material in the entire book. It is assumed that most students will have had introductory biology and general chemistry courses, but will not have had previous courses in organic chemistry, biochemistry, or molecular biology. Several aspects of the organization and features of the book will help students to approach and understand its subject matter.

Organization

The Cell is divided into four parts, each of which is self-contained, so that the order and emphasis of topics can be easily varied according to the needs of individual courses.

Part I provides background chapters on the evolution of cells, methods for studying cells, the chemistry of cells, and the fundamentals of modern molecular biology. For those students who have a strong background from either a comprehensive introductory biology course or a previous course in molecular biology, various parts of these chapters can be skipped or used for review.

Part II focuses on the molecular biology of cells and contains chapters dealing with genome organization and sequences; DNA replication, repair, and recombination; transcription and RNA processing; and the synthesis, processing, and regulation of proteins. The order of chapters follows the flow of genetic information (DNA → RNA → protein) and provides a concise but up-to-date overview of these topics.

Part III contains the core block of chapters on cell structure and function, including chapters on the nucleus, cytoplasmic organelles, the cytoskeleton, the plasma membrane, and the extracellular matrix. This part of the book starts with coverage of the nucleus, which puts the molecular biology of Part II within the context of the eukaryotic cell, and then works outward through cytoplasmic organelles and the cytoskeleton to the plasma membrane and the exterior of the cell. These chapters are relatively self-contained, however, and could be used in a different order should that be more appropriate for a particular course.

Finally, Part IV focuses on the exciting and fast-moving area of cell regulation, including coverage of topics such as cell signaling, the cell cycle, programmed cell death, and stem cells. This part of the book concludes with a chapter on cancer, which synthesizes the consequences of defects in basic cell regulatory mechanisms.

Features

Several pedagogical features have been incorporated into *The Cell* in order to help students master and integrate its contents. These features are reviewed below as a guide to students studying from this book.

Chapter organization. Each chapter is divided into three to five major sections, which are further divided into a similar number of subsections. An outline listing the major sections at the beginning of each chapter provides a brief overview of its contents.

Key Terms and Glossary. Key terms are identified as boldfaced words when they are introduced in each chapter. These key terms are reiterated in the chapter summary and defined in the glossary at the end of the book.

Illustrations and micrographs. An illustration program of full-color art and micrographs has been carefully developed to complement and visually reinforce the text.

Key Experiment and Molecular Medicine Essays. Each chapter contains either two Key Experiment essays or one Key Experiment and one Molecular Medicine essay. These features are designed to provide the student with a sense of both the experimental basis of cell and molecular biology and its applications to modern medicine. We have also found these essays to be a useful basis for student discussion sections, which can be accompanied with a review of the original paper upon which the Key Experiments are based.

Sidebars. Each chapter contains several sidebars that provide brief descriptive highlights of points of interest related to material covered in the text. The sidebars supplement the text and provide starting points for class discussion.

Chapter Summaries. Chapter summaries are organized in outline form corresponding to the major sections and subsections of each chapter. This section-by-section format is coupled with a list of the key terms introduced in each section, providing a succinct but comprehensive review of the material.

Questions and Answers. An expanded set of questions at the end of each chapter (with answers in the back of the book) are designed to further facilitate review of the material presented in the chapter and to encourage students to use this material to predict or interpret experimental results.

References. Comprehensive lists of references at the end of each chapter provide access to both reviews and selected papers from the primary literature. In order to help the student identify articles of interest, the references are organized according to chapter sections. Review articles and primary papers are distinguished by [R] and [P] designations, respectively.

Companion Website icons. New icons in the margin direct students to the website's animations, videos, quizzes, problems, and other review material.

Media & Supplements to Accompany
The Cell, Fourth Edition

For the Student

Companion Website

(*www.sinauer.com/cooper*)
The Fourth Edition of *The Cell* features a new companion website, available free of charge. This robust site features a wealth of study and review material coupled with rich multimedia resources including detailed animations, video microscopy, and micrographs. The companion site includes online quizzes, problem sets, animations, videos, chapter summaries, Web links, and a glossary. (See the inside front cover for details.)

For the Instructor*

Instructor's Resource Library

The Fourth Edition Instructor's Resource Library includes an expanded collection of digital resources to aid in planning your course, presenting lectures, and assessing your students. Included are:

- A convenient browser interface
- All textbook figures (art and photos) and tables in both high- and low-resolution JPEG formats
- PowerPoint® presentations of all figures, tables, supplemental photos, videos, and animations
- Complete, ready-to-use lecture PowerPoint® presentations
- An expanded collection of video microscopy
- The entire collection of animations from the student website, for use in lecture
- Over 100 supplemental micrographs
- Problems and quizzes from the student website, with answers
- Textbook end-of-chapter questions, with answers
- The complete Test File, in Microsoft® Word® format
- The computerized Test File (includes Brownstone Diploma® software)
- Chapter outlines

Test File By Dennis Goode

Expanded for the Fourth Edition, the Test File includes a collection of over 1300 multiple-choice, true/false, and short answer questions covering the full range of content covered in every chapter. Questions are organized by chapter heading, making it easy for the instructor to find questions on specific topics. The Test File also includes the quiz questions and problems from the companion website, as well as the textbook end-of-chapter discussion questions. The Test File is included in the Instructor's Resource Library and is provided both as Microsoft Word® files and in the Brownstone Diploma® exam-creation software (software included). Diploma® enables the instructor to easily create exams from the bank of questions provided, as well as add their own questions, publish secure Internet exams, and more.

Online Quizzing By Robert E. McGehee and Brian Storrie

New for the Fourth Edition, online quizzes and problems are available for student use on the companion website. These quizzes are linked to a new online grade book that makes it easy to track student performance. In addition, instructors can add their own questions to the online quizzes, control quiz availability, and more. Instructor registration is required in order for students to access the quizzes. Instructors can register online at the companion website (www.sinauer.com/cooper).

Assessment via Course Management Systems

Instructors using course management systems such as WebCT®, Blackboard®, and Angel® can easily create and export quizzes and exams (or the entire test bank) for integration into their online course. The entire test file is provided in WebCT® and Blackboard® formats, and other formats can be easily generated from the Brownstone Diploma® software (included).

Overhead Transparencies

A set of 150 full-color figures from the textbook is available as overhead transparencies. These have been formatted and color-adjusted for optimal projection in the classroom.

Other Media Resources

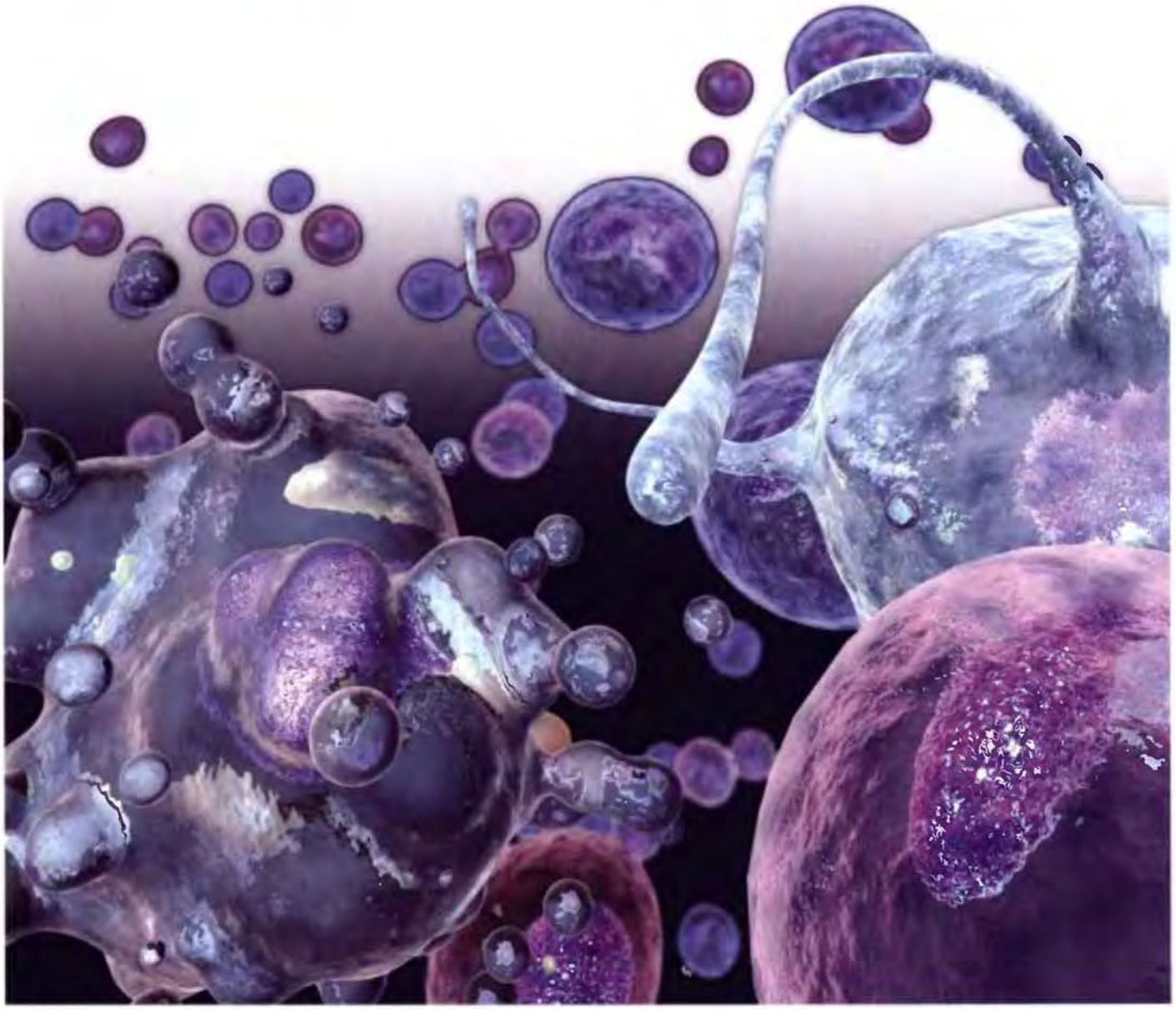
The following videos and CD-ROMs are available to qualified adopters of the text:

- Fink, *CELLebration* (VHS, ISBN 0-87893-166-X)
- Pickett-Heaps and Pickett-Heaps, *Diatoms: Life in Glass Houses* (DVD, ISBN 0-9586081-7-2)
- Pickett-Heaps and Pickett-Heaps, *The Dynamics and Mechanics of Mitosis* (VHS, ISBN 0-9586081-5-6)
- Pickett-Heaps and Pickett-Heaps, *Living Cells: Structure and Diversity* (VHS, ISBN 0-646-29291-9)
- Pickett-Heaps and Pickett-Heaps, *Remarkable Plants: The Oedogoniales (Green Algae)* (DVD, ISBN 0-9586081-8-0)
- Sardet, Larsonneur, and Koch, *Voyage Inside the Cell* (DVD, ISBN 0-87893-755-2; VHS, ISBN 0-87893-763-3)

*Available to qualified adopters

PART I

Introduction



CHAPTER 1 ■ An Overview of Cells and Cell Research

CHAPTER 2 ■ The Composition of Cells

CHAPTER 3 ■ Cell Metabolism

CHAPTER 4 ■ Fundamentals of Molecular Biology

CHAPTER 1

An Overview of Cells and Cell Research

- ***The Origin and Evolution of Cells 4***
- ***Cells as Experimental Models 16***
- ***Tools of Cell Biology 21***
- ***KEY EXPERIMENT:*** Animal Cell Culture 35
- ***MOLECULAR MEDICINE:*** Viruses and Cancer 37

UNDERSTANDING THE MOLECULAR BIOLOGY OF CELLS is an active area of research that is fundamental to all of the biological sciences. This is true not only from the standpoint of basic science, but also with respect to a growing number of applications in medicine, agriculture, biotechnology, and biomedical engineering. Especially with the completion of the sequence of the human genome, progress in cell and molecular biology is opening new horizons in the practice of medicine. Striking examples include the development of new drugs specifically targeted to interfere with the growth of cancer cells and the potential use of stem cells to replace damaged tissues and treat patients suffering from conditions like diabetes, Parkinson's disease, Alzheimer's disease, spinal cord injuries, and heart disease.

Because cell and molecular biology is a rapidly growing field of research, it is important to understand its experimental basis as well as the current state of our knowledge. This chapter will therefore focus on how cells are studied, as well as reviewing some of their basic properties. Appreciating the similarities and differences between cells is particularly important to understanding cell biology. The first section of this chapter therefore discusses both the unity and the diversity of present-day cells in terms of their evolution from a common ancestor. On the one hand, all cells share common fundamental properties that have been conserved throughout evolution. For example, all cells employ DNA as their genetic material, are surrounded by plasma membranes, and use the same basic mechanisms for energy metabolism. On the other hand, present-day cells have evolved a variety of different lifestyles. Many organisms, such as bacteria, amoebas, and yeasts, consist of single cells that are capable of independent self-replication. More complex organisms are composed of collections of cells that function in a

coordinated manner, with different cells specialized to perform particular tasks. The human body, for example, is composed of more than 200 different kinds of cells, each specialized for such distinctive functions as memory, sight, movement, and digestion. The diversity exhibited by the many different kinds of cells is striking; for example, consider the differences between bacteria and the cells of the human brain.

The fundamental similarities between different types of cells provide a unifying theme to cell biology, allowing the basic principles learned from experiments with one kind of cell to be extrapolated and generalized to other cell types. Several kinds of cells and organisms are widely used to study different aspects of cell and molecular biology; the second section of this chapter discusses some of the properties of these cells that make them particularly valuable as experimental models. Finally, it is important to recognize that progress in cell biology depends heavily on the availability of experimental tools that allow scientists to make new observations or conduct novel kinds of experiments. This introductory chapter therefore concludes with a discussion of some of the experimental approaches used to study cells, as well as a review of some of the major historical developments that have led to our current understanding of cell structure and function.

The Origin and Evolution of Cells

Cells are divided into two main classes, initially defined by whether they contain a nucleus. **Prokaryotic cells** (bacteria) lack a nuclear envelope; **eukaryotic cells** have a nucleus in which the genetic material is separated from the cytoplasm. Prokaryotic cells are generally smaller and simpler than eukaryotic cells; in addition to the absence of a nucleus, their genomes are less complex and they do not contain cytoplasmic organelles or a cytoskeleton (Table 1.1). In spite of these differences, the same basic molecular mechanisms govern the lives of both prokaryotes and eukaryotes, indicating that all present-day cells are descended from a single primordial ancestor. How did this first cell develop? And how did the complexity and diversity exhibited by present-day cells evolve?

The First Cell

It appears that life first emerged at least 3.8 billion years ago, approximately 750 million years after Earth was formed. How life originated and how the first cell came into being are matters of speculation, since these events cannot be reproduced in the laboratory. Nonetheless, several types of experiments provide important evidence bearing on some steps of the process.

TABLE 1.1 Prokaryotic and Eukaryotic Cells

Characteristic	Prokaryote	Eukaryote
Nucleus	Absent	Present
Diameter of a typical cell	≈1 μm	10–100 μm
Cytoskeleton	Absent	Present
Cytoplasmic organelles	Absent	Present
DNA content (base pairs)	1×10^6 to 5×10^6	1.5×10^7 to 5×10^9
Chromosomes	Single circular DNA molecule	Multiple linear DNA molecules

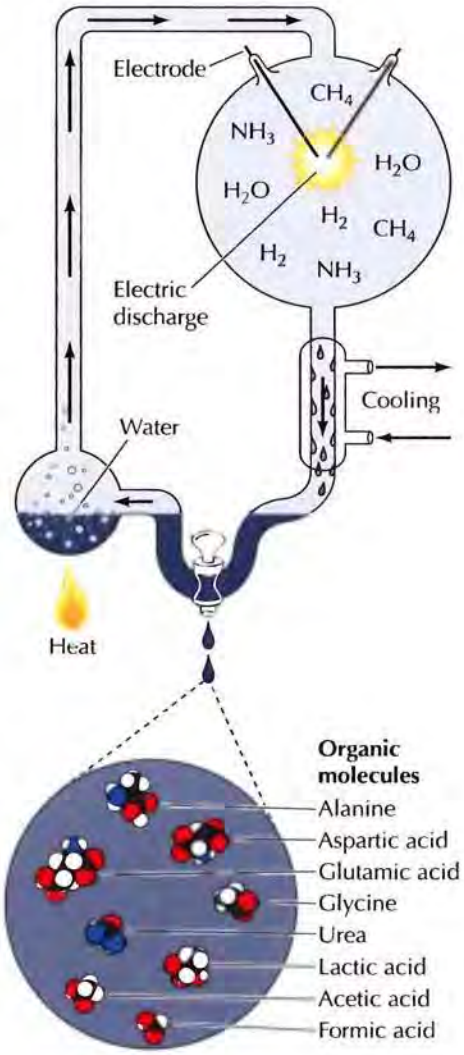


FIGURE 1.1 Spontaneous formation of organic molecules Water vapor was refluxed through an atmosphere consisting of CH_4, NH_3, and H_2, into which electric sparks were discharged. Analysis of the reaction products revealed the formation of a variety of organic molecules, including the amino acids alanine, aspartic acid, glutamic acid, and glycine.

It was first suggested in the 1920s that simple organic molecules could form and spontaneously polymerize into macromolecules under the conditions thought to exist in primitive Earth's atmosphere. At the time life arose, the atmosphere of Earth is thought to have contained little or no free oxygen, instead consisting principally of CO_2 and N_2 in addition to smaller amounts of gases such as H_2, H_2S, and CO. Such an atmosphere provides reducing conditions in which organic molecules, given a source of energy such as sunlight or electrical discharge, can form spontaneously. The spontaneous formation of organic molecules was first demonstrated experimentally in the 1950s, when Stanley Miller (then a graduate student) showed that the discharge of electric sparks into a mixture of H_2, CH_4, and NH_3, in the presence of water, leads to the formation of a variety of organic molecules, including several amino acids (Figure 1.1). Although Miller's experiments did not precisely reproduce the conditions of primitive Earth, they clearly demonstrated the plausibility of the spontaneous synthesis of organic molecules, providing the basic materials from which the first living organisms arose.

The next step in evolution was the formation of macromolecules. The monomeric building blocks of macromolecules have been demonstrated to polymerize spontaneously under plausible prebiotic conditions. Heating dry mixtures of amino acids, for example, results in their polymerization to form polypeptides. But the critical characteristic of the macromolecule from which life evolved must have been the ability to replicate itself. Only a macromolecule capable of directing the synthesis of new copies of itself would have been capable of reproduction and further evolution.

Of the two major classes of informational macromolecules in present-day cells (nucleic acids and proteins), only the nucleic acids are capable of directing their own self-replication. Nucleic acids can serve as templates for their own synthesis as a result of specific base pairing between complementary nucleotides (Figure 1.2). A critical step in understanding molecular evo-

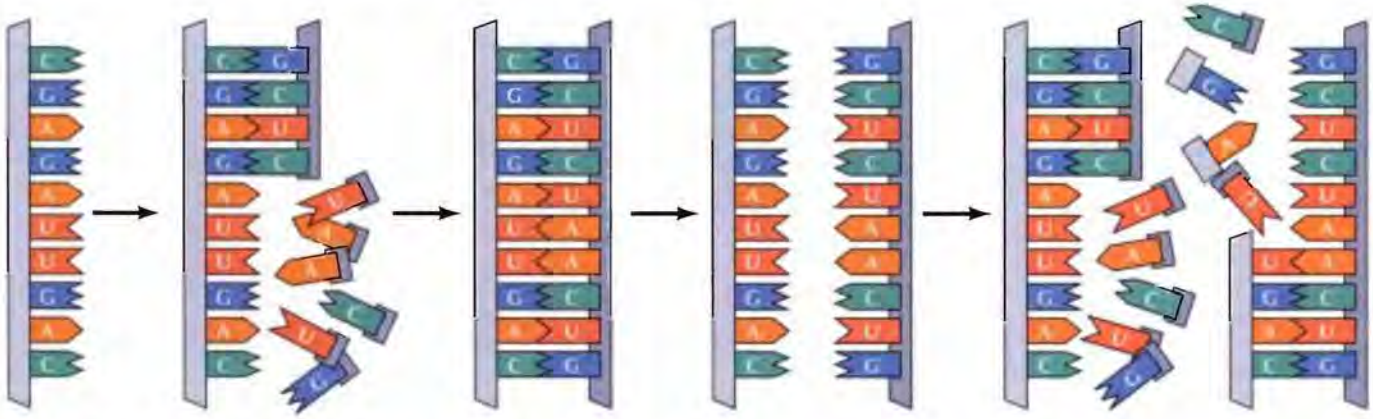


FIGURE 1.2 Self-replication of RNA Complementary pairing between nucleotides (adenine [A] with uracil [U] and guanine [G] with cytosine [C]) allows one strand of RNA to serve as a template for the synthesis of a new strand with the complementary sequence.

lution was thus reached in the early 1980s, when it was discovered in the laboratories of Sid Altman and Tom Cech that RNA is capable of catalyzing a number of chemical reactions, including the polymerization of nucleotides. Further studies have extended the known catalytic activities of RNA, including the description of RNA molecules that direct the synthesis of a new RNA strand from an RNA template. RNA is thus uniquely able to both serve as a template for and to catalyze its own replication. Consequently, RNA is generally believed to have been the initial genetic system, and an early stage of chemical evolution is thought to have been based on self-replicating RNA molecules—a period of evolution known as the **RNA world**. Ordered interactions between RNA and amino acids then evolved into the present-day genetic code, and DNA eventually replaced RNA as the genetic material.

The first cell is presumed to have arisen by the enclosure of self-replicating RNA in a membrane composed of **phospholipids** (Figure 1.3). As discussed in detail in the next chapter, phospholipids are the basic components of all present-day biological membranes, including the plasma membranes of both prokaryotic and eukaryotic cells. The key characteristic of the phospholipids that form membranes is that they are **amphipathic** molecules, meaning that one portion of the molecule is soluble in water and another portion is not. Phospholipids have long, water-insoluble (**hydrophobic**) hydrocarbon chains joined to water-soluble (**hydrophilic**) head groups that contain phosphate. When placed in water, phospholipids spontaneously aggregate into a bilayer with their phosphate-containing head groups on the outside in contact with water and their hydrocarbon tails in the interior in contact with each other. Such a phospholipid bilayer forms a stable barrier between two aqueous compartments—for example, separating the interior of the cell from its external environment.

The enclosure of self-replicating RNA and associated molecules in a phospholipid membrane would thus have maintained them as a unit, capa-

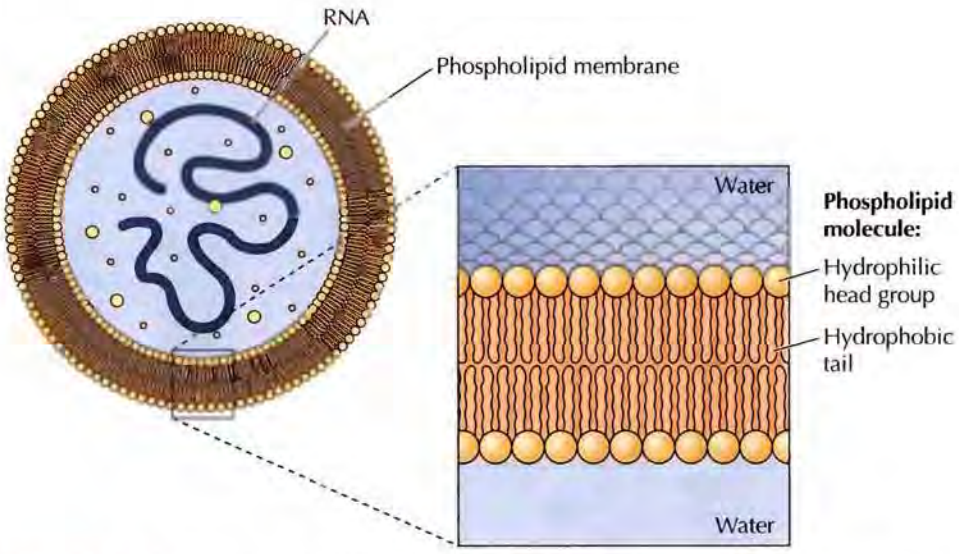


FIGURE 1.3 Enclosure of self-replicating RNA in a phospholipid membrane
The first cell is thought to have arisen by the enclosure of self-replicating RNA and associated molecules in a membrane composed of phospholipids. Each phospholipid molecule has two long hydrophobic tails attached to a hydrophilic head group. The hydrophobic tails are buried in the lipid bilayer; the hydrophilic heads are exposed to water on both sides of the membrane.

ble of self-reproduction and further evolution. RNA-directed protein synthesis may already have evolved by this time, in which case the first cell would have consisted of self-replicating RNA and its encoded proteins.

The Evolution of Metabolism

Because cells originated in a sea of organic molecules, they were able to obtain food and energy directly from their environment. But such a situation is self-limiting, so cells needed to evolve their own mechanisms for generating energy and synthesizing the molecules necessary for their replication. The generation and controlled utilization of metabolic energy is central to all cell activities, and the principal pathways of energy metabolism (discussed in detail in Chapter 3) are highly conserved in present-day cells. All cells use **adenosine 5′-triphosphate (ATP)** as their source of metabolic energy to drive the synthesis of cell constituents and carry out other energy-requiring activities, such as movement (e.g., muscle contraction). The mechanisms used by cells for the generation of ATP are thought to have evolved in three stages, corresponding to the evolution of glycolysis, photosynthesis, and oxidative metabolism (Figure 1.4). The development of these metabolic pathways changed Earth's atmosphere, thereby altering the course of further evolution.

In the initially anaerobic atmosphere of Earth, the first energy-generating reactions presumably involved the breakdown of organic molecules in the absence of oxygen. These reactions are likely to have been a form of present-day **glycolysis**—the anaerobic breakdown of glucose to lactic acid, with the net energy gain of two molecules of ATP. In addition to using ATP as their source of intracellular chemical energy, all present-day cells carry out glycolysis, consistent with the notion that these reactions arose very early in evolution.

Glycolysis provided a mechanism by which the energy in preformed organic molecules (e.g., glucose) could be converted to ATP, which could then be used as a source of energy to drive other metabolic reactions. The

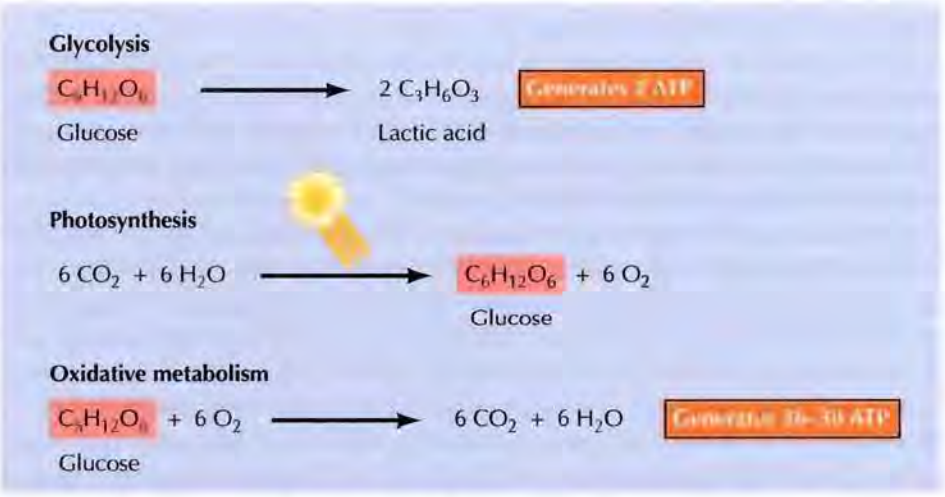


FIGURE 1.4 Generation of metabolic energy Glycolysis is the anaerobic breakdown of glucose to lactic acid. Photosynthesis utilizes energy from sunlight to drive the synthesis of glucose from CO_2 and H_2O, with the release of O_2 as a by-product. The O_2 released by photosynthesis is used in oxidative metabolism, in which glucose is broken down to CO_2 and H_2O, releasing much more energy than is obtained from glycolysis.

development of **photosynthesis** is generally thought to have been the next major evolutionary step, which allowed the cell to harness energy from sunlight and provided independence from the utilization of preformed organic molecules. The first photosynthetic bacteria, which evolved more than 3 billion years ago, probably utilized H_2S to convert CO_2 to organic molecules—a pathway of photosynthesis still used by some bacteria. The use of H_2O as a donor of electrons and hydrogen for the conversion of CO_2 to organic compounds evolved later and had the important consequence of changing Earth's atmosphere. The use of H_2O in photosynthetic reactions produces the by-product free O_2; this mechanism is thought to have been responsible for making O_2 abundant in Earth's atmosphere.

The release of O_2 as a consequence of photosynthesis changed the environment in which cells evolved and is commonly thought to have led to the development of **oxidative metabolism**. Alternatively, oxidative metabolism may have evolved before photosynthesis, with the increase in atmospheric O_2 then providing a strong selective advantage for organisms capable of using O_2 in energy-producing reactions. In either case, O_2 is a highly reactive molecule, and oxidative metabolism, utilizing this reactivity, has provided a mechanism for generating energy from organic molecules that is much more efficient than anaerobic glycolysis. For example, the complete oxidative breakdown of glucose to CO_2 and H_2O yields energy equivalent to that of 36 to 38 molecules of ATP, in contrast to the 2 ATP molecules formed by anaerobic glycolysis. With few exceptions, present-day cells use oxidative reactions as their principal source of energy.

Present-Day Prokaryotes

Present-day prokaryotes, which include all the various types of bacteria, are divided into two groups—the **archaebacteria** and the **eubacteria**—which diverged early in evolution. Some archaebacteria live in extreme environments, which are unusual today but may have been prevalent in primitive Earth. For example, thermoacidophiles live in hot sulfur springs with temperatures as high as 80°C and pH values as low as 2. The eubacteria include the common forms of present-day bacteria—a large group of organisms that live in a wide range of environments, including soil, water, and other organisms (e.g., human pathogens).

Most bacterial cells are spherical, rod-shaped, or spiral, with diameters of 1 to 10 μm. Their DNA contents range from about 0.6 million to 5 million base pairs, an amount sufficient to encode about 5000 different proteins. The largest and most complex prokaryotes are the **cyanobacteria**, bacteria in which photosynthesis evolved.

■ **Existence of organisms in extreme conditions has led to the hypothesis that life could exist in similar environments elsewhere in the solar system. The field of astrobiology (or exobiology) seeks to find signs of this extraterrestrial life.**

The structure of a typical prokaryotic cell is illustrated by ***Escherichia coli (E. coli)***, a common inhabitant of the human intestinal tract (Figure 1.5). The cell is rod-shaped, about 1 μm in diameter and about 2 μm long. Like most other prokaryotes, *E. coli* is surrounded by a rigid **cell wall** composed of polysaccharides and peptides. Beneath the cell wall is the **plasma membrane**, which is a bilayer of phospholipids and associated proteins. Whereas the cell wall is porous and readily penetrated by a variety of molecules, the plasma membrane provides the functional separation between the inside of the cell and its external environment. The DNA of *E. coli* is a single circular molecule in the nucleoid, which, in contrast to the nucleus of eukaryotes, is not surrounded by a membrane separating it from the cytoplasm. The cytoplasm contains approximately 30,000 **ribosomes** (the sites of protein synthesis), which account for its granular appearance.

FIGURE 1.5 Electron micrograph of *E. coli* The cell is surrounded by a cell wall, beneath which is the plasma membrane. DNA is located in the nucleoid. (Menge and Wurtz/Biozentrum, University of Basel/Science Photo Library/Photo Researchers, Inc.)

Eukaryotic Cells

Like prokaryotic cells, all eukaryotic cells are surrounded by a plasma membrane and contain ribosomes. However, eukaryotic cells are much more complex and contain a nucleus, a variety of cytoplasmic organelles, and a cytoskeleton (Figure 1.6). The largest and most prominent organelle of eukaryotic cells is the **nucleus**, with a diameter of approximately 5 μm. The nucleus contains the genetic information of the cell, which in eukaryotes is organized as linear rather than circular DNA molecules. The nucleus is the site of DNA replication and of RNA synthesis; the translation of RNA into proteins takes place on ribosomes in the cytoplasm.

In addition to a nucleus, eukaryotic cells contain a variety of membrane-enclosed organelles within their cytoplasm. These organelles provide compartments in which different metabolic activities are localized. Eukaryotic cells are generally much larger than prokaryotic cells, frequently having a cell volume at least a thousandfold greater. The compartmentalization provided by cytoplasmic organelles is what allows eukaryotic cells to function efficiently. Two of these organelles, **mitochondria** and **chloroplasts**, play critical roles in energy metabolism. Mitochondria, which are found in almost all eukaryotic cells, are the sites of oxidative metabolism and are thus responsible for generating most of the ATP derived from the breakdown of organic molecules. Chloroplasts are the sites of photosynthesis and are found only in the cells of plants and green algae. **Lysosomes** and **peroxisomes** also provide specialized metabolic compartments for the digestion of macromolecules and for various oxidative reactions, respectively. In addition, most plant cells contain large **vacuoles** that perform a variety of functions, including the digestion of macromolecules and the storage of both waste products and nutrients.

Because of the size and complexity of eukaryotic cells, the transport of proteins to their correct destinations within the cell is a formidable task. Two cytoplasmic organelles, the **endoplasmic reticulum** and the **Golgi apparatus**, are specifically devoted to the sorting and transport of proteins destined for secretion, incorporation into the plasma membrane, and incorporation into lysosomes. The endoplasmic reticulum is an extensive network of intracellular membranes, extending from the nuclear membrane throughout the cytoplasm. It functions not only in the processing and transport of proteins, but also in the synthesis of lipids. From the endoplasmic reticulum, proteins are transported within small membrane vesicles to the Golgi apparatus, where they are further processed and sorted for transport to their final destinations. In addition to this role in protein transport, the Golgi apparatus serves as a site of lipid synthesis and (in plant cells) as the site of synthesis of some of the polysaccharides that compose the cell wall.

Eukaryotic cells have another level of internal organization: the **cytoskeleton**, a network of protein filaments extending throughout the cytoplasm. The cytoskeleton provides the structural framework of the cell, determining cell shape and the general organization of the cytoplasm. In

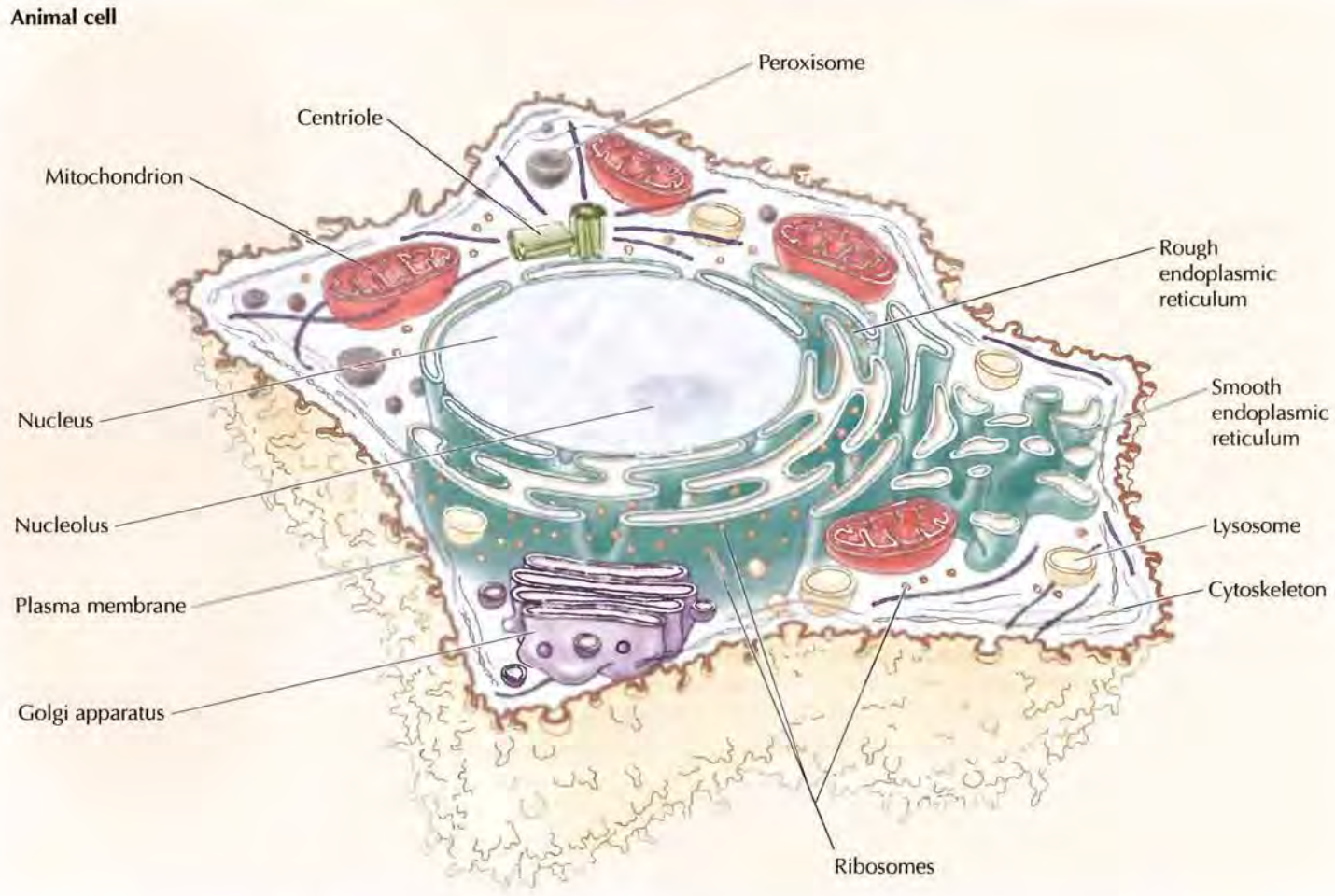


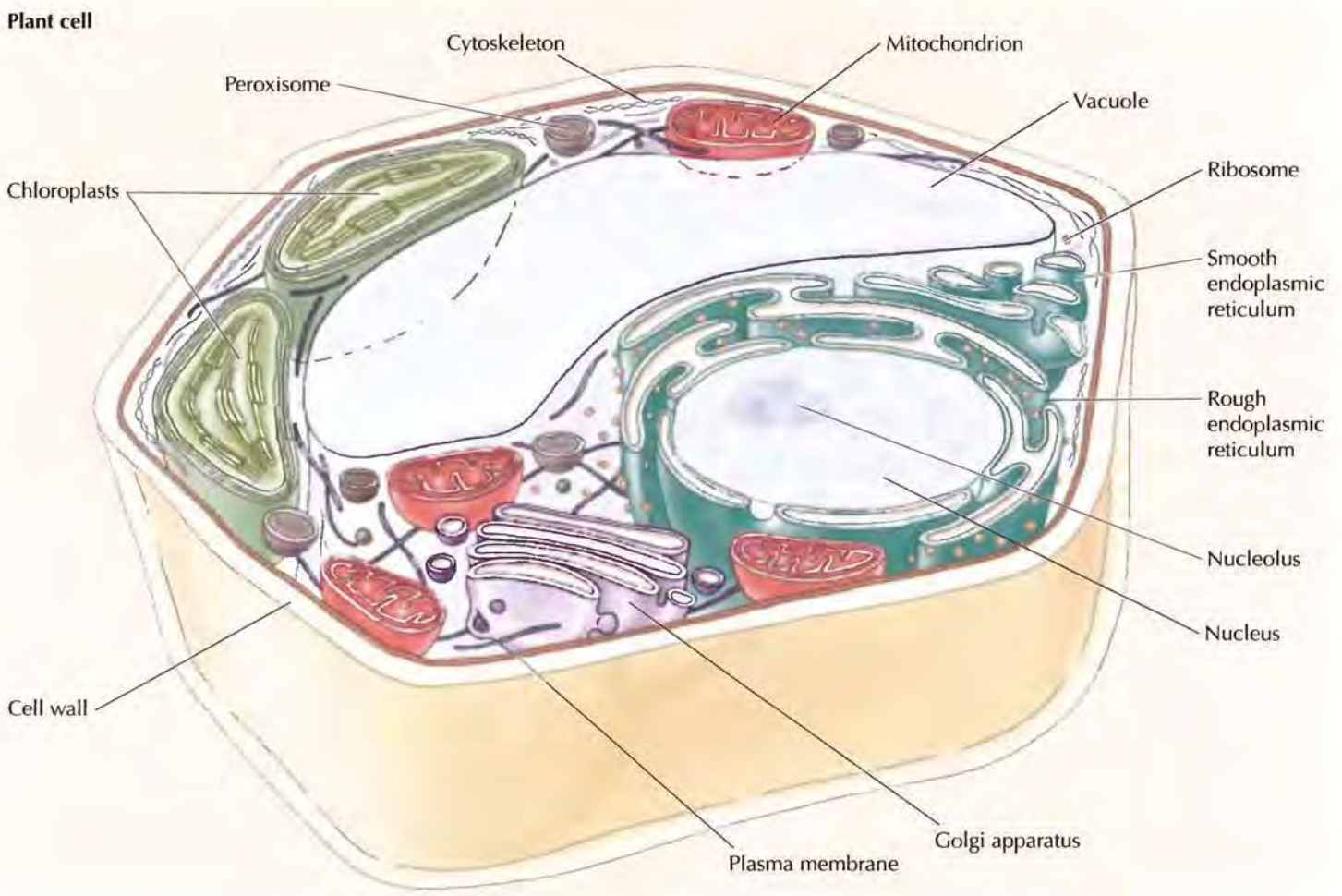
FIGURE 1.6 Structures of animal and plant cells Both animal and plant cells are surrounded by a plasma membrane and contain a nucleus, a cytoskeleton, and many cytoplasmic organelles in common. Plant cells are also surrounded by a cell wall and contain chloroplasts and large vacuoles.

addition, the cytoskeleton is responsible for the movements of entire cells (e.g., the contraction of muscle cells) and for the intracellular transport and positioning of organelles and other structures, including the movements of chromosomes during cell division.

The Origin of Eukaryotes

A critical step in the evolution of eukaryotic cells was the acquisition of membrane-enclosed subcellular organelles, allowing the development of the complexity characteristic of these cells. The organelles of eukaryotes are thought to have arisen by **endosymbiosis**—one cell living inside another. In particular, eukaryotic organelles are thought to have evolved from prokaryotic cells living inside the ancestors of eukaryotes.

The hypothesis that eukaryotic cells evolved by endosymbiosis is particularly well supported by studies of mitochondria and chloroplasts, which are thought to have evolved from eubacteria living in larger cells. Both mitochondria and chloroplasts are similar to bacteria in size, and like bacteria, they reproduce by dividing in two. Most important, both mitochondria and chloroplasts contain their own DNA, which encodes some of their components. The mitochondrial and chloroplast DNAs are replicated each time the organelle divides, and the genes they encode are transcribed within the organelle and translated on organelle ribosomes. Mitochondria and chloroplasts thus contain their own genetic systems, which are distinct from the nuclear genome of the cell. Furthermore, the ribosomes and ribosomal



RNAs of these organelles are more closely related to those of bacteria than to those encoded by the nuclear genomes of eukaryotes.

An endosymbiotic origin for these organelles is now generally accepted, with mitochondria thought to have evolved from aerobic eubacteria and chloroplasts from photosynthetic eubacteria, such as the cyanobacteria. The acquisition of aerobic bacteria would have provided an anaerobic cell with the ability to carry out oxidative metabolism. The acquisition of photosynthetic bacteria would have provided the nutritional independence afforded by the ability to perform photosynthesis. Thus, these endosymbiotic associations were highly advantageous to their partners and were selected for in the course of evolution. Through time, most of the genes originally present in these bacteria apparently became incorporated into the nuclear genome of the cell, so only a few components of mitochondria and chloroplasts are still encoded by the organelle genomes.

■ **Certain present day marine protists engulf algae to serve as endosymbionts that carry out photosynthesis for their hosts.**

The precise origin of eukaryotic cells remains an unsettled issue in our understanding of early evolution. Studies of their DNA sequences indicate that the archaebacteria and eubacteria are as different from each other as either is from present-day eukaryotes. Therefore, a very early event in evolution appears to have been the divergence of two lines of descent from a common prokaryotic ancestor, giving rise to present-day archaebacteria and eubacteria. However, it has proven difficult to determine whether eukaryotes evolved from eubacteria or from archaebacteria. Surprisingly, some eukaryotic genes are more similar to eubacterial genes whereas others

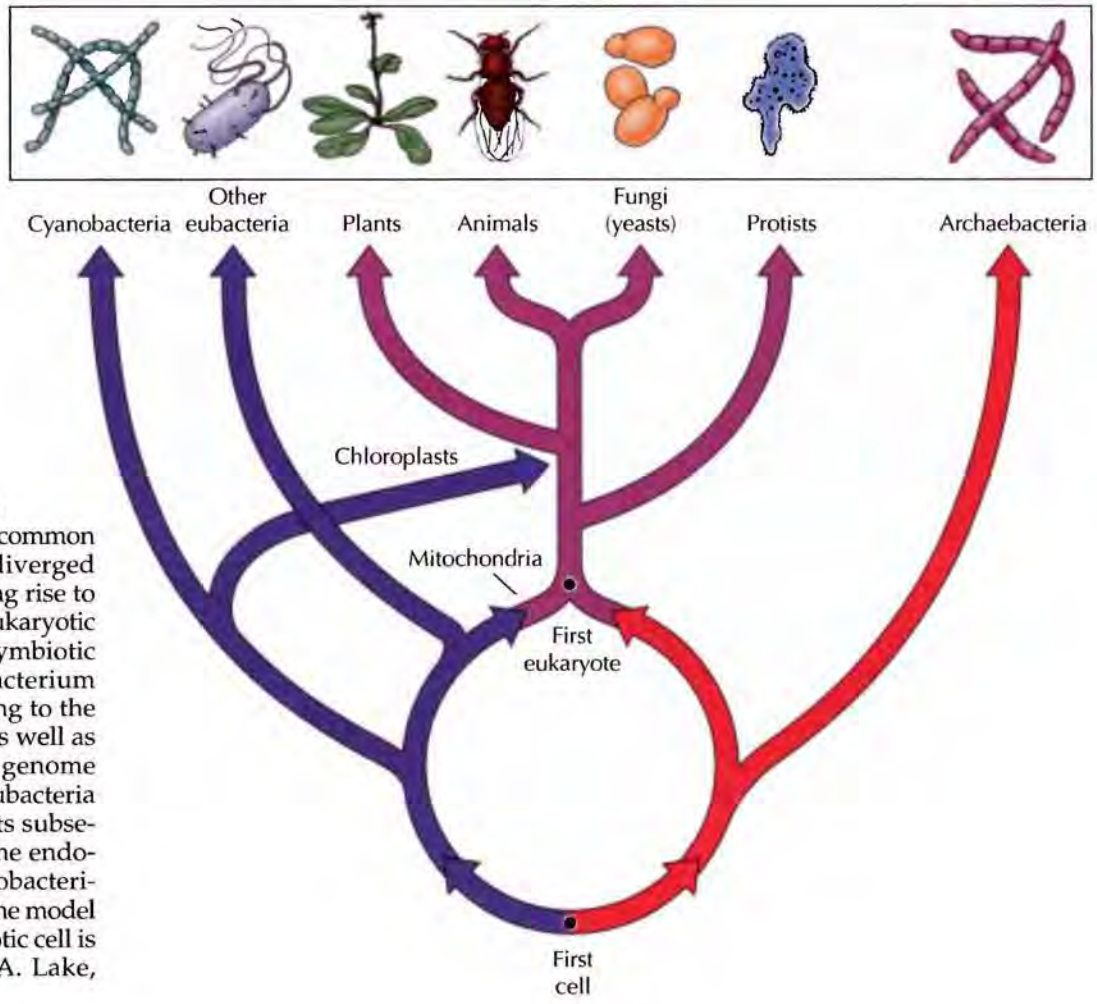


FIGURE 1.7 Evolution of cells
Present-day cells evolved from a common prokaryotic ancestor which diverged along two lines of descent, giving rise to archaebacteria and eubacteria. Eukaryotic cells may have arisen by endosymbiotic association of an aerobic eubacterium with an archaebacterium, leading to the development of mitochondria as well as the formation of a eukaryotic genome with genes derived from both eubacteria and archaebacteria. Chloroplasts subsequently evolved as a result of the endosymbiotic association of a cyanobacterium with the ancestor of plants. The model for formation of the first eukaryotic cell is based on M. C. Rivera and J. A. Lake, 2004. *Nature* 431: 152.

are more similar to archaebacterial genes. The genome of eukaryotes thus appears to consist of some genes derived from eubacteria and others from archaebacteria, rather than reflecting the genome of either a eubacterial or archaebacterial ancestor.

A recent hypothesis explains the mosaic nature of eukaryotic genomes by proposing that the genome of eukaryotes arose from a fusion of archaebacterial and eubacterial genomes (Figure 1.7). According to this proposal, an endosymbiotic association between a eubacterium and an archaebacterium was followed by fusion of the two prokaryotic genomes, giving rise to an ancestral eukaryotic genome with contributions from both eubacteria and archaebacteria. The simplest version of this hypothesis is that an initial endosymbiotic relationship of a eubacterium living inside an archaebacterium gave rise not only to mitochondria but also to the genome of eukaryotic cells, containing genes derived from both prokaryotic ancestors.

The Development of Multicellular Organisms

Many eukaryotes are unicellular organisms that, like bacteria, consist of only single cells capable of self-replication. The simplest eukaryotes are the **yeasts**. Yeasts are more complex than bacteria, but much smaller and simpler than the cells of animals or plants. For example, the commonly studied yeast ***Saccharomyces cerevisiae*** is about 6 μm in diameter and contains 12 million base pairs of DNA (Figure 1.8). Other unicellular eukaryotes, how-

FIGURE 1.8 Scanning electron micrograph of *Saccharomyces cerevisiae* Artificial color has been added to the micrograph. (Andrew Syed/Science Photo Library/Photo Researchers, Inc.)

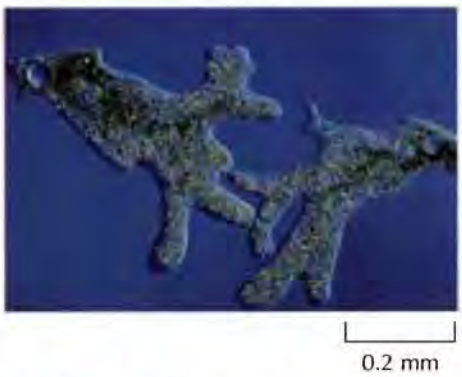


FIGURE 1.9 Light micrograph of *Amoeba proteus* (M. I. Walker/ Photo Researchers, Inc.)

ever, are far more complex cells, some containing as much DNA as human cells have. They include organisms specialized to perform a variety of tasks, including photosynthesis, movement, and the capture and ingestion of other organisms as food. *Amoeba proteus*, for example, is a large, complex cell. Its volume is more than 100,000 times that of *E. coli*, and its length can exceed 1 mm when the cell is fully extended (Figure 1.9). Amoebas are highly mobile organisms that use cytoplasmic extensions, called **pseudopodia**, to move and to engulf other organisms, including bacteria and yeasts, as food. Other unicellular eukaryotes (the green algae) contain chloroplasts and are able to carry out photosynthesis.

Multicellular organisms evolved from unicellular eukaryotes more than 1 billion years ago. Some unicellular eukaryotes form multicellular aggregates that appear to represent an evolutionary transition from single cells to multicellular organisms. For instance, the cells of many algae (e.g., the green alga *Volvox*) associate with each other to form multicellular colonies (Figure 1.10), which are thought to have been the evolutionary precursors of present-day plants. Increasing cell specialization then led to the transition from colonial aggregates to truly multicellular organisms. Continuing cell specialization and division of labor among the cells of an organism have led to the complexity and diversity observed in the many types of cells that make up present-day plants and animals, including human beings.

Plants are composed of fewer cell types than are animals, but each different kind of plant cell is specialized to perform specific tasks required by the organism as a whole (Figure 1.11). The cells of plants are organized into three main tissue systems: ground tissue, dermal tissue, and vascular tissue. The ground tissue contains parenchyma cells, which carry out most of the metabolic reactions of the plant, including photosynthesis. Ground tissue also contains two specialized cell types (collenchyma cells and sclerenchyma cells) that are characterized by thick cell walls and provide structural support to the plant. Dermal tissue covers the surface of the plant and is composed of epidermal cells, which form a protective coat and allow the absorption of nutrients. Finally, several types of elongated cells form the

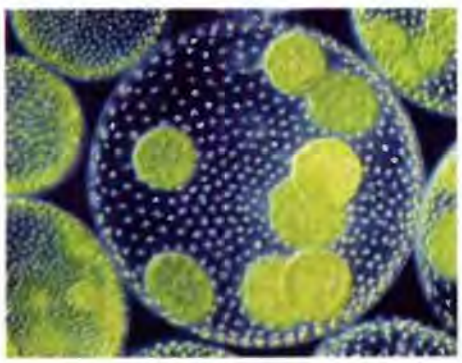


FIGURE 1.10 Colonial green algae Individual cells of *Volvox* form colonies consisting of hollow balls in which hundreds or thousands of cells are embedded in a gelatinous matrix. (Cabisco/ Visuals Unlimited.)

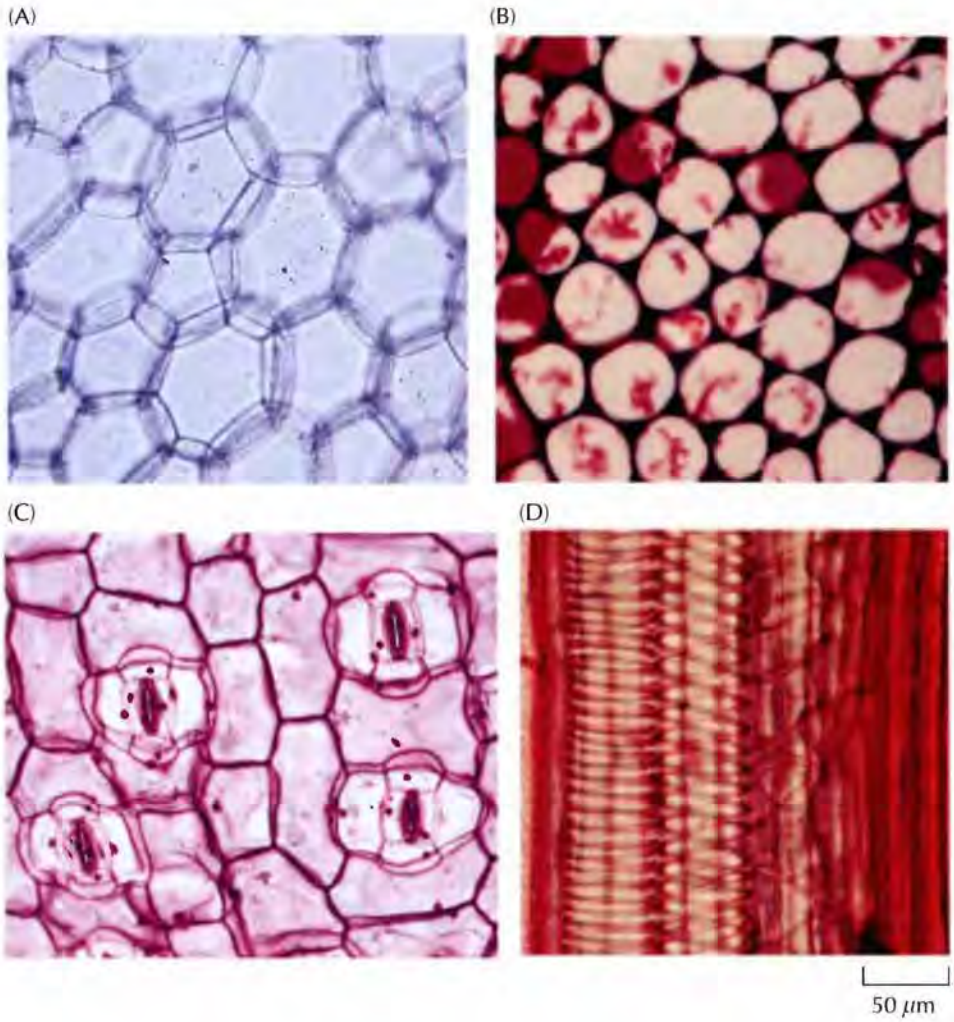


FIGURE 1.11 Light micrographs of representative plant cells
(A) Parenchyma cells, which are responsible for photosynthesis and other metabolic reactions. (B) Collenchyma cells, which are specialized for support and have thickened cell walls. (C) Epidermal cells on the surface of a leaf. Tiny pores (stomata) are flanked by specialized cells called guard cells. (D) Vessel elements and tracheids are elongated cells that are arranged end to end to form vessels of the xylem. (A, Jack M. Bastsack/Visuals Unlimited; B, A. J. Karpoff/Visuals Unlimited; C, Alfred Owczarzak/Biological Photo Service; D, Biophoto Associates/Science Source/Photo Researchers Inc.)

vascular system (the xylem and phloem), which is responsible for the transport of water and nutrients throughout the plant.

The cells found in animals are considerably more diverse than those of plants. The human body, for example, is composed of more than 200 different kinds of cells, which are generally considered to be components of five main types of tissues: epithelial tissue, connective tissue, blood, nervous tissue, and muscle (Figure 1.12). **Epithelial cells** form sheets that cover the surface of the body and line the internal organs. There are many different types of epithelial cells, each specialized for a specific function, including protection (the skin), absorption (e.g., the cells lining the small intestine), and secretion (e.g., cells of the salivary gland). Connective tissues include bone, cartilage, and adipose tissue, each of which is formed by different types of cells (osteoblasts, chondrocytes, and adipocytes, respectively). The loose connective tissue that underlies epithelial layers and fills the spaces between organs and tissues in the body is formed by another cell type, the **fibroblast**. Blood contains several different types of cells: red blood cells (**erythrocytes**) function in oxygen transport and white blood cells (**granulocytes**, **monocytes**, **macrophages,** and **lymphocytes**) function in inflammatory reactions and the immune response. Nervous tissue is composed of supporting cells and nerve cells, or **neurons**, which are highly specialized to transmit signals throughout the body. Various types of sensory cells, such as cells of the eye and ear, are further specialized to receive external signals

from the environment. Finally, several different types of muscle cells are responsible for the production of force and movement.

The evolution of animals clearly involved the development of considerable diversity and specialization at the cellular level. Understanding the mechanisms that control the growth and differentiation of such a complex array of specialized cells, starting from a single fertilized egg, is one of the major challenges facing contemporary cell and molecular biology.

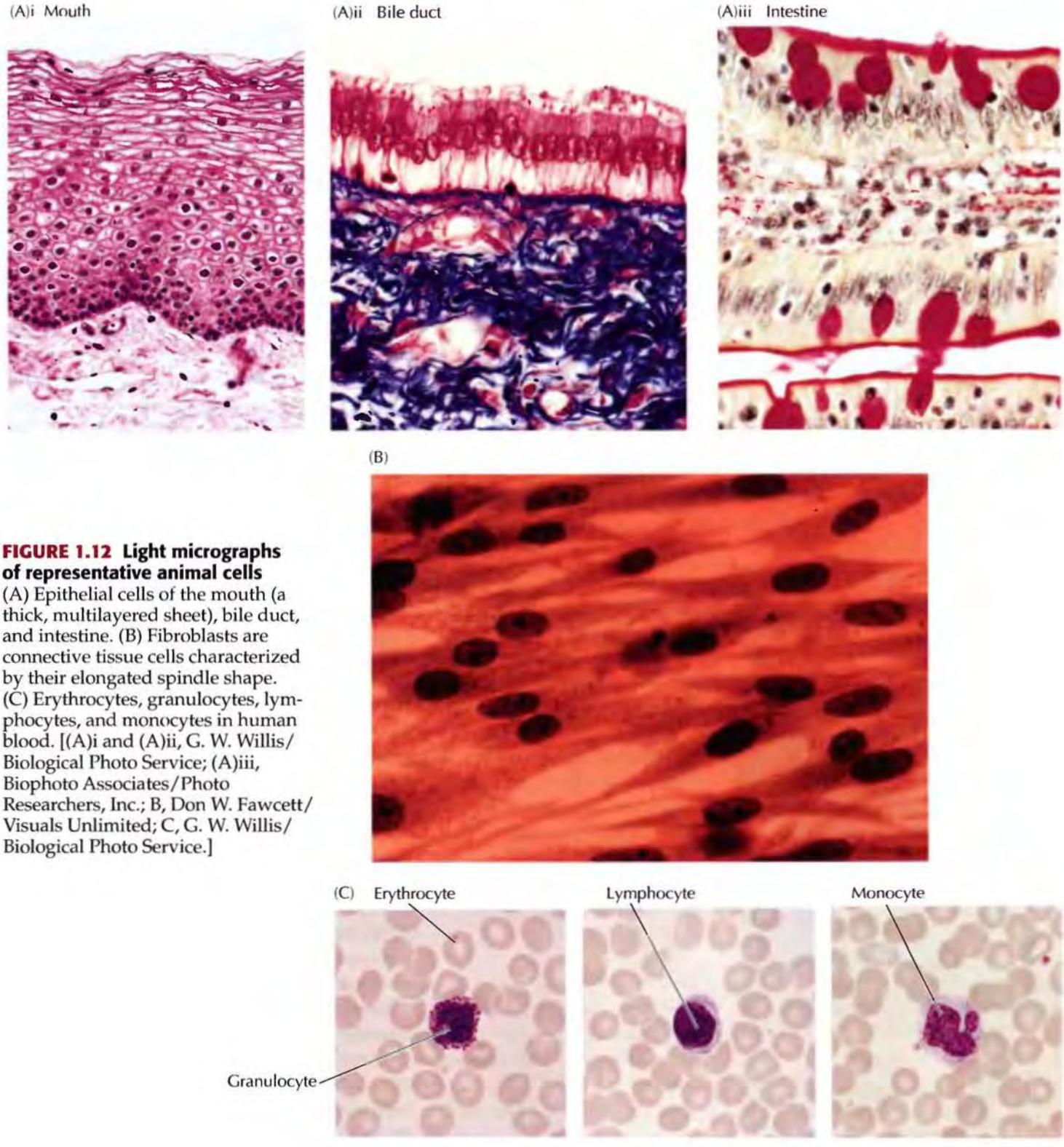


FIGURE 1.12 Light micrographs of representative animal cells
(A) Epithelial cells of the mouth (a thick, multilayered sheet), bile duct, and intestine. (B) Fibroblasts are connective tissue cells characterized by their elongated spindle shape. (C) Erythrocytes, granulocytes, lymphocytes, and monocytes in human blood. [(A)i and (A)ii, G. W. Willis/Biological Photo Service; (A)iii, Biophoto Associates/Photo Researchers, Inc.; B, Don W. Fawcett/Visuals Unlimited; C, G. W. Willis/Biological Photo Service.]

Cells as Experimental Models

The evolution of present-day cells from a common ancestor has important implications for cell and molecular biology as an experimental science. Because the fundamental properties of all cells have been conserved during evolution, the basic principles learned from experiments performed with one type of cell are generally applicable to other cells. On the other hand, because of the diversity of present-day cells, many kinds of experiments can be more readily undertaken with one type of cell than with another. Several different kinds of cells and organisms are commonly used as experimental models to study various aspects of cell and molecular biology. The features of some of these cells that make them particularly advantageous as experimental models are discussed in the sections that follow. In many cases, the availability of complete genome sequences further enhances the value of these organisms as model systems in understanding the molecular biology of cells.

E. coli

Because of their comparative simplicity, prokaryotic cells (bacteria) are ideal models for studying many fundamental aspects of biochemistry and molecular biology. The most thoroughly studied species of bacteria is *E. coli*, which has long been the favored organism for investigation of the basic mechanisms of molecular genetics. Most of our present concepts of molecular biology—including our understanding of DNA replication, the genetic code, gene expression, and protein synthesis—derive from studies of this humble bacterium.

E. coli has been especially useful to molecular biologists because of both its relative simplicity and the ease with which it can be propagated and studied in the laboratory. The genome of *E. coli*, for example, consists of approximately 4.6 million base pairs and contains about 4300 genes. The human genome is nearly a thousand times larger (approximately 3 billion base pairs) and is thought to contain 20,000 to 25,000 genes (Table 1.2). The small size of the *E. coli* genome (which was completely sequenced in 1997) provides obvious advantages for genetic analysis.

TABLE 1.2 DNA Content of Cells

Organism	Haploid DNA content (millions of base pairs)	Number of genes
Bacteria		
Mycoplasma	0.6	470
E. coli	4.6	4300
Unicellular eukaryotes		
Saccharomyces cerevisiae (yeast)	12	6000
Dictyostelium discoideum	70	Not known
Euglena	3000	Not known
Plants		
Arabidopsis thaliana	125	26,000
Zea mays (corn)	5000	Not known
Animals		
Caenorhabditis elegans (nematode)	97	19,000
Drosophila melanogaster (fruit fly)	180	14,000
Chicken	1200	20–23,000
Zebrafish	1700	Not known
Mouse	3000	20–25,000
Human	3000	20–25,000

Molecular genetic experiments are further facilitated by the rapid growth of *E. coli* under well-defined laboratory conditions. Under optimal culture conditions, *E. coli* divide every 20 minutes. Moreover, a clonal population of *E. coli*, in which all cells are derived by division of a single cell of origin, can be readily isolated as a colony grown on semisolid agar-containing medium (Figure 1.13). Because bacterial colonies containing as many as 10^8 cells can develop overnight, selecting genetic variants of an *E. coli* strain—for example, mutants that are resistant to an antibiotic, such as penicillin—is easy and rapid. The ease with which such mutants can be selected and analyzed was critical to the success of

experiments that defined the basic principles of molecular genetics, discussed in Chapter 4.

The nutrient mixtures in which *E. coli* divide most rapidly include glucose, salts, and various organic compounds, such as amino acids, vitamins, and nucleic acid precursors. However, *E. coli* can also grow in much simpler media consisting only of salts, a source of nitrogen (such as ammonia), and a source of carbon and energy (such as glucose). In such a medium, the bacteria grow a little more slowly (with a division time of about 40 minutes) because they must synthesize all their own amino acids, nucleotides, and other organic compounds. The ability of *E. coli* to carry out these biosynthetic reactions in simple defined media has made them extremely useful in elucidating the biochemical pathways involved. Thus, the rapid growth and simple nutritional requirements of *E. coli* have greatly facilitated fundamental experiments in both molecular biology and biochemistry.

FIGURE 1.13 Bacterial colonies
Photograph of colonies of *E. coli* growing on the surface of an agar-containing medium. (A. M. Siegelman/Visuals Unlimited.)

Yeasts

Although bacteria have been an invaluable model for studies of many conserved properties of cells, they obviously cannot be used to study aspects of cell structure and function that are unique to eukaryotes. Yeasts, the simplest eukaryotes, have a number of experimental advantages similar to those of *E. coli*. Consequently, yeasts have provided a crucial model for studies of many fundamental aspects of eukaryotic cell biology.

The genome of the most frequently studied yeast, *Saccharomyces cerevisiae*, consists of 12 million base pairs of DNA and contains about 6000 genes. Although the yeast genome is approximately three times larger than that of *E. coli*, it is far more manageable than the genomes of more complex eukaryotes, such as humans. Yet, even in its simplicity, the yeast cell exhibits the typical features of eukaryotic cells (Figure 1.14): It contains a distinct nucleus surrounded by a nuclear membrane, its genomic DNA is organized as 16 linear chromosomes, and its cytoplasm contains a cytoskeleton and subcellular organelles.

Yeasts can be readily grown in the laboratory and can be studied by many of the same molecular genetic approaches that have proved so successful with *E. coli*. Although yeasts do not replicate as rapidly as bacteria, they still divide as frequently as every 2 hours and they can easily be grown as colonies from a single cell. Consequently, yeasts can be used for a variety of genetic manipulations similar to those that can be performed using bacteria.

These features have made yeast cells the most approachable eukaryotic cells from the standpoint of molecular biology. Yeast mutants have been important in understanding many fundamental processes in eukaryotes, including DNA replication, transcription, RNA processing, protein sorting, and the regulation of cell division, as will be discussed in subsequent chapters. The unity of molecular cell biology is made abundantly clear by the fact that the general principles of cell structure and function revealed by studies of yeasts apply to all eukaryotic cells.

FIGURE 1.14 Electron micrograph of *Saccharomyces cerevisiae*
(David Scharf/Peter Arnold, Inc.)

Caenorhabditis elegans

The unicellular yeasts are important models for studies of eukaryotic cells, but understanding the development of multicellular organisms requires the experimental analysis of plants and animals, organisms that are more complex. The nematode ***Caenorhabditis elegans*** (Figure 1.15) possesses several notable features that make it one of the most widely used models for studies of animal development and cell differentiation.

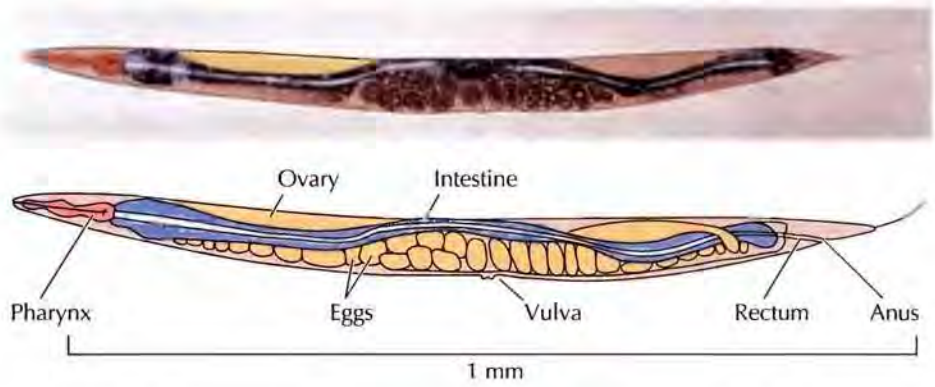


FIGURE 1.15 *Caenorhabditis elegans* (From J. E. Sulston and H. R. Horvitz, 1977. *Dev. Biol.* 56: 110.)

Although the genome of *C. elegans* (approximately 100 million base pairs) is larger than those of unicellular eukaryotes, it is simpler and more manageable than the genomes of most animals. Its complete sequence has been determined, revealing that the genome of *C. elegans* contains approximately 19,000 genes—more than three times the number of genes in yeast, and nearly the same number of genes in humans. Biologically, *C. elegans* is a relatively simple multicellular organism: Adult worms consist of only 959 somatic cells, plus 1000 to 2000 germ cells. In addition, *C. elegans* can be easily grown and subjected to genetic manipulations in the laboratory.

The simplicity of *C. elegans* has enabled the course of its development to be studied in detail by microscopic observation. Such analyses have successfully traced the embryonic origin and lineage of all the cells in the adult worm. Genetic studies have also identified many of the mutations responsible for developmental abnormalities, leading to the isolation and characterization of critical genes that control nematode development and differentiation. Importantly, similar genes have also been found to function in complex animals (including humans), making *C. elegans* an important model for studies of animal development.

Drosophila melanogaster

Like *C. elegans*, the fruit fly ***Drosophila melanogaster*** (Figure 1.16) has been a crucial model organism in developmental biology. The genome of *Drosophila* is 180 million base pairs, larger than that of *C. elegans*, but the *Drosophila* genome only contains about 14,000 genes. Furthermore, *Drosophila* can be easily maintained and bred in the laboratory, and the short reproductive cycle of *Drosophila* (about 2 weeks) makes it a very useful organism for genetic experiments. Many fundamental concepts of genetics—such as the relationship between genes and chromosomes—were derived from studies of *Drosophila* early in the twentieth century (see Chapter 4).

FIGURE 1.16 *Drosophila melanogaster* (Darwin Dale/ Photo Researchers, Inc.)

Extensive genetic analysis of *Drosophila* has uncovered many genes that control development and differentiation, and current methods of molecular biology have allowed the functions of these genes to be analyzed in detail. Consequently, studies of *Drosophila* have led to striking advances in understanding the molecular mechanisms that govern animal development, particularly with respect to formation of the body plan of complex multicellular organisms. As with *C. elegans*, similar genes and mechanisms exist in vertebrates, validating the use of *Drosophila* as a major experimental model in contemporary developmental biology.

Arabidopsis thaliana

The study of plant molecular biology and development is an active and expanding field of considerable economic importance as well as intellectual interest. Since the genomes of plants cover a range of complexity comparable to that of animal genomes (see Table 1.2), an optimal model for studies of plant development would be a relatively simple organism with some of the advantageous properties of *C. elegans* and *Drosophila*. The small flowering plant ***Arabidopsis thaliana*** (Figure 1.17) meets these criteria and is therefore widely used as a model to study the molecular biology of plants.

Arabidopsis is notable for its genome of only about 125 million base pairs. Although *Arabidopsis* contains a total of about 26,000 genes, many of these are repeated, so the number of unique genes in *Arabidopsis* is approximately 15,000—a complexity similar to that of *C. elegans* and *Drosophila*. In addition, *Arabidopsis* is relatively easy to grow in the laboratory, and methods for molecular genetic manipulations of this plant have been developed. These studies have led to the identification of genes involved in various aspects of plant development, such as the development of flowers. Analysis of these genes points to many similarities, but also to striking differences, between the mechanisms that control the development of plants and animals.

FIGURE 1.17 ***Arabidopsis thaliana*** (Jeremy Burgess/Photo Researchers, Inc.)

Vertebrates

The most complex animals are the vertebrates, including humans and other mammals. The human genome is approximately 3 billion base pairs—about 20–30 times larger than the genomes of *C. elegans*, *Drosophila*, or *Arabidopsis* – and contains 20,000 to 25,000 genes. Moreover, the human body is composed of more than 200 different kinds of specialized cell types. This complexity makes the vertebrates difficult to study from the standpoint of cell and molecular biology, but much of the interest in biological sciences nonetheless stems from the desire to understand the human organism. Moreover, an understanding of many questions of immediate practical importance (e.g., in medicine) must be based directly on studies of human (or closely related) cell types.

One important approach to studying human and other mammalian cells is to grow isolated cells in culture, where they can be manipulated under controlled laboratory conditions. The use of cultured cells has allowed studies of many aspects of mammalian cell biology, including experiments that have elucidated the mechanisms of DNA replication, gene expression, protein synthesis and processing, and cell division. Moreover, the ability to culture cells in chemically defined media has allowed studies of the signaling mechanisms that normally control cell growth and differentiation within the intact organism.

The specialized properties of some highly differentiated cell types have made them important models for studies of particular aspects of cell biology. Muscle cells, for example, are highly specialized to undergo contraction, producing force and movement. Because of this specialization, muscle cells are a crucial model for studying cell movement at the molecular level. Another example is provided by nerve cells (neurons), which are specialized to conduct electrochemical signals over long distances. In humans, nerve cell axons may be more than a meter long, and some invertebrates, such as the squid, have giant neurons with axons as large as 1 mm in diameter. Because of their highly specialized structure and function, these giant neurons have provided important models for studies of ion transport across

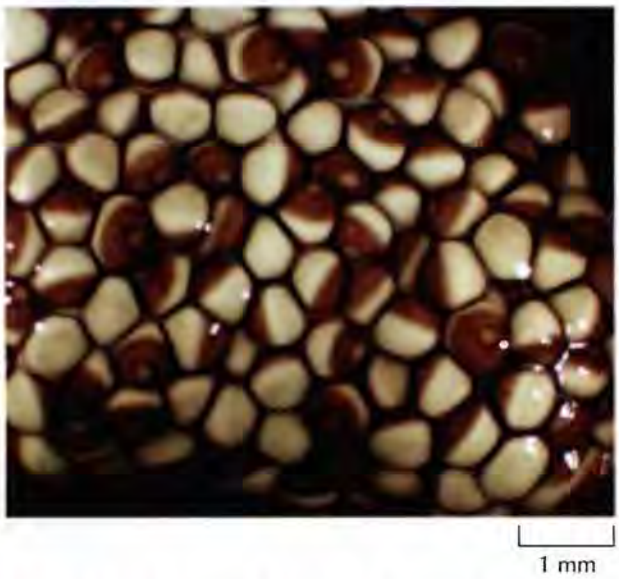


FIGURE 1.18 Eggs of the frog *Xenopus laevis* (Courtesy of Michael Danilchik and Kimberly Ray.)

the plasma membrane, and of the role of the cytoskeleton in the transport of cytoplasmic organelles.

The frog ***Xenopus laevis*** is an important model for studies of early vertebrate development. *Xenopus* eggs are unusually large cells, with a diameter of approximately 1 mm (Figure 1.18). Because those eggs develop outside of the mother, all stages of development from egg to tadpole can be readily studied in the laboratory. In addition, *Xenopus* eggs can be obtained in large numbers, facilitating biochemical analysis. Because of these technical advantages, *Xenopus* has been widely used in studies of developmental biology and has provided important insights into the molecular mechanisms that control development, differentiation, and embryonic cell division.

The **zebrafish** (Figure 1.19) possesses a number of advantages for genetic studies of vertebrate development. These small fish are easy to maintain in the laboratory and they reproduce rapidly. In addition, the embryos develop outside of the mother and are transparent, so that early stages of development can be easily observed. Powerful methods have been developed to facilitate the isolation of mutations affecting zebrafish development, and several thousand such mutations have now been identified. Because the zebrafish is an easily studied vertebrate, it promises to bridge the gap between humans and the simpler invertebrate systems, such as *C. elegans* and *Drosophila*. Its utility as a model organism will be further enhanced with the completion of its genome sequence.

Among mammals, the mouse is the most suitable for genetic analysis, which is facilitated by the availability of its complete genome sequence. Although the technical difficulties in studying mouse genetics (compared, for example, to the genetics of yeasts or *Drosophila*) are formidable, many mutations affecting mouse development have been identified. Most important, recent advances in molecular biology have enabled the production of genetically engineered mice in which specific mutant genes have been introduced into the mouse germ line, allowing the functions of these genes to be studied in the context of the whole animal. The suitability of the mouse as a model for human development is indicated not only by the similarity of the mouse and human genomes but also by the fact that mutations

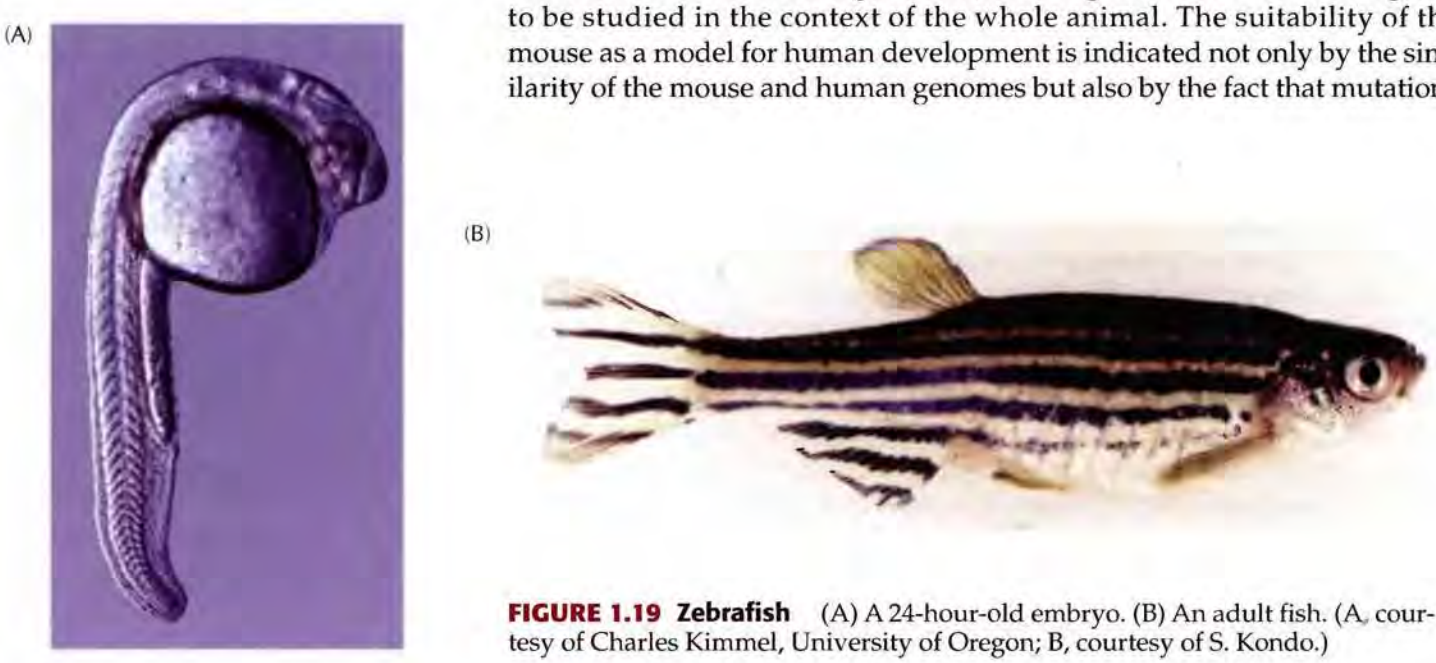


FIGURE 1.19 Zebrafish (A) A 24-hour-old embryo. (B) An adult fish. (A, courtesy of Charles Kimmel, University of Oregon; B, courtesy of S. Kondo.)

FIGURE 1.20 The mouse as a model for human development A child and a mouse show similar defects in pigmentation (piebaldism) as a result of mutations in a gene required for normal migration of melanocytes (the cells responsible for skin pigmentation) during embryonic development. (Courtesy of R. A. Fleischman, Markey Cancer Center, University of Kentucky.)

in homologous genes result in similar developmental defects in both species; piebaldism is a striking example (Figure 1.20).

Tools of Cell Biology

As in all experimental sciences, research in cell biology depends on the laboratory methods that can be used to study cell structure and function. Many important advances in understanding cells have directly followed the development of new methods that have opened novel avenues of investigation. An appreciation of the experimental tools available to the cell biologist is thus critical to understanding both the current status and future directions of this rapidly moving area of science. Some of the important general methods of cell biology are described in the sections that follow. Other experimental approaches, including the methods of biochemistry and molecular biology, will be discussed in later chapters.

Light Microscopy

Because most cells are too small to be seen by the naked eye, the study of cells has depended heavily on the use of microscopes. Indeed, the very discovery of cells arose from the development of the microscope: Robert Hooke first coined the term "cell" following his observations of a piece of cork with a simple light microscope in 1665 (Figure 1.21). Using a microscope that magnified objects up to about 300 times their actual size, Antony van Leeuwenhoek, in the 1670s, was able to observe a variety of different types of cells, including sperm, red blood cells, and bacteria. The proposal of the cell theory by Matthias Schleiden and Theodor Schwann in 1838 may be seen as the birth of contemporary cell biology. Microscopic studies of plant tissues by Schleiden and of animal tissues by Schwann led to the same conclusion: All organisms are composed of cells. Shortly thereafter, it was recognized that cells are not formed *de novo* but arise only from division of preexisting cells. Thus, the cell achieved its current recognition as the

FIGURE 1.21 The cellular structure of cork A reproduction of Robert Hooke's drawing of a thin slice of cork examined with a light microscope. The "cells" that Hooke observed were actually only the cell walls remaining from cells that had long since died.

fundamental unit of all living organisms because of observations made with the light microscope.

The light microscope remains a basic tool of cell biologists, with technical improvements allowing the visualization of ever-increasing details of cell structure. Contemporary light microscopes are able to magnify objects up to about a thousand times. Since most cells are between 1 and 100 μm in diameter, they can be observed by light microscopy, as can some of the larger subcellular organelles, such as nuclei, chloroplasts, and mitochondria. However, the light microscope is not sufficiently powerful to reveal fine details of cell structure, for which **resolution**—the ability of a microscope to distinguish objects separated by small distances—is even more important than magnification. Images can be magnified as much as desired (for example, by projection onto a large screen), but such magnification does not increase the level of detail that can be observed.

The limit of resolution of the light microscope is approximately 0.2 μm; two objects separated by less than this distance appear as a single image, rather than being distinguished from one another. This theoretical limitation of light microscopy is determined by two factors—the wavelength (λ) of visible light and the light-gathering power of the microscope lens (numerical aperture, *NA*)—according to the following equation:

$$\text{Resolution} = \frac{0.61\lambda}{NA}$$

The wavelength of visible light is 0.4 to 0.7 μm, so the value of λ is fixed at approximately 0.5 μm for the light microscope. The numerical aperture can be envisioned as the size of the cone of light that enters the microscope lens after passing through the specimen (Figure 1.22). It is given by the equation

$$NA = \eta \sin \alpha$$

where η is the refractive index of the medium through which light travels between the specimen and the lens. The value of η for air is 1.0, but it can be

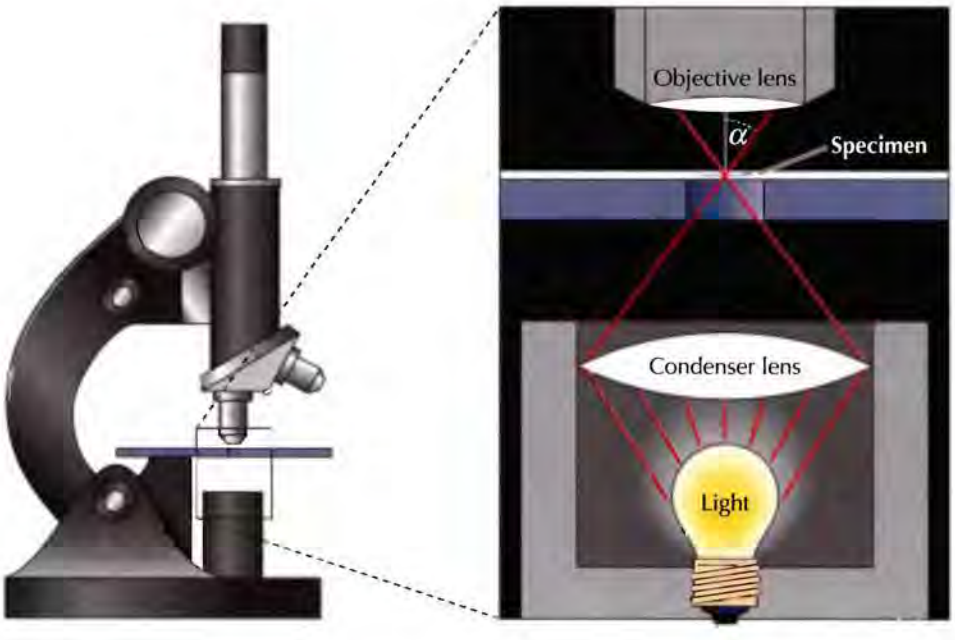


FIGURE 1.22 Numerical aperture Light is focused on the specimen by the condenser lens and then collected by the objective lens of the microscope. The numerical aperture is determined by the angle of the cone of light entering the objective lens (α) and by the refractive index of the medium (usually air or oil) between the lens and the specimen.

increased to a maximum of approximately 1.4 by using an oil-immersion lens to view the specimen through a drop of oil. The angle α corresponds to half the width of the cone of light collected by the lens. The maximum value of α is 90°, at which $\sin \alpha = 1$, so the highest possible value for the numerical aperture is 1.4.

The theoretical limit of resolution of the light microscope can therefore be calculated as follows:

$$\text{Resolution} = \frac{0.61 \times 0.5}{1.4} = 0.22\ \mu\text{m}$$

Microscopes capable of achieving this level of resolution had already been made by the end of the nineteenth century; further improvements in this aspect of light microscopy cannot be expected.

Several different types of light microscopy are routinely used to study various aspects of cell structure. The simplest is **bright-field microscopy**, in which light passes directly through the cell and the ability to distinguish different parts of the cell depends on contrast resulting from the absorption of visible light by cell components. In many cases, cells are stained with dyes that react with proteins or nucleic acids in order to enhance the contrast between different parts of the cell. Prior to staining, specimens are usually treated with fixatives (such as alcohol, acetic acid, or formaldehyde) to stabilize and preserve their structures. The examination of fixed and stained tissues by bright-field microscopy is the standard approach for the analysis of tissue specimens in histology laboratories (Figure 1.23). Such staining procedures kill the cells, however, and therefore are not suitable for many experiments in which the observation of living cells is desired.

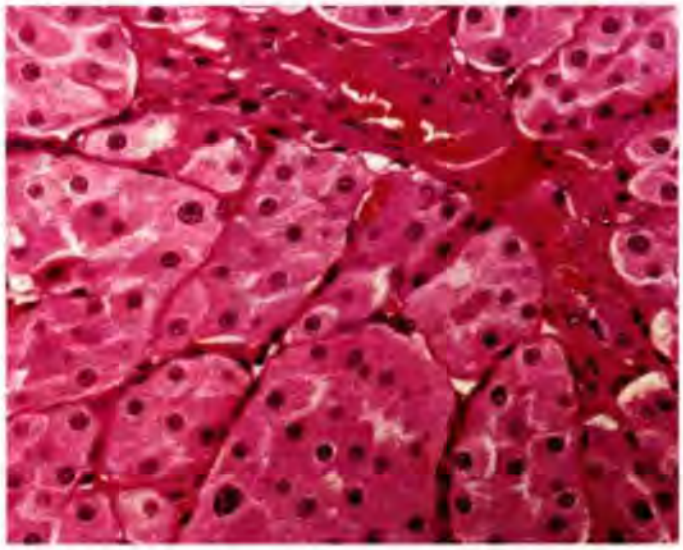


FIGURE 1.23 Bright-field micrograph of stained tissue Section of a benign kidney tumor. (G. W. Willis/Biological Photo Service.)

Without staining, the direct passage of light does not provide sufficient contrast to distinguish many parts of the cell, limiting the usefulness of bright-field microscopy. However, optical variations of the light microscope can be used to enhance the contrast between light waves passing through regions of the cell with different densities. The two most common methods for visualizing living cells are **phase-contrast microscopy** and **differential interference-contrast microscopy** (Figure 1.24). Both kinds of microscopy use optical systems that convert variations in density or thickness between different parts of the cell to differences in contrast that can be seen in the final image. In bright-field microscopy, transparent structures (such as the nucleus) have little contrast because they absorb light poorly. However, light is slowed down as it passes through these structures so that its phase is altered compared to light that has passed through the surrounding cytoplasm. Phase-contrast and differential interference-contrast microscopy convert these differences in phase to differences in contrast, thereby yielding improved images of live, unstained cells.

The power of the light microscope has been considerably expanded by the use of video cameras and computers for image analysis and processing. Such electronic image-processing systems can substantially enhance the

FIGURE 1.24 Microscopic observation of living cells Photomicrographs of human cheek cells obtained with (A) bright-field, (B) phase-contrast, and (C) differential interference-contrast microscopy. (Courtesy of Mort Abramowitz, Olympus America, Inc.)

FIGURE 1.25 Video-enhanced differential interference-contrast microscopy Electronic image processing allows the visualization of single microtubules. (Courtesy of E. D. Salmon, University of North Carolina, Chapel Hill.)

contrast of images obtained with the light microscope, allowing the visualization of small objects that otherwise could not be detected. For example, **video-enhanced differential interference-contrast microscopy** has allowed visualization of the movement of organelles along microtubules, which are cytoskeletal protein filaments with a diameter of only 0.025 μm (Figure 1.25). However, this enhancement does not overcome the theoretical limit of resolution of the light microscope, approximately 0.2 μm. Thus, although video enhancement allows the visualization of microtubules, the microtubules appear as blurred images at least 0.2 μm in diameter and an individual microtubule cannot be distinguished from a bundle of adjacent structures.

Light microscopy has been brought to the level of molecular analysis by methods for labeling specific molecules so that they can be visualized within cells. Specific genes or RNA transcripts can be detected by hybridization with nucleic acid probes of complementary sequence, and proteins can be detected using appropriate antibodies (see Chapter 4). Both nucleic acid probes and antibodies can be labeled with a variety of tags that allow their visualization in the light microscope, making it possible to determine the location of specific molecules within individual cells.

Fluorescence microscopy is a widely used and very sensitive method for studying the intracellular distribution of molecules (Figure 1.26). A fluorescent dye is used to label the molecule of interest within either fixed or living cells. The fluorescent dye is a molecule that absorbs light at one wavelength and emits light at a second wavelength. This fluorescence is detected by illuminating the specimen with a wavelength of light that excites the fluorescent dye and then using appropriate filters to detect the specific wavelength of light that the dye emits. Fluorescence microscopy can be used to

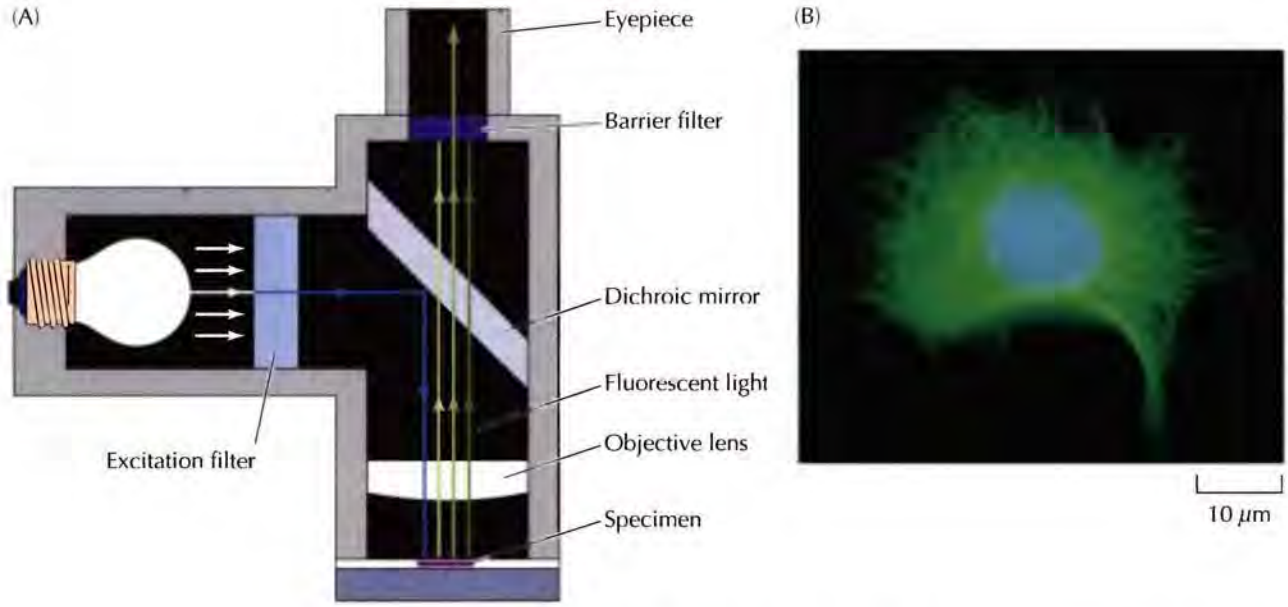


FIGURE 1.26 Fluorescence microscopy (A) Light passes through an excitation filter to select light of the wavelength (e.g., blue) that excites the fluorescent dye. A dichroic mirror then deflects the excitation light down to the specimen. The fluorescent light emitted by the specimen (e.g., green) then passes through the dichroic mirror and a second filter (the barrier filter) to select light of the wavelength emitted by the dye. (B) Fluorescence micrograph of a newt lung cell in which the DNA is stained blue and microtubules in the cytoplasm are stained green. (Conly S. Rieder/Biological Photo Service.)

study a variety of molecules within cells. One frequent application is to label antibodies directed against a specific protein with fluorescent dyes, so that the intracellular distribution of the protein can be determined.

An important recent advance in fluorescence microscopy has been the use of the **green fluorescent protein (GFP)** of jellyfish to visualize proteins within living cells. GFP can be fused to any protein of interest using standard methods of recombinant DNA, and the GFP-tagged protein can then be expressed in cells and detected by fluorescence microscopy, without the need to fix and stain the cell as would be required for the detection of proteins with antibodies. Because of its versatility, the use of GFP has become extremely widespread in cell biology, and has been used to study the localization of a wide variety of proteins within living cells (Figure 1.27). Several related fluorescent proteins with blue, yellow or red emissions are also available, further expanding the utility of this technique.

A variety of methods have been developed to follow the movement and interactions of GFP-labeled proteins within living cells. One widely used method for studying the movements of GFP-labeled proteins is **fluorescence recovery after photobleaching (FRAP)** (Figure 1.28). In this technique, a region of interest in a cell expressing a GFP-labeled protein is bleached by exposure to high-intensity light. Fluorescence recovers over time due to the movement of unbleached GFP-labeled molecules into the bleached region, allowing the rate at which the protein moves within the cell to be determined.

The interactions of two proteins with one another within a cell can be analyzed by a technique called **fluorescence resonance energy transfer (FRET)** (Figure 1.29). In FRET experiments, the two proteins of interest are coupled to different fluorescent dyes, such as two variants of GFP. The GFP variants are chosen to absorb and emit distinct wavelengths of light, such that the light emitted by one GFP variant excites the second. Interaction between the two proteins can then be detected by illuminating the cell with

■ **Semiconducting nanocrystals (termed quantum dots) are increasingly being use in place of fluorescent dyes for many applications in fluorescence microscopy. Quantum dots fluoresce brighter and are more stable than traditional fluorescent dyes.**

■ **GFP is derived from the Pacific jellyfish *Aequoria victoria*. Proteins that fluoresce in different colors have been isolated from other marine organisms.**

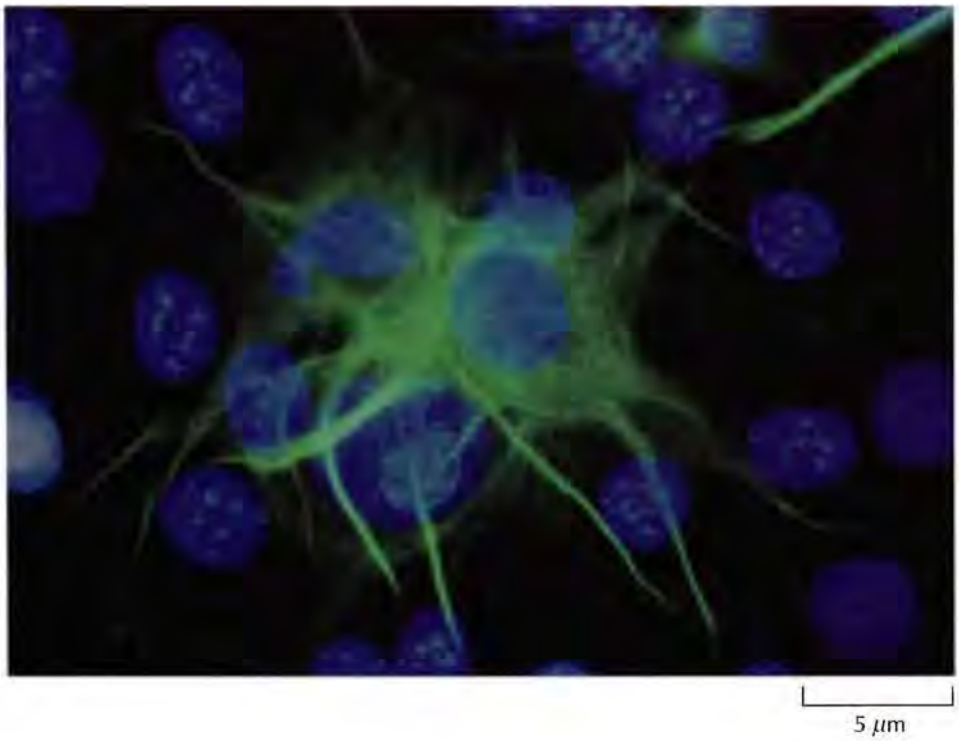


FIGURE 1.27 Fluorescence microscopy of a protein labeled with GFP A microtubule-associated protein fused to GFP was introduced into mouse neurons in culture and visualized by fluorescence microscopy. Nuclei are stained blue. (From A. Cariboni, 2004. *Nature Cell Biol.* 6:929.)

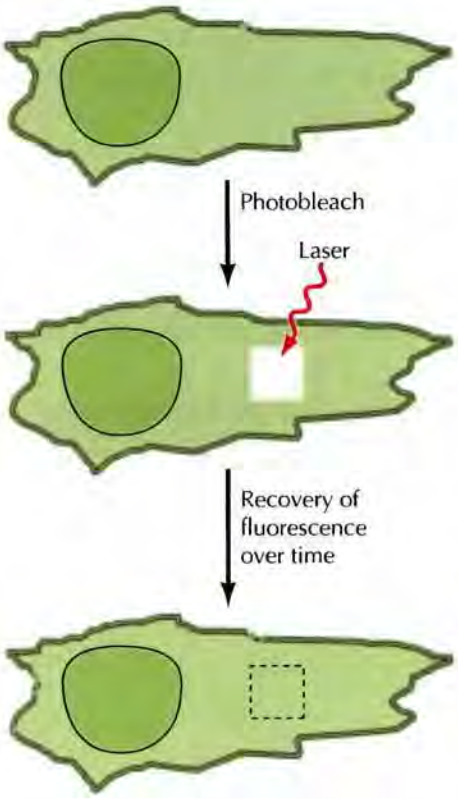


FIGURE 1.28 Fluorescence recovery after photobleaching (FRAP) Region of a cell expressing a GFP-labeled protein is bleached with a laser. Fluorescence recovers over time as unbleached GFP-labeled molecules diffuse into the bleached region. The rate of recovery of fluorescence therefore provides a measurement of the rate of protein movement within the cell.

a wavelength of light that excites the first GFP variant and analyzing the wavelength of emitted light. If the proteins coupled to these GFP variants interact within the cell, the fluorescent molecules will be brought close together and the light emitted by the first GFP variant will excite the second, resulting in emission of light at the wavelength characteristic of the second variant of GFP.

The images obtained by conventional fluorescence microscopy are blurred as a result of out-of-focus fluorescence. These images can be improved by a computational approach called image deconvolution, in which a computer analyzes images obtained from different depths of focus and generates a sharper image as would have been expected from a single focal point. Alternatively, **confocal microscopy** allows images of increased contrast and detail to be obtained by analyzing fluorescence from only a single point in the specimen. A small point of light, usually supplied by a laser, is focused on the specimen at a particular depth. The emitted fluorescent light is then collected using a detector, such as a video camera. Before the emitted light reaches the detector, however, it must pass through a pinhole aperture (called a confocal aperture) placed at precisely the point

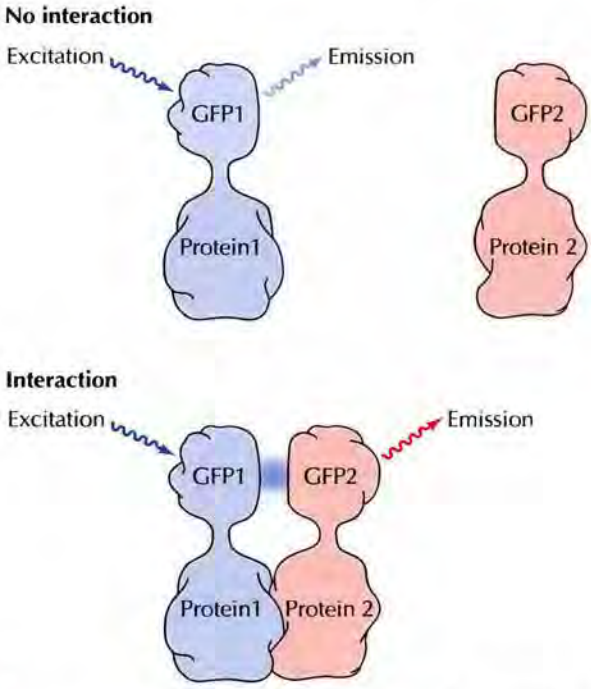


FIGURE 1.29 Fluorescence resonance energy transfer (FRET) Two proteins are fused to different variants of GFP (GFP1 and GFP2) with distinct wavelengths for excitation and emission, chosen such that the light emitted by GFP1 excites GFP2. Cells are then illuminated with a wavelength of light that excites GFP1. If the proteins do not interact, light emitted by GFP1 will be detected. However, if the proteins do interact, GFP1 will excite GFP2, and light emitted by GFP2 will be detected.

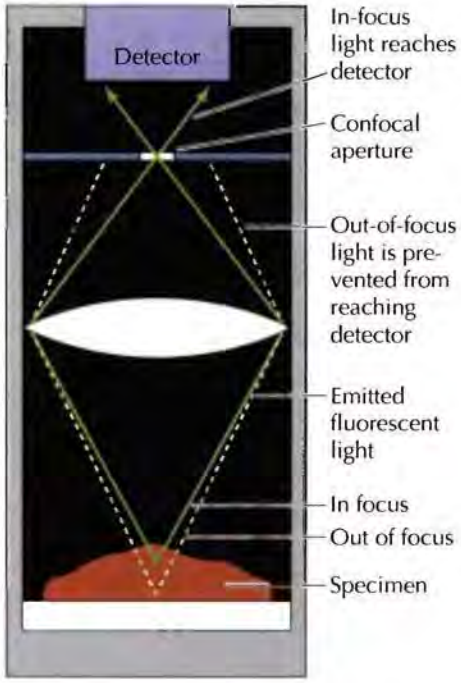


FIGURE 1.30 Confocal microscopy A pinpoint of light is focused on the specimen at a particular depth, and emitted fluorescent light is collected by a detector. Before reaching the detector, the fluorescent light emitted by the specimen must pass through a confocal aperture placed at the point where light emitted from the chosen depth of the specimen comes into focus. As a result, only in-focus light emitted from the chosen depth of the specimen is detected.

where light emitted from the chosen depth of the specimen comes to a focus (Figure 1.30). Consequently, only light emitted from the plane of focus is able to reach the detector. Scanning across the specimen generates a two-dimensional image of the plane of focus, a much sharper image than that obtained with standard fluorescence microscopy (Figure 1.31). Moreover, a series of images obtained at different depths can be used to reconstruct a three-dimensional image of the sample.

Multi-photon excitation microscopy is an alternative to confocal microscopy that can be applied to living cells. The specimen is illuminated with a wavelength of light such that excitation of the fluorescent dye requires the simultaneous absorption of two or more photons (Figure 1.32). The probability of two photons simultaneously exciting the fluorescent dye is only significant at the point in the specimen upon which the input laser beam is focused, so fluorescence is only emitted from the plane of focus of the input light. This highly localized excitation automatically provides three-dimensional resolution, without the need for passing the emitted light through a pinhole aperture, as in confocal microscopy. Moreover, the localization of excitation minimizes damage to the specimen, allowing three-dimensional imaging of living cells.

Electron Microscopy

Because of the limited resolution of the light microscope, analysis of the details of cell structure has required the use of more powerful microscopic techniques—namely electron microscopy, which was developed in the 1930s and first applied to biological specimens by Albert Claude, Keith Porter, and George Palade in the 1940s and 1950s. The electron microscope can achieve a much greater resolution than that obtained with the light microscope because the wavelength of electrons is shorter than that of light. The wavelength of electrons in an electron microscope can be as short as 0.004 nm—about 100,000 times shorter than the wavelength of visible light. Theoretically, this wavelength could yield a resolution of 0.002 nm, but such a resolution cannot be obtained in practice, because resolution is determined not only by wavelength, but also by the numerical aperture of the microscope lens. Numerical aperture is a limiting factor for electron microscopy because inherent properties of electromagnetic lenses limit their aperture angles to about 0.5 degrees, corresponding to numerical apertures of only about 0.01. Thus, under optimal conditions, the resolving power of the electron microscope is approximately

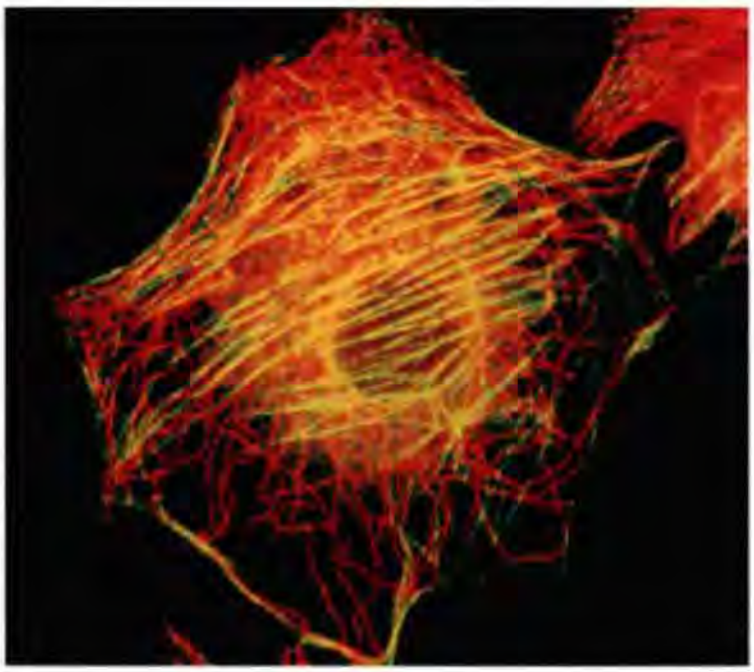


FIGURE 1.31 Confocal micrograph of human cells Microtubules and actin filaments are stained with red and green fluorescent dyes, respectively. (K. G. Murti/Visuals Unlimited.)

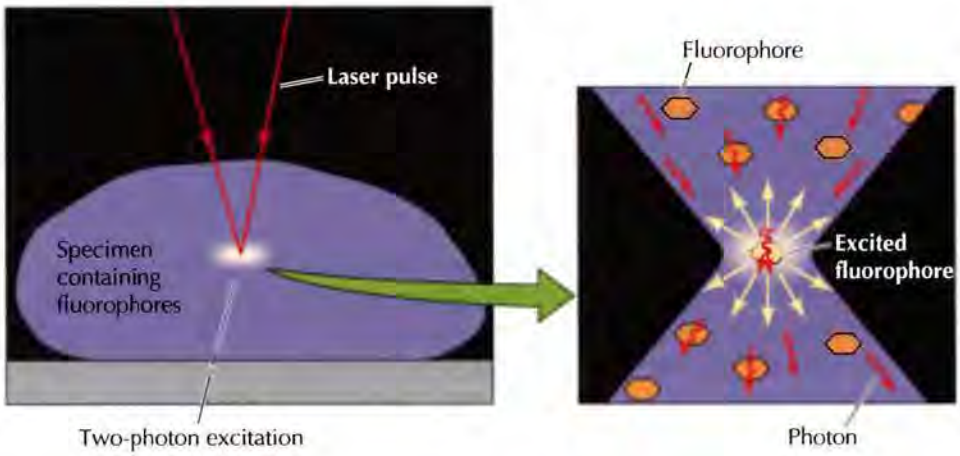


FIGURE 1.32 Two-photon excitation microscopy Simultaneous absorption of two photons is required to excite the fluorescent dye. This only occurs at the point in the specimen upon which the input light is focused, so fluorescent light is only emitted from the chosen depth of the specimen.

0.2 nm. Moreover, the resolution that can be obtained with biological specimens is further limited by their lack of inherent contrast. Consequently, for biological samples the practical limit of resolution of the electron microscope is 1 to 2 nm. Although this resolution is much less than that predicted simply from the wavelength of electrons, it represents more than a hundredfold improvement over the resolving power of the light microscope.

Two types of electron microscopy—transmission and scanning—are widely used to study cells. In principle, **transmission electron microscopy** is similar to the observation of stained cells with the bright-field light microscope. Specimens are fixed and stained with salts of heavy metals, which provide contrast by scattering electrons. A beam of electrons is then passed through the specimen and focused to form an image on a fluorescent screen. Electrons that encounter a heavy metal ion as they pass through the sample are deflected and do not contribute to the final image, so stained areas of the specimen appear dark.

Specimens to be examined by transmission electron microscopy can be prepared by either positive or negative staining. In positive staining, tissue specimens are cut into thin sections and stained with heavy metal salts (such as osmium tetroxide, uranyl acetate, and lead citrate) that react with lipids, proteins, and nucleic acids. These heavy metal ions bind to a variety of cell structures, which consequently appear dark in the final image (Figure 1.33). Alternative positive-staining procedures can also be used to identify specific macromolecules within cells. For example, antibodies labeled with electron-dense heavy metals (such as gold particles) are frequently used to determine the subcellular location of specific proteins in the electron microscope. This method is similar to the use of antibodies labeled with fluorescent dyes in fluorescence microscopy. Three-dimensional views of structures with resolutions of 2–10 nm can also be obtained using the technique of **electron tomography**, which generates three-dimensional images by computer analysis of multiple two-dimensional images obtained over a range of viewing directions.

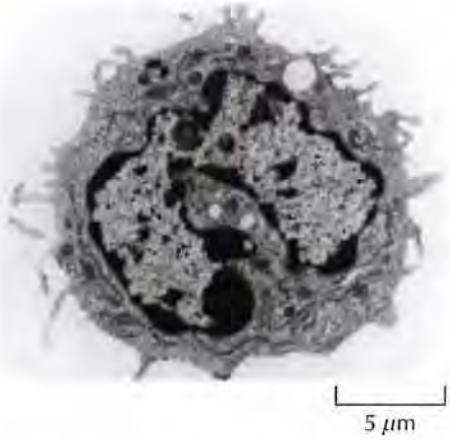


FIGURE 1.33 Positive staining Transmission electron micrograph of a positively stained white blood cell. (Don W. Fawcett/Visuals Unlimited.)

Negative staining is useful for the visualization of intact biological structures, such as bacteria, isolated subcellular organelles, and macromolecules (Figure 1.34). In this method, the biological specimen is deposited on a sup-

FIGURE 1.34 Negative staining
Transmission electron micrograph of negatively stained actin filaments. (Courtesy of Roger Craig, University of Massachusetts Medical Center.)

porting film, and a heavy metal stain is allowed to dry around its surface. The unstained specimen is then surrounded by a film of electron-dense stain, producing an image in which the specimen appears light against a stained dark background.

Metal shadowing is another technique used to visualize the surface of isolated subcellular structures or macromolecules in the transmission electron microscope (Figure 1.35). The specimen is coated with a thin layer of evaporated metal, such as platinum. The metal is sprayed onto the specimen from an angle so that surfaces of the specimen that face the source of evaporated metal molecules are coated more heavily than others. This differential coating creates a shadow effect, giving the specimen a three-dimensional appearance in electron micrographs.

The preparation of samples by **freeze fracture**, in combination with metal shadowing, has been particularly important in studies of membrane structure. Specimens are frozen in liquid nitrogen (at –196°C) and then fractured with a knife blade. This process frequently splits the lipid bilayer, revealing the interior faces of a cell membrane (Figure 1.36). The specimen is then shadowed with platinum, and the biological material is dissolved with acid, producing a metal replica of the surface of the sample. Examination of such replicas in the electron microscope reveals many surface bumps, corresponding to proteins that span the lipid bilayer. A variation of freeze frac-

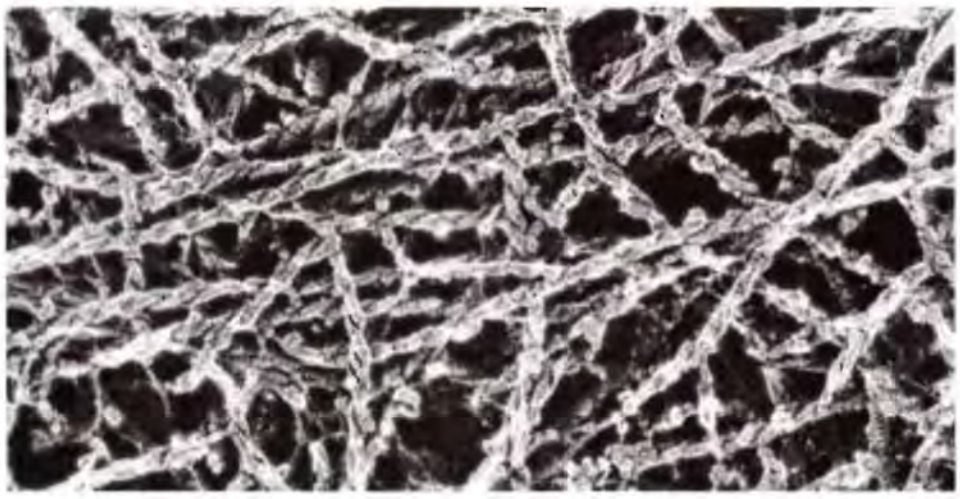


FIGURE 1.35 Metal shadowing
Electron micrograph of actin/myosin filaments of the cytoskeleton prepared by metal shadowing. (Don W. Fawcett, J. Heuser/Photo Researchers, Inc.)

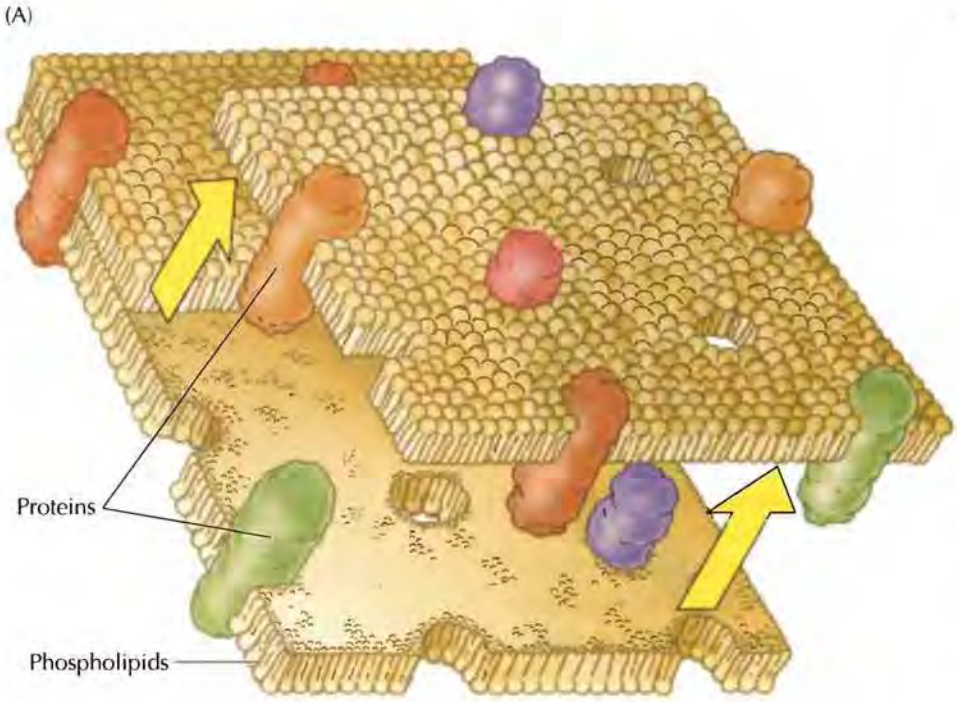


FIGURE 1.36 Freeze fracture (A) Freeze fracture splits the lipid bilayer, leaving proteins embedded in the membrane associated with one of the two membrane halves. (B) Micrograph of freeze-fractured plasma membranes of two adjacent cells. Proteins that span the bilayer appear as intramembranous particles (arrow). (Don W. Fawcett/ Photo Researchers, Inc.)

1.1 WEBSITE ANIMATION

Cell Fractionation

After cells are broken apart by sonication, the subcellular constituents are fractionated using different types of centrifugation.

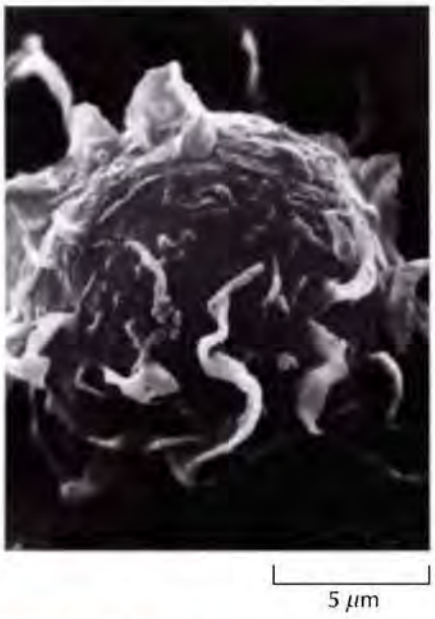


FIGURE 1.37 Scanning electron microscopy Scanning electron micrograph of a macrophage. (David Phillips/Visuals Unlimited.)

ture called freeze etching allows visualization of the external surfaces of cell membranes in addition to their interior faces.

The second type of electron microscopy, **scanning electron microscopy**, is used to provide a three-dimensional image of cells (Figure 1.37). In scanning electron microscopy the electron beam does not pass through the specimen. Instead, the surface of the cell is coated with a heavy metal, and a beam of electrons is used to scan across the specimen. Electrons that are scattered or emitted from the sample surface are collected to generate a three-dimensional image as the electron beam moves across the cell. Because the resolution of scanning electron microscopy is only about 10 nm, its use is generally restricted to studying whole cells rather than subcellular organelles or macromolecules.

Subcellular Fractionation

Although the electron microscope has allowed detailed visualization of cell structure, microscopy alone is not sufficient to define the functions of the various components of eukaryotic cells. To address many of the questions concerning the function of subcellular organelles, it has proven necessary to isolate the organelles of eukaryotic cells in a form that can be used for biochemical studies. This is usually accomplished by **differential centrifugation**—a method developed largely by Albert Claude, Christian de Duve, and their colleagues in the 1940s and 1950s to separate the components of cells on the basis of their size and density.

The first step in subcellular fractionation is the disruption of the plasma membrane under conditions that do not destroy the internal components of the cell. Several methods are used, including sonication (exposure to high-frequency sound), grinding in a mechanical homogenizer, or treatment with a high-speed blender. All these procedures break the plasma membrane and the endoplasmic reticulum into small fragments while leaving other components of the cell (such as nuclei, lysosomes, peroxisomes, mitochondria, and chloroplasts) intact.

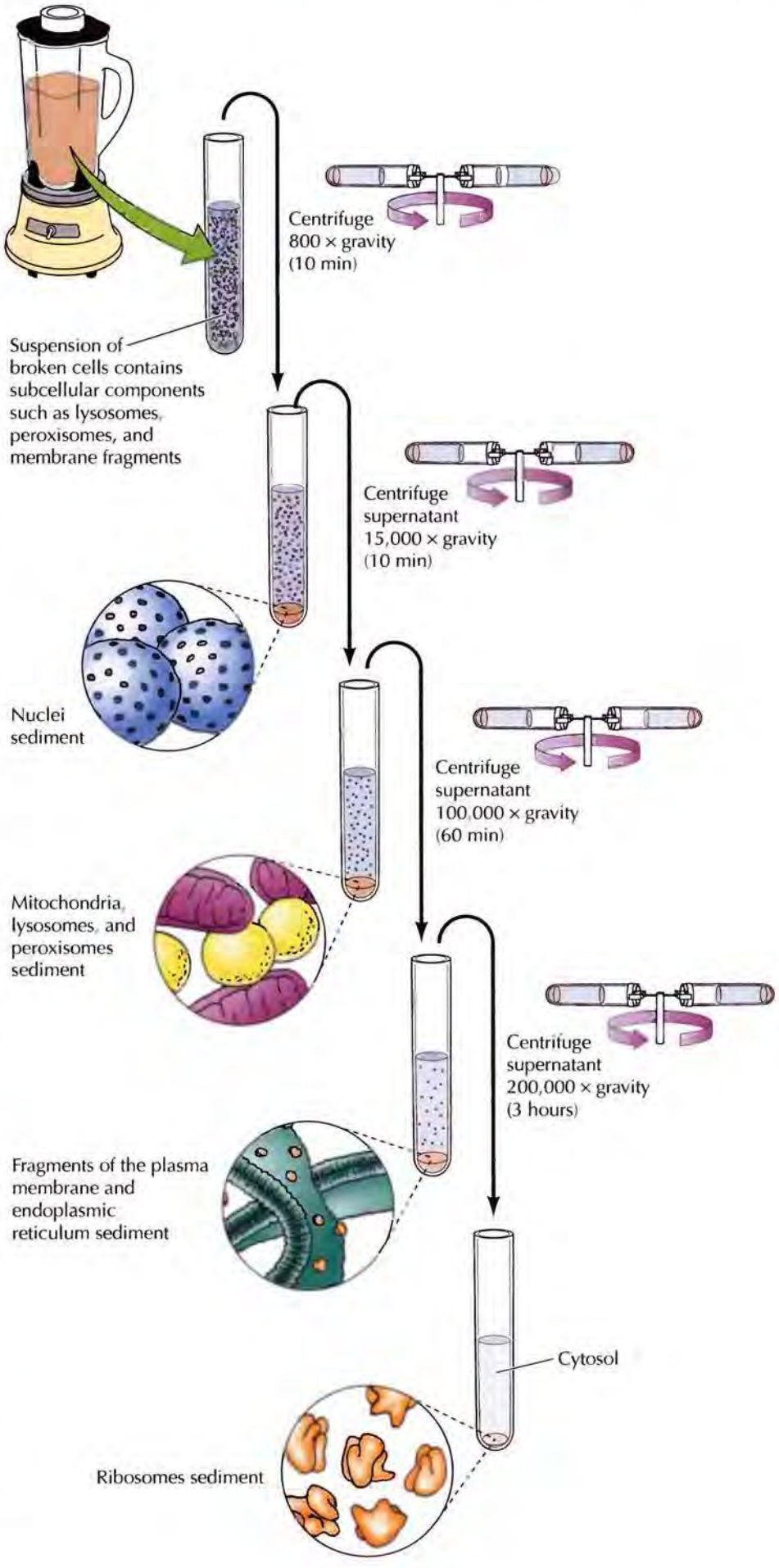


FIGURE 1.38 Subcellular fractionation Cells are lysed and subcellular components are separated by a series of centrifugations at increasing speeds. Following each centrifugation, the organelles that have sedimented to the bottom of the tube are recovered in the pellet. The supernatant is then recentrifuged at a higher speed to sediment the next-largest organelles.

The suspension of broken cells (called a lysate or homogenate) is then fractionated into its components by a series of centrifugations in an **ultracentrifuge**, which rotates samples at very high speeds (over 100,000 rpm) to produce forces up to 500,000 times greater than gravity. This force causes cell components to move toward the bottom of the centrifuge tube and form a pellet (a process called sedimentation) at a rate that depends on their size and density, with the largest and heaviest structures sedimenting most rapidly (Figure 1.38). Usually the cell homogenate is first centrifuged at a low speed, which sediments only unbroken cells and the largest subcellular structures—the nuclei. Thus, an enriched fraction of nuclei can be recovered from the pellet of such a low-speed centrifugation while the other cell components remain suspended in the supernatant (the remaining solution). The supernatant is then centrifuged at a higher speed to sediment mitochondria, chloroplasts, lysosomes, and peroxisomes. Recentrifugation of the supernatant at an even higher speed sediments fragments of the plasma membrane and the endoplasmic reticulum. A fourth centrifugation at a still higher speed sediments ribosomes, leaving only the soluble portion of the cytoplasm (the cytosol) in the supernatant.

The fractions obtained from differential centrifugation correspond to enriched, but still not pure, organelle preparations. A greater degree of purification can be achieved by **density-gradient centrifugation**, in which organelles are separated by sedimentation through a gradient of a dense substance, such as sucrose. In **velocity centrifuga-**

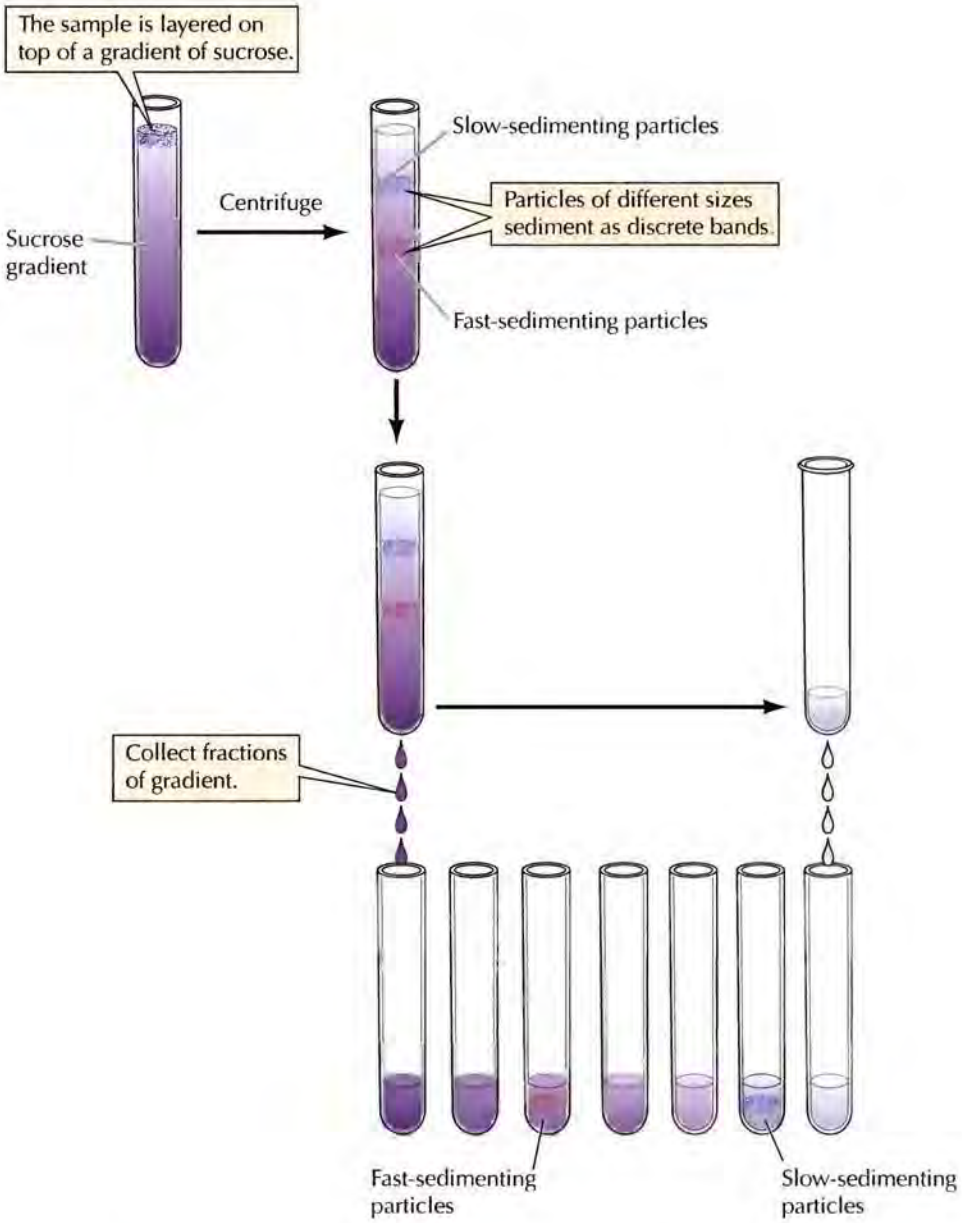


FIGURE 1.39 Velocity centrifugation in a density gradient The sample is layered on top of a gradient of sucrose, and particles of different sizes sediment through the gradient as discrete bands. The separated particles can then be collected in individual fractions of the gradient, which can be obtained simply by puncturing the bottom of the centrifuge tube and collecting drops.

tion, the starting material is layered on top of the sucrose gradient (Figure 1.39). Particles of different sizes sediment through the gradient at different rates, moving as discrete bands. Following centrifugation, the collection of individual fractions of the gradient provides sufficient resolution to separate organelles of similar size, such as mitochondria, lysosomes, and peroxisomes.

Equilibrium centrifugation in density gradients can be used to separate subcellular components on the basis of their buoyant density, independent of their size and shape. In this procedure, the sample is centrifuged in a gradient containing a high concentration of sucrose or cesium chloride. Rather than being separated on the basis of their sedimentation velocity, the sample particles are centrifuged until they reach an equilibrium position at which their buoyant density is equal to that of the surrounding sucrose or cesium chloride solution. Such equilibrium centrifugations are useful in separating different types of membranes from one another and are sufficiently sensitive to separate macromolecules that are labeled with different isotopes. A classic example, discussed in Chapter 4, is the analysis of DNA replication by separating DNA molecules containing heavy and light iso-

topes of nitrogen (^{15}N and ^{14}N) by equilibrium centrifugation in cesium chloride gradients.

Growth of Animal Cells in Culture

The ability to study cells depends largely on how readily they can be grown and manipulated in the laboratory. Although the process is technically far more difficult than the culture of bacteria or yeasts, a wide variety of animal and plant cells can be grown and manipulated in culture. Such *in vitro* cell culture systems have enabled scientists to study cell growth and differentiation, as well as to perform genetic manipulations required to understand gene structure and function.

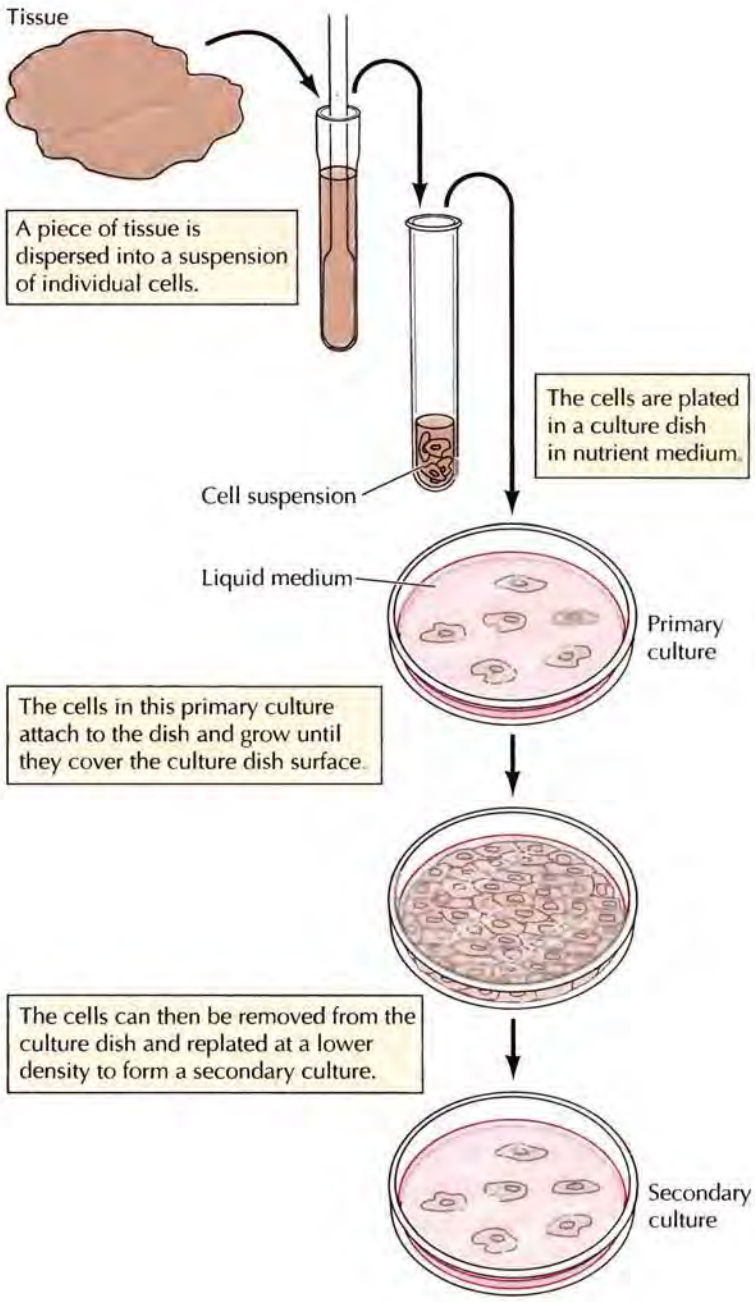
Animal cell cultures are initiated by the dispersion of a piece of tissue into a suspension of its component cells, which is then added to a culture dish containing nutrient media. Most animal cell types, such as fibroblasts and epithelial cells, attach and grow on the plastic surface of dishes used for cell culture (Figure 1.40). Because they contain rapidly growing cells, embryos or tumors are frequently used as starting material. Embryo fibroblasts grow particularly well in culture and consequently are one of the most widely studied types of animal cells. Under appropriate conditions, however, many specialized cell types can also be grown in culture, allowing their differentiated properties to be studied in a controlled experimental environment. **Embryonic stem cells** are a particularly notable example. These cells are established in culture from early embryos and maintain their ability to differentiate into all of the cell types present in adult organisms. Consequently, embryonic stem cells have played an important role in studying the functions of a variety of genes in mouse development, as well as offering the possibility of contributing to the treatment of human diseases by providing a source of tissue for transplantation therapies.

FIGURE 1.40 Animal cells in culture Scanning electron micrograph of human fibroblasts attached to the surface of a culture dish. (David M. Phillips/Visuals Unlimited.)

The culture media required for the propagation of animal cells are much more complex than the minimal media sufficient to support the growth of bacteria and yeasts. Early studies of cell culture utilized media consisting of undefined components, such as plasma, serum, and embryo extracts. A major advance was thus made in 1955, when Harry Eagle described the first defined media that supported the growth of animal cells. In addition to salts and glucose, the media used for animal cell cultures contain various amino acids and vitamins, which the cells cannot make for themselves. The growth media for most animal cells in culture also include serum, which serves as a source of polypeptide growth factors that are required to stimulate cell division. Several such growth factors have been identified. They serve as critical regulators of cell growth and differentiation in multicellular organisms, providing signals by which different cells communicate with each other. For example, an important function of skin fibroblasts in the intact animal is to proliferate when needed to repair damage resulting from a cut or wound. Their division is triggered by a growth factor released from platelets during blood clotting, thereby stimulating proliferation of fibroblasts in the neighborhood of the damaged tissue. The identification of individual growth factors has made possible the culture of a variety of cells in serum-free media (media in which serum has been replaced by the specific growth factors required for proliferation of the cells in question).

The initial cell cultures established from a tissue are called **primary cultures** (Figure 1.41). The cells in a primary culture usually grow until they cover the culture dish surface. They can then be removed from the dish and replated at a lower density to form secondary cultures. This process can be

FIGURE 1.41 Culture of animal cells



repeated many times, but most normal cells cannot be grown in culture indefinitely. For example, normal human fibroblasts can usually be cultured for 50 to 100 population doublings, after which they stop growing and die. In contrast, embryonic stem cells and cells derived from tumors frequently proliferate indefinitely in culture and are referred to as permanent or immortal **cell lines**. In addition, a number of immortalized rodent cell lines have been isolated from cultures of normal fibroblasts. Instead of dying as most of their counterparts do, a few cells in these cultures continue proliferating indefinitely, forming cell lines like those derived from tumors. Such permanent cell lines have been particularly useful for many types of experiments because they provide a continuous and uniform source of cells that can be manipulated, cloned, and indefinitely propagated in the laboratory.

Even under optimal conditions, the division time of most actively growing animal cells is on the order of 20 hours—ten times longer than the divi-

KEY EXPERIMENT

Animal Cell Culture

Nutrition Needs of Mammalian Cells in Tissue Culture

Harry Eagle
National Institutes of Health, Bethesda, MD

Science, Volume 122, 1955, pages 501–504

Harry Eagle

The Context

The earliest cell cultures involved the growth of cells from fragments of tissue that were embedded in clots of plasma—a culture system that was far from suitable for experimental analysis. In the late 1940s, a major advance was the establishment of cell lines that grew from isolated cells attached to the surface of culture dishes. But these cells were still grown in undefined media consisting of varying combinations of serum and embryo extracts. For example, a widely used human cancer cell line (called HeLa cells) was initially established in 1952 by growth in a medium consisting of chicken plasma, bovine embryo extract, and human placental cord serum. The use of such complex and undefined culture media made analysis of the specific growth requirements of animal cells impossible. Harry Eagle was the first to solve this problem, by carrying out a systematic analysis of the nutrients needed to support the growth of animal cells in culture.

The Experiments

Eagle studied the growth of two established cell lines: HeLa cells and a mouse fibroblast line called L cells. He was able to grow these cells in a medium consisting of a mixture of salts, carbohydrates, amino acids, and vitamins, supplemented with serum protein. By systematically varying the components of this medium, Eagle was able to determine the specific nutrients required for cell growth. In addition to salts and glucose, these nutrients included 13 amino acids and several vitamins. A small amount of serum protein was also required.

The basal medium developed by Eagle is described in the accompanying table, reprinted from his 1955 paper.

The Impact

The medium developed by Eagle is still the basic medium used for animal cell culture. Its use has enabled scientists to grow a wide variety of cells under defined experimental conditions, which has been critical to studies of animal cell growth and differentiation, including identification of the growth factors present in serum—now known to include polypeptides that control the behavior of individual cells within intact animals.

Table 4. Basal media for cultivation of the HeLa cell and mouse fibroblast (*10*)

L-Amino acids* (m*M*)			Vitamins‡ (m*M*)		Miscellaneous	
Arginine	0.1		Biotin	10^{-3}	Glucose	5m*M*§
Cystine	0.05	(0.02)†	Choline	10^{-3}	Penicillin	0.005%#
Glutamine	2.0	(1.0)‖	Folic acid	10^{-3}	Streptomycin	0.005%#
Histidine	0.05	(0.02)†	Nicotinamide	10^{-3}	Phenol red	0.0005%#
Isoleucine	0.2		Pantothenic acid	10^{-3}		
Leucine	0.2	(0.1)†	Pyridoxal	10^{-3}		
Lysine	0.2	(0.1)†	Thiamine	10^{-3}	For studies of cell nutrition	
Methionine	0.05		Riboflavin	10^{-4}	Dialyzed horse serum, 1%†	
Phenylalanine	0.1	(0.05)†			Dialyzed human serum, 5%	
Threonine	0.2	(0.1)†	Salts§ (m*M*)			
Tryptophan	0.02	(0.01)†			For stock cultures	
Tyrosine	0.1				Whole horse serum, 5%†	
Valine	0.2	(0.1)†	NaCl	100	Whole human serum, 10%	
			KCl	5		
			$NaH_2PO_4 \cdot H_2O$	1		
			$NaHCO_3$	20		
			$CaCl_2$	1		
			$MgCl_2$	0.5		

* Conveniently stored in the refrigerator as a single stock solution containing 20 times the indicated concentration of each amino acid.
† For mouse fibroblast.
‡ Conveniently stored as a single stock solution containing 100 or 1000 times the indicated concentration of each vitamin; kept frozen.
§ Conveniently stored in the refrigerator in two stock solutions, one containing NaCl, KCl, NaH_2PO_4, $NaHCO_3$, and glucose at 10 times the indicated concentration of each, and the second containing $CaCl_2$ and $MgCl_2$ at 20 times the indicated concentration.
‖ Conveniently stored as a 100m*M* stock solution; frozen when not in use.
Conveniently stored as a single stock solution containing 100 times the indicated concentrations of penicillin, streptomycin, and phenol red.

FIGURE 1.42 Plant cells in culture An undifferentiated mass of plant cells (a callus) growing on a solid medium. (John N. A. Lott/Biological Photo Service.)

sion time of yeasts. Consequently, experiments with cultured animal cells are more difficult and take much longer than those with bacteria or yeasts. For example, the growth of a visible colony of animal cells from a single cell takes a week or more, whereas colonies of *E. coli* or yeast develop from single cells overnight. Nonetheless, genetic manipulations of animal cells in culture have been indispensable to our understanding of cell structure and function.

Culture of Plant Cells

Plant cells can also be cultured in nutrient media containing appropriate growth regulatory molecules. In contrast to the polypeptide growth factors that regulate the proliferation of most animal cells, the growth regulators of plant cells are small molecules that can pass through the plant cell wall. When provided with appropriate mixtures of these growth regulatory molecules, many types of plant cells proliferate in culture, producing a mass of undifferentiated cells called a **callus** (Figure 1.42).

It is noteworthy that many plant cells are capable of forming any of the different cell types and tissues ultimately needed to regenerate an entire plant. Consequently, by appropriate manipulation of nutrients and growth regulatory molecules, undifferentiated plant cells in culture can be induced to form a variety of plant tissues, including roots, stems, and leaves. In many cases, even an entire plant can be regenerated from a single cultured cell. In addition to its theoretical interest, the ability to produce a new plant from a single cell that has been manipulated in culture makes it easy to introduce genetic alterations into plants, opening important possibilities for agricultural genetic engineering.

Viruses

Viruses are intracellular parasites that cannot replicate on their own. They reproduce by infecting host cells and usurping the cellular machinery to produce more virus particles. In their simplest forms, viruses consist only of genomic nucleic acid (either DNA or RNA) surrounded by a protein coat (Figure 1.43). Viruses are important in molecular and cellular biology because

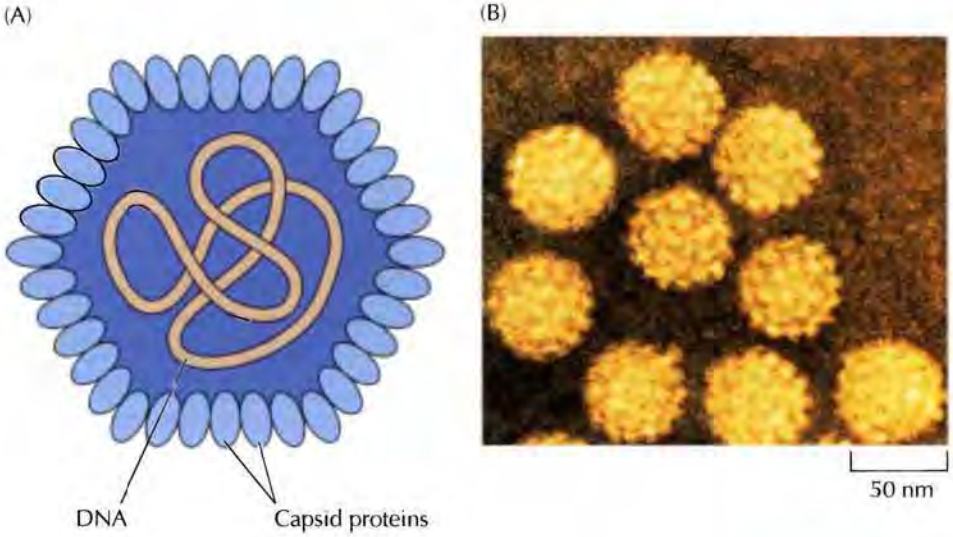


FIGURE 1.43 Structure of an animal virus (A) Papillomavirus particles contain a small circular DNA molecule enclosed in a protein coat (the capsid). (B) Electron micrograph of human papillomavirus particles. Artificial color has been added. (B, Linda Stannard/Science Photo Library/Photo Researchers, Inc.)

they provide simple systems that can be used to investigate the functions of cells. Because virus replication depends on the metabolism of the infected cells, studies of viruses have revealed many fundamental aspects of cell biology. Studies of bacterial viruses contributed substantially to our under-

MOLECULAR MEDICINE

Viruses and Cancer

The Disease

Cancer is a family of diseases characterized by uncontrolled cell proliferation. The growth of normal animal cells is carefully regulated to meet the needs of the complete organism. In contrast, cancer cells grow in an unregulated manner, ultimately invading and interfering with the function of normal tissues and organs. Cancer is the second most common cause of death (next to heart disease) in the United States. Approximately one out of every three Americans will develop cancer at some point in life and, in spite of major advances in treatment, nearly one out of every four Americans ultimately die of this disease. Understanding the causes of cancer and developing more effective methods of cancer treatment therefore represent major goals of medical research.

Molecular and Cellular Basis

Cancer is now known to result from mutations in the genes that normally control cell proliferation. The major insights leading to identification of these genes came from studies of viruses that cause cancer in animals, the prototype of which was isolated by Peyton Rous in 1911. Rous found that sarcomas (cancers of connective tissues) in chickens could be transmitted by a virus, now known as Rous sarcoma virus, or RSV. Because RSV is a retrovirus with a genome of only 10,000 base pairs, it can be subjected to molecular analysis much more readily than the complex genomes of chickens or other animal cells can. Such studies eventually led to identification of a specific cancer-causing gene (oncogene) carried by the virus, and to the discovery of related genes in normal cells of all vertebrate species, including humans. Some cancers in humans are now known to be caused by viruses; others result from mutations in normal cell genes similar to the oncogene first identified in RSV.

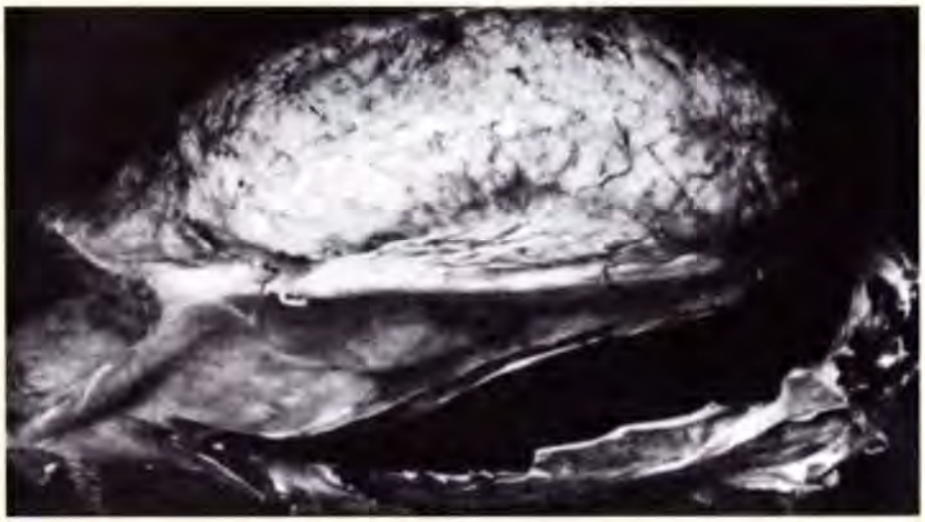
Prevention and Treatment

The human cancers that are caused by viruses include cervical and other anogenital cancers (papilloma viruses), liver cancer (hepatitis B and C viruses), and some types of lymphomas (Epstein-Barr virus and human T-cell lymphotropic virus). Together, these virus-induced cancers account for about 20% of worldwide cancer incidence. In principle, these cancers could be prevented by vaccination against the responsible viruses, and considerable progress in this area has been made by the development of an effective vaccine against hepatitis B virus.

Other human cancers are caused by mutations in normal cell genes, most of which occur during the lifetime of the individual rather than being inherited. Studies of cancer-causing viruses have led to the identification of many of the genes responsible for non-virus-induced cancers, and to an understanding of the molecular mechanisms responsible for cancer development. Major efforts are now under way to use these insights into the molecular and cellular biology of cancer to develop new approaches to cancer treatment. Indeed, the first designer drug effective in treating a human cancer (the drug STI-571 or Gleevec, discussed in chapter 18) was developed against a gene very similar to the oncogene of RSV.

Reference

Rous, P. 1911. A sarcoma of the fowl transmissible by an agent separable from the tumor cells. *J. Exp. Med.* 13: 397–411.



The transplantable tumor from which Rous sarcoma virus was isolated.

FIGURE 1.44 Bacteriophage plaques T4 plaques are visible on a lawn of *E. coli*. Each plaque arises by the replication of a single virus particle. (E. C. S. Chen/Visuals Unlimited.)

standing of the basic mechanisms of molecular genetics, and experiments with a plant virus (tobacco mosaic virus) first demonstrated the genetic potential of RNA. Animal viruses have provided particularly sensitive probes for investigations of various activities of eukaryotic cells.

The rapid growth and small genome size of bacteria make them excellent subjects for experiments in molecular biology, and bacterial viruses (**bacteriophages**) have simplified the study of bacterial genetics even further. One of the most important bacteriophages is T4, which infects and replicates in *E. coli*. Infection with a single particle of T4 leads to the formation of approximately 200 progeny virus particles in 20 to 30 minutes. The initially infected cell then bursts (lyses), releasing progeny virus particles into the medium, where they can infect new cells. In a culture of bacteria growing on agar medium, the replication of T4 leads to the formation of a clear area of lysed cells (a plaque) in the lawn of bacteria (Figure 1.44). Just as infectious virus particles are easy to grow and assay, viral mutants—for example, viruses that will grow in one strain of *E. coli* but not another—are easy to isolate. Thus, T4 is manipulated even more readily than *E. coli* for studies of molecular genetics. Moreover, the genome of T4 is 23 times smaller than that of *E. coli*—approximately 0.2 million base pairs—further facilitating genetic analysis. Some other bacteriophages have even smaller genomes—the simplest consisting of RNA molecules of only about 3600 nucleotides. Bacterial viruses have thus provided extremely facile experimental systems for molecular genetics. Studies of these viruses are largely what have led to the elucidation of many fundamental principles of molecular biology.

■ **Animal viruses are often used in gene therapy as carriers of genes to be introduced into cells.**

Because of the increased complexity of the animal cell genome, viruses have been even more important in studies of animal cells than in studies of bacteria. Many animal viruses replicate and can be assayed by plaque formation in cell cultures, much as bacteriophages can. Moreover, the genomes of animal viruses are similar in complexity to those of bacterial viruses (ranging from approximately 3000 to 300,000 base pairs), so animal viruses are far more manageable than are their host cells.

There are many diverse animal viruses, each containing either DNA or RNA as their genetic material (Table 1.3). Most viruses with RNA genomes replicate by synthesizing new RNA copies of their genomes from RNA tem-

TABLE 1.3 Examples of Animal Viruses

Virus family	Representative member	Genome size (thousands of base pairs)
RNA genomes		
Picornaviruses	Poliovirus	7–8
Togaviruses	Rubella virus	12
Flaviviruses	Yellow fever virus	10
Paramyxoviruses	Measles virus	16–20
Orthomyxoviruses	Influenza virus	14
Retroviruses	Human immunodeficiency virus	9
DNA genomes		
Hepadnaviruses	Hepatitis B virus	3.2
Papovaviruses	Human papillomavirus	5–8
Adenoviruses	Adenovirus	36
Herpesviruses	Herpes simplex virus	120–200
Poxviruses	Vaccinia virus	130–280

plates in infected cells. However, one family of animal viruses—the **retroviruses**—contain RNA genomes in their virus particles but synthesize a DNA copy of their genome in infected cells. These viruses provide a good example of the importance of viruses as models, because studies of the retroviruses are what first demonstrated the synthesis of DNA from RNA templates—a fundamental mode of genetic information transfer now known to occur in all eukaryotic cells. Other examples in which animal viruses have provided important models for investigations of their host cells include studies of DNA replication, transcription, RNA processing, and protein transport and secretion.

It is particularly noteworthy that infection by some animal viruses, rather than killing the host cell, converts a normal cell into a cancer cell. Studies of such cancer-causing viruses, first described by Peyton Rous in 1911, not only have provided the basis for our current understanding of cancer at the level of cell and molecular biology, but also have led to the elucidation of many of the molecular mechanisms that control animal cell growth and differentiation.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

THE ORIGIN AND EVOLUTION OF CELLS

The First Cell: All present-day cells, both prokaryotes and eukaryotes, are descended from a single ancestor. The first cell is thought to have arisen at least 3.8 billion years ago as a result of enclosure of self-replicating RNA in a phospholipid membrane.

prokaryotic cell, eukaryotic cell, RNA world, phospholipid, amphipathic, hydrophobic, hydrophilic

The Evolution of Metabolism: The earliest reactions for the generation of metabolic energy were a form of anaerobic glycolysis. Photosynthesis then evolved, followed by oxidative metabolism.

adenosine 5′-triphosphate (ATP), glycolysis, photosynthesis, oxidative metabolism

Present-Day Prokaryotes: Present-day prokaryotes are divided into two groups, the archaebacteria and the eubacteria, which diverged early in evolution.

archaebacteria, eubacteria, cyanobacteria, *Escherichia coli* (*E. coli*), cell wall, plasma membrane, ribosome

Eukaryotic Cells: Eukaryotic cells, which are larger and more complex than prokaryotic cells, contain a nucleus, cytoplasmic organelles, and a cytoskeleton.

nucleus, mitochondria, chloroplasts, lysosome, peroxisome, vacuole, endoplasmic reticulum, Golgi apparatus, cytoskeleton

The Origin of Eukaryotes: Eukaryotic cells are thought to have evolved from symbiotic associations of prokaryotes. The genome of eukaryotes may have arisen from a fusion of eubacterial and archaebacterial genomes.

endosymbiosis

The Development of Multicellular Organisms: The simplest eukaryotes are unicellular organisms, such as yeasts and amoebas. Multicellular organisms evolved from associations between such unicellular eukaryotes, and division of labor led to the development of the many kinds of specialized cells that make up present-day plants and animals.

yeast, *Saccharomyces cerevisiae*, pseudopodium, epithelial cell, fibroblast, erythrocyte, granulocyte, monocyte, macrophage, lymphocyte, neuron

KEY TERMS

KEY TERMS

SUMMARY

CELLS AS EXPERIMENTAL MODELS

E. coli: Because of their genetic simplicity and ease of study, bacteria such as *E. coli* are particularly useful for investigation of fundamental aspects of biochemistry and molecular biology.

Yeasts: As the simplest eukaryotic cells, yeasts are an important model for studying various aspects of eukaryotic cell biology.

Caenorhabditis elegans

Caenorhabditis elegans: The nematode *C. elegans* is a simple multicellular organism that serves as an important model in developmental biology.

Drosophila melanogaster

Drosophila melanogaster: Because of extensive genetic analysis, studies of the fruit fly *Drosophila* have led to major advances in understanding animal development.

Arabidopsis thaliana

Arabidopsis thaliana: The small flowering plant *Arabidopsis* is widely used as a model for studies of plant molecular biology and development.

***Xenopus laevis*, zebrafish**

Vertebrates: Many kinds of vertebrate cells can be grown in culture, where they can be studied under controlled laboratory conditions. Specialized cell types, such as neurons and muscle cells, provide useful models for investigating particular aspects of cell biology. The frog *Xenopus laevis* and zebrafish are important models for studies of early vertebrate development, and the mouse is a mammalian species suitable for genetic analysis.

TOOLS OF CELL BIOLOGY

resolution, bright-field microscopy, phase-contrast microscopy, differential interference-contrast microscopy, video-enhanced differential interference-contrast microscopy, fluorescence microscopy, green fluorescent protein (GFP), fluorescence recovery after photobleaching (FRAP), fluorescence resonance energy transfer (FRET), confocal microscopy, multi-photon excitation microscopy

Light Microscopy: A variety of methods are used to visualize cells and subcellular structures and to determine the intracellular localization of specific molecules using the light microscope.

transmission electron microscopy, electron tomography, metal shadowing, freeze fracture, scanning electron microscopy

Electron Microscopy: Electron microscopy, with a resolution that is approximately a hundredfold greater than that of light microscopy, is used to analyze details of cell structure.

differential centrifugation, ultracentrifuge, density-gradient centrifugation, velocity centrifugation, equilibrium centrifugation

Subcellular Fractionation: The organelles of eukaryotic cells can be isolated for biochemical analysis by differential centrifugation.

SUMMARY

Growth of Animal Cells in Culture: The propagation of animal cells in culture has allowed studies of the mechanisms that control cell growth and differentiation.

Culture of Plant Cells: Cultured plant cells can differentiate to form specialized cell types and, in some cases, can regenerate entire plants.

Viruses: Viruses provide simple models for studies of cell function.

KEY TERMS

embryonic stem cell, primary cultures, cell line

callus

bacteriophage, retrovirus

Questions

1. What did Stanley Miller's experiments show about the formation of organic molecules?

2. What kind of macromolecule is capable of directing its own replication?

3. Discuss the evidence that mitochondria and chloroplasts originated from bacteria that were engulfed by the precursor of eukaryotic cells.

4. Why is the evolution of photosynthesis thought to have favored the subsequent evolution of oxidative metabolism?

5. Given that the diameter of a *S. aureus* bacterial cell is 1 μm and the diameter of a human macrophage is 50 μm, how many *S. aureus* cells can fit inside a single human macrophage? Assume that the cells are spherical.

6. Which model organism provides the simplest system for studying eukaryotic DNA replication?

7. You are studying a gene involved in mammalian embryonic development. Which model organism would be best suited for your studies?

8. What resolution can be obtained with a light microscope if the specimen is viewed through air rather than through oil? Assume that the wavelength of visible light is 0.5 μm.

9. You are about to purchase a research microscope for your laboratory, and you have a choice between two different objective lenses. One has a magnification of 100× and a numerical aperture (NA) of 1.1. The other has a magnification of 60× and a numerical aperture of 1.3. Assuming that price is not a concern, which of these objectives would you choose?

10. What advantage does the use of green fluorescent protein (GFP) have over the use of fluorescent-labeled antibodies for studying the location and movement of a protein in cells?

11. Identify the different characteristics or properties of organelles that allow separation by velocity centrifugation as compared to equilibrium centrifugation in a sucrose gradient.

12. Why is serum usually required in the media used to grow animal cells in culture?

13. Distinguish between primary cell cultures and immortal cell lines.

14. Why is the ability to culture embryonic stem cells important?

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CHAPTER 2

The Composition of Cells

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- ***Cell Membranes 58***
- ***Proteomics: Large-Scale Analysis of Cell Proteins 65***
- ***KEY EXPERIMENT:*** The Folding of Polypeptide Chains 54
- ***KEY EXPERIMENT:*** The Structure of Cell Membranes 59

CELLS ARE INCREDIBLY COMPLEX AND DIVERSE STRUCTURES, capable not only of self-replication—the very essence of life—but also of performing a wide range of specialized tasks in multicellular organisms. Yet cells obey the same laws of chemistry and physics that determine the behavior of nonliving systems. Consequently, modern cell biology seeks to understand cellular processes in terms of chemical and physical reactions.

This chapter considers the chemical composition of cells and the properties of the molecules that are ultimately responsible for all cellular activities. Proteins are given particular emphasis because of their diverse roles within the cell, including acting as enzymes that catalyze almost all biological reactions and serving as key components of cell membranes. In recent years, a variety of new technologies have been developed that allow the analysis of cell proteins on a large scale. This emerging field of proteomics is expected to make a major impact on our understanding of the expression and interactions of proteins within cells, much like the impact that large scale genome sequencing has made on our understanding of the genetic content of cells and organisms.

The Molecules of Cells

Cells are composed of water, inorganic ions, and carbon-containing (organic) molecules. Water is the most abundant molecule in cells, accounting for 70% or more of total cell mass. Consequently, the interactions between water and the other constituents of cells are of central importance in biological chemistry. The critical property of water in this respect is that it is a polar molecule, in which the hydrogen atoms have a slight positive charge and the oxygen has a slight negative charge (Figure 2.1). Because of their polar nature, water molecules can form hydrogen bonds with each other or with other polar molecules, as well as interact

2.1 WEBSITE ANIMATION

Bond Formation The polymerization of sugars, amino acids, and nucleotides to form polysaccharides, polypeptides, and nucleic acids, respectively, occurs through covalent bonding.

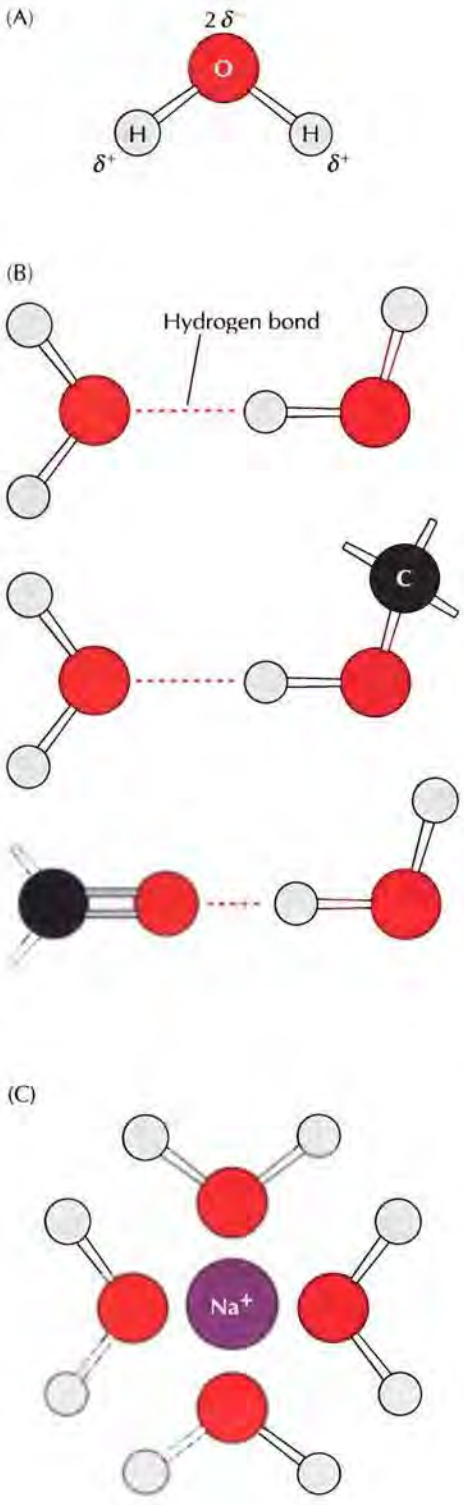


FIGURE 2.1 Characteristics of water (A) Water is a polar molecule, with a slight negative charge (δ^-) on the oxygen atom and a slight positive charge (δ^+) on the hydrogen atoms. Because of this polarity, water molecules can form hydrogen bonds (dashed lines) either with each other or with other polar molecules (B), in addition to interacting with charged ions (C).

with positively or negatively charged ions. As a result of these interactions, ions and polar molecules are readily soluble in water (hydrophilic). In contrast, nonpolar molecules, which cannot interact with water, are poorly soluble in an aqueous environment (hydrophobic). Consequently, nonpolar molecules tend to minimize their contact with water by associating closely with each other instead. As discussed later in this chapter, such interactions of polar and nonpolar molecules with water and with each other play crucial roles in the formation of biological structures, such as cell membranes. The inorganic ions of the cell, including sodium (Na^+), potassium (K^+), magnesium (Mg^{2+}), calcium (Ca^{2+}), phosphate (HPO_4^{2-}), chloride (Cl^-), and bicarbonate (HCO_3^-), constitute 1% or less of the cell mass. These ions are involved in a number of aspects of cell metabolism, and thus play critical roles in cell function.

It is, however, the organic molecules that are the unique constituents of cells. Most of these organic compounds belong to one of four classes of molecules: carbohydrates, lipids, proteins, and nucleic acids. Proteins, nucleic acids, and most carbohydrates (the polysaccharides) are macromolecules formed by the joining (polymerization) of hundreds or thousands of low-molecular-weight precursors: amino acids, nucleotides, and simple sugars, respectively. Such macromolecules constitute 80 to 90% of the dry weight of most cells. Lipids are the other major constituent of cells. The remainder of the cell mass is composed of a variety of small organic molecules, including macromolecular precursors. The basic chemistry of cells can thus be understood in terms of the structures and functions of four major classes of organic molecules.

Carbohydrates

The **carbohydrates** include simple sugars as well as polysaccharides. These simple sugars, such as glucose, are the major nutrients of cells. As discussed in Chapter 3, their breakdown provides both a source of cellular energy and the starting material for the synthesis of other cell constituents. Polysaccharides are storage forms of sugars and form structural components of the cell. In addition, polysaccharides and shorter polymers of sugars act as markers for a variety of cell recognition processes, including the adhesion of cells to their neighbors and the transport of proteins to appropriate intracellular destinations.

The structures of representative simple sugars (**monosaccharides**) are illustrated in Figure 2.2. The basic formula for these molecules is $(CH_2O)_n$, from which the name carbohydrate is derived (C = "carbo" and H_2O = "hydrate"). The six-carbon ($n = 6$) sugar glucose ($C_6H_{12}O_6$) is especially important in cells, since it provides the principal source of cellular energy. Other simple sugars have between three and seven carbons, with three- and five-carbon sugars being the most common. Sugars containing five or more carbons can cyclize to form ring structures, which are the predominant forms of these molecules within cells. As illustrated in Figure 2.2, the cyclized sugars exist in two alternative forms (called α or β), depending on the configuration of carbon 1.

■ **The artificial sweetener sucralose (Splenda) is a synthetic derivative of common sugar in which some of the hydroxyl groups of the sugar are replaced with chlorine.**

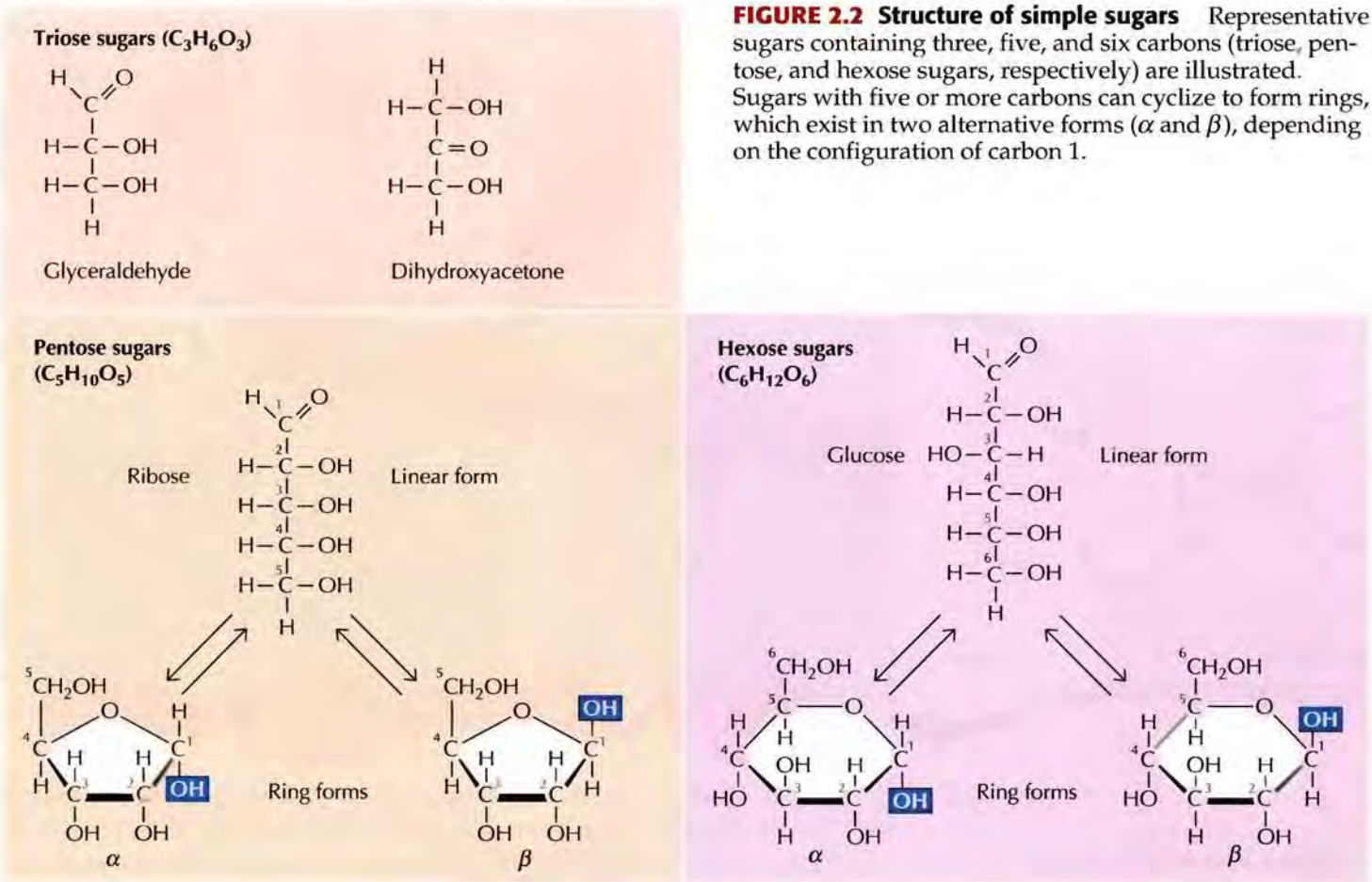


FIGURE 2.2 Structure of simple sugars Representative sugars containing three, five, and six carbons (triose, pentose, and hexose sugars, respectively) are illustrated. Sugars with five or more carbons can cyclize to form rings, which exist in two alternative forms (α and β), depending on the configuration of carbon 1.

Monosaccharides can be joined together by dehydration reactions in which H_2O is removed and the sugars are linked by a **glycosidic bond** between two of their carbons (Figure 2.3). If only a few sugars are joined together, the resulting polymer is called an **oligosaccharide**. If a large number (hundreds or thousands) of sugars are involved, the resulting polymers are macromolecules called **polysaccharides**.

Two common polysaccharides—**glycogen** and **starch**—are the storage forms of carbohydrates in animal and plant cells, respectively. Both glycogen and starch are composed entirely of glucose molecules in the α configuration (Figure 2.4). The principal linkage is between carbon 1 of one glucose and carbon 4 of a second. In addition, both glycogen and one form of starch (amylopectin) contain occasional $\alpha(1\rightarrow6)$ linkages, in which carbon 1 of one glucose is joined to carbon 6 of a second. As illustrated in Figure 2.4, these linkages lead to the formation of branches resulting from the joining of two separate $\alpha(1\rightarrow4)$ linked chains. Such branches are present in glycogen and amylopectin, although another form of starch (amylose) is an unbranched molecule.

The structures of glycogen and starch are thus basically similar, as is their function: to store glucose. **Cellulose**, in contrast, has a quite distinct function as the principal structural component of the plant cell wall. Perhaps surprisingly, then, cellulose

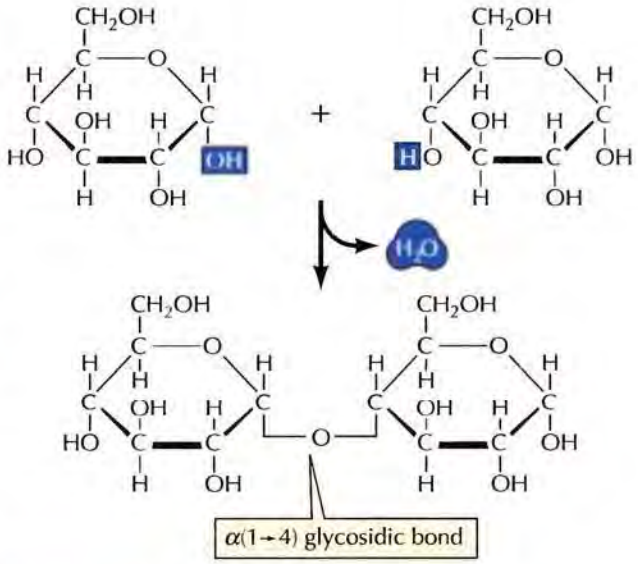


FIGURE 2.3 Formation of a glycosidic bond Two simple sugars are joined by a dehydration reaction (a reaction in which water is removed). In the example shown, two glucose molecules in the α configuration are joined by a bond between carbons 1 and 4, which is therefore called an $\alpha(1\rightarrow4)$ glycosidic bond.

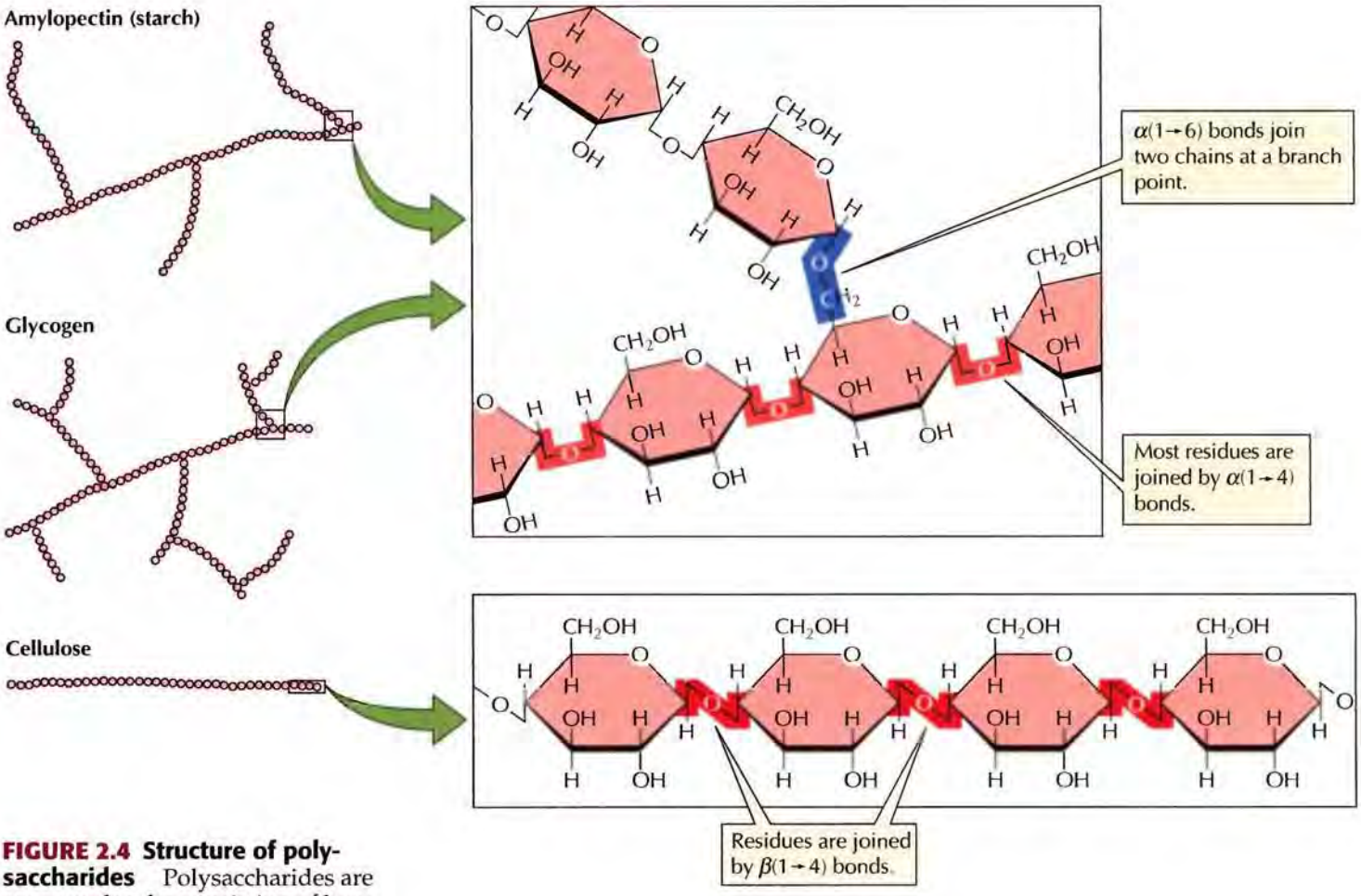


FIGURE 2.4 Structure of polysaccharides Polysaccharides are macromolecules consisting of hundreds or thousands of simple sugars. Glycogen, starch, and cellulose are all composed entirely of glucose residues, which are joined by $\alpha(1\rightarrow4)$ glycosidic bonds in glycogen and starch but by $\beta(1\rightarrow4)$ bonds in cellulose. Glycogen and one form of starch (amylopectin) also contain occasional $\alpha(1\rightarrow6)$ bonds, which serve as branch points by joining two separate $\alpha(1\rightarrow4)$ chains.

is also composed entirely of glucose molecules. The glucose residues in cellulose, however, are in the β rather than the α configuration, and cellulose is an unbranched polysaccharide (see Figure 2.4). The linkage of glucose residues by $\beta(1\rightarrow4)$ rather than $\alpha(1\rightarrow4)$ bonds causes cellulose to form long extended chains that pack side by side to form fibers of great mechanical strength.

In addition to their roles in energy storage and cell structure, oligosaccharides and polysaccharides are important in a variety of informational processes. For example, oligosaccharides are frequently linked to proteins, where they play important roles in protein folding and serve as markers to target proteins for transport to the cell surface or incorporation into different subcellular organelles. Oligosaccharides and polysaccharides also serve as markers on the surface of cells, playing important roles in cell recognition and the interactions between cells in tissues of multicellular organisms.

Lipids

Lipids have three major roles in cells. First, they provide an important form of energy storage. Second, and of great importance in cell biology, lipids are the major components of cell membranes. Third, lipids play important roles in cell signaling, both as steroid hormones (e.g., estrogen and testosterone) and as messenger molecules that convey signals from cell surface receptors to targets within the cell.

The simplest lipids are **fatty acids**, which consist of long hydrocarbon chains, most frequently containing 16 or 18 carbon atoms, with a carboxyl group (COO^-) at one end (Figure 2.5). Unsaturated fatty acids contain one or

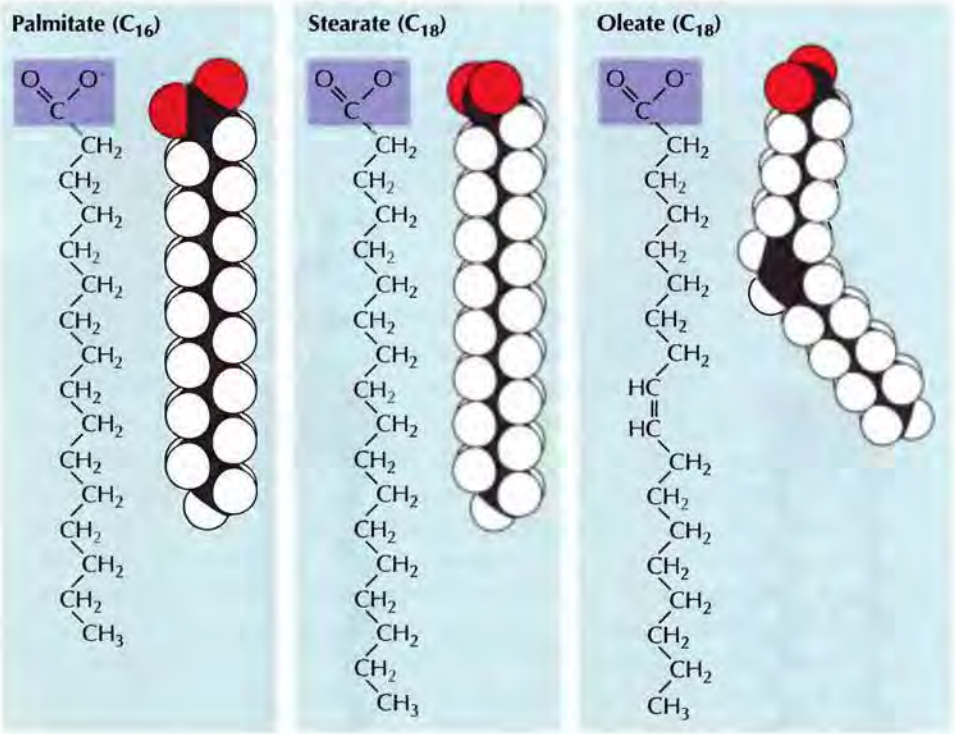


FIGURE 2.5 Structure of fatty acids Fatty acids consist of long hydrocarbon chains terminating in a carboxyl group (COO^-). Palmitate and stearate are saturated fatty acids consisting of 16 and 18 carbons, respectively. Oleate is an unsaturated 18-carbon fatty acid containing a double bond between carbons 9 and 10. Note that the double bond introduces a kink in the hydrocarbon chain.

more double bonds between carbon atoms; in saturated fatty acids all of the carbon atoms are bonded to the maximum number of hydrogen atoms. The long hydrocarbon chains of fatty acids contain only nonpolar C—H bonds, which are unable to interact with water. The hydrophobic nature of these fatty acid chains is responsible for much of the behavior of complex lipids, particularly in the formation of biological membranes.

Fatty acids are stored in the form of **triacylglycerols,** or **fats**, which consist of three fatty acids linked to a glycerol molecule (Figure 2.6). Triacylglycerols are insoluble in water and therefore accumulate as fat droplets in the cytoplasm. When required, they can be broken down for use in energy-yielding reactions (see Chapter 3). It is noteworthy that fats are a more efficient form of energy storage than carbohydrates, yielding more than twice as much energy per weight of material broken down. Fats therefore allow energy to be stored in less than half the body weight that would be required to store the same amount of energy in carbohydrates—a particularly important consideration for animals because of their mobility.

Phospholipids, the principal components of cell membranes, consist of two fatty acids joined to a polar head group (Figure 2.7). In the **glycerol phospholipids**, the two fatty acids are bound to carbon atoms in glycerol, as in triacylglycerols. The third carbon of glycerol, however, is bound to a phosphate group, which is in turn frequently attached to another small polar molecule, such as choline, serine, inositol, or ethanolamine. **Sphingomyelin**, the only nonglycerol phospholipid in cell membranes, contains two hydrocarbon chains linked to a polar head group formed from serine rather than from glycerol. All phospholipids have hydrophobic tails, con-

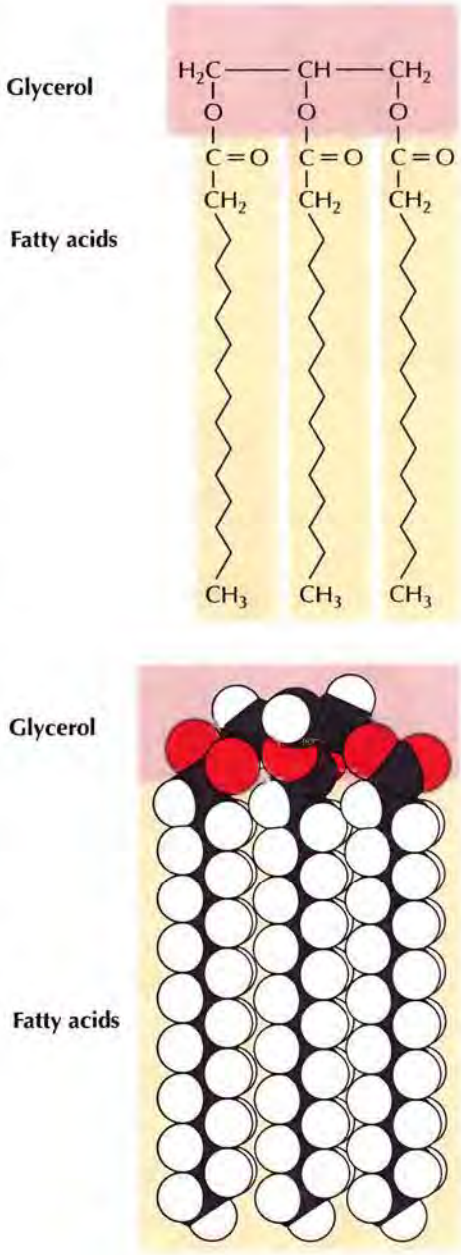
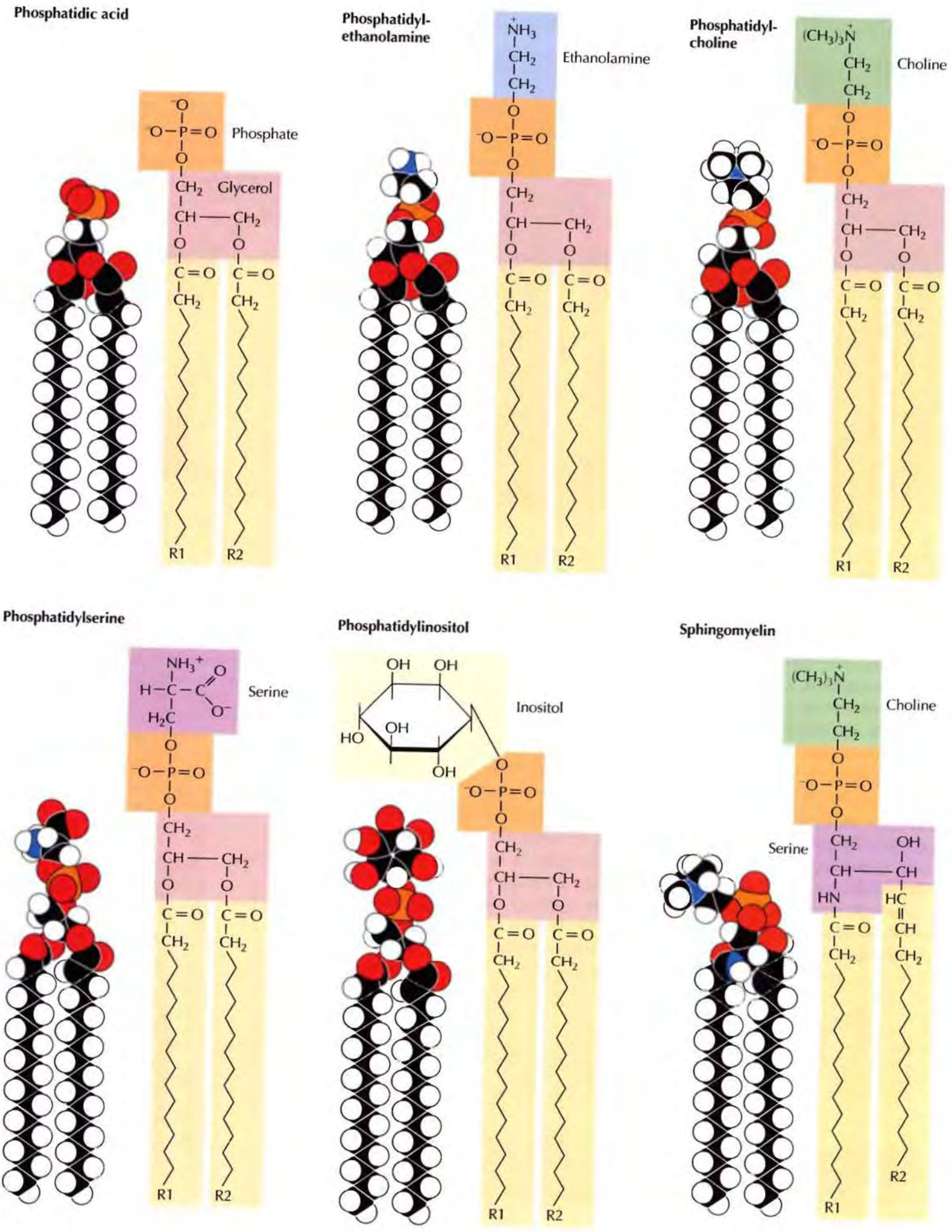


FIGURE 2.6 Structure of triacylglycerols Triacylglycerols (fats) contain three fatty acids joined to glycerol. In this example, all three fatty acids are palmitate, but triacylglycerols often contain a mixture of different fatty acids.


Phosphatidic acid
Phosphate
Glycerol
R1
R2
Phosphatidyl-ethanolamine
Ethanolamine
R1
R2
Phosphatidyl-choline
Choline
R1
R2
Phosphatidylserine
Serine
R1
R2
Phosphatidylinositol
Inositol
R1
R2
Sphingomyelin
Choline
Serine
R1
R2

◀ **FIGURE 2.7 Structure of phospholipids** Glycerol phospholipids contain two fatty acids joined to glycerol. The fatty acids may be different from each other and are designated R1 and R2. The third carbon of glycerol is joined to a phosphate group (forming phosphatidic acid), which in turn is frequently joined to another small polar molecule (forming phosphatidylethanolamine, phosphatidylcholine, phosphatidylserine, or phosphatidylinositol). In sphingomyelin, two hydrocarbon chains are bound to a polar head group formed from serine instead of glycerol.

sisting of the two hydrocarbon chains, and hydrophilic head groups, consisting of the phosphate group and its polar attachments. Consequently, phospholipids are **amphipathic** molecules, part water-soluble and part water-insoluble. This property of phospholipids is the basis for the formation of biological membranes, as discussed later in this chapter.

In addition to phospholipids, many cell membranes contain **glycolipids** and **cholesterol**. Glycolipids consist of two hydrocarbon chains linked to polar head groups that contain carbohydrates (Figure 2.8). They are thus similar to the phospholipids in their general organization as amphipathic molecules. Cholesterol, in contrast, consists of four hydrocarbon rings rather than linear hydrocarbon chains (Figure 2.9). The hydrocarbon rings are strongly hydrophobic, but the hydroxyl (OH) group attached to one end of cholesterol is weakly hydrophilic, so cholesterol is also amphipathic.

In addition to their roles as components of cell membranes, lipids function as signaling molecules, both within and between cells. The **steroid hormones** (such as estrogens and testosterone) are derivatives of cholesterol (see Figure 2.9). These hormones are a diverse group of chemical messengers, all of which contain four hydrocarbon rings to which distinct functional groups are attached. Derivatives of phospholipids also serve as messenger molecules within cells, acting to convey signals from cell surface receptors to intracellular targets that regulate a wide range of cellular processes, including cell proliferation, movement, survival, and differentiation (see Chapter 15).

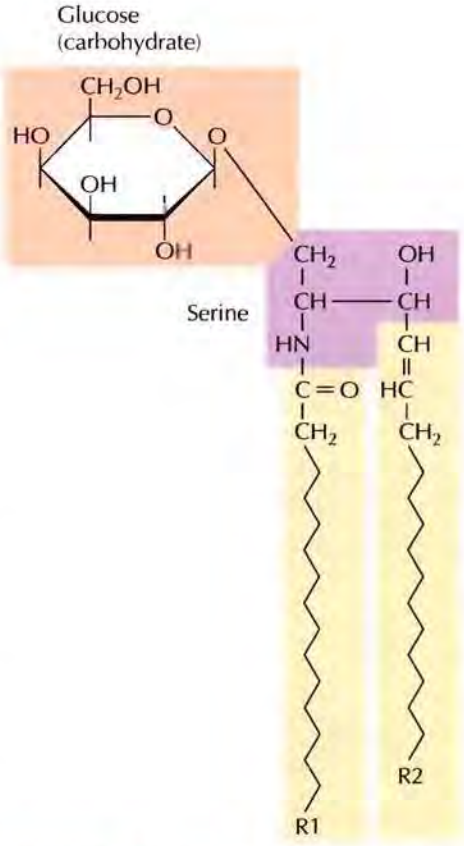
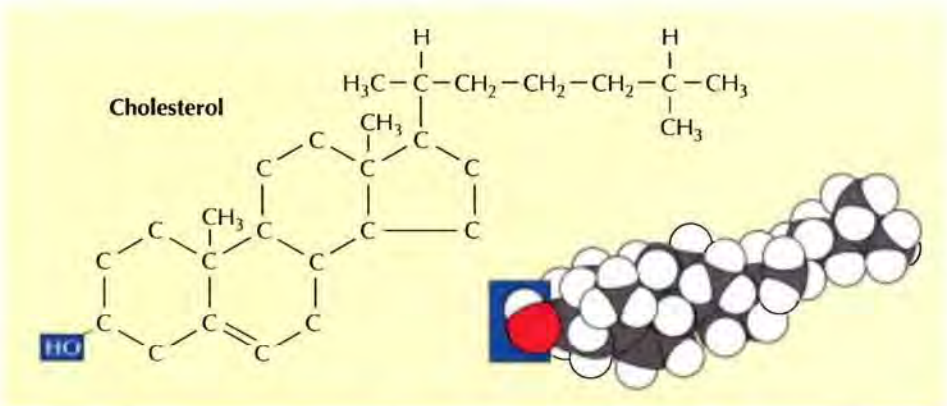


FIGURE 2.8 Structure of glycolipids Two hydrocarbon chains are joined to a polar head group formed from serine and containing carbohydrates (e.g., glucose).



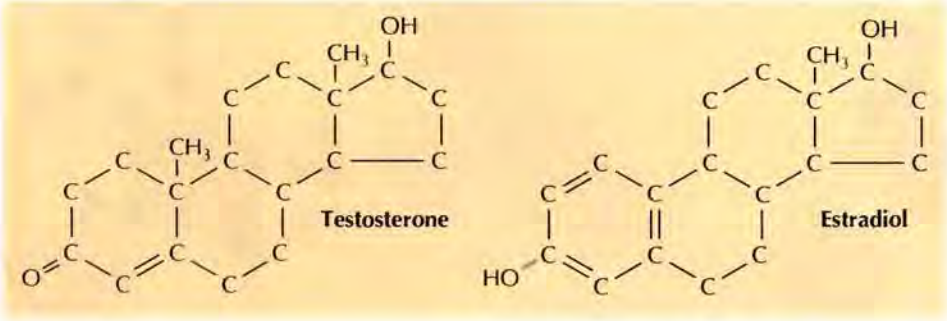


FIGURE 2.9 Cholesterol and steroid hormones Cholesterol, an important component of cell membranes, is an amphipathic molecule because of its polar hydroxyl group. Cholesterol is also a precursor to the steroid hormones, such as testosterone and estradiol (a form of estrogen). The hydrogen atoms bonded to the ring carbons are not shown in this figure.

Nucleic Acids

The nucleic acids—DNA and RNA—are the principal informational molecules of the cell. **Deoxyribonucleic acid (DNA)** has a unique role as the genetic material, which in eukaryotic cells is located in the nucleus. Different types of **ribonucleic acid (RNA)** participate in a number of cellular activities. **Messenger RNA (mRNA)** carries information from DNA to the ribosomes, where it serves as a template for protein synthesis. Two other types of RNA (**ribosomal RNA** and **transfer RNA**) are involved in protein synthesis. Still other kinds of RNAs are involved in the processing and transport of both RNAs and proteins. In addition to acting as an informational molecule, RNA is also capable of catalyzing a number of chemical reactions. In present-day cells, these include reactions involved in both protein synthesis and RNA processing.

DNA and RNA are polymers of **nucleotides**, which consist of **purine** and **pyrimidine** bases linked to phosphorylated sugars (Figure 2.10). DNA con-

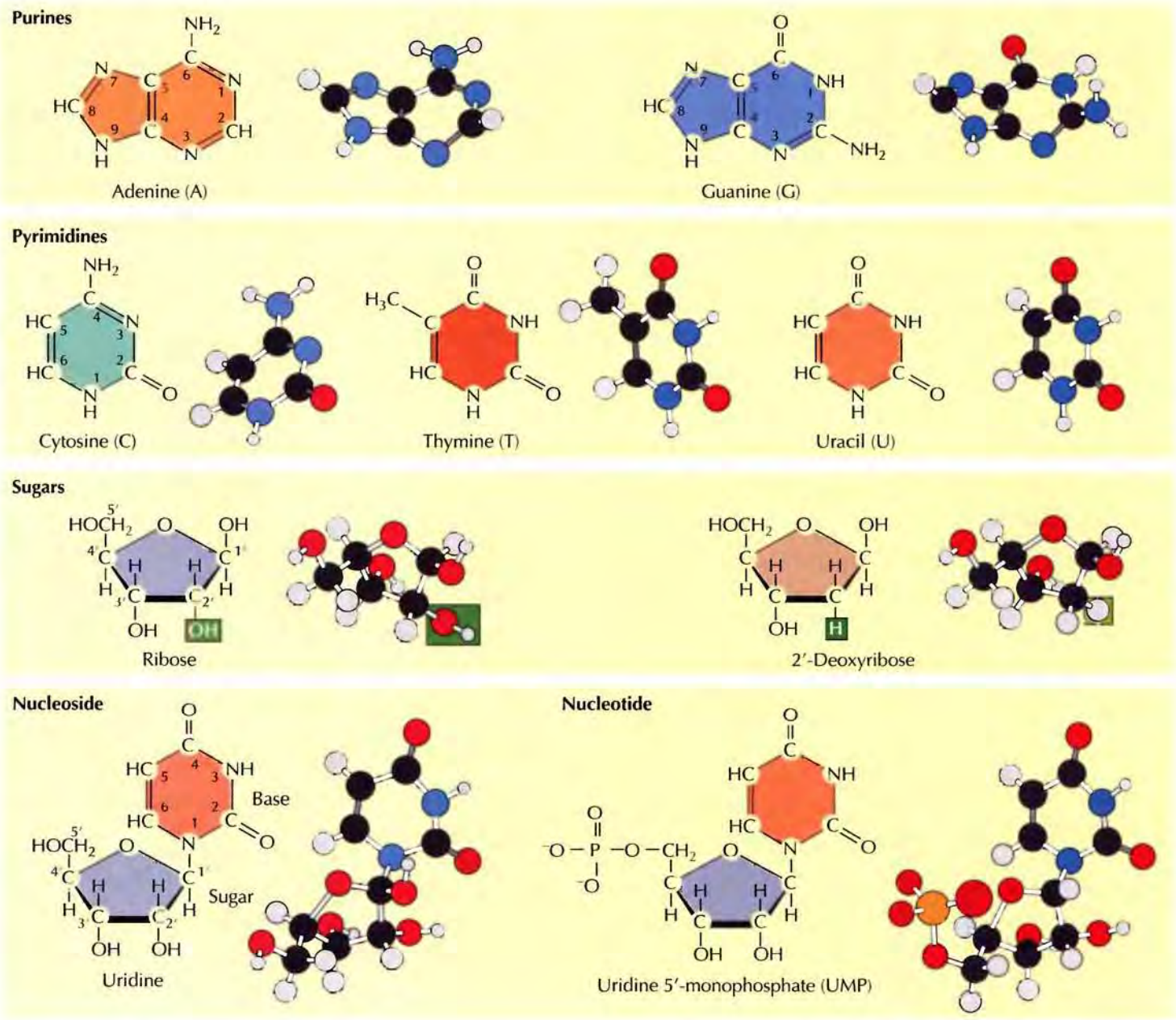


FIGURE 2.10 Components of nucleic acids Nucleic acids contain purine and pyrimidine bases linked to phosphorylated sugars. A nucleic acid base linked to a sugar alone is a nucleoside. Nucleotides additionally contain one or more phosphate groups.

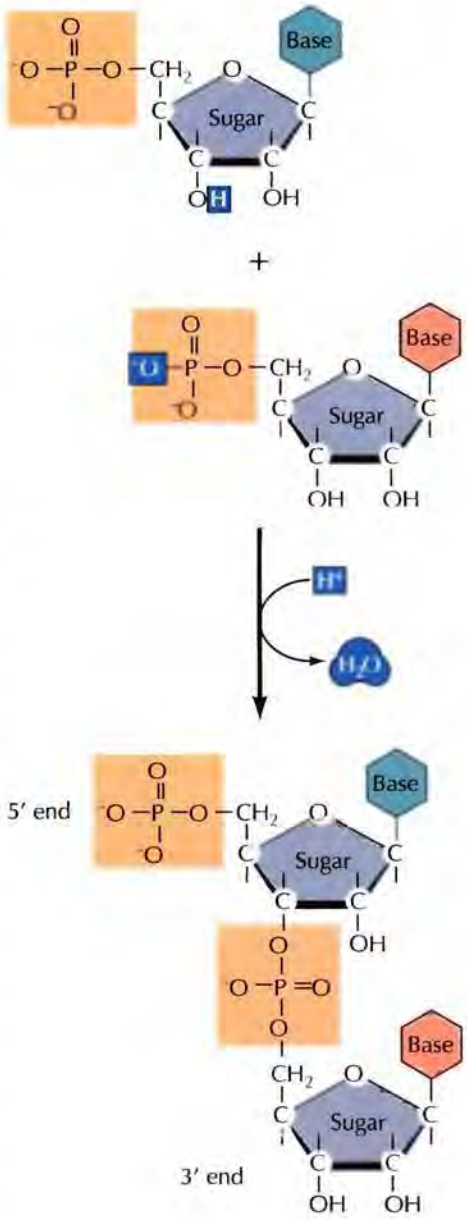


FIGURE 2.11 Polymerization of nucleotides A phosphodiester bond is formed between the 3′ hydroxyl group of one nucleotide and the 5′ phosphate group of another. A polynucleotide chain has a sense of direction, one end terminating in a 5′ phosphate group (the 5′ end) and the other end terminating in a 3′ hydroxyl group (the 3′ end).

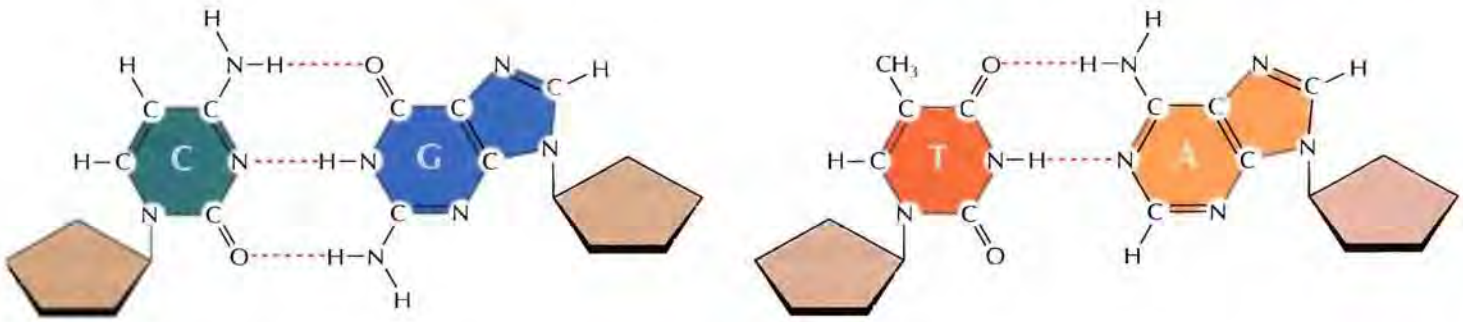
tains two purines (**adenine** and **guanine**) and two pyrimidines (**cytosine** and **thymine**). Adenine, guanine, and cytosine are also present in RNA, but RNA contains **uracil** in place of thymine. The bases are linked to sugars (**2′-deoxyribose** in DNA, or **ribose** in RNA) to form **nucleosides**. Nucleotides additionally contain one or more phosphate groups linked to the 5′ carbon of nucleoside sugars.

The polymerization of nucleotides to form nucleic acids involves the formation of **phosphodiester bonds** between the 5′ phosphate of one nucleotide and the 3′ hydroxyl of another (Figure 2.11). **Oligonucleotides** are small polymers containing only a few nucleotides; the large **polynucleotides** that make up cellular RNA and DNA may contain thousands or millions of nucleotides, respectively. It is important to note that a polynucleotide chain has a sense of direction, with one end of the chain terminating in a 5′ phosphate group and the other in a 3′ hydroxyl group. Polynucleotides are always synthesized in the 5′ to 3′ direction, with a free nucleotide being added to the 3′ OH group of a growing chain. By convention, the sequence of bases in DNA or RNA is also written in the 5′ to 3′ direction.

The information in DNA and RNA is conveyed by the order of the bases in the polynucleotide chain. DNA is a double-stranded molecule consisting of two polynucleotide chains running in opposite directions (see Chapter 4). The bases are on the inside of the molecule, and the two chains are joined by hydrogen bonds between complementary base pairs—adenine pairing with thymine and guanine with cytosine (Figure 2.12). The important consequence of such complementary base pairing is that one strand of DNA (or RNA) can act as a template to direct the synthesis of a complementary strand. Nucleic acids are thus uniquely capable of directing their own self-replication, allowing them to function as the fundamental informational molecules of the cell. The information carried by DNA and RNA directs the synthesis of specific proteins, which control most cellular activities.

Nucleotides are not only important as the building blocks of nucleic acids; they also play critical roles in other cell processes. Perhaps the most prominent example is adenosine 5′-triphosphate (ATP), which is the principal form of chemical energy within cells. Other nucleotides similarly function as carriers of either energy or reactive chemical groups in a wide variety of metabolic reactions. In addition, some nucleotides (e.g., cyclic AMP) are important signaling molecules within cells (see Chapter 15).

FIGURE 2.12 Complementary pairing between nucleic acid bases The formation of hydrogen bonds between bases on opposite strands of DNA leads to the specific pairing of guanine (G) with cytosine (C) and adenine (A) with thymine (T).



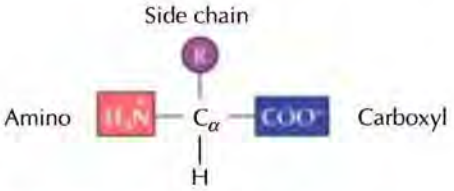


FIGURE 2.13 Structure of amino acids Each amino acid consists of a central carbon atom (the α carbon) bonded to a hydrogen atom, a carboxyl group, an amino group, and a specific side chain (designated R). At physiological pH, both the carboxyl and amino groups are ionized, as shown.

Proteins

While nucleic acids carry the genetic information of the cell, the primary responsibility of **proteins** is to execute the tasks directed by that information. Proteins are the most diverse of all macromolecules, and each cell contains several thousand different proteins, which perform a wide variety of functions. The roles of proteins include serving as structural components of cells and tissues, acting in the transport and storage of small molecules (e.g., the transport of oxygen by hemoglobin), transmitting information between cells (e.g., protein hormones), and providing a defense against infection (e.g., antibodies). The most fundamental property of proteins, however, is their ability to act as enzymes, which, as discussed in the next chapter, catalyze nearly all the chemical reactions in biological systems. Thus, proteins direct virtually all activities of the cell. The central importance of proteins in biological chemistry is indicated by their name, which is derived from the Greek word *proteios*, meaning "of the first rank."

Proteins are polymers of 20 different **amino acids**. Each amino acid consists of a carbon atom (called the α carbon) bonded to a carboxyl group (COO^-), an amino group (NH_3^+), a hydrogen atom, and a distinctive side chain (Figure 2.13). The specific chemical properties of the different amino acid side chains determine the roles of each amino acid in protein structure and function.

The amino acids can be grouped into four broad categories according to the properties of their side chains (Figure 2.14). Ten amino acids have nonpolar side chains that do not interact with water. Glycine is the simplest amino acid, with a side chain consisting of only a hydrogen atom. Alanine, valine, leucine, and isoleucine have hydrocarbon side chains consisting of up to four carbon atoms. The side chains of these amino acids are hydrophobic and therefore tend to be located in the interior of proteins, where they are not in contact with water. Proline similarly has a hydrocarbon side chain, but it is unique in that its side chain is bonded to the nitrogen of the amino group as well as to the α carbon, forming a cyclic structure. The side chains of two amino acids, cysteine and methionine, contain sulfur atoms. Methionine is quite hydrophobic, but cysteine is less so because of its sulfhydryl (SH) group. As discussed later, the sulfhydryl group of cysteine plays an important role in protein structure because disulfide bonds can form between the side chains of different cysteine residues. Finally, two nonpolar amino acids, phenylalanine and tryptophan, have side chains containing very hydrophobic aromatic rings.

■ **Besides being the building blocks of proteins, amino acids serve another important biological function as molecules used by cells to communicate. The amino acids glycine and glutamate, along with several other amino acids that are not found in proteins, act as neurotransmitters (see Chapter 15).**

Five amino acids have uncharged but polar side chains. These include serine, threonine, and tyrosine, which have hydroxyl groups on their side chains, as well as asparagine and glutamine, which have polar amide ($O{=}C{-}NH_2$) groups. Because the polar side chains of these amino acids can form hydrogen bonds with water, these amino acids are hydrophilic and tend to be located on the outside of proteins.

The amino acids lysine, arginine, and histidine have side chains with charged basic groups. Lysine and arginine are very basic amino acids, and their side chains are positively charged in the cell. Consequently, they are very hydrophilic and are found in contact with water on the surface of proteins. Histidine can be either uncharged or positively charged at physiological pH, so it frequently plays an active role in enzymatic reactions involving the exchange of hydrogen ions, as discussed in Chapter 3.

Finally, two amino acids, aspartic acid and glutamic acid, have acidic side chains terminating in carboxyl groups. These amino acids are negatively charged within the cell and are therefore frequently referred to as aspartate

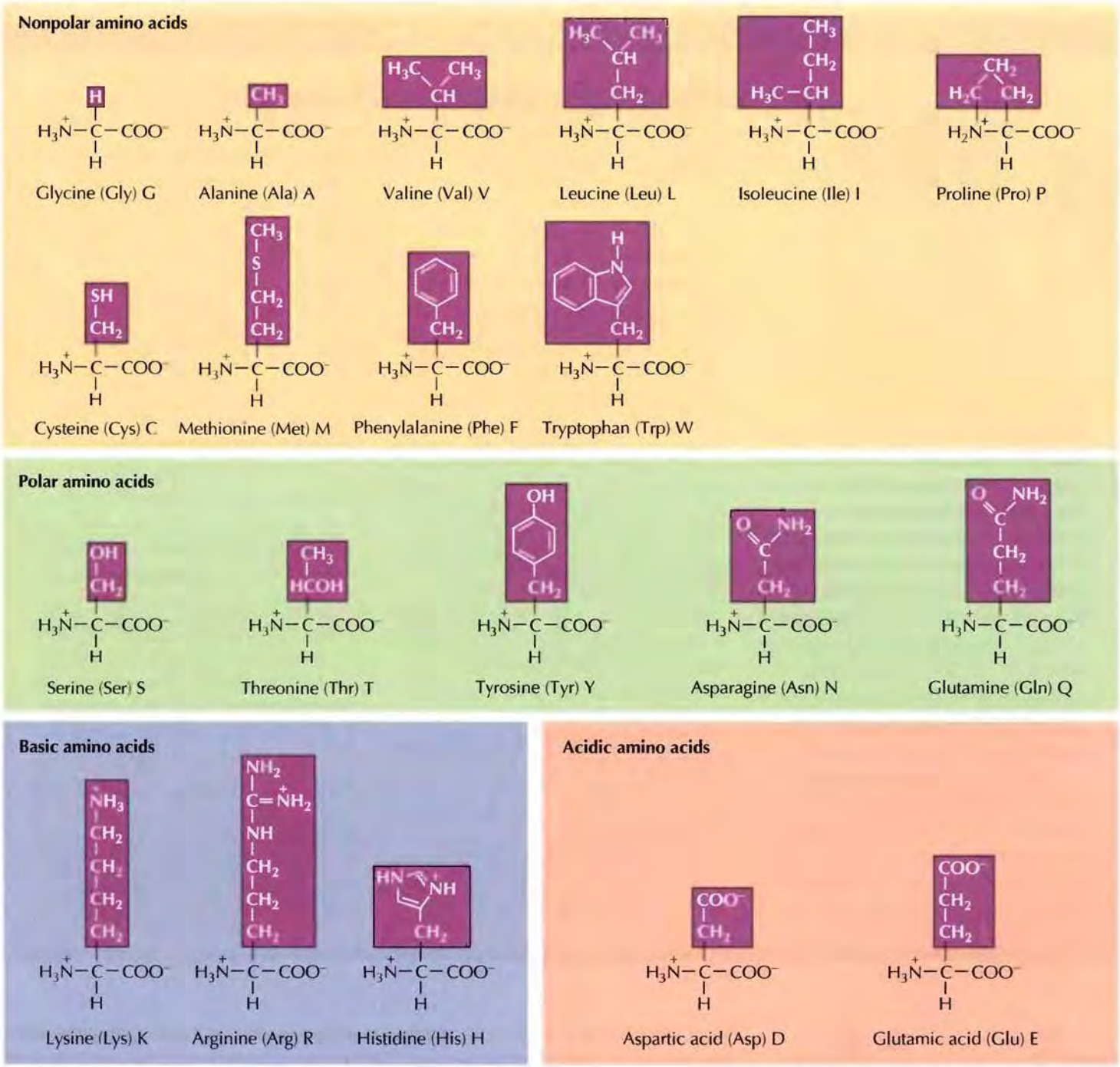


FIGURE 2.14 The amino acids
The three-letter and one-letter abbreviations for each amino acid are indicated. The amino acids are grouped into four categories according to the properties of their side chains: nonpolar, polar, basic, and acidic.

and glutamate. Like the basic amino acids, these acidic amino acids are very hydrophilic and are usually located on the surface of proteins.

Amino acids are joined together by **peptide bonds** between the α amino group of one amino acid and the α carboxyl group of a second (Figure 2.15). **Polypeptides** are linear chains of amino acids, usually hundreds or thousands of amino acids in length. Each polypeptide chain has two distinct ends, one terminating in an α amino group (the amino, or N, terminus) and the other in an α carboxyl group (the carboxy, or C, terminus). Polypeptides are synthesized from the amino to the carboxy terminus, and the sequence of amino acids in a polypeptide is written (by convention) in the same order.

KEY EXPERIMENT

The Folding of Polypeptide Chains

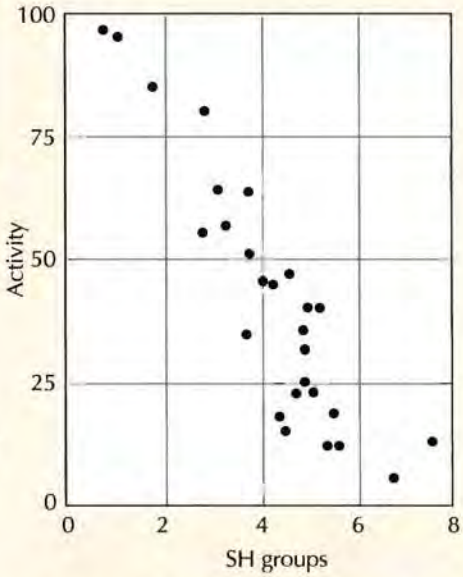
Reductive Cleavage of Disulfide Bridges in Ribonuclease

Michael Sela, Frederick H. White, Jr., and Christian B. Anfinsen
National Institutes of Health, Bethesda, MD
Science, Volume 125, 1957, pages 691–692

The Context

Functional proteins are structurally far more complex than linear chains of amino acids. The formation of active enzymes or other proteins requires the folding of polypeptide chains into precise three-dimensional conformations. This difference between proteins and polypeptide chains raises critical questions with respect to understanding protein structure and function. How is the correct protein conformation chosen from the many possible conformations that could be adopted by a polypeptide, and what is the nature of the information that directs protein folding?

Classic experiments done by Christian Anfinsen and his colleagues provided the answers to these questions. Studying the enzyme ribonuclease, Anfinsen and his collaborators were able to demonstrate that denatured proteins can spontaneously refold into an active conformation. Therefore, all the information required to specify the correct three-dimensional conformation of a protein is contained in its primary amino acid sequence. A series of such experiments led Anfinsen to the conclusion that the native three-dimensional structure of a protein corresponds to its thermodynamically most stable conformation, as determined by interactions between its constituent amino acids. The original observations leading to the formulation of this critical principle were reported in this 1957 paper by Christian Anfinsen, Michael Sela, and Fred White.



A summary of results of the renaturation experiments. The enzymatic activity of ribonuclease is plotted as a function of the number of sulfhydryl groups present after various treatments. Activity is expressed as percent activity of the native enzyme.

The Experiments

The protein studied by Sela, White, and Anfinsen was bovine ribonuclease, a small protein of 124 amino acids that contains four disulfide (S—S) bonds between side chains of cysteine residues. The enzymatic activity of ribonuclease could be

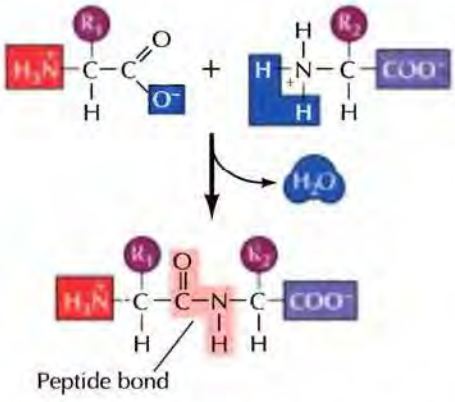


FIGURE 2.15 Formation of a peptide bond The carboxyl group of one amino acid is linked to the amino group of a second.

The defining characteristic of proteins is that they are polypeptides with specific amino acid sequences. In 1953 Frederick Sanger was the first to determine the complete amino acid sequence of a protein, the hormone insulin. Insulin was found to consist of two polypeptide chains, joined by disulfide bonds between cysteine residues (Figure 2.16). Most important, Sanger's experiment revealed that each protein consists of a specific amino acid sequence. Protein sequences are now usually deduced from the sequences of mRNAs, and the complete amino acid sequences of over 100,000 proteins have been established. Each consists of a unique sequence of amino acids, determined by the order of nucleotides in a gene (see Chapter 4).

The amino acid sequence of a protein is only the first element of its structure. Rather than being extended chains of amino acids, proteins adopt distinct three-dimensional conformations that are critical to their function. These three-dimensional conformations of proteins are the result of interactions between their constituent amino acids, so the shapes of proteins are determined by their amino acid sequences. This was first demonstrated by

KEY EXPERIMENT

determined by its ability to degrade RNA to nucleotides, providing a ready assay for function of the native protein. This enzymatic activity was completely lost when ribonuclease was subjected to treatments that both disrupted the noncovalent bonds (e.g., hydrogen bonds) and cleaved the disulfide bonds by reducing them to sulfhydryl (SH) groups. The denatured protein thus appeared to be in a random inactive conformation.

Sela, White, and Anfinsen made the crucial observation that enzymatic activity reappears if the denatured protein is incubated under conditions that allow the polypeptide chain to refold and disulfide bonds to re-form. In these experiments the denaturing agents were removed, and the inactive enzyme was incubated at room temperature in a physiological buffer in the presence of O_2. This procedure led to oxidation of the sulfhydryl groups and the re-formation of disulfide bonds. During this process, the enzyme regained its catalytic activity, indicating that it had refolded to its native conformation. Because no other cellular components were present, all the information required for proper protein folding appeared to be contained in the primary amino acid sequence of the polypeptide chain.

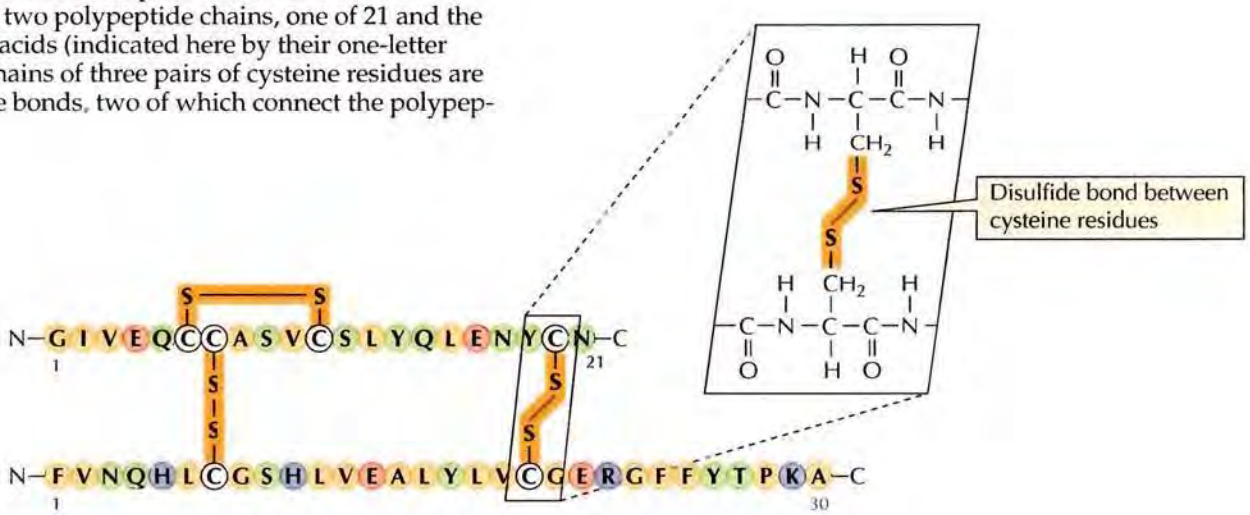
The Impact

Further experiments defined conditions under which denatured ribonuclease completely regains its native structure and enzymatic activity, establishing the "thermodynamic hypothesis" of protein folding—that is, the native three-dimensional structure of a protein corresponds to the thermodynamically most stable state under physiological conditions. Thermodynamic stability is governed by interactions of the constituent amino acids, so the three-dimensional conformations of proteins are determined directly by their primary amino acid sequences. Since the order of nucleotides in DNA specifies the amino acid sequence of a polypeptide, it follows that the nucleotide sequence of a gene contains all the information needed to determine the three-dimensional structure of its protein product.

Although Anfinsen's work established the thermodynamic basis of protein folding, understanding the mechanism of this process remains an active area of research. Protein folding is extremely complex, and we are still unable to deduce the three-dimensional structure of a protein directly from its amino acid sequence. It is also important to note that the spontaneous folding of proteins *in vitro* is much slower than protein folding within the cell, which is assisted by enzymes (see Chapter 8). The protein folding problem thus remains a central challenge in biological chemistry.

Christian B. Anfinsen

FIGURE 2.16 Amino acid sequence of insulin
Insulin consists of two polypeptide chains, one of 21 and the other of 30 amino acids (indicated here by their one-letter codes). The side chains of three pairs of cysteine residues are joined by disulfide bonds, two of which connect the polypeptide chains.



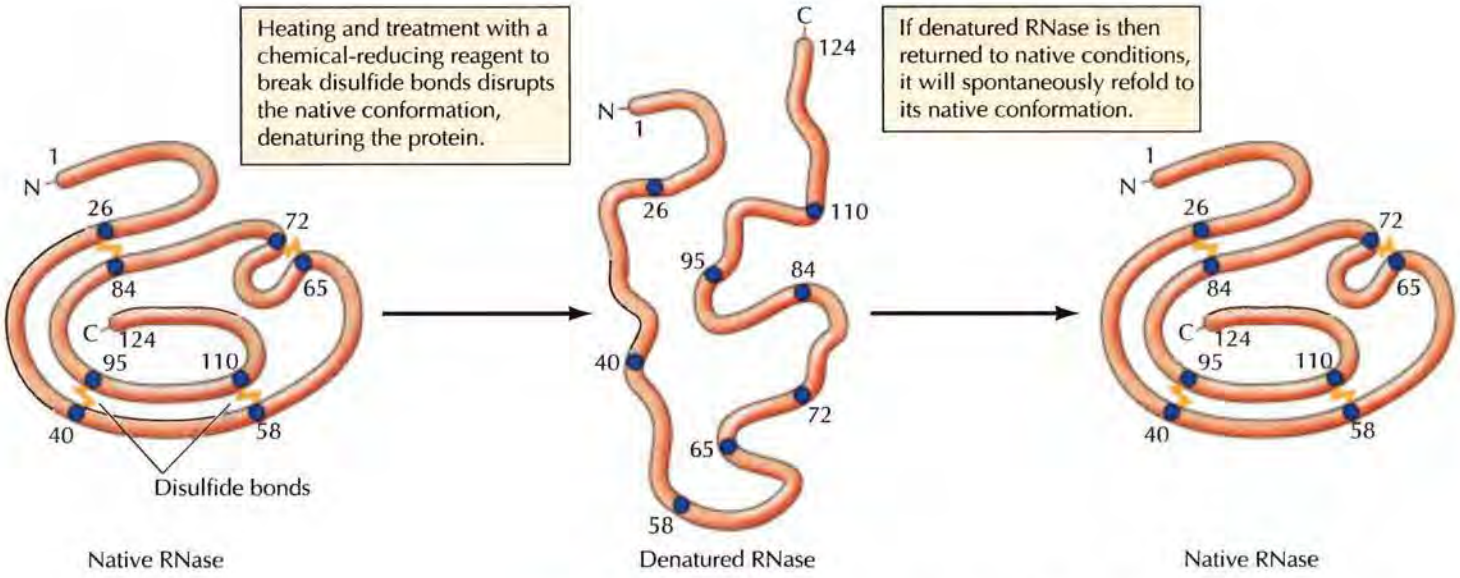


FIGURE 2.17 Protein denaturation and refolding Ribonuclease (RNase) is a protein of 124 amino acids (indicated by numbers). The protein is normally folded into its native conformation, which contains four disulfide bonds (indicated as paired circles representing the cysteine residues).

experiments of Christian Anfinsen in which he disrupted the three-dimensional structures of proteins by treatments, such as heating, that break noncovalent bonds—a process called denaturation (Figure 2.17). Following incubation under milder conditions, such denatured proteins often spontaneously returned to their native conformations, indicating that these conformations were directly determined by the amino acid sequence.

The three-dimensional structure of proteins is most frequently analyzed by **X-ray crystallography,** a high-resolution technique that can determine the arrangement of individual atoms within a molecule. A beam of X-rays is directed at crystals of the protein to be analyzed, and the pattern of X-rays that pass through the protein crystal is detected on X-ray film. As the X-rays strike the crystal, they are scattered in characteristic patterns determined by the arrangement of atoms in the molecule. The structure of the molecule can therefore be deduced from the pattern of scattered X-rays (the diffraction pattern).

In 1958 John Kendrew was the first person to determine the three-dimensional structure of a protein, myoglobin—a relatively simple protein of 153 amino acids (Figure 2.18). Since then, the three-dimensional structures of several thousand proteins have been analyzed. Most, like myoglobin, are globular proteins with polypeptide chains folded into compact structures, although some (such as the structural proteins of connective tissues) are long fibrous molecules. Analysis of the three-dimensional structures of these proteins has revealed several basic principles that govern protein folding, although protein structure is so complex that predicting the three-dimensional structure of a protein directly from its amino acid sequence is impossible.

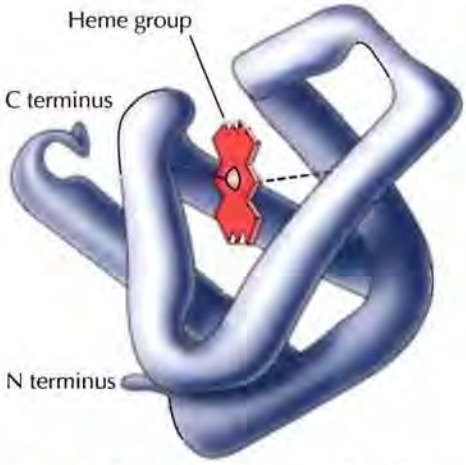


FIGURE 2.18 Three-dimensional structure of myoglobin Myoglobin is a protein of 153 amino acids that is involved in oxygen transport. The polypeptide chain is folded around a heme group that serves as the oxygen-binding site.

Protein structure is generally described as having four levels. The **primary structure** of a protein is the sequence of amino acids in its polypeptide chain. The **secondary structure** is the regular arrangement of amino acids within localized regions of the polypeptide. Two types of secondary struc-

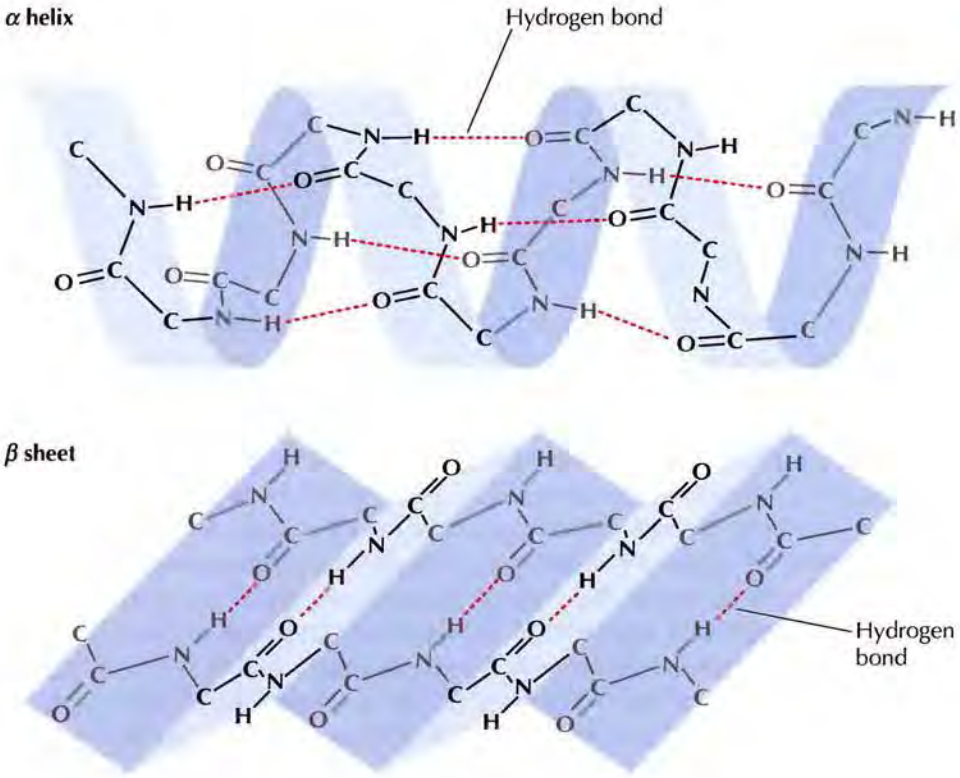


FIGURE 2.19 Secondary structure of proteins The most common types of secondary structure are the α helix and the β sheet. In an α helix, hydrogen bonds form between CO and NH groups of peptide bonds separated by four amino acid residues. In a β sheet, hydrogen bonds connect two parts of a polypeptide chain lying side by side. The amino acid side chains are not shown.

ture, which were first proposed by Linus Pauling and Robert Corey in 1951, are particularly common: the **α helix** and the **β sheet**. Both of these secondary structures are held together by hydrogen bonds between the CO and NH groups of peptide bonds. An α helix is formed when a region of a polypeptide chain coils around itself, with the CO group of one peptide bond forming a hydrogen bond with the NH group of a peptide bond located four residues downstream in the linear polypeptide chain (Figure 2.19). In contrast, a β sheet is formed when two parts of a polypeptide chain lie side by side with hydrogen bonds between them. Such β sheets can be formed between several polypeptide strands, which can be oriented either parallel or antiparallel to each other.

Tertiary structure is the folding of the polypeptide chain as a result of interactions between the side chains of amino acids that lie in different regions of the primary sequence (Figure 2.20). In most proteins, combinations of α helices and β sheets, connected by loop regions of the polypeptide chain, fold into compact globular structures called **domains**,

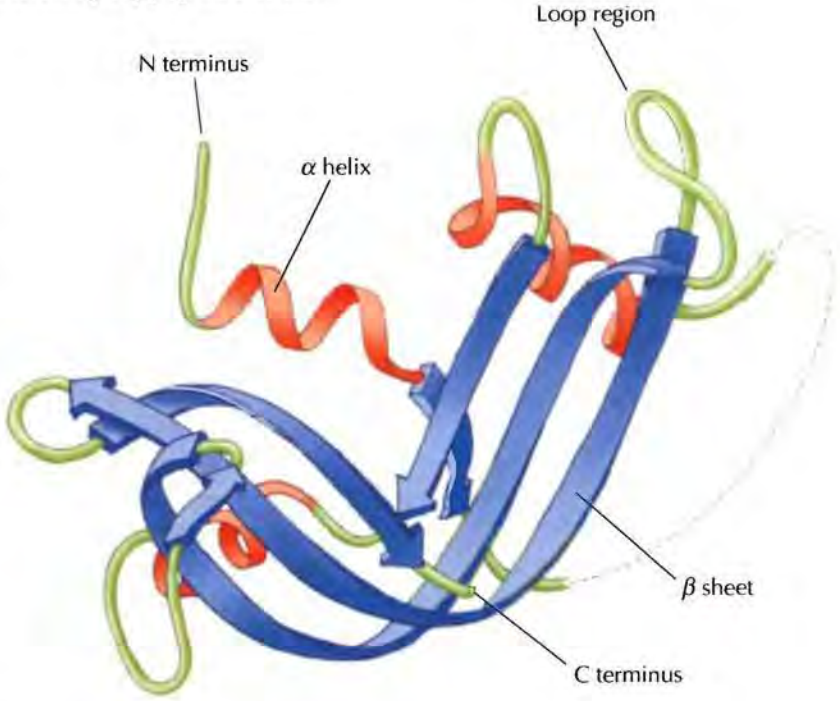


FIGURE 2.20 Tertiary structure of ribonuclease Regions of α-helix and β-sheet secondary structures, connected by loop regions, are folded into the native conformation of the protein. In this schematic representation of the polypeptide chain as a ribbon model, α helices are represented as spirals and β sheets as wide arrows.

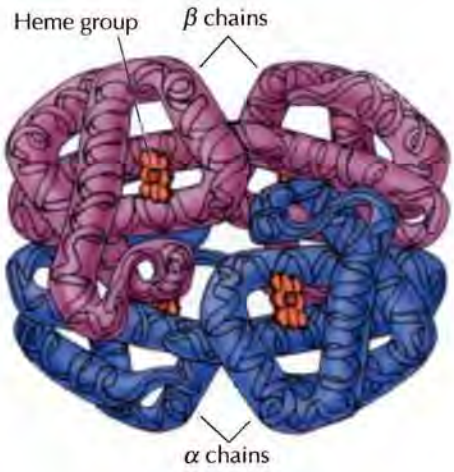


FIGURE 2.21 Quaternary structure of hemoglobin Hemoglobin is composed of four polypeptide chains, each of which is bound to a heme group. The two α chains and the two β chains are identical.

which are the basic units of tertiary structure. Small proteins, such as ribonuclease or myoglobin, contain only a single domain; larger proteins may contain a number of different domains, which are frequently associated with distinct functions.

A critical determinant of tertiary structure is the localization of hydrophobic amino acids in the interior of the protein and of hydrophilic amino acids on the surface, where they interact with water. The interiors of folded proteins thus consist mainly of hydrophobic amino acids arranged in α helices and β sheets; these secondary structures are found in the hydrophobic cores of proteins because hydrogen bonding neutralizes the polar character of the CO and NH groups of the polypeptide backbone. The loop regions connecting the elements of secondary structure are found on the surface of folded proteins, where the polar components of the peptide bonds form hydrogen bonds with water or with the polar side chains of hydrophilic amino acids. Interactions between polar amino acid side chains (hydrogen bonds and ionic bonds) on the protein surface are also important determinants of tertiary structure. In addition, the covalent disulfide bonds between the sulfhydryl groups of cysteine residues stabilize the folded structures of many cell-surface or secreted proteins.

The fourth level of protein structure, **quaternary structure**, consists of the interactions between different polypeptide chains in proteins composed of more than one polypeptide. Hemoglobin, for example, is composed of four polypeptide chains held together by the same types of interactions that maintain tertiary structure (Figure 2.21).

The distinct chemical characteristics of the 20 different amino acids thus lead to considerable variation in the three-dimensional conformations of folded proteins. Consequently, proteins constitute an extremely complex and diverse group of macromolecules, suited to the wide variety of tasks they perform in cell biology.

■ **Proteins exhibit incredible diversity. Some unusual proteins include:**

Cement proteins that are used by barnacles to attach to underwater surfaces. Scientists are trying to develop these proteins for industrial uses as glues or anticorrosive coatings.

Silk from the Golden Orb spider is extremely light and flexible but almost as strong as Kevlar, which is used to make bullet-proof vests.

Cell Membranes

The structure and function of cells are critically dependent on membranes, which not only separate the interior of the cell from its environment but also define the internal compartments of eukaryotic cells, including the nucleus and cytoplasmic organelles. The formation of biological membranes is based on the properties of lipids, and all cell membranes share a common structural organization: bilayers of phospholipids with associated proteins. These membrane proteins are responsible for many specialized functions; some act as receptors that allow the cell to respond to external signals, some are responsible for the selective transport of molecules across the membrane, and others participate in electron transport and oxidative phosphorylation. In addition, membrane proteins control the interactions between cells of multicellular organisms. The common structural organization of membranes thus underlies a variety of biological processes and specialized membrane functions, which will be discussed in detail in later chapters.

KEY EXPERIMENT

The Structure of Cell Membranes

The Fluid Mosaic Model of the Structure of Cell Membranes

S. J. Singer and Garth L. Nicolson

University of California at San Diego and the Salk Institute for Biological Sciences, La Jolla, CA

Science, Volume 175, 1972, pages 720–731

S. J. Singer

Garth L. Nicolson

The Context

Cell membranes play central roles in virtually all aspects of cell biology, serving as boundaries of the subcellular organelles of eukaryotic cells as well as defining the cell itself by separating its contents from the external environment. Understanding the structure and organization of cell membranes was thus a critical step in unraveling the molecular basis of cell behavior. In the 1960s, it was known that cell membranes were composed of proteins and lipids, but it was not clear how these molecules were organized. Moreover, there was a large amount of variation in the protein and lipid compositions of different cell membranes, leading some investigators to believe that different membranes might not share a common structural organization.

In contrast, Jonathan Singer and Garth Nicolson approached the problem of understanding membrane structure with the assumption that the same general principles would describe the organization of lipids and proteins in all cell membranes. They applied the principles of thermodynamics as well as integrating a variety of experimental results to develop the fluid mosaic model—a model that has stood the test of time and provided the structural basis for understanding the diverse functions of membranes in cell biology.

The Experiments

Considerations of thermodynamics were basic to Singer and Nicolson's development of the fluid mosaic model. In particular, they reasoned that the overall structure of membranes would be determined by a combination of hydrophobic and hydrophilic interactions. These interactions were recognized as determinants of the organization of phospholipid bilayers, and Singer and Nicolson assumed that similar interactions would govern the organization of membrane proteins. They therefore proposed that the nonpolar amino acid residues of membrane proteins would be sequestered within the membrane, like the fatty acid chains of the phospholipids, whereas polar groups of proteins would be exposed to the aqueous environment. These considerations argued strongly against an earlier model of membrane structure in which it was proposed that membranes had a three-layer structure consisting of an inner lipid bilayer sandwiched between two protein layers.

Singer and Nicolson instead proposed a mosaic structure, in which globular integral membrane proteins were inserted into a phospholipid bilayer (see figure). The proteins, like the phospholipids, were postulated to be amphipathic, with nonpolar regions inserted into the lipid bilayer and polar regions exposed to the aqueous environment. The lipid bilayer was thought to constitute the structural matrix of the membrane in which the proteins were embedded.

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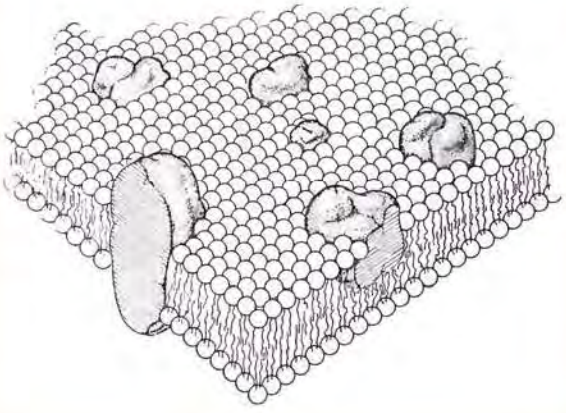

The lipid-globular protein mosaic model. Globular proteins are distributed in a matrix of phospholipids.

KEY EXPERIMENT

Several lines of experimental evidence were cited in support of the fluid mosaic model. First, electron microscopy of membranes using the freeze fracture technique provided direct evidence suggesting that proteins were embedded in the lipid bilayer (see Figure 1.36). Also as predicted by the fluid mosaic model, proteins were found to be randomly distributed in the plane of the membrane when detected by staining the surface of cells. And finally, as discussed in Chapter 13, experiments of Larry Frye and Michael Edidin demonstrated that proteins could move laterally through the membrane. Multiple experimental approaches thus indicated that membrane proteins were inserted into and able to move laterally through the lipid bilayer, providing strong support for the fluid mosaic model of membrane structure.

The Impact

Singer and Nicolson's fluid mosaic model has been abundantly supported by further experimentation, and continues to represent our basic understanding of the structure of cell membranes. Given the diverse roles of membranes in cell biology, the fluid mosaic model is thus fundamental to understanding virtually all aspects of cell behavior, including phenomena as diverse as energy metabolism, hormone action, and the transmission of information between nerve cells at a synapse. Initially based on thermodynamic considerations of the properties of lipids and proteins, Singer and Nicolson developed a model of membrane structure that has impacted all aspects of cell biology.

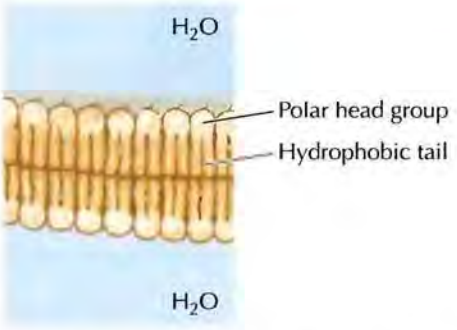

FIGURE 2.22 A phospholipid bilayer Phospholipids spontaneously form stable bilayers, with their polar head groups exposed to water and their hydrophobic tails buried in the interior of the membrane.

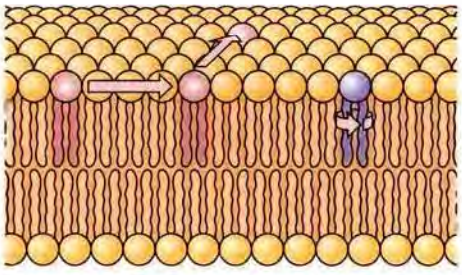

FIGURE 2.23 Mobility of phospholipids in a membrane Individual phospholipids can rotate and move laterally within a bilayer.

Membrane Lipids

The fundamental building blocks of all cell membranes are phospholipids, which are amphipathic molecules, consisting of two hydrophobic fatty acid chains linked to a phosphate-containing hydrophilic head group (see Figure 2.7). Because their fatty acid tails are poorly soluble in water, phospholipids spontaneously form bilayers in aqueous solutions, with the hydrophobic tails buried in the interior of the membrane and the polar head groups exposed on both sides, in contact with water (Figure 2.22). Such **phospholipid bilayers** form a stable barrier between two aqueous compartments and represent the basic structure of all biological membranes.

Lipids constitute approximately 50% of the mass of most cell membranes, although this proportion varies depending on the type of membrane. Plasma membranes, for example, are approximately 50% lipid and 50% protein. The inner membrane of mitochondria, on the other hand, contains an unusually high fraction (about 75%) of protein, reflecting the abundance of protein complexes involved in electron transport and oxidative phosphorylation. The lipid composition of different cell membranes also varies (Table 2.1). The plasma membrane of *E. coli* consists predominantly of phosphatidylethanolamine, which constitutes 80% of total membrane lipid. Mammalian plasma membranes are more complex, containing four major phospholipids—phosphatidylcholine, phosphatidylserine, phosphatidylethanolamine, and sphingomyelin—which together constitute 50 to 60% of total membrane lipid. In addition to the phospholipids, the plasma membranes of animal cells contain glycolipids and cholesterol, which generally correspond to about 40% of the total membrane lipid molecules.

An important property of lipid bilayers is that they behave as two-dimensional fluids in which individual molecules (both lipids and proteins) are free to rotate and move in lateral directions (Figure 2.23). Such fluidity is a critical property of membranes and is determined by both temperature and lipid composition. For example, the interactions between shorter fatty acid chains are weaker than those between longer chains, so membranes containing shorter fatty acid chains are less rigid and remain fluid at lower temperatures. Lipids containing unsaturated fatty acids similarly increase

TABLE 2.1 Lipid Composition of Cell Membranes[a]

Lipid	Plasma membrane		Rough endoplasmic reticulum	Outer mitochondrial membranes
	E. coli	Erythrocyte		
Phosphatidylcholine	0	17	55	50
Phosphatidylserine	0	6	3	2
Phosphatidyl-ethanolamine	80	16	16	23
Sphingomyelin	0	17	3	5
Glycolipids	0	2	0	0
Cholesterol	0	45	6	<5

Source: Data from P. L. Yeagle, 1993. *The Membranes of Cells*, 2nd ed. San Diego, CA: Academic Press.
[a] Membrane compositions are indicated as the mole percentages of major lipid constituents.

membrane fluidity because the presence of double bonds introduces kinks in the fatty acid chains, making them more difficult to pack together.

Because of its hydrocarbon ring structure (see Figure 2.9), cholesterol plays a distinct role in determining membrane fluidity. Cholesterol molecules insert into the bilayer with their polar hydroxyl groups close to the hydrophilic head groups of the phospholipids (Figure 2.24). The rigid hydrocarbon rings of cholesterol therefore interact with the regions of the fatty acid chains that are adjacent to the phospholipid head groups. This interaction decreases the mobility of the outer portions of the fatty acid chains, making this part of the membrane more rigid. On the other hand, insertion of cholesterol interferes with interactions between fatty acid chains, thereby maintaining membrane fluidity at lower temperatures.

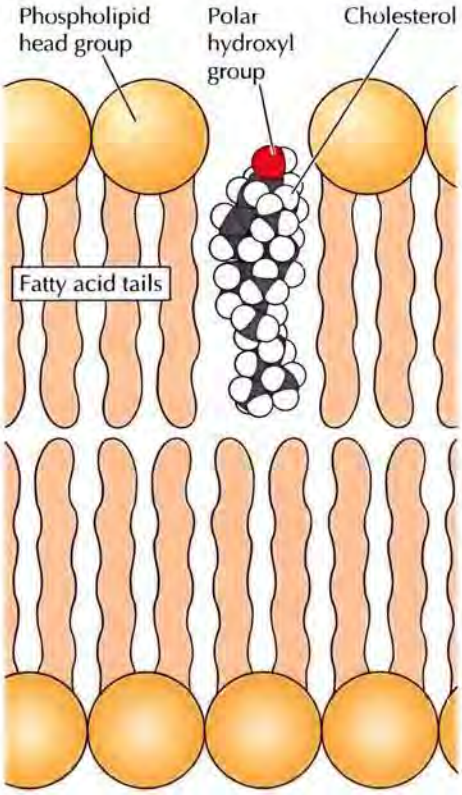

FIGURE 2.24 Insertion of cholesterol in a membrane Cholesterol inserts into the membrane with its polar hydroxyl group close to the polar head groups of the phospholipids.

Membrane Proteins

Proteins are the other major constituent of cell membranes, constituting 25 to 75% of the mass of the various membranes of the cell. The current model of membrane structure, proposed by Jonathan Singer and Garth Nicolson in 1972, views the membrane as a **fluid mosaic** in which proteins are inserted into a lipid bilayer (Figure 2.25). While phospholipids provide the basic structural organization of membranes, membrane proteins carry out the specific functions of the different membranes of the cell. These proteins are divided into two general classes, based on the nature of their association with the membrane. **Integral membrane proteins** are embedded directly within the lipid bilayer. **Peripheral membrane proteins** are not inserted into the lipid bilayer but are associated with the membrane indirectly, generally by interactions with integral membrane proteins.

Most integral membrane proteins (called **transmembrane proteins**) span the lipid bilayer, with portions exposed on both sides of the membrane. The membrane-spanning portions of these proteins are usually α-helical regions of 20 to 25 nonpolar amino acids. The hydrophobic side chains of these amino acids interact with the fatty acid chains of membrane lipids, and the formation of an α helix neutralizes the polar character of the peptide bonds, as discussed earlier in this chapter with respect to protein folding. The only other protein structure known to span lipid bilayers is the **β-barrel**, formed by the folding of β sheets into a barrel-like structure, which is found in some transmembrane proteins of bacteria, chloroplasts, and mitochondria (Figure 2.26). Like the phospholipids, transmembrane proteins are amphipathic molecules,

■ **How do plants sense cold? Evidence suggests that plants detect drops in temperature in part by sensing changes in the fluidity of their membranes.**

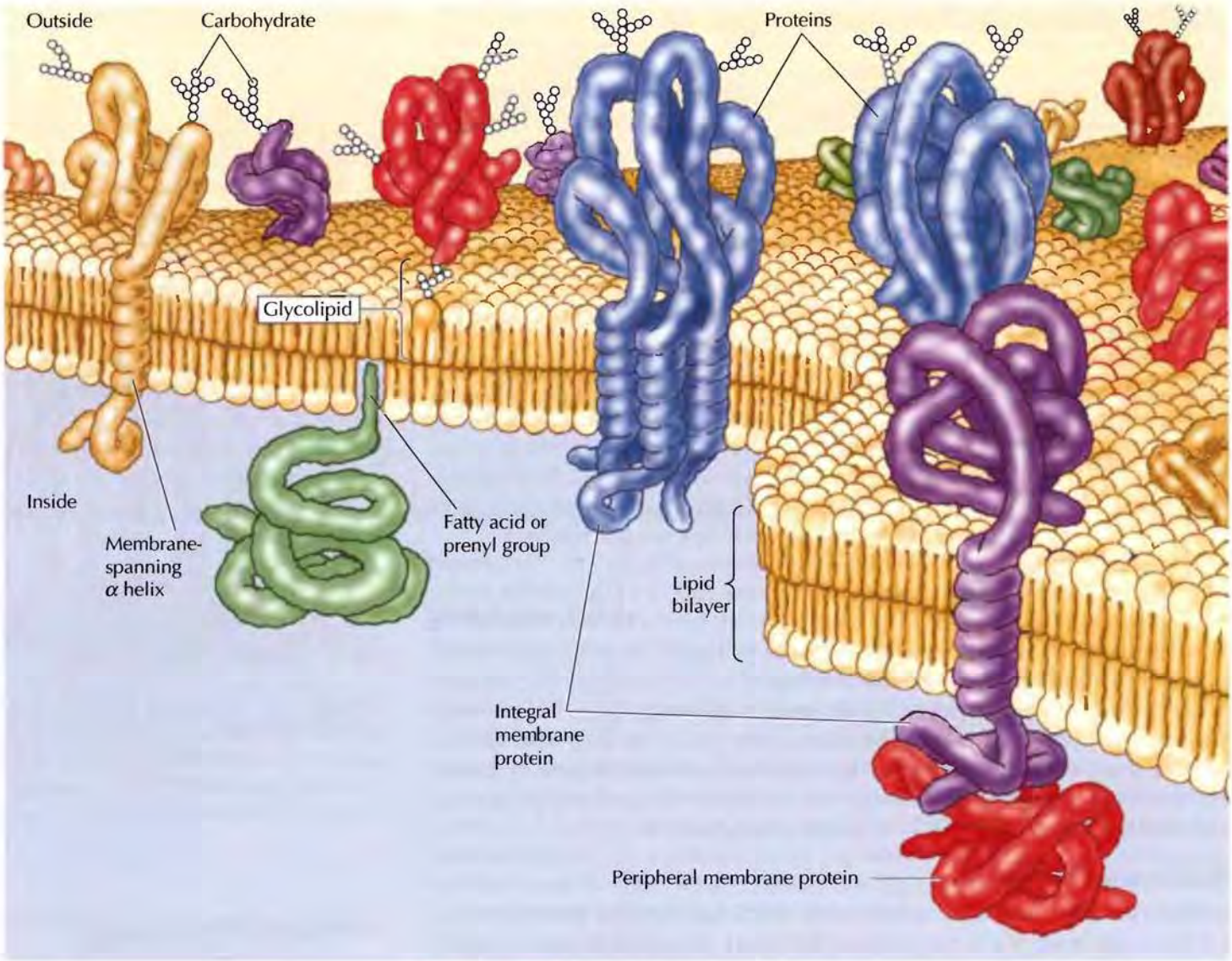

FIGURE 2.25 Fluid mosaic model of membrane structure Biological membranes consist of proteins inserted into a lipid bilayer. Integral membrane proteins are embedded in the membrane, usually via α-helical regions of 20 to 25 hydrophobic amino acids. Some transmembrane proteins span the membrane only once; others have multiple membrane-spanning regions. In addition, some proteins are anchored in the membrane by lipids that are covalently attached to the polypeptide chain. These proteins can be anchored to the extracellular face of the plasma membrane by glycolipids and to the cytosolic face by fatty acids or prenyl groups (see Chapter 8 for structures). Peripheral membrane proteins are not inserted in the membrane but are attached via interactions with integral membrane proteins.

with their hydrophilic portions exposed to the aqueous environment on both sides of the membrane. Some transmembrane proteins span the membrane only once; others have multiple membrane-spanning regions. Most transmembrane proteins of eukaryotic plasma membranes have been modified by the addition of carbohydrates, which are exposed on the surface of the cell and may participate in cell-cell interactions.

Proteins can also be anchored in membranes by lipids that are covalently attached to the polypeptide chain (see Chapter 8). Distinct lipid modifications anchor proteins to the cytosolic and extracellular faces of the plasma

membrane. Proteins can be anchored to the cytosolic face of the membrane either by the addition of a 14-carbon fatty acid (myristic acid) to their amino terminus or by the addition of either a 16-carbon fatty acid (palmitic acid) or 15- or 20-carbon prenyl groups to the side chains of cysteine residues. Alternatively, proteins are anchored to the extracellular face of the plasma membrane by the addition of glycolipids to their carboxy terminus.

Transport across Cell Membranes

The selective permeability of biological membranes to small molecules allows the cell to control and maintain its internal composition. Only small uncharged molecules can diffuse freely through phospholipid bilayers (Figure 2.27). Small nonpolar molecules, such as O_2 and CO_2, are soluble in the lipid bilayer and therefore can readily cross cell membranes. Small uncharged polar molecules, such as H_2O, also can diffuse through membranes, but larger uncharged polar molecules, such as glucose, cannot. Charged molecules, such as ions, are unable to diffuse through a phospholipid bilayer regardless of size; even H^+ ions cannot cross a lipid bilayer by free diffusion.

Although ions and most polar molecules cannot diffuse across a lipid bilayer, many such molecules (such as glucose) are able to cross cell membranes. These molecules pass across membranes via the action of specific transmembrane proteins, which act as transporters. Such transport proteins determine the selective permeability of cell membranes and thus play a critical role in membrane function. They contain multiple membrane-spanning regions that form a passage through the lipid bilayer, allowing polar or charged molecules to cross the membrane through a protein pore without interacting with the hydrophobic fatty acid chains of the membrane phospholipids.

As discussed in detail in Chapter 13, there are two general classes of membrane transport proteins (Figure 2.28). **Channel proteins** form open pores through the membrane, allowing the free passage of any molecule of the appropriate size. Ion channels, for example, allow the passage of inor-

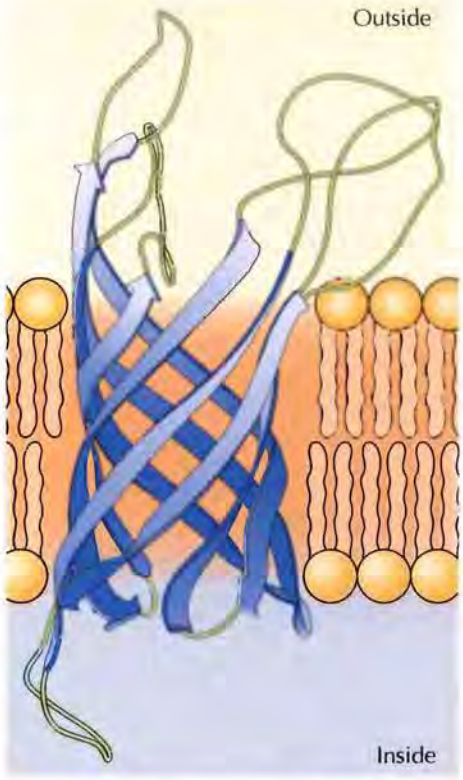

FIGURE 2.26 Structure of a β-barrel Some transmembrane proteins span the phospholipid bilayer as β-sheets folded into a barrel-like structure.

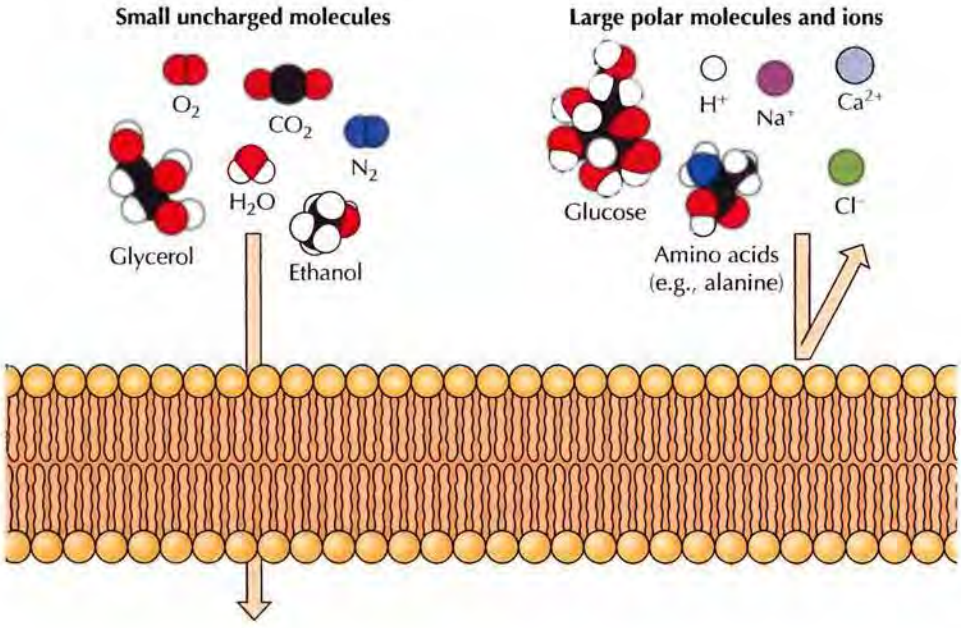

FIGURE 2.27 Permeability of phospholipid bilayers Small uncharged molecules can diffuse freely through a phospholipid bilayer. However, the bilayer is impermeable to larger polar molecules (such as glucose and amino acids) and to ions.

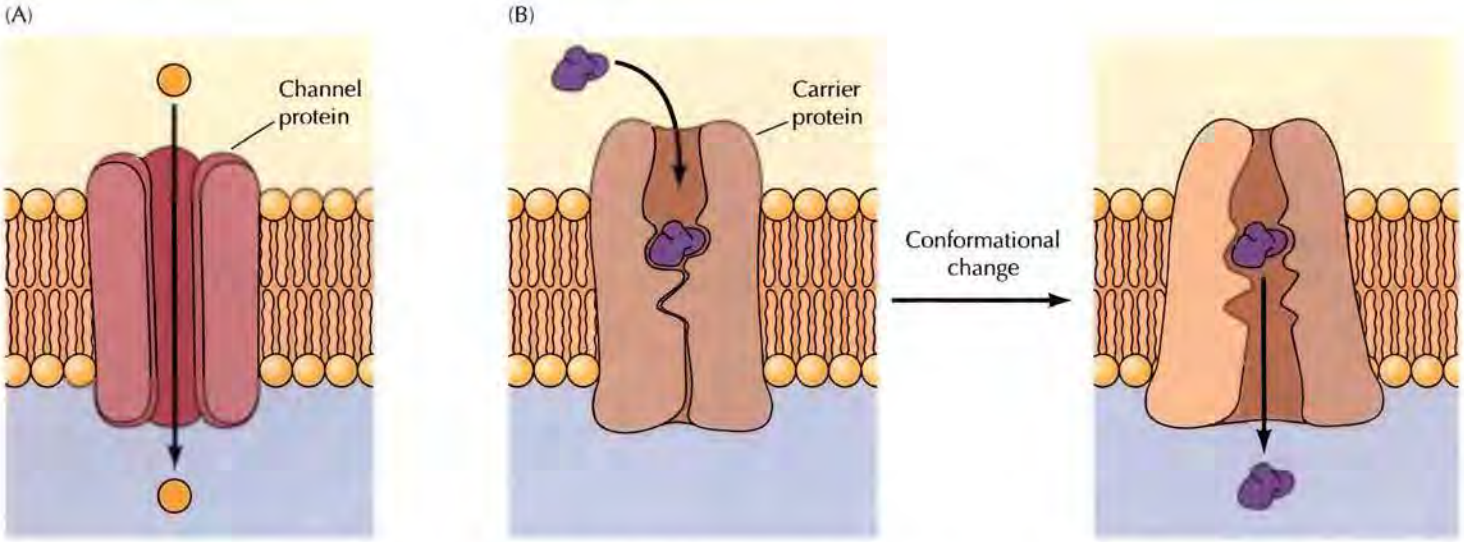

FIGURE 2.28 Channel and carrier proteins (A) Channel proteins form open pores through which molecules of the appropriate size (e.g., ions) can cross the membrane. (B) Carrier proteins selectively bind the small molecule to be transported and then undergo a conformational change to release the molecule on the other side of the membrane.

2.2 WEBSITE ANIMATION

Passive transport Simple diffusion and facilitated diffusion are types of passive transport by which certain molecules cross biological membranes.

ganic ions such as Na^+, K^+, Ca^{2+}, and Cl^- across the plasma membrane. Once open, channel proteins form small pores through which ions of the appropriate size and charge can cross the membrane by free diffusion. The pores formed by these channel proteins are not permanently open; rather, they can be selectively opened and closed in response to extracellular signals, allowing the cell to control the movement of ions across the membrane. Such regulated ion channels have been particularly well studied in nerve and muscle cells, where they mediate the transmission of electrochemical signals.

In contrast to channel proteins, **carrier proteins** selectively bind and transport specific small molecules, such as glucose. Rather than forming open channels, carrier proteins act like enzymes to facilitate the passage of specific molecules across membranes. In particular, carrier proteins bind specific molecules and then undergo conformational changes that open channels through which the molecule to be transported can pass across the membrane and be released on the other side.

As described so far, molecules transported by either channel or carrier proteins cross membranes in the energetically favorable direction, as determined by concentration and electrochemical gradients—a process known as **passive transport**. However, carrier proteins also provide a mechanism through which the energy changes associated with transporting molecules across a membrane can be coupled to the use or production of other forms of metabolic energy, just as enzymatic reactions can be coupled to the hydrolysis or synthesis of ATP. For example, molecules can be transported in an energetically unfavorable direction across a membrane (e.g., against a concentration gradient) if their transport in that direction is coupled to ATP hydrolysis as a source of energy—a process called **active transport** (Figure 2.29). Membrane proteins can thus use the free energy stored as ATP to control the internal composition of the cell.

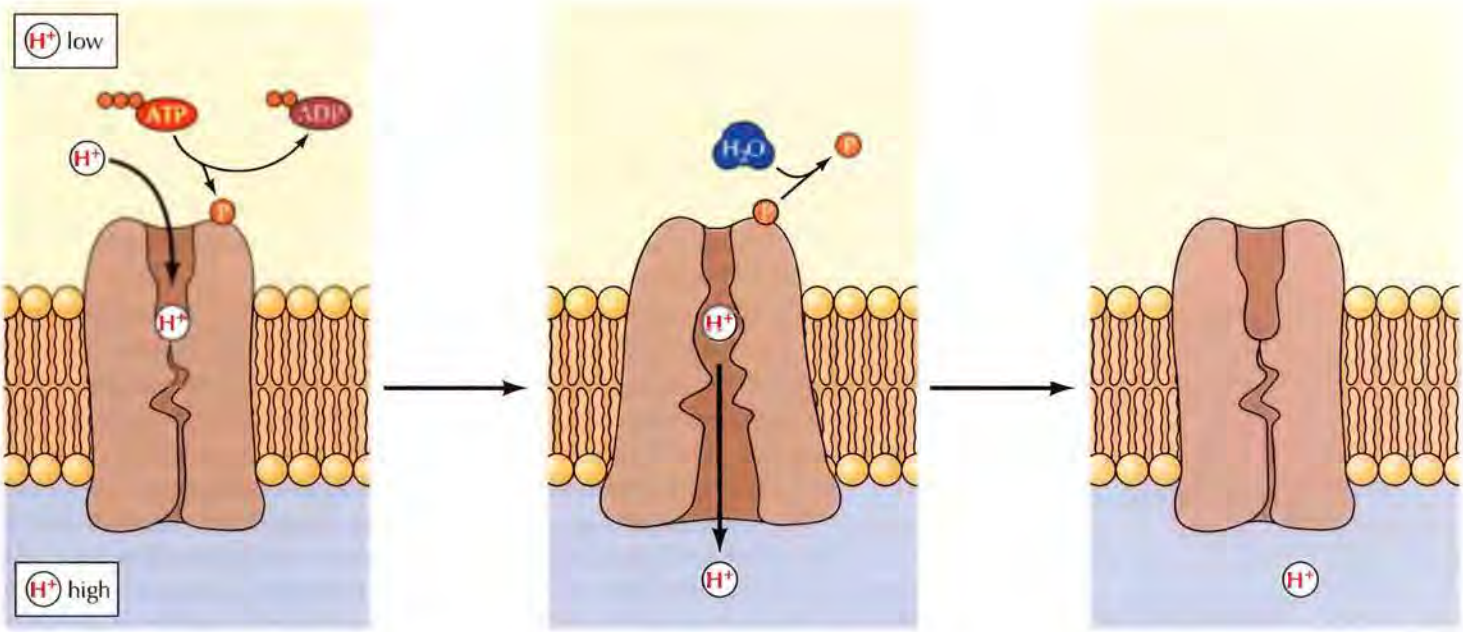

FIGURE 2.29 Model of active transport Energy derived from the hydrolysis of ATP is used to transport H^+ against the electrochemical gradient (from low to high H^+ concentration). Binding of H^+ is accompanied by phosphorylation of the carrier protein, which induces a conformational change that drives H^+ transport against the electrochemical gradient. Release of H^+ and hydrolysis of the bound phosphate group then restores the carrier to its original conformation.

Proteomics: Large-Scale Analysis of Cell Proteins

The last few years have seen a number of major changes in the way scientists approach cell and molecular biology, with large-scale experimental approaches being applied to understand the complexities of biological systems. Most dramatic have been the genome sequencing projects, which have yielded the complete sequence of the human genome, as well as the genome sequences of a number of model organisms. As will be discussed in later chapters, this systematic analysis of cell genomes (**genomics**) is making a major impact on our understanding of cell function.

However, the analysis of cell genomes is only the first step in understanding the workings of a cell. Since proteins are directly responsible for carrying out almost all cell activities, it is necessary to understand not only what proteins can be encoded by a cell's genome but also what proteins are actually expressed and how they function within the cell. A complete understanding of cell function therefore requires not only the sequence of its genome, but also a systematic analysis of its protein complement. This large-scale analysis of cell proteins (**proteomics**) has the goal of identifying and quantifying all of the proteins expressed in a given cell (the **proteome**), as well as establishing the localization of these proteins to different subcellular organelles and elucidating the networks of interactions between proteins that govern cell activities.

Identification of Cell Proteins

The number of distinct species of proteins in eukaryotic cells is typically far greater than the number of genes. This arises because many genes can

be expressed to yield several distinct mRNAs, which encode different polypeptides as a result of alternative splicing (discussed in Chapter 5). In addition, proteins can be modified in a variety of different ways, including the addition of phosphate groups, carbohydrates, and lipid molecules. The human genome, for example, contains approximately 20,000 to 25,000 different genes, and the number of genes expressed in any given cell is thought to be around 10,000. However, because of alternative splicing and protein modifications, it is estimated that these genes give rise to more than 100,000 different proteins. In addition, these proteins can be expressed at a wide range of different levels. Characterization of the complete protein complement of a cell, the goal of proteomics, thus represents a considerable challenge. Although major progress has been made in the last several years, substantial technological hurdles remain to be overcome before a complete characterization of cell proteomes can be achieved.

The first technology developed for the large scale separation of cell proteins was **two-dimensional gel electrophoresis** (Figure 2.30), which remains a well-established strategy for proteomic experiments. A mixture of cell proteins is first subjected to electrophoresis in a tube with a pH gradient running from end to end. Each protein migrates according to charge until it reaches a pH at which the charge of the protein is neutralized, as determined by the protein's content of acidic and basic amino acids. The proteins are then subjected to electrophoresis in a second dimension under conditions where they separate according to mass, with lower molecular weight proteins moving more rapidly through a gel. This approach is capable of resolving several thousand protein species from a cell extract. However, this is much less than the total number of proteins expressed in mammalian cells, and it is important to note that two-dimensional gel electrophoresis is biased towards the detection of the most abundant cell proteins.

FIGURE 2.30 Two-dimensional gel electrophoresis A mixture of cell proteins is first subjected to electrophoresis in a pH gradient so that they separate according to charge (horizontal axis). The proteins then undergo another electrophoresis in a second dimension under conditions where they separate by size (vertical axis). The gel is stained to reveal spots corresponding to distinct protein species. The example shown is a gel of proteins from *E. coli*. (From P. H. O'Farrell. 1975. *J. Biol. Chem.* 250: 4007; courtesy of Patrick H. O'Farrell.)

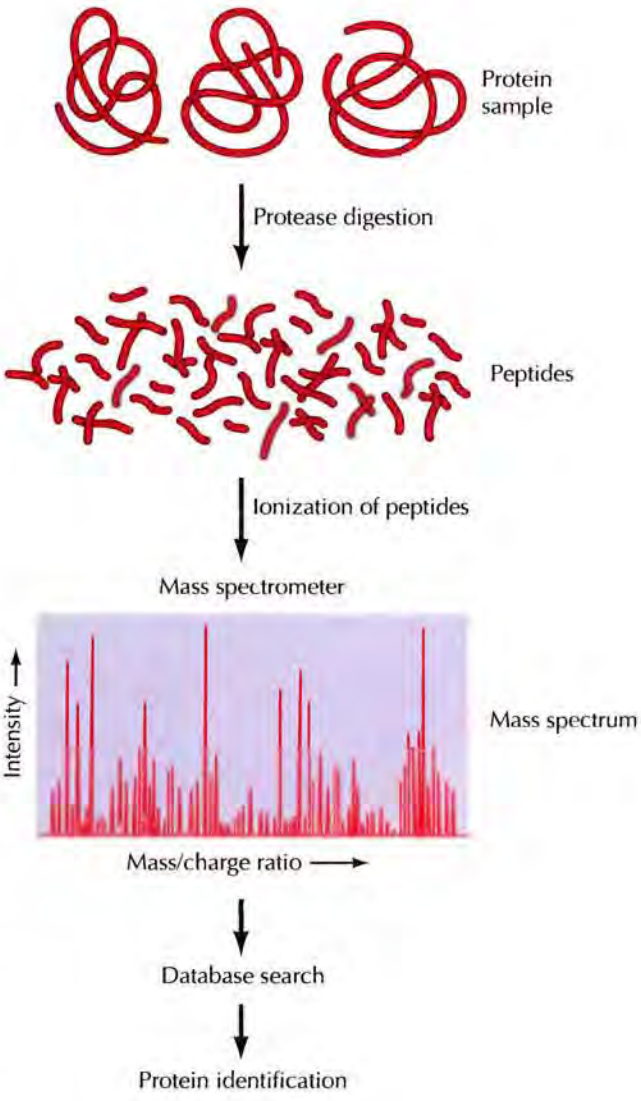

FIGURE 2.31 Identification of proteins by mass spectrometry A protein sample (for example, a spot excised from a two-dimensional gel) is digested with a protease that cleaves the protein into small peptides. The peptides are then ionized and analyzed in a mass spectrometer, which determines the mass-to-charge ratio of each peptide. The results are displayed as a mass spectrum, which is compared to a database of theoretical mass spectra of all known proteins for protein identification.

Proteins detected as spots on two-dimensional gels can be excised from the gels and identified by **mass spectrometry**, which was developed in the 1990s as a powerful tool for protein identification (Figure 2.31). The protein spot excised from the gel is digested with a protease to cleave the protein into small fragments (peptides) in the range of approximately 20 amino acid residues long. A commonly used protease is trypsin, which cleaves proteins at the carboxy-terminal side of lysine and arginine residues. The peptides are then ionized by irradiation with a laser or by passage through a field of high electrical potential and introduced into a mass spectrometer, which measures the mass-to-charge ratio of each peptide. This generates a mass spectrum in which individual tryptic peptides are indicated by a peak corresponding to their mass-to-charge ratio. Computer algorithms can then be used to compare the experimentally determined mass spectrum with a database of theoretical mass spectra representing tryptic peptides of all known proteins, allowing identification of the unknown protein. Alternatively, more detailed sequence information can be obtained by tandem mass spectrometry. In this technique, individual peptides from the initial mass spectrum are automatically selected to enter a "collision cell" in which they are partially degraded by random breakage of peptide bonds. A second mass spectrum of the partial degradation products of each peptide is then determined, and the amino acid sequence of the peptide can be deduced. Protein modifications, such as phosphorylation, can also be identified because they alter the mass of the modified amino acid.

■ **Mass spectrometry has many uses, including testing the blood samples of athletes for the presence of certain performance-enhancing drugs.**

Although powerful, the two-dimensional gel/mass spectrometry approach is limited. As noted above, two-dimensional gels favor detection of the most abundant cell proteins, and membrane proteins are characteristically difficult to resolve by this approach. Because of these limitations, it appears that two-dimensional gels are capable of resolving proteins corresponding to only several hundred genes, representing less than 10% of all cell proteins. An alternative "shotgun" approach has therefore been developed to eliminate the initial separation of proteins by two-dimensional gel electrophoresis. In this approach, an unfractionated mixture of cell proteins is digested with a protease (e.g., trypsin), and the complex mixture of peptides is subjected to sequencing by tandem mass spectrometry. The sequences of individual peptides are then used for database searching to identify the proteins present in the starting mixture. Additional methods have been developed to compare the amounts of proteins in two different samples, allowing a quantitative analysis of protein levels in different types

of cells or in cells that have been subjected to different treatments. Although several problems with the sensitivity and accuracy of these methods remain to be solved, the analysis of complex mixtures of proteins by shotgun mass spectrometry provides a powerful approach to the systematic analysis of cell proteins.

Global Analysis of Protein Localization

Understanding the function of eukaryotic cells will require not only the identification of the proteins expressed in a given cell type, but also characterization of the locations of those proteins within the cell. As reviewed in Chapter 1, eukaryotic cells contain a nucleus and a variety of subcellular organelles. Systematic analysis of the proteins present in these organelles has become an important goal of proteomic approaches to cell biology.

One approach to global analysis of protein localization is to isolate the organelle of interest by subcellular fractionation, as discussed in Chapter 1. The proteins present in isolated organelles can then be analyzed by mass spectrometry, which is simplified since any given organelle contains fewer proteins than would be present in total cell extracts. Mitochondria, for example, are estimated to contain about 1,000 different proteins—more than 600 of these have been identified by mass spectrometry of isolated mitochondria. Similar analysis of the proteome of a variety of subcellular organelles is currently a major research effort in a number of laboratories and is expected to provide a database of the protein constituents of these compartments of eukaryotic cells.

An alternative approach has recently been applied to a systematic analysis of protein localization in yeast. In these experiments, investigators constructed a collection of yeast strains in which each protein encoded in the yeast genome was expressed as a fusion with green fluorescent protein (GFP), allowing the subcellular localization of the GFP-tagged protein to be determined by fluorescence microscopy (see Figure 1.27). Using this approach, the subcellular localizations of over 4000 proteins was deter-

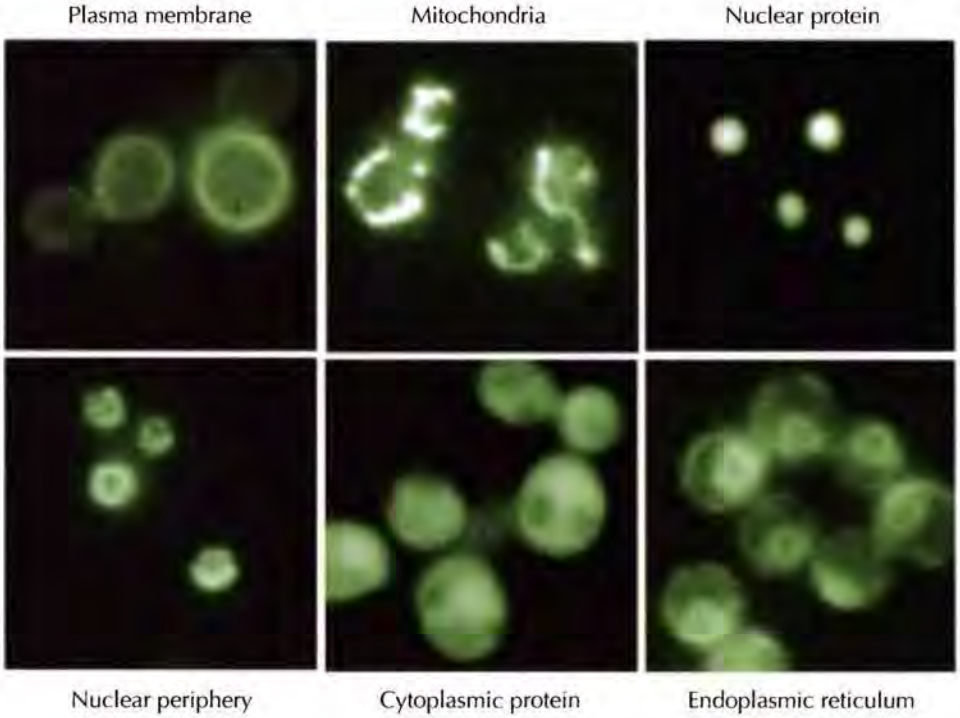

FIGURE 2.32 Subcellular localization of yeast proteins
The protein-coding regions of over 4000 yeast genes were expressed in a collection of yeast strains as fusions with green fluorescent protein (GFP). The subcellular localization of each fusion protein was then determined by fluorescence microscopy. The micrographs shown illustrate the indicated localization of representative fusion proteins. (After W. -K. Huh, J. V. Falvo, L. C. Gerke, A. S. Carroll, R. W. Howson, J. S. Weissman and E. K. O'Shea, 2003. *Nature* 425: 686. Images courtesy of James Falvo.)

mined (Figure 2.32). This represents 75% of the total number of yeast proteins, including many proteins whose subcellular localization was previously unknown, so this global analysis of yeast protein localization provides a comprehensive catalog of protein localization in a model eukaryotic cell.

Protein Interactions

Proteins almost never act alone within the cell. Instead, they generally function by interacting with other proteins in protein complexes and networks. Elucidating the interactions between proteins therefore provides important clues as to the function of novel proteins, as well as helping to understand the complex networks of protein interactions that govern cell behavior. Along with global studies of subcellular localization, the systematic analysis of protein complexes and interactions has therefore become an important goal of proteomics.

One approach to analysis of protein complexes is to isolate a protein from cells under gentle conditions, so that it remains associated with the proteins it normally interacts with inside the cell. The isolated protein complexes can then be analyzed by mass spectrometry to identify all of the proteins present in the complex. This approach to analysis of protein interactions has been used in large-scale investigations of protein complexes in yeast, and has identified numerous interactions between proteins involved in processes such as cell signaling or gene expression.

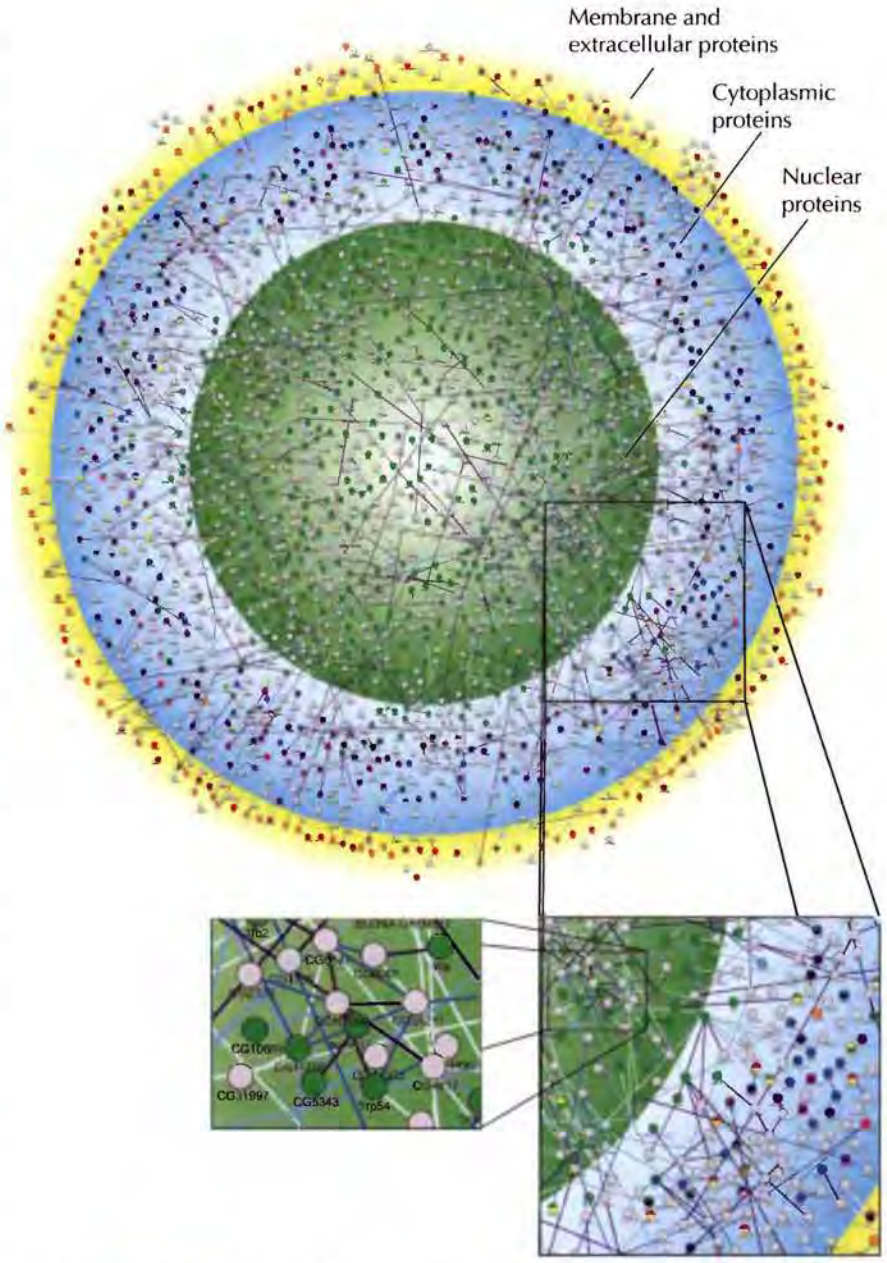

FIGURE 2.33 A protein interaction map of *Drosophila melanogaster* Interactions among 2346 proteins are depicted, with each protein represented as a circle placed according to its subcellular localization. (From L. Glot et al., 2003. *Science* 302: 1727).

Alternative approaches to systematic analyses of protein complexes include screens for protein interactions *in vitro* as well as genetic screens that detect interactions between pairs of proteins that are introduced into yeast cells. The latter approach (called the yeast two-hybrid method) was initially used for large scale studies of interactions between yeast proteins, but has more recently been applied to systematic screens of interactions between proteins of higher eukaryotes, such as *Drosophila* and *C. elegans*. The screens have identified thousands of protein-protein interactions, which can be presented as maps that depict an extensive network of interacting proteins within the cell (Figure 2.33). Continuing elucidation of these protein networks is expected to extend our understanding of the complexities of cell regulation, as well as illuminating the functions of many so-far unidentified proteins.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

KEY TERMS

SUMMARY

THE MOLECULES OF CELLS

carbohydrate, monosaccharide, glycosidic bond, oligosaccharide, polysaccharide, glycogen, starch, cellulose

Carbohydrates: Carbohydrates include simple sugars and polysaccharides. Polysaccharides serve as storage forms of sugars, structural components of cells, and markers for cell recognition processes.

lipid, fatty acid, triacylglycerol, fat, phospholipid, glycerol phospholipid, sphingomyelin, amphipathic, glycolipid, cholesterol, steroid hormone

Lipids: Lipids are the principal components of cell membranes, and they serve as energy storage and signaling molecules. Phospholipids consist of two hydrophobic fatty acid chains linked to a hydrophilic phosphate-containing head group.

deoxyribonucleic acid (DNA), ribonucleic acid (RNA), messenger RNA (mRNA), ribosomal RNA, transfer RNA, nucleotide, purine, pyrimidine, adenine, guanine, cytosine, thymine, uracil, 2′-deoxyribose, ribose, nucleoside, phosphodiester bond, oligonucleotide, polynucleotide

Nucleic Acids: Nucleic acids are the principal informational molecules of the cell. Both DNA and RNA are polymers of purine and pyrimidine nucleotides. Hydrogen bonding between complementary base pairs allows nucleic acids to direct their self-replication.

protein, amino acid, peptide bond, polypeptide, X-ray crystallography, primary structure, secondary structure, α helix, β sheet, tertiary structure, domain, quaternary structure

Proteins: Proteins are polymers of 20 different amino acids, each of which has a distinct side chain with specific chemical properties. Each protein has a unique amino acid sequence, which determines its three-dimensional structure. In most proteins, combinations of α helices and β sheets fold into globular domains with hydrophobic amino acids in the interior and hydrophilic amino acids on the surface.

CELL MEMBRANES

phospholipid bilayer

Membrane Lipids: The basic structure of all cell membranes is a phospholipid bilayer. Membranes of animal cells also contain glycolipids and cholesterol.

fluid mosaic, integral membrane protein, peripheral membrane protein, transmembrane protein, β-barrel

Membrane Proteins: Proteins can either be inserted into the lipid bilayer or associated with the membrane indirectly, by protein-protein interactions. Some proteins span the lipid bilayer; others are anchored to one side of the membrane.

channel protein, carrier protein, passive transport, active transport

Transport across Cell Membranes: Lipid bilayers are permeable only to small uncharged molecules. Ions and most polar molecules are transported across cell membranes by specific transport proteins, the action of which can be coupled to the hydrolysis or synthesis of ATP.

PROTEOMICS: LARGE-SCALE ANALYSIS OF CELL PROTEINS

genomics, proteomics, proteome

Identification of Cell Proteins: Characterization of the complete protein complement of cells is a major goal of proteomics. One well-established method for proteomic studies is two-dimensional gel electrophoresis, which can be used to separate thousands of cell proteins. Mass spectrometry provides a powerful tool for protein identification, which can

SUMMARY

be used to analyze protein samples separated by two-dimensional gel electrophoresis or to identify proteins in unfractionated cell extracts.

Global Analysis of Protein Localization: Isolated subcellular organelles can be analyzed by mass spectrometry to determine their protein constituents. Alternatively, a large fraction of yeast proteins have been tagged with green fluorescent protein for global studies of protein localization.

Protein Interactions: A variety of large-scale approaches are being applied to systematically identify protein-protein interactions and complexes with the goal of elucidating the complex networks of protein interactions that regulate cell behavior.

KEY TERMS

two-dimensional gel electrophoresis, mass spectrometry

Questions

1. What characteristics of water contribute to its function as the most abundant molecule in cells?

2. Although glycogen and cellulose are both composed of glucose monomers, they are chemically distinct. In what way?

3. What are the major functions of fats and phospholipids in cells?

4. In addition to serving as the building blocks for nucleic acids, what other important functions do nucleotides have in cells?

5. If you removed all the bases from an mRNA molecule, could it still function as an information-carrying molecule?

6. What was the experimental evidence that demonstrated that the information necessary for the folding of proteins is contained in the sequence of their amino acids?

7. How does the side chain of cysteine help in the folding of certain proteins?

8. What are the biological roles of cholesterol?

9. Why are the membrane-spanning regions of transmembrane proteins frequently α helical? What other protein structure is capable of spanning membranes?

10. Which of the following amino acids would most likely be found in a membrane-spanning α helix?

a. Lysine
b. Glutamine
c. Aspartic acid
d. Alanine
e. Arginine

11. What properties of proteins are exploited for the separation of individual proteins from a complex mixture by two-dimensional electrophoresis?

12. List and briefly explain the large-scale proteomic approaches used to determine the subcellular location of proteins.

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CHAPTER 3

Cell Metabolism

- ***The Central Role of Enzymes as Biological Catalysts 73***
- ***Metabolic Energy 81***
- ***The Biosynthesis of Cell Constituents 91***
- ***KEY EXPERIMENT:*** Antimetabolites and Chemotherapy 98
- ***MOLECULAR MEDICINE:*** Phenylketonuria 96

CELLS CARRY OUT A VAST ARRAY OF CHEMICAL REACTIONS, which are responsible both for the breakdown of molecules that serve as food and for the synthesis of cell constituents. This chapter presents an overview of this complex network of chemical reactions, which constitute the metabolism of the cell. It is intended neither to be a comprehensive discussion of biochemistry nor to chart all the metabolic reactions within cells. Rather, the chapter will focus on three major topics: the role of proteins as biological catalysts, the generation and utilization of metabolic energy, and the biosynthesis of major cell constituents. An appreciation of these key aspects of cell metabolism is basic for understanding the diverse aspects of cell structure and function that will be discussed throughout the rest of this text.

The Central Role of Enzymes as Biological Catalysts

Enzymes are remarkable catalysts. Some enzymes are capable of increasing the rate of a reaction by as much as 10^{18}-fold as compared to an uncatalyzed reaction, such that a process that would normally take billions of years can be carried out in less than a second.

A fundamental task of proteins is to act as enzymes—catalysts that increase the rate of virtually all the chemical reactions within cells. Although RNAs are capable of catalyzing some reactions, most biological reactions are catalyzed by proteins. In the absence of enzymatic catalysis, most biochemical reactions are so slow that they would not occur under the mild conditions of temperature and pressure that are compatible with life. Enzymes accelerate the rates of such reactions by well over a million-fold, so reactions that would take years in the absence of catalysis can occur in fractions of seconds if catalyzed by the appropriate enzyme. Cells contain thousands of different enzymes, and their activities determine which of the many possible chemical reactions actually take place within the cell.

The Catalytic Activity of Enzymes

Like all other catalysts, **enzymes** are characterized by two fundamental properties. First, they increase the rate of chemical reactions without

themselves being consumed or permanently altered by the reaction. Second, they increase reaction rates without altering the chemical equilibrium between reactants and products.

These principles of enzymatic catalysis are illustrated in the following example in which a molecule acted upon by an enzyme (referred to as a **substrate** [*S*]) is converted to a **product** (*P*) as a result of the reaction. In the absence of the enzyme, the reaction can be written as follows:

$$S \rightleftharpoons P$$

The chemical equilibrium between *S* and *P* is determined by the laws of thermodynamics (as discussed further in the next section of this chapter) and is represented by the ratio of the forward and reverse reaction rates ($S \rightarrow P$ and $P \rightarrow S$, respectively). In the presence of the appropriate enzyme, the conversion of *S* to *P* is accelerated, but the equilibrium between *S* and *P* is unaltered. Therefore, the enzyme must accelerate both the forward and reverse reactions equally. The reaction can be written as follows:

$$S \overset{E}{\rightleftharpoons} P$$

Note that the enzyme (*E*) is not altered by the reaction, so the chemical equilibrium remains unchanged, determined solely by the thermodynamic properties of *S* and *P*.

The effect of the enzyme on such a reaction is best illustrated by the energy changes that must occur during the conversion of *S* to *P* (Figure 3.1). The equilibrium of the reaction is determined by the final energy states of *S* and *P*, which are unaffected by enzymatic catalysis. In order for the reaction to proceed, however, the substrate must first be converted to a higher energy state, called the **transition state**. The energy required to reach the transition state (the **activation energy**) constitutes a barrier to the progress of the reaction, limiting the rate of the reaction. Enzymes (and other catalysts) act by reducing the activation energy, thereby increasing the rate of reaction. The increased rate is the same in both the forward and reverse directions, since both must pass through the same transition state.

The catalytic activity of enzymes involves the binding of their substrates to form an enzyme-substrate complex (*ES*). The substrate binds to a specific region of the enzyme, called the **active site**. While bound to the active site, the substrate is converted into the product of the reaction, which is then released from the enzyme. The enzyme-catalyzed reaction can thus be written as follows:

$$S + E \rightleftharpoons ES \rightleftharpoons E + P$$

Note that *E* appears unaltered on both sides of the equation, so the equilibrium is unaffected. However, the enzyme provides a surface upon which the reactions converting *S* to *P* can occur more readily. This is a result of interactions between the enzyme and substrate that lower the energy of activation and favor formation of the transition state.

Mechanisms of Enzymatic Catalysis

The binding of a substrate to the active site of an enzyme is a very specific interaction. Active sites are clefts or grooves on the surface of an enzyme, usually composed of amino acids from different parts of the polypeptide chain that are brought together in the tertiary structure of the folded protein. Substrates initially bind to the active site by noncovalent interactions, including hydrogen bonds, ionic bonds, and hydrophobic interactions.

3.1 WEBSITE ANIMATION

Catalysts and Activation Energy In a chemical reaction, an enzyme lowers the amount of energy required to reach the transition state, allowing the reaction to proceed at an accelerated rate.

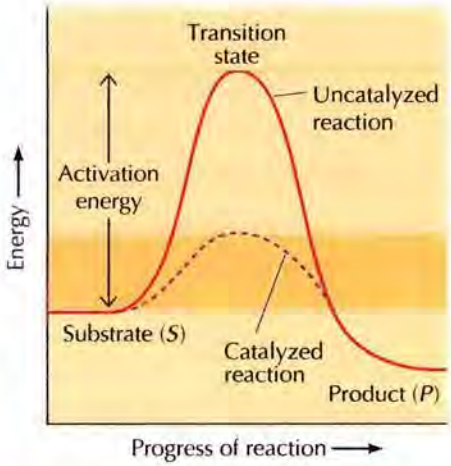


FIGURE 3.1 Energy diagrams for catalyzed and uncatalyzed reactions The reaction illustrated is the simple conversion of a substrate *S* to a product *P*. Because the final energy state of *P* is lower than that of *S*, the reaction proceeds from left to right. For the reaction to occur, however, *S* must first pass through a higher energy transition state. The energy required to reach this transition state (the activation energy) represents a barrier to the progress of the reaction and thereby determines the rate at which the reaction proceeds. In the presence of a catalyst (e.g., an enzyme), the activation energy is lowered and the reaction proceeds at an accelerated rate.

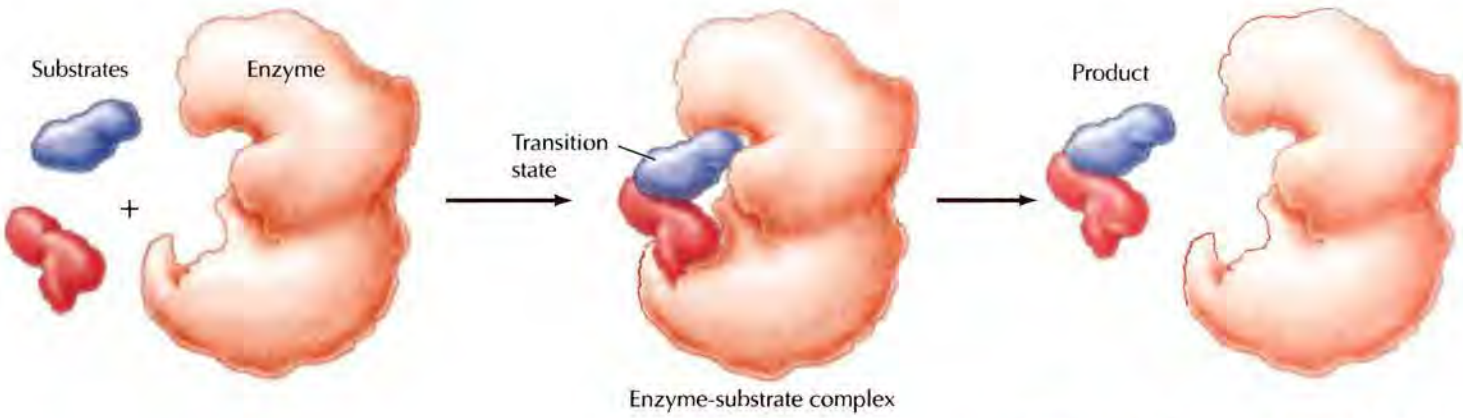


FIGURE 3.2 Enzymatic catalysis of a reaction between two substrates The enzyme provides a template upon which the two substrates are brought together in the proper position and orientation to react with each other.

Once a substrate is bound to the active site of an enzyme, multiple mechanisms can accelerate its conversion to the product of the reaction.

Although the simple example discussed in the previous section involved only a single substrate molecule, most biochemical reactions involve interactions between two or more different substrates. For example, the formation of a peptide bond involves the joining of two amino acids. For such reactions, the binding of two or more substrates to the active site in the proper position and orientation accelerates the reaction (Figure 3.2). The enzyme provides a template upon which the reactants are brought together and properly oriented to favor the formation of the transition state in which they interact.

3.2 WEBSITE ANIMATION

Enzyme-Catalyzed Reactions In this example, an enzyme provides a template upon which two substrates are brought together in the proper position and orientation to react with each other.

Enzymes accelerate reactions also by altering the conformation of their substrates to approach that of the transition state. The simplest model of enzyme-substrate interaction is the **lock-and-key model** in which the substrate fits precisely into the active site (Figure 3.3). In many cases, however, the configurations of both the enzyme and substrate are modified by substrate binding—a process called **induced fit**. In such cases the conformation of the substrate is altered so that it more closely resembles that of the transition state. The stress produced by such distortion of the substrate can further facilitate its conversion to the transition state by weakening critical bonds. Moreover, the transition state is stabilized by its tight binding to the enzyme, thereby lowering the required energy of activation.

In addition to bringing multiple substrates together and distorting the conformation of substrates to approach the transition state, many enzymes participate directly in the catalytic process. In such cases, specific amino acid side chains in the active site may react with the substrate and form bonds with reaction intermediates. The acidic and basic amino acids are often involved in these catalytic mechanisms, as illustrated in the following discussion of chymotrypsin as an example of enzymatic catalysis.

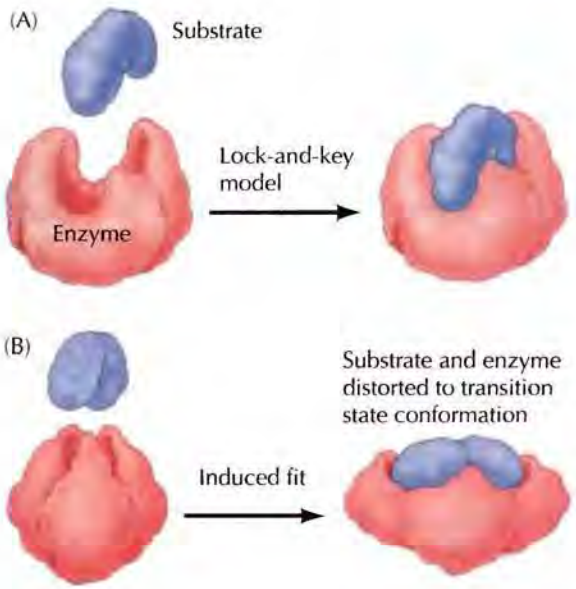


FIGURE 3.3 Models of enzyme-substrate interaction (A) In the lock-and-key model, the substrate fits precisely into the active site of the enzyme. (B) In the induced-fit model, substrate binding distorts the conformations of both substrate and enzyme. This distortion brings the substrate closer to the conformation of the transition state, thereby accelerating the reaction.

Chymotrypsin is a member of a family of enzymes (serine proteases) that digests proteins by catalyzing the hydrolysis of peptide bonds. The reaction can be written as follows:

$$\text{Protein} + H_2O \rightarrow \text{Peptide}_1 + \text{Peptide}_2$$

The different members of the serine protease family (including chymotrypsin, trypsin, elastase, and thrombin) have distinct substrate specificities; they preferentially cleave peptide bonds adjacent to different amino acids. For example, whereas chymotrypsin digests bonds adjacent to hydrophobic amino acids, such as tryptophan and phenylalanine, trypsin digests bonds next to basic amino acids, such as lysine and arginine. All the serine proteases, however, are similar in structure and use the same mechanism of catalysis. The active sites of these enzymes contain three critical amino acids—serine, histidine, and aspartate—that drive hydrolysis of the peptide bond. Indeed, these enzymes are called serine proteases because of the central role of the serine residue.

■ Proteases play a variety of roles in cells and viruses. Some HIV proteins are synthesized as precursors that need to be cleaved by a viral protease to make infectious viral particles. Inhibition of the protease blocks HIV replication, and drugs designed to inhibit the protease are currently used in the treatment of AIDS.

Substrates bind to the serine proteases by insertion of the amino acid adjacent to the cleavage site into a pocket at the active site of the enzyme (Figure 3.4). The nature of this pocket determines the substrate specificity of the different members of the serine protease family. For example, the binding pocket of chymotrypsin contains hydrophobic amino acids that interact with the hydrophobic side chains of its preferred substrates. In contrast, the binding pocket of trypsin contains a negatively charged acidic amino acid (aspartate), which is able to form an ionic bond with the lysine or arginine residues of its substrates.

Substrate binding positions the peptide bond to be cleaved adjacent to the active site serine (Figure 3.5). The proton of this serine is then transferred to the active site histidine. The conformation of the active site favors this proton transfer because the histidine interacts with the negatively charged aspartate residue. The serine reacts with the substrate, forming a tetrahedral transition state. The peptide bond is then cleaved, and the C-terminal portion of the substrate is released from the enzyme. However, the N-terminal

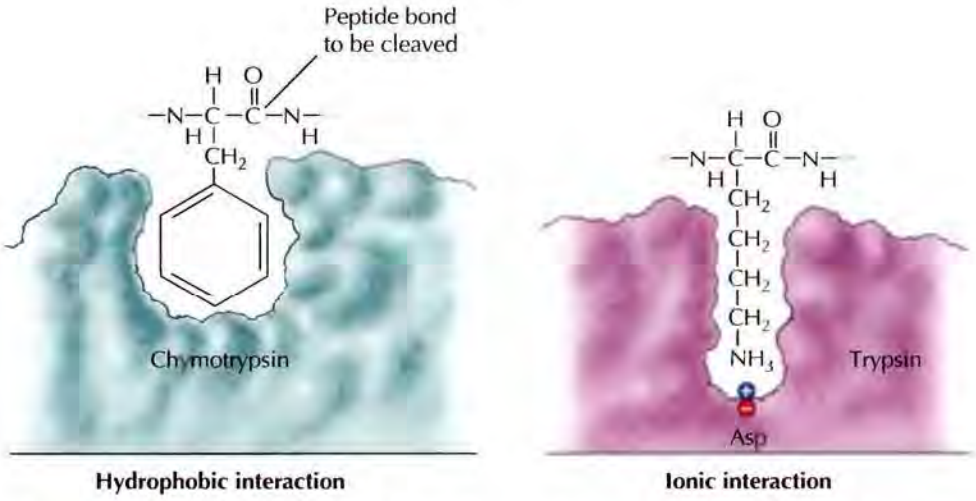


FIGURE 3.4 Substrate binding by serine proteases The amino acid adjacent to the peptide bond to be cleaved is inserted into a pocket at the active site of the enzyme. In chymotrypsin, the pocket binds hydrophobic amino acids; the binding pocket of trypsin contains a negatively charged aspartate residue that binds basic amino acids via an ionic interaction.

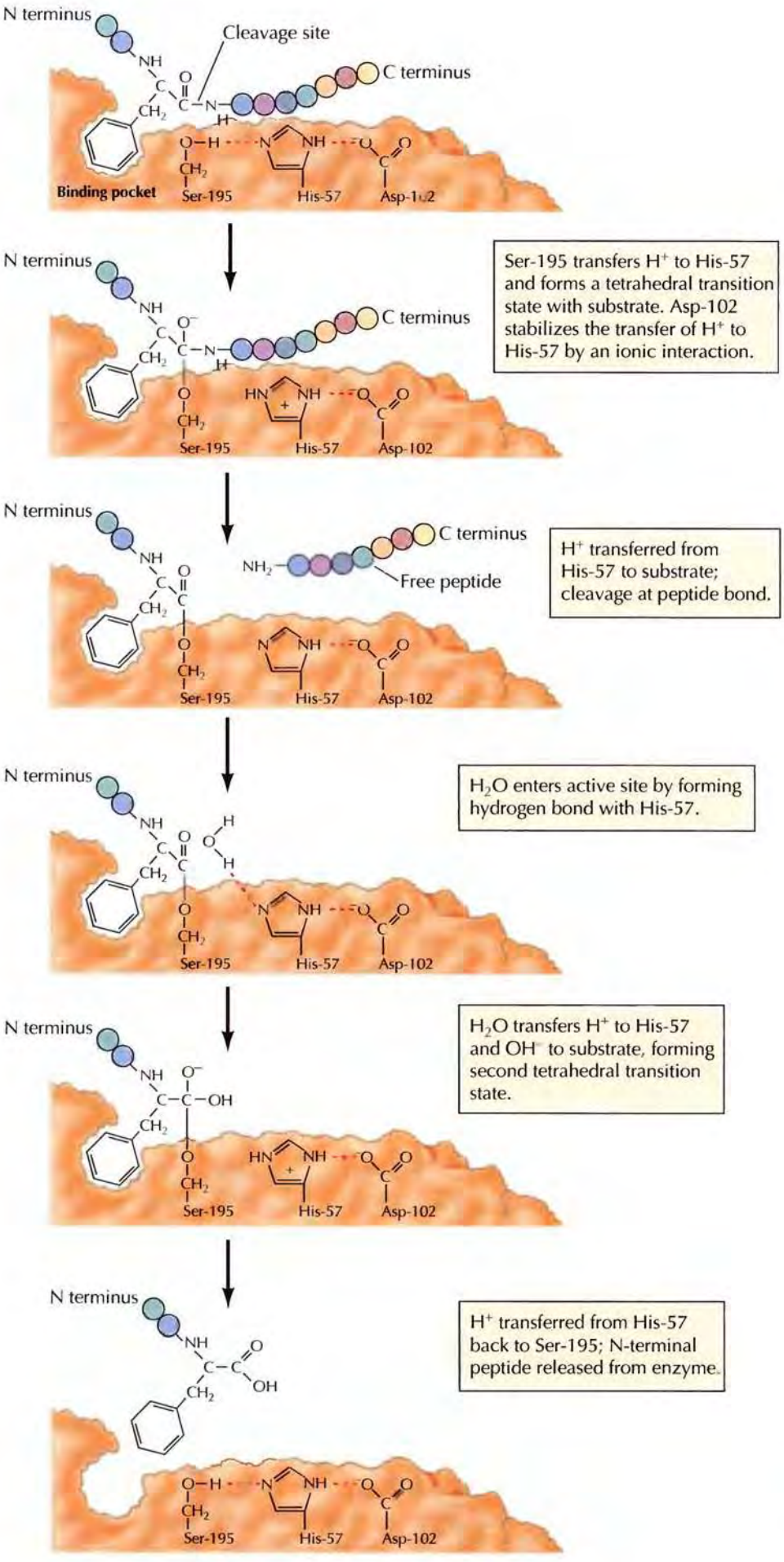


FIGURE 3.5 Catalytic mechanism of chymotrypsin Three amino acids at the active site (Ser-195, His-57, and Asp-102) play critical roles in catalysis.

peptide remains bound to serine. This situation is resolved when a water molecule (the second substrate) enters the active site and reverses the preceding reactions. The proton of the water molecule is transferred to histidine, and its hydroxyl group is transferred to the peptide, forming a second tetrahedral transition state. The proton is then transferred from histidine back to serine, and the peptide is released from the enzyme, completing the reaction.

This example illustrates several features of enzymatic catalysis: the specificity of enzyme-substrate interactions, the positioning of different substrate molecules in the active site, and the involvement of active-site residues in the formation and stabilization of the transition state. Although the thousands of enzymes in cells catalyze many different types of chemical reactions, the same basic principles apply to their operation.

Coenzymes

In addition to binding their substrates, the active sites of many enzymes bind other small molecules that participate in catalysis. **Prosthetic groups** are small molecules bound to proteins in which they play critical functional roles. For example, the oxygen carried by myoglobin and hemoglobin is bound to heme, a prosthetic group of these proteins. In many cases metal ions (such as zinc or iron) are bound to enzymes and play central roles in the catalytic process. In addition, various low-molecular-weight organic molecules participate in specific types of enzymatic reactions. These molecules are called **coenzymes** because they work together with enzymes to enhance reaction rates. In contrast to substrates, coenzymes are not irreversibly altered by the reactions in which they are involved. Rather, they are recycled and can participate in multiple enzymatic reactions.

Coenzymes serve as carriers of several types of chemical groups. A prominent example of a coenzyme is **nicotinamide adenine dinucleotide (NAD^+)**, which functions as a carrier of electrons in oxidation–reduction reactions (Figure 3.6). NAD^+ can accept a hydrogen ion (H^+) and two electrons (e^-) from one substrate, forming NADH. NADH can then donate these electrons to a second substrate, re-forming NAD^+. Thus, NAD^+ transfers electrons from the first substrate (which becomes oxidized) to the second substrate (which becomes reduced).

Several other coenzymes also act as electron carriers, and still others are involved in the transfer of a variety of additional chemical groups (e.g., carboxyl groups and acyl groups; Table 3.1). The same coenzymes function together with a variety of different enzymes to catalyze the transfer of spe-

TABLE 3.1 Examples of Coenzymes and Vitamins

Coenzyme	Related vitamin	Chemical reaction
NAD^+, $NADP^+$	Niacin	Oxidation–reduction
FAD	Riboflavin (B_2)	Oxidation–reduction
Thiamine pyrophosphate	Thiamine (B_1)	Aldehyde group transfer
Coenzyme A	Pantothenate	Acyl group transfer
Tetrahydrofolate	Folate	Transfer of one-carbon groups
Biotin	Biotin	Carboxylation
Pyridoxal phosphate	Pyridoxal (B_6)	Transamination

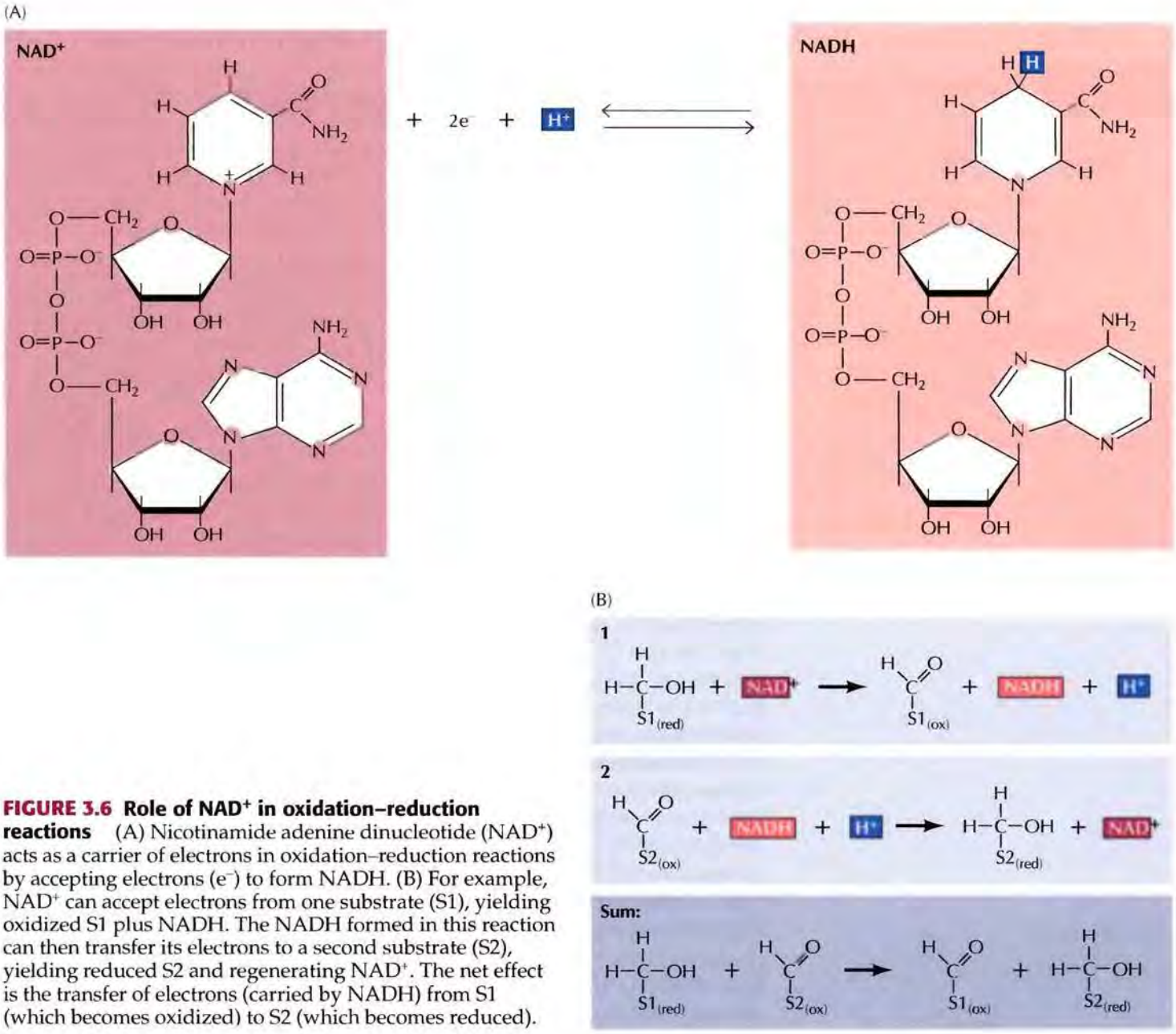


FIGURE 3.6 Role of NAD⁺ in oxidation–reduction reactions (A) Nicotinamide adenine dinucleotide (NAD^+) acts as a carrier of electrons in oxidation–reduction reactions by accepting electrons (e^-) to form NADH. (B) For example, NAD^+ can accept electrons from one substrate (S1), yielding oxidized S1 plus NADH. The NADH formed in this reaction can then transfer its electrons to a second substrate (S2), yielding reduced S2 and regenerating NAD^+. The net effect is the transfer of electrons (carried by NADH) from S1 (which becomes oxidized) to S2 (which becomes reduced).

cific chemical groups between a wide range of substrates. Many coenzymes are closely related to vitamins, which contribute part or all of the structure of the coenzyme. Vitamins are not required by bacteria such as *E. coli* but are necessary components of the diets of humans and other higher animals, which have lost the ability to synthesize these compounds.

Regulation of Enzyme Activity

An important feature of most enzymes is that their activities are not constant but instead can be modulated. That is, the activities of enzymes can be regulated so that they function appropriately to meet the varied physiological needs that may arise during the life of the cell.

One common type of enzyme regulation is **feedback inhibition** in which the product of a metabolic pathway inhibits the activity of an enzyme involved in its synthesis. For example, the amino acid isoleucine is synthe-

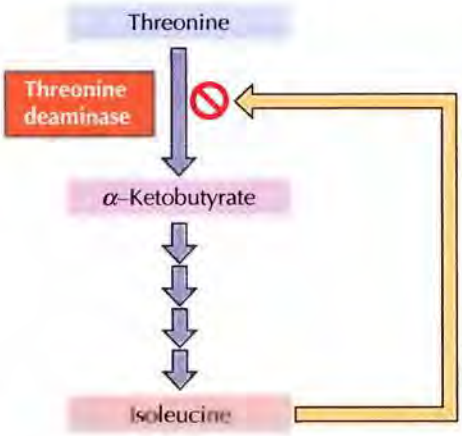


FIGURE 3.7 Feedback inhibition
The first step in the conversion of threonine to isoleucine is catalyzed by the enzyme threonine deaminase. The activity of this enzyme is inhibited by isoleucine, the end product of the pathway.

sized by a series of reactions starting from the amino acid threonine (Figure 3.7). The first step in the pathway is catalyzed by the enzyme threonine deaminase, which is inhibited by isoleucine, the end product of the pathway. Thus, an adequate amount of isoleucine in the cell inhibits threonine deaminase, blocking further synthesis of isoleucine. If the concentration of isoleucine decreases, feedback inhibition is relieved, threonine deaminase is no longer inhibited, and additional isoleucine is synthesized. By regulating the activity of threonine deaminase, the cell synthesizes the necessary amount of isoleucine but avoids wasting energy on the synthesis of more isoleucine than is needed.

Feedback inhibition is one example of **allosteric regulation** in which enzyme activity is controlled by the binding of small molecules to regulatory sites on the enzyme (Figure 3.8). The term "allosteric regulation" derives from the fact that the regulatory molecules bind not to the catalytic site but to a distinct site on the protein (*allo* = "other" and *steric* = "site"). Binding of the regulatory molecule changes the conformation of the protein, this in turn alters the shape of the active site and the catalytic activity of the enzyme. In the case of threonine deaminase, binding of the regulatory molecule (isoleucine) inhibits enzymatic activity. In other cases, regulatory molecules serve as activators, stimulating rather than inhibiting their target enzymes.

The activities of enzymes can also be regulated by their interactions with other proteins and by covalent modifications, such as the addition of phosphate groups to serine, threonine, or tyrosine residues. **Phosphorylation** is a particularly common mechanism for regulating enzyme activity; the addition of phosphate groups either stimulates or inhibits the activities of many different enzymes (Figure 3.9). For example, muscle cells respond to epinephrine (adrenaline) by breaking down glycogen into glucose, thereby providing a source of energy for increased muscular activity. The breakdown of glycogen is catalyzed by the enzyme glycogen phosphorylase, which is activated by phosphorylation in response to the binding of epinephrine to a receptor on the surface of the muscle cell. Protein phosphorylation plays a central role in controlling not only metabolic reactions but also many other cellular functions, including cell growth and differentiation.

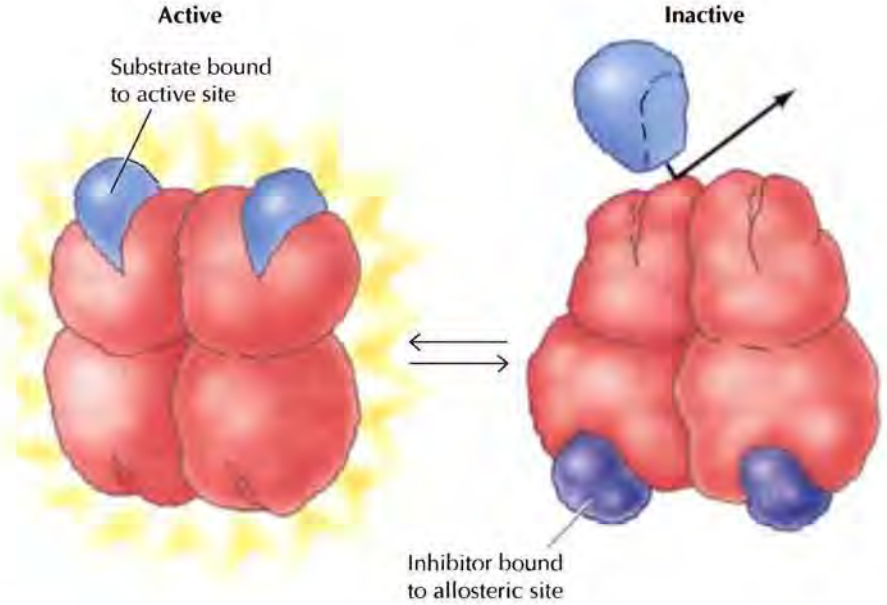


FIGURE 3.8 Allosteric regulation
In this example, enzyme activity is inhibited by the binding of a regulatory molecule to an allosteric site. In the absence of an inhibitor, the substrate binds to the active site of the enzyme and the reaction proceeds. The binding of an inhibitor to the allosteric site induces a conformational change in the enzyme and prevents substrate binding. Most allosteric enzymes consist of multiple subunits.

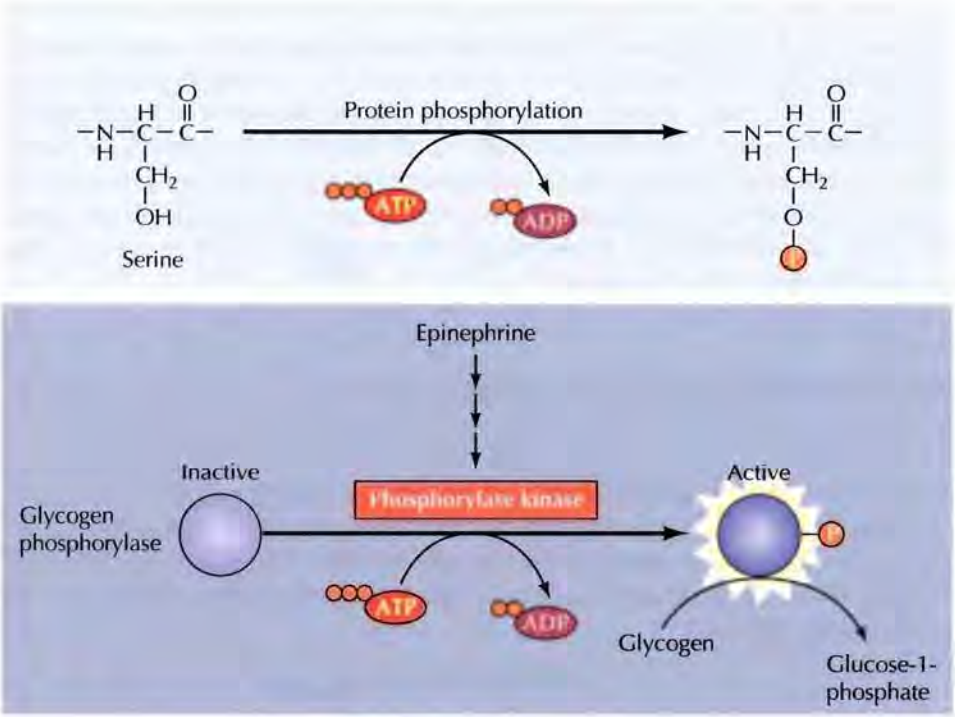


FIGURE 3.9 Protein phosphorylation Some enzymes are regulated by the addition of phosphate groups to the side-chain OH groups of serine (as shown here), threonine, or tyrosine residues. For example, the enzyme glycogen phosphorylase, which catalyzes the conversion of glycogen into glucose-1-phosphate, is activated by phosphorylation in response to the binding of epinephrine to muscle cells.

Metabolic Energy

Many tasks that a cell must perform, such as movement and the synthesis of macromolecules, require energy. A large portion of the cell's activities is therefore devoted to obtaining energy from the environment and using that energy to drive energy-requiring reactions. Although enzymes control the rates of virtually all chemical reactions within cells, the equilibrium position of chemical reactions is not affected by enzymatic catalysis. The laws of thermodynamics govern chemical equilibria and determine the energetically favorable direction of all chemical reactions. Many of the reactions that must take place within cells are energetically unfavorable, and are therefore able to proceed only at the cost of additional energy input. Consequently, cells must constantly expend energy derived from the environment. The generation and utilization of metabolic energy is thus fundamental to all of cell biology.

Free Energy and ATP

The energetics of biochemical reactions are best described in terms of the thermodynamic function called **Gibbs free energy (*G*)**, named for Josiah Willard Gibbs. The change in free energy (ΔG) of a reaction combines the effects of changes in enthalpy (the heat that is released or absorbed during a chemical reaction) and entropy (the degree of disorder resulting from a reaction) to predict whether or not a reaction is energetically favorable. All chemical reactions spontaneously proceed in the energetically favorable direction, accompanied by a decrease in free energy ($\Delta G < 0$). For example, consider a hypothetical reaction in which A is converted to B:

$$A \rightleftharpoons B$$

If $\Delta G < 0$, this reaction will proceed in the forward direction, as written. If $\Delta G > 0$, however, the reaction will proceed in the reverse direction, and B will be converted to A.

The ΔG of a reaction is determined not only by the intrinsic properties of reactants and products, but also by their concentrations and other reaction conditions (e.g., temperature). It is thus useful to define the free-energy change of a reaction under standard conditions. (Standard conditions are considered to be a 1-*M* concentration of all reactants and products, and 1 atm of pressure). The standard free-energy change ($\Delta G°$) of a reaction is directly related to its equilibrium position because the actual ΔG is a function of both $\Delta G°$ and the concentrations of reactants and products. For example, consider the reaction

$$A \rightleftharpoons B$$

The free-energy change can be written as follows:

$$\Delta G = \Delta G° + RT \ln [B]/[A]$$

where R is the gas constant and T is the absolute temperature.

At equilibrium, $\Delta G = 0$ and the reaction does not proceed in either direction. The equilibrium constant for the reaction ($K = [B]/[A]$ at equilibrium) is thus directly related to $\Delta G°$ by the above equation, which can be expressed as follows:

$$0 = \Delta G° + RT \ln K$$

or

$$\Delta G° = -RT \ln K$$

If the actual ratio [B]/[A] is greater than the equilibrium ratio (K), $\Delta G > 0$ and the reaction proceeds in the reverse direction (conversion of B to A). On the other hand, if the ratio [B]/[A] is less than the equilibrium ratio, $\Delta G < 0$ and A is converted to B.

The standard free-energy change ($\Delta G°$) of a reaction therefore determines its chemical equilibrium and predicts in which direction the reaction will proceed under any given set of conditions. For biochemical reactions, the standard free-energy change is usually expressed as $\Delta G°'$, which is the standard free-energy change of a reaction in aqueous solution at pH = 7, approximately the conditions within a cell.

Many biological reactions (such as the synthesis of macromolecules) are thermodynamically unfavorable ($\Delta G > 0$) under cellular conditions. In order for such reactions to proceed, an additional source of energy is required. For example, consider the reaction

$$A \rightleftharpoons B \qquad \Delta G = +10 \text{ kcal/mol}$$

The conversion of A to B is energetically unfavorable, so the reaction proceeds in the reverse rather than the forward direction. However, the reaction can be driven in the forward direction by coupling the conversion of A to B with an energetically favorable reaction, such as:

$$C \rightleftharpoons D \qquad \Delta G = -20 \text{ kcal/mol}$$

If these two reactions are combined, the coupled reaction can be written as follows:

$$A + C \rightleftharpoons B + D \qquad \Delta G = -10 \text{ kcal/mol}$$

The ΔG of the combined reaction is the sum of the free-energy changes of its individual components, so the coupled reaction is energetically favorable and will proceed as written. Thus, the energetically unfavorable conversion of A to B is driven by coupling it to a second reaction associated with a large

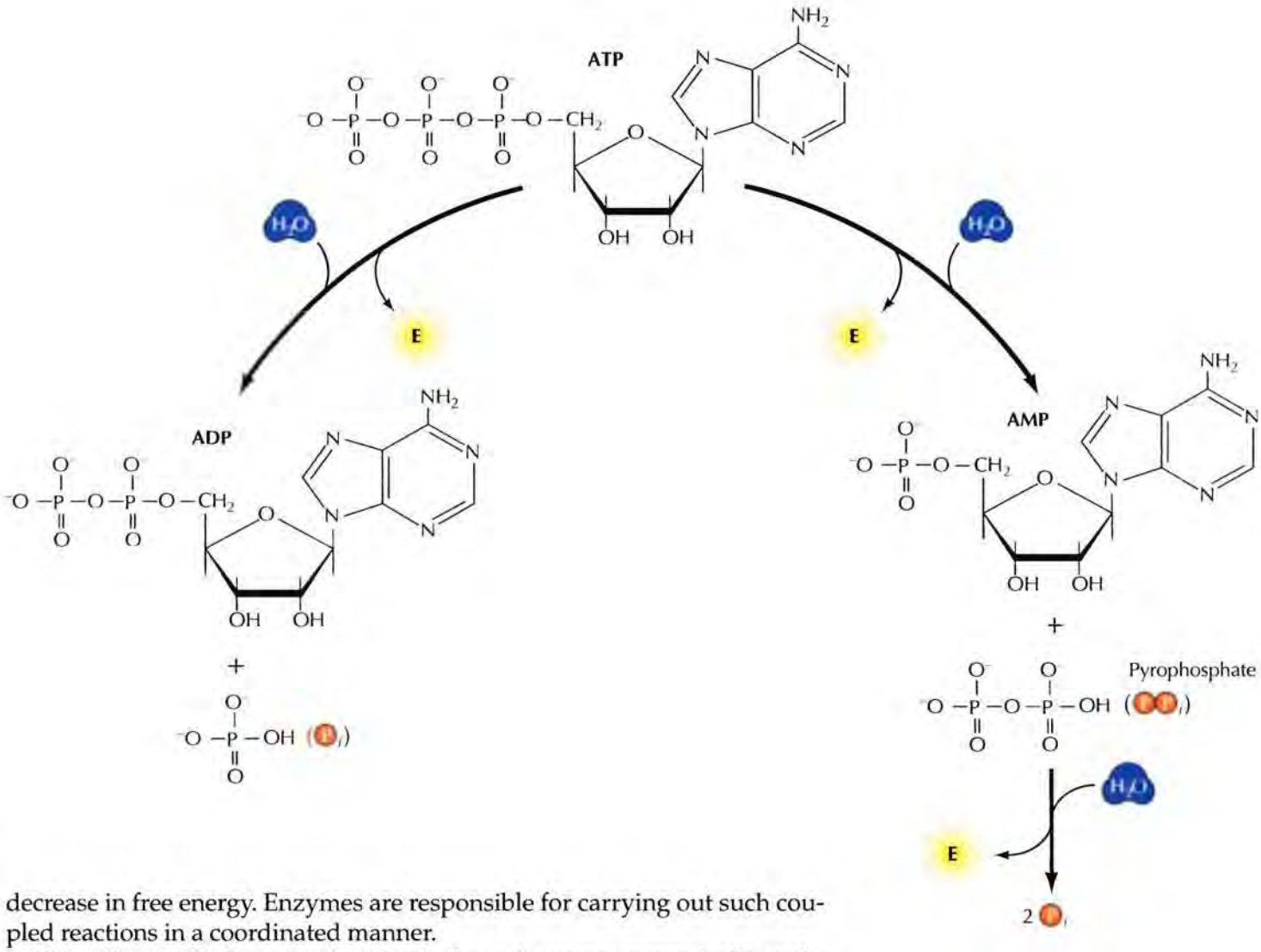


FIGURE 3.10 ATP as a store of free energy The bonds between the phosphate groups of ATP are called high-energy bonds because their hydrolysis results in a large decrease in free energy. ATP can be hydrolyzed either to ADP plus a phosphate group (HPO_4^{2-}) or to AMP plus pyrophosphate. In the latter case, pyrophosphate is itself rapidly hydrolyzed, releasing additional free energy.

decrease in free energy. Enzymes are responsible for carrying out such coupled reactions in a coordinated manner.

The cell uses this basic mechanism to drive the many energetically unfavorable reactions that must take place in biological systems. **Adenosine 5′-triphosphate (ATP)** plays a central role in this process by acting as a store of free energy within the cell (Figure 3.10). The bonds between the phosphates in ATP are known as **high-energy bonds** because their hydrolysis is accompanied by a relatively large decrease in free energy. There is nothing special about the chemical bonds themselves; they are called high-energy bonds only because a large amount of free energy is released when they are hydrolyzed within the cell. In the hydrolysis of ATP to ADP plus phosphate (P_i), $\Delta G^{\circ\prime} = -7.3$ kcal/mol. Recall, however, that $\Delta G^{\circ\prime}$ refers to "standard conditions" in which the concentrations of all products and reactants are 1 *M*. Actual intracellular concentrations of P_i are approximately 10^{-2} *M*, and intracellular concentrations of ATP are higher than those of ADP. These differences between intracellular concentrations and those of the standard state favor ATP hydrolysis, so for ATP hydrolysis within a cell, ΔG is approximately –12 kcal/mol.

Alternatively, ATP can be hydrolyzed to AMP plus pyrophosphate (PP_i). This reaction yields about the same amount of free energy as the hydrolysis of ATP to ADP. However, the pyrophosphate produced by this reaction is then itself rapidly hydrolyzed, with a ΔG similar to that of ATP hydrolysis. Thus, the total free-energy change resulting from the hydrolysis of ATP to AMP is approximately twice that obtained by the hydrolysis of ATP to ADP.

For comparison, the bond between the sugar and phosphate group of AMP, rather than having high energy, is typical of covalent bonds; for the hydrolysis of AMP, $\Delta G^{\circ\prime} = -3.3$ kcal/mol.

Because of the accompanying decrease in free energy, the hydrolysis of ATP can be used to drive other energy-requiring reactions within the cell. For example, the first reaction in glycolysis (discussed in the next section) is the conversion of glucose to glucose-6-phosphate. The reaction can be written as follows:

$$\text{Glucose} + \text{Phosphate } (HPO_4^{2-}) \rightarrow \text{Glucose-6-phosphate} + H_2O$$

Because this reaction is energetically unfavorable as written ($\Delta G^{\circ\prime} = +3.3$ kcal/mol), it must be driven in the forward direction by being coupled to ATP hydrolysis ($\Delta G^{\circ\prime} = -7.3$ kcal/mol):

$$ATP + H_2O \rightarrow ADP + HPO_4^{2-}$$

The combined reaction can be written as follows:

$$\text{Glucose} + ATP \rightarrow \text{Glucose-6-phosphate} + ADP$$

The free-energy change for this reaction is the sum of the free-energy changes for the individual reactions, so for the coupled reaction $\Delta G^{\circ\prime} = -4.0$ kcal/mol, favoring glucose-6-phosphate formation.

Other molecules, including other nucleoside triphosphates (e.g., GTP), also have high-energy bonds and can be used as ATP is used to drive energy-requiring reactions. For most reactions, however, ATP provides the free energy. The energy-yielding reactions within the cell are therefore coupled to ATP synthesis, while the energy-requiring reactions are coupled to ATP hydrolysis. The high-energy bonds of ATP thus play a central role in cell metabolism by serving as a usable storage form of free energy.

The Generation of ATP from Glucose

The breakdown of carbohydrates, particularly glucose, is a major source of cellular energy. The complete oxidative breakdown of glucose to CO_2 and H_2O can be written as follows:

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O$$

The reaction yields a large amount of free energy: $\Delta G^{\circ\prime} = -686$ kcal/mol. To harness this free energy in usable form, glucose is oxidized within cells in a series of steps coupled to the synthesis of ATP.

3.3 WEBSITE ANIMATION

Glycolysis Glycolysis is the initial stage in the breakdown of glucose, resulting in two molecules of pyruvate and a net gain of two molecules of ATP.

Glycolysis, the initial stage in the breakdown of glucose, is common to virtually all cells. Glycolysis occurs in the absence of oxygen and can provide all the metabolic energy of anaerobic organisms. In aerobic cells, however, glycolysis is only the first stage in glucose degradation.

The reactions of glycolysis result in the breakdown of glucose into pyruvate, with the net gain of two molecules of ATP (Figure 3.11). The initial reactions in the pathway actually consume energy, using ATP to phosphorylate glucose to glucose-6-phosphate and then fructose-6-phosphate to fructose-1,6-bisphosphate. The enzymes that catalyze these two reactions—hexokinase and phosphofructokinase, respectively—are important regulatory points of the glycolytic pathway. The key control element is phosphofructokinase, which is inhibited by high levels of ATP. Inhibition of phosphofructokinase results in an accumulation of glucose-6-phosphate, which in turn inhibits hexokinase. Thus, when the cell has an adequate supply of metabolic energy available in the form of ATP, the breakdown of glucose is inhibited.

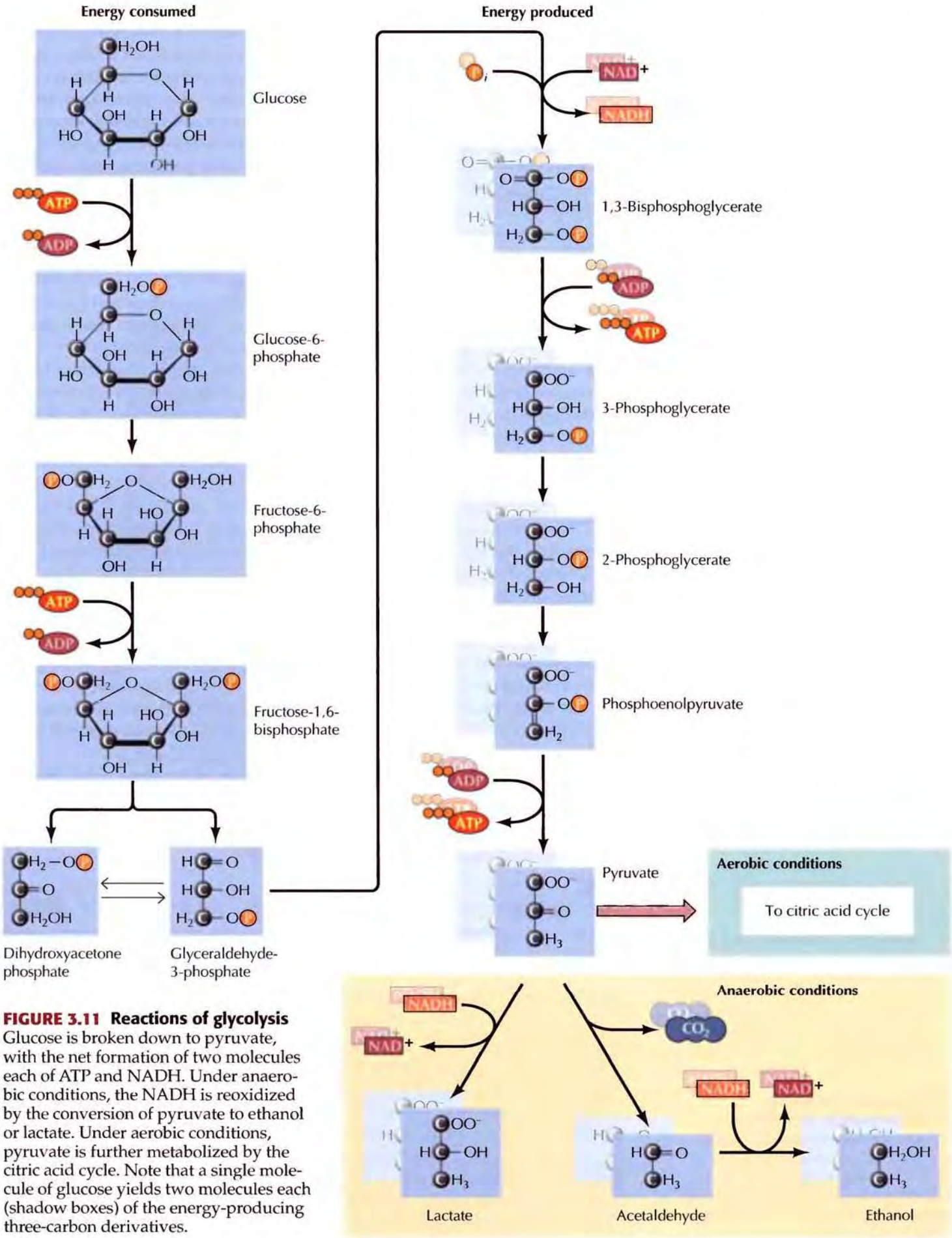


FIGURE 3.11 Reactions of glycolysis Glucose is broken down to pyruvate, with the net formation of two molecules each of ATP and NADH. Under anaerobic conditions, the NADH is reoxidized by the conversion of pyruvate to ethanol or lactate. Under aerobic conditions, pyruvate is further metabolized by the citric acid cycle. Note that a single molecule of glucose yields two molecules each (shadow boxes) of the energy-producing three-carbon derivatives.

The reactions following the formation of fructose-1,6-bisphosphate constitute the energy-producing part of the glycolytic pathway. Cleavage of fructose-1,6-bisphosphate yields two molecules of the three-carbon sugar glyceraldehyde-3-phosphate, which is oxidized to 1,3-bisphosphoglycerate. The phosphate group of this compound has a very high free energy of hydrolysis ($\Delta G°' = -11.5$ kcal/mol), so it is used in the next reaction of glycolysis to drive the synthesis of ATP from ADP. The product of this reaction, 3-phosphoglycerate, is then converted to phosphoenolpyruvate, the second high-energy intermediate in glycolysis. In the hydrolysis of the high-energy phosphate of phosphoenolpyruvate, $\Delta G°' = -14.6$ kcal/mol, its conversion to pyruvate is coupled to the synthesis of ATP. Each molecule of glyceraldehyde-3-phosphate converted to pyruvate is thus coupled to the generation of two molecules of ATP; in total, four ATPs are synthesized from each starting molecule of glucose. Since two ATPs were required to prime the initial reactions, the net gain from glycolysis is two ATP molecules.

In addition to producing ATP, glycolysis converts two molecules of the coenzyme NAD^+ to NADH. In this reaction, NAD^+ acts as an oxidizing agent that accepts electrons from glyceraldehyde-3-phosphate. The NADH formed as a product must be recycled by serving as a donor of electrons for other oxidation–reduction reactions within the cell. In anaerobic conditions, the NADH formed during glycolysis is reoxidized to NAD^+ by the conversion of pyruvate to lactate or ethanol. In aerobic organisms, however, the NADH serves as an additional source of energy by donating its electrons to the electron transport chain, where they are ultimately used to reduce O_2 to H_2O, coupled to the generation of additional ATP.

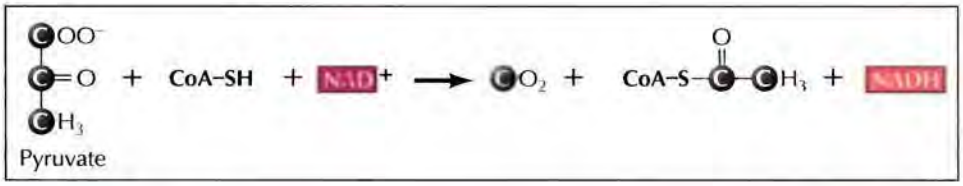
In eukaryotic cells, glycolysis takes place in the cytosol. Pyruvate is then transported into mitochondria, where its complete oxidation to CO_2 and H_2O yields most of the ATP derived from glucose breakdown. The next step in the metabolism of pyruvate is its oxidative decarboxylation in the presence of **coenzyme A (CoA)**, which serves as a carrier of acyl groups in various metabolic reactions (Figure 3.12). One carbon of pyruvate is released as CO_2, and the remaining two carbons are added to CoA to form acetyl CoA. In the process, one molecule of NAD^+ is reduced to NADH.

The acetyl CoA formed by this reaction enters the **citric acid cycle** or **Krebs cycle** (Figure 3.13), which is the central pathway in oxidative metabo-

3.4 WEBSITE ANIMATION

The Citric Acid Cycle

The citric acid cycle is the central pathway in oxidative metabolism and completes the oxidation of glucose to six molecules of carbon dioxide.



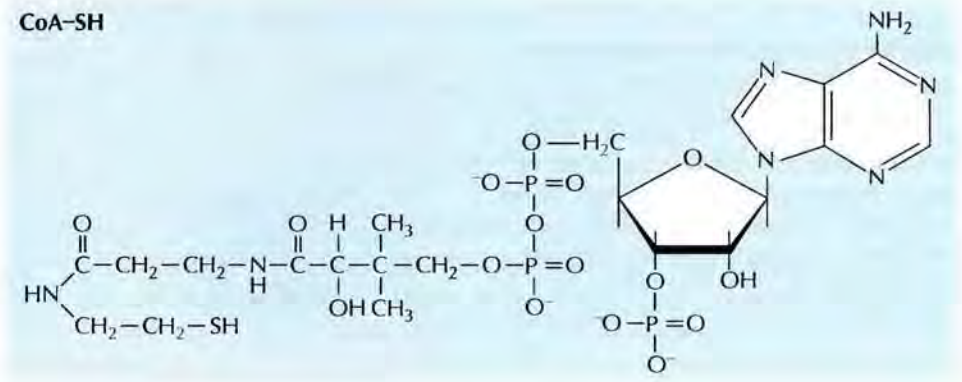


FIGURE 3.12 Oxidative decarboxylation of pyruvate Pyruvate is converted to CO_2 and acetyl CoA, and one molecule of NADH is produced in the process. Coenzyme A (CoA–SH) is a general carrier of activated acyl groups in a variety of reactions.

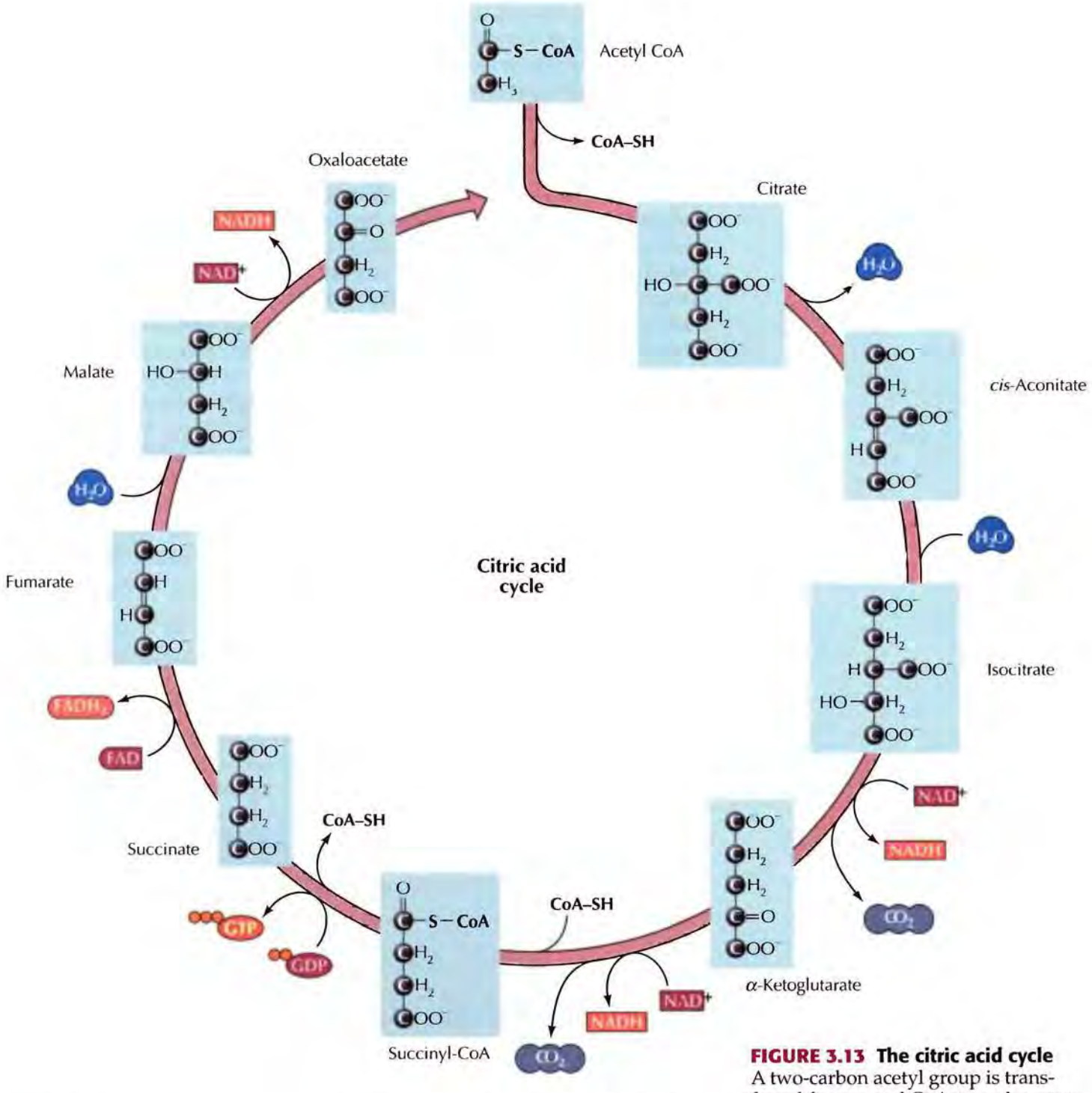


FIGURE 3.13 The citric acid cycle
A two-carbon acetyl group is transferred from acetyl CoA to oxaloacetate, forming citrate. Two carbons of citrate are then oxidized to CO_2 and oxaloacetate is regenerated. Each turn of the cycle yields one molecule of GTP, three molecules of NADH, and one molecule of $FADH_2$.

lism. The two-carbon acetyl group combines with oxaloacetate (four carbons) to yield citrate (six carbons). Through eight further reactions, two carbons of citrate are completely oxidized to CO_2 and oxaloacetate is regenerated. During the cycle, one high-energy phosphate bond is formed in GTP, which is used directly to drive the synthesis of one ATP molecule. In addition, each turn of the cycle yields three molecules of NADH and one molecule of reduced **flavin adenine dinucleotide ($FADH_2$)**, which is another carrier of electrons in oxidation–reduction reactions.

The citric acid cycle completes the oxidation of glucose to six molecules of CO_2. Four molecules of ATP are obtained directly from each glucose molecule—two from glycolysis and two from the citric acid cycle (one for each

molecule of pyruvate). In addition, ten molecules of NADH (two from glycolysis, two from the conversion of pyruvate to acetyl CoA, and six from the citric acid cycle) and two molecules of $FADH_2$ are formed. The remaining energy derived from the breakdown of glucose comes from the reoxidation of NADH and $FADH_2$, with their electrons being transferred through the electron transport chain to (eventually) reduce O_2 to H_2O.

During **oxidative phosphorylation**, the electrons of NADH and $FADH_2$ combine with O_2, and the energy released from the process drives the synthesis of ATP from ADP. The transfer of electrons from NADH to O_2 releases a large amount of free energy: $\Delta G^{\circ\prime} = -52.5$ kcal/mol for each pair of electrons transferred. So that this energy can be harvested in usable form, the process takes place gradually by the passage of electrons through a series of carriers, which constitute the **electron transport chain** (Figure 3.14). The components of the electron transport chain are located in the inner

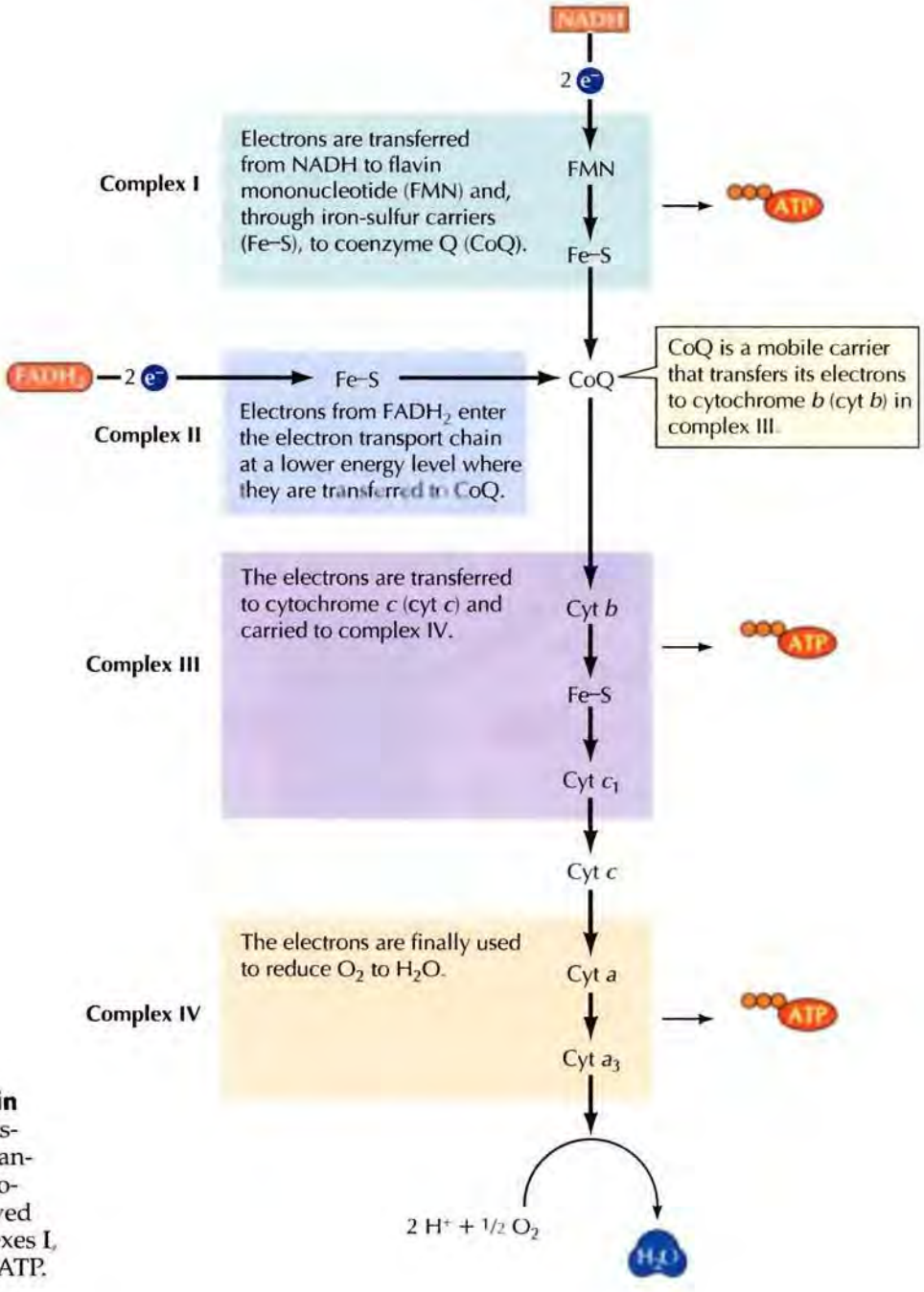


FIGURE 3.14 The electron transport chain Electrons from NADH and $FADH_2$ are transferred to O_2 through a series of carriers organized into four protein complexes in the mitochondrial membrane. The free energy derived from electron transport reactions at complexes I, III, and IV is used to drive the synthesis of ATP.

mitochondrial membrane of eukaryotic cells, and oxidative phosphorylation is considered in more detail when mitochondria are discussed in Chapter 11. In aerobic bacteria, which use a comparable system, components of the electron transport chain are located in the plasma membrane. In either case, the transfer of electrons from NADH to O_2 yields sufficient energy to drive the synthesis of approximately three molecules of ATP. Electrons from $FADH_2$ enter the electron transport chain at a lower energy level, so their transfer to O_2 yields less usable free energy—only two ATP molecules.

It is now possible to calculate the total yield of ATP from the oxidation of glucose. The net gain from glycolysis is two molecules of ATP and two molecules of NADH. The conversion of pyruvate to acetyl CoA and its metabolism via the citric acid cycle yields two additional molecules of ATP, eight of NADH, and two of $FADH_2$. Assuming that three molecules of ATP are derived from the oxidation of each NADH and two from each $FADH_2$, the total yield is 38 molecules of ATP per molecule of glucose. However, this yield is lower in some cells because the two molecules of NADH generated by glycolysis in the cytosol are unable to enter mitochondria directly. Instead, their electrons must be transferred into the mitochondrion via a shuttle system. Depending on the system used, this transfer may result in these electrons entering the electron transport chain at the level of $FADH_2$. In such cases, the two molecules of NADH derived from glycolysis give rise to two rather than three molecules of ATP, reducing the total yield to 36 rather than 38 ATPs per molecule of glucose.

The Derivation of Energy from Other Organic Molecules

Energy in the form of ATP can be derived from the breakdown of other organic molecules, with the pathways involved in glucose degradation again playing a central role. Nucleotides, for example, can be broken down to sugars, which then enter the glycolytic pathway, and amino acids are degraded via the citric acid cycle. The two principal storage forms of energy within cells, polysaccharides and lipids, can also be broken down to produce ATP. Polysaccharides are broken down into free sugars, which are then metabolized as discussed in the previous section. Lipids, however, are an even more efficient energy storage molecule. Because lipids are more reduced than carbohydrates, which consist primarily of hydrocarbon chains, the oxidation of lipids yields substantially more energy per weight of starting material.

Fats (triacylglycerols) are the major storage form of lipids. The first step in their utilization is their breakdown to glycerol and free fatty acids. Each fatty acid is joined to coenzyme A, yielding a fatty acyl-CoA at the cost of one molecule of ATP (Figure 3.15). The fatty acids are then degraded in a stepwise oxidative process, two carbons at a time, yielding acetyl CoA plus a fatty acyl-CoA shorter by one two-carbon unit. Each round of oxidation also yields one molecule of NADH and one of $FADH_2$. The acetyl CoA then enters the citric acid cycle, and degradation of the remainder of the fatty acid continues in the same manner.

The breakdown of a 16-carbon fatty acid thus yields seven molecules of NADH, seven of $FADH_2$, and eight of acetyl CoA. In terms of ATP generation, this yield corresponds to 21 molecules of ATP derived from NADH (3×7), 14 ATPs from $FADH_2$ (2×7), and 96 from acetyl CoA (8×12). Since one ATP was used to start the process, the net gain is 130 ATPs per molecule of a 16-carbon fatty acid. Compare this yield with the net gain of 38 ATPs per molecule of glucose. Since the molecular weight of a saturated 16-carbon fatty acid is 256 and that of glucose is 180, the yield of ATP is approximately

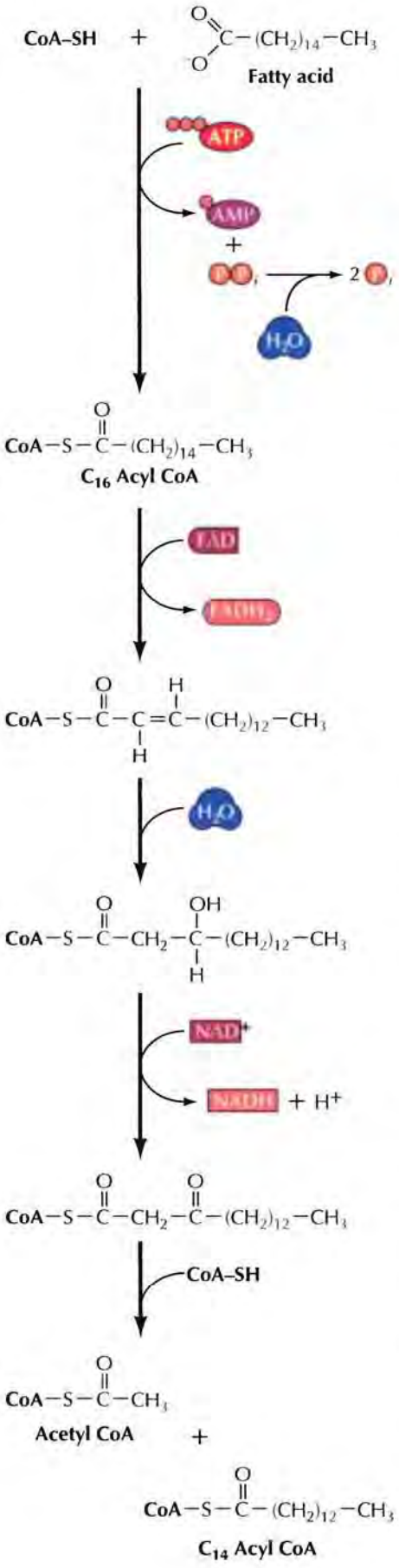


FIGURE 3.15 Oxidation of fatty acids The fatty acid (e.g., the 16-carbon saturated fatty acid palmitate) is initially joined to coenzyme A at the cost of one molecule of ATP. Oxidation of the fatty acid then proceeds by stepwise removal of two-carbon units as acetyl CoA, coupled to the formation of one molecule each of NADH and $FADH_2$.

2.5 times greater per gram of the fatty acid—hence, the advantage of lipids over polysaccharides as energy storage molecules.

Photosynthesis

The generation of energy from oxidation of carbohydrates and lipids relies on the degradation of preformed organic compounds. The energy required for the synthesis of these compounds is ultimately derived from sunlight, which is harvested and used by plants and photosynthetic bacteria to drive the synthesis of carbohydrates. By converting the energy of sunlight to a usable form of chemical energy, photosynthesis is the source of virtually all metabolic energy in biological systems.

The overall equation of photosynthesis can be written as follows:

$$6\,CO_2 + 6\,H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 6\,O_2$$

The process is much more complex, however, and takes place in two distinct stages. In the first, called the **light reactions**, energy absorbed from sunlight drives the synthesis of ATP and NADPH (a coenzyme similar to NADH), coupled to the oxidation of H_2O to O_2. The ATP and NADPH generated by the light reactions drive the synthesis of carbohydrates from CO_2 and H_2O in a second set of reactions, called the **dark reactions** because they do not require sunlight. In eukaryotic cells, both the light and dark reactions occur in chloroplasts.

Photosynthetic pigments capture energy from sunlight by absorbing photons. Absorption of light by these pigments causes an electron to move from its normal molecular orbital to one of higher energy, thus converting energy from sunlight into chemical energy. In plants the most abundant photosynthetic pigments are the **chlorophylls** (Figure 3.16), which together absorb visible light of all wavelengths other than green. Additional pigments absorb light of other wavelengths, so essentially the entire spectrum of visible light can be captured and utilized for photosynthesis.

The energy captured by the absorption of light is used to convert H_2O to O_2 (Figure 3.17). The high-energy electrons derived from this process then enter an electron transport chain in which their transfer through a series of carriers is coupled to the synthesis of ATP. In addition, these high energy electrons reduce $NADP^+$ to NADPH.

In the dark reactions, the ATP and NADPH produced from the light reactions drive the synthesis of carbohydrates from CO_2 and H_2O. One molecule of CO_2 at a time is added to a cycle of reactions—known as the **Calvin cycle** after its discoverer, Melvin Calvin—that leads to the formation of carbohydrates (Figure 3.18). Overall, the Calvin cycle consumes 18 molecules of ATP and 12 of NADPH for each molecule of glucose synthesized. Two electrons are needed to convert each molecule of $NADP^+$ to NADPH, so 24 electrons must pass through the electron transport chain to generate sufficient

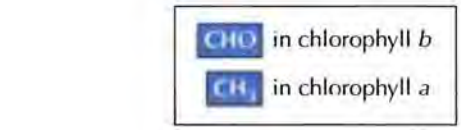


FIGURE 3.16 The structure of chlorophyll Chlorophylls consist of porphyrin ring structures linked to hydrocarbon tails. Chlorophylls *a* and *b* differ by a single functional group in the porphyrin ring.

NADPH to synthesize one molecule of glucose. These electrons are obtained by the conversion of 12 molecules of H_2O to six molecules of O_2, consistent with the formation of six molecules of O_2 for each molecule of glucose. It is not clear, however, whether the passage of the same 24 electrons through the electron transport chain is also sufficient to generate the 18 ATPs that are required by the Calvin cycle. Some of these ATP molecules may instead be generated by alternative electron transport chains that use the energy derived from sunlight to synthesize ATP without the synthesis of NADPH (see Chapter 11).

The Biosynthesis of Cell Constituents

The preceding section reviewed the major metabolic reactions by which the cell obtains and stores energy in the form of ATP. This metabolic energy is then used to accomplish various tasks, including the synthesis of macromolecules and other cell constituents. Thus, energy derived from the breakdown of organic molecules (catabolism) is used to drive the synthesis of other required components of the cell. Most catabolic pathways involve the oxidation of organic molecules coupled to the generation of both energy (ATP) and reducing power (NADH). In contrast, biosynthetic (anabolic) pathways generally involve the use of both ATP and reducing power (usually in the form of NADPH) for the production of new organic compounds. One major biosynthetic pathway, the synthesis of carbohydrates from CO_2 and H_2O during the dark reactions of photosynthesis, was discussed in the preceding section. Additional pathways leading to the biosynthesis of

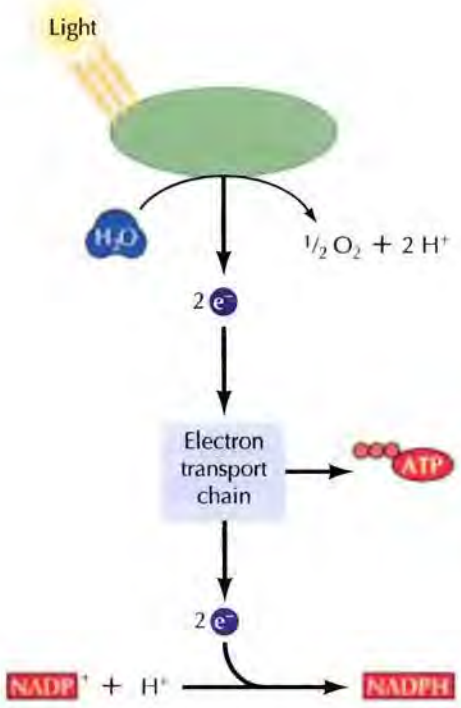


FIGURE 3.17 The light reactions of photosynthesis Energy from sunlight is used to split H_2O to O_2. The high-energy electrons derived from this process are then transported through a series of carriers and used to convert $NADP^+$ to NADPH. Energy derived from the electron transport reactions also drives the synthesis of ATP. The details of these reactions are discussed in Chapter 11.

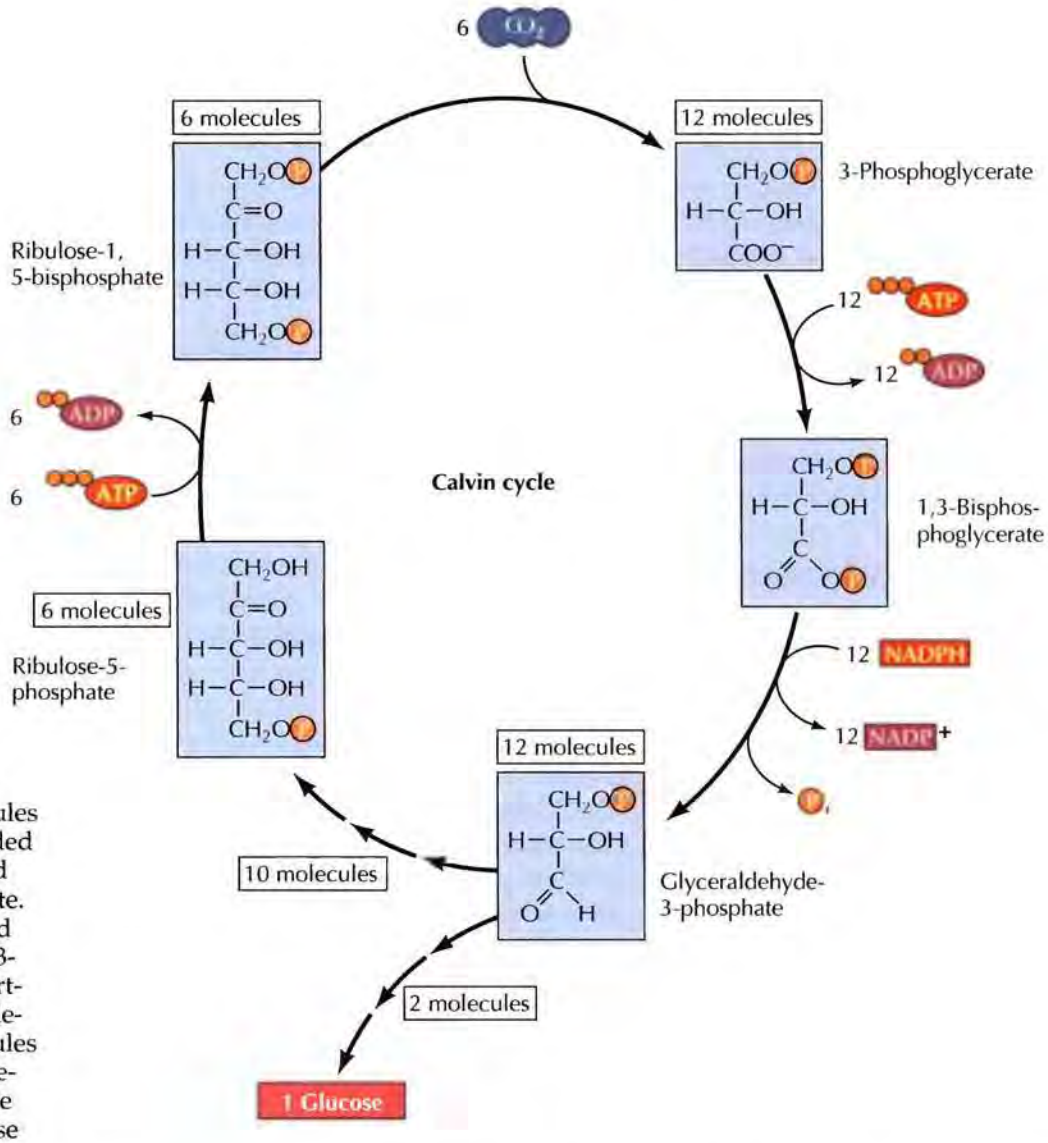


FIGURE 3.18 The Calvin cycle Shown here is the synthesis of one molecule of glucose from six molecules of CO_2. Each molecule of CO_2 is added to ribulose-1,5-bisphosphate to yield two molecules of 3-phosphoglycerate. These six molecules of CO_2 thus lead to the formation of 12 molecules of 3-phosphoglycerate, which are converted to 12 molecules of glyceraldehyde-3-phosphate at the cost of 12 molecules each of ATP and NADPH. Two molecules of glyceraldehyde-3-phosphate are then used for synthesis of glucose and ten molecules continue in the Calvin cycle to form six molecules of ribulose-5-phosphate. The cycle is then completed by the use of six additional ATP molecules for the synthesis of ribulose-1,5-bisphosphate.

3.5 WEBSITE ANIMATION

The Calvin Cycle In the Calvin cycle of photosynthesis, six molecules of carbon dioxide are used to generate a molecule of glucose.

major cellular constituents (carbohydrates, lipids, proteins, and nucleic acids) are reviewed in the sections that follow.

Carbohydrates

In addition to being obtained directly from food or generated by photosynthesis, glucose can be synthesized from other organic molecules. In animal cells, glucose synthesis (**gluconeogenesis**) usually starts with lactate (produced by anaerobic glycolysis), amino acids (derived from the breakdown of proteins), or glycerol (produced by the breakdown of lipids). Plants (but not animals) are also able to synthesize glucose from fatty acids—a process that is particularly important during the germination of seeds, when energy stored as fats must be converted to carbohydrates to support growth of the plant. In both animal and plant cells, simple sugars are polymerized and stored as polysaccharides.

Gluconeogenesis involves the conversion of pyruvate to glucose—essentially the reverse of glycolysis (Figure 3.19). However, as discussed earlier, the glycolytic conversion of glucose to pyruvate is an energy-yielding pathway, generating two molecules each of ATP and NADH. Although some reactions of glycolysis are readily reversible, others will proceed only in the

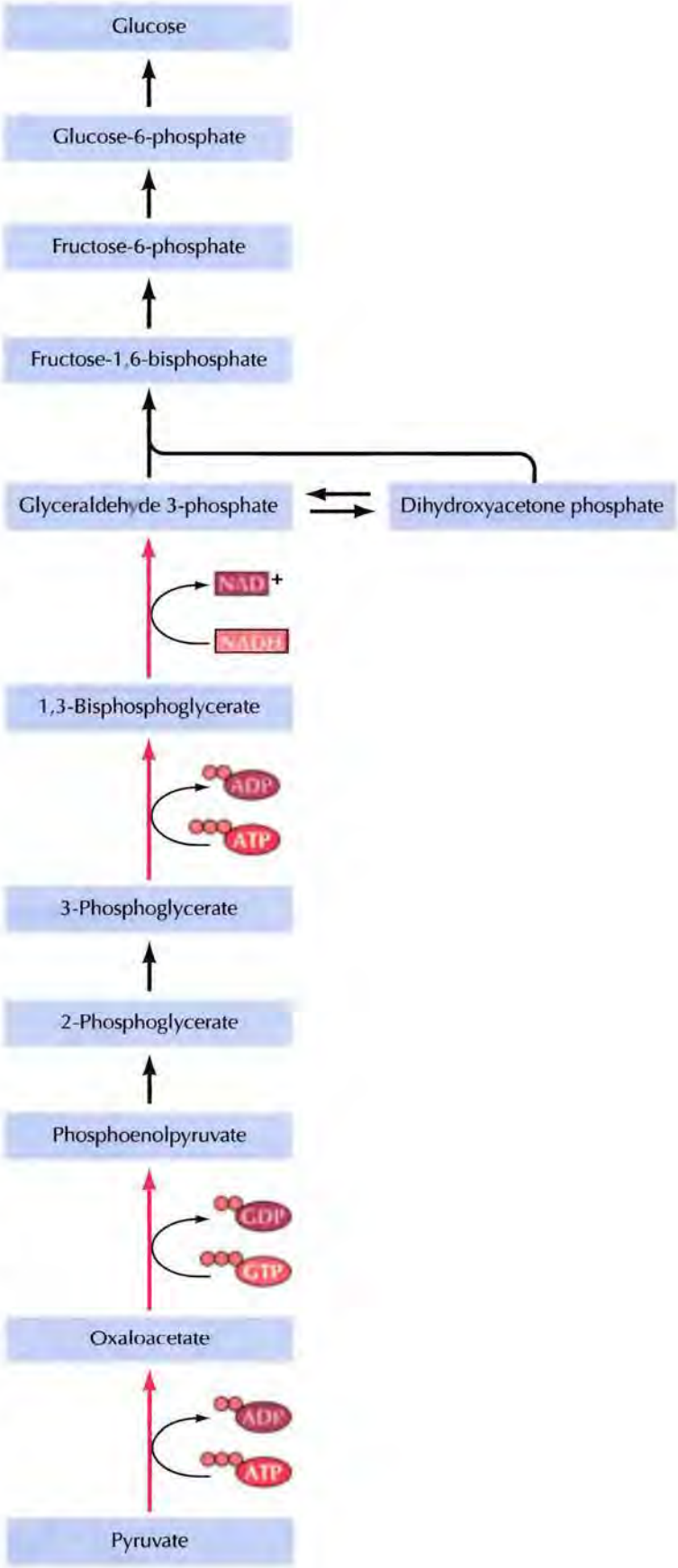


FIGURE 3.19 Gluconeogenesis Glucose is synthesized from two molecules of pyruvate, at the cost of 4 molecules of ATP, 2 molecules of GTP, and 2 molecules of NADH. The energy-requiring steps of gluconeogenesis are indicated by red arrows.

direction of glucose breakdown, because they are associated with a large decrease in free energy. These energetically favorable reactions of glycolysis are bypassed during gluconeogenesis by other reactions (catalyzed by different enzymes) that are coupled to the expenditure of ATP and NADH in order to drive them in the direction of glucose synthesis. Overall, the generation of glucose from two molecules of pyruvate requires four molecules of ATP, two molecules of GTP, and two molecules of NADH. This process is considerably more costly than the simple reversal of glycolysis (which would require two molecules of ATP and two of NADH), illustrating the additional energy required to drive the pathway in the direction of biosynthesis.

In both plant and animal cells, glucose is stored in the form of polysaccharides (starch and glycogen, respectively). The synthesis of polysaccharides, like that of all other macromolecules, is an energy-requiring reaction. As discussed in Chapter 2, the linkage of two sugars by a glycosidic bond can be written as a dehydration reaction, in which H_2O is removed (see Figure 2.3). Such a reaction, however, is energetically unfavorable and therefore unable to proceed in the forward direction. Consequently, the formation of a glycosidic bond must be coupled to an energy-yielding reaction, which is accomplished by the use of nucleotide sugars as intermediates in polysaccharide synthesis (Figure 3.20). Glucose is first phosphorylated in an ATP-driven reaction to glucose-6-phosphate, which is then converted to glucose-1-phosphate. Glucose-1-phosphate reacts with UTP (uridine triphosphate), yielding UDP-glucose plus pyrophosphate, which is hydrolyzed to phosphate with the release of additional free energy. UDP-glucose is an activated intermediate that then donates its glucose residue to a growing polysaccharide chain in an energetically favorable reaction. Thus, chemical energy in the form of ATP and UTP drives the synthesis of polysaccharides from simple sugars.

Lipids

Lipids are important energy storage molecules and the major constituent of cell membranes. They are synthesized from acetyl CoA, which is formed from the breakdown of carbohydrates, in a series of reac-

tions that resemble the reverse of fatty acid oxidation. As with carbohydrate biosynthesis, however, the reactions leading to the synthesis of fatty acids differ from those involved in their degradation and are driven in the biosynthetic direction by being coupled to the expenditure of both energy in the form of ATP and reducing power in the form of NADPH. Fatty acids are synthesized by the stepwise addition of two-carbon units derived from acetyl CoA to a growing chain. The addition of each of these two-carbon units requires the expenditure of one molecule of ATP and two molecules of NADPH.

The major product of fatty acid biosynthesis, which occurs in the cytosol of eukaryotic cells, is the 16-carbon fatty acid palmitate. The principal constituents of cell membranes (phospholipids, sphingomyelin, and glycolipids) are then synthesized from free fatty acids in the endoplasmic reticulum and Golgi apparatus (see Chapter 10).

Proteins

In contrast to carbohydrates and lipids, proteins (as well as nucleic acids) contain nitrogen in addition to carbon, hydrogen, and oxygen. Nitrogen is incorporated into organic compounds from different sources in different organisms (Figure 3.21). Some bacteria can use atmospheric N_2 by a process called **nitrogen fixation** in which N_2 is reduced to NH_3 at the expense of energy in the form of ATP. Although relatively few species of bacteria are capable of nitrogen fixation, most bacteria, fungi, and plants can use nitrate (NO_3^-), which is a common constituent of soil, by reducing it to NH_3 via electrons derived from NADH or NADPH. Finally, all organisms are able to incorporate ammonia (NH_3) into organic compounds.

NH_3 is incorporated into organic molecules primarily during the synthesis of the amino acids glutamate and glutamine, which are derived from the citric acid cycle intermediate α-ketoglutarate. These amino acids then serve as donors of amino groups during the synthesis of the other amino acids, which are also derived from central metabolic pathways, such as glycolysis and the citric acid cycle (Figure 3.22). The raw material for amino acid syn-

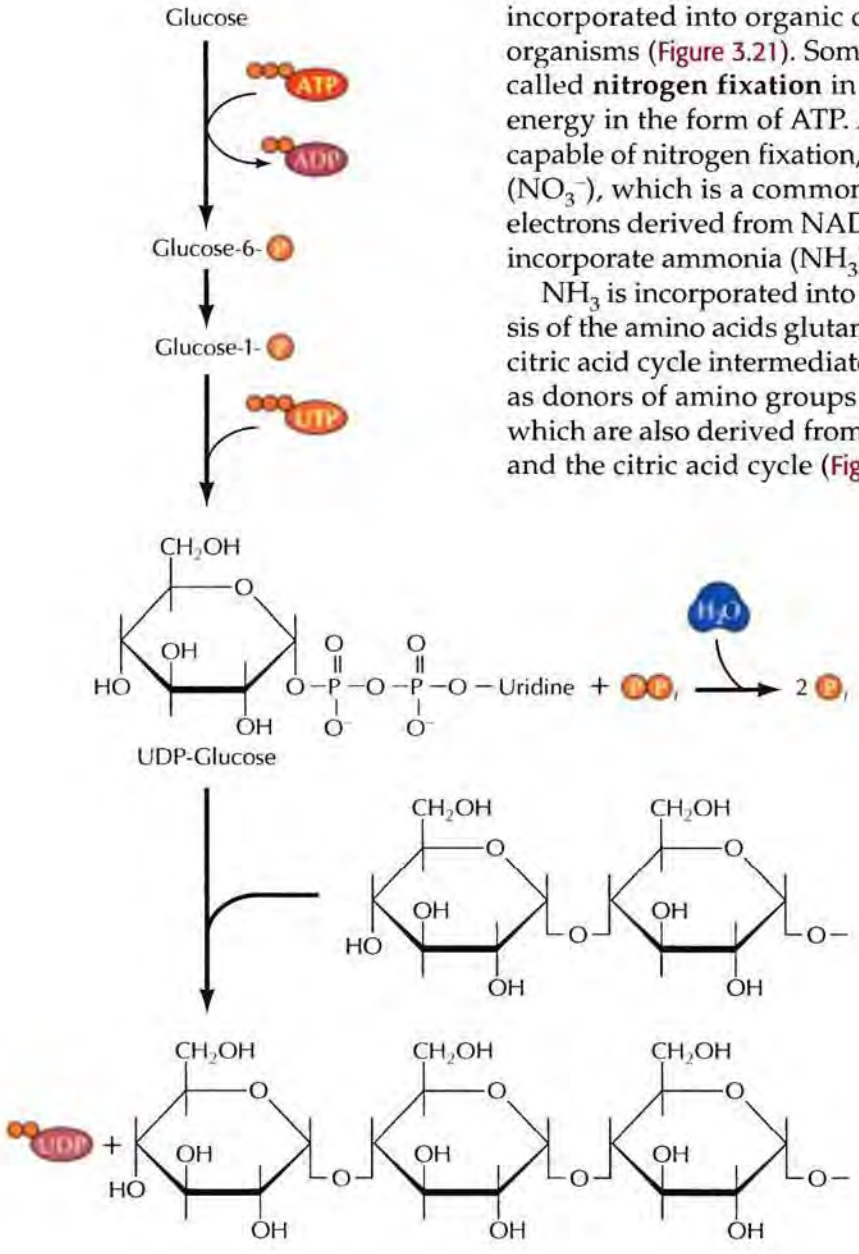


FIGURE 3.20 Synthesis of polysaccharides Glucose is first converted to an activated form, UDP-glucose, at the cost of one molecule each of ATP and UTP. The glucose residue can then be transferred from UDP-glucose to a growing polysaccharide chain in an energetically favorable reaction.

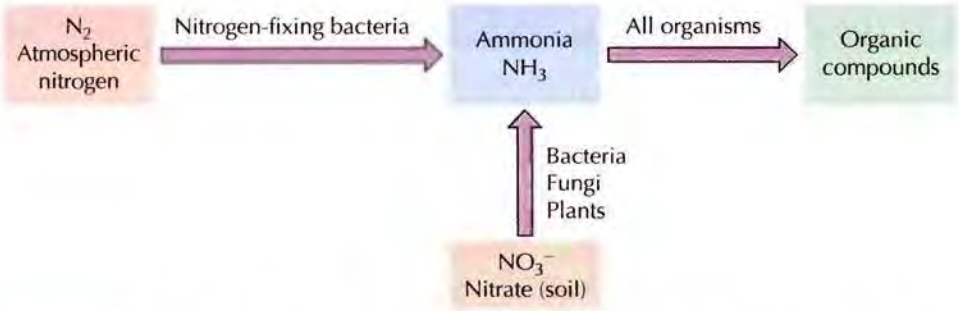


FIGURE 3.21 Assimilation of nitrogen into organic compounds Ammonia is incorporated into organic compounds by all organisms. Some bacteria are capable of converting atmospheric nitrogen to ammonia, and most bacteria, fungi, and plants can utilize nitrate from soil.

thesis is thus obtained from glucose, and the amino acids are synthesized at the cost of both energy (ATP) and reducing power (NADPH). Many bacteria and plants can synthesize all 20 amino acids. Humans and other mammals, however, can synthesize only about half of the required amino acids; the remainder must be obtained from dietary sources (Table 3.2).

The polymerization of amino acids to form proteins also requires energy. Like the synthesis of polysaccharides, the formation of the peptide bond can be considered a dehydration reaction, which must be driven in the

■ **A comparison of catalysts: The industrial production of ammonia is carried out by an iron catalyst under high pressure and temperature. The enzyme nitrogenase produces ammonia under physiological conditions with the expenditure of ATP and reducing power in the form of NADH.**

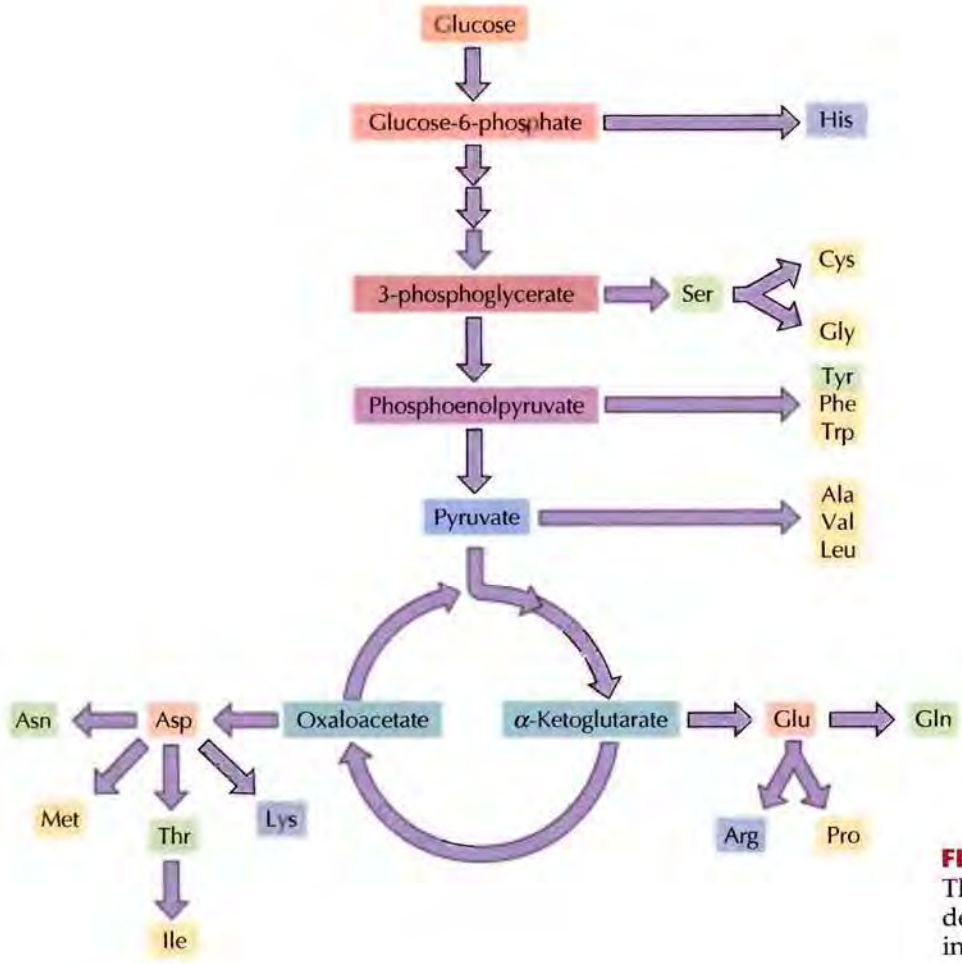
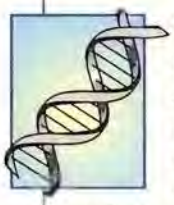


FIGURE 3.22 Biosynthesis of amino acids The carbon skeletons of the amino acids are derived from intermediates in glycolysis and in the citric acid cycle.

MOLECULAR MEDICINE

Phenylketonuria



The Disease

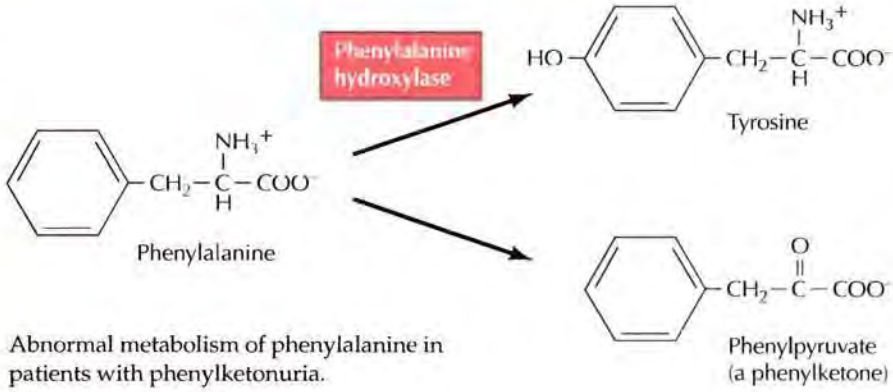
Phenylketonuria, or PKU, is an inborn error of amino acid metabolism with devastating effects. It afflicts approximately one in 10,000 newborn infants and, if untreated, results in severe mental retardation. Fortunately, understanding the nature of the defect responsible for phenylketonuria has allowed early diagnosis and effective treatment.

Molecular and Cellular Basis

Phenylketonuria is caused by a deficiency of the enzyme phenylalanine hydroxylase, which converts phenylalanine to tyrosine. This deficiency causes phenylalanine to accumulate to very high levels and undergo other reactions, such as conversion to phenylpyruvate. Phenylalanine, phenylpyruvate, and other abnormal metabolites accumulate in the blood and are secreted at high levels in the urine (the name of the disease derives from the high levels of phenylpyruvate, a phenylketone found in the urine of affected children). Although the precise biochemical cause is unknown, mental retardation is a critical consequence of the accumulation of these abnormal phenylalanine metabolites.

Prevention and Treatment

The enzyme deficiency causes no difficulties while the fetus is within the uterus, so children with phenylketonuria are normal at birth. If untreated, however, affected children become permanently and severely retarded within the first year of life. Fortunately, newborns with phenylketonuria can be readily identified by routine screening tests that detect elevated levels of phenylalanine in blood. Mental retardation can be prevented by feeding affected infants a synthetic diet that is low in phenylalanine. This dietary treatment eliminates the accumulation of toxic phenylalanine metabolites and effectively prevents the permanent mental deficiency that would otherwise ensue. Routine screening for phenylketonuria is thus a critical test for all newborn infants.



Abnormal metabolism of phenylalanine in patients with phenylketonuria.

TABLE 3.2 Dietary Requirements for Amino Acids in Humans

Essential	Nonessential
Histidine	Alanine
Isoleucine	Arginine[a]
Leucine	Asparagine
Lysine	Aspartate
Methionine	Cysteine
Phenylalanine	Glutamate
Threonine	Glutamine
Tryptophan	Glycine
Valine	Proline
	Serine
	Tyrosine

The essential amino acids must be obtained from dietary sources; the nonessential amino acids can be synthesized by human cells.

[a]Although arginine is classified as a nonessential amino acid, growing children must obtain additional arginine from their diet.

direction of protein synthesis by being coupled to another source of metabolic energy. In the biosynthesis of polysaccharides, this coupling is accomplished through the conversion of sugars to activated intermediates, such as UDP-glucose. Amino acids are similarly activated before being used for protein synthesis.

A critical difference between the synthesis of proteins and that of polysaccharides is that the amino acids are incorporated into proteins in a unique order, specified by a gene. The order of nucleotides in a gene specifies the amino acid sequence of a protein via translation in which messenger RNA (mRNA) acts as a template for protein synthesis (see Chapter 4). Each amino acid is first attached to a specific transfer RNA (tRNA) molecule in a reaction coupled to ATP hydrolysis (Figure 3.23). The aminoacyl tRNAs then align on the mRNA template bound to ribosomes, and each amino acid is added to the C terminus of a growing peptide chain through a series of reactions that will be discussed in detail in Chapter 8. During the process, two additional molecules of GTP are hydrolyzed, so the incorporation of each amino acid into a protein is coupled to the hydrolysis of one ATP and two GTP molecules.

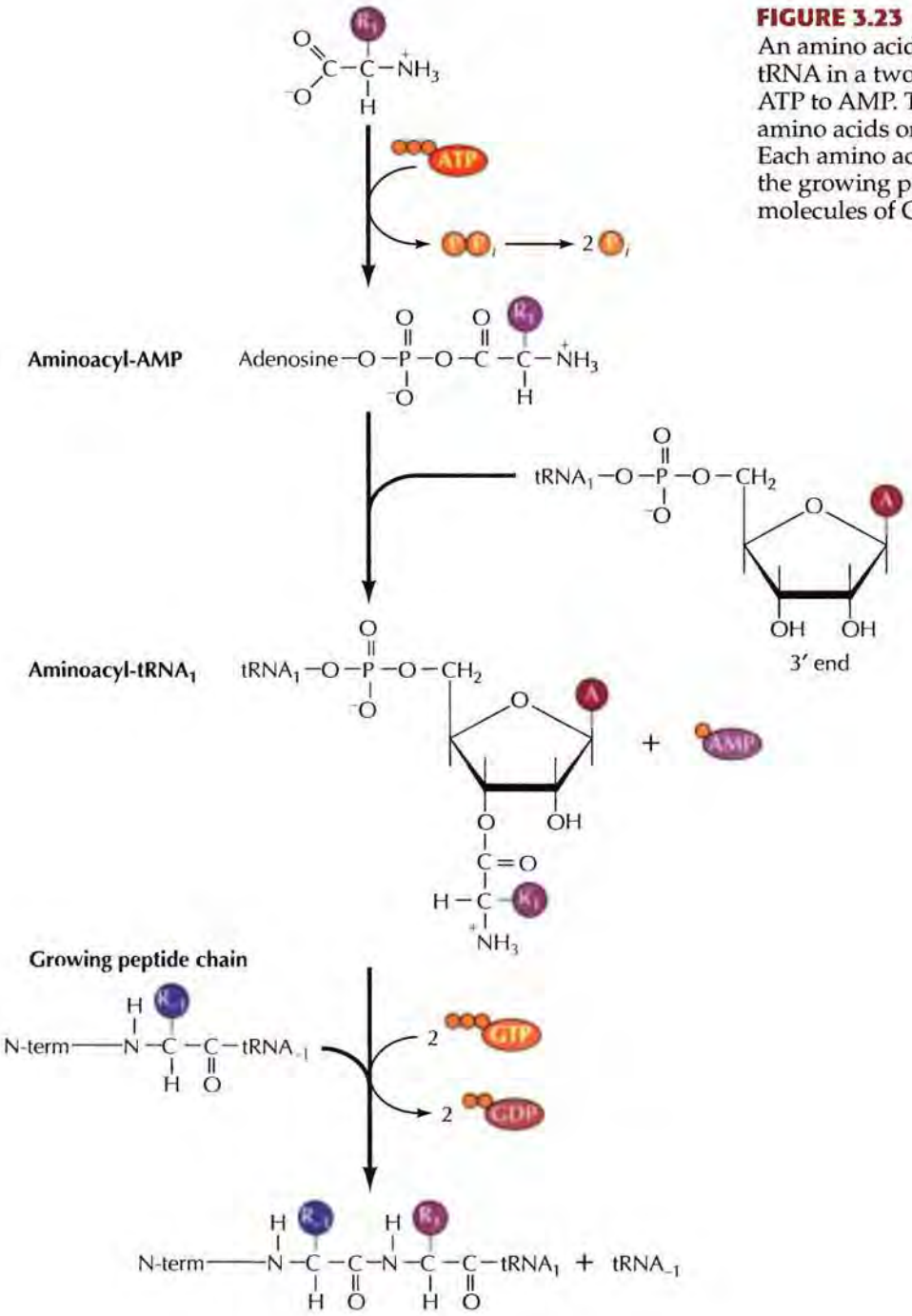


FIGURE 3.23 Formation of the peptide bond
An amino acid is first activated by attachment to its tRNA in a two-step reaction involving the hydrolysis of ATP to AMP. The tRNAs serve as adaptors to align the amino acids on an mRNA template bound to ribosomes. Each amino acid is then transferred to the C terminus of the growing peptide chain at the cost of two additional molecules of GTP.

Nucleic Acids

The precursors of nucleic acids, the nucleotides, are composed of phosphorylated five-carbon sugars joined to nucleic acid bases. Nucleotides can be synthesized from carbohydrates and amino acids; they can also be obtained from dietary sources or reused following nucleic acid breakdown. The starting point for nucleotide biosynthesis is the phosphorylated sugar ribose-5-phosphate, which is derived from glucose-6-phosphate. Divergent pathways then lead to the synthesis of purine and pyrimidine ribonucleotides, which are the immediate precursors for RNA synthesis (Figure 3.24). These ribonucleotides are converted to deoxyribonucleotides, which serve as the monomeric building blocks of DNA.

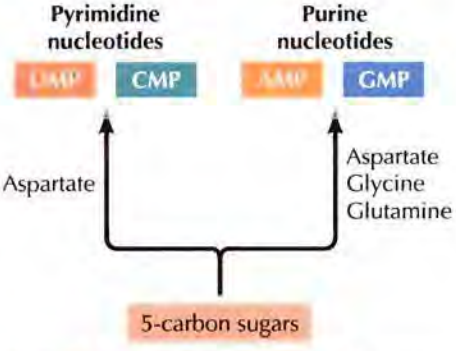


FIGURE 3.24 Biosynthesis of purine and pyrimidine nucleotides
Purine and pyrimidine nucleotides are synthesized from 5-carbon sugars and amino acids.

KEY EXPERIMENT

Antimetabolites and Chemotherapy

Antagonists of Nucleic Acid Derivatives VI. Purines
Gertrude B. Elion, George H. Hitchings, and Henry Vanderwerff
Wellcome Research Laboratories, Tuckahoe, NY
Journal of Biological Chemistry, Volume 192, 1951, pages 505–518

The Context

Gertrude Elion and George Hitchings began a collaboration in 1944 that lasted over 20 years and led to the development of drugs that have proven effective for the treatment of cancer, gout, viruses, and parasitic infections. The principle of their approach to drug development was based on the antimetabolite theory, which initially proposed that some drugs active against bacteria functioned to prevent the utilization of essential nutrients (metabolites) by the bacterial cells. Elion and Hitchings suggested that the growth of rapidly dividing cells, such as cancer cells, might be inhibited by analogs of the nucleic acid bases that interfered with the normal synthesis of DNA. They proceeded to test this hypothesis by synthesizing a large number of compounds that were related to purines and testing their biological effects. These studies led to the finding that 6-mercaptopurine was a potent inhibitor of purine utilization in bacteria, which was rapidly followed by studies showing that it was an effective drug for treatment of acute leukemia in children.

The Experiments

Elion and Hitchings chose to test the potential activities of purine analogs on the growth of the bacterium *Lactobacillus casei*. In 1948, they found that the compound 2,6-diaminopurine inhibited the growth of *L. casei* by interfering with normal purine metabolism. They extended these findings in their 1951 paper by testing the effects of 100 different purines, bearing substitutions of amino, hydroxyl, methyl, chloro, sulfhydryl and other chemical groups at various positions of the purine ring. Two compounds, 6-mercaptopurine and 6-thioguanine, in which oxygen was substituted by sulfur at the 6-position of guanine and hypoxanthine, were found to be potent inhibitors of bacterial growth (see figure). These compounds were then found to be active inhibitors of the growth of a variety of rodent cancers, leading to the first trial of 6-mercaptopurine as a treatment of childhood leukemia. The results of this trial were a spectacular success, and 6-mercaptopurine was approved for treatment of childhood

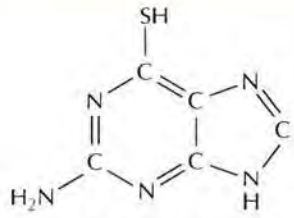
RNA and DNA are polymers of nucleoside monophosphates. As for other macromolecules, however, direct polymerization of nucleoside monophosphates is energetically unfavorable, and the synthesis of polynucleotides instead uses nucleoside triphosphates as activated precursors (Figure 3.25). A nucleoside 5′-triphosphate is added to the 3′ hydroxyl group of a growing polynucleotide chain, with the release and subsequent hydrolysis of pyrophosphate serving to drive the reaction in the direction of polynucleotide synthesis.

COMPANION WEBSITE

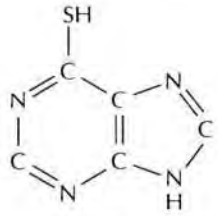
Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

FIGURE 3.25 Synthesis of polynucleotides Nucleoside triphosphates ▶ are joined to the 3′ end of a growing polynucleotide chain with the release of pyrophosphate.

KEY EXPERIMENT



6-Thioguanine



6-Mercaptopurine

Structures of 6-thioguanine and 6-mercaptopurine.

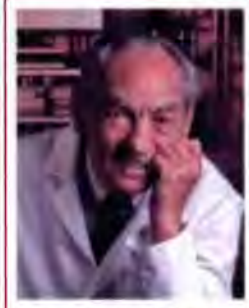
leukemia by the Food and Drug Administration in 1953, just over two years after its synthesis and initial demonstration of its activity as a purine antimetabolite in bacteria.

The Impact

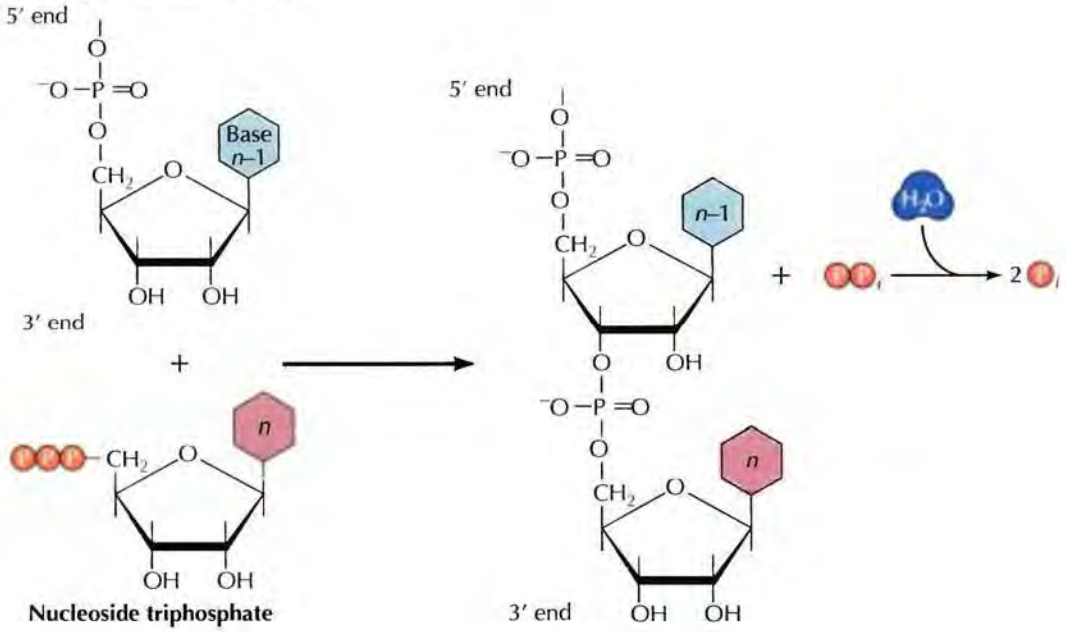
The success of 6-mercaptopurine in treatment of childhood leukemia provided convincing evidence that inhibitors of nucleic acid metabolism could be effective drugs against cancer. This remains the case today, and 6-mercaptopurine is still one of the drugs being successfully used for leukemia treatment. In addition, a number of other antimetabolites that interfere with nucleic acid metabolism have been found to be useful in cancer therapy.

Gertrude Elion and George Hitchings continued their collaborative work on purine antimetabolites until Hitchings' retirement in 1967. In addition to 6-mercaptopurine, they developed drugs that are used for immunosuppression after tissue transplantation, for treatment of rheumatoid arthritis, for treatment of gout, and for treatment of parasitic infections. Following Hitchings' retirement, Elion focused her research on treatment of viral infections and developed the first drug effective against human viral infections—the guanine analog acyclovir, which is highly active against herpes simplex and other herpes viruses. Subsequent success in development of nucleic acid antimetabolites as antiviral drugs includes the thymine analog AZT, which is widely used as an HIV inhibitor for treatment of AIDS. The pioneering work of Elion and Hitchings has thus opened many new areas of research and had a profound impact on the treatment of a wide range of diseases.

Gertrude B. Elion



George H. Hitchings



KEY TERMS

SUMMARY

THE CENTRAL ROLE OF ENZYMES AS BIOLOGICAL CATALYSTS

enzyme, substrate, product, transition state, activation energy, active site

The Catalytic Activity of Enzymes: Virtually all chemical reactions within cells are catalyzed by enzymes.

lock-and-key model, induced fit

Mechanisms of Enzymatic Catalysis: Enzymes increase reaction rates by binding substrates in the proper position, by altering the conformation of substrates to approach the transition state, and by participating directly in chemical reactions.

prosthetic group, coenzyme, nicotinamide adenine dinucleotide (NAD^+)

Coenzymes: Coenzymes function in conjunction with enzymes to carry chemical groups between substrates.

feedback inhibition, allosteric regulation, phosphorylation

Regulation of Enzyme Activity: The activities of enzymes are regulated to meet the physiological needs of the cell. Enzyme activity can be controlled by the binding of small molecules, by interactions with other proteins, and by covalent modifications.

METABOLIC ENERGY

Gibbs free energy (*G*), adenosine 5′-triphosphate (ATP), high-energy bond

Free Energy and ATP: ATP serves as a store of free energy, which is used to drive energy-requiring reactions within cells.

glycolysis, coenzyme A (CoA), citric acid cycle, Krebs cycle, flavin adenine dinucleotide ($FADH_2$), oxidative phosphorylation, electron transport chain

The Generation of ATP from Glucose: The breakdown of glucose provides a major source of cellular energy. In aerobic cells, the complete oxidation of glucose yields 36 to 38 molecules of ATP. Most of this ATP is derived from electron transport reactions in which O_2 is reduced to H_2O.

The Derivation of Energy from Other Organic Molecules: ATP can also be derived from the breakdown of organic molecules other than glucose. Because fats are more reduced than carbohydrates, they provide a more efficient form of energy storage.

light reactions, dark reactions, photosynthetic pigment, chlorophyll, Calvin cycle

Photosynthesis: The energy required for the synthesis of organic molecules is ultimately derived from sunlight, which is harvested by plants and photosynthetic bacteria. In the first stage of photosynthesis, energy from sunlight is used to drive the synthesis of ATP and NADPH, coupled to the oxidation of H_2O to O_2. The ATP and NADPH produced by these reactions are then used to synthesize glucose from CO_2 and H_2O.

THE BIOSYNTHESIS OF CELL CONSTITUENTS

gluconeogenesis

Carbohydrates: Glucose can be synthesized from other organic molecules, using energy and reducing power in the forms of ATP and NADH, respectively. Additional ATP is then needed to drive the synthesis of polysaccharides from simple sugars.

Lipids: Lipids are synthesized from acetyl CoA, which is formed from the breakdown of carbohydrates.

SUMMARY

Proteins: The amino acids are synthesized from intermediates in glycolysis and the citric acid cycle. Their polymerization to form proteins requires additional energy in the form of ATP and GTP.

Nucleic Acids: Purine and pyrimidine nucleotides are synthesized from carbohydrates and amino acids. Their polymerization to DNA and RNA is driven by the use of nucleoside triphosphates as activated precursors.

KEY TERMS

nitrogen fixation

Questions

1. The binding pocket of trypsin contains an aspartate residue. How would changing this amino acid to lysine affect the enzyme's activity?

2. What characteristic of histidine allows it to play a role in enzymatic reactions involving the exchange of hydrogen ions?

3. The following is a pathway for the biosynthesis of molecule D catalyzed by the enzymes E1, E2, and E3.

$$A \xrightarrow{E1} B \xrightarrow{E2} C \xrightarrow{E3} D$$

You are studying the reaction catalyzed by E1 in the absence of the other enzymes. You find that the rate of the reaction decreases as you add increasing concentrations of D. What does this tell you about the mechanism of regulation of enzyme E1?

4. Many biochemical reactions are energetically unfavorable under physiological conditions ($\Delta G^{o\prime} > 0$). How does the cell carry out these reactions?

5. Consider the reaction:

Fructose-6-phosphate + $HPO_4^{2-} \rightleftharpoons$ Fructose-1,6-bisphosphate + H_20, $\Delta G^{o\prime} = +4$ kcal/mol

Knowing the standard free energy change for ATP hydrolysis (–7.3 kcal/mol), calculate the standard free energy change for the reaction catalyzed by phosphofructokinase.

6. Consider the reaction A $\rightleftharpoons$ B + C, in which $\Delta G^{o\prime} = +3.5$ kcal/mol. Calculate ΔG under intracellular conditions, given that the concentration of A is 10^{-2} *M* and the concentrations of B and C are each 10^{-3} *M*. In which direction will the reaction proceed in the cell? For your calculation, $R = 1.98 \times 10^{-3}$ kcal/mol/degree and T = 298 K (25°C). Note, $\ln(x) = 2.3 \log_{10}(x)$.

7. How would glycolysis be affected by an increase in the cellular concentration of ATP?

8. Yeast can grow under anaerobic as well as aerobic conditions. For every molecule of glucose consumed, how many molecules of ATP will be generated by yeast grown in anaerobic compared to aerobic conditions?

9. How do anaerobic organisms regenerate NAD^+ from NADH produced during glycolysis?

10. Why are lipids more efficient energy storage molecules than carbohydrates?

11. Differentiate between the light and dark reactions of photosynthesis.

12. Why is gluconeogenesis not a simple reversal of glycolysis?

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CHAPTER 4

Fundamentals of Molecular Biology

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CONTEMPORARY MOLECULAR BIOLOGY is concerned principally with understanding the mechanisms responsible for transmission and expression of the genetic information that governs cell structure and function. As reviewed in Chapter 1, all cells share a number of basic properties, and this underlying unity of cell biology is particularly apparent at the molecular level. Such unity has allowed scientists to choose simple organisms (such as bacteria) as models for many fundamental experiments, with the expectation that similar molecular mechanisms are operative in organisms as diverse as *E. coli* and humans. Numerous experiments have established the validity of this assumption, and it is now clear that the molecular biology of cells provides a unifying theme to understanding diverse aspects of cell behavior.

Initial advances in molecular biology were made by taking advantage of the rapid growth and readily manipulable genetics of simple bacteria, such as *E. coli*, and their viruses. The development of recombinant DNA then allowed both the fundamental principles and many of the experimental approaches first developed in prokaryotes to be extended to eukaryotic cells. The application of recombinant DNA technology has had a tremendous impact, initially allowing individual eukaryotic genes to be isolated and characterized in detail and more recently allowing the determination of the complete genome sequences of complex plants and animals, including humans.

Heredity, Genes, and DNA

Perhaps the most fundamental property of all living things is the ability to reproduce. All organisms inherit the genetic information specifying their structure and function from their parents. Likewise, all cells arise from preexisting cells, so the genetic material must be replicated and passed from parent to progeny cell at each cell division. How genetic information is replicated and transmitted from cell to cell and organism to organism

thus represents a question that is central to all of biology. Consequently, elucidation of the mechanisms of genetic transmission and identification of the genetic material as DNA were discoveries that formed the foundation of our current understanding of biology at the molecular level.

Genes and Chromosomes

The classical principles of genetics were deduced by Gregor Mendel in 1865, on the basis of the results of breeding experiments with peas. Mendel studied the inheritance of a number of well-defined traits, such as seed color, and was able to deduce general rules for their transmission. In all cases, he could correctly interpret the observed patterns of inheritance by assuming that each trait is determined by a pair of inherited factors, which are now called **genes**. One gene copy (called an **allele**) specifying each trait is inherited from each parent. For example, breeding two strains of peas—one having yellow seeds, and the other green seeds—yields the following results (Figure 4.1). The parental strains each have two identical copies of the gene specifying yellow (*Y*) or green (*y*) seeds, respectively. The progeny plants are therefore hybrids, having inherited one gene for yellow seeds (*Y*) and one for green seeds (*y*). All these progeny plants (the first filial, or F_1, generation) have yellow seeds, so yellow (*Y*) is said to be **dominant** and green (*y*) **recessive**. The **genotype** (genetic composition) of the F_1 peas is thus *Yy*, and their **phenotype** (physical appearance) is yellow. If one F_1 offspring is bred with another, giving rise to F_2 progeny, the genes for yellow and green seeds segregate in a characteristic manner such that the ratio between F_2 plants with yellow seeds and those with green seeds is 3:1.

Mendel's findings, apparently ahead of their time, were largely ignored until 1900, when Mendel's laws were rediscovered and their importance recognized. Shortly thereafter, the role of **chromosomes** as the carriers of genes was proposed. It was realized that most cells of higher plants and

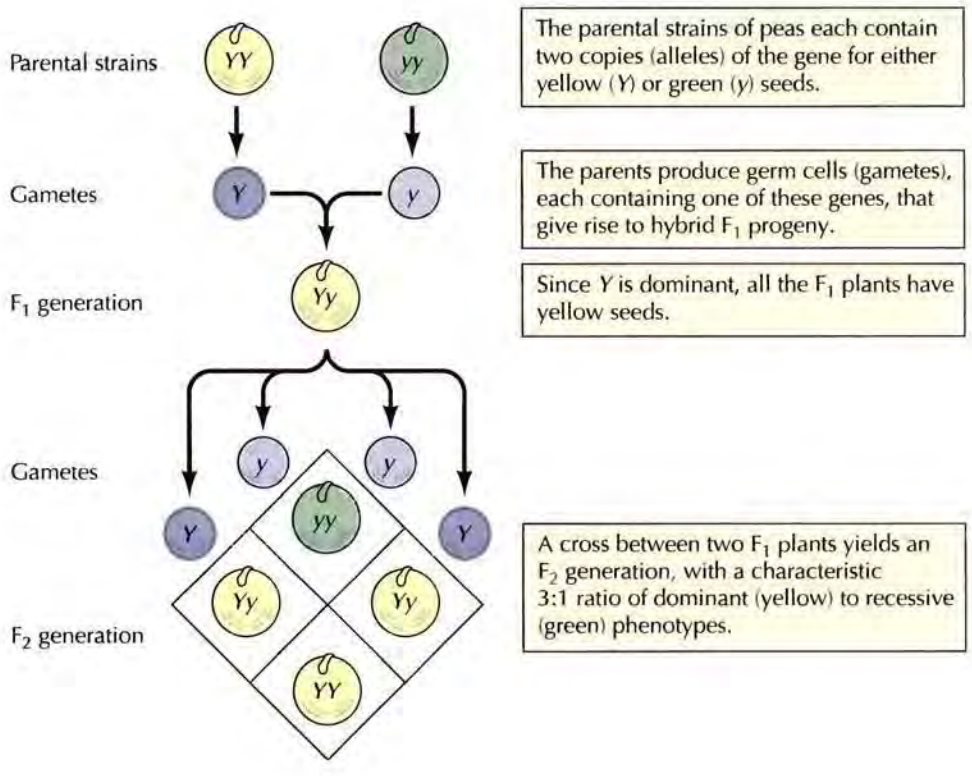


FIGURE 4.1 Inheritance of dominant and recessive genes

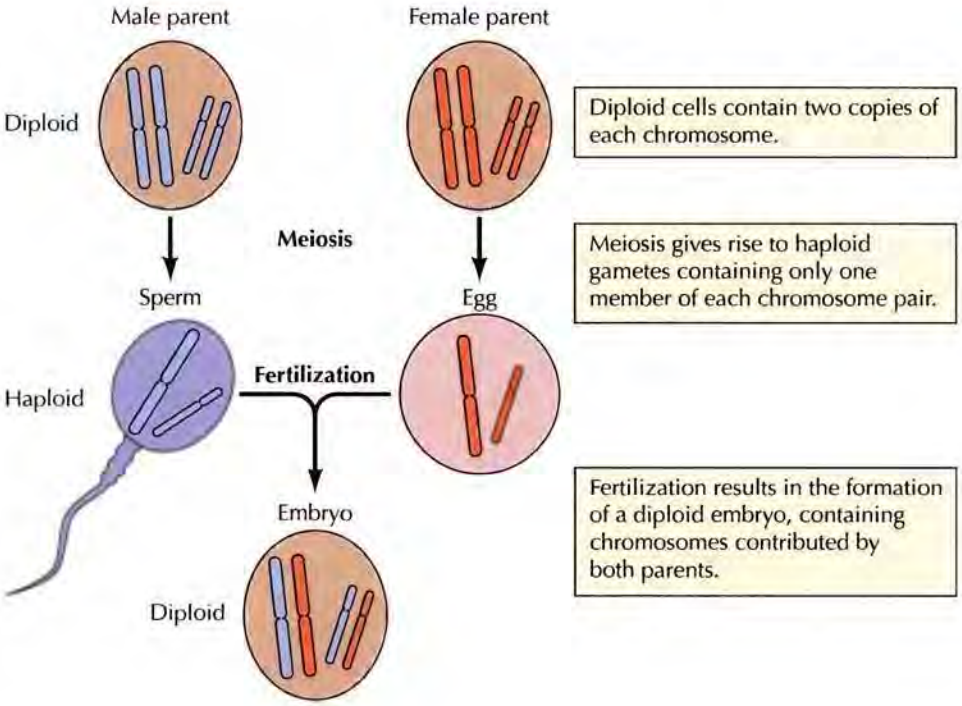


FIGURE 4.2 Chromosomes at meiosis and fertilization
Two chromosome pairs of a hypothetical organism are illustrated.

animals are **diploid**—containing two copies of each chromosome. Formation of the germ cells (the sperm and egg), however, involves a unique type of cell division (**meiosis**) in which only one member of each chromosome pair is transmitted to each progeny cell (Figure 4.2). Consequently, the sperm and egg are **haploid**, containing only one copy of each chromosome. The union of these two haploid cells at fertilization creates a new diploid organism, now containing one member of each chromosome pair derived from the male and one from the female parent. The behavior of chromosome pairs thus parallels that of genes, leading to the conclusion that genes are carried on chromosomes.

The fundamentals of mutation, genetic linkage, and the relationships between genes and chromosomes were largely established by experiments performed with the fruit fly, *Drosophila melanogaster*. *Drosophila* can be easily maintained in the laboratory, and they reproduce about every two weeks, which is a considerable advantage for genetic experiments. Indeed, these features continue to make *Drosophila* an organism of choice for genetic studies of animals, particularly the genetic analysis of development and differentiation.

In the early 1900s, a number of genetic alterations (**mutations**) were identified in *Drosophila*, usually affecting readily observable characteristics, such as eye color or wing shape. Breeding experiments indicated that some of the genes governing these traits are inherited independently of each other, suggesting that these genes are located on different chromosomes that segregate independently during meiosis (Figure 4.3). Other genes, however, are frequently inherited together as paired characteristics. Such genes are said to be linked to each other by virtue of being located on the same chromosome. The number of groups of linked genes is the same as the number of chromosomes (four in *Drosophila*), supporting the idea that chromosomes are carriers of the genes. By 1915, nearly a hundred genes had been defined and mapped onto the four chromosomes of *Drosophila*, leading to general acceptance of the chromosomal basis of heredity.

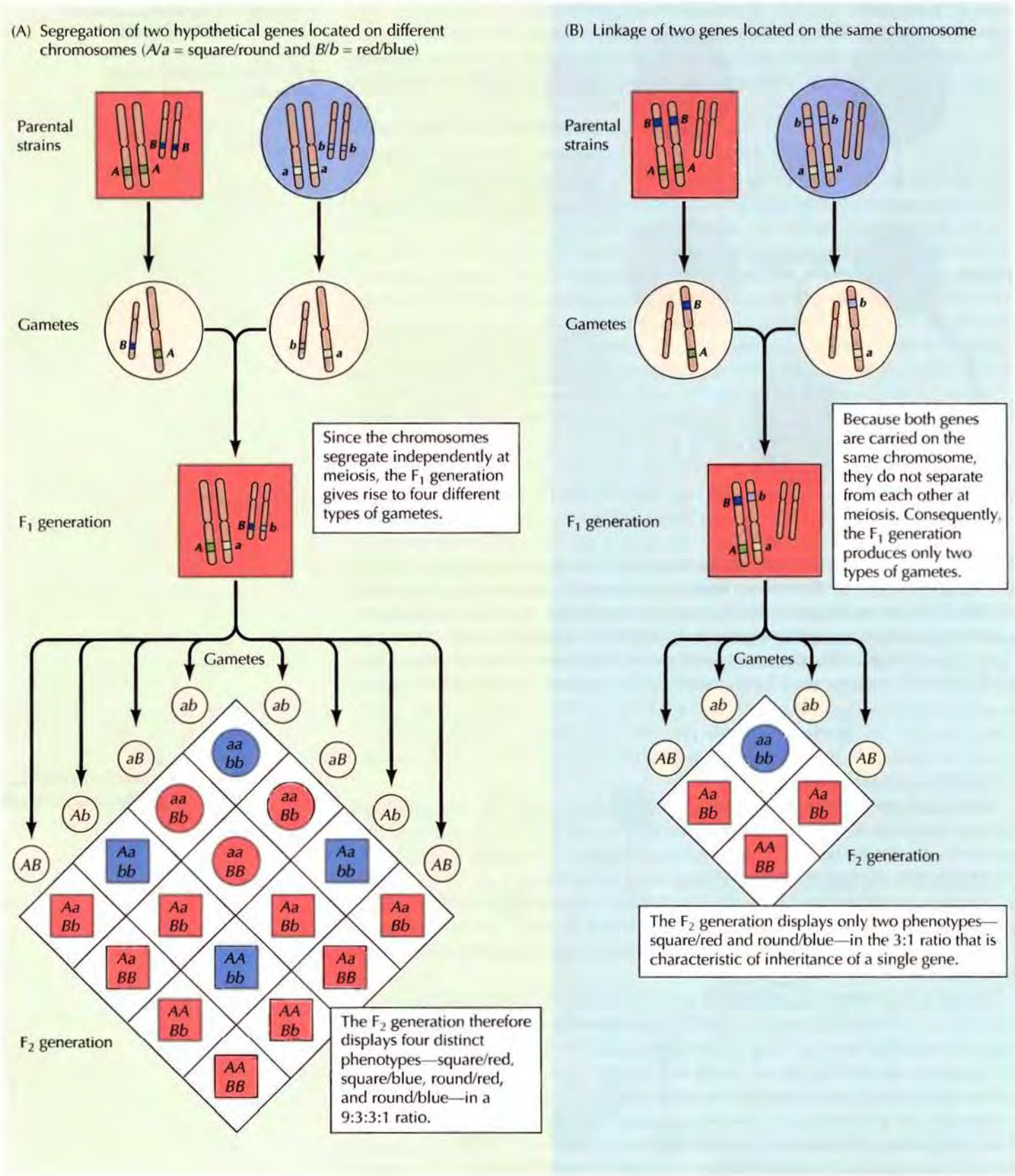


FIGURE 4.3 Gene segregation and linkage (A) Segregation of two hypothetical genes for shape (A/a = square/round) and color (B/b = red/blue) located on different chromosomes. (B) Linkage of two genes located on the same chromosome.

Genes and Enzymes

Early genetic studies focused on the identification and chromosomal localization of genes that control readily observable characteristics, such as the eye color of *Drosophila*. How these genes lead to the observed phenotypes, however, was unclear. The first insight into the relationship between genes and enzymes came in 1909, when it was realized that the inherited human disease phenylketonuria (see Molecular Medicine in Chapter 3) results from a genetic defect in metabolism of the amino acid phenylalanine. This defect was hypothesized to result from a deficiency in the enzyme needed to catalyze the relevant metabolic reaction, leading to the general suggestion that genes specify the synthesis of enzymes.

Clearer evidence linking genes with the synthesis of enzymes came from experiments of George Beadle and Edward Tatum, performed in 1941 with the fungus *Neurospora crassa*. In the laboratory, *Neurospora* can be grown on minimal or rich media similar to those discussed in Chapter 1 for the growth of *E. coli*. For *Neurospora*, minimal media consist only of salts, glucose, and biotin; rich media are supplemented with amino acids, vitamins, purines, and pyrimidines. Beadle and Tatum isolated mutants of *Neurospora* that grew normally on rich media but could not grow on minimal media. Each mutant was found to require a specific nutritional supplement, such as a particular amino acid, for growth. Furthermore, the requirement for a specific nutritional supplement correlated with the failure of the mutant to synthesize that particular compound. Thus each mutation resulted in a deficiency in a specific metabolic pathway. Since such metabolic pathways were known to be governed by enzymes, the conclusion from these experiments was that each gene specified the structure of a single enzyme—the **one gene–one enzyme hypothesis**. Many enzymes are now known to consist of multiple polypeptides, so the currently accepted statement of this hypothesis is that each gene specifies the structure of a single polypeptide chain.

Identification of DNA as the Genetic Material

Understanding the chromosomal basis of heredity and the relationship between genes and enzymes did not in itself provide a molecular explanation of the gene. Chromosomes contain proteins as well as DNA, and it was initially thought that genes were proteins. The first evidence leading to the identification of DNA as the genetic material came from studies in bacteria. These experiments represent a prototype for current approaches to defining the function of genes by introducing new DNA sequences into cells, as discussed later in this chapter.

The experiments that defined the role of DNA were derived from studies of the bacterium that causes pneumonia (*Pneumococcus*). Virulent strains of *Pneumococcus* are surrounded by a polysaccharide capsule that protects the bacteria from attack by the immune system of the host. Because the capsule gives bacterial colonies a smooth appearance in culture, encapsulated strains are denoted S. Mutant strains that have lost the ability to make a capsule (denoted R) form rough-edged colonies in culture and are no longer lethal when inoculated into mice. In 1928 it was observed that mice inoculated with nonencapsulated (R) bacteria plus heat-killed encapsulated (S) bacteria developed pneumonia and died. Importantly, the bacteria that were then isolated from these mice were of the S type. Subsequent experiments showed that a cell-free extract of S bacteria was similarly capable of converting (or transforming) R bacteria to the S state. Thus a substance in the S extract (called the transforming principle) was responsible for inducing the genetic **transformation** of R to S bacteria.

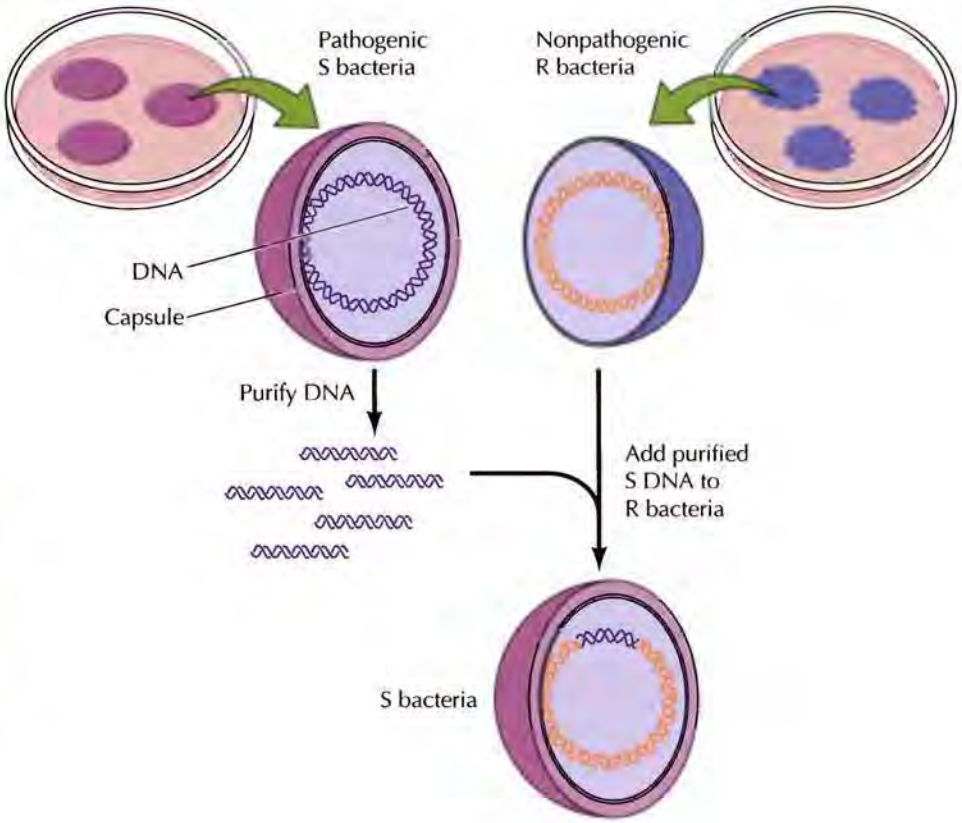


FIGURE 4.4 Transfer of genetic information by DNA DNA is extracted from a pathogenic strain of *Pneumococcus*, which is surrounded by a capsule and forms smooth colonies (S). Addition of the purified S DNA to a culture of nonpathogenic, nonencapsulated bacteria (R for "rough" colonies) results in the formation of S colonies. The purified DNA therefore contains the genetic information responsible for transformation of R to S bacteria.

4.2 WEBSITE ANIMATION

Bacterial Transformation

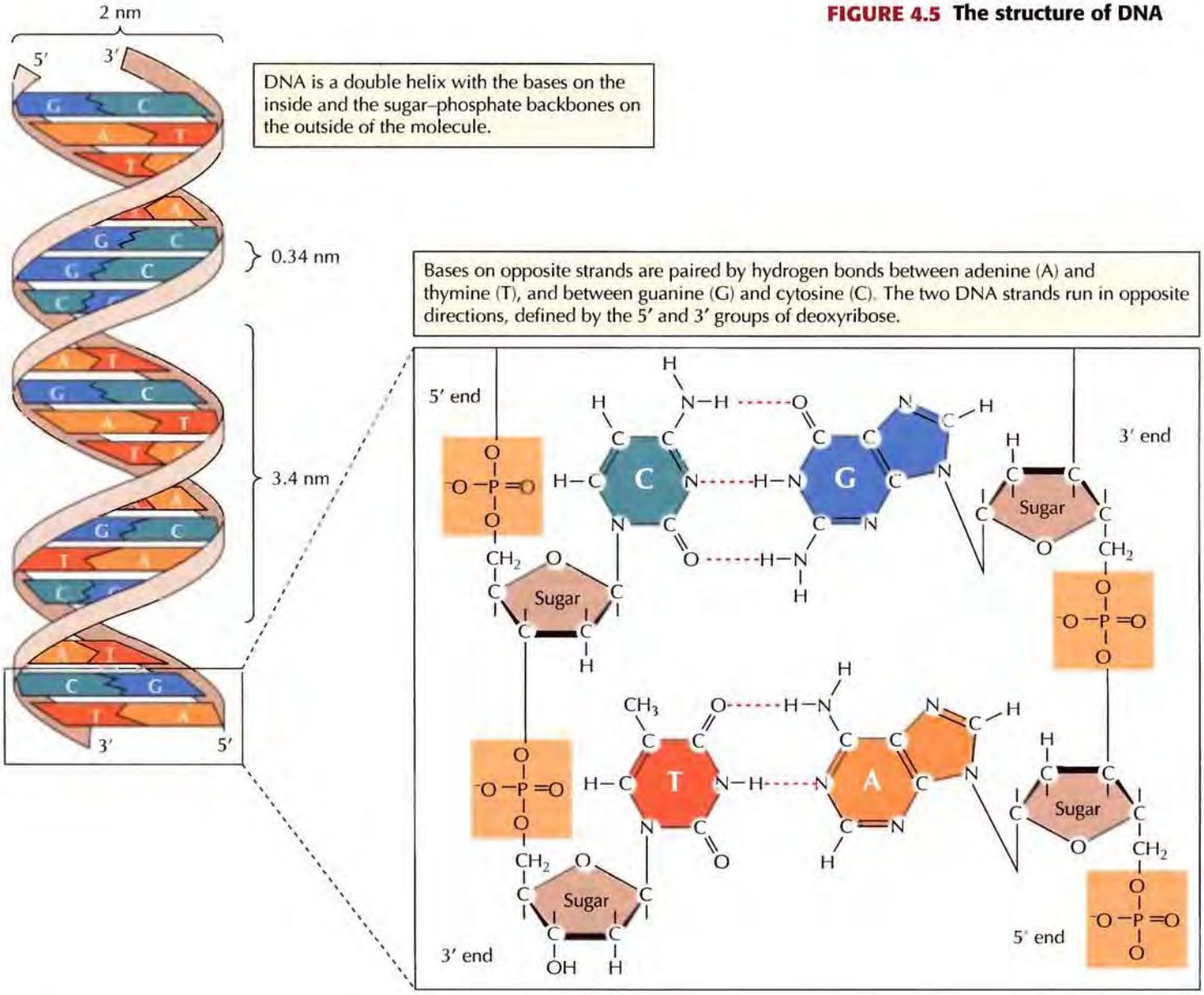
A nonpathogenic strain of *Pneumococcus* can be transformed into a pathogenic strain by taking up DNA fragments from the pathogenic strain and incorporating them into its chromosome.

In 1944 Oswald Avery, Colin MacLeod, and Maclyn McCarty established that the transforming principle was DNA, both by purifying it from bacterial extracts and by demonstrating that the activity of the transforming principle is abolished by enzymatic digestion of DNA but not by digestion of proteins (Figure 4.4). Although these studies did not immediately lead to the acceptance of DNA as the genetic material, they were extended within a few years by experiments with bacterial viruses. In particular, it was shown that, when a bacterial virus infects a cell, the viral DNA rather than the viral protein must enter the cell in order for the virus to replicate. Moreover, the parental viral DNA (but not the protein) is transmitted to progeny virus particles. The concurrence of these results with continuing studies of the activity of DNA in bacterial transformation led to acceptance of the idea that DNA is the genetic material.

The Structure of DNA

Our understanding of the three-dimensional structure of DNA, deduced in 1953 by James Watson and Francis Crick, has been the basis for present-day molecular biology. At the time of Watson and Crick's work, DNA was known to be a polymer composed of four nucleic acid bases—two purines (adenine [A] and guanine [G]) and two pyrimidines (cytosine [C] and thymine [T])—linked to phosphorylated sugars. Given the central role of DNA as the genetic material, elucidation of its three-dimensional structure appeared critical to understanding its function. Watson and Crick's consideration of the problem was heavily influenced by Linus Pauling's description of hydrogen bonding and the α helix, a common element of the secondary structure of proteins (see Chapter 2). Moreover, experimental data on the structure of DNA were available from X-ray crystallography studies by Maurice Wilkins and Rosalind Franklin. Analysis of these data revealed

FIGURE 4.5 The structure of DNA



that the DNA molecule is a helix that turns every 3.4 nm. In addition, the data showed that the distance between adjacent bases is 0.34 nm, so there are ten bases per turn of the helix. An important finding was that the diameter of the helix is approximately 2 nm, suggesting that it is composed of not one but two DNA chains.

From these data, Watson and Crick built their model of DNA (Figure 4.5). The central features of the model are that DNA is a double helix with the sugar–phosphate backbones on the outside of the molecule. The bases are on the inside, oriented such that hydrogen bonds are formed between purines and pyrimidines on opposite chains. The base pairing is very specific: A always pairs with T and G with C. This specificity accounts for the earlier results of Erwin Chargaff, who had analyzed the base composition of various DNAs and found that the amount of adenine was always equal to that of thymine, and the amount of guanine to that of cytosine. Because of this specific base pairing, the two strands of a DNA molecule are complementary: Each strand contains all the information required to specify the sequences of bases on the other.

■ **The structure of DNA described by Watson and Crick was a remarkable achievement, but not perfect. In their initial paper, they assumed that both A-T and G-C were paired by two hydrogen bonds, missing the third hydrogen bond in G-C base pairs.**

Replication of DNA

The discovery of complementary base pairing between DNA strands immediately suggested a molecular solution to the question of how the genetic material could direct its own replication—a process that is required each time a cell divides. It was proposed that the two strands of a DNA molecule could separate and serve as templates for synthesis of new complementary strands, the sequence of which would be dictated by the specificity of base pairing (Figure 4.6). The process is called **semiconservative replication** because one strand of parental DNA is conserved in each progeny DNA molecule.

Direct support for semiconservative DNA replication was obtained in 1958 as a result of elegant experiments, performed by Matthew Meselson and Frank Stahl, in which DNA was labeled with isotopes that altered its density (Figure 4.7). *E. coli* were first grown in media containing the heavy isotope of nitrogen (^{15}N) in place of the normal light isotope (^{14}N). The DNA of these bacteria consequently contained ^{15}N and was heavier than that of bacteria grown in ^{14}N. Such heavy DNA could be separated from DNA containing ^{14}N by equilibrium centrifugation in a density gradient of CsCl. This ability to separate heavy (^{15}N) DNA from light (^{14}N) DNA enabled the study of DNA synthesis. *E. coli* that had been grown in ^{15}N were transferred to media containing ^{14}N and allowed to replicate one more time. Their DNA was then extracted and analyzed by CsCl density gradient centrifugation. The results of this analysis indicated that all of the heavy DNA had been replaced by newly synthesized DNA with a density intermediate between that of heavy (^{15}N) and that of light (^{14}N) DNA molecules. The implication was that during replication, the two parental strands of heavy DNA separated and served as templates for newly synthesized progeny strands of light DNA, yielding double-stranded molecules of intermediate density. This experiment thus provided direct evidence for semiconservative DNA replication, clearly underscoring the importance of complementary base pairing between strands of the double helix.

The ability of DNA to serve as a template for its own replication was further established with the demonstration that an enzyme purified from *E. coli* (**DNA polymerase**) could catalyze DNA replication *in vitro*. In the presence of DNA to act as a template, DNA polymerase was able to direct the incorporation of nucleotides into a complementary DNA molecule.

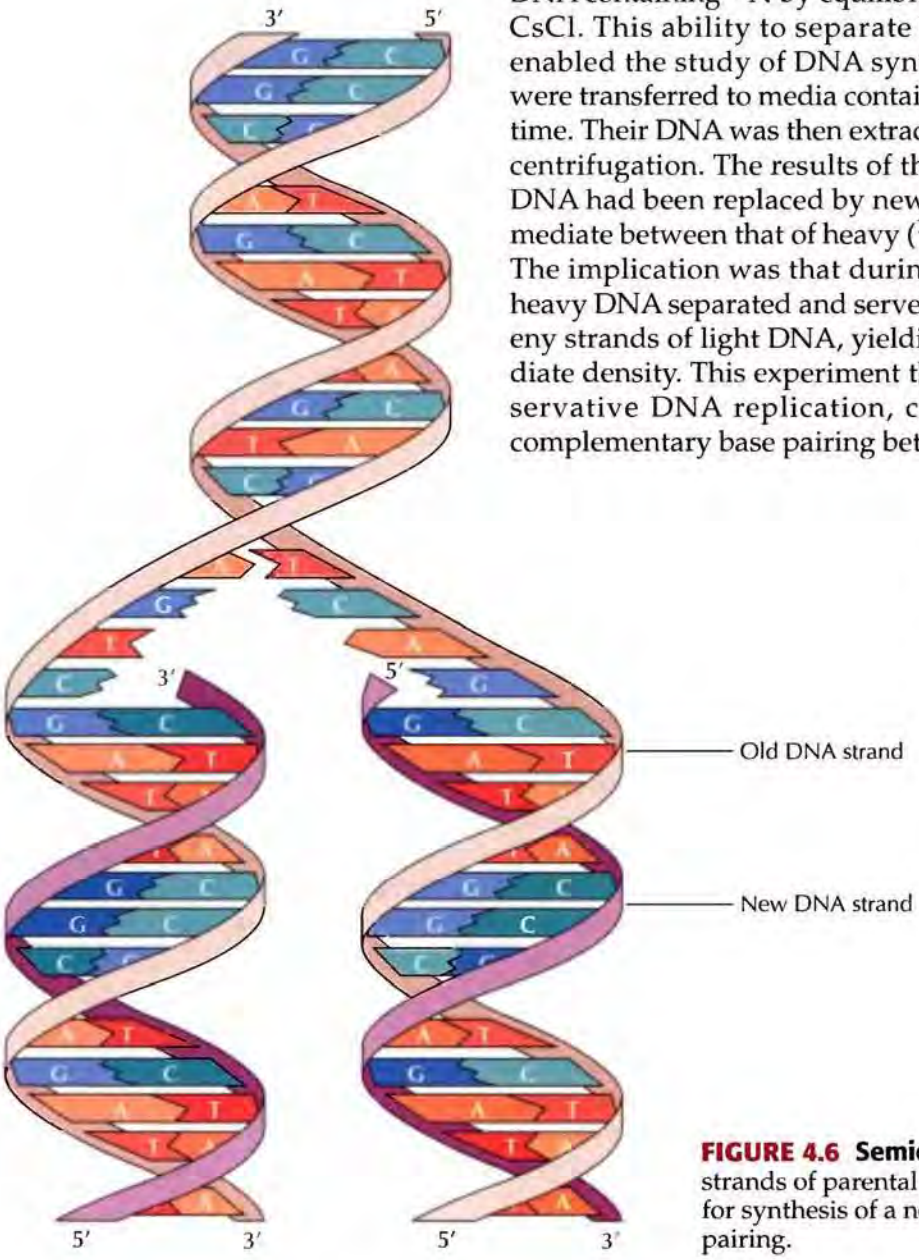


FIGURE 4.6 Semiconservative replication of DNA The two strands of parental DNA separate, and each serves as a template for synthesis of a new daughter strand by complementary base pairing.

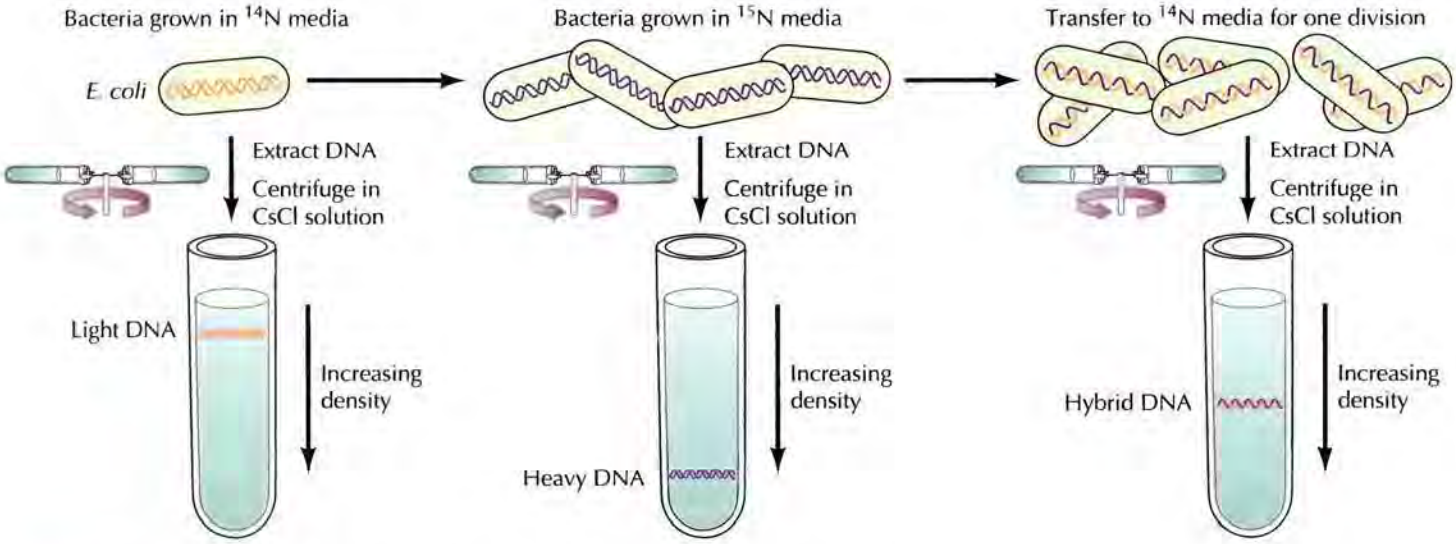


FIGURE 4.7 Experimental demonstration of semiconservative replication
Bacteria grown in medium containing the normal isotope of nitrogen (^{14}N) are transferred into medium containing the heavy isotope (^{15}N) and grown in this medium for several generations. They are then transferred back to medium containing ^{14}N and grown for one additional generation. DNA is extracted from these bacteria and analyzed by equilibrium ultracentrifugation in a CsCl solution. The CsCl sediments to form a density gradient, and the DNA molecules band at a position where their density is equal to that of the CsCl solution. DNA of the bacteria transferred from ^{15}N to ^{14}N medium for a single generation bands at a density intermediate between that of ^{15}N DNA and that of ^{14}N DNA, indicating that it represents a hybrid molecule with one heavy and one light strand.

Expression of Genetic Information

Genes act by determining the structure of proteins, which are responsible for directing cell metabolism through their activity as enzymes. The identification of DNA as the genetic material and the elucidation of its structure revealed that genetic information must be specified by the order of the four bases (A, C, G, and T) that make up the DNA molecule. Proteins, in turn, are polymers of 20 amino acids, the sequence of which determines their structure and function. The first direct link between a genetic mutation and an alteration in the amino acid sequence of a protein was made in 1957, when it was found that patients with the inherited disease sickle-cell anemia had hemoglobin molecules that differed from normal ones by a single amino acid substitution. Deeper understanding of the molecular relationship between DNA and proteins came, however, from a series of experiments that took advantage of *E. coli* and its viruses as genetic models.

Colinearity of Genes and Proteins

The simplest hypothesis to account for the relationship between genes and enzymes was that the order of nucleotides in DNA specified the order of amino acids in a protein. Mutations in a gene would correspond to alterations in the sequence of DNA, which might result from the substitution of one nucleotide for another or from the addition or deletion of nucleotides. These changes in the nucleotide sequence of DNA would then lead to corresponding changes in the amino acid sequence of the protein encoded by the gene in question. This hypothesis predicted that different mutations within a single gene could alter different amino acids in the encoded protein, and

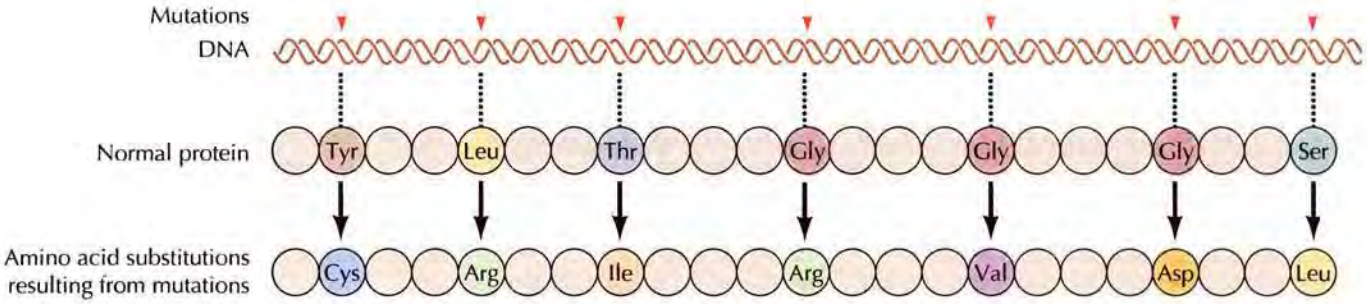


FIGURE 4.8 Colinearity of genes and proteins A series of mutations (arrowheads) were mapped in the *E. coli* gene encoding tryptophan synthetase (top line). The amino acid substitutions resulting from each of the mutations were then determined by sequence analysis of the proteins of mutant bacteria (bottom line). These studies revealed that the order of mutations in DNA was the same as the order of amino acid substitutions in the encoded protein.

FIGURE 4.9 Synthesis of RNA from DNA The two strands of DNA unwind, and one is used as a template for synthesis of a complementary strand of RNA.

that the positions of mutations in a gene should reflect the positions of amino acid alterations in its protein product.

The rapid replication and the simplicity of the genetic system of *E. coli* were of major help in addressing these questions. A variety of mutants of *E. coli* could be isolated, including nutritional mutants that (like the *Neurospora* mutants discussed earlier) require particular amino acids for growth. Importantly, the rapid growth of *E. coli* made feasible the isolation and mapping of multiple mutants in a single gene, leading to the first demonstration of the linear relationship between genes and proteins. In these studies, Charles Yanofsky and his colleagues mapped a series of mutations in the gene that encodes an enzyme required for synthesis of the amino acid tryptophan. Analysis of the enzymes encoded by the mutant genes indicated that the relative positions of the amino acid alterations were the same as those of the corresponding mutations (Figure 4.8). Thus the sequence of amino acids in the protein was colinear with that of mutations in the gene, as expected if the order of nucleotides in DNA specifies the order of amino acids in proteins.

The Role of Messenger RNA

Although the sequence of nucleotides in DNA appeared to specify the order of amino acids in proteins, it did not necessarily follow that DNA itself directs protein synthesis. Indeed, this appeared not to be the case, since DNA is located in the nucleus of eukaryotic cells, whereas protein synthesis takes place in the cytoplasm. Some other molecule was therefore needed to convey genetic information from DNA to the sites of protein synthesis (the ribosomes).

RNA appeared a likely candidate for such an intermediate because the similarity of its structure to that of DNA suggested that RNA could be synthesized from a DNA template (Figure 4.9). RNA differs from DNA in that it is single-stranded rather than double-stranded, its sugar component is ribose instead of deoxyribose, and it contains the pyrimidine base uracil (U) instead of thymine (T) (see Figure 2.10). However, neither the change in sugar nor the substitution of U for T alters base pairing, so the synthesis of RNA can be readily directed by a DNA template. Moreover, since RNA is located primarily in the cytoplasm, it appeared a logical intermediate to convey information from DNA to the ribosomes. These characteristics of

RNA suggested a pathway for the flow of genetic information that is known as the **central dogma** of molecular biology:

$$\text{DNA} \rightarrow \text{RNA} \rightarrow \text{Protein}$$

According to this concept, RNA molecules are synthesized from DNA templates (a process called **transcription**), and proteins are synthesized from RNA templates (a process called **translation**).

Experimental evidence for the RNA intermediates postulated by the central dogma was obtained by Sidney Brenner, Francois Jacob, and Matthew Meselson in studies of *E. coli* infected with the bacteriophage T4. The synthesis of *E. coli* RNA stops following infection by T4, and the only new RNA synthesized in infected bacteria is transcribed from T4 DNA. This T4 RNA becomes associated with bacterial ribosomes, thus conveying the information from DNA to the site of protein synthesis. Because of their role as intermediates in the flow of genetic information, RNA molecules that serve as templates for protein synthesis are called **messenger RNAs (mRNAs)**. They are transcribed by an enzyme (**RNA polymerase**) that catalyzes the synthesis of RNA from a DNA template.

In addition to mRNA, two other types of RNA molecules are important in protein synthesis. **Ribosomal RNA (rRNA)** is a component of ribosomes, and **transfer RNAs (tRNAs)** serve as adaptor molecules that align amino acids along the mRNA template. The structures and functions of these molecules are discussed in the following section and in more detail in Chapters 7 and 8.

The Genetic Code

How is the nucleotide sequence of mRNA translated into the amino acid sequence of a protein? In this step of gene expression, genetic information is transferred between chemically unrelated types of macromolecules—nucleic acids and proteins—raising two new types of problems in understanding the action of genes.

First, since amino acids are structurally unrelated to the nucleic acid bases, direct complementary pairing between mRNA and amino acids during the incorporation of amino acids into proteins seemed impossible. How then could amino acids align on an mRNA template during protein synthesis? This question was solved by the discovery that tRNAs serve as adaptors between amino acids and mRNA during translation (Figure 4.10). Prior to its use in protein synthesis, each amino acid is attached by a specific enzyme to its appropriate tRNA. Base pairing between a recognition sequence on each tRNA and a complementary sequence on the mRNA then directs the attached amino acid to its correct position on the mRNA template.

The second problem in the translation of nucleotide sequence to amino acid sequence was determination of the **genetic code**. How could the information contained in the sequence of four different nucleotides be converted to the sequences of 20 different amino acids in proteins? Because 20 amino acids must be specified by only four nucleotides, at least three nucleotides must be used to encode each amino acid. Used singly, four nucleotides could encode only four amino acids and, used in pairs, four nucleotides could encode only sixteen (4^2) amino acids. Used as triplets, however, four nucleotides could encode 64 (4^3) different amino acids—more than enough to account for the 20 amino acids actually found in proteins.

Direct experimental evidence for the triplet code was obtained by studies of bacteriophage T4 bearing mutations in an extensively studied gene called *rII*. Phages with mutations in this gene form abnormally large

4.3 WEBSITE ANIMATION

The "Central Dogma"

The central dogma of molecular biology, as first stated by Francis Crick, asserts that information flows from DNA to RNA, and then from RNA to protein.

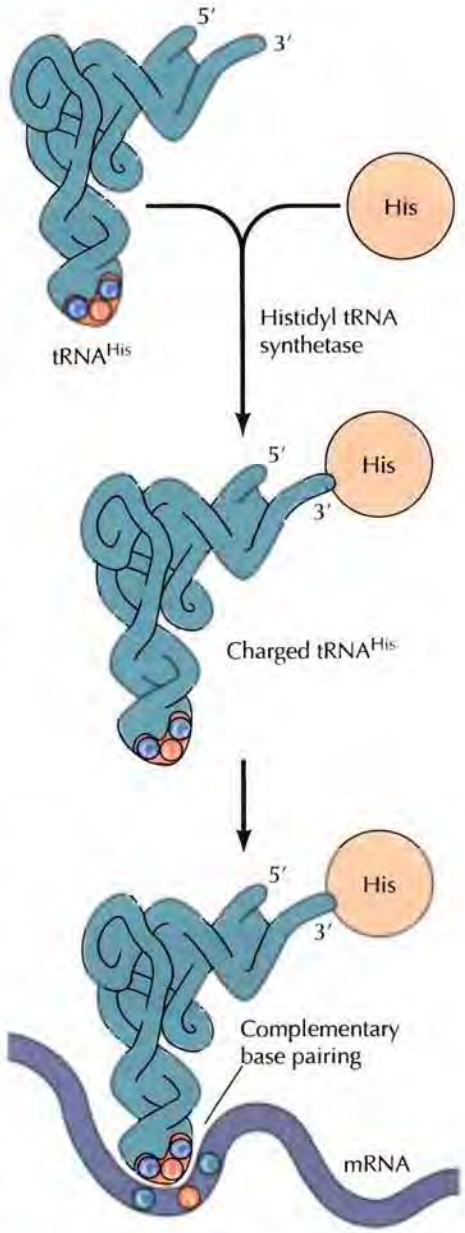


FIGURE 4.10 Function of transfer RNA Transfer RNA serves as an adaptor during protein synthesis. Each amino acid (e.g., histidine) is attached to the 3′ end of a specific tRNA by an appropriate enzyme (an aminoacyl tRNA synthetase). The charged tRNAs then align on an mRNA template by complementary base pairing.

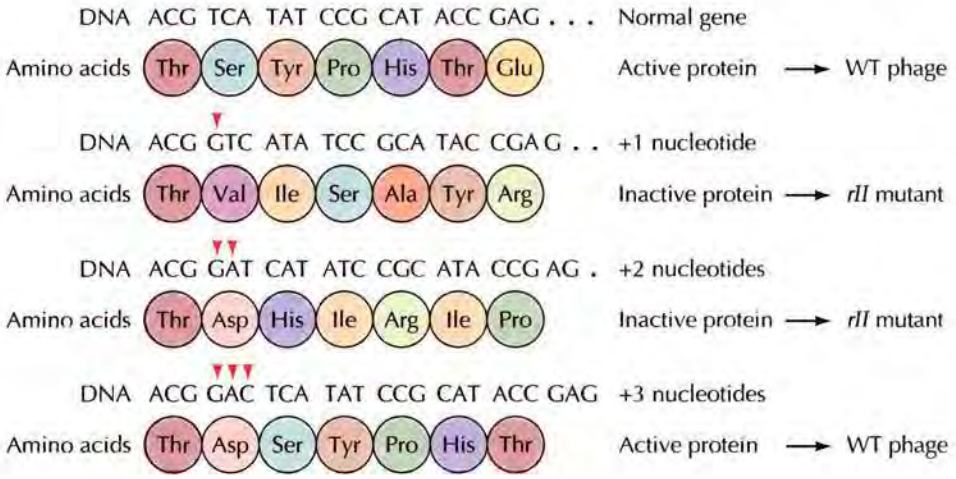


FIGURE 4.11 Genetic evidence for a triplet code A series of mutations consisting of additions of one, two, or three nucleotides were studied in the *rII* gene of bacteriophage T4. Additions of one or two nucleotides alter the reading frame of the remainder of the gene. Therefore all the subsequent amino acids are abnormal, and an inactive protein is produced, giving rise to mutant phage. Additions of three nucleotides, however, alter only a single amino acid. The reading frame of the remainder of the gene is normal, and an active protein giving rise to wild-type (WT) phage is produced.

4.4 WEBSITE ANIMATION

DNA Mutations A single base-pair mutation in the coding region of a gene can result in a prematurely terminated polypeptide or in a polypeptide with an incorrect amino acid in its amino acid chain, depending on the particular mutation.

plaques, which can be clearly distinguished from those formed by wild-type phages. Hence isolating and mapping a number of *rII* mutants was easy and led to the establishment of a detailed genetic map of this locus. Study of recombinants between *rII* mutants that had arisen by additions or deletions of nucleotides revealed that phages containing additions or deletions of one or two nucleotides always exhibited the mutant phenotype. Phages containing additions or deletions of three nucleotides, however, were frequently wild-type in function (Figure 4.11). These findings suggested that the gene is read in groups of three nucleotides, starting from a fixed point. Additions or deletions of one or two nucleotides would then alter the reading frame of the entire gene, leading to the coding of abnormal amino acids throughout the encoded protein. In contrast, additions or deletions of three nucleotides would lead to the addition or deletion of only a single amino acid; the rest of the amino acid sequence would remain unaltered, frequently yielding an active protein.

Deciphering the genetic code thus became a problem of assigning nucleotide triplets to their corresponding amino acids. This problem was approached using *in vitro* systems that could carry out protein synthesis (***in vitro* translation**). Cell extracts containing ribosomes, amino acids, tRNAs, and the enzymes responsible for attaching amino acids to the appropriate tRNAs (aminoacyl-tRNA synthetases) were known to catalyze the incorporation of amino acids into proteins. However, such protein synthesis depends on the presence of mRNA bound to the ribosomes, and can be greatly enhanced by the addition of purified mRNA. Since added mRNA directs protein synthesis in such systems, the genetic code could be deciphered by study of the translation of synthetic mRNAs of known base sequence.

The first such experiment, performed by Marshall Nirenberg and Heinrich Matthaei, involved the *in vitro* translation of a synthetic RNA polymer containing only uracil (Figure 4.12). This poly-U template was found to direct the incorporation of only a single amino acid—phenylalanine—into a polypeptide consisting of repeated phenylalanine residues. Therefore the triplet UUU encodes the amino acid phenylalanine. Similar experiments with RNA polymers containing only single nucleotides established that AAA encodes lysine and CCC encodes proline. The remainder of the code was deciphered using RNA polymers containing mixtures of nucleotides, leading to the coding assignment of all 64 possible triplets (called **codons**) (Table 4.1). Of the 64 codons, 61 specify an amino acid; the remaining three

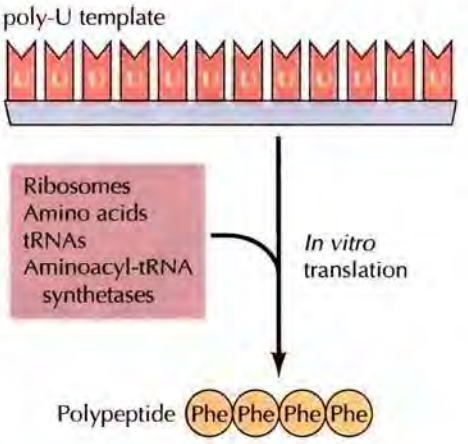


FIGURE 4.12 The triplet UUU encodes phenylalanine *In vitro* translation of a synthetic RNA consisting of repeated uracils (a poly-U template) results in the synthesis of a polypeptide containing only phenylalanine.

TABLE 4.1 The Genetic Code

First position	Second position U	C	A	G	Third position
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	stop	stop	A
	Leu	Ser	stop	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

(UAA, UAG, and UGA) are stop codons that signal the termination of protein synthesis. The code is degenerate; that is, many amino acids are specified by more than one codon. With few exceptions (discussed in Chapter 11), all organisms utilize the same genetic code, providing strong support for the conclusion that all present-day cells evolved from a common ancestor.

RNA Viruses and Reverse Transcription

With the elucidation of the genetic code, the fundamental principles of the molecular biology of cells appeared to have been established. According to the central dogma, the genetic material consists of DNA, which is capable of self-replication as well as being transcribed into mRNA, which serves in turn as the template for protein synthesis. However, as noted in Chapter 1, many viruses contain RNA rather than DNA as their genetic material, implying the use of other modes of information transfer.

RNA genomes were first discovered in plant viruses, many of which were found to be composed of only RNA and protein. Direct proof that RNA acts as the genetic material of these viruses was obtained in the 1950s by experiments demonstrating that RNA purified from tobacco mosaic virus could infect new host cells, giving rise to infectious progeny virus. The mode of replication of most viral RNA genomes was subsequently determined by studies of the RNA bacteriophages of *E. coli*. These viruses were found to encode a specific enzyme that could catalyze the synthesis of RNA from an RNA template (RNA-directed RNA synthesis), using the same mechanism of base pairing between complementary strands as is employed during DNA replication or transcription of RNA from DNA.

Although most animal viruses, such as poliovirus or influenza virus, were found to replicate by RNA-directed RNA synthesis, this mechanism did not appear to account for the replication of one family of animal viruses (the RNA tumor viruses), which were of particular interest because of their

KEY EXPERIMENT

The DNA Provirus Hypothesis

Nature of the Provirus of Rous Sarcoma
Howard M. Temin
McArdle Laboratory, University of Wisconsin, Madison, WI
National Cancer Institute Monographs, Volume 17, 1964, pages 557–570

Howard M. Temin

The Context

Rous sarcoma virus (RSV), the first cancer-causing virus to be described, was of considerable interest as an experimental system for studying the molecular biology of cancer. Howard Temin began his research in this area when, as a graduate student in 1958, he developed the first assay for the transformation of normal cells to cancer cells in culture following infection with RSV. The availability of such a quantitative *in vitro* assay provided the tool needed for further studies of both cell transformation and virus replication. As Temin proceeded with these studies, he made a series of unexpected observations indicating that the replication of RSV was fundamentally different from that of other RNA viruses. These experiments led to Temin's proposal of the DNA provirus hypothesis, which stated that the viral RNA was copied into DNA in infected cells—a proposal that ran directly counter to the universally accepted central dogma of molecular biology.

The Experiments

The DNA provirus hypothesis was based on several different types of experimental evidence. First, studies of cell transformation using mutants of RSV indicated that important characteristics of transformed cells were determined by genetic information of the virus. This information was regularly transmitted to daughter cells following cell division, even in the absence of virus replication. Temin therefore proposed that the viral genome was present in infected cells in a stably inherited form, which he called a provirus.

Evidence that the provirus was DNA was then derived from experiments with metabolic inhibitors. First, actinomycin D, which inhibits the synthesis of RNA from a DNA template, was found to inhibit virus production by RSV-infected cells (see figure). Second, inhibitors of DNA synthesis inhibited early stages of cell infection by RSV. Thus, DNA synthesis appeared to be required early in infection, and DNA-directed RNA synthesis appeared to be needed subsequently for the production of progeny viruses, leading to the proposal that the provirus was a DNA copy of the viral RNA genome. Temin sought further evidence for this proposal by using nucleic acid hybridization to detect viral sequences in infected cell DNA, but the sensitivity of the available techniques was limited and the data were unconvincing.

The Impact

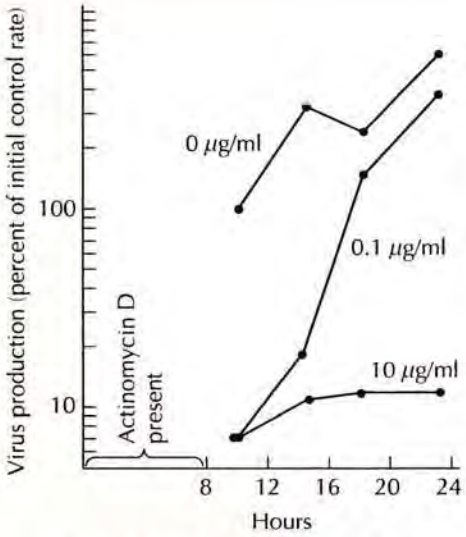
The DNA provirus hypothesis was thus proposed principally on the basis of genetic experiments and the effects of metabolic inhibitors. It was a radical proposal, which contradicted the accepted central dogma of molecular biology. In this setting, Temin's ability to cause cancer in infected animals. Although these viruses contain genomic RNA in their viral particles, experiments performed by Howard Temin in the early 1960s indicated that their replication requires DNA synthesis in infected cells, leading to the hypothesis that the RNA tumor viruses (now called **retroviruses**) replicate via synthesis of a DNA intermediate, called a DNA provirus (Figure 4.13). This hypothesis was initially met with widespread disbelief because it involves RNA-directed synthesis of DNA—a reversal of the central dogma. In 1970, however, Temin and David Baltimore independently discovered that the RNA tumor viruses contain a novel enzyme that catalyzes the synthesis of DNA from an RNA template. In addition, clear-cut evidence for the existence of viral DNA sequences in infected cells was obtained. The synthesis of DNA from RNA, now called

4.5 WEBSITE ANIMATION
HIV Reproduction
As part of a replication cycle within a host cell, a retrovirus uses the enzyme reverse transcriptase to copy its RNA genome into DNA.

KEY EXPERIMENT

hypothesis that RSV replicated by the transfer of information from RNA to DNA not only failed to win the acceptance of the scientific community, but was met with general derision. Nonetheless, Temin persevered through the 1960s, continuing with experiments to test his hypothesis and providing increasingly convincing evidence in its support. These efforts culminated in 1970 with the discovery by Temin and Satoshi Mizutani, and at the same time by David Baltimore, of a viral enzyme, now known as reverse transcriptase, that synthesizes DNA from an RNA template—an unambiguous biochemical demonstration that the central dogma could be reversed.

Temin concluded his 1970 paper with the statement that the results "constitute strong evidence that the DNA provirus hypothesis is correct and that RNA tumour viruses have a DNA genome when they are in cells and an RNA genome when they are in virions. This result would have strong implications for theories of viral carcinogenesis and, possibly, for theories of information transfer in other biological systems." As Temin predicted, the discovery of RNA-directed DNA synthesis has led to major advances in our understanding of cancer, human retroviruses, and gene rearrangements. Reverse transcriptase has further provided a critical tool for cDNA cloning, thereby impacting virtually all areas of contemporary cell and molecular biology.



Effect of actinomycin D on RSV replication. RSV-infected cells were cultured with the indicated concentrations of actinomycin D for 8 hours. Actinomycin D was then removed and the amount of virus produced was determined.

reverse transcription, was thus established as a mode of information transfer in biological systems.

Reverse transcription is important not only in the replication of retroviruses but also in at least two other broad aspects of molecular and cellular biology. First, reverse transcription is not restricted to retroviruses; it also occurs in cells and, as discussed in Chapters 5 and 6, is frequently responsible for the transposition of DNA sequences from one chromosomal location to another. Indeed, the sequence of the human genome has revealed that approximately 40% of human genomic DNA is derived from reverse transcription. Second, enzymes that catalyze RNA-directed DNA synthesis (**reverse transcriptases**) can be used experimentally to generate DNA copies of any RNA molecule. The use of reverse transcriptase has thus allowed mRNAs of eukaryotic cells to be studied using the molecular approaches that are currently applied to the manipulation of DNA, as discussed in the following section.

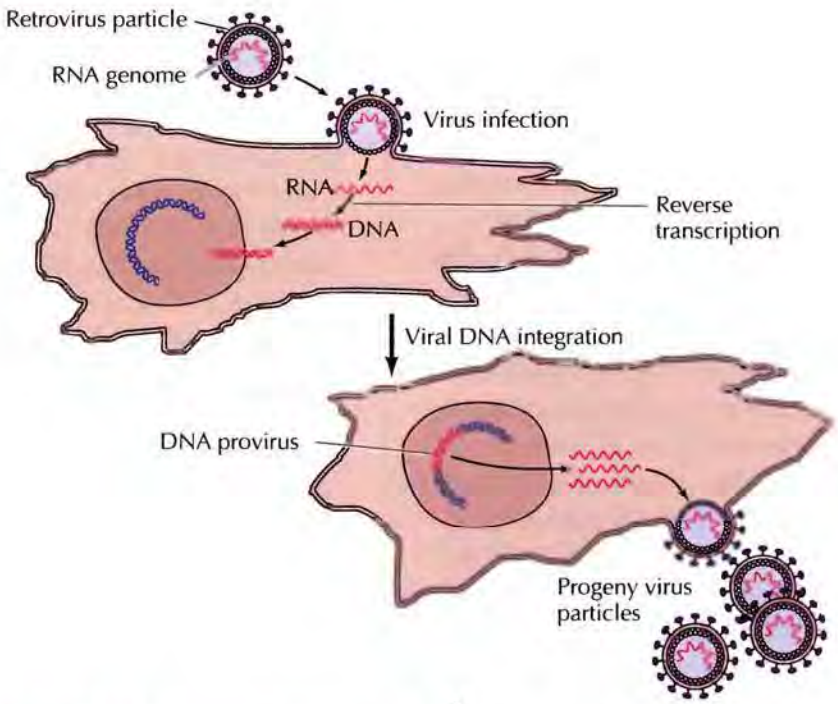


FIGURE 4.13 Reverse transcription and retrovirus replication Retroviruses contain RNA genomes in their viral particles. When a retrovirus infects a host cell, however, a DNA copy of the viral RNA is synthesized via reverse transcription. This viral DNA is then integrated into chromosomal DNA of the host to form a DNA provirus, which is transcribed to yield progeny virus RNA.

Recombinant DNA

Classical experiments in molecular biology were strikingly successful in developing our fundamental concepts of the nature and expression of genes. Since these studies were based primarily on genetic analysis, their success depended largely on the choice of simple, rapidly replicating organisms (such as bacteria and viruses) as models. It was not clear, however, how these fundamental principles could be extended to provide a molecular understanding of the complexities of eukaryotic cells, since the genomes of most eukaryotes (e.g., the human genome) are up to a thousand times more complex than that of *E. coli*. In the early 1970s, the possibility of studying such genomes at the molecular level seemed daunting. In particular, there appeared to be no way in which individual genes could be isolated and studied.

This obstacle to the progress of molecular biology was overcome by the development of recombinant DNA technology, which provided scientists with the ability to isolate, sequence, and manipulate individual genes derived from any type of cell. The application of recombinant DNA has thus enabled detailed molecular studies of the structure and function of eukaryotic genes and genomes, thereby revolutionizing our understanding of cell biology.

Restriction Endonucleases

The first step in the development of recombinant DNA technology was the characterization of **restriction endonucleases**—enzymes that cleave DNA at specific sequences. These enzymes were identified in bacteria, where they apparently provide a defense against the entry of foreign DNA (e.g., from a virus) into the cell. Bacteria have a variety of restriction endonucleases that cleave DNA at more than a hundred distinct recognition sites, each of which consists of a specific sequence of four to eight base pairs (examples are given in Table 4.2).

Since restriction endonucleases digest DNA at specific sequences, they can be used to cleave a DNA molecule at unique sites. For example, the restriction endonuclease *Eco*RI recognizes the six-base-pair sequence GAATTC. This sequence is present at five sites in DNA of the bacteriophage λ, so *Eco*RI digests λ DNA into six fragments ranging from 3.6 to 21.2 kilo-

4.6 WEBSITE ANIMATION

Restriction Endonucleases Restriction endonucleases cleave DNA at specific sequences, leaving staggered or blunt ends on the resulting DNA fragments.

■ **Why isn't the DNA of bacteria digested by their own restriction endonucleases? Restriction endonucleases do not work alone in bacteria. Instead, they usually work in conjunction with enzymes that modify the bacterial DNA so that it is no longer recognized by the restriction endonuclease. This restriction-modification system ensures that only unmodified foreign DNA (e.g., viral DNA) is digested by the restriction endonucleases expressed by any particular species of bacteria.**

TABLE 4.2 Recognition Sites of Representative Restriction Endonucleases

Enzyme[a]	Source	Recognition site[b]
*Bam*HI	*Bacillus amyloliquefaciens* H	GGATCC
*Eco*RI	*Escherichia coli* RY13	GAATTC
*Hae*III	*Haemophilus aegyptius*	GGCC
*Hind*III	*Haemophilus influenzae* Rd	AAGCTT
*Hpa*I	*Haemophilus parainfluenzae*	GTTAAC
*Hpa*II	*Haemophilus parainfluenzae*	CCGG
*Mbo*I	*Moraxella bovis*	GATC
*Not*I	*Nocardia otitidis-caviarum*	GCGGCCGC
*Sfi*I	*Streptomyces fimbriatus*	GGCCNNNNNGGCC
*Taq*I	*Thermus aquaticus*	TCGA

[a] Enzymes are named according to their species of isolation, followed by a number to distinguish different enzymes isolated from the same organism (e.g., *Hpa*I and *Hpa*II).

[b] Recognition sites show the sequence of only one strand of double-stranded DNA. "N" represents any base.

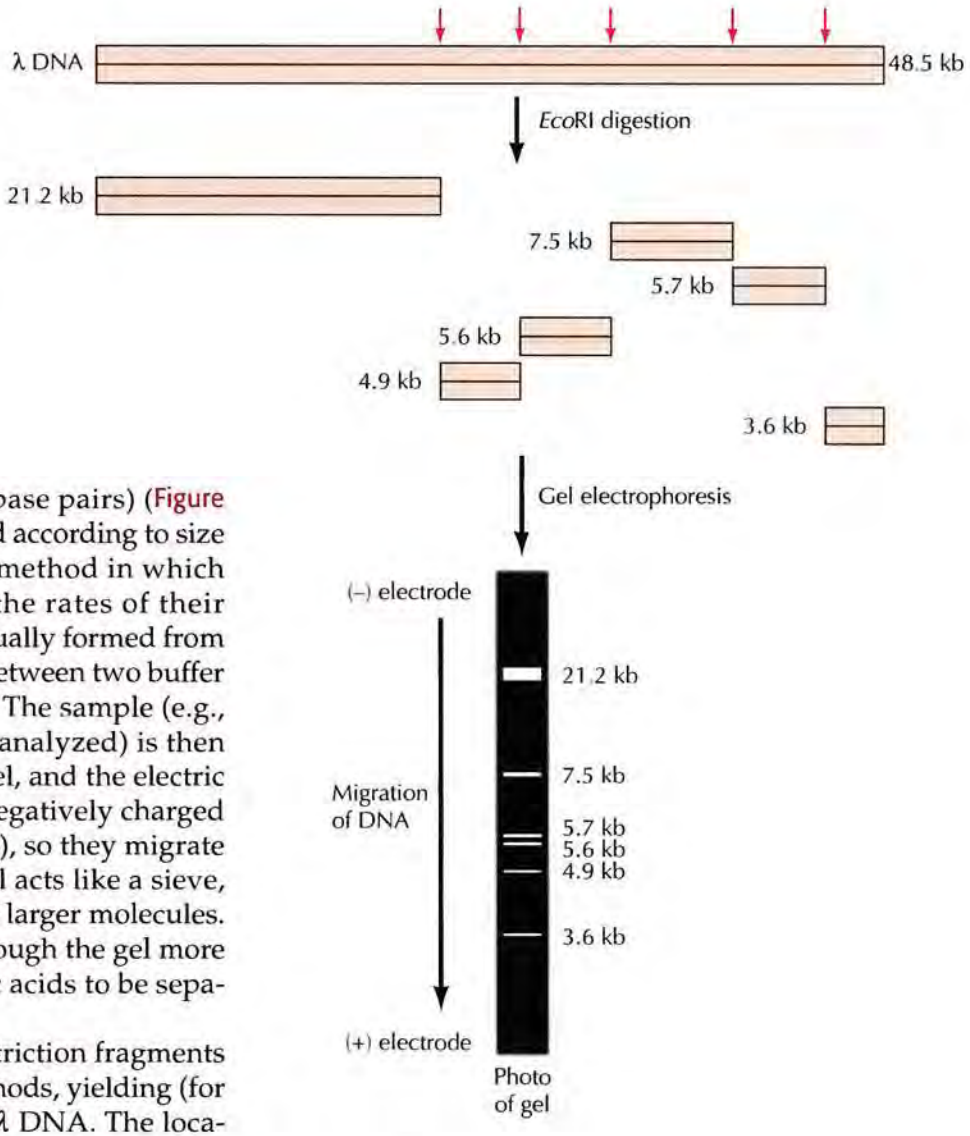


FIGURE 4.14 *Eco*RI digestion and gel electrophoresis of λ DNA
*Eco*RI cleaves λ DNA at five sites (arrows), yielding six DNA fragments. These fragments are then separated by electrophoresis in an agarose gel. The DNA fragments migrate toward the positive electrode, with smaller fragments moving more rapidly through the gel. Following electrophoresis, the DNA is stained with a fluorescent dye and photographed. The sizes of DNA fragments are indicated.

bases long (1 kilobase, or kb = 1000 base pairs) (Figure 4.14). These fragments can be separated according to size by **gel electrophoresis**—a common method in which molecules are separated based on the rates of their migration in an electric field. A gel, usually formed from agarose or polyacrylamide, is placed between two buffer compartments containing electrodes. The sample (e.g., the mixture of DNA fragments to be analyzed) is then pipetted into preformed slots in the gel, and the electric field is turned on. Nucleic acids are negatively charged (because of their phosphate backbone), so they migrate toward the positive electrode. The gel acts like a sieve, selectively retarding the movement of larger molecules. Smaller molecules therefore move through the gel more rapidly, allowing a mixture of nucleic acids to be separated on the basis of size.

In addition to size, the order of restriction fragments can be determined by a variety of methods, yielding (for example) a map of the *Eco*RI sites in λ DNA. The locations of cleavage sites for multiple different restriction endonucleases can be used to generate detailed **restriction maps** of DNA molecules, such as viral genomes (Figure 4.15). In addition, individual DNA fragments produced by restriction endonuclease digestion can be isolated following electrophoresis for further study—including determination of their DNA sequence. The DNAs of many viruses have been characterized by this approach.

Restriction endonuclease digestion alone, however, does not provide sufficient resolution for the analysis of larger DNA molecules, such as cellular genomes. A restriction endonuclease with a six-base-pair recognition site (such as *Eco*RI) cleaves DNA with a statistical frequency of once every 4096

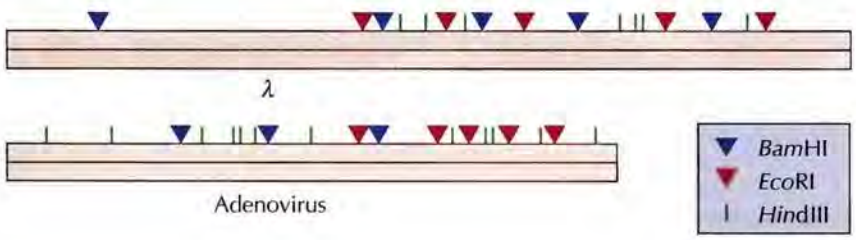
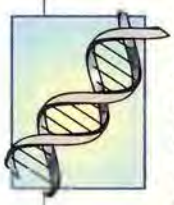


FIGURE 4.15 Restriction maps of λ and adenovirus DNAs The locations of cleavage sites for *Bam*HI, *Eco*RI, and *Hind*III are shown in the DNAs of *E. coli* bacteriophage λ (48.5 kb) and human adenovirus-2 (35.9 kb).

MOLECULAR MEDICINE

HIV and AIDS



The Disease

Acquired immune deficiency syndrome (AIDS) is a new disease, first described in 1981. It has now become a worldwide pandemic, with approximately 70 million people having been infected with HIV and nearly 30 million having died of AIDS. The clinical manifestations of AIDS result principally from failure of the immune system to function normally. In the absence of normal immunity, AIDS patients are sensitive to opportunistic infections by agents (viruses, bacteria, fungi, and protozoans) against which a healthy individual would be resistant. People with AIDS also suffer a high frequency of some types of cancers, particularly lymphomas and Kaposi's sarcoma, although it is the opportunistic infections that are responsible for most deaths.

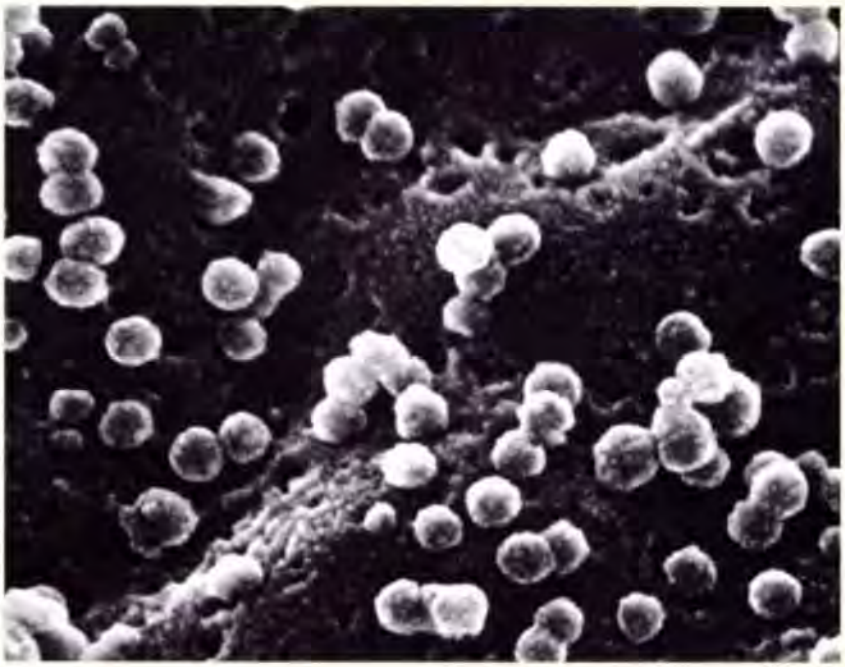
Molecular and Cellular Basis

AIDS is caused by a retrovirus (human immunodeficiency virus or HIV) that was discovered by the research groups of Robert Gallo and Luc Montagnier in 1983. HIV infects principally a specific type of lymphocyte (the T4 lymphocyte) that is required for a normal immune response. In contrast to many other retroviruses, such as Rous sarcoma virus, HIV does not cause the cells it infects to become cancerous. Instead, HIV eventually kills the cells in which it replicates, ultimately resulting in the depletion of the population of T4 lymphocytes and the failure of the immune system in infected individuals. This failure of the immune system in turn leads to the opportunistic infections and cancers that represent the clinical manifestations of AIDS.

Prevention and Treatment

At present, the only means of preventing AIDS is to avoid HIV infection. HIV is a fragile virus that quickly loses infectivity outside the body, so it cannot be transmitted by casual contact with an infected person. HIV can be transmitted in three ways: through sexual contact, through contaminated blood products, and from mother to child during pregnancy or breastfeeding. Following the isolation of HIV, screening tests were developed to ensure the safety of clotting factors and blood supplies used for transfusions. Prevention of HIV infection by other routes currently depends on individuals minimizing their personal risk of infection by adhering to safe sexual practices and avoiding sources of contaminated blood, such as shared needles used for intravenous drug injection.

Beyond modifying individual behavior to reduce the risk of infection, the identification of HIV as the cause of AIDS opens possibilities for prevention and treatment. A vaccine to prevent HIV infection is being actively pursued, although several features of the biology of HIV pose difficulties to this approach. Alternatively, drugs that inhibit virus replication are now providing effective therapies for HIV-infected individuals. These drugs either act as inhibitors of the HIV reverse transcriptase or of the HIV protease, which is an enzyme required for processing viral proteins. Combinations of such drugs are now prolonging the lives of AIDS patients, although further work is clearly needed to develop drugs that are not only more effective but also less expensive and more practical for use in developing countries.



Scanning electron micrograph of HIV budding from T lymphocytes (Cecil Fox/Photo Researchers, Inc.)

base pairs ($1/4^6$). A molecule the size of λ DNA (48.5 kb) would therefore be expected to yield about ten *Eco*RI fragments, consistent with the results illustrated in Figure 4.14. However, restriction endonuclease digestion of larger genomes yields quite different results. For example, the human genome is approximately 3×10^6 kb long and is therefore expected to yield more than 500,000 *Eco*RI fragments. Such a large number of fragments cannot be separated from one another, so agarose gel electrophoresis of *Eco*RI-digested human DNA yields a continuous smear rather than a discrete pattern of DNA fragments. Because it is impossible to isolate single restriction fragments from such digests, restriction endonuclease digestion alone does not yield a source of homogeneous DNA suitable for further analysis. Quantities of such purified DNA fragments, however, can be obtained through molecular cloning.

Generation of Recombinant DNA Molecules

The basic strategy in **molecular cloning** is to insert a DNA fragment of interest (e.g., a segment of human DNA) into a DNA molecule (called a **vector**) that is capable of independent replication in a host cell. The result is a **recombinant molecule** or **molecular clone**, composed of the DNA insert linked to vector DNA sequences. Large quantities of the inserted DNA can be obtained if the recombinant molecule is allowed to replicate in an appropriate host. For example, fragments of human DNA can be cloned in plasmid vectors (Figure 4.16). **Plasmids** are small circular DNA molecules that can replicate independently—without being associated with chromosomal DNA— in bacteria. Recombinant plasmids carrying human DNA inserts can be introduced into *E. coli*, where they replicate along with the bacteria to yield millions of copies of plasmid DNA. The DNA of these plasmids can then be isolated, yielding large quantities of recombinant molecules containing a single fragment of human DNA. Whereas a typical DNA fragment 1 kb in length would represent less than 1 part in a million of human genomic DNA, it would represent approximately 1 part in 5 after being cloned in a plasmid vector. Moreover, the fragment can be easily isolated from the rest of the vector DNA by restriction endonuclease digestion and gel electrophoresis, allowing a pure fragment of human DNA to be analyzed and further manipulated.

The DNA fragments used to create recombinant molecules are usually generated by digestion with restriction endonucleases. Many of these enzymes cleave their recognition sequences at staggered sites, leaving overhanging or cohesive single-stranded tails that can associate with each other by complementary base pairing (Figure 4.17). The association between such paired complementary ends can be established permanently by treatment with **DNA ligase**, an enzyme that seals breaks in DNA strands (see Chapter 6). Thus two different fragments of DNA (e.g., a human DNA insert and a plasmid DNA vector) prepared by digestion with the same restriction endonuclease can be readily joined to create a recombinant DNA molecule.

The fragments of DNA that can be cloned are not limited to those that terminate in restriction endonuclease cleavage sites. Synthetic DNA "linkers" containing desired restriction endonuclease sites can be added to the

4.7 WEBSITE ANIMATION

Recombinant DNA Molecules The basic strategy in molecular cloning is to insert a DNA fragment of interest into a DNA molecule (called a vector) that is capable of independent replication in a host cell.

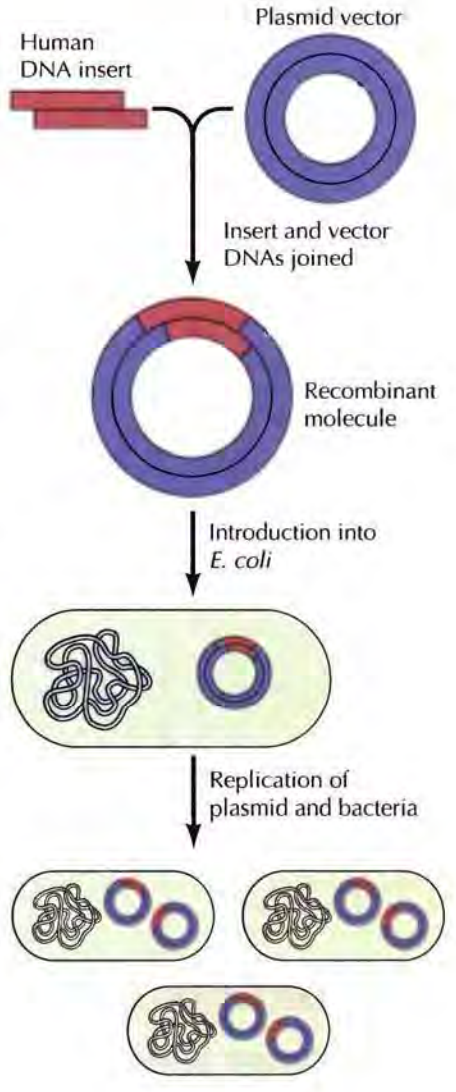


FIGURE 4.16 Generation of a recombinant DNA molecule A fragment of human DNA is inserted into a plasmid DNA vector. The resulting recombinant molecule is then introduced into *E. coli*, where it replicates along with the bacteria to yield a population of bacteria carrying plasmids with the human DNA insert.

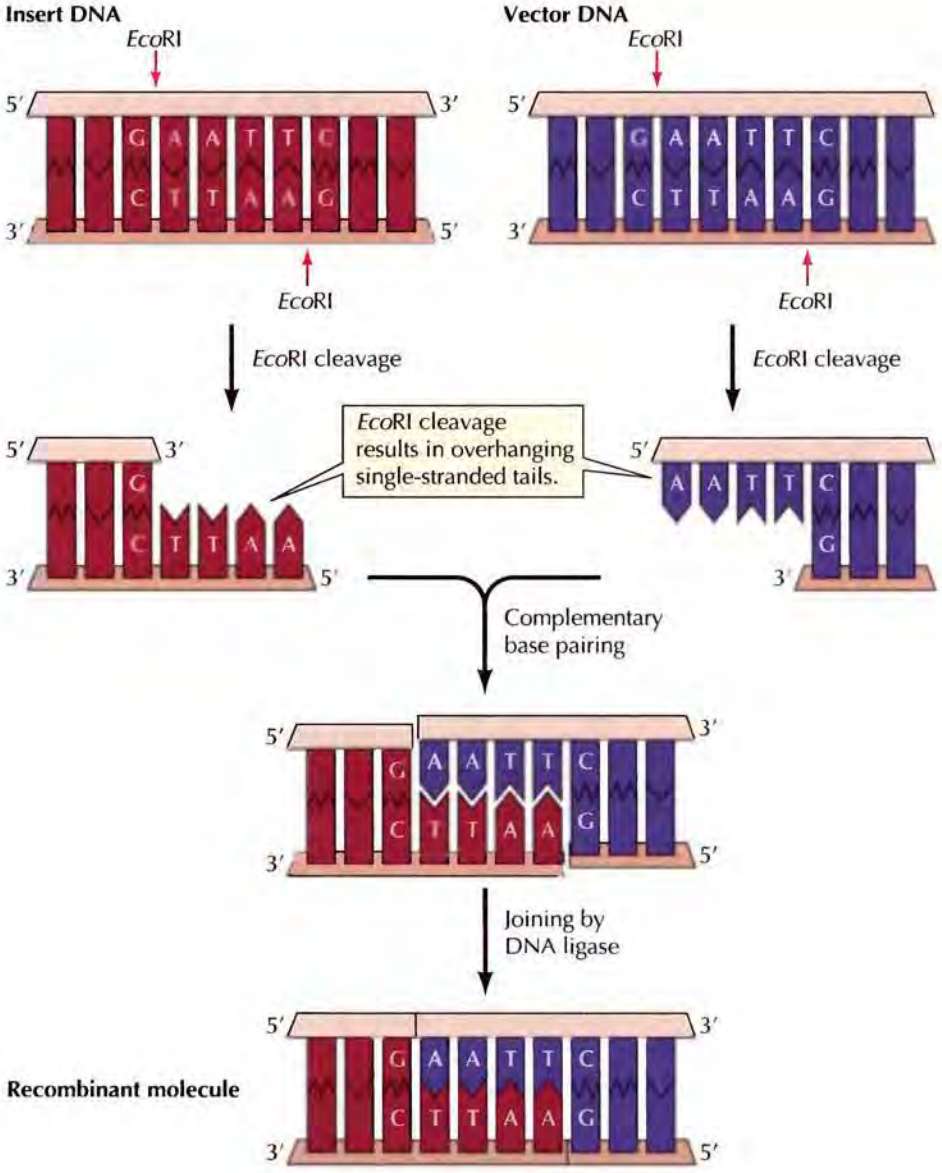


FIGURE 4.17 Joining of DNA molecules Vector and insert DNAs are digested with a restriction endonuclease (such as *Eco*RI), which cleaves at staggered sites leaving overhanging single-stranded tails. Vector and insert DNAs can then associate by complementary base pairing, and covalent joining of the DNA strands by DNA ligase yields a recombinant molecule.

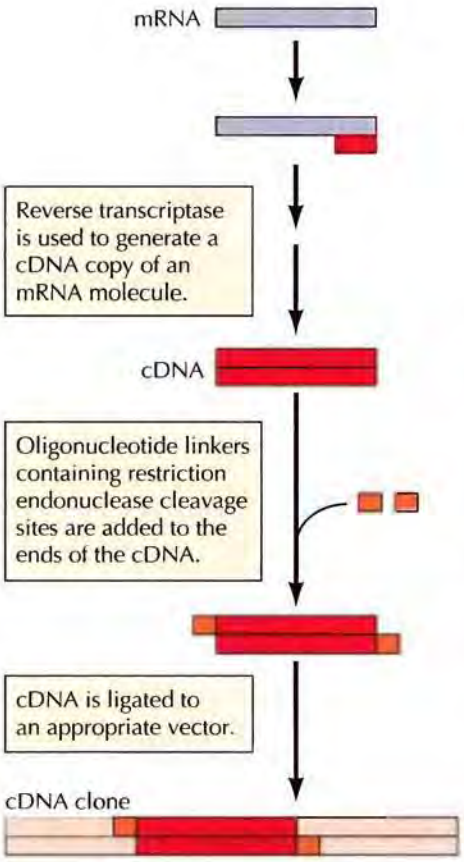


FIGURE 4.18 cDNA cloning

ends of any DNA fragment, allowing virtually any fragment of DNA to be ligated to a vector and isolated as a molecular clone.

Not only DNA, but also RNA sequences can be cloned (Figure 4.18). The first step is to synthesize a DNA copy of the RNA using the enzyme reverse transcriptase. The DNA product (called a **cDNA** because it is complementary to the RNA used as a template) can then be ligated to vector DNA as already described. Since eukaryotic genes are usually interrupted by noncoding sequences (introns; see Chapter 5), which are removed from mRNA by splicing, the ability to clone cDNA as well as genomic DNA has been critical for understanding gene structure and function. Moreover, cDNA cloning allows the mRNA corresponding to a single gene to be isolated as a molecular clone.

Vectors for Recombinant DNA

Depending on the size of the insert DNA and the purpose of the experiment, many different types of cloning vectors can be used for the generation of recombinant molecules. The basic vector systems used for the isolation and propagation of cloned DNAs are reviewed here. Other vectors

developed for the expression of cloned DNAs and the introduction of recombinant molecules into eukaryotic cells are discussed in subsequent sections.

Plasmids are commonly used for cloning genomic or cDNA inserts of up to a few thousand base pairs. Plasmid vectors usually consist of 2 to 4 kb of DNA, including an **origin of replication**—the DNA sequence that signals the host cell DNA polymerase to replicate the DNA molecule. In addition, plasmid vectors carry genes that confer resistance to antibiotics (e.g., ampicillin resistance), so bacteria carrying the plasmids can be selected. For example, Figure 4.19 illustrates the isolation of human cDNA clones in a plasmid vector. A pool of cDNA fragments are ligated to restriction endonuclease-digested plasmid DNA. The resulting recombinant DNA molecules are then used to transform *E. coli*. Antibiotic-resistant colonies, which contain plasmid DNA, are selected. Since each recombinant plasmid yields a single antibiotic-resistant colony, the bacteria present in any given colony will contain a unique cDNA insert. Plasmid-containing bacteria can then be grown in large quantities and their DNA extracted. The small circular plasmid DNA molecules, of which there are often hundreds of copies per cell, can be separated from the bacterial chromosomal DNA; the result is purified plasmid DNA that is suitable for analysis of the cloned insert.

Bacteriophage λ vectors are also used for the isolation of either genomic or cDNA clones from eukaryotic cells, and will accommodate larger fragments of insert DNA than plasmids. In λ cloning vectors, sequences of the bacteriophage genome that are dispensable for virus replication have been removed and replaced with unique restriction sites for insertion of cloned DNA. These recombinant molecules can be introduced into *E. coli*, where they replicate to yield millions of progeny phages containing a single DNA insert. The DNA of these phages can then be isolated, yielding large quantities of recombinant molecules containing a single fragment of cloned DNA. The DNA inserts can be as large as about 15 kb and still yield a recombinant genome that can be packaged into bacteriophage λ particles.

For many studies involving analysis of genomic DNA, it is desirable to clone larger fragments of DNA than are accommodated by plasmid or λ vectors. There are five major types of vectors that are used for this purpose (Table 4.3). **Cosmid** vectors accommodate inserts of approximately 45 kb. These vectors contain bacteriophage λ sequences that allow efficient packaging of the cloned DNA into phage particles. In addition, cosmids contain origins of replication and the genes for antibiotic resistance that are characteristic of plasmids, so they are able to replicate as plasmids in bacterial cells. Two other types of vectors are derived from bacteriophage P1, rather than from bacteriophage λ. Bacteriophage P1 vectors, which will accommodate DNA fragments of 70–100 kb, contain sequences that allow recombinant molecules to be packaged *in vitro* into P1 phage particles and then to be replicated as plasmids in *E. coli*. **P1 artificial chromosome (PAC)** vectors also contain sequences of bacteriophage P1 but are introduced directly as plasmids into *E. coli* and will accommodate larger inserts of 130–150 kb. **Bacterial artificial chromosome (BAC)** vectors are derived from a naturally occurring plasmid of *E. coli* (called the F factor). The replication origin and other F factor sequences allow BACs to replicate as stable plasmids carrying inserts of 120–300 kb. Even larger fragments of DNA (250–400 kb) can be cloned in **yeast artificial chromosome (YAC)** vectors. These vectors contain yeast origins of replication as well as other sequences (centromeres and telomeres, discussed in Chapter 5) that allow them to replicate as linear chromosome-like molecules in yeast cells.

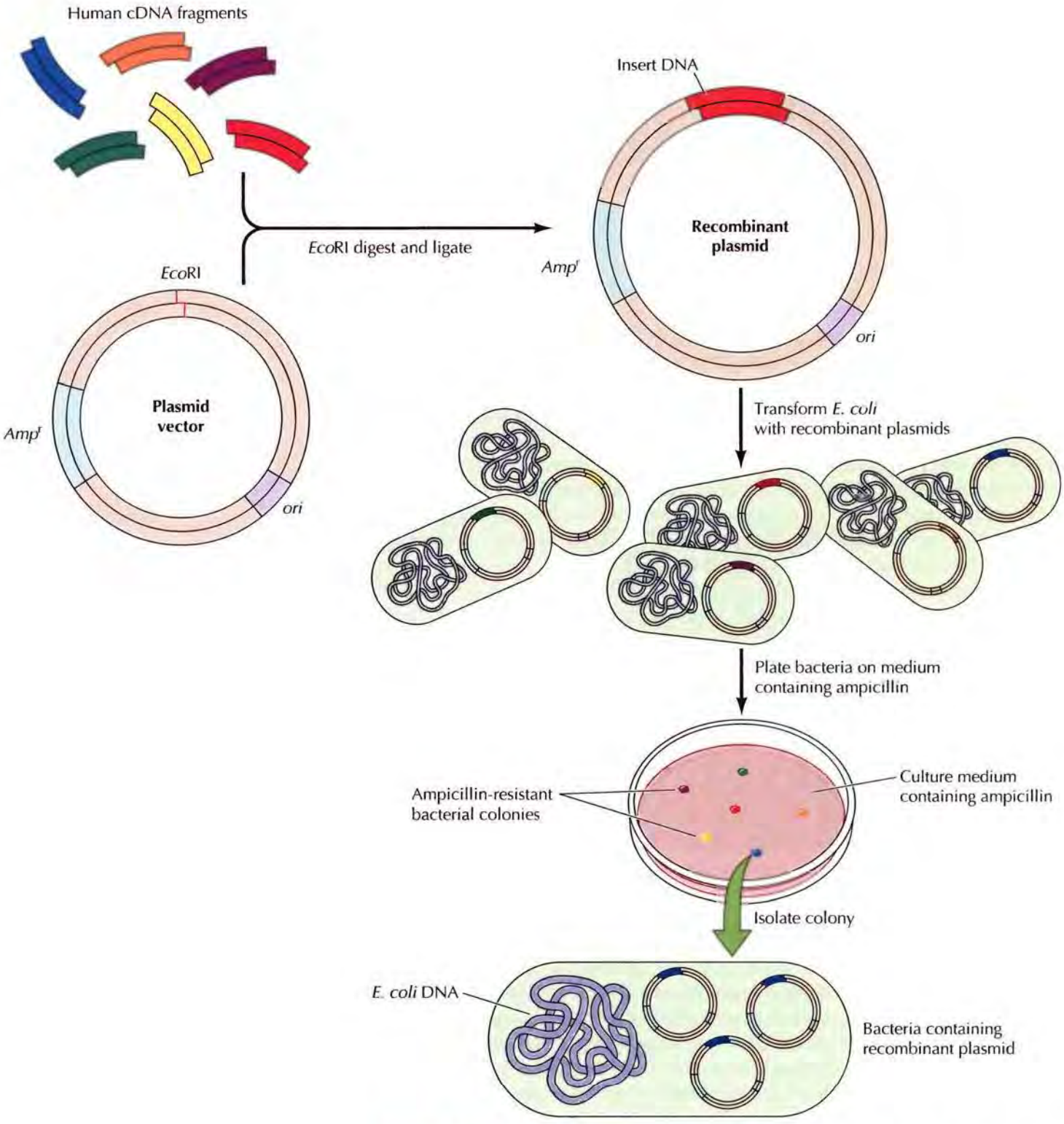


FIGURE 4.19 Cloning in plasmid vectors The vector is a small circular molecule that contains an origin of replication (*ori*), a gene conferring resistance to ampicillin (Amp^r), and a restriction site (e.g., *Eco*RI), which can be used to insert foreign DNA. Insert DNA (e.g., human cDNA) is ligated to the vector, and the recombinant plasmids are used to transform *E. coli*. The bacteria are plated on medium containing ampicillin, so that only the bacteria that are ampicillin-resistant because they carry plasmid DNAs are able to form colonies. Individual colonies of plasmid-containing bacteria can then be isolated and grown in large quantities for isolation of recombinant plasmids.

TABLE 4.3 Vectors for Cloning Large Fragments of DNA

Vector	DNA Insert (kb)	Host cell
Cosmids	30–45	*E. coli*
Bacteriophage P1	70–100	*E. coli*
P1 Artificial Chromosomes (PACs)	130–150	*E. coli*
Bacterial Artificial Chromosomes (BACs)	120–300	*E. coli*
Yeast Artificial Chromosomes (YACs)	250–400	Yeast

DNA Sequencing

Molecular cloning allows the isolation of individual fragments of DNA in quantities suitable for detailed characterization, including the determination of nucleotide sequence. Indeed, determination of the nucleotide sequences of many genes has elucidated not only the structure of their protein products but also the properties of DNA sequences that regulate gene expression. Furthermore, the coding sequences of novel genes are frequently related to those of previously studied genes, and the functions of newly isolated genes can often be correctly deduced on the basis of such sequence similarities.

DNA sequencing is usually performed with automated systems that are both rapid and accurate, so determining the sequence of several kilobases of DNA is a straightforward task. Thus it is far easier to clone and sequence DNA than it is to determine the amino acid sequence of a protein. Since the nucleotide sequence of a gene can be readily translated into the amino acid sequence of its encoded protein, the easiest way of determining protein sequence is the sequencing of a cloned gene or cDNA.

The most common method of DNA sequencing is based on premature termination of DNA synthesis resulting from the inclusion of chain-terminating **dideoxynucleotides** (which do not contain the deoxyribose 3′ hydroxyl group) in DNA polymerase reactions (Figure 4.20). DNA synthesis is initiated at a unique site on the cloned DNA from a synthetic primer. The DNA synthesis reaction includes each of the four dideoxynucleotides (A, C, G, and T) in addition to their normal counterparts. Each of the four dideoxynucleotides is labeled with a different fluorescent dye, so their incorporation into DNA can be monitored. Incorporation of a dideoxynucleotide stops further DNA synthesis because no 3′ hydroxyl group is available for addition of the next nucleotide. Thus a series of labeled DNA molecules is generated, each terminating at the base represented by a specific fluorescent dideoxynucleotide. These fragments of DNA are then separated according to size by gel electrophoresis. As the newly synthesized DNA strands are electrophoresed through the gel, they pass through a laser beam that excites the fluorescent labels. The resulting emitted light is then detected by a photomultiplier, and a computer collects and analyzes the data. The size of each fragment is determined by its terminal dideoxynucleotide, marked by a specific color fluorescence, so the DNA sequence can be read from the order of fluorescent-labeled fragments as they migrate through the gel. High throughput automated DNA sequencing of this type has enabled the large-scale analysis required for determination of the sequences of complete genomes, including that of humans.

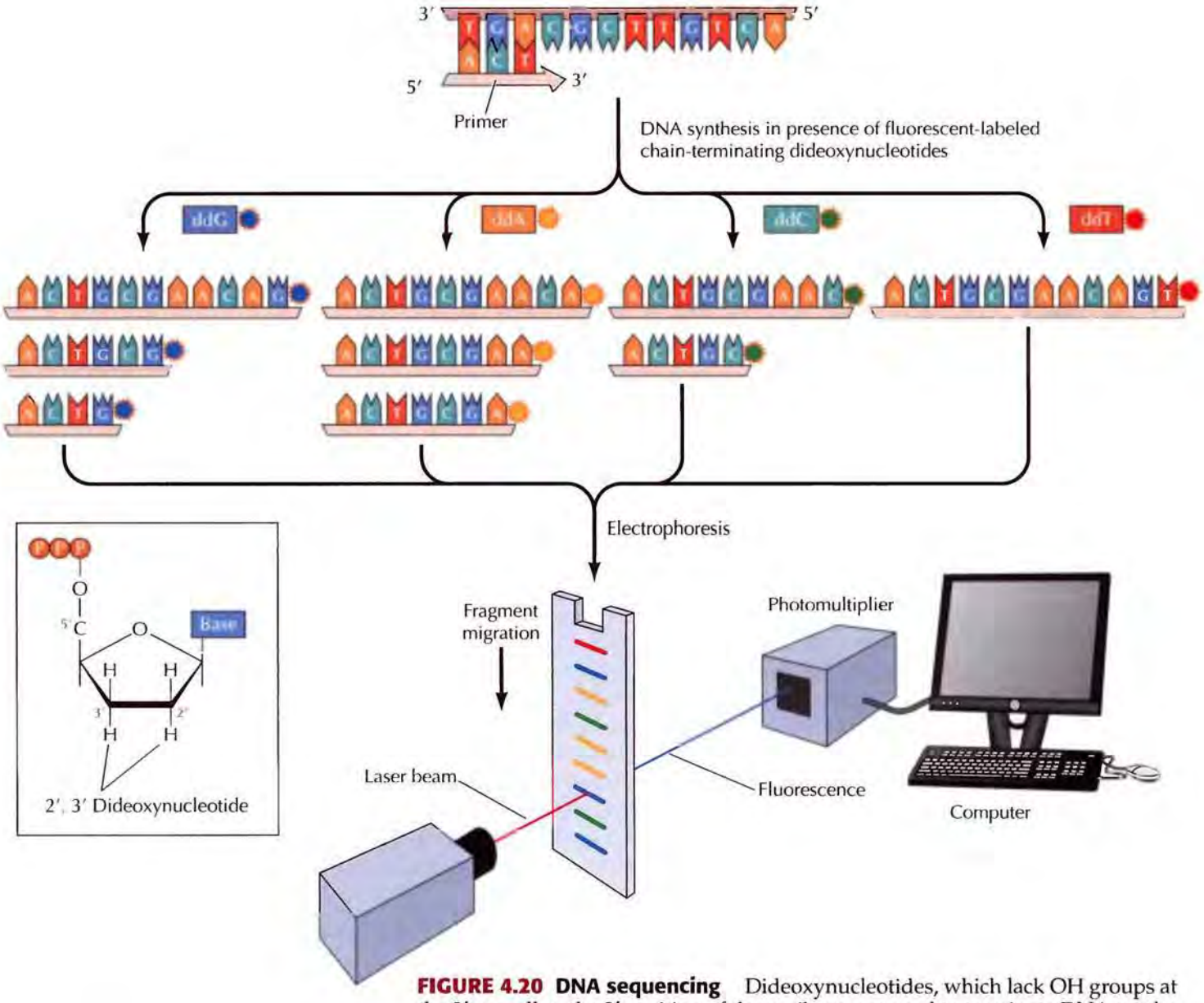


FIGURE 4.20 DNA sequencing Dideoxynucleotides, which lack OH groups at the 3′ as well as the 2′ position of deoxyribose, are used to terminate DNA synthesis at specific bases. These molecules are incorporated normally into growing DNA strands. Because they lack a 3′ OH, however, the next nucleotide cannot be added, so synthesis of that DNA strand terminates. DNA synthesis is initiated at a specific site with a primer. The reaction contains the four dideoxynucleotides, each labeled with a different fluorescent dye, as well as the four normal deoxynucleotides. When the dideoxynucleotide is incorporated, DNA synthesis stops, so the reaction yields a series of products extending from the primer to the base substituted by a fluorescent dideoxynucleotide. These products are then separated by gel electrophoresis. As the DNA strands migrate through the gel, they pass through a laser beam that excites the fluorescent labels on the dideoxynucleotides. The emitted light is detected by a photomultiplier, which is connected to a computer that collects and analyzes the data to determine the DNA sequence.

4.8 WEBSITE ANIMATION

Sequencing a DNA Strand

One method of DNA sequencing involves a modified DNA synthesis reaction using chain-terminating nucleotides and fluorescent primers, which can be identified by automated detection systems.

Expression of Cloned Genes

In addition to enabling determination of the nucleotide sequences of genes—and hence the amino acid sequences of their protein products—molecular cloning has provided new approaches to obtaining large amounts of proteins for structural and functional characterization. Many proteins of interest are present at only low levels in eukaryotic cells and therefore cannot be purified in significant amounts by conventional bio-

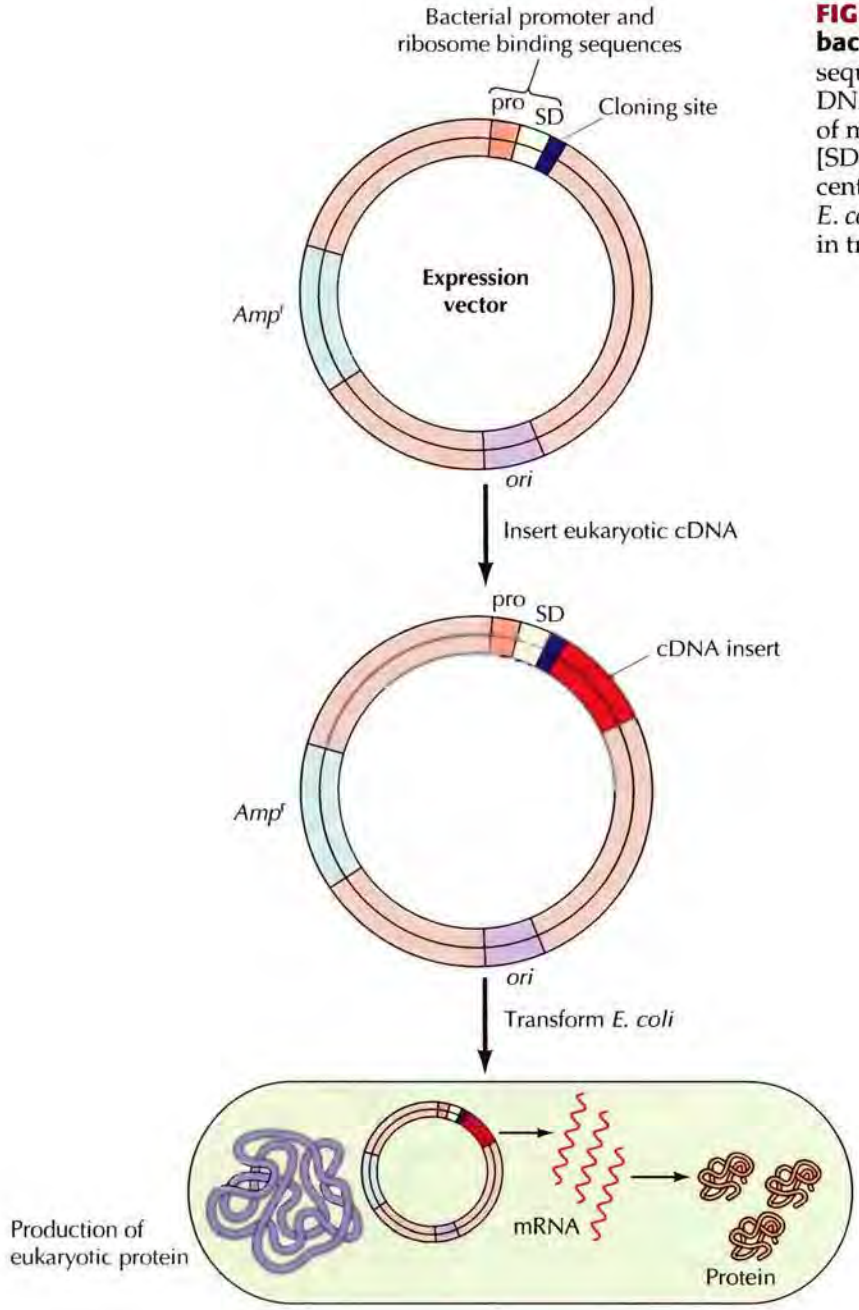


FIGURE 4.21 Expression of cloned genes in bacteria Expression vectors contain promoter sequences (pro) that direct transcription of inserted DNA in bacteria and sequences required for binding of mRNA to bacterial ribosomes (Shine-Dalgarno [SD] sequences). A eukaryotic cDNA inserted adjacent to these sequences can be efficiently expressed in *E. coli*, resulting in production of eukaryotic proteins in transformed bacteria.

chemical techniques. Given a cloned gene, however, this problem can be solved by the engineering of vectors that lead to high levels of gene expression in either bacteria or eukaryotic cells.

To express a eukaryotic gene in *E. coli*, the cDNA of interest is cloned into a plasmid or phage vector (called an **expression vector**) that contains sequences that drive transcription and translation of the inserted gene in bacterial cells (Figure 4.21). Inserted genes often can be expressed at levels high enough that the protein encoded by the cloned gene corresponds to as much as 10% of the total bacterial protein. Purifying the protein encoded by the cloned gene in quantities suitable for detailed biochemical or structural studies is then a straightforward matter.

It is frequently useful to express high levels of a cloned gene in eukaryotic cells rather than in bacteria. This mode of expression may be important, for example, to ensure that posttranslational modifications of the protein

(such as addition of carbohydrates or lipids) occur normally. Such protein expression in eukaryotic cells can be achieved, as in *E. coli*, by insertion of the cloned gene into a vector (usually derived from a virus) that directs high-level gene expression. One system frequently used for protein expression in eukaryotic cells is infection of insect cells by baculovirus vectors, which direct very high levels of expression of genes inserted in place of a viral structural protein. Alternatively, high levels of protein expression can be achieved using appropriate vectors in mammalian cells.

Expression of cloned genes in yeast is particularly useful because simple methods of yeast genetics can be employed to identify proteins that interact with one another. In this type of analysis, called the **yeast two-hybrid** system, two different cDNAs (for example, from human cells) are joined to two distinct domains of a protein that stimulates expression of a target gene in yeast (Figure 4.22). Yeast are then transformed with the hybrid cDNA clones to test for interactions between the two proteins. If the human proteins do interact with each other, they will bring the two domains of the yeast protein together, resulting in stimulation of target gene expression in the transformed yeast. Expression of the target gene can be easily detected by growth of the yeast in a specific medium or by production of an enzyme that produces a blue yeast colony, so the yeast two-hybrid system provides a straightforward method to test protein-protein interactions. Indeed, high throughput yeast two-hybrid screens have been used to construct large scale interaction maps of thousands of proteins in eukaryotic cells (see Figure 2.33).

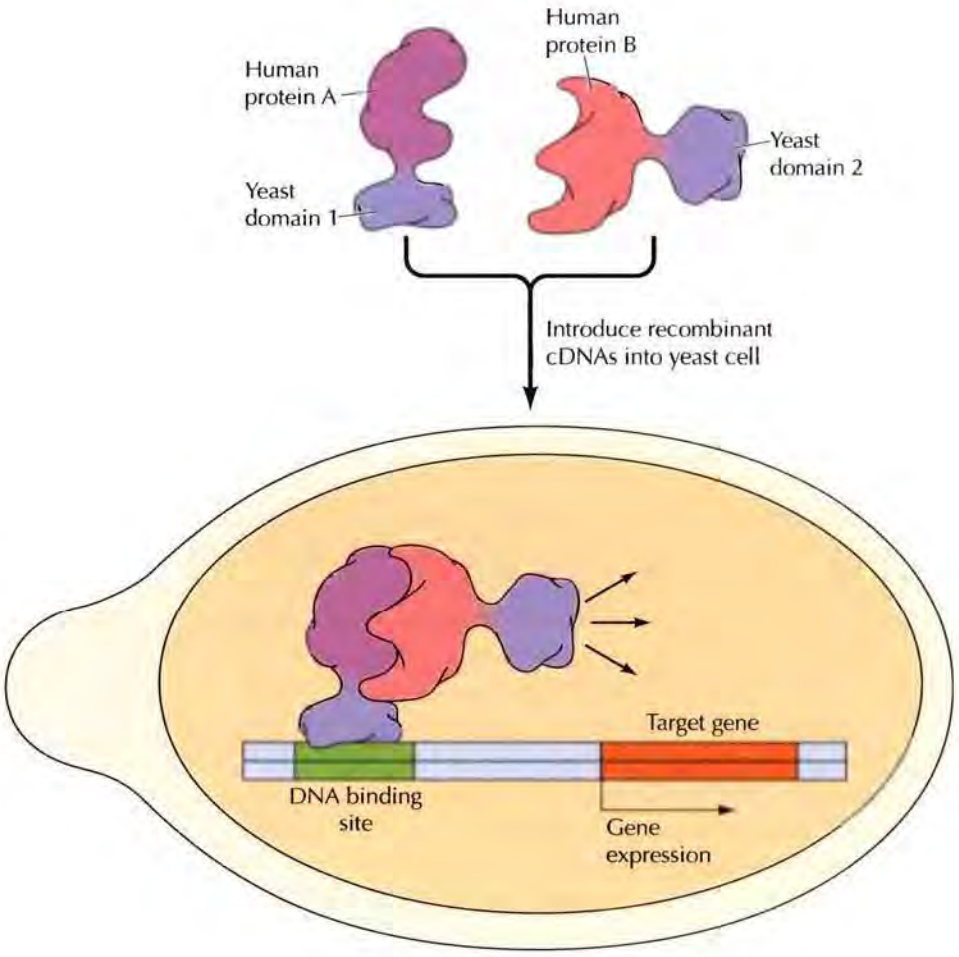


FIGURE 4.22 The yeast two-hybrid system cDNAs of two human proteins are cloned as fusions with two domains (designated 1 and 2) of a yeast protein that stimulates transcription of a target gene. The two recombinant cDNAs are introduced into a yeast cell. If the two human proteins interact with each other, they bring the two domains of the yeast protein together. Domain 1 binds DNA sequences at a site upstream of the target gene, and domain 2 stimulates target gene transcription. The interaction between the two human proteins can thus be detected by expression of the target gene in transformed yeast.

Detection of Nucleic Acids and Proteins

The advent of molecular cloning has enabled the isolation and characterization of individual genes from eukaryotic cells. Understanding the role of genes within cells, however, requires analysis of the intracellular organization and expression of individual genes and their encoded proteins. In this section, the basic procedures used for detection of specific nucleic acids and proteins are discussed. These approaches are important for a wide variety of studies, including the mapping of genes to chromosomes, the analysis of gene expression, and the localization of proteins to subcellular organelles.

Amplification of DNA by the Polymerase Chain Reaction

Molecular cloning allows individual DNA fragments to be propagated in bacteria and isolated in large amounts. An alternative method to isolating large amounts of a single DNA molecule is the **polymerase chain reaction (PCR)**, which was developed by Kary Mullis in 1988. Provided that some sequence of the DNA molecule is known, PCR can achieve a striking amplification of DNA via reactions carried out entirely *in vitro*. Essentially, DNA polymerase is used for repeated replication of a defined segment of DNA. The number of DNA molecules increases exponentially, doubling with each round of replication, so a substantial quantity of DNA can be obtained from a small number of initial template copies. For example, a single DNA molecule amplified through 30 cycles of replication would theoretically yield 2^{30} (approximately 1 billion) progeny molecules. Single DNA molecules can thus be amplified to yield readily detectable quantities of DNA that can be isolated by molecular cloning or further analyzed directly by restriction endonuclease digestion or nucleotide sequencing.

The general procedure for PCR amplification of DNA is illustrated in Figure 4.23. The starting material can be either a cloned DNA fragment or a mixture of DNA molecules—for example, total DNA from human cells. A specific region of DNA can be amplified from such a mixture, provided that the nucleotide sequence surrounding the region is known so that primers can be designed to initiate DNA synthesis at the desired point. Such primers are usually chemically synthesized oligonucleotides containing 15 to 20 bases of DNA. Two primers are used to initiate DNA synthesis in opposite directions from complementary DNA strands. The reaction is started by heating the template DNA to a high temperature (e.g., 95°C) so that the two strands separate. The temperature is then lowered to allow the primers to pair with their complementary sequences on the template strands. DNA polymerase then uses the primers to synthesize a new strand complementary to each template. Thus, in one cycle of amplification, two new DNA molecules are synthesized from one template molecule. The process can be repeated multiple times, with a twofold increase in DNA molecules resulting from each round of replication.

The multiple cycles of heating and cooling involved in PCR are performed by programmable heating blocks called thermocyclers. The DNA polymerases used in these reactions are heat-stable enzymes from bacteria such as *Thermus aquaticus*, which lives in hot springs at temperatures of about 75°C. These polymerases are stable even at the high temperatures used to separate the strands of double-stranded DNA, so PCR amplification can be performed rapidly and automatically. RNA sequences can also be amplified by this method if reverse transcriptase is used to synthesize a cDNA copy prior to PCR amplification.

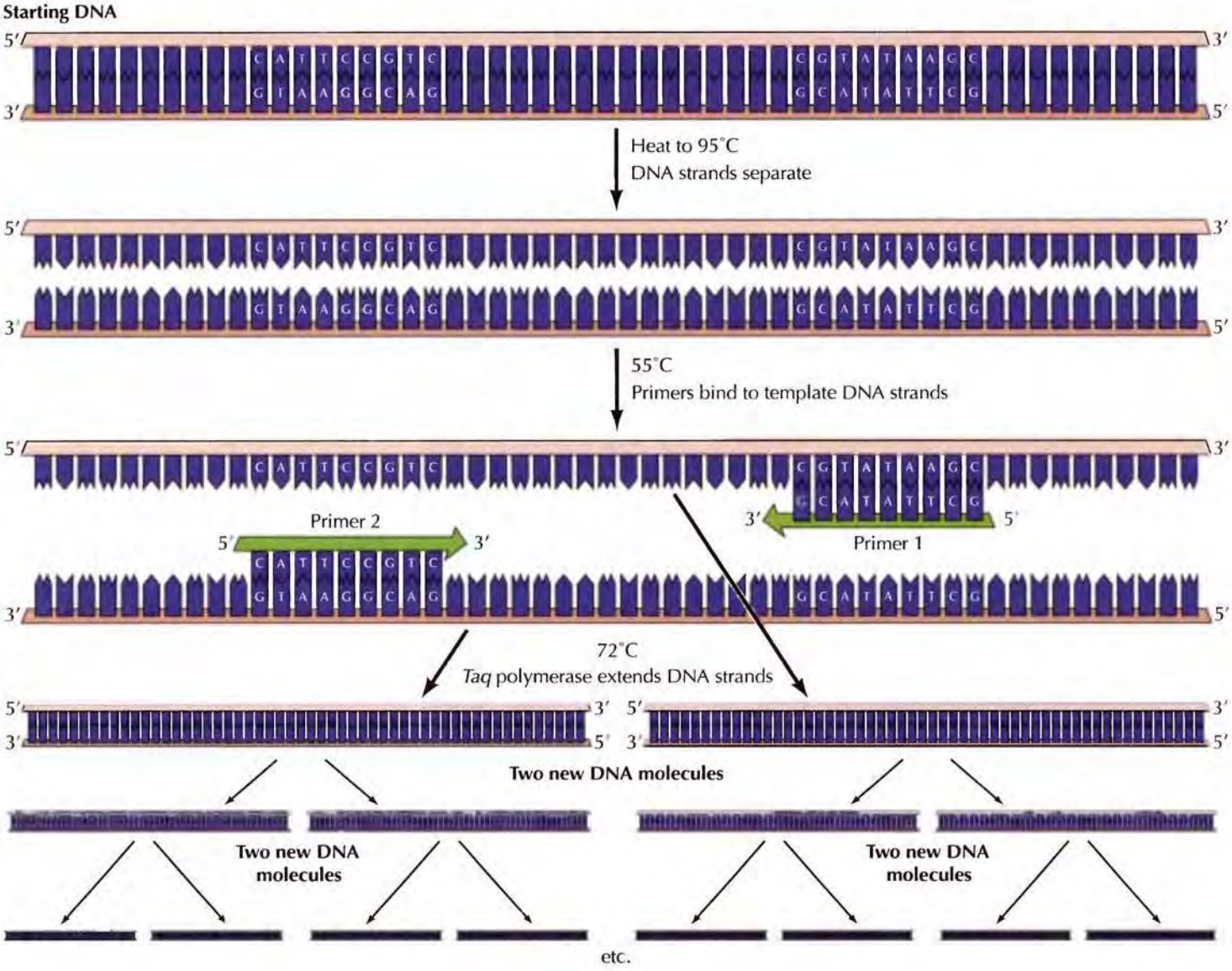


FIGURE 4.23 Amplification of DNA by PCR The region of DNA to be amplified is flanked by two sequences used to prime DNA synthesis. The starting double-stranded DNA is heated to separate the strands and then cooled to allow primers (usually oligonucleotides of 15 to 20 bases) to bind to each strand of DNA. DNA polymerase from *Thermus aquaticus* (*Taq* polymerase) is used to synthesize new DNA strands starting from the primers, resulting in the formation of two new DNA molecules. The process can be repeated for multiple cycles, each resulting in a twofold amplification of DNA.

■ **PCR has become an extremely powerful forensic method. Amplification by PCR allows investigators to obtain a DNA profile from small samples of DNA at a crime scene.**

If enough of the sequence of a gene is known so that primers can be specified, PCR amplification provides an extremely powerful method of detecting small amounts of specific DNA or RNA molecules in a complex mixture of other molecules. The only DNA molecules that will be amplified by PCR are those containing sequences complementary to the primers used in the reaction. Therefore PCR can selectively amplify a specific template from complex mixtures, such as total cell DNA or RNA. This extraordinary sensitivity has made PCR an important method for a variety of applications, including the analysis of gene expression in cells available in only limited quantities.

The DNA segments amplified by PCR can also be directly sequenced or ligated to vectors and propagated as molecular clones. PCR thus allows the amplification and cloning of any segment of DNA for which primers can be designed. Since the complete genome sequences of many organisms are now known, PCR can be used to amplify and clone a vast array of desired DNA fragments, making it a powerful addition to the repertoire of recombinant DNA techniques.

4.9 WEBSITE ANIMATION

Polymerase Chain Reaction The polymerase chain reaction allows the production of millions of copies of a DNA fragment from just one starting DNA molecule.

Nucleic Acid Hybridization

The key to detection of specific nucleic acid sequences is base pairing between complementary strands of RNA or DNA. At high temperatures (e.g., 90 to 100°C) the complementary strands of DNA separate (denature), yielding single-stranded molecules. If such denatured DNA strands are then incubated under appropriate conditions (e.g., 65°), they will renature to form double-stranded molecules as dictated by complementary base pairing—a process called **nucleic acid hybridization**. Nucleic acid hybrids can be formed between two strands of DNA, two strands of RNA, or one strand of DNA and one of RNA.

As discussed above, hybridization between the primers and the template DNA provides the specificity to PCR amplification. In addition, a variety of other methods use nucleic acid hybridization as a means for detecting DNA or RNA sequences that are complementary to any isolated nucleic acid, such as a cloned DNA sequence (Figure 4.24). The cloned DNA is labeled with either radioactive nucleotides or with modified nucleotides that can be detected by fluorescence or chemiluminescence. This labeled DNA is then used as a probe for hybridization to complementary DNA or RNA sequences, which are detected by virtue of the radioactivity, fluorescence, or luminescence of the resulting double-stranded hybrids.

4.10 WEBSITE ANIMATION

Nucleic Acid Hybridization At high temperatures, the complementary strands of DNA separate, and then when they are cooled, they re-form double-stranded molecules as dictated by complementary base pairing.

Southern blotting (a technique developed by E. M. Southern) is widely used for detection of specific genes in cellular DNA (Figure 4.25). The DNA to be analyzed is digested with a restriction endonuclease, and the digested DNA fragments are separated by gel electrophoresis. The gel is then overlaid with a nitrocellulose filter or nylon membrane to which the DNA fragments are transferred (blotted) to yield a replica of the gel. The filter is then incubated with a labeled probe, which hybridizes to the DNA fragments that contain the complementary sequence, allowing visualization of these specific fragments of cell DNA.

Northern blotting is a variation of the Southern blotting technique (hence its name) that is used for detection of RNA instead of DNA. In this method, total cellular RNAs are extracted and fractionated according to size by gel electrophoresis. As in Southern blotting, the RNAs are transferred to a filter and detected by hybridization with a cloned probe. Northern blotting is frequently used in studies of gene expression—for example, to determine whether specific mRNAs are present in different types of cells.

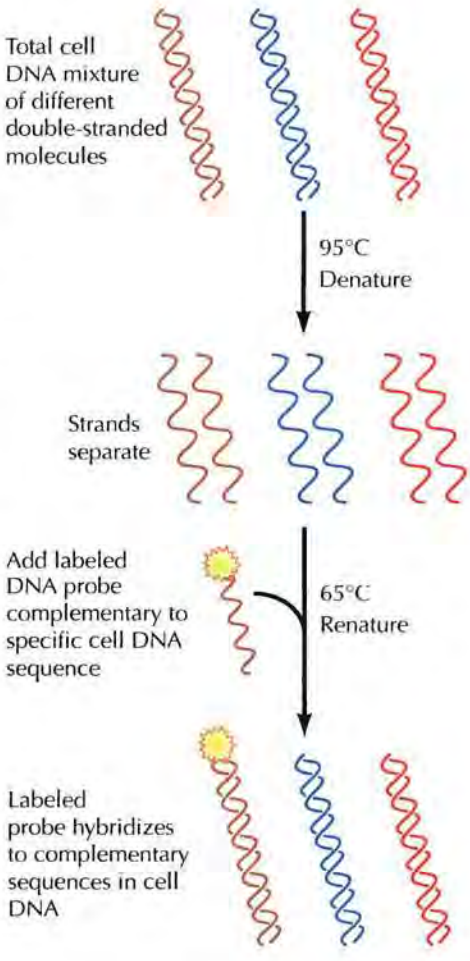


FIGURE 4.24 Detection of DNA by nucleic acid hybridization A specific sequence can be detected in total cell DNA by hybridization with a labeled DNA probe, containing radioactive nucleotides or modified nucleotides that can be detected by fluorescence or chemiluminescence. The DNA is denatured by heating to 95°C, yielding single-stranded molecules. The labeled probe is then added and the temperature is lowered to 65°C, allowing complementary DNA strands to renature by pairing with each other. The probe hybridizes to complementary sequences in cell DNA, which can then be detected by incorporation of the labeled probe into double-stranded molecules.

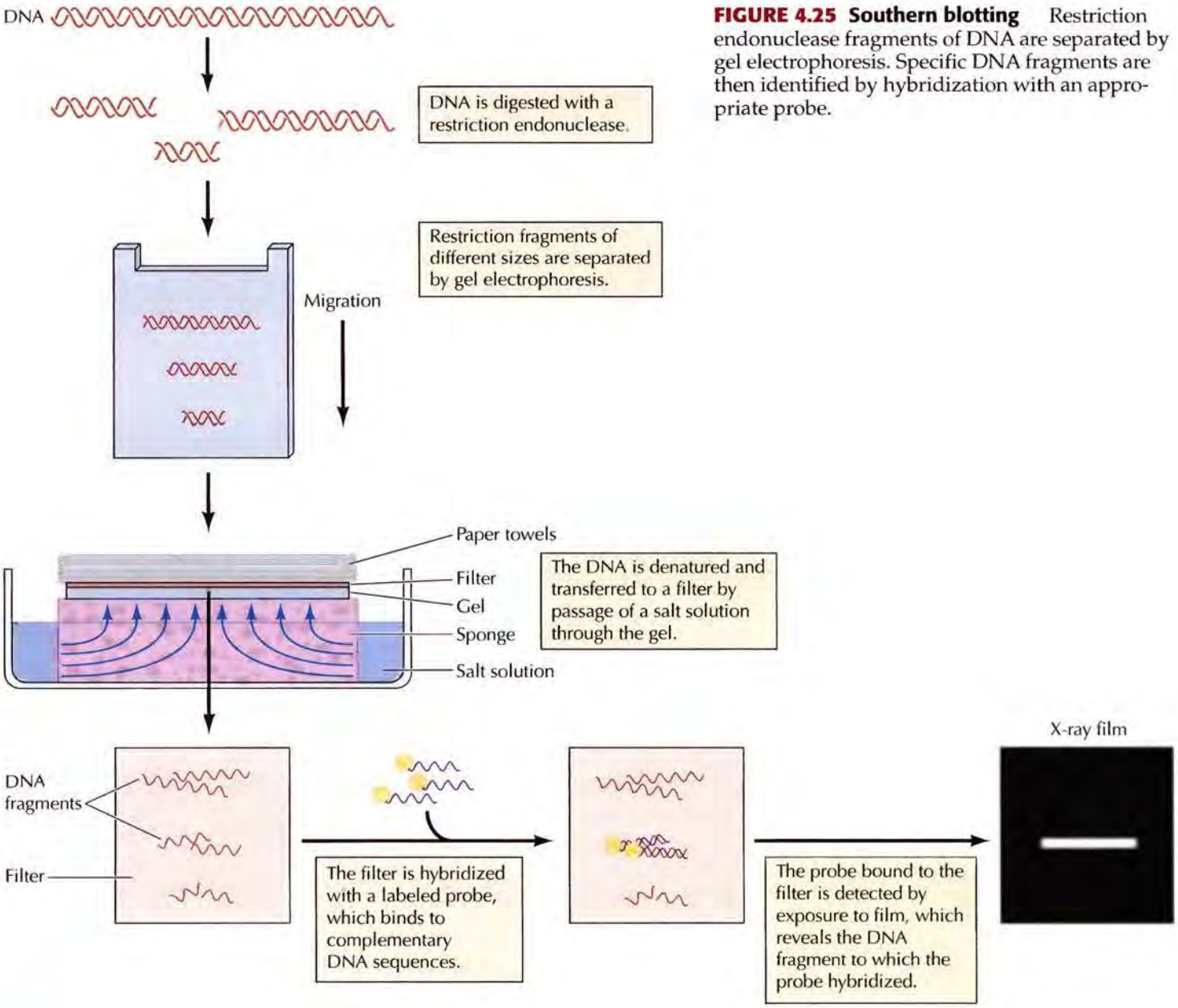


FIGURE 4.25 Southern blotting Restriction endonuclease fragments of DNA are separated by gel electrophoresis. Specific DNA fragments are then identified by hybridization with an appropriate probe.

4.11 WEBSITE ANIMATION

Southern Blotting

DNA fragments are separated by size using gel electrophoresis, and then—as part of the Southern blotting technique—they are incubated with a radioactive DNA probe to identify specific DNA fragments.

Nucleic acid hybridization can also be used to identify molecular clones that contain specific cellular DNA inserts. The first step in isolation of either genomic or cDNA clones is frequently the preparation of **recombinant DNA libraries**—collections of clones that contain all the genomic or mRNA sequences of a particular cell type (Figure 4.26). For example, a genomic library of human DNA might be prepared by cloning random DNA fragments of about 15 kb in a λ vector. Since the human genome is approximately 3 million kb, the complete human genome would be represented in a collection of approximately 500,000 such clones. Any gene for which a probe is available can then be isolated from such a recombinant library. The recombinant phages are plated on *E. coli*, and each phage replicates to produce a plaque on the lawn of bacteria. The plaques are then blotted onto filters in a process similar to the transfer of DNA from a gel to a filter during Southern blotting, and the filters are hybridized with a labeled probe to identify the phage plaques that contain the gene of interest. A variety of

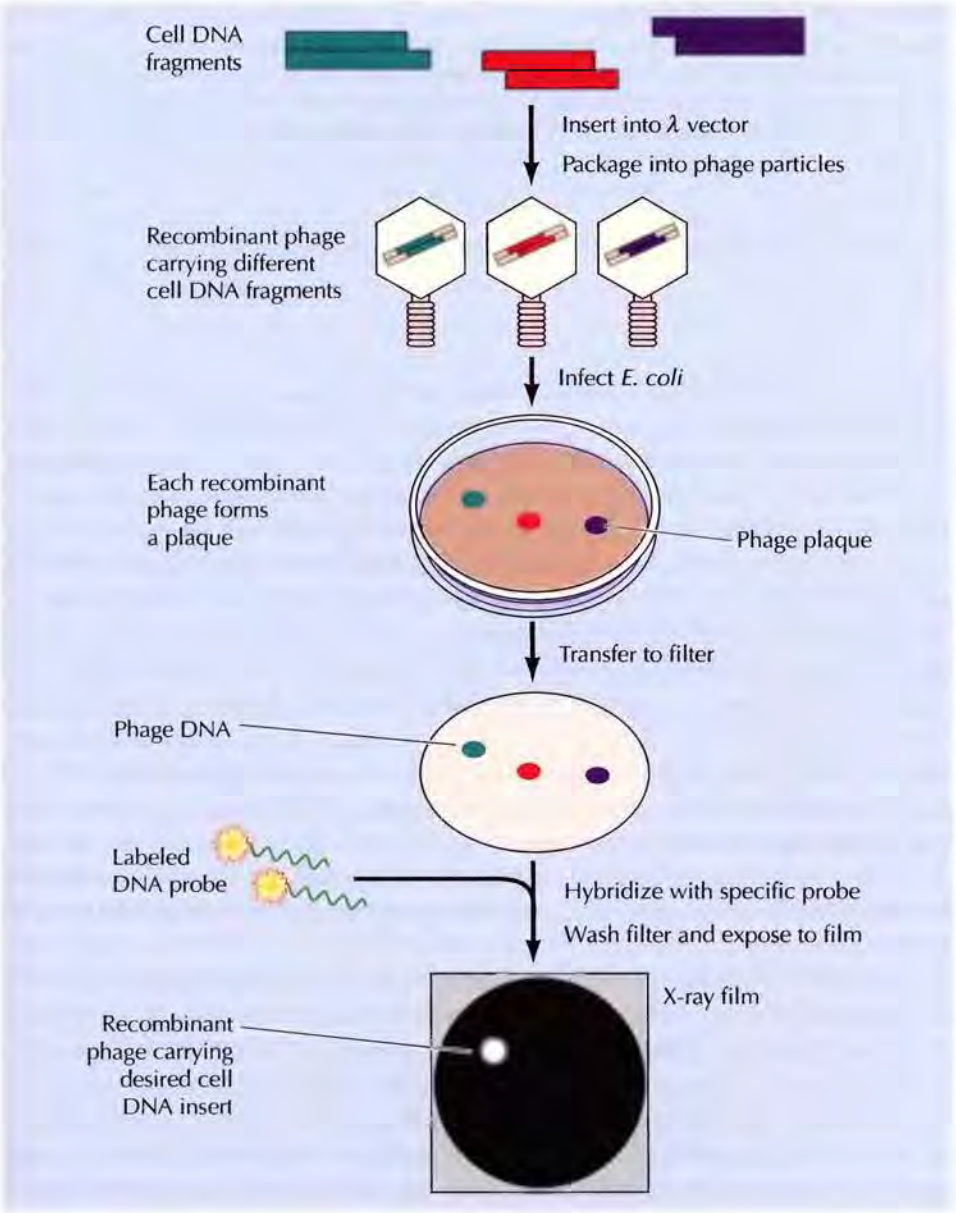


FIGURE 4.26 Screening a recombinant library by hybridization
Fragments of cell DNA are cloned in a bacteriophage λ vector and packaged into phage particles, yielding a collection of recombinant phage carrying different cell inserts. The phages are used to infect bacteria, and the culture is overlaid with a filter. Some of the phages in each plaque are transferred to the filter, which is then hybridized with a labeled probe to identify the phage plaque containing the desired gene. The appropriate phage plaque can then be isolated from the original culture plate.

4.12 WEBSITE ANIMATION

Colony Hybridization

Bacteria can house recombinant DNA libraries, and specific recombinant DNA molecules within the library can be identified by the colony hybridization procedure—a procedure in which the DNA is incubated with a specific radioactive DNA probe.

probes can be used for such experiments. For example, a cDNA clone can be used as a probe to isolate the corresponding genomic clone, or a gene cloned from one species (e.g., mouse) can be used to isolate a related gene from a different species (e.g., human). The appropriate plaque can then be isolated from the original plate in order to propagate the recombinant phage that carries the desired cell DNA insert. Similar procedures can be used to screen bacterial colonies carrying plasmid DNA clones, so specific clones can be isolated by hybridization from either phage or plasmid libraries.

Rather than analyzing one gene at a time, as in Southern or Northern blotting, hybridization to **DNA microarrays** allows tens of thousands of genes to be analyzed simultaneously. As the complete sequences of eukaryotic genomes have become available, hybridization to DNA microarrays has enabled researchers to undertake global analyses of sequences present in either cellular DNA or RNA samples. A DNA microarray consists of a glass slide or membrane filter onto which oligonucleotides or fragments of

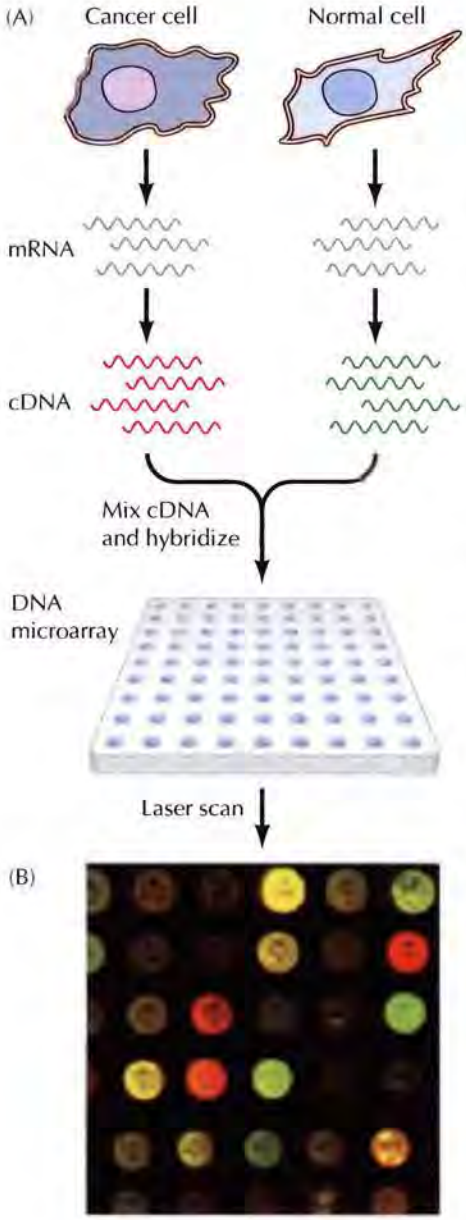


FIGURE 4.27 DNA microarrays (A) An example of comparative analysis of gene expression in cancer cells and normal cells. mRNAs extracted from cancer cells and normal cells are used as templates for synthesis of cDNA probes labeled with different fluorescent dyes (e.g., a red fluorescent label for cancer cell cDNA and green for normal cell cDNA). The two cDNA probes are mixed and hybridized to a DNA microarray containing spots of oligonucleotides corresponding to 10,000 or more distinct human genes. The relative level of expression of each gene in cancer cells compared to normal cells is indicated by the ratio of red to green fluorescence at each position on the microarray. (B) Photograph of a portion of a microarray.

cDNAs are printed by a robotic system in small spots at a high density (Figure 4.27). Each spot on the array consists of a single oligonucleotide or cDNA. More than 10,000 unique DNA sequences can be printed onto a typical glass microscope slide, so it is readily possible to produce DNA microarrays containing sequences representing all of the genes in cellular genomes. As illustrated in Figure 4.27, one widespread application of DNA microarrays is in studies of gene expression; for example, a comparison of the genes expressed by two different types of cells. In an experiment of this type, cDNA probes are synthesized from the mRNAs expressed in each of the two cell types (e.g., cancer cells and normal cells). The two cDNAs are labeled with different fluorescent dyes (typically red and green), and a mixture of the cDNAs is hybridized to a DNA microarray in which 10,000 or more human genes are represented as single spots. The array is then analyzed using a high-resolution laser scanner, and the relative extent of transcription of each gene in the cancer cells compared to the normal cells is indicated by the ratio of red to green fluorescence at the appropriate spot on the array.

Nucleic acid hybridization can be used to detect homologous DNA or RNA sequences not only in cell extracts but also in chromosomes or intact cells—a procedure called ***in situ* hybridization** (Figure 4.28). In this case, the hybridization of fluorescent probes to specific cells or subcellular structures is analyzed by microscopic examination. For example, labeled probes can be hybridized to intact chromosomes in order to identify the chromosomal regions that contain a gene of interest. *In situ* hybridization can also be used to detect specific mRNAs in different types of cells within a tissue.

Antibodies as Probes for Proteins

Studies of gene expression and function require the detection not only of DNA and RNA, but also of specific proteins. For these studies, **antibodies** take the place of nucleic acid probes as reagents that can selectively react with unique protein molecules. Antibodies are proteins produced by cells of the immune system (B lymphocytes) that react against molecules (**antigens**) that the host organism recognizes as foreign substances—for example, the protein coat of a virus. The immune systems of vertebrates are capable of producing millions of different antibodies, each of which specifically recognizes a unique antigen, which may be a protein, a carbohydrate, or a nonbiological molecule. An individual lymphocyte produces only a single type

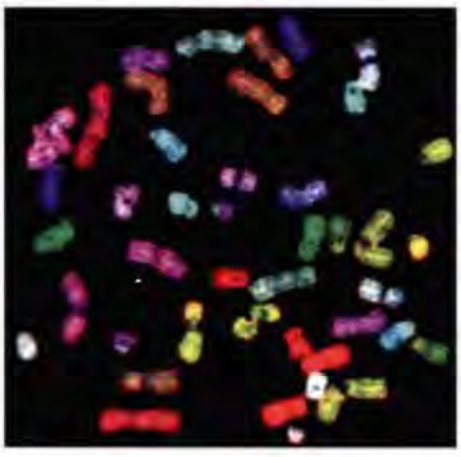


FIGURE 4.28 Fluorescence *in situ* hybridization Hybridization of human chromosomes with chromosome-specific fluorescent probes that label each of the 24 chromosomes a different color. (Courtesy of Thomas Reid and Hesed Padilla-Nash, National Cancer Institute.)

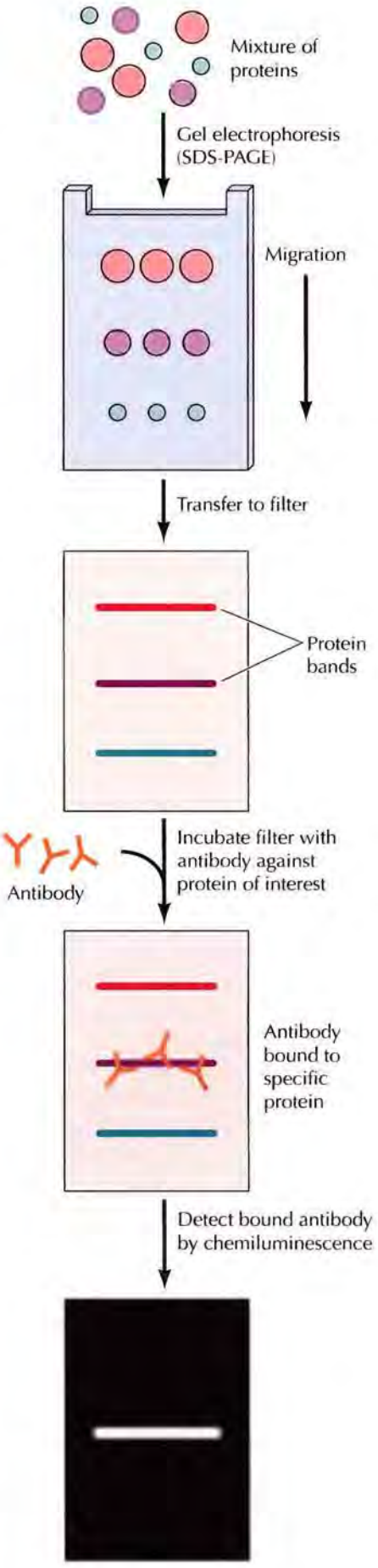


FIGURE 4.29 Western blotting Proteins are separated according to size by SDS-polyacrylamide gel electrophoresis and transferred from the gel to a filter. The filter is incubated with an antibody directed against a protein of interest. The antibody bound to the filter can then be detected by reaction with various reagents, such as a chemiluminescent probe that binds to the antibody.

of antibody, but the antibody genes of different lymphocytes vary as a result of programmed gene rearrangements during development of the immune system (see Chapter 6). This variation gives rise to an array of lymphocytes with distinct antibody genes, programmed to respond to different antigens.

Antibodies can be generated by inoculation of an animal with any foreign protein. For example, antibodies against human proteins are frequently raised in rabbits. The sera of such immunized animals contain a mixture of antibodies (produced by different lymphocytes) that react against multiple sites on the immunizing antigen. However, single species of antibodies (**monoclonal antibodies**) can also be produced by the culturing of clonal lines of B lymphocytes from immunized animals (usually mice). Because each lymphocyte is programmed to produce only a single antibody, a clonal line of lymphocytes produces a monoclonal antibody that recognizes only a single antigenic determinant, thereby providing a highly specific immunological reagent.

Although antibodies can be raised against proteins purified from cells, other materials may also be used for immunization. For example, animals may be immunized with intact cells to raise antibodies against unknown proteins expressed by a specific cell type (e.g., a cancer cell). Such antibodies may then be used to identify proteins specifically expressed by the cell type used for immunization. In addition, antibodies are frequently raised against proteins expressed in bacteria as recombinant clones. In this way, molecular cloning allows the production of antibodies against proteins that may be difficult to isolate from eukaryotic cells. Moreover, antibodies can be raised against synthetic peptides that consist of only 10 to 15 amino acids, rather than against intact proteins. Therefore, once the sequence of a gene is known, antibodies against peptides synthesized from part of its predicted protein sequence can be produced. Because antibodies against these synthetic peptides frequently react with the intact protein as well, it is possible to produce antibodies against a protein starting with only the sequence of a gene.

Antibodies can be used in a variety of ways to detect proteins in cell extracts. Two common methods are **immunoblotting** (also called **Western blotting**) and **immunoprecipitation**. Western blotting (Figure 4.29) is another variation of Southern blotting. Proteins in cell extracts are first separated according to size by gel electrophoresis. Because proteins have different shapes and charges, however, this process requires a modification of the methods used for electrophoresis of nucleic acids. Proteins are separated by a method known as **SDS-polyacrylamide gel electrophoresis (SDS-PAGE)** in which they are dissolved in a solution containing the negatively charged detergent sodium dodecyl sulfate (SDS). Each protein binds many detergent molecules, which denature the protein and give the protein an overall negative charge. Under these conditions, all proteins migrate toward the positive electrode—their rates of migration determined (like those of nucleic acids) only by size. Following electrophoresis, the proteins are transferred to a filter, which is then allowed to react with antibodies against the protein of interest. The antibody bound to the filter can be detected by a

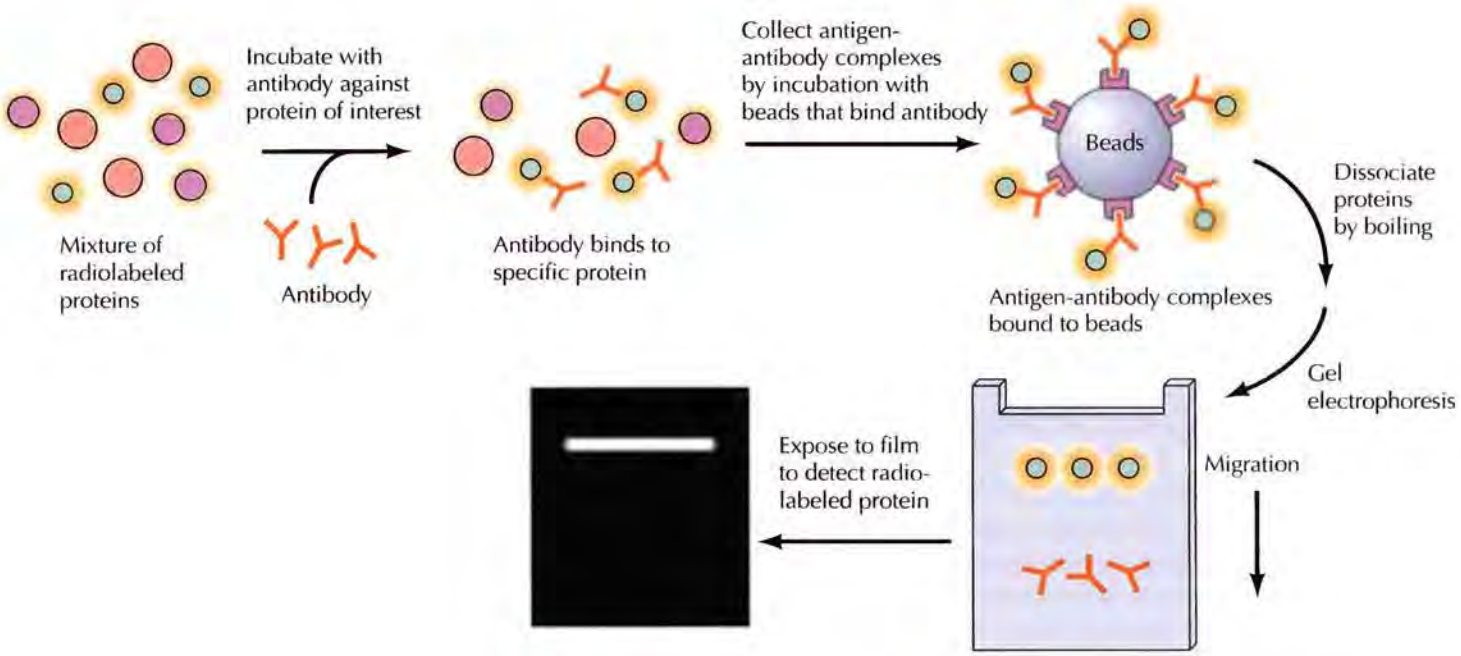


FIGURE 4.30 Immunoprecipitation Radiolabeled proteins are incubated with an antibody, which forms complexes with the protein against which it is directed (the antigen). These antigen-antibody complexes are collected on beads that bind the antibody. The beads are then boiled to dissociate the antigen-antibody complexes, and the recovered proteins are analyzed by SDS-polyacrylamide gel electrophoresis. The radioactive protein that was immunoprecipitated is detected by exposure to film.

4.13 WEBSITE ANIMATION

Monoclonal Antibodies

Monoclonal antibodies can be made by inoculating an animal (such as a mouse) with an antigen, providing the animal time to mount an immune response, collecting the B cells from the animal, and then fusing the B cells to long-lived myeloma cells—thereby creating hybridomas that can be tested for the production of useful antibodies.

variety of methods, such as chemiluminescence, to identify the protein against which the antibody is targeted.

In immunoprecipitation, antibodies are used to isolate the proteins against which they are directed (Figure 4.30). Typically, cells are incubated with radioactive amino acids to label their proteins. Such a radiolabeled cell extract is then incubated with an antibody, which binds to its antigenic target protein. The resulting antigen-antibody complexes are isolated and subjected to electrophoresis, allowing detection of the radioactive antigen.

Immunoprecipitation can also be used to detect protein-protein interactions within cells, by co-immunoprecipitation of two interacting proteins. As discussed in Chapter 2, one approach to identification of protein complexes is to immunoprecipitate a protein from cells under gentle conditions, so that it remains associated with the proteins it normally interacts with within the cell. The immunoprecipitated protein complexes can then be analyzed, for example by gel electrophoresis and mass spectrometry, to identify not only the protein against which the antibody was directed, but also other proteins with which it was associated in the cell extract.

As discussed in Chapter 1, antibodies can be used to visualize proteins in intact cells, as well as in cell lysates. For example, cells can be stained with antibodies labeled with fluorescent dyes, and the subcellular localization of the antigenic proteins visualized by fluorescence microscopy (Figure 4.31). Antibodies can also be labeled with tags that are visible in the electron microscope, such as heavy metals, allowing visualization of antigenic proteins at the ultrastructural level.

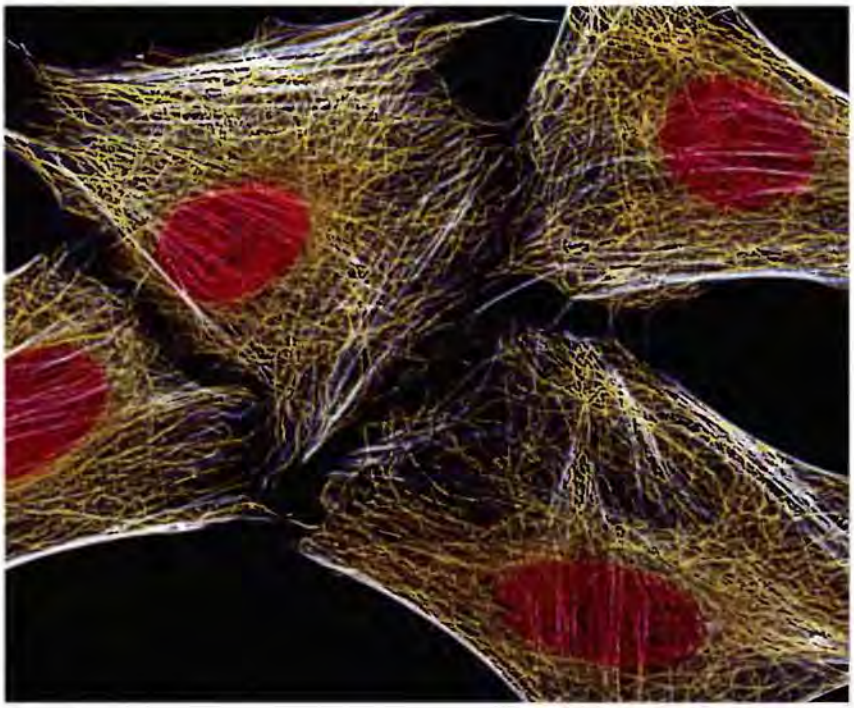


FIGURE 4.31 Immunofluorescence
Human cells in culture were stained with immunofluorescent antibodies against actin (blue) and tubulin (yellow). Nuclei are stained with a red fluorescent dye. (From Dr. Torsten Wittmann/Photo Researchers, Inc.)

Gene Function in Eukaryotes

The recombinant DNA techniques discussed in the preceding sections provide powerful approaches to the isolation and detailed characterization of the genes of eukaryotic cells. Understanding the function of a gene, however, requires analysis of the gene within cells or intact organisms—not simply as a molecular clone in bacteria. In classical genetics, the function of genes has generally been revealed by the altered phenotypes of mutant organisms. The advent of recombinant DNA has added a new dimension to studies of gene function. Namely, it has become possible to investigate the function of a cloned gene directly by reintroducing the cloned DNA into eukaryotic cells. In simpler eukaryotes, such as yeasts, this technique has made possible the isolation of molecular clones corresponding to virtually any mutant gene. In addition, there are several methods by which cloned genes can be introduced into cultured animal and plant cells, as well as intact organisms, for functional analysis. These approaches can be coupled with the ability to introduce mutations in cloned DNA *in vitro*, extending the power of recombinant DNA to allow functional studies of the genes of more complex eukaryotes.

Genetic Analysis in Yeasts

Yeasts are particularly advantageous for studies of eukaryotic molecular biology (see Chapter 1). The genome of *Saccharomyces cerevisiae*, which consists of approximately 1.2×10^7 base pairs, is 200 times smaller than the human genome. Moreover, yeasts can easily be grown in culture, reproducing with a division time of about 2 hours. Thus yeasts offer the same basic advantages—a small genome and rapid reproduction—that are afforded by bacteria.

Mutations in yeasts can be identified as readily as in *E. coli*. For example, yeast mutants that require a particular amino acid or other nutrient for growth can easily be isolated. In addition, yeasts with defects in genes required for fundamental cell processes (in contrast to metabolic defects) can

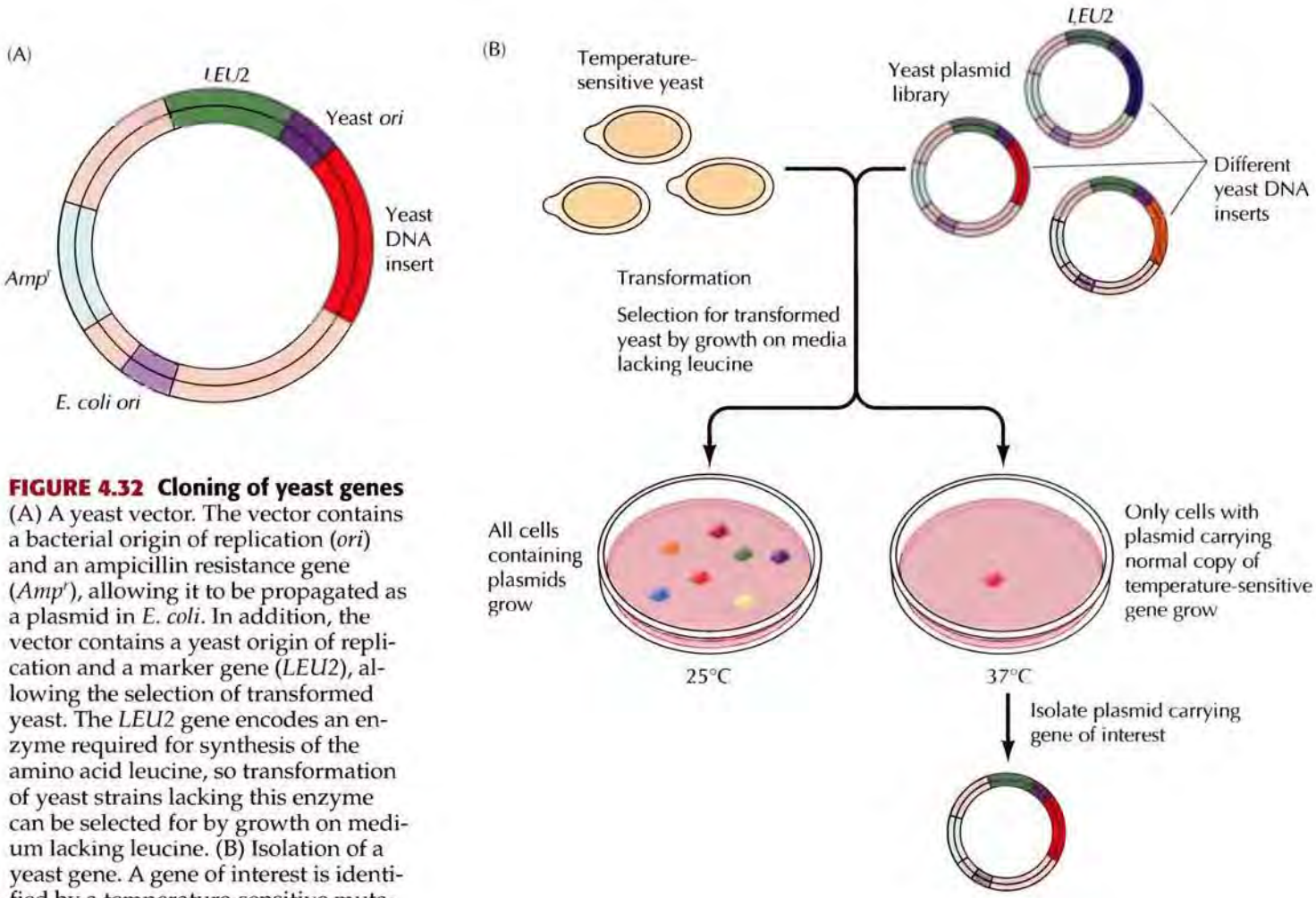


FIGURE 4.32 Cloning of yeast genes (A) A yeast vector. The vector contains a bacterial origin of replication (*ori*) and an ampicillin resistance gene (*Amp*^{r}), allowing it to be propagated as a plasmid in *E. coli*. In addition, the vector contains a yeast origin of replication and a marker gene (*LEU2*), allowing the selection of transformed yeast. The *LEU2* gene encodes an enzyme required for synthesis of the amino acid leucine, so transformation of yeast strains lacking this enzyme can be selected for by growth on medium lacking leucine. (B) Isolation of a yeast gene. A gene of interest is identified by a temperature-sensitive mutation, which allows yeast to grow at 25°C but not at 37°C. To isolate a clone of the gene, the temperature-sensitive yeasts are transformed with a plasmid library containing a collection of genes encompassing the entire yeast genome. All yeasts transformed by plasmid DNAs are able to grow on media lacking leucine at 25°C, but only those yeasts transformed by a plasmid carrying a normal copy of the gene of interest are able to grow at 37°C. The desired plasmid can be isolated from transformed yeasts that form colonies at the nonpermissive temperature.

be isolated as **temperature-sensitive mutants**. Such mutants encode proteins that are functional at one temperature (the permissive temperature) but not another (the nonpermissive temperature), whereas normal proteins are functional at both. A yeast with a temperature-sensitive mutation in an essential gene can be identified by its ability to grow only at the permissive temperature. The ability to isolate such temperature-sensitive mutants has allowed the identification of yeast genes controlling many fundamental cell processes, such as RNA synthesis and processing, progression through the cell cycle, and transport of proteins between cellular compartments.

The relatively simple genetics of yeast also enables a gene corresponding to any yeast mutation to be cloned, simply on the basis of its functional activity (Figure 4.32). First, a genomic library of normal yeast DNA is prepared in vectors that replicate as plasmids in yeasts as well as in *E. coli*. The small size of the yeast genome means that a complete library consists of only a few thousand plasmids. A mixture of such plasmids is then used to transform a temperature-sensitive yeast mutant, and transformants that are able to grow at the nonpermissive temperature are selected. Such transformants have acquired a normal copy of the gene of interest on plasmid DNA, which can then be easily isolated from the transformed yeast cells for further characterization.

Yeast genes encoding a wide variety of essential proteins have been identified in this manner. In many cases, such genes isolated from yeasts have

also been useful in identifying and cloning related genes from mammalian cells. Thus, the simple genetics of yeast has not only provided an important model for eukaryotic cells but has also led directly to the cloning of related genes from more complex eukaryotes.

Gene Transfer in Plants and Animals

Although the cells of complex eukaryotes are not amenable to the simple genetic manipulations possible in yeasts, gene function can still be assayed by the introduction of cloned DNA into plant and animal cells. Such experiments (generally called **gene transfer**) have proven critical to addressing a wide variety of questions, including studies of the mechanisms that regulate gene expression and protein processing. In addition, as discussed later in the book, gene transfer has enabled the identification and characterization of genes that control animal cell growth and differentiation, including a variety of genes responsible for the abnormal growth of human cancer cells.

The methodology for introduction of DNA into animal cells was initially developed for infectious viral DNAs and is therefore called **transfection** (a word derived from *trans*formation + in*fection*) (Figure 4.33). DNA can be introduced into animal cells in culture by a variety of methods, including direct microinjection into the cell nucleus, coprecipitation of DNA with calcium phosphate to form small particles that are taken up by the cells, incorporation of DNA into lipid vesicles (**liposomes**) that fuse with the plasma membrane, and exposure of cells to a brief electric pulse that transiently opens pores in the plasma membrane (**electroporation**). The DNA taken up by a high fraction of cells is transported to the nucleus, where it can be tran-

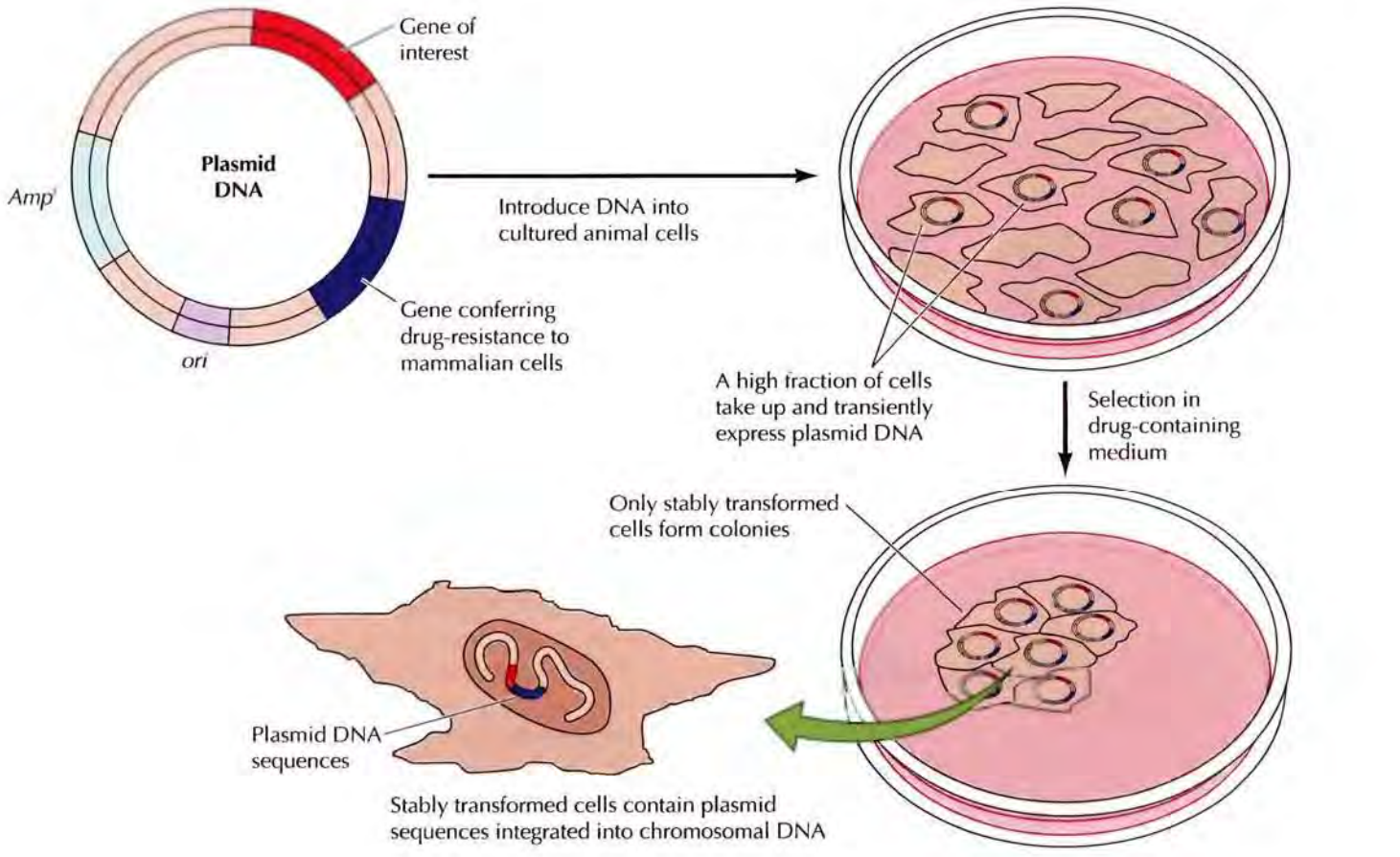


FIGURE 4.33 Introduction of DNA into animal cells A eukaryotic gene of interest is cloned in a plasmid containing a drug resistance marker that can be selected for in cultured animal cells. The plasmid DNA is taken up and expressed by a high fraction of the cells for a few days (transient expression). Stably transformed cells, in which the plasmid DNA becomes integrated into chromosomal DNA, can then be selected for their ability to grow in drug-containing medium.

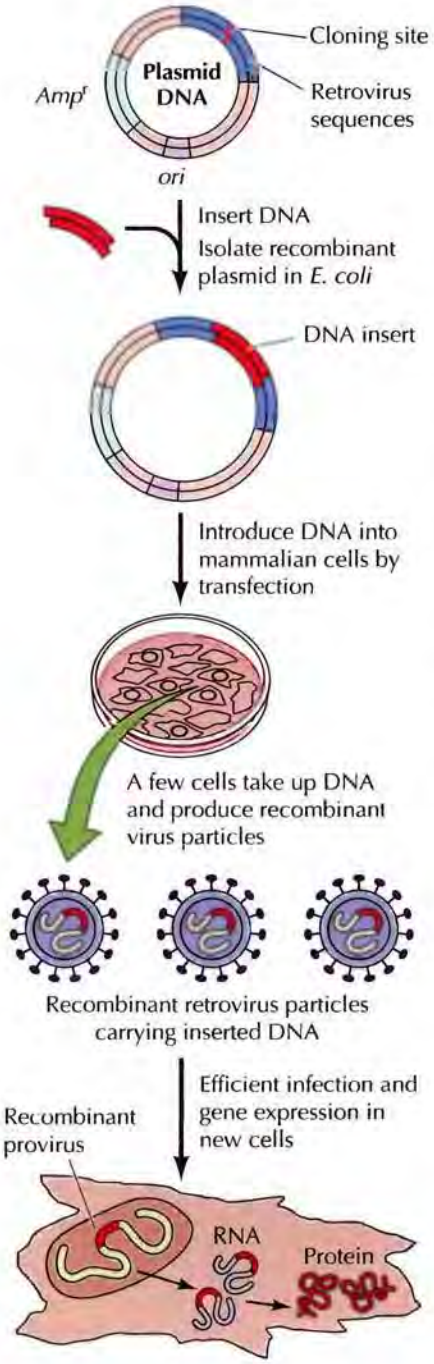


FIGURE 4.34 Retroviral vectors The vector consists of retroviral sequences cloned in a plasmid that can be propagated in *E. coli*. Foreign DNA is inserted into the viral sequences, and recombinant plasmids are isolated in bacteria. Animal cells in culture are then transfected with the recombinant DNA. The DNA is taken up by a small fraction of the cells, which produce recombinant retroviruses carrying the inserted DNA. These recombinant retroviruses can be used to efficiently infect new cells, where the viral genome carrying the inserted gene integrates into chromosomal DNA as a provirus.

scribed for several days—a phenomenon called **transient expression**. In a smaller fraction of cells (usually 1% or less), the foreign DNA becomes stably integrated into the cell genome and is transferred to progeny cells at cell division just as any other cell gene is. These stably transformed cells can be isolated if the transfected DNA contains a selectable marker, such as resistance to a drug that inhibits the growth of normal cells. Thus any cloned gene can be introduced into mammalian cells by being transferred together with a drug resistance marker that can be used to isolate stable transformants. The effects of such cloned genes on cell behavior—for example, cell growth or differentiation—can then be analyzed.

Animal viruses can also be used as vectors for more efficient introduction of cloned DNAs into cells. Retroviruses are particularly useful in this respect, since their life cycle involves the stable integration of viral DNA into the genome of infected cells (Figure 4.34). Consequently, retroviral vectors can be used to efficiently introduce cloned genes into a wide variety of cell types, making them an important vehicle for a broad range of applications.

Cloned genes can also be introduced into the germ line of multicellular organisms, allowing them to be studied in the context of the intact animal rather than in cultured cells. One method used to produce mice that carry such foreign genes (**transgenic mice**) is the direct microinjection of cloned DNA into the pronucleus of a fertilized egg (Figure 4.35). The injected eggs are then transferred to foster mothers and allowed to develop to term. In a fraction of the progeny mice (often about 10%), the foreign DNA will have integrated into the genome of the fertilized egg and is therefore present in all cells of the animal. Since the foreign DNA is present in germ cells as well as in somatic cells, it is transferred by breeding to new progeny mice just as any other cell gene would be.

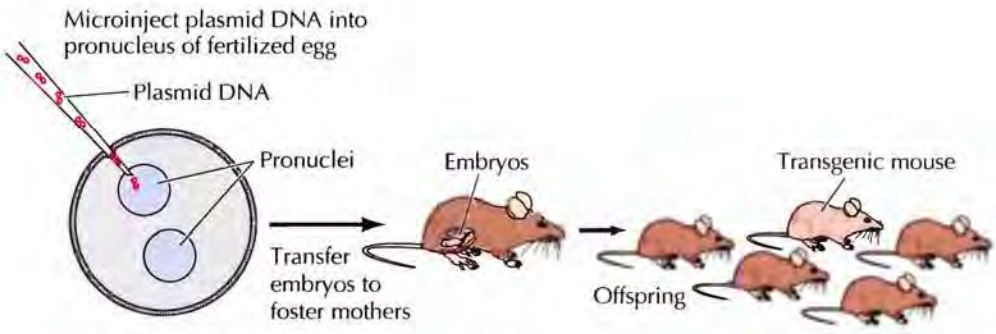


FIGURE 4.35 Production of transgenic mice DNA is microinjected into a pronucleus of a fertilized mouse egg (fertilized eggs contain two pronuclei, one from the egg and one from the sperm). The microinjected eggs are then transferred to foster mothers and allowed to develop. Some of the offspring (transgenic) have incorporated the injected DNA into their genome.

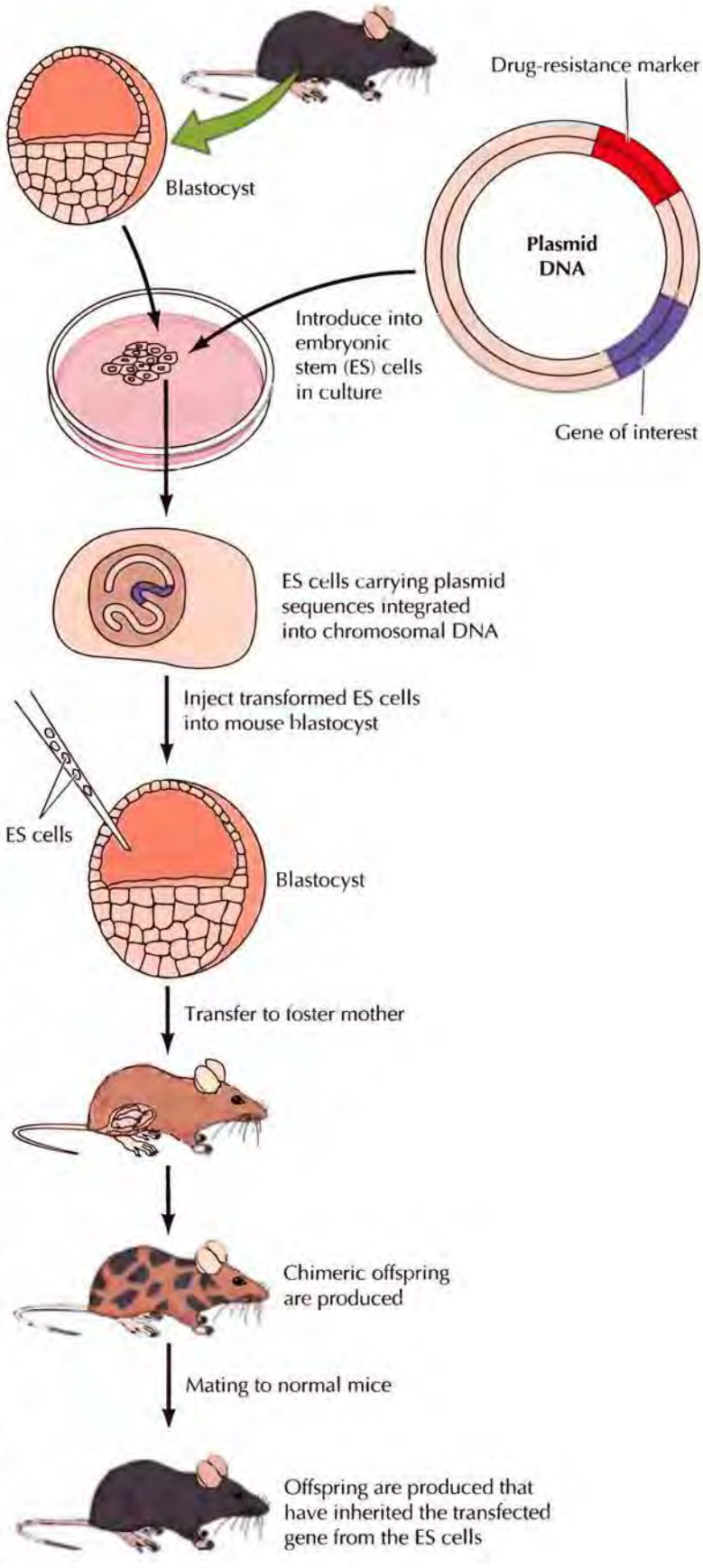


FIGURE 4.36 Introduction of genes into mice via embryonic stem cells Embryonic stem (ES) cells are cultured cells derived from early mouse embryos (blastocysts). DNA can be introduced into these cells in culture, and stably transformed ES cells can be isolated. These transformed ES cells can then be injected into a recipient blastocyst, where they are able to participate in normal development of the embryo. Some of the progeny mice that develop after transfer of injected embryos to foster mothers therefore contain cells derived from transformed ES cells, as well as from the normal cells of the blastocyst. Since these mice are a mixture of two different cell types, they are referred to as chimeric. Offspring carrying the transfected gene can then be produced by the breeding of chimeric mice in which descendants of the transformed ES cells have been incorporated into the germ line.

The properties of **embryonic stem (ES) cells** provide an alternative means of introducing cloned genes into mice (Figure 4.36). ES cells can be established in culture from early mouse embryos. They can also be reintroduced into early embryos, where they participate normally in development and can give rise to cells in all tissues of the mouse—including germ cells. It is thus possible to introduce cloned DNA into ES cells in culture, select stably transformed cells, and then introduce these cells back into mouse embryos. Such embryos give rise to chimeric offspring in which some cells are derived from the normal embryo cells and some from the transfected ES cells. In some such mice, the transfected ES cells are incorporated into the germ line. Breeding these mice therefore leads to the direct inheritance of the transfected gene by their progeny.

Cloned DNAs can also be introduced into plant cells. One approach is to bombard plant cells with DNA-coated microprojectiles, such as small particles of tungsten. The DNA-coated particles are shot directly into the plant cells; some of the cells are killed, but others survive and become stably transformed. In addition, a plasmid from the bacterium *Agrobacterium tumefaciens* (the **Ti plasmid**) provides a novel vehicle for the introduction of cloned DNAs into many species of plants (Figure 4.37). In nature, *Agrobacterium* attaches to the leaves of plants and the Ti plasmid is transferred into plant cells where it becomes incorporated into chromosomal DNA. Vectors developed from the Ti plasmid therefore provide an efficient means of introducing recombinant DNA into sensitive plant cells. Since many plants can be regenerated from single cultured cells (see Chapter 1), transgenic plants can be established directly from cells into which recombinant DNA has been introduced in culture—a much simpler proce-

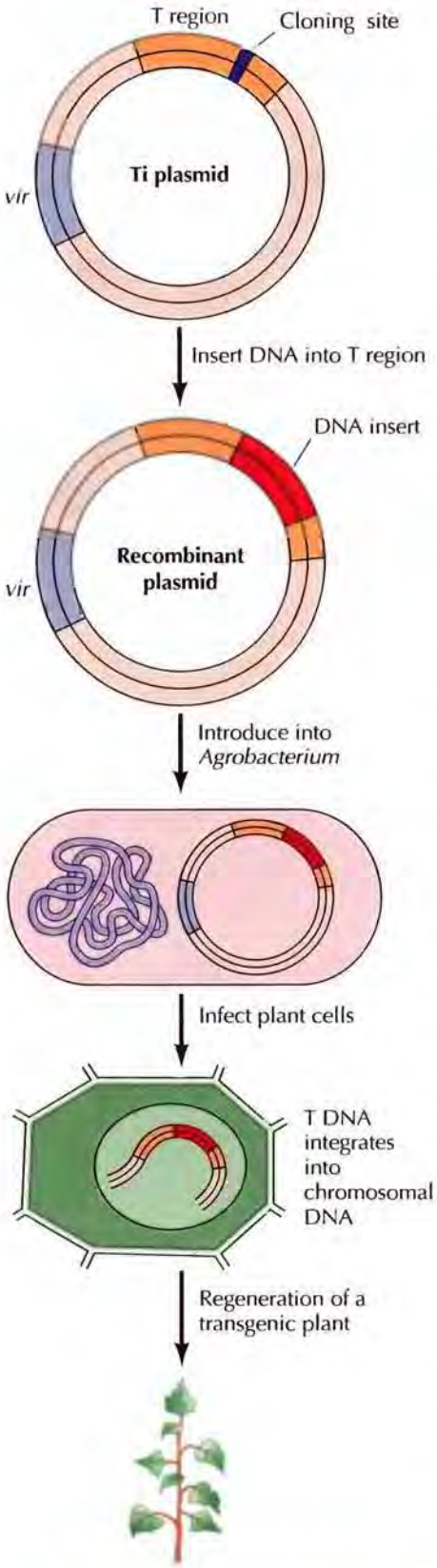


FIGURE 4.37 Introduction of genes into plant cells via the Ti plasmid The Ti plasmid contains the T region, which is transferred to infected plant cells, and virulence (*vir*) genes, which function in T DNA transfer. In Ti plasmid vectors, foreign DNA is inserted into the T region. The recombinant plasmid is introduced into *Agrobacterium tumefaciens*, which is then used to infect cultured cells. The T region of the plasmid (carrying the inserted DNA) is transferred to the plant cells and integrates into chromosomal DNA. A transgenic plant can then be generated from the transformed cells.

dure than the production of transgenic animals. Indeed, many economically important types of plants, including corn, tomatoes, soybeans and potatoes, are transgenic varieties.

Mutagenesis of Cloned DNAs

In classical genetic studies (e.g., in bacteria or yeasts), mutants are the key to identifying genes and understanding their function by observing the altered phenotype of mutant organisms. In such studies, mutant genes are detected because they result in observable phenotypic changes—for example, temperature-sensitive growth or a specific nutritional requirement. The isolation of genes by recombinant DNA, however, has opened a different approach to mutagenesis. It is now possible to introduce any desired alteration into a cloned gene and to determine the effect of the mutation on gene function. Such procedures have been called **reverse genetics**, since a mutation is introduced into a gene first and its functional consequence is determined second. The ability to introduce specific mutations into cloned DNAs (***in vitro* mutagenesis**) has proven to be a powerful tool in studying the expression and function of eukaryotic genes.

Cloned genes can be altered by many *in vitro* mutagenesis procedures, which can lead to the introduction of deletions, insertions, or single nucleotide alterations. The most common method of mutagenesis is the use of synthetic oligonucleotides to generate nucleotide changes in a DNA sequence (Figure 4.38). In this procedure a synthetic oligonucleotide bearing the desired mutation is used as a primer for DNA synthesis. Newly synthesized DNA molecules containing the mutation can then be isolated and characterized. For example, specific amino acids of a protein can be altered in order to characterize their role in protein function.

Variations of this approach, combined with the versatility of other methods for manipulating recombinant DNA molecules, can be used to introduce virtually any desired alteration in a cloned gene. The effects of such mutations on gene expression and function can then be determined by introduction of the gene into an appropriate cell type. *In vitro* mutagenesis has thus allowed detailed characterization of the functional roles of both the regulatory and protein-coding sequences of cloned genes.

Introducing Mutations into Cellular Genes

Although the transfer of cloned genes into cells (particularly in combination with *in vitro* mutagenesis) provides a powerful approach to studying gene structure and function, such experiments fall short of defining the role of an unknown gene in a cell or intact organism. The cells used as recipients for transfer of cloned genes usually already have normal copies of the gene in their chromosomal DNAs, and these normal copies continue to perform their roles in the cell. Determining the biological role of a cloned gene therefore requires that the activity of the normal cellular gene copies be eliminated. Several approaches can be used to either inac-

■ **The United States is the largest producer of genetically modified plants, with 49.8 million hectares growing genetically modified crops. The first genetically modified plant approved for consumption by the FDA was the Flavr Savr tomato, which had been modified to delay ripening.**

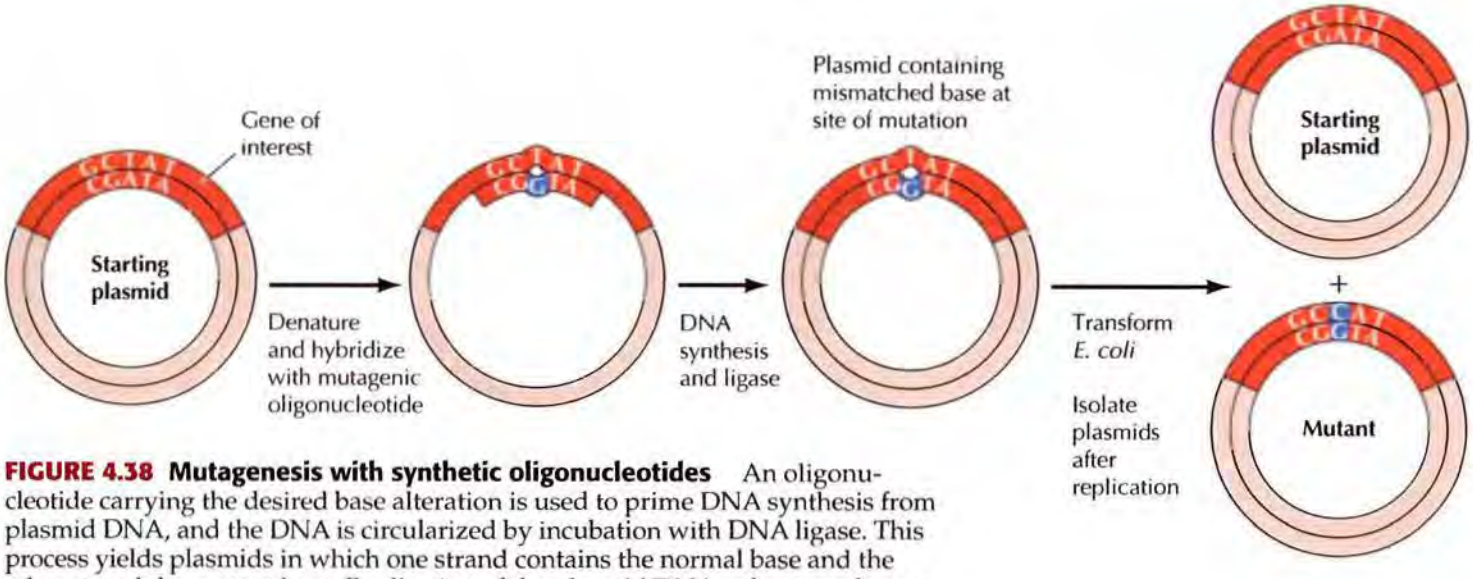


FIGURE 4.38 Mutagenesis with synthetic oligonucleotides An oligonucleotide carrying the desired base alteration is used to prime DNA synthesis from plasmid DNA, and the DNA is circularized by incubation with DNA ligase. This process yields plasmids in which one strand contains the normal base and the other strand the mutant base. Replication of the plasmid DNAs after transformation of *E. coli* therefore yields a mixture of both starting and mutant plasmids.

tivate the chromosomal copies of a cloned gene or inhibit normal gene function, both in cultured cells and in transgenic mice.

Mutating chromosomal genes is based on the ability of a cloned gene introduced into a cell to undergo **homologous recombination** with its chromosomal copy (Figure 4.39). In homologous recombination, the cloned gene replaces the normal allele, so mutations introduced into the cloned gene *in vitro* become incorporated into the chromosomal copy of the gene. In the simplest case, mutations that inactivate the cloned gene can be introduced in place of the normal gene copy in order to determine its role in cellular processes.

Recombination between transferred DNA and the homologous chromosomal gene occurs frequently in yeast but is a rare event in mammalian cells, so inactivation of mammalian genes by this approach is technically difficult. Possibly because the genomes of mammalian cells are so much larger than that of yeasts, most transfected DNA that integrates into the recipient cell genome does so at random sites by recombination with unrelated sequences. However, various procedures have been developed both to increase the frequency of homologous recombination and to select and isolate the transformed cells in which homologous recombination has occurred, so it is feasible to inactivate any desired gene in mammalian cells by this approach. Importantly, genes can be readily inactivated in mouse embryonic stem cells, which can then be used to generate transgenic mice (Figure 4.40). These mice can be bred to yield progeny containing mutated copies of the targeted gene on both homologous chromosomes, so the effects of inactivation of a gene can be investigated in the context of the intact

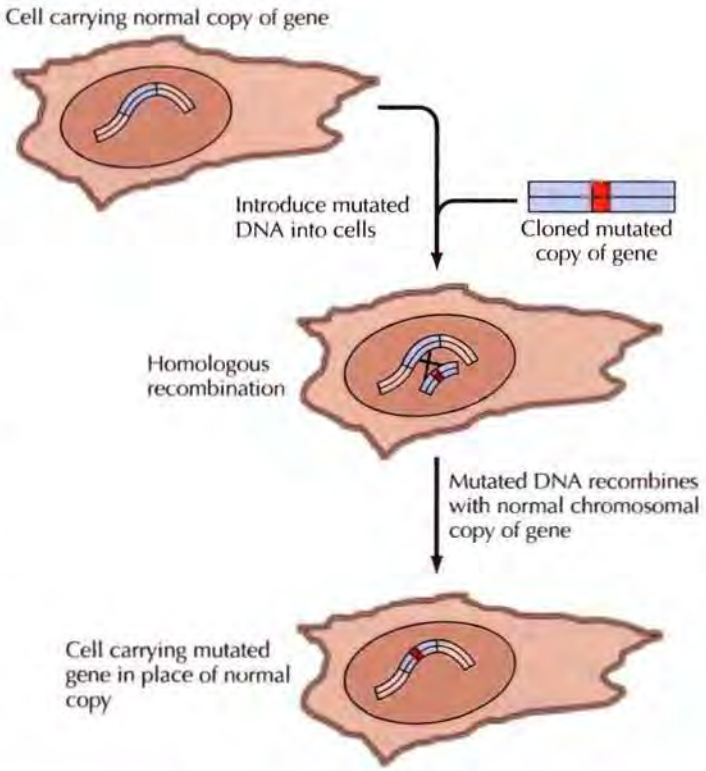


FIGURE 4.39 Gene inactivation by homologous recombination A mutated copy of the cloned gene is introduced into cultured animal cells. The cloned gene may then replace the normal gene copy by homologous recombination, yielding a cell carrying the desired mutation in its chromosomal DNA.

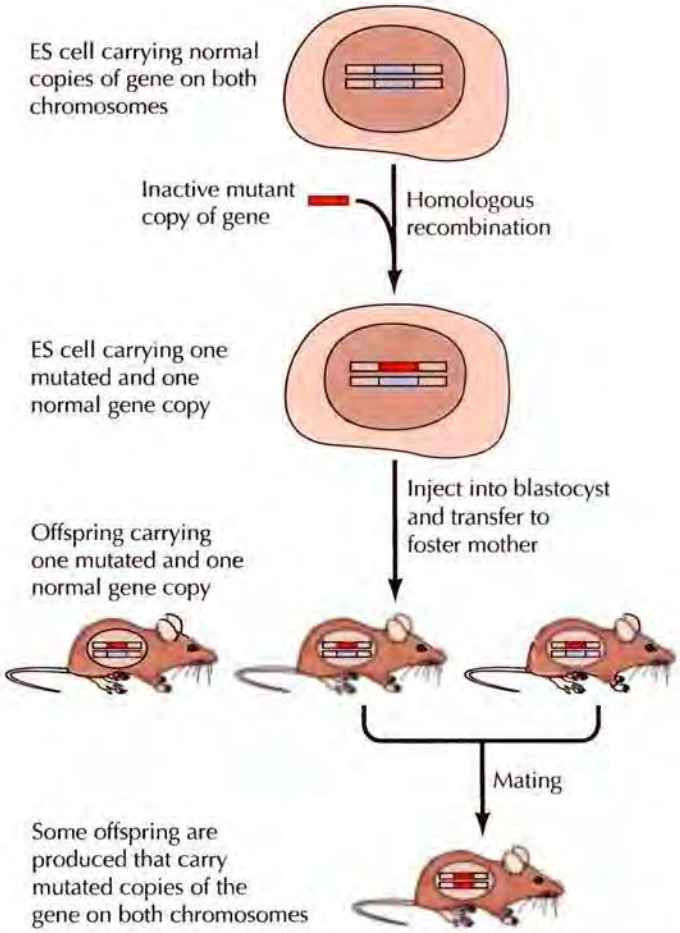


FIGURE 4.40 Production of mutant mice by homologous recombination in ES cells Homologous recombination is used to replace a normal gene with an inactive mutant copy in embryonic stem cells (ES cells), resulting in ES cells carrying one normal and one inactive copy of the gene on homologous chromosomes. These ES cells can be transferred to blastocysts and used to generate mice as described in Figure 4.36. These mice, carrying one mutant and one normal gene copy, can then be mated to produce offspring carrying mutant copies of the gene on both chromosomes.

animal. In addition, cells can be cultured from mouse embryos containing the mutated gene copies, so the functions of target genes can also be studied in cell culture. The biological activities of over 2000 mouse genes have been investigated in this way, and such studies have been critically important in revealing the roles of many genes in mouse development.

Homologous recombination has been used to systematically inactivate (or **knockout**) every gene in yeast, so a collection of genome-wide yeast mutants is available for scientists to use to study the function of any desired gene. In mice, approximately 10% of genes have currently been knocked out by homologous recombination. However, a large scale project has been proposed to systematically knockout all mouse genes, potentially yielding a genome-wide collection of mutant mice that would provide a major resource for researchers interested in mammalian development and cell function. Methods have also been developed to conditionally knockout genes in specific mouse tissues, allowing the function of a gene to be studied in a defined cell type (for example, in nerve cells) rather than in all cells of the organism.

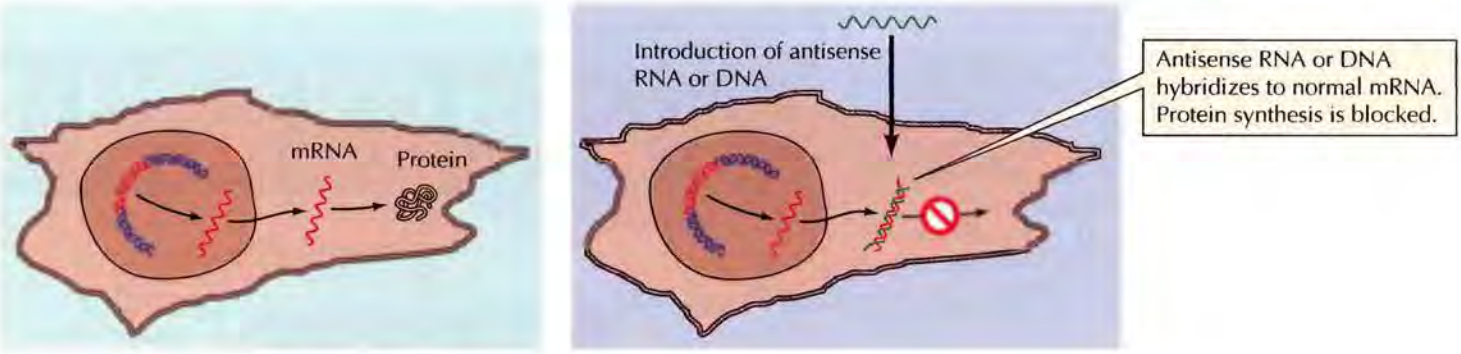


FIGURE 4.41 Inhibition of gene expression by antisense RNA or DNA
Antisense RNA or single-stranded DNA is complementary to the mRNA of a gene of interest. Antisense nucleic acids therefore form hybrids with their target mRNAs, blocking the translation of mRNA into protein and leading to mRNA degradation.

Interfering with Cellular Gene Expression

As an alternative to gene inactivation by homologous recombination, a variety of approaches can be used to specifically interfere with gene expression or function. One method that has been used to inhibit expression of a desired target gene is the introduction of **antisense nucleic acids** into cultured cells (Figure 4.41). RNA or single-stranded DNA complementary to the mRNA of the gene of interest (antisense) hybridizes with the mRNA and blocks its translation into protein. Moreover, the RNA-DNA hybrids resulting from the introduction of antisense DNA molecules are usually degraded within the cell. Antisense RNAs can be introduced directly into cells, or cells can be transfected with vectors that have been engineered to express antisense RNA. Antisense DNA is usually in the form of short oligonucleotides (about 20 bases long), which can either be transfected into cells or, in many cases, taken up by cells directly from the culture medium.

In the last several years, **RNA interference (RNAi)** has emerged as an extremely effective and widely used method for interfering with gene expression at the mRNA level. RNA interference was first discovered in *C. elegans* in 1998, when Andrew Fire, Craig Mello and their colleagues found that injection of double-stranded RNA inhibited expression of a gene with a complementary mRNA sequence. This unexpected observation demonstrated an unanticipated role for double-stranded RNAs in gene regulation, which has subsequently been developed into a powerful experimental tool for inhibiting expression of target genes. When double-stranded RNAs are introduced into cells, they are cleaved into short double-stranded molecules (21–23 nucleotides) by an enzyme called Dicer (Figure 4.42). These short double-stranded molecules, called short interfering RNAs (siRNAs), then associate with a complex of proteins known as the RNA-induced silencing complex (RISC). Within this complex, the two strands of siRNA separate and the strand complementary to the mRNA (the antisense strand) guides the complex to the target mRNA by complementary base pairing. The mRNA is then cleaved by one of the RISC proteins. The RISC-siRNA complex is released following degradation of the mRNA and can continue to participate in multiple rounds of mRNA cleavage, leading to effective destruction of the targeted mRNA.

RNAi has been established as a potent method for interfering with gene expression in *C. elegans*, *Drosophila*, *Arabidopsis*, and mammalian cells, and provides a relatively straightforward approach to investigating the function

■ **Both antisense oligonucleotides and siRNAs are being investigated as potential therapeutic agents. The antisense oligonucleotide fomiversen has been approved by the FDA for use in the treatment of eye infections caused by a herpesvirus. An siRNA is currently in clinical trial for the treatment of age related macular degeneration.**

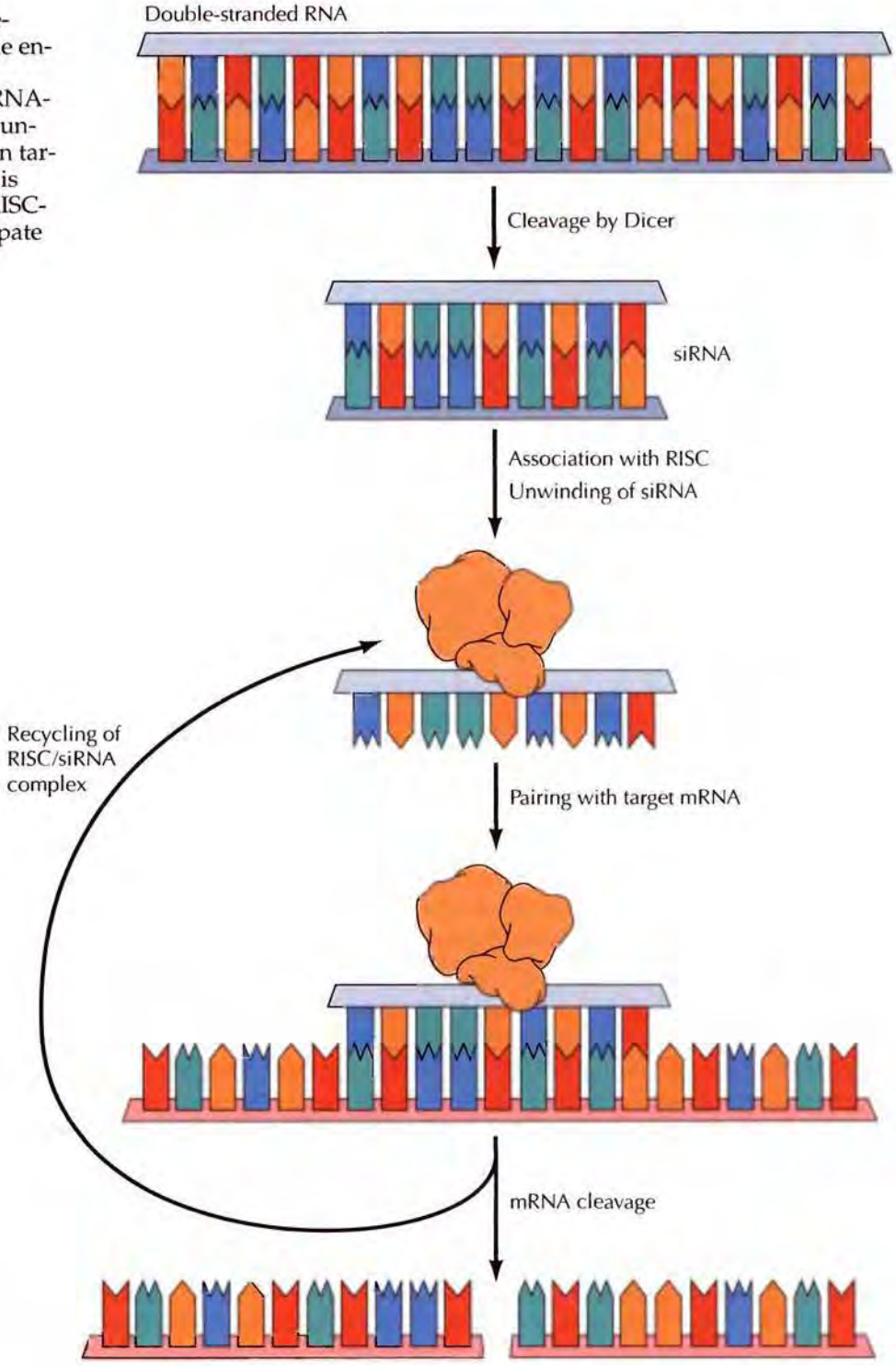


FIGURE 4.42 RNA interference Double-stranded RNA molecules are cleaved by the enzyme Dicer into short interfering RNAs (siRNAs). The siRNAs associate with the RNA-induced silencing complex (RISC) and are unwound. The antisense strand of siRNA then targets RISC to a homologous mRNA, which is cleaved by one of the RISC proteins. The RISC-siRNA complex is released and can participate in further cycles of mRNA degradation.

of any gene whose sequence is known. In addition, libraries of double-stranded RNAs or siRNAs that cover a large fraction of genes in the genome are being used to screen *C. elegans*, *Drosophila*, and human cells to identify novel genes involved in specific biological functions, such as cell growth or cell survival. Finally, RNAi is not only an experimental tool: as will be discussed in Chapters 7 and 8, RNA interference is also a major regulatory mechanism used by cells to control gene expression at both the transcriptional and translational levels.

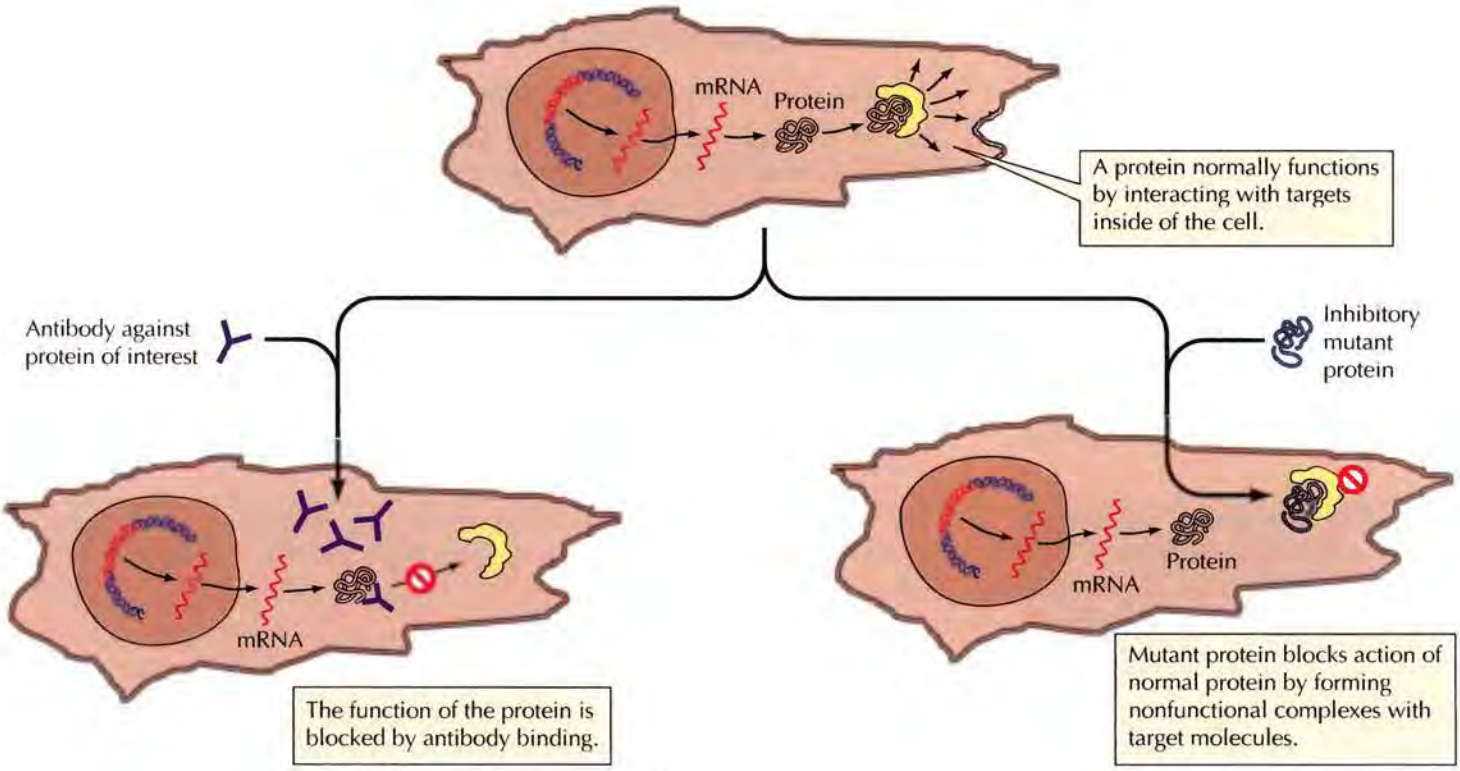


FIGURE 4.43 Direct inhibition of protein function Microinjected antibodies can bind to proteins within cells, thereby inhibiting their normal function. In addition, some mutant proteins interfere with the function of normal proteins—for example, by competing with the normal protein for interaction with target molecules.

In addition to inactivating a gene or inducing degradation of an mRNA, it is sometimes possible to interfere directly with the function of proteins within cells (Figure 4.43). One approach is to microinject antibodies that block the activity of the protein against which they are directed. Alternatively, some mutant proteins interfere with the function of their normal counterparts when they are expressed within the same cell—for example, by competing with the normal protein for binding to its target molecule. Cloned DNAs encoding such mutant proteins (called **dominant inhibitory mutants**) can be introduced into cells by gene transfer and used to study the effects of blocking normal gene function.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

HEREDITY, GENES, AND DNA

Genes and Chromosomes: Chromosomes are the carriers of genes.

Genes and Enzymes: A gene specifies the amino acid sequence of a polypeptide chain.

Identification of DNA as the Genetic Material: DNA was identified as the genetic material by bacterial transformation experiments.

KEY TERMS

gene, allele, dominant, recessive, genotype, phenotype, chromosome, diploid, meiosis, haploid, mutation

one gene–one enzyme hypothesis

transformation

KEY TERMS

SUMMARY

The Structure of DNA: DNA is a double helix in which hydrogen bonds form between purines and pyrimidines on opposite strands. Because of specific base pairing—A with T and G with C—the two strands of a DNA molecule are complementary in sequence.

semiconservative replication, DNA polymerase

Replication of DNA: DNA replicates by semiconservative replication in which the two strands separate and each serves as a template for synthesis of a new progeny strand.

EXPRESSION OF GENETIC INFORMATION

Colinearity of Genes and Proteins: The order of nucleotides in DNA specifies the order of amino acids in proteins.

central dogma, transcription, translation, messenger RNA (mRNA), RNA polymerase, ribosomal RNA (rRNA), transfer RNA (tRNA)

The Role of Messenger RNA: Messenger RNA functions as an intermediate to convey information from DNA to the ribosomes, where it serves as a template for protein synthesis.

genetic code, *in vitro* translation, codon

The Genetic Code: Transfer RNAs serve as adaptors between amino acids and mRNA during translation. Each amino acid is specified by a codon consisting of three nucleotides.

retrovirus, reverse transcription, reverse transcriptase

RNA Viruses and Reverse Transcription: DNA can be synthesized from RNA templates, as first discovered in retroviruses.

RECOMBINANT DNA

restriction endonuclease, gel electrophoresis, restriction map

Restriction Endonucleases: Restriction endonucleases cleave specific DNA sequences, yielding defined fragments of DNA molecules.

molecular cloning, vector, recombinant molecule, molecular clone, plasmid, DNA ligase, cDNA

Generation of Recombinant DNA Molecules: Recombinant DNA molecules consist of a DNA fragment of interest ligated to a vector that is able to replicate independently in an appropriate host cell.

origin of replication, cosmid, P1 artificial chromosome (PAC), bacterial artificial chromosome (BAC), yeast artificial chromosome (YAC)

Vectors for Recombinant DNA: A variety of vectors are used to clone different sizes of DNA fragments.

dideoxynucleotide

DNA Sequencing: The nucleotide sequences of cloned DNA fragments can be readily determined.

expression vector, yeast two-hybrid

Expression of Cloned Genes: The proteins encoded by cloned genes can be expressed at high levels in either bacteria or eukaryotic cells.

SUMMARY

DETECTION OF NUCLEIC ACIDS AND PROTEINS

Amplification of DNA by the Polymerase Chain Reaction: PCR allows the amplification and isolation of specific fragments of DNA *in vitro*, providing a sensitive method for detecting small amounts of specific DNA or RNA molecules.

polymerase chain reaction (PCR)

Nucleic Acid Hybridization: Nucleic acid hybridization allows the detection of specific DNA or RNA sequences by base pairing between complementary strands.

nucleic acid hybridization, Southern blotting, Northern blotting, recombinant DNA library, DNA microarray, *in situ* hybridization

Antibodies as Probes for Proteins: Antibodies are used to detect specific proteins in cells or cell extracts.

antibody, antigen, monoclonal antibody, immunoblotting, Western blotting, immunoprecipitation, SDS-polyacrylamide gel electrophoresis (SDS-PAGE)

GENE FUNCTION IN EUKARYOTES

Genetic Analysis in Yeasts: The simple genetics and rapid replication of yeasts facilitate the molecular cloning of a gene corresponding to any yeast mutation.

temperature-sensitive mutant

Gene Transfer in Plants and Animals: Cloned genes can be introduced into complex eukaryotic cells and multicellular organisms for functional analysis.

gene transfer, transfection, liposome, electroporation, transient expression, transgenic mice, embryonic stem (ES) cell, Ti plasmid

Mutagenesis of Cloned DNAs: *In vitro* mutagenesis of cloned DNAs is used to study the effect of engineered mutations on gene function.

reverse genetics, *in vitro* mutagenesis

Introducing Mutations into Cellular Genes: Mutations can be introduced into chromosomal gene copies by homologous recombination with cloned DNA sequences.

homologous recombination, knockout, RNA interference (RNAi), dominant inhibitory mutant

Interfering with Cellular Gene Expression: The expression or function of specific genes can be blocked by antisense nucleic acids, RNA interference, or dominant inhibitory mutants.

antisense nucleic acid, RNA interference

KEY TERMS

Questions

1. How would you determine whether two different genes are linked in *Drosophila*?

2. You grow *E. coli* for several generations in media containing ^{15}N. The cells are then transferred to media containing ^{14}N and grown for two additional generations. What proportion of the DNA isolated from these cells will be heavy, light, or intermediate in density?

3. Addition or deletion of one or two nucleotides in the coding sequence of a gene produces a nonfunctioning protein, whereas addition or deletion of three nucleotides often produces a protein with nearly normal function. Explain.

4. Describe the features that a yeast artificial chromosome must have in order to be used to clone a piece of human DNA cut with *Eco*RI.

5. You are studying an enzyme in which an active-site cysteine residue is encoded by the triplet UGU. How would mutating the third base to a C affect enzyme function? How about mutating it to an A?

6. Digestion of a 4-kb DNA molecule with *Eco*RI yields two fragments of 1 kb and 3 kb each. Digestion of the same molecule with *Hind*III yields fragments of 1.5 kb and 2.5 kb. Finally, digestion with *Eco*RI and *Hind*III in combination yields fragments of 0.5 kb, 1 kb, and 2.5 kb. Draw a restriction map indicating the positions of the *Eco*RI and *Hind*III cleavage sites.

7. Starting with DNA from a single sperm, how many copies of a specific gene sequence will be obtained after 10 cycles of PCR amplification? After 30 cycles?

8. You have cloned a fragment of human genomic DNA in a cosmid vector; approximately how many times would you expect the insert to be cut by the restriction enzyme *Bam*HI?

9. What is the minimum number of BAC clones required to make a genomic library of human DNA?

10. How would you expect actinomycin D to affect the replication of influenza virus?

11. What is the critical feature of a cloning vector that would allow you to isolate stably transfected mammalian cells?

12. Nucleic acids have a net negative charge and can be separated by gel electrophoresis on the basis of their size. In contrast, different proteins have different charges. How then can proteins be separated on the basis of size by electrophoresis?

References and Further Reading

Heredity, Genes, and DNA

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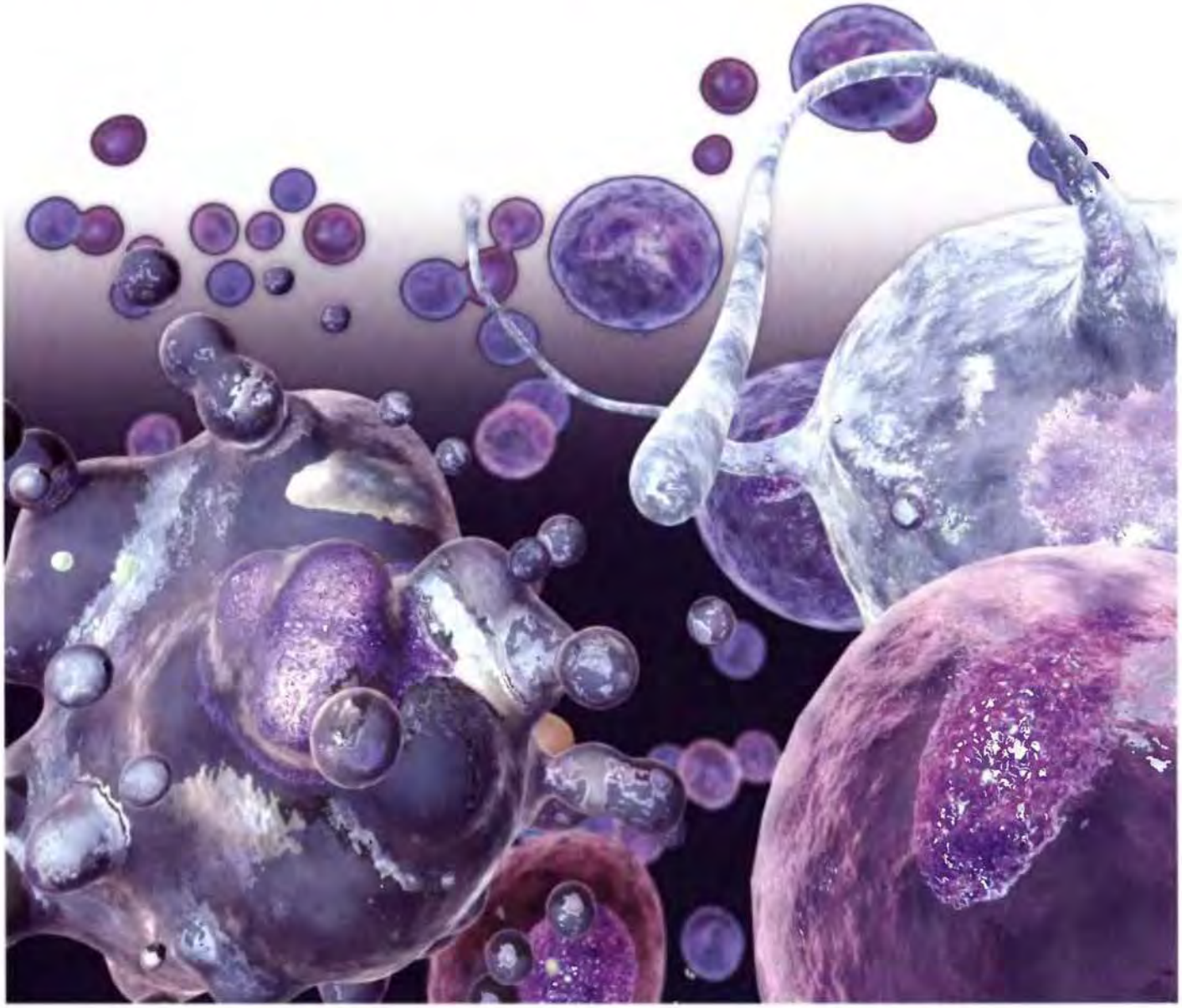
PART II

The Flow of Genetic Information



CHAPTER 5 ■ The Organization and Sequences of Cellular Genomes

CHAPTER 6 ■ Replication, Maintenance, and Rearrangements of Genomic DNA

CHAPTER 7 ■ RNA Synthesis and Processing

CHAPTER 8 ■ Protein Synthesis, Processing, and Regulation

CHAPTER 5

The Organization and Sequences of Cellular Genomes

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AS THE GENETIC MATERIAL, DNA PROVIDES A BLUEPRINT that directs all cellular activities and specifies the developmental plan of multicellular organisms. An understanding of gene structure and function is therefore fundamental to an appreciation of the molecular biology of cells. The development of gene cloning represented a major step toward this goal, enabling scientists to dissect complex eukaryotic genomes and probe the functions of eukaryotic genes. Continuing advances in recombinant DNA technology have now brought us to the exciting point of determining the sequences of entire genomes, providing a new approach to deciphering the genetic basis of cell behavior.

As reviewed in Chapter 4, the initial applications of recombinant DNA were directed toward the isolation and analysis of individual genes. More recently, large scale sequencing projects have yielded the complete genome sequences of many bacteria, of yeast, and of several species of plants and animals, including humans. The sequences of these complete cellular genomes provide a rich harvest of information, enabling the identification of many hitherto unknown genes and regulatory sequences. The results of these genome sequencing projects can be expected to stimulate many years of future research in molecular and cellular biology, and to have a profound impact on our understanding and treatment of human disease.

The Complexity of Eukaryotic Genomes

The genomes of most eukaryotes are larger and more complex than those of prokaryotes (Figure 5.1). This larger size of eukaryotic genomes is not inherently surprising, since one would expect to find more genes in organisms that are more complex. However, the genome size of many eukaryotes does not appear to be related to genetic complexity. For example, the genomes of salamanders and lilies contain more than ten times the

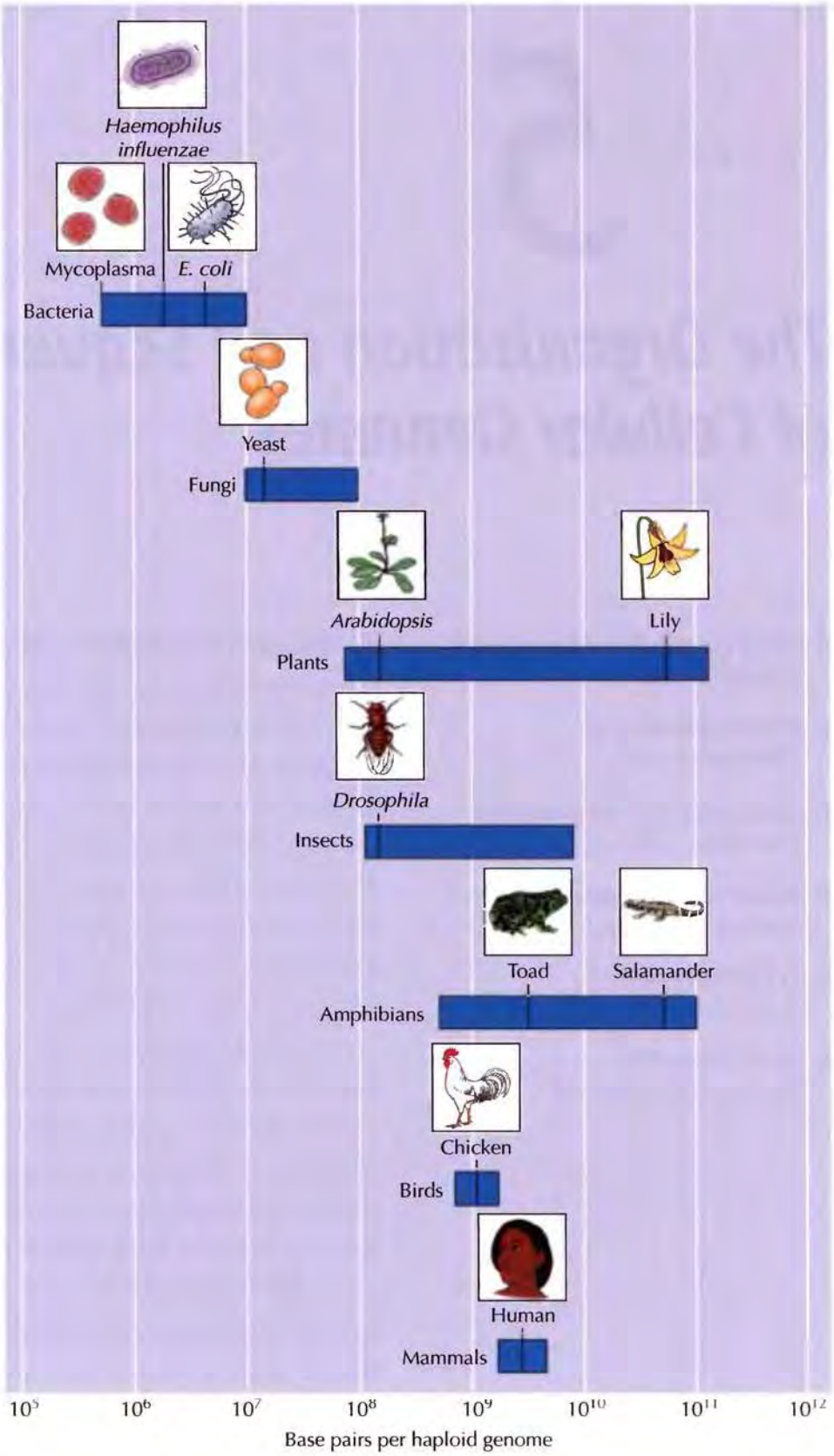


FIGURE 5.1 Genome size The range of sizes of the genomes of representative groups of organisms is shown on a logarithmic scale.

amount of DNA that is in the human genome, yet these organisms are clearly not ten times more complex than humans.

This apparent paradox was resolved by the discovery that the genomes of most eukaryotic cells contain not only functional genes but also large amounts of DNA sequences that do not code for proteins. The difference in the sizes of the salamander and human genomes thus reflects larger

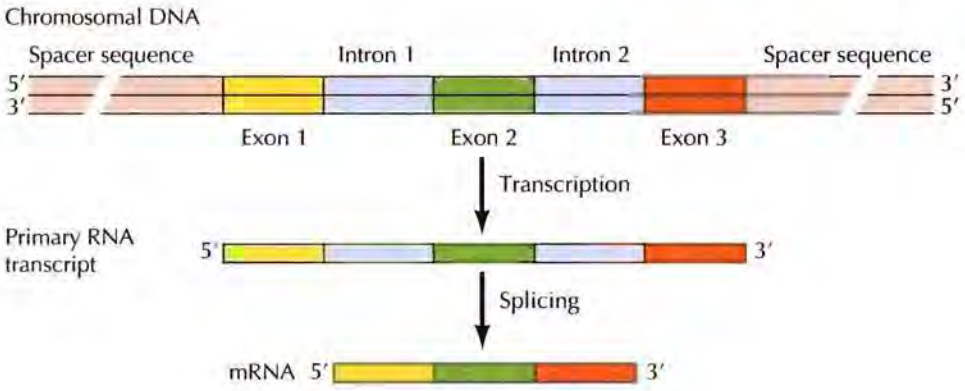


FIGURE 5.2 The structure of eukaryotic genes Most eukaryotic genes contain segments of coding sequences (exons) interrupted by noncoding sequences (introns). Both exons and introns are transcribed to yield a long primary RNA transcript. The introns are then removed by splicing to form the mature mRNA.

amounts of noncoding DNA, rather than more genes, in the genome of the salamander. The presence of large amounts of noncoding sequences is a general property of the genomes of complex eukaryotes. Thus the thousandfold greater size of the human genome compared to that of *E. coli* is not due solely to a larger number of human genes. The human genome is thought to contain 20,000–25,000 genes—only about 5 times more than *E. coli* has. Much of the complexity of eukaryotic genomes thus results from the abundance of several different types of noncoding sequences, which constitute most of the DNA of higher eukaryotic cells.

Introns and Exons

In molecular terms, a **gene** can be defined as a segment of DNA that is expressed to yield a functional product, which may be either an RNA (e.g., ribosomal and transfer RNAs) or a polypeptide. Some of the noncoding DNA in eukaryotes is accounted for by long DNA sequences that lie between genes (**spacer sequences**). However, large amounts of noncoding DNA are also found within most eukaryotic genes. Such genes have a split structure in which segments of coding sequence (called **exons**) are separated by noncoding sequences (intervening sequences, or **introns**) (Figure 5.2). The entire gene is transcribed to yield a long RNA molecule and the introns are then removed by splicing, so only exons are included in the mRNA. Although most introns have no known function, they account for a substantial fraction of DNA in the genomes of higher eukaryotes.

Introns were first discovered in 1977, independently in the laboratories of Phillip Sharp and Richard Roberts, during studies of the replication of adenovirus in cultured human cells. Adenovirus is a useful model for studies of gene expression, both because the viral genome is only about 3.5×10^4 base pairs long and because adenovirus mRNAs are produced at high levels in infected cells. One approach used to characterize the adenovirus mRNAs was to determine the locations of the corresponding viral genes by examination of RNA-DNA hybrids in the electron microscope. Because RNA-DNA hybrids are distinguishable from single-stranded DNA, the positions of RNA transcripts on a DNA molecule can be determined. Surprisingly, such experiments revealed that adenovirus mRNAs do not hybridize to only a single region of viral DNA (Figure 5.3). Instead, a single mRNA molecule hybridizes to several separated regions of the viral genome. Thus the adenovirus mRNA does not correspond to an uninterrupted transcript of the template DNA; rather the mRNA is assembled from several distinct blocks of sequences that originated from different parts of the viral DNA. This was subsequently shown to occur by **RNA splicing**, which will be discussed in detail in Chapter 7.

KEY EXPERIMENT

The Discovery of Introns

Spliced Segments at the 5′ Terminus of Adenovirus 2 Late mRNA

Susan M. Berget, Claire Moore, and Phillip A. Sharp
Massachusetts Institute of Technology, Cambridge, Massachusetts
Proceedings of the National Academy of Sciences USA, Volume 74, 1977, pages 3171–3175

Phillip Sharp

Richard Roberts

The Context

Prior to molecular cloning, little was known about mRNA synthesis in eukaryotic cells. However, it was clear that this process is more complex in eukaryotes than in bacteria. The synthesis of eukaryotic mRNAs appeared to require not only transcription but also processing reactions that modify the structure of primary transcripts. Most notably, eukaryotic mRNAs appeared to be synthesized as long primary transcripts, found in the nucleus, which were then cleaved to yield much shorter mRNA molecules that were exported to the cytoplasm.

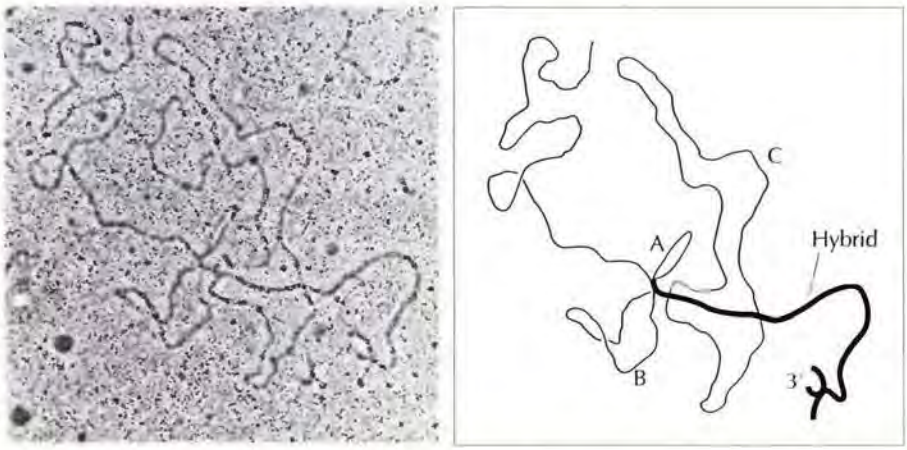
These processing steps were generally assumed to involve the removal of sequences from the 5′ and 3′ ends of the primary transcripts. In this model, mRNAs embedded within long primary transcripts would be encoded by uninterrupted DNA sequences. This view of eukaryotic mRNA was changed radically by the discovery of splicing, made independently by Berget, Moore, and Sharp, and by Louise Chow, Richard Gelinas, Tom Broker, and Richard Roberts (An amazing sequence arrangement at the 5′ ends of adenovirus 2 messenger RNA, 1977. *Cell* 12: 1–8).

The Experiments

Both of the research groups that discovered splicing used adenovirus 2 to investigate mRNA synthesis in human cells. The major advantage of the virus is that it provides a model that is much simpler than the host cell. Viral DNA can be isolated directly from virus particles, and mRNAs encoding the viral structural proteins are present in such high amounts that they can be purified directly from infected cells. Berget, Moore, and Sharp focused their experiments on an abundant mRNA that encodes a viral structural polypeptide known as the hexon.

To map the hexon mRNA on the viral genome, purified mRNA was hybridized to adenovirus DNA and the hybrid molecules were examined by electron microscopy. As expected, the body of the hexon mRNA formed hybrids with restriction fragments of adenovirus DNA that had previously been shown to contain the hexon gene. Surprisingly, however, sequences at the 5′ end of hexon mRNA failed to hybridize to DNA sequences adjacent to those encoding the body of the message, suggesting that the 5′ end of the mRNA had arisen from sequences located elsewhere in the viral genome.

This possibility was tested by hybridization of hexon mRNA to a restriction fragment extending upstream of the hexon gene. The mRNA-DNA hybrids formed in this experiment displayed a complex loop structure (see figure). The body of the mRNA formed a long hybrid region with the previously identified hexon DNA sequences. Strikingly, the 5′ end of the hexon mRNA hybridized to three short upstream regions of DNA, which were separated from each other and from the body of the message by large single-stranded DNA loops. The sequences at the 5′ end of hexon mRNA thus appeared to be transcribed from three separate regions of the viral genome, which were spliced to the body of the mRNA during the processing of a long primary transcript.



An electron micrograph and tracing of hexon mRNA hybridized to adenovirus DNA. The single-stranded loops designated A, B, and C, correspond to introns.

KEY EXPERIMENT

The Impact

The discovery of splicing in adenovirus mRNA was quickly followed by similar experiments with cellular mRNAs, demonstrating that eukaryotic genes had a previously unexpected structure. Rather than being continuous, their coding sequences were interrupted by introns, which were removed from primary transcripts by splicing. Introns are now known to account for much of the DNA in eukaryotic genomes, and the roles of introns in the evolution and regulation of gene expression continue to be active areas of investigation. The discovery of splicing also stimulated intense interest in the mechanism of this unexpected RNA processing reaction. As discussed in Chapter 7, these studies have not only illuminated new mechanisms of regulating gene expression; they have also revealed novel catalytic activities of RNA and provided critical evidence supporting the hypothesis that early evolution was based on self-replicating RNA molecules. The unexpected structure of adenovirus mRNAs has thus had a major impact on diverse areas of cellular and molecular biology.

Soon after the discovery of introns in adenovirus, similar observations were made on cloned genes of eukaryotic cells. For example, electron microscopic analysis of RNA-DNA hybrids and subsequent nucleotide sequencing of cloned genomic DNAs and cDNAs indicated that the coding region of the mouse β-globin gene (which encodes the β subunit of hemoglobin) is interrupted by two introns that are removed from the mRNA by splicing (Figure 5.4). The intron-exon structure of many eukaryotic genes is quite complicated, and the amount of DNA in the intron sequences is often

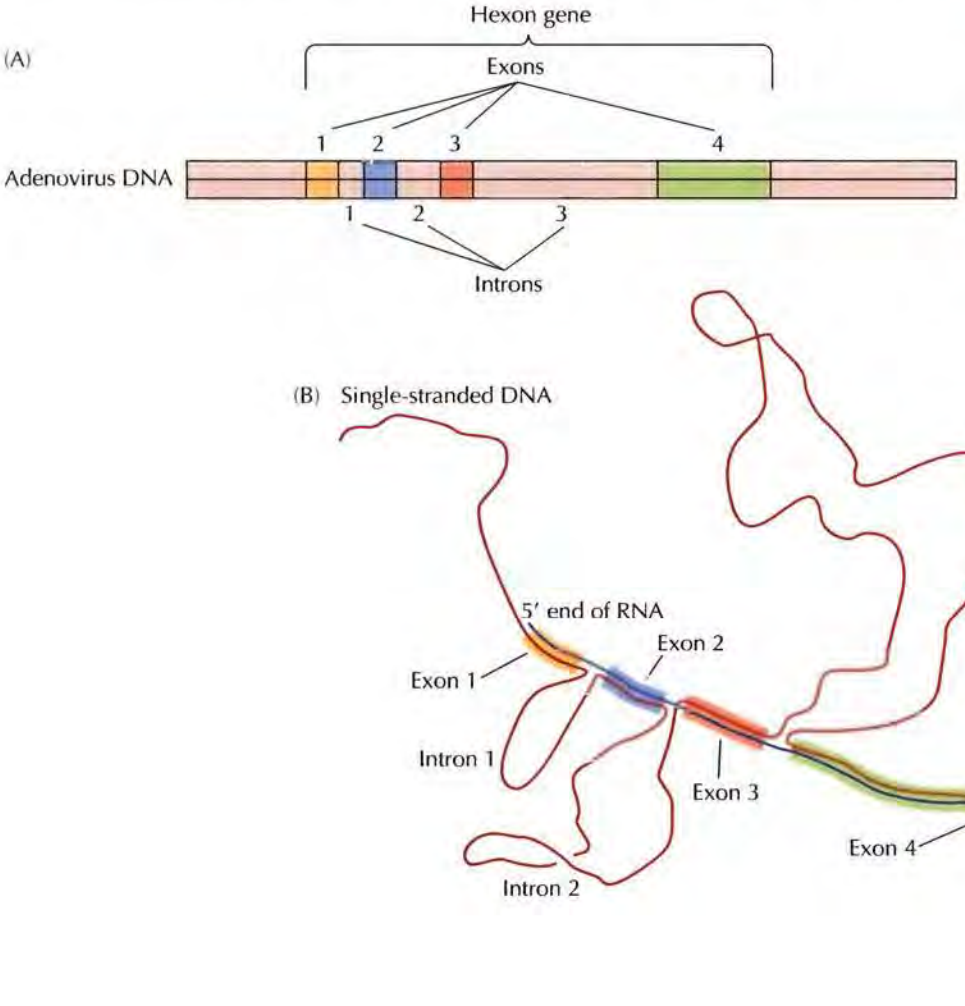


FIGURE 5.3 Identification of introns in adenovirus mRNA (A) The gene encoding the adenovirus hexon (a major structural protein of the viral particle) consists of four exons, interrupted by three introns. (B) This tracing illustrates an electron micrograph of a hypothetical hybrid between hexon mRNA and a portion of adenovirus DNA. The exons are seen as regions of RNA-DNA hybrid, which are separated by single-stranded DNA loops corresponding to the introns.

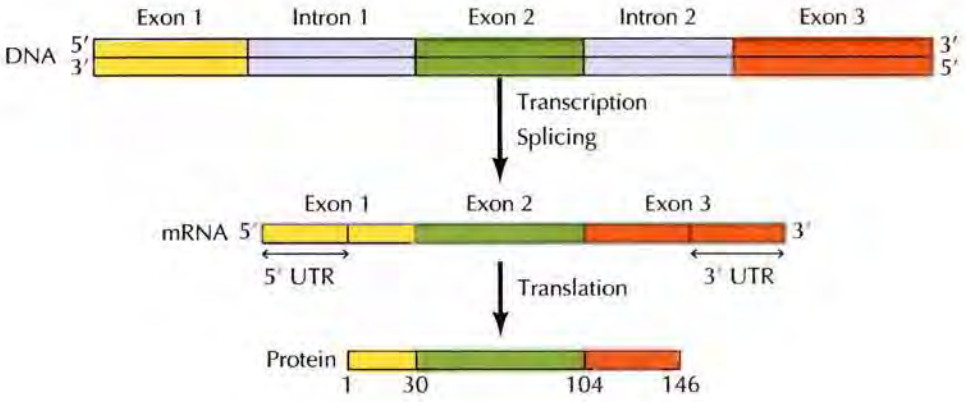


FIGURE 5.4 The mouse β-globin gene This gene contains two introns, which divide the coding region among three exons. Exon 1 encodes amino acids 1 to 30, exon 2 encodes amino acids 31 to 104, and exon 3 encodes amino acids 105 to 146. Exons 1 and 3 also contain untranslated regions (UTRs) at the 5′ and 3′ ends of the mRNA, respectively.

greater than that in the exons. For example, an average human gene contains approximately 9 exons, interrupted by 8 introns and distributed over approximately 30,000 base pairs (30 **kilobases**, or **kb**) of genomic DNA (Table 5.1). The exons generally total only about 2.5 kb, including regions at both the 5′and 3′ ends of the mRNA that are not translated into protein (5′ and 3′ untranslated regions or UTRs). Introns thus comprise more than 90% of the average human gene.

Introns are present in most genes of complex eukaryotes, although they are not universal. Almost all histone genes, for example, lack introns, so introns are clearly not required for gene function in eukaryotic cells. In addition, introns are not found in most genes of simple eukaryotes, such as yeasts. Conversely, introns *are* present in rare genes of prokaryotes. The presence or absence of introns is therefore not an absolute distinction between prokaryotic and eukaryotic genes, although introns are much more prevalent in higher eukaryotes (both plants and animals), where they account for a substantial amount of total genomic DNA. Many introns are conserved in genes of both plants and animals, indicating that they arose early in evolution, prior to the plant-animal divergence.

Most introns do not specify the synthesis of a cellular product, although a few do encode functional RNAs or proteins. However, introns play important roles in controlling gene expression. For example, the presence of introns allows the exons of a gene to be joined in different combinations, resulting in the synthesis of different proteins from the same gene. This process, called **alternative splicing** (Figure 5.5), occurs frequently in the genes of complex eukaryotes and is thought to be important in extending the functional repertoire of the 20,000–25,000 genes of the human genome.

Introns are also thought to have played an important role in evolution by facilitating recombination between protein-coding regions (exons) of differ-

TABLE 5.1 Characteristics of the Average Human Gene

Number of exons	9
Number of introns	8
Exon Sequence:	
5′ untranslated region	300 base pairs
coding sequence	1400 base pairs
3′ untranslated region	800 base pairs
TOTAL	2500 base pairs
Intron Sequence:	27,000 base pairs

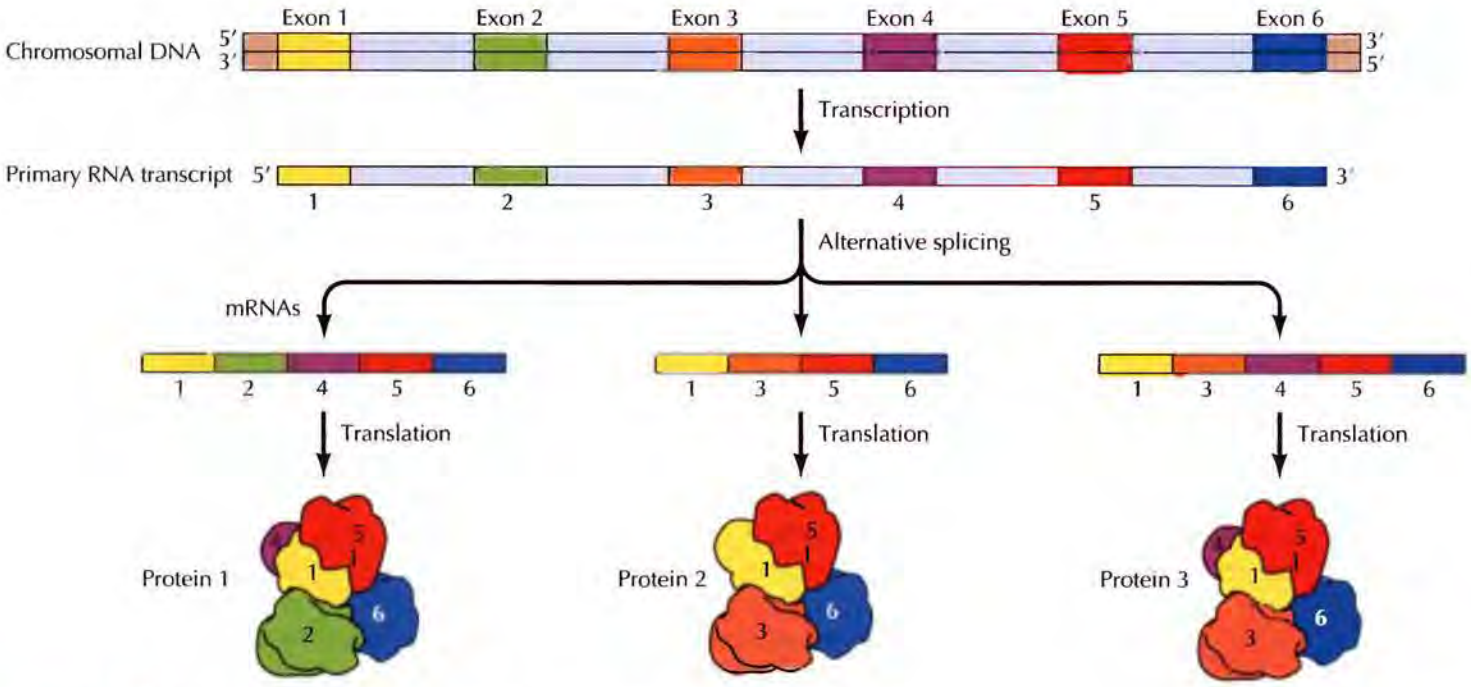


FIGURE 5.5 Alternative splicing The gene illustrated contains six exons, separated by five introns. Alternative splicing allows these exons to be joined in different combinations, resulting in the formation of three distinct mRNAs and proteins from the single primary transcript.

ent genes—a process known as exon shuffling. Exons frequently encode functionally distinct protein domains, so recombination between introns of different genes would result in new genes containing novel combinations of protein-coding sequences. As predicted by this hypothesis, DNA sequencing studies have demonstrated that some genes are chimeras of exons derived from several other genes, providing direct evidence that new genes can be formed by recombination between intron sequences.

Repetitive DNA Sequences

Introns make a substantial contribution to the large size of higher eukaryotic genomes. In humans, for example, introns account for approximately 20% of the total genomic DNA. However, an even larger portion of complex eukaryotic genomes consists of highly repeated noncoding DNA sequences. These sequences, sometimes present in hundreds of thousands of copies per genome, were first demonstrated by Roy Britten and David Kohne during studies of the rates of reassociation of denatured fragments of cellular DNAs (Figure 5.6). Denatured strands of DNA hybridize to each other (reassociate), re-forming double-stranded molecules (see Figure 4.24). Since DNA reassociation is a bimolecular reaction (two separated strands of denatured DNA must collide with each other in order to hybridize), the rate of reassociation depends on the concentration of DNA strands. When fragments of *E. coli* DNA were denatured and allowed to hybridize with each other, all of the DNA reassociated at the same rate, as expected if each DNA sequence were represented once per genome. However, reassociation of fragments of DNA extracted from mammalian cells showed a very different pattern. Approximately 50% of the DNA fragments reassociated at the rate expected for sequences present once per genome, but the remainder reasso-

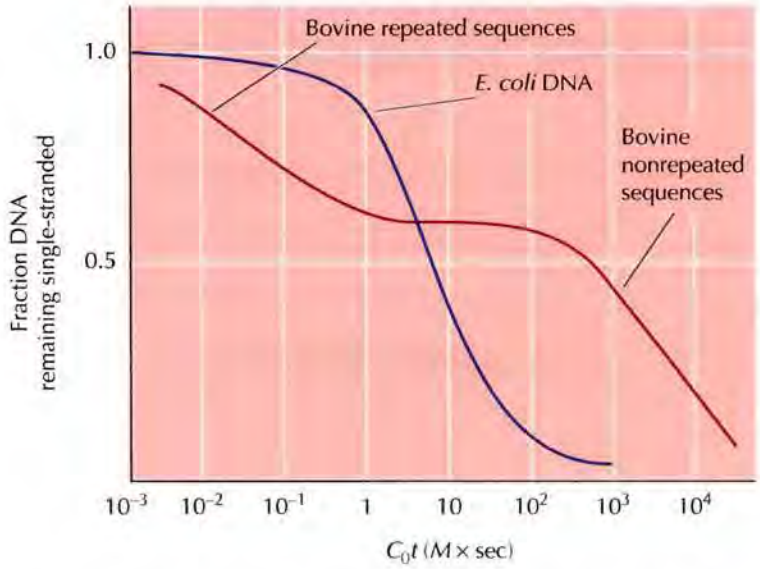


FIGURE 5.6 Identification of repetitive sequences by DNA reassociation The kinetics of the reassociation of fragments of *E. coli* and bovine DNAs are illustrated as a function of C_0t, which is the initial concentration of DNA multiplied by the time of incubation. The *E. coli* DNA reassociates at a uniform rate, consistent with each fragment of DNA being represented once in a genome of 4.6×10^6 base pairs. In contrast, the bovine DNA fragments exhibit two distinct steps in their reassociation. About 60% of the DNA fragments (the nonrepeated sequences) reassociate more slowly than *E. coli* DNA, as expected for sequences represented as single copies in the larger bovine genome (3×10^9 base pairs). However, the other 40% of the bovine DNA fragments (the repeated sequences) reassociate more rapidly than *E. coli* DNA, indicating that multiple copies of these sequences are present.

ciated much more rapidly than expected. The interpretation of these results was that some sequences were present in multiple copies and therefore reassociated more rapidly than those sequences that were represented only once per genome. In particular, these experiments indicated that approximately 50% of mammalian DNA consists of highly repetitive sequences, some of which are repeated 10^5 to 10^6 times.

Further analysis, culminating in the sequencing of complete genomes, has identified several types of these highly repeated sequences (Table 5.2). One class (called **simple-sequence repeats**) consists of tandem arrays of up to thousands of copies of short sequences, ranging from 1 to 500 nucleotides. For example, one type of simple-sequence repeat in *Drosophila* consists of tandem repeats of the seven nucleotide unit ACAAACT. Because of their distinct base compositions, many simple-sequence DNAs can be separated from the rest of the genomic DNA by equilibrium centrifugation in CsCl density gradients. The density of DNA is determined by its base composition, with AT-rich sequences being less dense than GC-rich sequences. Therefore an AT-rich simple-sequence DNA bands in CsCl gradients at a lower density than the bulk of *Drosophila* genomic DNA (Figure 5.7). Since such repeat-sequence DNAs band as "satellites" separate from the main band of DNA, they are frequently referred to as **satellite DNAs**. These sequences are repeated millions of times per genome, accounting for about 10% of the DNA of most higher eukaryotes. Simple-sequence DNAs are not transcribed and do not convey functional genetic information.

TABLE 5.2 Repetitive Sequences in the Human Genome

Type of sequence	Number of copies	Fraction of genome
Simple-sequence repeats[a]	>1,000,000	~10%
Retrotransposons		
LINEs	850,000	21%
SINEs	1,500,000	13%
Retrovirus-like elements	450,000	8%
DNA transposons	300,000	3%

[a] The content of simple-sequence repeats is estimated from the fraction of heterochromatin in the human genome.

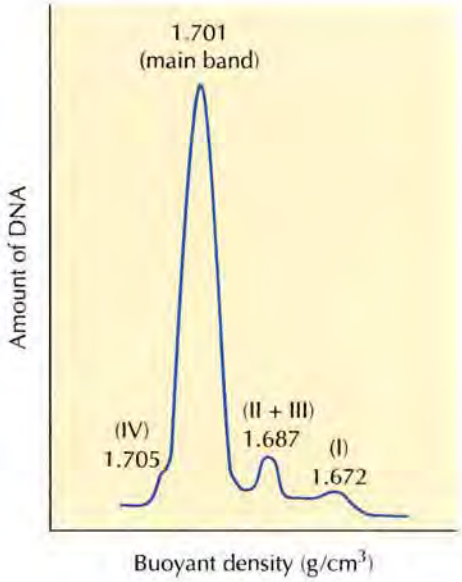


FIGURE 5.7 Satellite DNA Equilibrium centrifugation of *Drosophila* DNA in a CsCl gradient separates satellite DNAs (designated I–IV) with buoyant densities (in g/cm^3) of 1.672, 1.687, and 1.705 from the main band of genomic DNA (buoyant density 1.701).

Some, however, play important roles in chromosome structure, as discussed in the next section of this chapter.

Other repetitive DNA sequences are scattered throughout the genome rather than being clustered as tandem repeats. These interspersed repetitive elements are a major contributor to genome size, accounting for approximately 45% of human genomic DNA. The two most prevalent classes of these sequences are called **SINEs** (short interspersed elements) and **LINEs** (long interspersed elements). SINEs are 100–300 base pairs long. About 1.5 million such sequences are dispersed throughout the genome, accounting for approximately 13% of the total human DNA. Although SINES are transcribed into RNA, they do not encode proteins and their function is unknown. The major human LINEs are 4–6 kb long, although many repeated sequences derived from LINEs are shorter, with an average size of about 1 kb. There are approximately 850,000 repeats of LINE sequences in the genome, accounting for about 21% of human DNA. LINEs are transcribed and at least some encode proteins, but like SINEs, they have no known function in cell physiology.

Both SINEs and LINEs are examples of transposable elements, which are capable of moving to different sites in genomic DNA. As discussed in detail in Chapter 6, both SINEs and LINEs are **retrotransposons**, meaning that their transposition is mediated by reverse transcription (Figure 5.8). An RNA copy of a SINE or LINE is converted to DNA by reverse transcriptase within the cell, and the new DNA copy is integrated at a new site in the genome. A third class of interspersed repetitive sequences, which closely resemble retroviruses and are called **retrovirus-like elements**, also move within the genome by reverse transcription. Human retrovirus-like elements range from approximately 2–10 kb in length. There are approximately 450,000 retrovirus-like elements in the human genome, accounting for approximately 8% of human DNA. In contrast, the fourth class of inter-

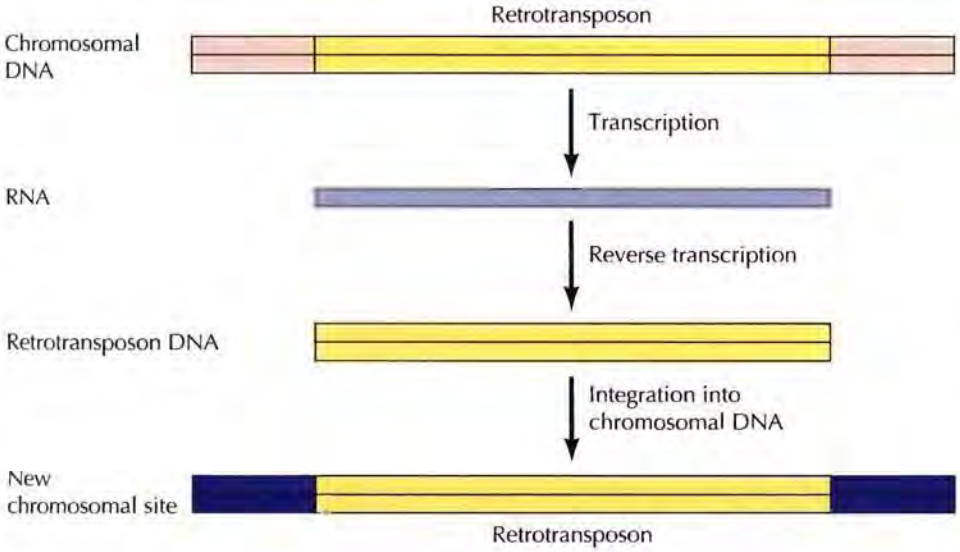


FIGURE 5.8 Movement of retrotransposons A retrotransposon present at one site in chromosomal DNA is transcribed into RNA, and then converted back into DNA by reverse transcription. The retrotransposon DNA can then integrate into a new chromosomal site.

spersed repetitive elements (**DNA transposons**) moves through the genome by being copied and reinserted as DNA sequences, rather than moving by reverse transcription. In the human genome, there are about 300,000 copies of DNA transposons, ranging from 80–3000 base pairs in length and accounting for approximately 3% of human DNA.

Nearly half of the human genome thus consists of interspersed repetitive elements that have replicated and moved through the genome by either RNA or DNA intermediates. It is noteworthy that the vast majority of these elements transpose via RNA intermediates, so reverse transcription has been responsible for generating more than 40% of the human genome. Some of these sequences may help regulate gene expression, but most interspersed repetitive sequences appear not to make a useful contribution to the cell. Instead, they appear to represent "selfish DNA elements" that have been selected for their own ability to replicate within the genome rather than conferring a selective advantage to their host. In some cases, however, transposable elements have played important evolutionary roles by stimulating gene rearrangements and contributing to the generation of genetic diversity.

Gene Duplication and Pseudogenes

Another factor contributing to the large size of eukaryotic genomes is that many genes are present in multiple copies, some of which are frequently nonfunctional. In some cases, multiple copies of genes are needed to produce RNAs or proteins required in large quantities, such as ribosomal RNAs or histones. In other cases, distinct members of a group of related genes (called a **gene family**) may be transcribed in different tissues or at different stages of development. For example, the α and β subunits of hemoglobin are both encoded by gene families in the human genome, with different members of these families being expressed in embryonic, fetal, and adult tissues (Figure 5.9). Members of many gene families (e.g., the globin genes) are clustered within a region of DNA; members of other gene families are dispersed to different chromosomes.

Gene families are thought to have arisen by duplication of an original ancestral gene, with different members of the family then diverging as a consequence of mutations during evolution. Such divergence can lead to the evolution of related proteins that are optimized to function in different tissues or at different stages of development. For example, fetal globins

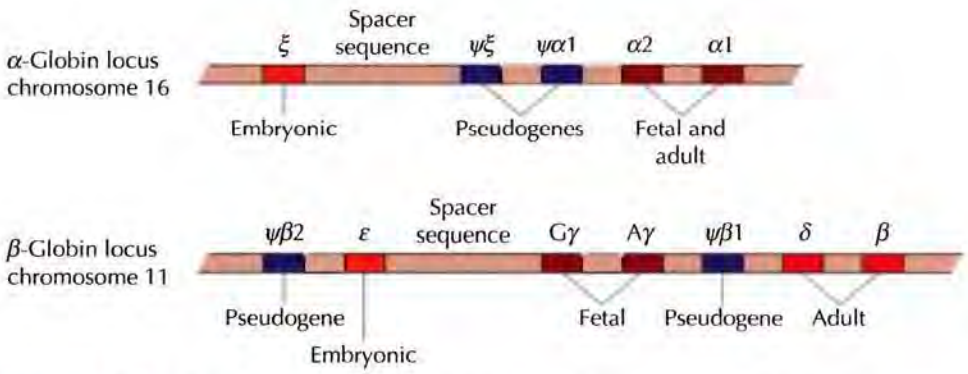


FIGURE 5.9 Globin gene families Members of the human α- and β-globin gene families are clustered on chromosomes 16 and 11, respectively. Each family contains genes that are specifically expressed in embryonic, fetal, and adult tissues, in addition to nonfunctional gene copies (pseudogenes).

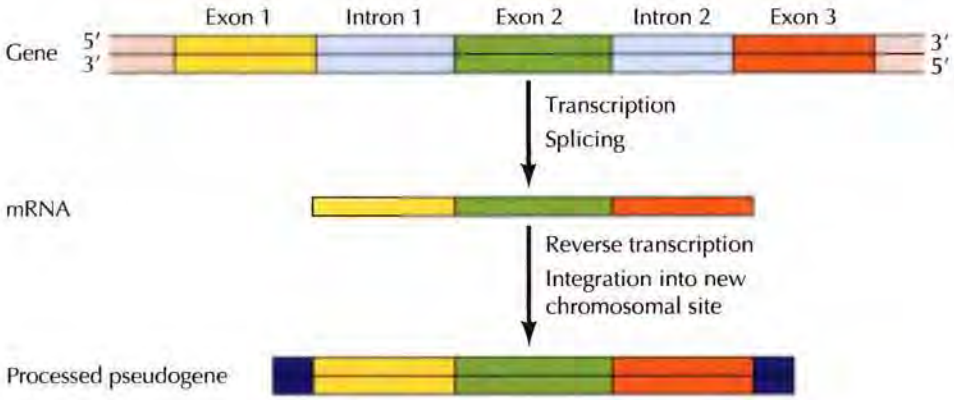


FIGURE 5.10 Formation of a processed pseudogene A gene is transcribed and spliced to yield an mRNA from which the introns have been removed. The mRNA is copied by reverse transcriptase, yielding a cDNA copy lacking introns. Integration into chromosomal DNA results in formation of a processed pseudogene.

have a higher affinity for O_2 than do adult globins—a difference that allows the fetus to obtain O_2 from the maternal circulation.

As might be expected, however, not all mutations enhance gene function. Some gene copies have instead sustained mutations that result in their loss of ability to produce a functional gene product. For example, the human α- and β-globin gene families each contain two genes that have been inactivated by mutations. Such nonfunctional gene copies (called **pseudogenes**) represent evolutionary relics that increase the size of eukaryotic genomes without making a functional genetic contribution. Recent studies have identified more than 20,000 pseudogenes in the human genome. Since this is generally assumed to be an underestimate, it is likely that our genome contains many more pseudogenes than functional genes.

Gene duplications can arise by two distinct mechanisms. The first is duplication of a segment of DNA, which can result in the transfer of a block of DNA sequence to a new location in the genome. Such duplications of DNA segments ranging from 1 kb to more than 50 kb are estimated to account for approximately 5% of the human genome. Alternatively, genes can be duplicated by reverse transcription of an mRNA, followed by integration of the cDNA copy into a new chromosomal site (Figure 5.10). This mode of gene duplication, analogous to the transposition of repetitive elements that move via RNA intermediates, results in the formation of gene copies that lack introns and also lack the normal chromosomal sequences that direct transcription of the gene into mRNA. As a result, duplication of a gene by reverse transcription usually yields an inactive gene copy called a **processed pseudogene**. Processed pseudogenes account for about two-thirds of the pseudogenes that have been identified in the human genome.

The Composition of Higher Eukaryotic Genomes

Having discussed several kinds of noncoding DNA that contribute to the genomic complexity of higher eukaryotes, it is of interest to overview the composition of cell genomes. In bacterial genomes, most of the DNA encodes proteins. For example, the genome of *E. coli* is approximately 4.6×10^6 base pairs long and contains about 4000 genes, with nearly 90% of the DNA used as protein-coding sequence. The yeast genome, which consists of 12×10^6 base pairs, is about 2.5 times the size of the genome of *E. coli*, but is still extremely compact. Only 4% of the genes of *Saccharomyces cerevisiae* contain introns, and these usually have only a single small intron near the start of the coding sequence. Approximately 70% of the yeast genome is used as protein-coding sequence, specifying a total of about 6000 proteins.

The relatively simple animal genomes of *C. elegans* and *Drosophila* are about 10 times larger than the yeast genome, but contain only 2–3 times more genes. Instead, these simple animal genomes contain more introns and more repetitive sequence, so that protein-coding sequences correspond to only about 25% of the *C. elegans* genome and about 13% of the genome of *Drosophila*. The genome of the model plant *Arabidopsis* contains a similar number of genes, with approximately 26% of the genome corresponding to protein-coding sequence.

The genomes of higher animals (such as humans) are approximately 20–30 times larger than those of *C. elegans* and *Drosophila*. However, a major surprise from deciphering the human genome sequence was the discovery that the human genome contains only 20,000 to 25,000 genes. It appears that only about 1.2% of the human genome consists of protein-coding sequence. Approximately 20% of the genome consists of introns, and more than 60% is composed of various types of repetitive and duplicated DNA sequences, with the remainder corresponding to pseudogenes, to nonrepetitive spacer sequences between genes, and to exon sequences that are present at the 5′ and 3′ ends of mRNAs but are not translated into protein. The increased size of the genomes of higher eukaryotes is thus due far more to the presence of large amounts of repetitive sequences and introns than to an increased number of genes.

Chromosomes and Chromatin

5.1 WEBSITE ANIMATION

Chromatin and Chromosomes In a eukaryotic cell, DNA is wrapped tightly around histone proteins (forming chromatin), and when a cell prepares for division, the chromatin coils upon itself multiple times to form compact chromosomes.

Not only are the genomes of most eukaryotes much more complex than those of prokaryotes, but the DNA of eukaryotic cells is also organized differently from that of prokaryotic cells. The genomes of prokaryotes are contained in single chromosomes, which are usually circular DNA molecules. In contrast, the genomes of eukaryotes are composed of multiple chromosomes, each containing a linear molecule of DNA. Although the numbers and sizes of chromosomes vary considerably between different species (Table 5.3), their basic structure is the same in all eukaryotes. The DNA of eukaryotic cells is tightly bound to small basic proteins (histones) that package the DNA in an orderly way in the cell nucleus. This task is substantial, given the DNA content of most eukaryotes. For example, the total extended length of DNA in a human cell is nearly 2 meters, but this DNA must fit into a nucleus with a diameter of only 5 to 10 μm.

Chromatin

The complexes between eukaryotic DNA and proteins are called **chromatin**, which typically contains about twice as much protein as DNA. The major proteins of chromatin are the **histones**—small proteins containing a high proportion of basic amino acids (arginine and lysine) that facilitate binding to the negatively charged DNA molecule. There are five major types of histones—called H1, H2A, H2B, H3, and H4—which are very similar among different species of eukaryotes (Table 5.4). The histones are extremely abundant proteins in eukaryotic cells; together their mass is approximately equal to that of the cell's DNA. In addition, chromatin contains an approximately equal mass of a wide variety of nonhistone chromosomal proteins. There are more than a thousand different types of these proteins, which are involved in a range of activities, including DNA replication and gene expression.

TABLE 5.3 Chromosome Numbers of Eukaryotic Cells

Organism	Genome size (Mb)[a]	Chromosome number[a]
Yeast (*Saccharomyces cerevisiae*)	12	16
Slime mold (*Dictyostelium*)	70	7
Arabidopsis thaliana	125	5
Corn	5000	10
Onion	15,000	8
Lily	50,000	12
Nematode (*Caenorhabditis elegans*)	97	6
Fruit fly (*Drosophila*)	180	4
Toad (*Xenopus laevis*)	3000	18
Lungfish	50,000	17
Chicken	1200	39
Mouse	3000	20
Cow	3000	30
Dog	3000	39
Human	3000	23

[a] Both genome size and chromosome number are for haploid cells.
Mb = millions of base pairs.

The basic structural unit of chromatin, the **nucleosome**, was described by Roger Kornberg in 1974 (Figure 5.11). Two types of experiments led to Kornberg's proposal of the nucleosome model. First, partial digestion of chromatin with micrococcal nuclease (an enzyme that degrades DNA) was found to yield DNA fragments approximately 200 base pairs long. In contrast, a similar digestion of naked DNA (not associated with proteins) yielded a continuous smear of randomly sized fragments. These results suggested that the binding of proteins to DNA in chromatin protects regions of the DNA from nuclease digestion, so that the enzyme can attack DNA only at sites separated by approximately 200 base pairs. Consistent with this notion, electron microscopy revealed that chromatin fibers have a beaded appearance, with the beads spaced at intervals of approximately 200 base pairs. Thus both the nuclease digestion and the electron microscopic studies suggested that chromatin is composed of repeating 200-base-pair units, which were called nucleosomes.

TABLE 5.4 The Major Histone Proteins

Histone[a]	Molecular weight	Number of amino acids	Percentage lysine + arginine
H1	22,500	244	30.8
H2A	13,960	129	20.2
H2B	13,774	125	22.4
H3	15,273	135	22.9
H4	11,236	102	24.5

[a] Data are for rabbit (H1) and bovine histones.

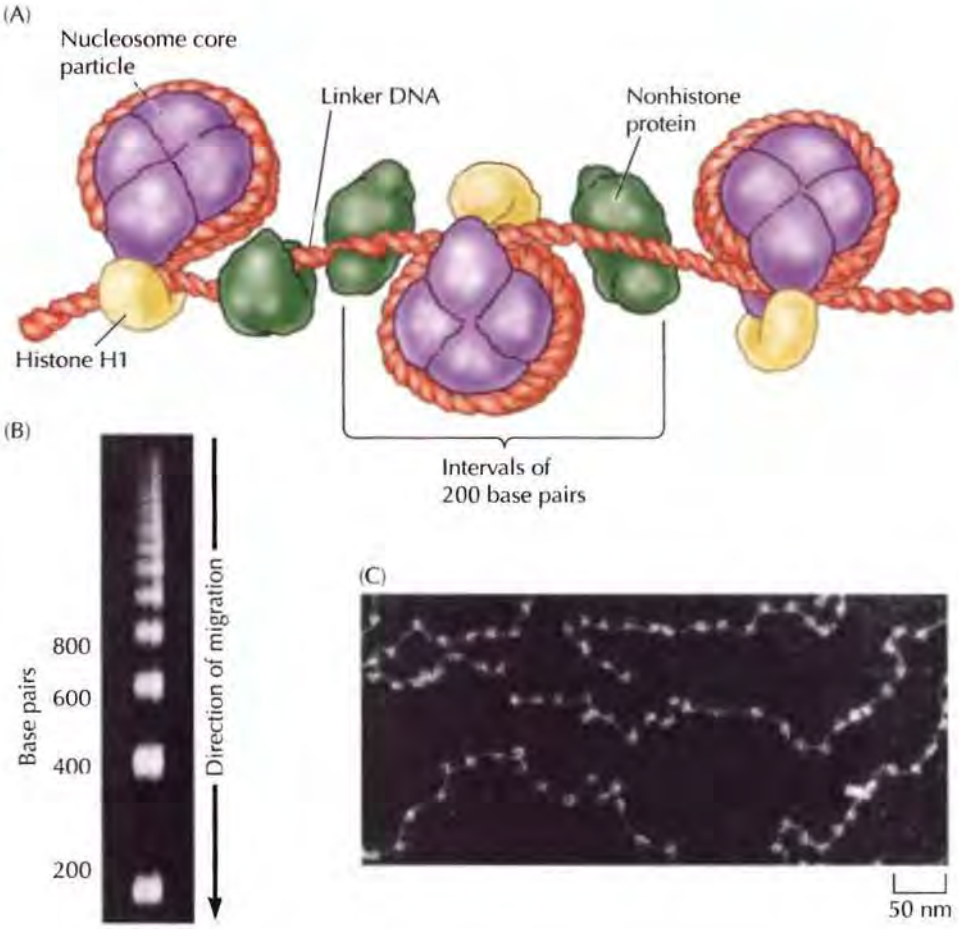
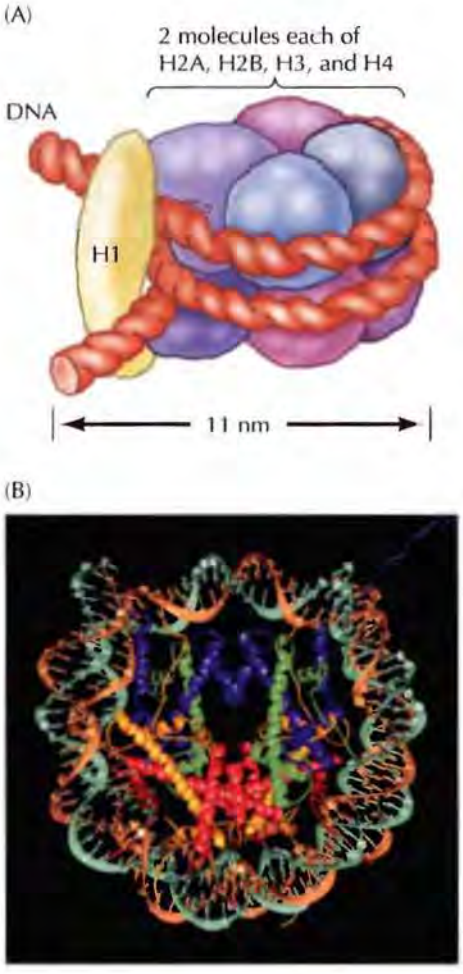


FIGURE 5.11 The organization of chromatin in nucleosomes (A) The DNA is wrapped around histones in nucleosome core particles and sealed by histone H1. Nonhistone proteins bind to the linker DNA between nucleosome core particles. (B) Gel electrophoresis of DNA fragments obtained by partial digestion of chromatin with micrococcal nuclease. The linker DNA between the nucleosome core particles is preferentially sensitive, so limited digestion of chromatin yields fragments corresponding to multiples of 200 base pairs. (C) An electron micrograph of an extended chromatin fiber, illustrating its beaded appearance. (B, courtesy of Roger Kornberg, Stanford University; C, courtesy of Ada L. Olins and Donald E. Olins, Oak Ridge National Laboratory.)



More extensive digestion of chromatin with micrococcal nuclease was found to yield particles (called **nucleosome core particles**) that correspond to the beads visible by electron microscopy. Detailed analysis of these particles has shown that they contain 147 base pairs of DNA wrapped 1.67 times around a histone core consisting of two molecules each of H2A, H2B, H3, and H4 (the core histones) (Figure 5.12). One molecule of the fifth histone, H1, is bound to the DNA as it enters each nucleosome core particle. This forms a chromatin subunit known as a **chromatosome**, which consists of 166 base pairs of DNA wrapped around the histone core and held in place by H1 (a linker histone).

The packaging of DNA with histones yields a chromatin fiber approximately 10 nm in diameter that is composed of chromatosomes separated by linker DNA segments averaging about 50 base pairs in length (Figure 5.13). In the electron microscope, this 10-nm fiber has the beaded appearance that suggested the nucleosome model. Packaging of DNA into such a 10-nm chromatin fiber shortens its length approximately sixfold. The chromatin

FIGURE 5.12 Structure of a chromatosome (A) The nucleosome core particle consists of 147 base pairs of DNA wrapped 1.67 turns around a histone octamer consisting of two molecules each of H2A, H2B, H3, and H4. A chromatosome contains two full turns of DNA (166 base pairs) locked in place by one molecule of H1. (B) Model of the nucleosome core particle. The DNA backbones are shown in brown and turquoise. The histones are shown in blue (H3), green (H4), yellow (H2A), and red (H2B). (B, from K. Luger et al., 1997. *Nature* 389: 251.)

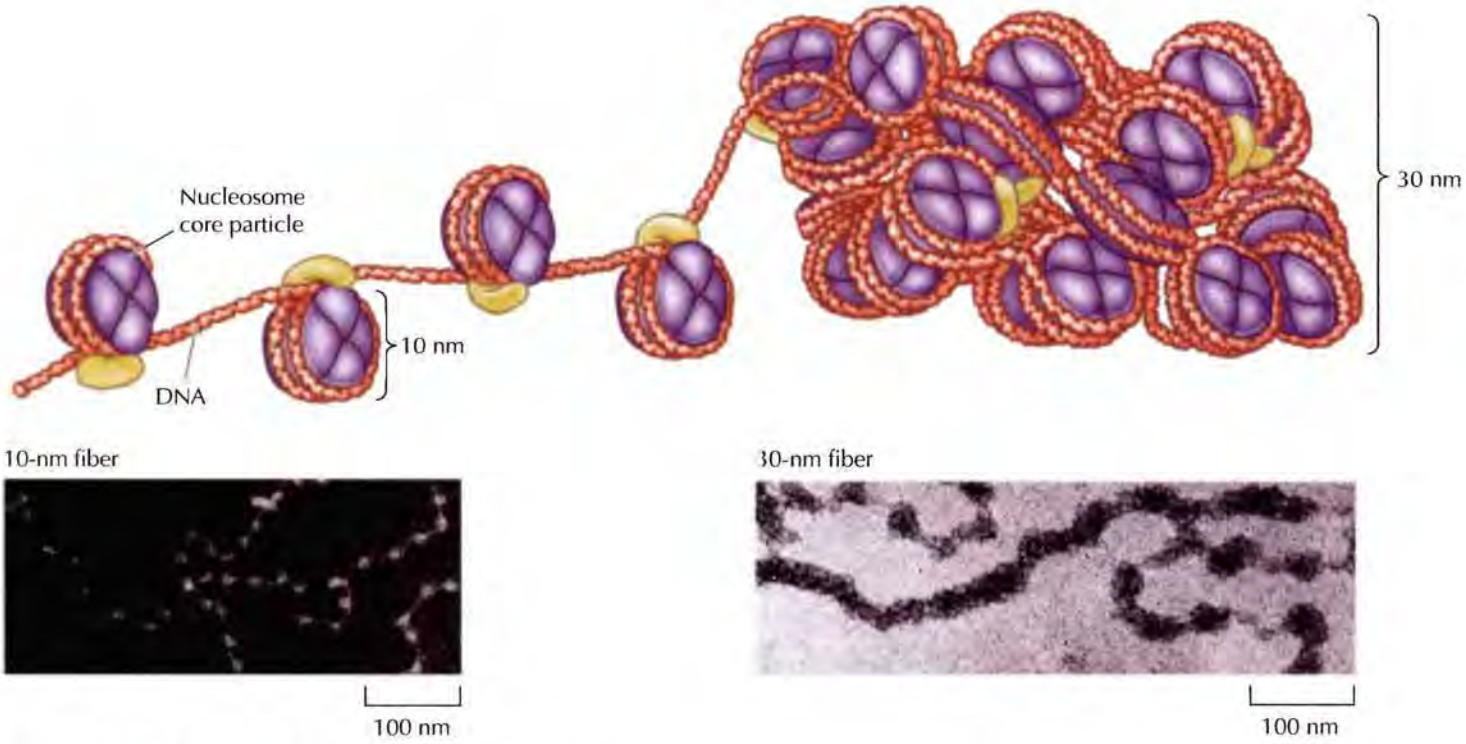


FIGURE 5.13 Chromatin fibers The packaging of DNA into nucleosomes yields a chromatin fiber approximately 10 nm in diameter. The chromatin is further condensed by coiling into a 30-nm fiber, containing about six nucleosomes per turn. (Photographs courtesy of Ada L. Olins and Donald E. Olins, Oak Ridge National Laboratory.)

can then be further condensed by coiling into 30-nm fibers, resulting in a total condensation of about fiftyfold. Interactions between histone H1 molecules appear to play an important role in this stage of chromatin condensation, which is critical to determining the accessibility of chromosomal DNA for processes such as DNA replication and transcription. Despite its importance, the structure of the 30-nm fiber remained unknown until 2005, when X-ray studies by Timothy Richmond and his colleagues revealed that the fiber is formed by two stacks of nucleosomes, with linker DNA zigzagging back and forth between them. Folding of 30-nm fibers upon themselves can lead to further condensation of chromatin within the cell.

The extent of chromatin condensation varies during the life cycle of the cell and plays an important role in regulating gene expression, as will be discussed in Chapter 7. In interphase (nondividing) cells, most of the chromatin (called **euchromatin**) is relatively decondensed and distributed throughout the nucleus (Figure 5.14). During this period of the cell cycle, genes are transcribed and the DNA is replicated in preparation for cell division. Most of the euchromatin in interphase nuclei appears to be in the form of 30-nm, or somewhat more condensed 60- to 130-nm, chromatin fibers. Genes that are actively transcribed are in a more decondensed state that makes the DNA accessible to the transcription machinery. In contrast to euchromatin, about 10% of interphase chromatin (called **heterochromatin**) is in a very highly condensed state that resembles the chromatin of cells undergoing mitosis. Heterochromatin is transcriptionally inactive and contains highly repeated DNA sequences, such as those present at centromeres and telomeres.

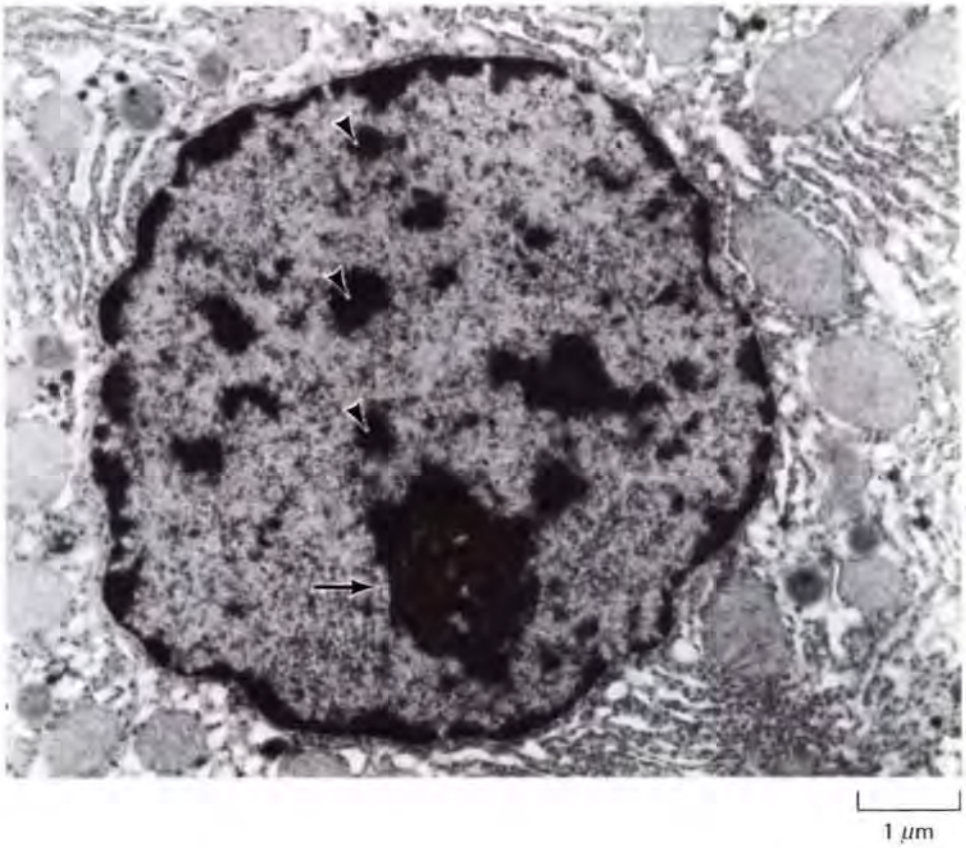


FIGURE 5.14 Interphase chromatin Electron micrograph of an interphase nucleus. The euchromatin is distributed throughout the nucleus. The heterochromatin is indicated by arrowheads and the nucleolus by an arrow. (Courtesy of Ada L. Olins and Donald E. Olins, Oak Ridge National Laboratory.)

As cells enter mitosis, their chromosomes become highly condensed so that they can be distributed to daughter cells. Loops of 30-nm chromatin fibers are thought to fold upon themselves to form the compact metaphase chromosomes of mitotic cells in which the DNA has been condensed nearly ten thousandfold (Figure 5.15). Such condensed chromatin can no longer be used as a template for RNA synthesis, so transcription ceases during mitosis. Electron micrographs indicate that the DNA in metaphase chromosomes is organized into large loops attached to a protein scaffold (Figure 5.16), but we currently understand neither the detailed structure of this highly condensed chromatin nor the mechanism of chromatin condensation.

Metaphase chromosomes are so highly condensed that their morphology can be studied using the light microscope (Figure 5.17). Several staining techniques yield characteristic patterns of alternating light and dark chromosome bands, which result from the preferential binding of stains or fluores-

FIGURE 5.15 Chromatin condensation during mitosis Scanning electron micrograph of metaphase chromosomes. Artificial color has been added. (Biophoto Associates/Photo Researchers Inc.)

FIGURE 5.16 Structure of metaphase chromosomes An electron micrograph of DNA loops attached to the protein scaffold of metaphase chromosomes that have been depleted of histones. (From J. R. Paulson and U. K. Laemmli, 1977. *Cell* 12: 817.)

cent dyes to AT-rich versus GC-rich DNA sequences. These bands are specific for each chromosome and appear to represent distinct chromosome regions. Genes can be localized to specific chromosome bands by *in situ* hybridization, indicating that the packaging of DNA into metaphase chromosomes is a highly ordered and reproducible process.

Centromeres

The **centromere** is a specialized region of the chromosome that plays a critical role in ensuring the correct distribution of duplicated chromosomes to daughter cells during mitosis (Figure 5.18). The cellular DNA is replicated during interphase, resulting in the formation of two copies of each chromosome prior to the beginning of mitosis. As the cell enters mitosis, chromatin condensation leads to the formation of metaphase chromosomes consisting of two identical sister chromatids. These sister chromatids are held together at the centromere, which is seen as a constricted chromosomal region. As mitosis proceeds, microtubules of the mitotic spindle attach to the centromere, and the two sister chromatids separate and move to opposite poles of the spindle. At the end of mitosis, nuclear membranes re-form and the chromosomes decondense, resulting in the formation of daughter nuclei containing one copy of each parental chromosome.

The centromeres thus serve both as the sites of association of sister chromatids and as the attachment sites for microtubules of the mitotic spindle. They consist of specific DNA sequences to which a number of centromere-associated proteins bind, forming a specialized structure called the **kinetochore** (Figure 5.19). The binding of microtubules to kinetochore proteins

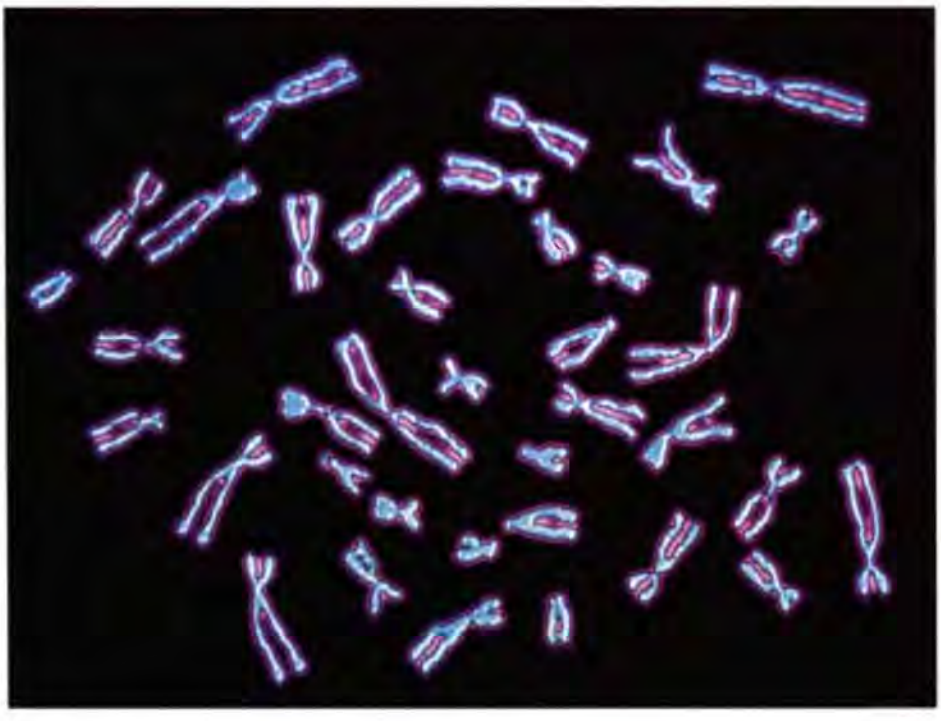


FIGURE 5.17 Human metaphase chromosomes A micrograph of human chromosomes spread from a metaphase cell. (Leonard Lessin/Peter Arnold, Inc.)

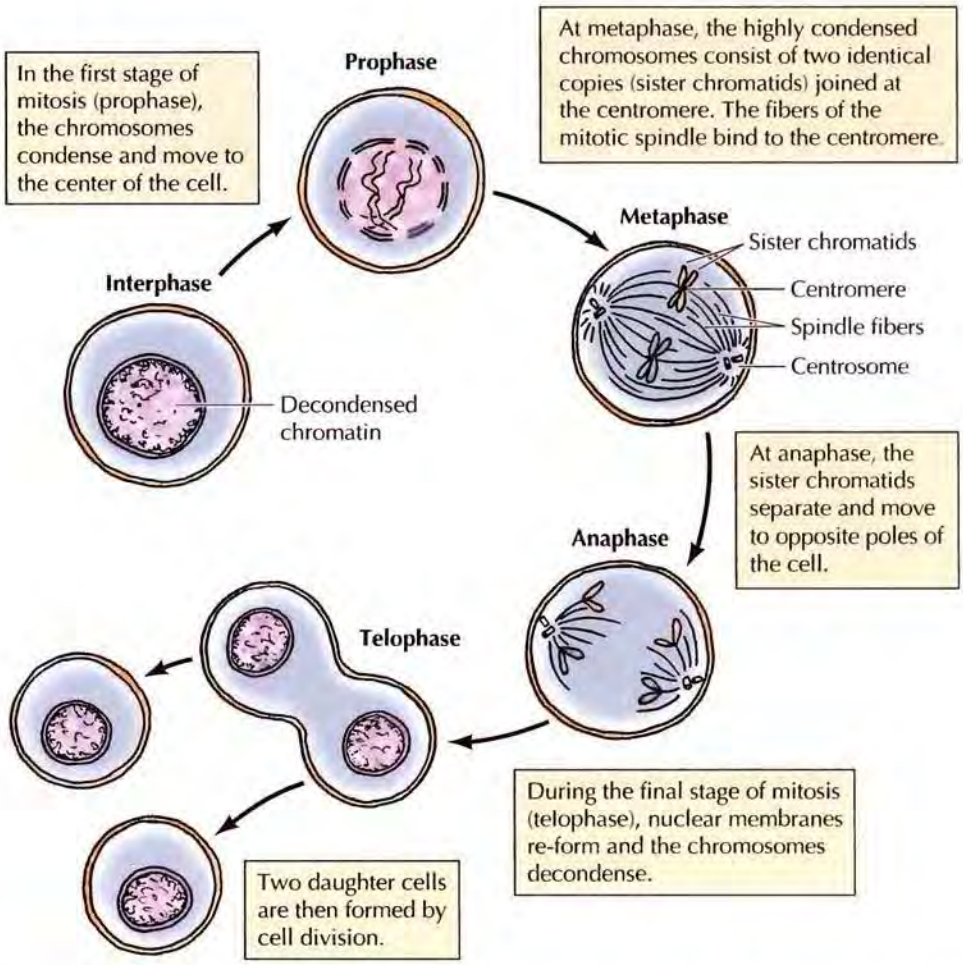
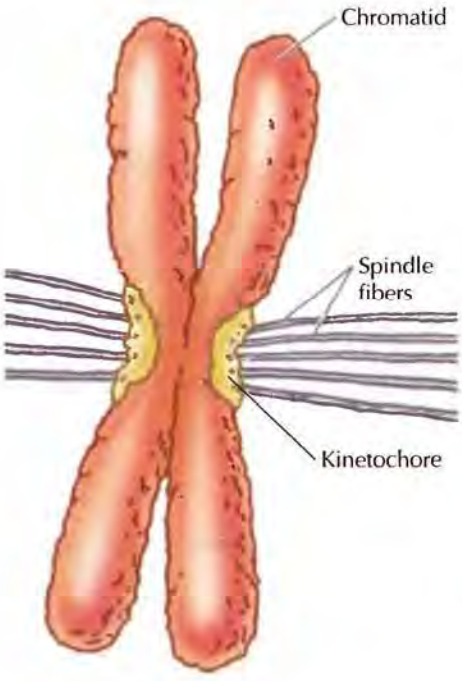


FIGURE 5.18 Chromosomes during mitosis Since DNA replicates during interphase, the cell contains two identical duplicated copies of each chromosome prior to entering mitosis.



mediates the attachment of chromosomes to the mitotic spindle. Proteins associated with the kinetochore then act as "molecular motors" that drive the movement of chromosomes along the spindle fibers, segregating the chromosomes to daughter nuclei.

Centromeric DNA sequences were initially defined in yeasts, where their function can be assayed by following the segregation of plasmids at mitosis (Figure 5.20). Plasmids that contain functional centromeres segregate like chromosomes and are equally distributed to daughter cells following mitosis. In the absence of a functional centromere, however, the plasmid does not segregate properly, and many daughter cells fail to inherit plasmid DNA. Assays of this type have enabled determination of the sequences required for centromere function. Such experiments first showed that the centromere sequences of the well-studied yeast *Saccharomyces cerevisiae* are contained in approximately 125 base pairs consisting of three sequence ele-

FIGURE 5.19 The centromere of a metaphase chromosome The centromere is the region at which the two sister chromatids remain attached at metaphase. Specific proteins bind to centromeric DNA, forming the kinetochore, which is the site of spindle fiber attachment.

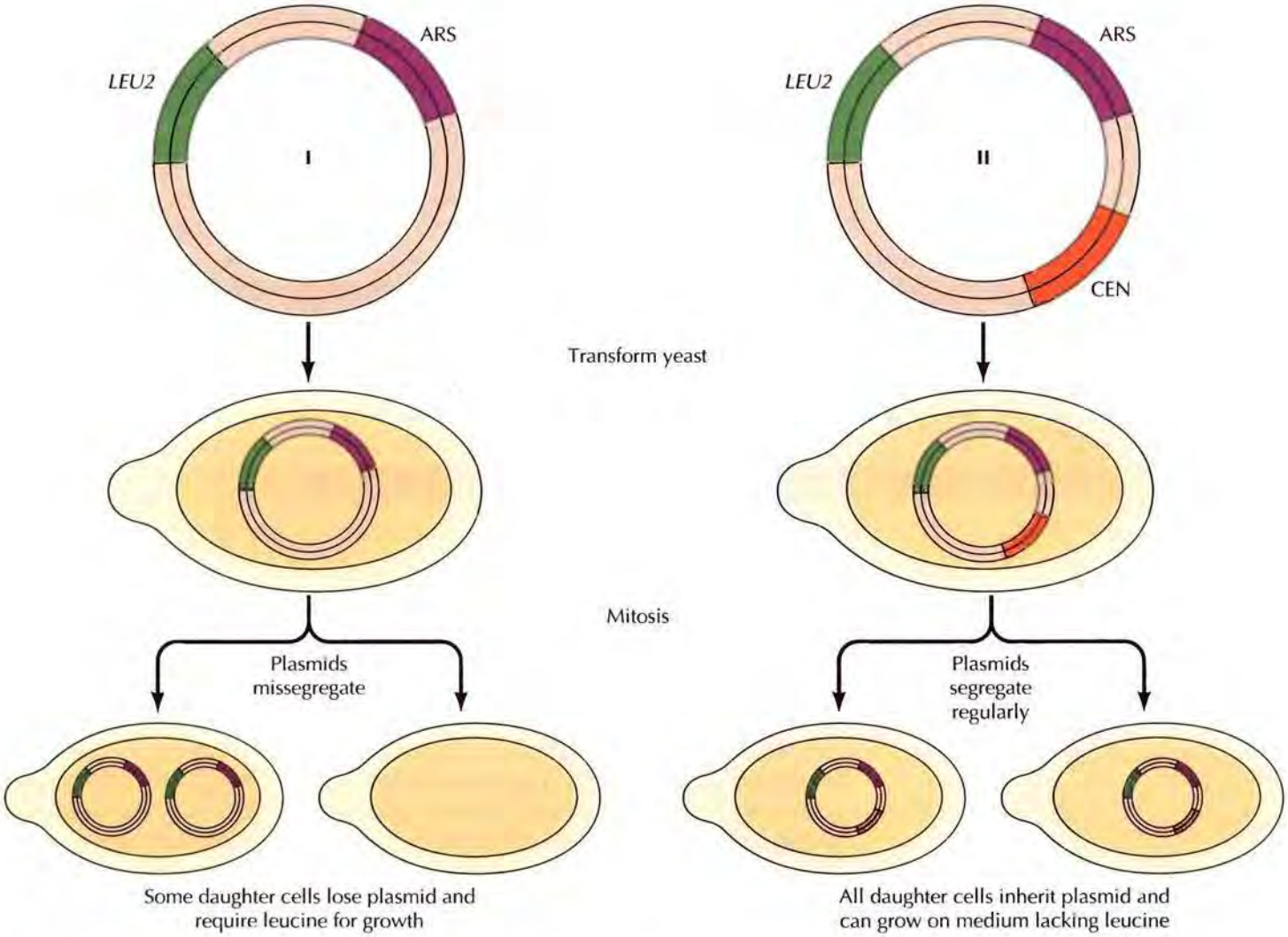


FIGURE 5.20 Assay of a centromere in yeast Both plasmids shown contain a selectable marker (*LEU2*) and DNA sequences that serve as origins of replication in yeast (ARS, which stands for autonomously replicating sequence). However, plasmid I lacks a centromere and is therefore frequently lost as a result of missegregation during mitosis. In contrast, the presence of a centromere (CEN) in plasmid II ensures its regular transmission to daughter cells.

ments: two short sequences of 8 and 25 base pairs separated by 78 to 86 base pairs of very AT-rich DNA (Figure 5.21A).

The short centromere sequences defined in *S. cerevisiae*, however, do not appear to reflect the situation in other eukaryotes. More recent studies have defined the centromeres of the fission yeast *Schizosaccharomyces pombe* by a similar functional approach. Although *S. cerevisiae* and *S. pombe* are both yeasts, they appear to be as divergent from each other as either is from humans and are quite different in many aspects of their cell biology. These two yeast species thus provide complementary models for simple and easily studied eukaryotic cells. The centromeres of *S. pombe* span 40 to 100 kb of DNA; they are approximately a thousand times larger than those of *S. cerevisiae*. They consist of a central core of 4 to 7 kb of single-copy DNA flanked by repetitive sequences (Figure 5.21B). Not only the central core but also the flanking repeated sequences are required for centromere function, so the

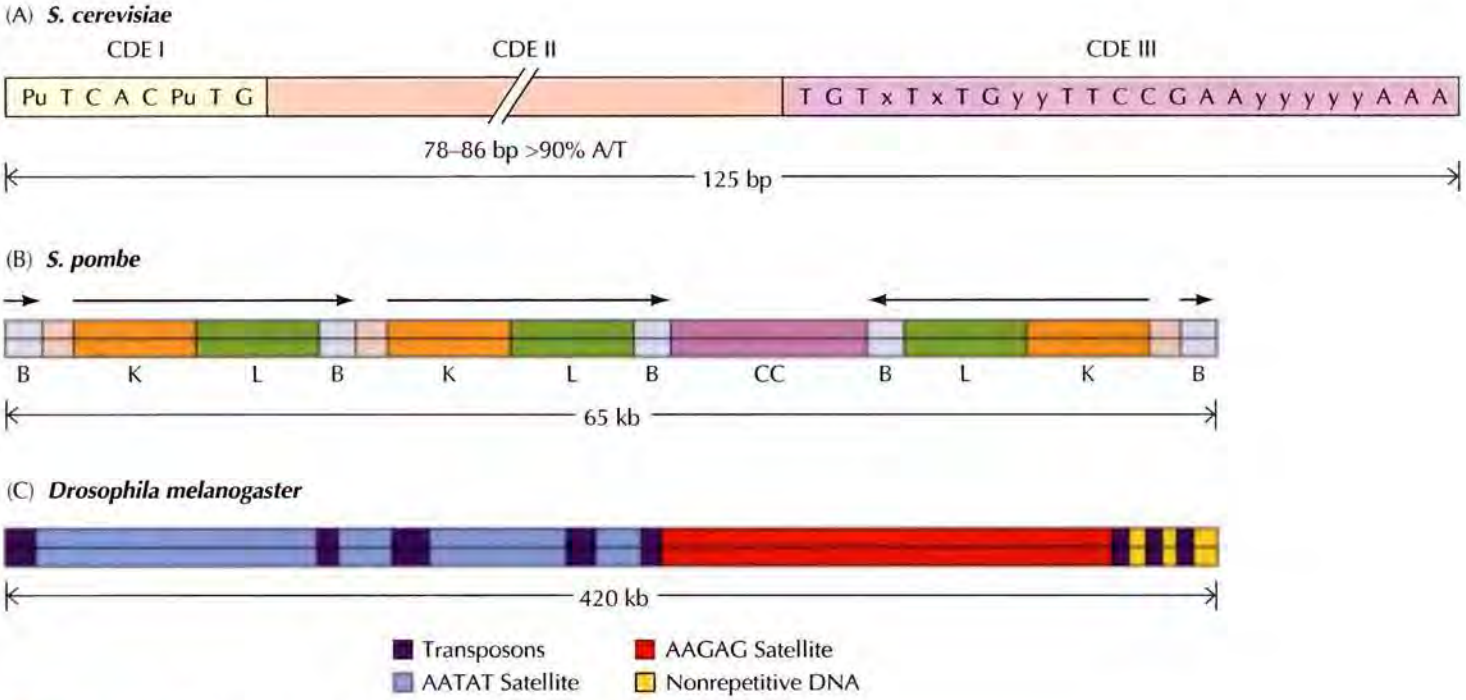


FIGURE 5.21 Centromeres of *S. cerevisiae*, *S. pombe*, and *Drosophila melanogaster* (A) The *S. cerevisiae* centromere (CEN) sequences consist of two short conserved sequences (CDE I and CDE III) separated by 78 to 86 base pairs (bp) of AT-rich DNA (CDE II). The sequences shown are consensus sequences derived from analysis of the centromere sequences of individual yeast chromosomes. Pu = A or G; x = A or T; y = any base. (B) The arrangement of sequences at the centromere of *S. pombe* chromosome II is illustrated. The centromere consists of a central core (CC) of unique-sequence DNA, flanked by tandem repeats of three repetitive sequence elements (B, K, and L). (C) The *Drosophila* centromere consists of two satellite sequences, transposable elements, and nonrepetitive DNA.

centromeres of *S. pombe* appear to be considerably more complex than those of *S. cerevisiae*.

Studies of a *Drosophila* chromosome provided the first characterization of a centromere in higher eukaryotes (Figure 5.21C). The *Drosophila* centromere spans 420 kb, most of which (more than 85%) consists of two highly repeated satellite DNAs with the sequences AATAT and AAGAG. The remainder of the centromere consists of interspersed transposable elements, which are also found at other sites in the *Drosophila* genome, in addition to a nonrepetitive region of AT-rich DNA. Deletion of the satellite sequences and transposable elements, as well as the nonrepetitive DNA, reduced the activity of the centromere in functional assays. Thus both repetitive and nonrepetitive sequences appear to contribute to kinetochore formation and centromere function. However, there do not appear to be any sequences that are specific to the centromere or that define centromere activity.

Centromeres of other plants and animals are characterized by heterochromatin containing extensive arrays of highly repetitive sequences. In *Arabidopsis*, centromeres consist of 3 million base pairs of an AT-rich 178-base pair satellite DNA. In humans and other primates the primary centromeric sequence is α satellite DNA, which is a 171-base-pair AT-rich sequence arranged in tandem repeats spanning 1–5 million base pairs. The α satellite DNA has been found to bind centromere-associated proteins, and recent experiments have shown that the centromeric α satellite array of the human X chromosome is sufficient to serve as a functional centromere. However, abnormal human chromosomes have also been described with functional centromeres that lack α satellite DNA, so the precise requirements for centromere function in higher eukaryotes remain unclear.

Although specific DNA sequences have not been associated with centromere function, it has been shown that the chromatin at centromeres has a unique structure. In particular, histone H3 is replaced in centromeric chromatin by an H3-like variant histone called centromeric H3 (CenH3). CenH3 is uniformly present at the centromeres of all organisms that have been

studied and CenH3-containing nucleosomes are required for assembly of the other kinetochore proteins needed for centromere function. It thus appears that chromatin structure rather than a specific DNA sequence may be the primary determinant of the identity and function of centromeres. However, we still do not understand how centromeric chromatin is specified and stably maintained following cell division, so fundamental questions about the nature of centromeres in higher eukaryotes remain to be answered.

Telomeres

The sequences at the ends of eukaryotic chromosomes, called **telomeres**, play critical roles in chromosome replication and maintenance. Telomeres were initially recognized as distinct structures because broken chromosomes were highly unstable in eukaryotic cells, implying that specific sequences are required at normal chromosomal termini. This was subsequently demonstrated by experiments in which telomeres from the protozoan *Tetrahymena* were added to the ends of linear molecules of yeast plasmid DNA. The addition of these telomeric DNA sequences allowed these plasmids to replicate as linear chromosome-like molecules in yeasts, demonstrating directly that telomeres are required for the replication of linear DNA molecules.

The telomere DNA sequences of a variety of eukaryotes are similar, consisting of repeats of a simple-sequence DNA containing clusters of G residues on one strand (Table 5.5). For example, the sequence of telomere repeats in humans and other mammals is AGGGTT, and the telomere repeat in *Tetrahymena* is GGGGTT. These sequences are repeated hundreds or thousands of times and terminate with a 3′ overhang of single-stranded DNA. The repeated sequences of telomere DNA of some organisms (including humans) form loops at the ends of chromosomes as well as binding a number of proteins that protect the chromosome termini from degradation or from being joined together (Figure 5.22).

TABLE 5.5 Telomeric DNAs

Organism	Telomeric repeat sequence
Yeasts	
Saccharomyces cerevisiae	$G_{1\text{-}3}T$
Schizosaccharomyces pombe	$G_{2\text{-}5}TTAC$
Protozoans	
Tetrahymena	GGGGTT
Dictyostelium	$G_{1\text{-}8}A$
Plant	
Arabidopsis	AGGGTTT
Mammal	
Human	AGGGTT

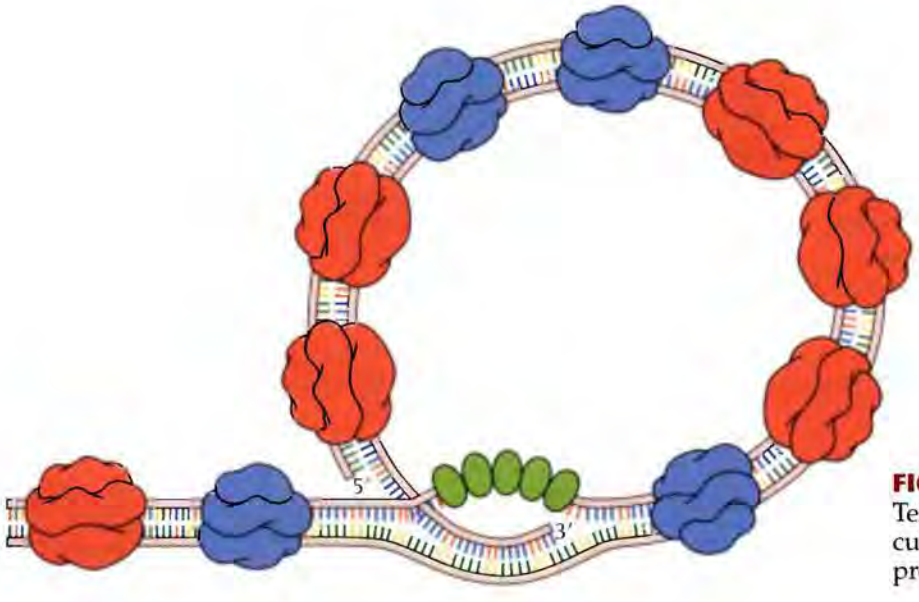


FIGURE 5.22 Structure of a telomere
Telomere DNA loops back on itself to form a circular structure and associates with a number of proteins that protect the ends of chromosomes.

■ **Cancer cells have high levels of telomerase, allowing them to maintain the ends of their chromosomes through indefinite divisions. Since normal somatic cells lack telomerase activity and do not divide indefinitely, drugs that inhibit telomerase are being developed as anti-cancer agents.**

Telomeres play a critical role in replication of the ends of linear DNA molecules (see Chapter 6). DNA polymerase is able to extend a growing DNA chain but cannot initiate synthesis of a new chain at the terminus of a linear DNA molecule. Consequently, the ends of linear chromosomes cannot be replicated by the normal action of DNA polymerase. This problem has been solved by the evolution of a special enzyme, **telomerase**, which uses reverse transcriptase activity to replicate telomeric DNA sequences. Maintenance of telomeres appears to be an important factor in determining the lifespan and reproductive capacity of cells, so studies of telomeres and telomerase have the promise of providing new insights into conditions such as aging and cancer.

The Sequences of Complete Genomes

Some of the most exciting recent advances in molecular biology have been the results of analyzing the complete nucleotide sequences of both the human genome and the genomes of several model organisms, including *E. coli*, *Saccharomyces cerevisiae*, *Caenorhabditis elegans*, *Drosophila*, *Arabidopsis*, and the mouse (Table 5.6). The results of whole genome sequencing have taken us beyond the characterization of individual genes to a global view of the organization and gene content of entire genomes. In principle, this approach has the potential of identifying all the genes in an organism, which then become accessible for investigations of their structure and function. Moreover, the availability of complete genome sequences opens the exciting possibility of identifying the sequences that regulate gene expression by genome-wide analysis. While much remains to be learned, the available genome sequences have provided scientists with a unique data-

TABLE 5.6 Representative Sequenced Genomes

Organism	Genome size (Mb)[a]	Number of genes	Protein-coding sequence
Bacteria			
Mycoplasma genitalium	0.58	470	88%
H. influenzae	1.8	1743	89%
E. coli	4.6	4288	88%
Yeasts			
S. cerevisiae	12	6000	70%
S. pombe	12	4800	60%
Invertebrates			
C. elegans	97	19,000	25%
Drosophila	180	13,600	13%
Plants			
Arabidopsis thaliana	125	26,000	25%
Rice	390	37,000	12%
Fish			
Pufferfish	370	20,000–23,000	10%
Birds			
Chicken	1000	20,000–23,000	3%
Mammals			
Human	3200	20,000–25,000	1.2%

[a]Mb = millions of base pairs

base, consisting of the nucleotide sequences of complete sets of genes and their regulatory sequences. Since many of these genes have not been previously identified, determination of their functions will form the basis of many future studies in cell biology.

Prokaryotic Genomes

We now know the complete genome sequences of more than 100 different bacteria, and still more are in the process of being determined. The first complete sequence of a cellular genome, reported in 1995 by a team of researchers led by Craig Venter, was that of the bacterium *Haemophilus influenzae*, a common inhabitant of the human respiratory tract. The genome of *H. influenzae* is approximately 1.8×10^6 base pairs (1.8 **megabases,** or **Mb**), slightly less than half the size of the *E. coli* genome. The complete nucleotide sequence indicated that the *H. influenzae* genome is a circular molecule containing 1,830,137 base pairs of DNA. The sequence was then analyzed to identify the genes encoding rRNAs, tRNAs, and proteins. Potential protein-coding regions were identified by computer analysis of the DNA sequence to detect **open-reading frames**—long stretches of nucleotide sequence that can encode polypeptides because they contain none of the three chain-terminating codons (UAA, UAG, and UGA). Since these chain-terminating codons occur randomly once in every 21 codons (3 chain-terminating codons out of 64 total), open-reading frames that extend for more than a hundred codons usually represent functional genes.

This analysis identified six copies of rRNA genes, 54 different tRNA genes, and 1743 potential protein-coding regions in the *H. influenzae* genome (Figure 5.23). More than a thousand of these could be assigned a biological role (e.g., an enzyme of the citric acid cycle) on the basis of their relationships to known protein sequences, but the others represent genes of

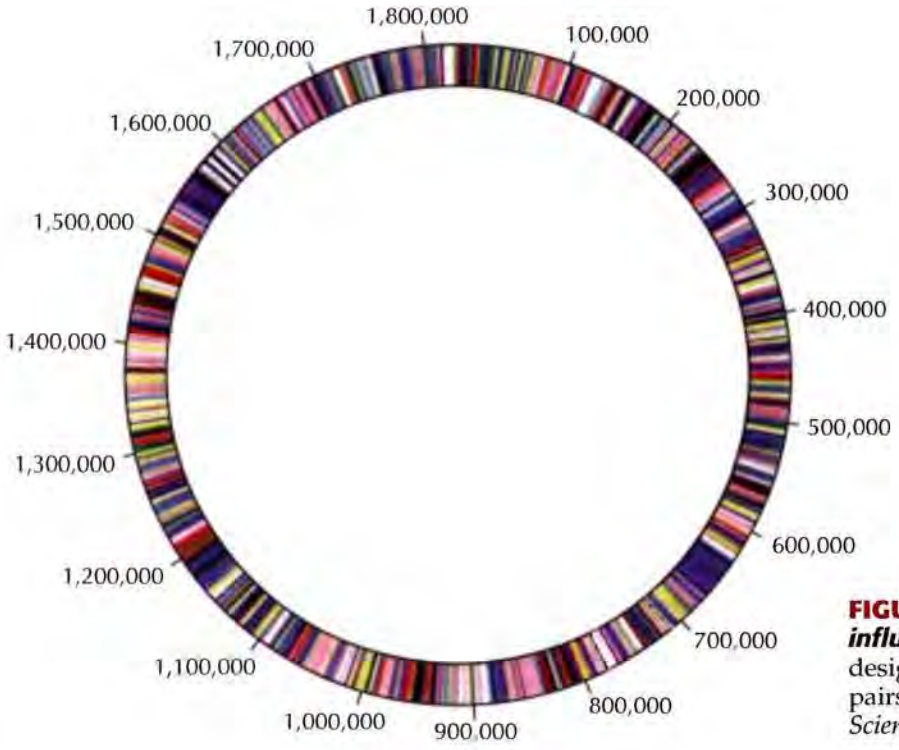


FIGURE 5.23 The genome of *Haemophilus influenzae* Predicted protein-coding regions are designated by colored bars. Numbers indicate base pairs of DNA. (From R. D. Fleischmann et al., 1995. *Science* 269: 496.)

unknown function. The predicted coding sequences have an average size of approximately 900 base pairs, so they cover about 1.6 Mb of DNA, corresponding to nearly 90% of the genome of *H. influenzae*.

The complete sequence of the genome of *Mycoplasma genitalium* is of particular interest because mycoplasmas are the simplest present-day bacteria and contain the smallest genomes of all known cells. The genome of *M. genitalium* is only 580 kb (0.58 Mb) long and may represent the minimal set of genes required to maintain a self-replicating organism. Analysis of its DNA sequence indicates that *M. genitalium* contains only 470 predicted protein-coding sequences, which correspond to approximately 88% of genomic DNA. Many of these sequences were identified as genes encoding proteins involved in DNA replication, transcription, translation, membrane transport, and energy metabolism. However, *M. genitalium* contains many fewer genes for metabolic enzymes than does *H. influenzae*, reflecting its more limited metabolism. For example, many genes known to encode components of biosynthetic pathways are lacking in the genome of *M. genitalium*, consistent with its need to obtain amino acids and nucleotide precursors from a host organism. Interestingly, the *Mycoplasma* genome also includes approximately 150 genes of currently unknown function. Thus, even in the simplest of cells, the biological roles of many genes remain to be determined.

The sequence of the genome of the archaebacterium *Methanococcus jannaschii*, reported in 1996, provided major insights into the evolutionary relationships between the archaebacteria, eubacteria, and eukaryotes. The genome of *M. jannaschii* is 1.7 Mb and contains 1738 predicted protein-coding sequences—similar in size to the genome of *H. influenzae*. However, only about one-third of the protein-coding sequences identified in *M. jannaschii* were related to known genes of either eubacteria or eukaryotes, indicating the distinct genetic composition of the archaebacteria. The genes of *M. jannaschii* encoding proteins involved in energy production and biosynthesis of cell constituents are related to those of eubacteria, suggesting that basic metabolic processes evolved in a common ancestor of both the archaebacteria and the eubacteria. Importantly, however, the *M. jannaschii* genes encoding proteins involved in DNA replication, transcription, and translation are more closely related to those of eukaryotes than to those of eubacteria. Genomic sequencing of this archaebacterium thus indicates that the archaebacteria and eukaryotes are as closely related to each other as either is to the eubacteria (see Figure 1.7).

Although the relative simplicity and facile genetics of *E. coli* have made it a favored organism of molecular biologists, the 4.6-Mb *E. coli* genome was not completely sequenced until 1997. Analysis of the *E. coli* sequence revealed a total of 4288 genes, with protein-coding sequences accounting for approximately 88% of the *E. coli* genome. Of the 4288 genes revealed by sequencing, 1835 had been previously identified and the functions of an additional 821 could be deduced by comparisons to the sequences of characterized genes of other organisms. However, the functions of 1632 *E. coli* genes (nearly 40% of the genome) could not be determined. Thus, even for an organism as thoroughly studied as *E. coli*, genomic sequencing demonstrates that a great deal remains to be learned about prokaryotic cell biology.

The Yeast Genome

As noted already, the simplest eukaryotic genome (1.2×10^7 base pairs of DNA) is found in the yeast *Saccharomyces cerevisiae*. Moreover, yeasts grow rapidly and are subject to simple genetic manipulations. Thus in many

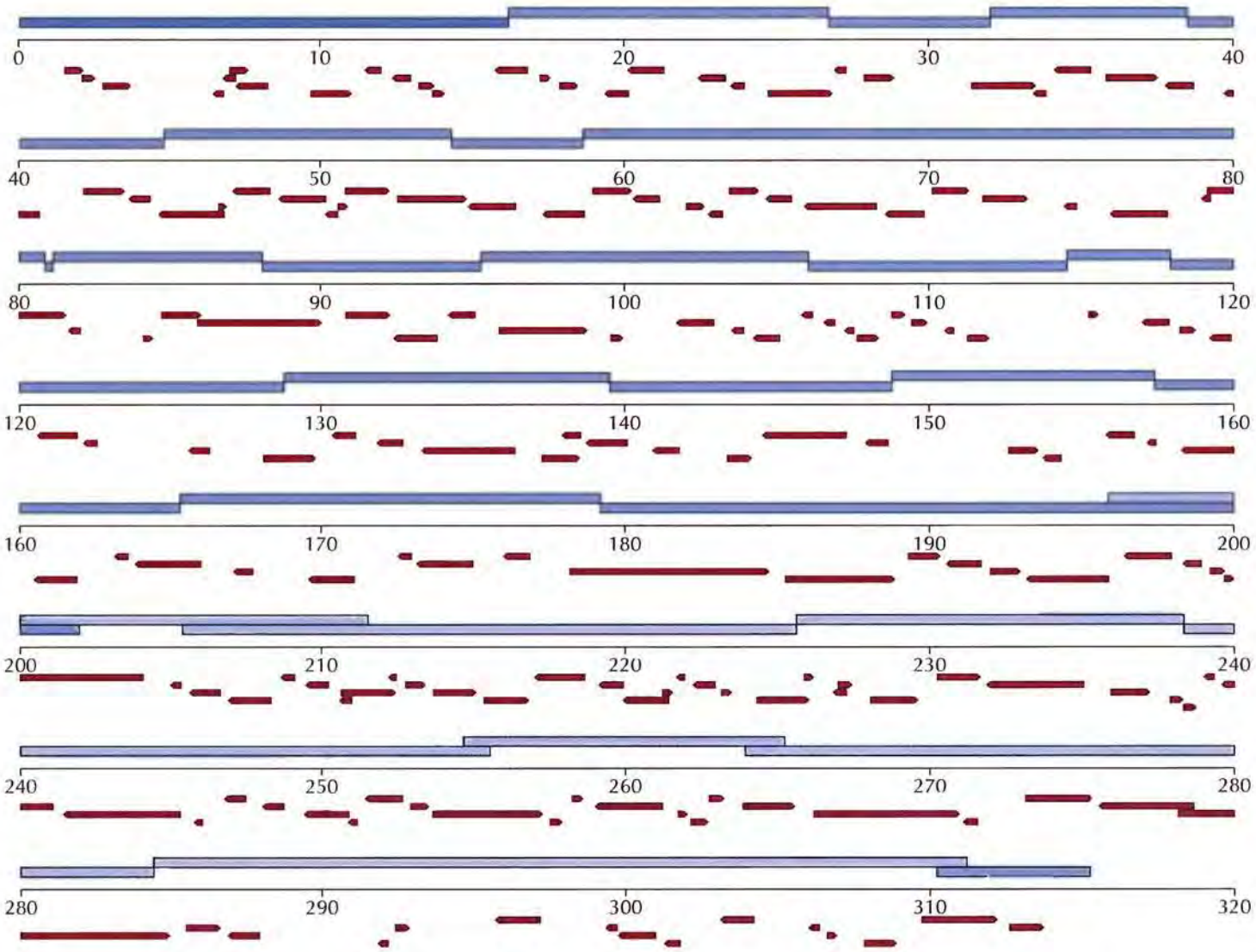


FIGURE 5.24 Yeast chromosome III The upper blue bars designate the clones used for DNA sequencing. Open-reading frames are indicated by arrows. (From S. G. Oliver et al., 1992. *Nature* 357: 38.)

ways yeasts are model eukaryotic cells that can be studied much more readily than the cells of mammals or other higher eukaryotes. Consequently, the complete sequencing of an entire yeast chromosome in 1992 (Figure 5.24), followed by determination of the sequence of the complete *S. cerevisiae* genome in 1996, were major steps in understanding the molecular biology of eukaryotic cells.

The *S. cerevisiae* genome contains about 6000 genes, including 5885 predicted protein-coding sequences, 140 ribosomal RNA genes, 275 transfer RNA genes, and 40 genes encoding small nuclear RNAs involved in RNA processing (discussed in Chapter 7). Yeasts thus have a high density of protein-coding sequences, similar to bacterial genomes, with protein-coding sequences accounting for approximately 70% of total yeast DNA. Consistent with this, only 4% of yeast genes were found to contain introns. Moreover, those *S. cerevisiae* genes that do contain introns usually have only a single small intron near the beginning of the gene.

Computer analysis was able to assign a predicted function to approximately 3000 of the *S. cerevisiae* protein-coding sequences based on similarities to the sequences of known genes. Based on analysis of these genes, it appears that approximately 11% of yeast proteins function in metabolism,

3% in the production and storage of metabolic energy, 3% in DNA replication, repair, and recombination, 7% in transcription, 6% in translation, and 14% in protein sorting and transport. However, the functions of many of these genes are only known in general terms (such as "transcription factor"), so their precise roles within the cell still need to be determined. Moreover, since half of the proteins encoded by the yeast genome were unrelated to previously described genes, the functions of an additional 3000 unknown proteins remain to be elucidated by genetic and biochemical analyses.

The sequence of the *S. cerevisiae* genome has been more recently followed by the sequence of the genome of the fission yeast, *S. pombe*, as well as the genomes of several other yeast and fungi. As discussed earlier in this chapter, *S. cerevisiae* and *S. pombe* are quite divergent and differ in many aspects of their biology, including the structure of their centromeres (see Figure 5.21). Interestingly, their genomes also display considerable differences. Although both *S. cerevisiae* and *S. pombe* have approximately the same amount of unique sequence DNA (12.5 Mb), *S. pombe* appears to contain only about 4800 genes. Introns are much more prevalent in *S. pombe* than in *S cerevisiae*. Approximately 43% of *S. pombe* genes contain introns and the introns in *S. pombe* are larger than those in *S cerevisiae*, so protein-coding sequence accounts for only about 60% of the *S. pombe* genome. The majority of *S. pombe* genes have homologs in the *S cerevisiae* genome, but approximately 700 genes are unique to *S. pombe*.

Now that yeast genome sequences have been completed, determination of the functions of the many new genes described in both *S. cerevisiae* and *S. pombe* is a major goal. Fortunately, yeasts are particularly amenable to functional analyses of unknown genes because of the facility with which normal chromosomal loci can be inactivated by homologous recombination with cloned sequences (discussed in Chapter 4). Therefore direct functional analysis of yeast genes that were initially identified only on the basis of their nucleotide sequence can be systematically undertaken. Sequencing the yeast genomes has thus opened the door to studying many new areas of the biology of a simple eukaryotic cell. Such studies are expected to reveal the functions of many new genes that are not restricted to yeasts but are common to all eukaryotes, including humans.

The Genomes of *Caenorhabditis elegans* and *Drosophila melanogaster*

The genomes of *C. elegans* and *Drosophila* are relatively simple animal genomes, intermediate in size and complexity between those of yeasts and humans. Distinctive features of each of these organisms make them important models for genome analysis: *C. elegans* is widely used for studies of animal development, and *Drosophila* has been especially well analyzed genetically. The genomes of these organisms, however, are about tenfold larger than those of yeasts, introducing a new order of difficulty in genome mapping and sequencing. Determination of the sequence of *C. elegans* in 1998 therefore represented an important milestone in genome analysis, which extended genome sequencing from unicellular organisms (bacteria and yeast) to a multicellular organism recognized as an important model for animal development.

The initial phases of analysis of the *C. elegans* genome used DNA fragments cloned in cosmids, which accommodate DNA inserts of approximately 30–45 kb (see Table 4.3). This approach, however, was unable to cover the complete genome, which was accomplished by the cloning of much larger pieces of DNA in **yeast artificial chromosome (YAC)** vectors.

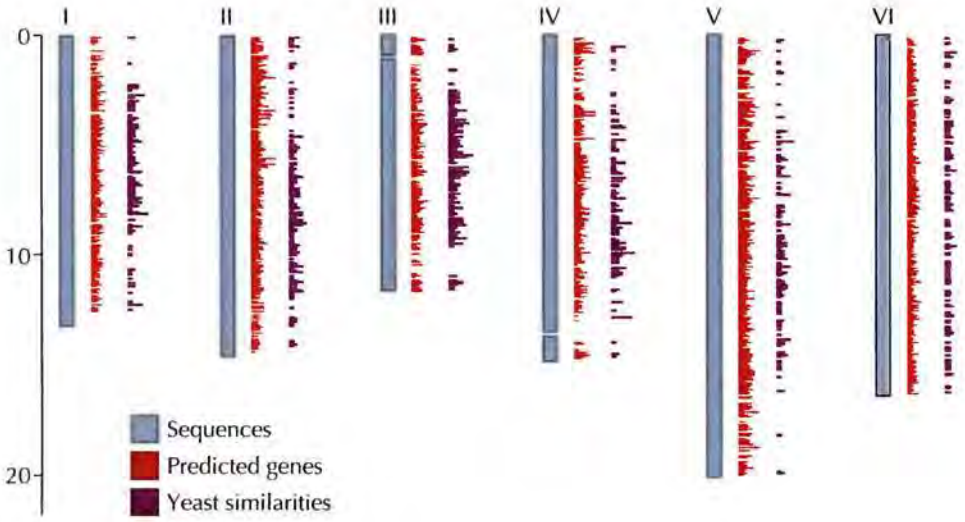


FIGURE 5.25 The *C. elegans* genome The positions of the predicted genes of *C. elegans* on each chromosome are indicated by red bars. Those that are similar to genes of yeast are indicated by purple. (From The *C. elegans* Sequencing Consortium, 1998. *Science* 282: 2012.)

As noted in Chapter 4, the unique feature of YACs is that they contain centromeres and telomeres, allowing them to replicate as linear chromosome-like molecules in yeasts. They can therefore be used to clone DNA fragments the size of yeast chromosomal DNAs, up to thousands of kilobases in length. The large DNA inserts that can be cloned in YACs and other high-capacity vectors are critically important for analysis of complex genomes.

The *C. elegans* genome is 97×10^6 base pairs and contains about 19,000 predicted protein-coding sequences—approximately three times the number of genes in yeast (Figure 5.25). In contrast to the compact genome organization of yeast, genes in *C. elegans* span about 5 kilobases and contain an average of five introns. Protein-coding sequences thus account for only about 25% of the *C. elegans* genome, as compared to 60–70% of *S. pombe* and *S. cerevisiae* and nearly 90% of bacterial genomes.

Approximately 40% of the predicted *C. elegans* proteins displayed significant similarity to known proteins of other organisms. As expected, there are substantially more similarities between the proteins of *C. elegans* and humans than between *C. elegans* and either yeast or bacteria. Proteins that are common between *C. elegans* and yeast may function in the basic cellular processes shared by these organisms, such as metabolism, DNA replication, transcription, translation, and protein sorting. These core biological processes appear to be carried out by a similar number of genes in both organisms, and it is likely that these genes will be shared by all eukaryotic cells. In contrast, the majority of *C. elegans* genes are not found in yeast and may function in the more intricate regulatory activities required for the development of multicellular organisms. Elucidating the functions of these genes is likely to be particularly exciting in terms of understanding animal development. Although adult *C. elegans* contain only 959 somatic cells in the entire body, they have all of the specialized cell types found in more complicated animals. Moreover, the complete pattern of cell divisions leading to *C. elegans* development has been described, including analysis of the connections made by all 302 neurons in the adult animal. Many of the genes involved in *C. elegans* development and differentiation have already been found to be related to genes involved in controlling the proliferation and differentiation of mammalian cells, substantiating the validity of *C. elegans* as a model for more complex animals. With little doubt, many more critical developmental control genes will be uncovered from studies of the *C. elegans* genomic sequence.

FIGURE 5.26 Polytene chromosomes of *Drosophila* A light micrograph of stained salivary gland chromosomes. The four chromosomes (X, 2, 3, and 4) are joined at their centromeres. (Peter J. Bryant/Biological Photo Service.)

Drosophila is another key model for animal development, which has been particularly well-characterized genetically. The advantages of *Drosophila* for genetic analysis include its relatively simple genome and the fact that it can be easily maintained and bred in the laboratory. In addition, a special tool for genetic analysis in *Drosophila* is provided by the giant **polytene chromosomes** that are found in some tissues, such as the salivary glands of larvae. These chromosomes arise in nondividing cells as a consequence of repeated replication of DNA strands that fail to separate from each other. Thus each polytene chromosome contains hundreds of identical DNA molecules aligned in parallel. Because of their size, these polytene chromosomes are visible in the light microscope, and appropriate staining procedures reveal a distinct banding pattern (Figure 5.26). The banding of polytene chromosomes provides a much greater degree of resolution than that achieved with metaphase chromosomes (e.g., see Figure 5.17). The polytene chromosomes are decondensed interphase chromosomes that contain actively expressed genes. More than 5000 bands are visible, each corresponding to an average length of approximately 20 kb of DNA. In contrast, the bands identified in human metaphase chromosomes contain several megabases of DNA.

The banding pattern of polytene chromosomes thus provides a high-resolution physical map of the *Drosophila* genome. Gene deletions can often be correlated with the loss of a specific chromosomal band, thereby defining the physical location of the gene on the chromosome. In addition, cloned DNAs can be mapped by *in situ* hybridization to polytene chromosomes, often with sufficient resolution to localize cloned genes to specific bands (Figure 5.27). Thus the map positions of cosmid or YAC clones (which span many bands) can readily be determined, providing the base for genomic sequence analysis.

Because of the power of *Drosophila* genetics, the sequencing of the *Drosophila* genome early in 2000 was an important advance in genomic analysis. The genome of *Drosophila* consists of approximately 180×10^6 base pairs, of which about one-third is heterochromatin. The heterochromatin consists principally of simple sequence satellite repeats, in addition to inter-

spersed transposable elements, and was not included in the genomic sequence. The remaining 120×10^6 base pairs of euchromatin was sequenced using a combination of **bacterial artificial chromosome (BAC)** clones, which carry large inserts of DNA (see Table 4.3), and a shotgun approach in which small fragments of DNA were randomly cloned and sequenced in plasmid vectors. The sequences of these small fragments of DNA were then assembled into a large contiguous sequence by identification of overlaps between fragments, and these sequence assemblies were aligned with the BAC clones to yield a complete sequence of the euchromatic portion of the *Drosophila* genome.

The *Drosophila* genome contains approximately 13,600 genes; surprisingly fewer than the number of genes in *C. elegans*, even though *Drosophila* is a more complex organism. However, it is important to note that this difference in gene number does not correspond to a difference in genetic complexity, because many genes are duplicated in both *Drosophila* and *C. elegans*. When these duplications are taken into account, it appears that both *Drosophila* and *C. elegans* contain a similar number of distinct genes, estimated between 10,000 and 15,000. Like *C. elegans*, *Drosophila* genes contain an average of 4 introns, and the total amount of intron sequence is similar to the amount of exon sequence. In total, protein-coding sequence accounts for about 13% of the *Drosophila* genome.

It is especially striking that a complex animal like *Drosophila* has only about twice the number of unique genes found in yeast, which appears to be a much simpler organism. Apparently, the complexity of multicellular organisms is not simply related to a greater number of genes. Part of the increased biological complexity of *Drosophila* and *C. elegans* may arise from the fact that their proteins are generally larger and contain more functional domains than the proteins of yeast. Further studies and functional analysis of the genes that have been uncovered by sequencing the *Drosophila* and *C. elegans* genomes will undoubtedly play a major role in understanding the ways in which these genes act to direct the complex process of animal development.

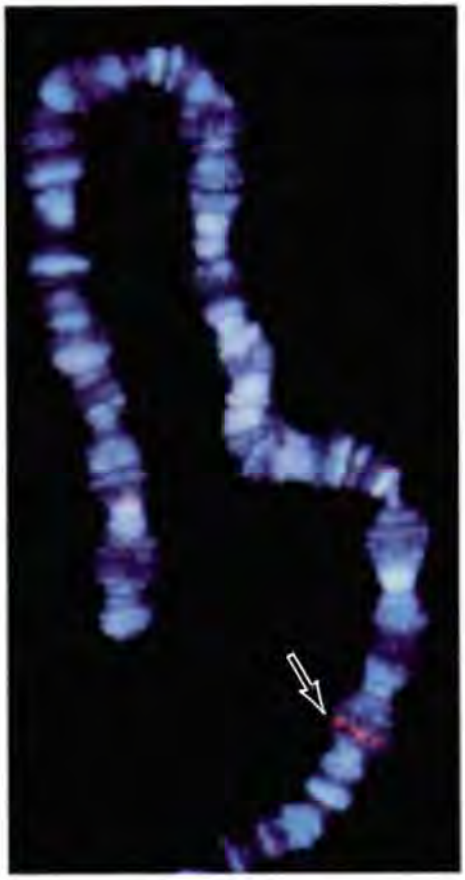


FIGURE 5.27 *In situ* hybridization to a *Drosophila* polytene chromosome Hybridization of a YAC clone to a polytene chromosome is illustrated. The region of hybridization is indicated by an arrow. (Courtesy of Daniel L. Hartl, Harvard University.)

Plant Genomes

The completion of the genome sequence of *Arabidopsis thaliana* in 2000 extended genome sequencing from animals to plants, and was thus a major event in plant biology. *Arabidopsis thaliana* is a simple flowering plant, which has been widely used as a model for studies of plant molecular biology and development. Its advantages as a model organism for molecular biology and genetics include its relatively small genome of approximately 125×10^6 base pairs, similar in size to the genomes of *C. elegans* and *Drosophila*. Like the *Drosophila* genome, the *Arabidopsis* genome was sequenced principally using BAC vectors to accommodate large DNA inserts.

Surprisingly, analysis of the *Arabidopsis* genome indicated that it contained approximately 26,000 protein-coding genes—significantly more genes than were found in either *C. elegans* or *Drosophila*. However, this unexpectedly large number of genes does not reflect a greater diversity of proteins encoded by the *Arabidopsis* genome. Instead, it appears that the large number of genes in *Arabidopsis* is the result of duplications of large segments of the *Arabidopsis* genome. These duplications involve approximately 60% of the genome, so the number of distinct protein-coding genes in *Arabidopsis* is estimated to be about 15,000—similar to the number of genes in *C. elegans* or *Drosophila*.

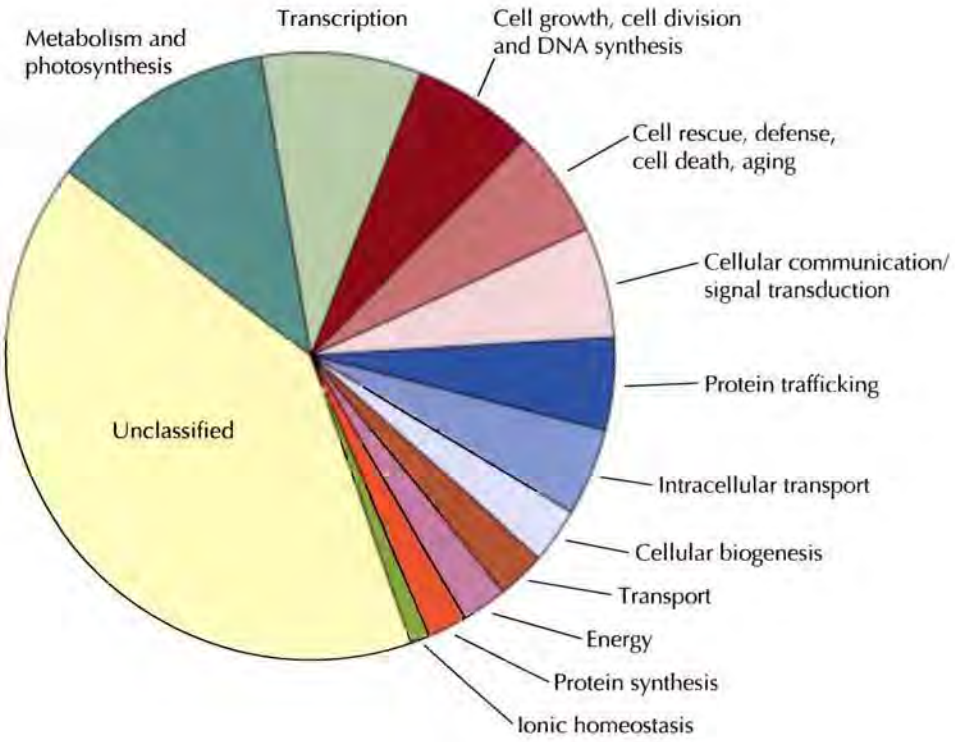


FIGURE 5.28 Functions of predicted genes of *Arabidopsis thaliana* The chart illustrates the proportion of *Arabidopsis* in different functional categories. (From The *Arabidopsis* Genome Initiative, 2000. *Nature* 408: 796.)

The gene density in *Arabidopsis* is also similar to that of *C. elegans*, with protein-coding sequences accounting for about 25% of the *Arabidopsis* genome. On the average, *Arabidopsis* genes have approximately 4 introns, and the total length of intron sequences is about the same as the total length of exon sequences. Transposable elements account for about 10% of the *Arabidopsis* genome. As in *Drosophila*, transposable element repeats are clustered at the centromeres together with satellite repetitive sequences.

Comparative analysis of the functions of the *Arabidopsis* genes has revealed both interesting similarities and differences between the genes of plants and animals. *Arabidopsis* genes involved in fundamental cellular processes such as DNA replication, repair, transcription, translation, and protein trafficking are similar to those in yeast, *C. elegans*, and *Drosophila*, reflecting the common evolutionary origins of all eukaryotic cells. In contrast, the *Arabidopsis* genes encoding proteins involved in processes such as cell signaling and membrane transport are quite different from those in animals, consistent with the major differences in physiology and development between plants and animals. About one-third of all *Arabidopsis* genes appear unique to plants, as they are not found in yeast or animal genomes. The largest functional group of *Arabidopsis* genes, corresponding to 22% of the genome, encodes proteins involved in metabolism and photosynthesis (Figure 5.28). Another large group of genes (12% of the genome) encodes proteins involved in plant defense. It is also noteworthy that *Arabidopsis* encodes more than 3000 proteins that regulate transcription (accounting for approximately 17% of the genome). This number of gene regulatory proteins (transcription factors) is two or three times more than are found in *Drosophila* and *C. elegans*, respectively. Many of the *Arabidopsis* transcription factors are unique to plants, presumably reflecting distinct features of gene expression in plant development and in the response of plants to the environment.

The sequence of *Arabidopsis* was followed in 2002 by publication of two draft sequences of the rice genome. Rice is of major importance as a cereal crop and is the staple food for more than half the world's population, so sequencing the rice genome has the potential of leading to very significant applications in agriculture and biotechnology. Two groups of researchers reported draft sequences of the genomes of two subspecies of rice: the *indica* subspecies, which is the most widely cultivated subspecies in China and most of the rest of Asia; and the *japonica* subspecies, which is the variety preferred in Japan. These initial draft sequences of the rice genome were followed by a high quality complete sequence of the *japonica* subspecies in 2005.

The rice genome consists of about 390×10^6 base pairs of DNA—about 3 times larger than the genome of *Arabidopsis*. At least 35% of the rice genome consists of transposable elements, in part accounting for its larger size. In addition, rice contain a surprisingly high number of predicted protein-coding genes, estimated at approximately 37,000. Like *Arabidopsis*, rice contains many duplicated genes, which have arisen as a result of duplication of large segments (approximately 60%) of the genome. Nonetheless, the rice genome contains more genes than either *Arabidopsis* or humans, underscoring the fact that gene number does not directly correlate with biological complexity in eukaryotes. Interestingly, approximately 70% of the genes predicted in rice are also found in *Arabidopsis*, and almost 90% of the genes that have been identified in *Arabidopsis* are found in rice. Most of the genes shared between *Arabidopsis* and rice are not found in yeast or animal genomes and therefore appear to be specific for plants.

The Human Genome

For many scientists, the ultimate goal of genome analysis was determination of the complete nucleotide sequence of the human genome: approximately 3×10^9 base pairs of DNA. To understand the magnitude of this undertaking, recall that the human genome is more than ten times larger than that of *Drosophila;* that the smallest human chromosome is several times larger than the entire yeast genome; and that the extended length of DNA that makes up the human genome is about 1 m long. From all of these perspectives, determination of the human genome sequence was a phenomenal undertaking, and its publication in draft form in 2001 was heralded as a scientific achievement of historic magnitude.

The human genome is distributed among 24 chromosomes (22 autosomes and the 2 sex chromosomes), each containing between 45 and 280 Mb of DNA (Figure 5.29). Prior to determination of the genome sequence, several thousand human genes had been identified and mapped to positions on the human chromosomes. One commonly used method to localize genes is *in situ* hybridization of probes labeled with fluorescent dyes to chromosomes—a method generally referred to as **fluorescence *in situ* hybridization,** or **FISH** (Figure 5.30). *In situ* hybridization to metaphase chromosomes allows the mapping of a cloned gene to a locus defined by a chromosome band. Because each band of human metaphase chromosomes contains thousands of kilobases of DNA, *in situ* hybridization to human metaphase chromosomes does not provide the detailed mapping information obtained by hybridization to the polytene chromosomes of *Drosophila*, which allows the localization of genes to interphase chromosome bands containing only 10 to 20 kb of DNA. Higher resolution can be obtained, however, by hybridization to more extended human chromosomes from

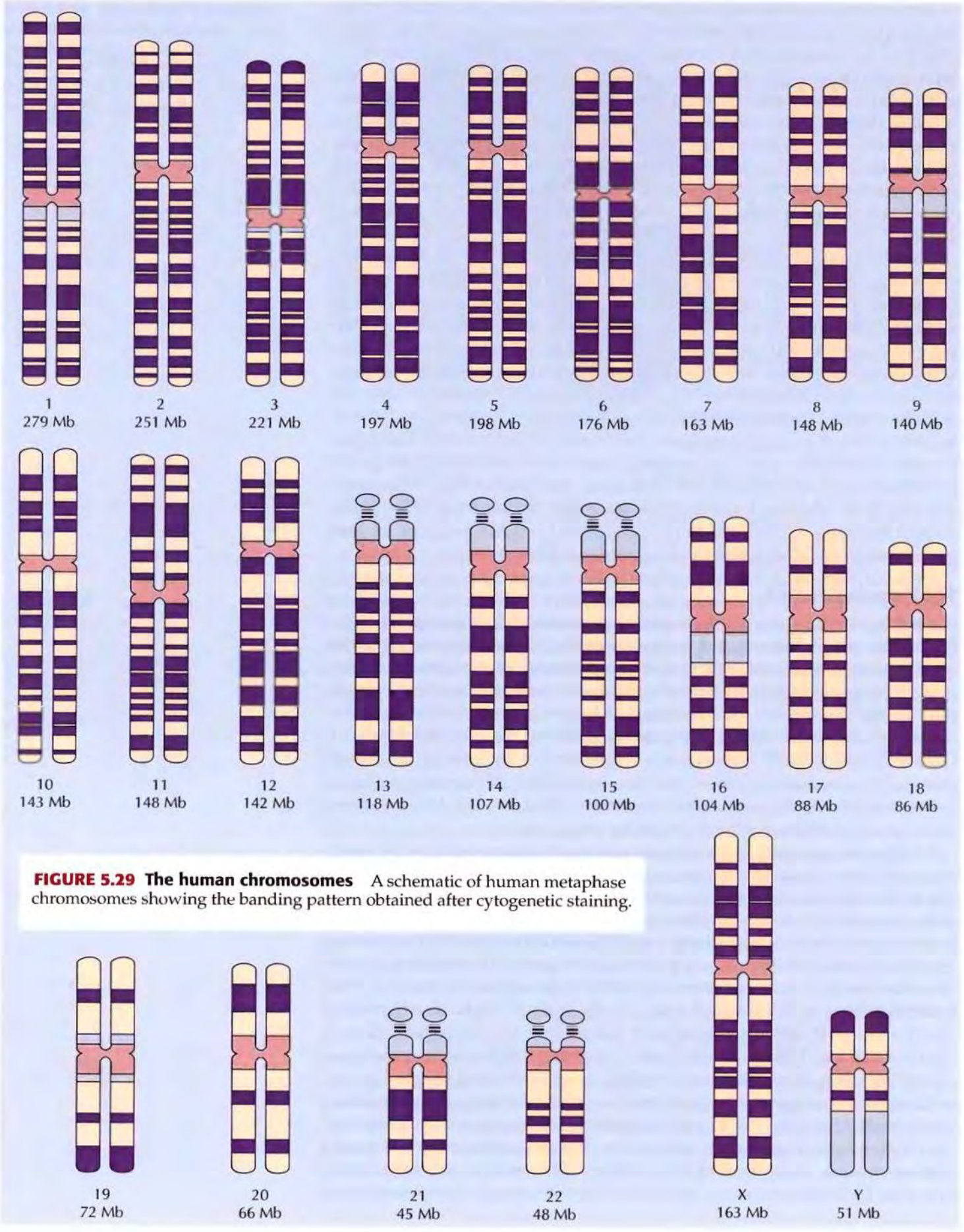


FIGURE 5.29 The human chromosomes A schematic of human metaphase chromosomes showing the banding pattern obtained after cytogenetic staining.

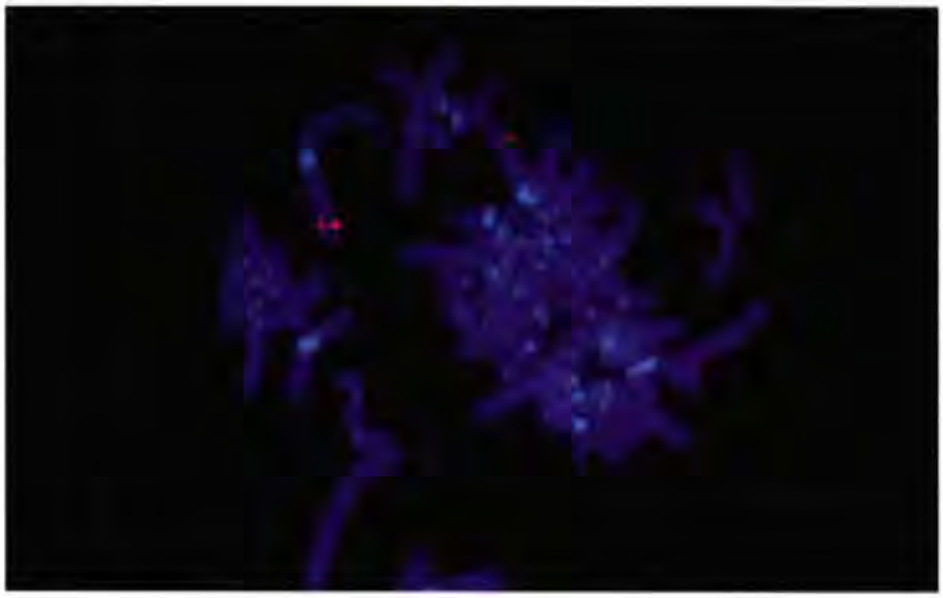


FIGURE 5.30 Fluorescence *in situ* hybridization A fluorescent probe for the gene encoding lamin B receptor is hybridized to stained human metaphase chromosomes (blue). Single gene hybridization signals are detected as red fluorescence. (Courtesy of K. L. Wydner and J. B. Lawrence, University of Massachusetts Medical Center.)

prometaphase or interphase cells, allowing the use of fluorescence *in situ* hybridization to map cloned genes to regions of about 100 kb. In addition to FISH, genetic linkage analysis and the physical mapping of cloned genomic and cDNA sequences were used to establish physical and genetic maps of the human genome, which provided a background for genomic sequencing.

The draft sequences of the human genome published in 2001 were produced by two independent teams of researchers, who used different approaches. One research team, The International Human Genome Sequencing Consortium, used BAC clones that had been mapped to sites on the human chromosomes as the substrates for sequencing. The other team, led by Craig Venter of Celera Genomics, used a shotgun approach in which small fragments were cloned and sequenced, and overlaps between the sequences of these fragments were then used to assemble the sequence of the genome. Both of these sequences were initially incomplete drafts in which approximately 90% of the euchromatin portion of the genome had been sequenced and assembled. Continuing efforts have closed the gaps and improved the accuracy of the draft sequences, leading to publication of a high-quality human genome sequence in 2004.

The sequenced euchromatin portion of the genome encompasses approximately 2.9×10^6 kb of DNA (Figure 5.31). The total size of the genome is approximately 3.2×10^6 kb, with the remaining 10% of the genome (0.3×10^6

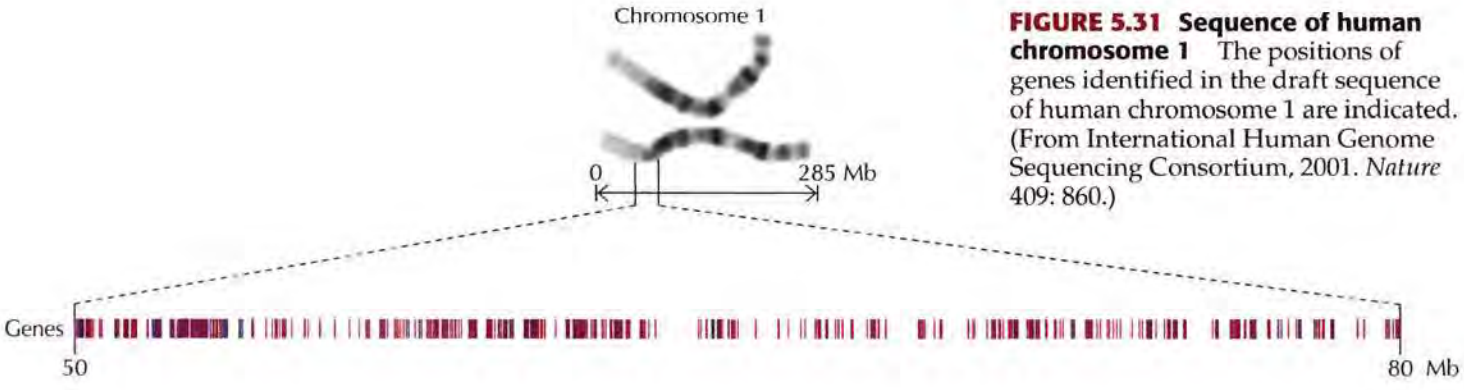


FIGURE 5.31 Sequence of human chromosome 1 The positions of genes identified in the draft sequence of human chromosome 1 are indicated. (From International Human Genome Sequencing Consortium, 2001. *Nature* 409: 860.)

KEY EXPERIMENT

The Human Genome

Initial Sequencing and Analysis of the Human Genome
International Human Genome Sequencing Consortium
Nature, Volume 409, 2001, pages 860–921

The Sequence of the Human Genome
J. Craig Venter and 273 others
Science, Volume 291, 2001, pages 1304–1351

The Context

The idea of sequencing the entire human genome was first conceived in the mid-1980s. It was initially met with broad skepticism among biologists, most of whom felt it was simply not a feasible undertaking. At the time, the largest genome that had been completely sequenced was that of Epstein-Barr virus, which totaled approximately 180,000 base pairs of DNA. From this perspective, sequencing the human genome, which was almost 20,000 times larger, seemed inconceivable to many. However, the idea of such a massive project in biology captivated the imagination of others, including Charles DeLisi who was then head of the Office of Health and Environmental Research at the Department of Energy. In 1986 DeLisi succeeded in launching the Human Genome Initiative as a project within the Department of Energy.

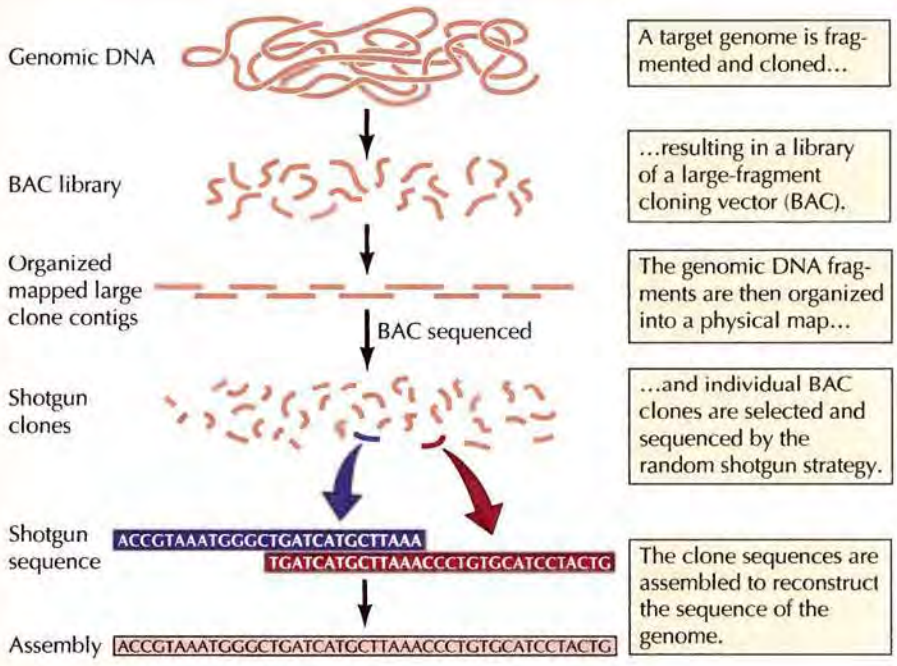
The project gained broader support in 1988 when it was endorsed by a committee of the National Research Council. This committee recommended a broader effort, including sequencing the genomes of several model organisms and the parallel development of detailed genetic and physical maps of the human chromosomes. This effort was centered at the National Institutes of Health, initially under the direction of James Watson (codiscoverer of the structure of DNA), and then under the leadership of Frances Collins.

The first complete genome to be sequenced was that of the bacterium *Haemophilus influenzae*, reported by Craig Venter and colleagues in 1995. Venter had been part of the genome sequencing effort at the National Institutes of Health but had left to head a nonprofit company, The Institute for Genomic Research, in 1991. In the meantime, considerable progress had been made in mapping the human genome, and the initial sequence of *H. influenzae* was followed by the sequences of other bacteria, yeast, and *C. elegans* in 1998.

In 1998 Venter formed a new company, Celera Genomics, and announced plans to use advanced sequencing technologies to obtain the entire human genome sequence in 3 years. Collins and other leaders of the publicly funded genome project responded by accelerating their efforts, resulting in a race that eventually led to the publication of two draft sequences of the human genome in February, 2001.

The Experiments

The two groups of scientists used different approaches to obtain the human genome sequence. The publicly funded team, The International Human Genome Sequencing Consortium, headed by Eric Lander, sequenced DNA fragments derived from BAC clones that had been previously mapped to human chromosomes, similar to the approach used to determine the sequence of the yeast and *C. elegans* genomes (see figure). In contrast, the Celera Genomics team used a whole-genome shotgun sequencing approach that Venter and colleagues had first used to sequence the genome of *H. influenzae*. In this approach, DNA fragments were sequenced at random, and overlaps between fragments were then used to reassemble a complete genome sequence. Both sequences covered only the euchromatin portion of the human genome—approximately 2900



Strategy for genome sequencing using BAC clones that had been organized into overlapping clusters (contigs) and mapped to human chromosomes.

KEY EXPERIMENT

Mb of DNA—with the heterochromatin repeat-rich portion of the genome (approximately 300 Mb) remaining unsequenced.

Both of these initially published versions were draft, rather than completed, sequences. Subsequent efforts completed the sequence, leading to publication of a highly accurate sequence of the human genome in 2004.

The Impact

Several important conclusions immediately emerged from the human genome sequences. First, the number of human genes was surprisingly small and appears to be between 20,000 and 25,000 in the completed sequence. Interestingly, however, alternative splicing appears to be common in the human genome, so many genes may encode more than 1 protein. Introns account for about 20% of the human genome and repetitive sequences for about 60%. It is noteworthy that over 40% of human DNA is composed of sequences derived by reverse transcription, emphasizing the importance of this mode of information transfer in shaping our genome.

Beyond these immediate conclusions, the sequence of the human genome, together with the genome sequences of other organisms, will provide a new basis for biology and medicine in the years to come. The impact of the genome sequence will be felt in discovering new genes and their functions, understanding gene regulation, elucidating the basis of human diseases, and developing new strategies for prevention and treatment based on the genetic makeup of individuals. Knowledge of the human genome may ultimately contribute to meeting what Venter and colleagues refer to as "The real challenge of human biology…to explain how our minds have come to organize thoughts sufficiently well to investigate our own existence."

Eric Lander

Craig Venter

kb) corresponding to highly repetitive sequences in heterochromatin. As discussed earlier in this chapter, interspersed repetitive sequences, the majority of which are transposable elements that have moved throughout the genome by reverse transcription of RNA intermediates, account for approximately 45% of the human euchromatin sequence. Another 5% of the genome consists of duplicated segments of DNA, so about 60% of the human genome consists of repetitive DNA sequences.

A major surprise from the genome sequence is the unexpectedly low number of human genes. The human genome consists of only 20,000–25,000 genes, which is not much larger than the number of genes in simpler animals like *C. elegans* and *Drosophila*. In fact, humans have fewer genes than rice, emphasizing one of the major conclusions that has emerged from the results of genome sequencing: the biological complexity of an organism is not simply a function of the number of genes in its genome. On the other hand, there appears to be a significant amount of alternative splicing in human genes, allowing a single gene to specify more than one protein (see Figure 5.5). Although the extent of alternative splicing in humans is not yet clear, it may substantially expand the number of proteins that can be encoded by the human genome.

Human genes are spread over much larger distances and contain more intron sequence than genes in *Drosophila* or *C. elegans*. The average protein-coding sequence in human genes is approximately 1400 base pairs, similar to that in *Drosophila* and *C. elegans*. However, the average human gene spans about 30 kb of DNA, with more than 90% of the gene corresponding to introns. Approximately 20% of the genome thus consists of introns, with only about 1.2% of the human genome corresponding to protein-coding sequences.

For many years, scientists generally accepted an estimate of approximately 100,000 genes in the human genome. On publication of the draft genome sequence in 2001, the number was drastically reduced to between 30,000 and 40,000. Current estimates, based on the high quality sequence published in 2004 and using improved computational tools to identify genes, reduce the number of human genes still further, to approximately 20,000 to 25,000.

Over 40% of the predicted human proteins are related to proteins in other sequenced organisms, including *Drosophila* and *C. elegans*. Many of these conserved proteins function in basic cellular processes, such as metabolism, DNA replication and repair, transcription, translation, and protein trafficking. Most of the proteins that are unique to humans are made up of protein domains that are also found in other organisms, but these domains are arranged in novel combinations to yield distinct proteins in humans. Compared to *Drosophila* and *C. elegans*, the human genome contains expanded numbers of genes involved in functions related to the greater complexity of vertebrates, such as the immune response, the nervous system, and blood clotting, as well as increased numbers of genes involved in development, cell signaling, and the regulation of transcription.

The Genomes of Other Vertebrates

In addition to the human genome, a large and growing number of vertebrate genomes have been sequenced in the last few years, including the genomes of fish, chickens, and other mammals (Figure 5.32). These sequences provide interesting comparisons to that of the human genome and are expected to facilitate the identification of a variety of different types of functional sequences, including regulatory elements that control gene expression.

■ **Pufferfish contain a very powerful neurotoxin, called tetrodotoxin, in some of their tissues. In Japan, pufferfish are considered a delicacy and prepared by specially trained chefs in licensed restaurants.**

The genome of the pufferfish *Fugu rubripes* was chosen for sequencing because it is unusually compact for a vertebrate genome. Consisting of only 3.7×10^8 base pairs of DNA, the pufferfish genome is only about one-eighth the size of the human genome. Although the pufferfish and human genomes contain a similar number of genes, the pufferfish has far less repetitive sequence and smaller introns. In particular, repetitive sequences account for only about 15% of the pufferfish genome (corresponding to approximately 50 million base pairs of DNA) as compared to about 60% of the human genome (approximately 2 billion base pairs). Because of this reduced amount of repetitive sequence, genes are more closely packed in the pufferfish and occupy about one-third of its genome. Pufferfish and human genes contain similar numbers of introns, but introns are shorter in the pufferfish, so that protein coding sequence corresponds to approximately one-third of the average gene or about 10% of the pufferfish genome (as compared to 1.2% of the human genome). The pufferfish thus provides a compact model of a vertebrate genome in which genes and critical regulatory sequences are highly concentrated, facilitating efforts to focus continuing studies on these functional genomic elements.

The chicken is intermediate between the pufferfish and mammals, both in evolutionary divergence and in the size of its genome. Consisting of approximately 10^9 base pairs, the chicken genome is about one-third the size of the human genome. However, it is estimated to contain 20,000 to 23,000 genes, similar to the gene content of humans. The smaller size of the chicken genome is largely the result of a substantial reduction in the amount of repetitive sequences and pseudogenes compared to mammalian genomes.

The mammalian genomes that have been sequenced, in addition to the human genome, include the genomes of the mouse, rat, dog, and chimpanzee. These genomes are all similar in size to the human genome and contain similar numbers of genes. However, each offers particular advantages for further understanding gene regulation and function. As discussed in earlier chapters, the mouse is the key model system for experimental studies of mammalian genetics and development, so the availability of the

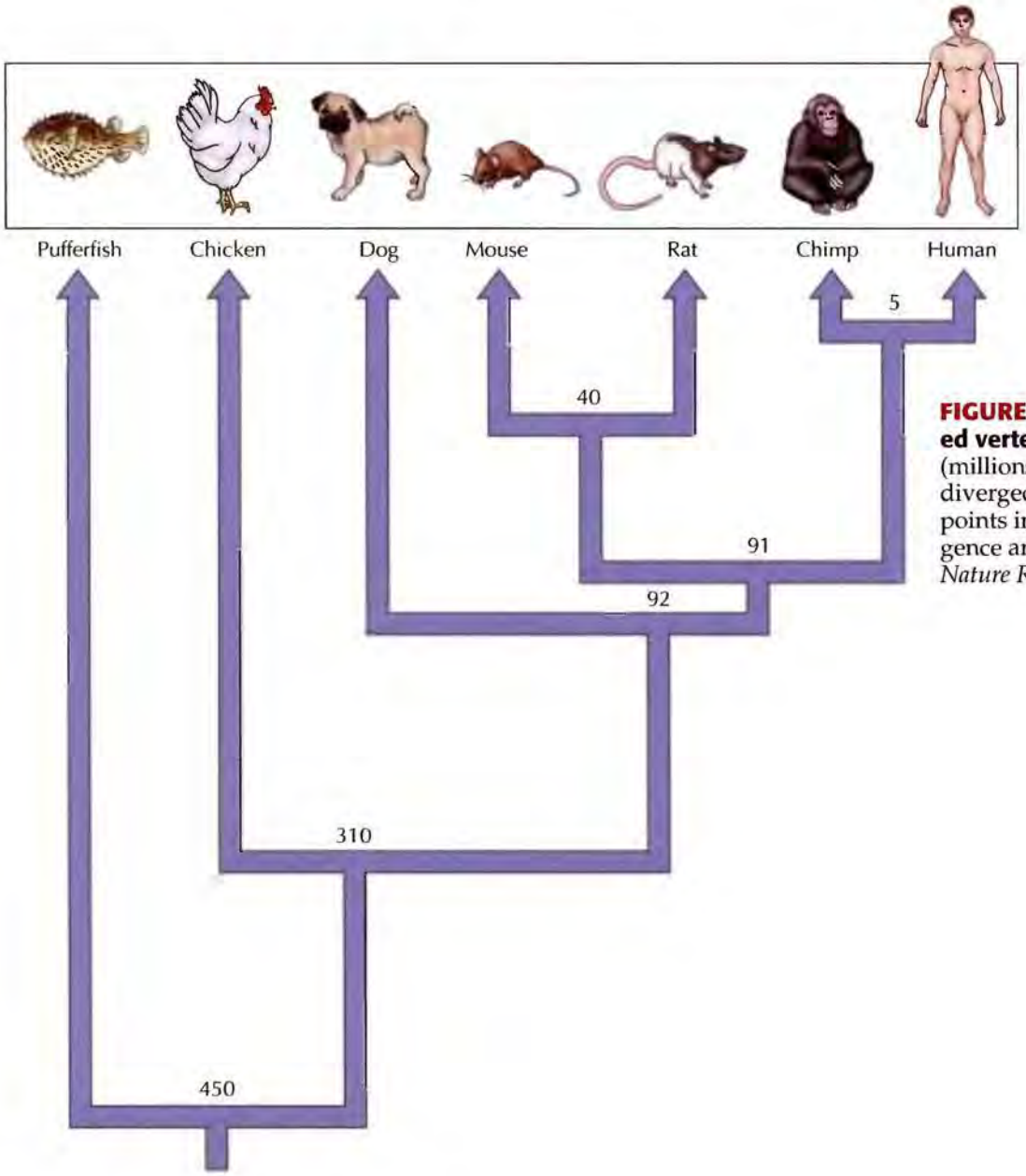


FIGURE 5.32 Evolution of sequenced vertebrates The estimated times (millions of years ago) when species diverged are indicated at branch points in the diagram. (Times of divergence are from S. B. Hedges, 2002. *Nature Rev. Genet.* 3: 838.)

mouse genome sequence provides an essential database for research in these areas. Likewise, the rat is an important model for human physiology and medicine, and these studies will be facilitated by the availability of the rat genome sequence. Mice, rats, and humans have 90% of their genes in common, providing a clear genetic foundation for the use of the mouse and rat as models for human development and disease.

The many distinct breeds of pet dogs make the sequence of the dog genome particularly important in understanding the genetic basis of morphology, behavior, and a variety of complex diseases that afflict both dogs and humans. There are approximately 300 breeds of dogs, which differ in their physical and behavioral characteristics as well as in their susceptibility to a variety of diseases, including several types of cancer, blindness, deafness, and metabolic disorders. Susceptibility to particular diseases is a highly specific property of different breeds, greatly facilitating identification of the responsible genes. Since many of these diseases are common to both dogs and humans, genetic studies in dogs can be expected to impact

human health as well as veterinary medicine. An interesting example is provided by studies of sleep disorders in which the gene responsible for a rare inherited form of narcolepsy was identified in Doberman pinschers. Subsequent studies implicated related defects in human narcolepsy and possibly other sleep disorders. Similar types of genetic analysis are underway to understand the genetic basis of other complex diseases, such as hip dysplasia and rheumatoid arthritis, that are common in some breeds of dogs, and the results of these studies will undoubtedly benefit both dogs and humans. In the future, we can also expect genetic analysis of behavior in dogs. Since many canine behaviors, such as separation anxiety, are also common in humans, psychologists may have much to learn from the species that has been our closest companion for thousands of years.

The sequence of the genome of the chimpanzee, our nearest evolutionary relative, is expected to help pinpoint the unique features of our genome that distinguish humans from other primates. Interestingly however, comparison of the chimpanzee and human genome sequences does not suggest an easy answer to the question of what makes us human. The nucleotide sequences of the chimpanzee and human genomes are nearly 99% identical. The difference between the sequences of these closely related species (approximately 1 nucleotide in 100) is about ten times greater than the difference between the genomes of individual humans (approximately 1 nucleotide in 1000). Perhaps surprisingly, the sequence differences between humans and chimpanzees are not restricted to noncoding sequences. Instead, they frequently alter the coding sequences of genes, leading to changes in the amino acid sequences of most of the proteins encoded by chimpanzees and humans. Although many of these amino acid changes may not affect protein function, it appears that there are changes in the structure as well as in the expression of thousands of genes between chimpanzees and humans, so identifying those differences that are key to the origin of humans will not be a simple task.

Bioinformatics and Systems Biology

The human genome sequence, together with the sequences of other genomes, provides a wealth of information that forms a new framework for studies of cell and molecular biology and opens new possibilities in medical practice. In addition, the genome sequencing projects have raised new questions and substantially changed the way in which many problems in biology are being approached. Traditionally, molecular biologists have studied one or a few genes or proteins at a time. This has been changed by the genome sequencing projects, which introduced new large-scale experimental approaches in which vast amounts of data were generated. Handling the enormous amounts of data generated by whole genome sequencing required sophisticated computational analysis and spawned the new field of **bioinformatics**, which lies at the interface between biology and computer science and is focused on developing the computational methods needed to analyze and extract useful biological information from the sequence of billions of bases of DNA. The development of such computational methods has also led to other types of large-scale biological experimentation, including simultaneous analysis of the expression of thousands of mRNAs or proteins and the development of high-throughput methods to determine gene function using RNA interference. These large-scale experimental approaches form the basis of the new field of **systems biology**,

which seeks a quantitative understanding of the integrated dynamic behavior of complex biological systems and processes. Systems biology thus combines large-scale biological experimentation with quantitative analysis and the development of testable models for complex biological processes. The global analysis of cell proteins (proteomics), discussed in Chapter 2, is one example of these new large-scale experimental/computational approaches. Some of the additional research areas that are amenable to large-scale experimentation, bioinformatics, and systems biology are discussed below.

Systematic Screens of Gene Function

The identification of all of the genes in an organism opens the possibility for a large-scale systematic analysis of gene function. One approach is to systematically inactivate (or knockout) each gene in the genome by homologous recombination with an inactive mutant allele (see Figure 4.39). As noted in Chapter 4, this has been done in yeast to produce a collection of yeast strains with mutations in all known genes, which can then be analyzed to determine which genes are involved in any biological property of interest. Alternatively, large-scale screens based on RNA interference (RNAi) are being used to systematically dissect gene function in a variety of organisms, including *Drosophila*, *C. elegans*, and mammalian cells in culture.

In RNAi screens, double-stranded RNAs are used to induce degradation of the homologous mRNAs in cells (see Figure 4.42). With the availability of complete genome sequences, libraries of double-stranded RNAs can be designed and used in genome-wide screens to identify all of the genes involved in any biological process that can be assayed in a high-throughput manner. For example, genome-wide RNAi analysis has been used to identify genes required for the growth and viability of *Drosophila* cells in culture (Figure 5.33). Individual double-stranded RNAs from the genome-wide library are tested in microwells in a high-throughput format to identify those that interfere with the growth of cultured cells, thereby characterizing the entire set of genes in the *Drosophila* genome that are required for cell growth or survival. Similar RNAi screens have been used to identify genes involved in a variety of biological processes, including cell signaling path-

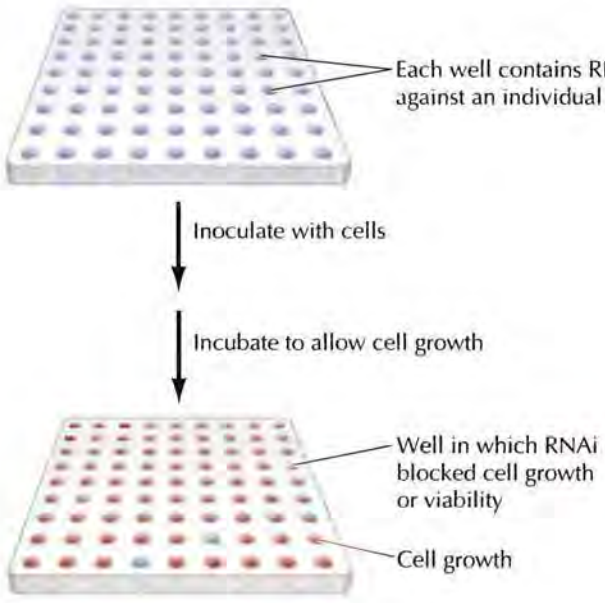


FIGURE 5.33 Genome-wide RNAi screen for cell growth and viability Each microwell contains RNAi corresponding to an individual gene in the *Drosophila* genome. *Drosophila* tissue culture cells are added to each well and incubated to allow cell growth. Those wells in which cells fail to grow identify genes required for cell growth or viability.

ways, protein degradation, and transmission at synapses in the nervous system.

Regulation of Gene Expression

Genome sequences can in principle reveal not only the protein-coding sequences of genes, but also the regulatory elements that control gene expression. As discussed in subsequent chapters, regulation of gene expression is critical to many aspects of cell function, including the development of complex multicellular organisms. Understanding the mechanisms that control gene expression is therefore a central undertaking in contemporary cell and molecular biology, and it is expected that the availability of genome sequences will contribute substantially to this task. Unfortunately, it is far more difficult to identify gene regulatory sequences than it is to identify protein-coding sequences. Most regulatory elements are short sequences of DNA, typically spanning only about 10 base pairs. Consequently, sequences resembling regulatory elements occur frequently by chance in genomic DNA, so physiologically significant elements can not be identified from DNA sequence alone. The identification of functional regulatory elements and elucidation of the signaling networks that control gene expression therefore represent major challenges in bioinformatics and systems biology.

The availability of genomic sequences has enabled scientists to undertake global studies of gene expression in which the expression levels of all genes in a cell can be assayed simultaneously. These experiments employ DNA microarrays in which each gene is represented by an oligonucleotide corresponding to a small dot on a slide (see Figure 4.27). Hybridization of fluorescent-labeled cDNA copies of cellular mRNAs to such a microarray allows simultaneous determination of the mRNA levels of all cellular genes. This approach has been particularly valuable in revealing global changes in gene regulation associated with discrete cell behaviors, such as cell differentiation or the response of cells to a particular hormone or growth factor. Since genes that are coordinately regulated within a cell may be controlled by similar mechanisms, analyzing changes in the expression of multiple genes can help to pinpoint shared regulatory elements.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

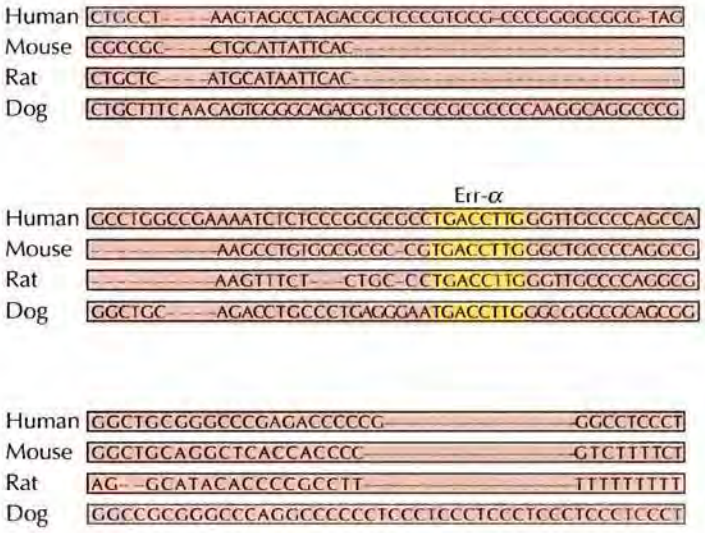


FIGURE 5.34 Conservation of functional gene regulatory elements Human, mouse, rat, and dog sequences near the transcription start site of a gene contain a functional regulatory element that binds the transcriptional regulatory protein Err-α. These sequences (highlighted in yellow) are conserved in all four genomes, whereas the surrounding sequences are not. (From X. Xie et al., 2005. *Nature* 434: 338.)

A variety of computational approaches are also being used to characterize functional regulatory elements. One approach is comparative analysis of the genome sequences of related organisms. This is based on the assumption that functionally important sequences are conserved in evolution, whereas nonfunctional segments of DNA diverge more rapidly. Computational analysis based on this approach has recently identified gene regulatory sequences that are conserved between the mouse, rat, dog, and human genomes (Figure 5.34). In addition, functional regulatory elements often occur in clusters, reflecting the fact that genes are generally regulated by the interactions of multiple transcription factors (see Chapter 7). Computer algorithms designed to detect clusters of transcription factor binding sites in genomic DNA have also proven useful in identifying sequences that regulate gene expression.

The combination of large-scale experimental methods and computational analysis has been successful in providing at least an initial indication of the transcriptional regulatory elements that govern expression of genes in yeast. However, extending these approaches to the far more complicated genomes of humans and other mammals remains a major challenge for future research.

Variation among Individuals and Genomic Medicine

Comparisons of genome sequences of related species is helpful in understanding the basis of differences between species, as well as in identifying genes and regulatory sequences that have been conserved in evolution. A different type of information can be gained by comparing the genome sequences of different individuals. Variations between individual genomes underlie differences in physical and mental characteristics, including susceptibility to many diseases. One of the major applications of the human genome sequence will be helping to uncover new genes involved in many of the diseases that afflict mankind, including cancer, heart disease, and degenerative diseases of the nervous system such as Parkinson's and Alzheimer's disease. In addition, understanding our unique genetic makeup as individuals is expected to lead to the development of new tailor-made strategies for disease prevention and treatment.

The genomes of two unrelated people differ in about one of every thousand bases. Most of this variation is in the form of single base changes, known as single nucleotide polymorphisms (SNPs), which are found at about 10 million positions in the genome. Over a million commonly occurring SNPs have been mapped in the human genome. These SNPs are distributed relatively uniformly in genomic DNA (Figure 5.35), and it is noteworthy that more than 90% of protein-coding genes contain at least one SNP. It is likely that these SNPs are responsible for most genetic differences in individual characteristics, so a substantial effort is being directed towards using SNPs to map the genes responsible for inherited differences in disease susceptibility. Analysis of these variations among individuals will not only allow specific genes to be associated with susceptibility to different diseases but will also enable physicians to tailor strategies for disease prevention and treatment to match the genetic makeup of individual patients. Comparisons between the genomes of different individuals may also help to elucidate the contribution of our genes to other unique characteristics, such as athletic ability or intelligence, and to better understand the interactions between genes and environment that lead to complex human behaviors.

FIGURE 5.35 Single nucleotide polymorphisms (SNPs) in human chromosome 1 The distribution of SNPs (frequency per kilobase) is indicated. (From D. A. Hinds et al., 2005. *Science* 307: 1072.)

■ **Current efforts aim to develop technologies that would be capable of sequencing the genome of individuals at very low cost. Such affordable 'personal genome projects' might provide better health care options tailored to the needs of individual patients.**

KEY TERMS

SUMMARY

THE COMPLEXITY OF EUKARYOTIC GENOMES

gene, spacer sequence, exon, intron, RNA splicing, kilobase (kb), alternative splicing

Introns and Exons: Most eukaryotic genes have a split structure in which segments of coding sequence (exons) are interrupted by noncoding sequences (introns). In complex eukaryotes, introns account for more than ten times as much DNA as exons.

simple-sequence repeat, satellite DNA, SINE, LINE, retrotransposon, DNA transposon, retrovirus-like element

Repetitive DNA Sequences: Over 50% of mammalian DNA consists of highly repetitive DNA sequences, some of which are present in 10^5 to 10^6 copies per genome. These sequences include simple-sequence repeats as well as repetitive elements that have moved throughout the genome by either RNA or DNA intermediates.

gene family, pseudogene, processed pseudogene

Gene Duplications and Pseudogenes: Many eukaryotic genes are present in multiple copies, called gene families, which have arisen by duplication of ancestral genes. Some members of gene families function in different tissues or at different stages of development. Other members of gene families (pseudogenes) have been inactivated by mutations and no longer represent functional genes. Gene duplications can occur either by duplication of a segment of DNA or by reverse transcription of an mRNA, giving rise to a processed pseudogene. Approximately 5% of the human genome consists of duplicated DNA segments. In addition, there are more than 10,000 processed pseudogenes in the human genome.

The Composition of Higher Eukaryotic Genomes: Only a small fraction of the genome in complex eukaryotes corresponds to protein-coding sequences. The human genome is estimated to contain 20,000–25,000 genes, with protein-coding sequence corresponding to only about 1.2% of the DNA. Approximately 20% of the human genome consists of introns, and more than 60% is composed of repetitive and duplicated DNA sequences.

CHROMOSOMES AND CHROMATIN

chromatin, histone, nucleosome, nucleosome core particle, chromatosome, euchromatin, heterochromatin

Chromatin: The DNA of eukaryotic cells is wrapped around histones to form nucleosomes. Chromatin can be further compacted by the folding of nucleosomes into higher-order structures, including the highly condensed metaphase chromosomes of cells undergoing mitosis.

centromere, kinetochore

Centromeres: Centromeres are specialized regions of eukaryotic chromosomes that serve as the sites of association between sister chromatids and the sites of spindle fiber attachment during mitosis.

telomere, telomerase

Telomeres: Telomeres are specialized sequences required to maintain the ends of eukaryotic chromosomes.

THE SEQUENCES OF COMPLETE GENOMES

megabase (Mb), open-reading frame

Prokaryotic Genomes: The genomes of more than 100 different bacteria, including *E. coli*, have been completely sequenced. The *E. coli* genome contains 4288 genes, with protein-coding sequences accounting for nearly 90% of the DNA.

SUMMARY

The Yeast Genome: The first eukaryotic genome to be sequenced was that of the yeast *S. cerevisiae*. The *S. cerevisiae* genome contains about 6000 genes, and protein-coding sequences account for approximately 70% of the genome. The genome of the fission yeast *S. pombe* contains fewer genes (about 5000) and more introns than *S. cerevisiae*, with protein-coding sequence corresponding to about 60% of the *S. pombe* genome.

***The Genomes of* Caenorhabditis elegans *and* Drosophila melanogaster:** The genome of *C. elegans* was the first sequenced genome of a multicellular organism. The *C. elegans* genome contains about 19,000 protein-coding sequences, which account for only about 25% of the genome. The genome of *Drosophila* contains approximately 14,000 genes, with protein-coding sequences accounting for about 13% of the genome. Although *Drosophila* contains fewer genes than *C. elegans*, many genes in both species are duplicated, and it appears that both species contain 10,000–15,000 unique genes. Some of these genes are shared between *Drosophila*, *C. elegans*, and yeast—these genes may encode proteins with common functions in all eukaryotic cells. However, the majority of *Drosophila* and *C. elegans* genes are not found in yeast and are likely to function in the regulation and development of multicellular animals.

Plant Genomes: The genome of the small flowering plant *Arabidopsis thaliana* contains approximately 26,000 genes—surprisingly more genes than were found in either *Drosophila* or *C. elegans*. However, many of these genes are the result of duplications of large segments of the *Arabidopsis* genome, so the number of unique genes in *Arabidopsis* is about 15,000. Many of these genes are unique to plants, including genes involved in plant physiology, development, and defense. The sequence of the rice genome is of particular agricultural interest because rice is the staple food for more than half the world's population. The draft sequence of the rice genome is estimated to contain approximately 37,000 genes, many of which are duplicated and may have arisen by duplication of large genome segments.

The Human Genome: The human genome appears to contain 20,000–25,000 genes—not much more than the number of genes found in simpler animals like *Drosophila* and *C. elegans*. Over 40% of the predicted human proteins are related to proteins found in other sequenced organisms, including *Drosophila* and *C. elegans*. In addition, the human genome contains expanded numbers of genes involved in the nervous system, the immune system, blood clotting, development, cell signaling, and the regulation of gene expression.

The Genomes of Other Vertebrates: The genomes of fish, chickens, mice, rats, dogs, and chimpanzees provide important comparisons to the human genome. All of these vertebrates contain similar numbers of genes but in some cases differ substantially in their content of repetitive sequences.

KEY TERMS

yeast artificial chromosome (YAC), polytene chromosome, bacterial artificial chromosome (BAC)

fluorescence *in situ* hybridization (FISH)

KEY TERMS

bioinformatics, systems biology

SUMMARY

BIOINFORMATICS AND SYSTEMS BIOLOGY

Systematic Screens of Gene Function: The genome sequencing projects have introduced large-scale experimental and computational approaches to research in cell and molecular biology. Genome-wide screens using RNA interference can systematically identify all of the genes in an organism that are involved in any biological process that can be assayed in a high-throughput format.

Regulation of Gene Expression: The identification of gene regulatory sequences and elucidation of the signaling networks that control gene expression are major challenges in bioinformatics and systems biology. These problems are being approached by genome-wide studies of gene expression combined with the development of computational approaches to identify functional regulatory elements.

Variation among Individuals and Genomic Medicine: Variations in our genomes are responsible for the characteristics of individual people, including susceptibility to many diseases. Analysis of these variations will allow the identification of genes responsible for disease susceptibility and enable the development of new strategies for disease prevention and treatment that match the genetic makeup of different individuals.

Questions

1. Many eukaryotic organisms have genome sizes that are much larger than their complexity would seem to require. Explain this paradox.

2. How were introns discovered during studies of adenovirus mRNAs?

3. How do intron sequences in the human genome increase the diversity of proteins expressed from the limited number of 20,000–25,000 genes?

4. How can simple-sequence repetitive DNA be separated from the bulk of the nuclear DNA?

5. Yeast (*S. cerevisiae*) centromeres form a kinetochore that attaches to a single microtubule, whereas multiple microtubules are attached to the kinetochores of most animal cells. How does the structure of *S. cerevisiae* centromeres reflect this difference?

6. When circular plasmids are provided with a centromere sequence and inserted into yeast cells, they reproduce and segregate normally each cell division. However, if a linear chromosome is generated by cutting the plasmid at a single site with a restriction endonuclease, the plasmid genes are quickly lost from the yeast. Explain. What additional experiment could you perform to test your hypothesized explanation?

7. What is the average distance between genes in the human genome?

8. Approximately how many molecules of histone H1 are bound to yeast genomic DNA?

9. What is the average length of an intron in a human gene?

10. You have made a library in a plasmid vector containing complete human cDNAs. What is the expected average size of an insert?

11. How was the approach used by Celera Genomics to sequence the human genome different from that used by the International Human Genome Sequencing Consortium?

12. Why is it more difficult to identify regulatory sequences than it is to identify protein coding sequences? What are the different approaches used to identify functional regulatory sequences?

13. What is a SNP? What results are expected from the study of SNPs?

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CHAPTER 6

Replication, Maintenance, and Rearrangements of Genomic DNA

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THE FUNDAMENTAL BIOLOGICAL PROCESS OF REPRODUCTION requires the faithful transmission of genetic information from parent to offspring. Thus, the accurate replication of genomic DNA is essential to the lives of all cells and organisms. Each time a cell divides, its entire genome must be duplicated, and complex enzymatic machinery is required to copy the large DNA molecules that make up both prokaryotic and eukaryotic chromosomes. In addition, cells have evolved mechanisms to correct mistakes that sometimes occur during DNA replication and to repair DNA damage that can result from the action of environmental agents, such as radiation. Abnormalities in these processes result in a failure of accurate replication and maintenance of genomic DNA—a failure that can have disastrous consequences, such as the development of cancer.

Despite the importance of accurate DNA replication and maintenance, cell genomes are far from static. In order for species to evolve, mutations and gene rearrangements are needed to maintain genetic variation between individuals. Recombination between homologous chromosomes during meiosis plays an important role in this process by allowing parental genes to be rearranged into new combinations in the next generation. Rearrangements of DNA sequences within the genome are also thought to contribute to evolution by creating novel combinations of genetic information. In addition, some DNA rearrangements are programmed to regulate gene expression during the differentiation and development of individual cells and organisms. In humans, a prominent example is the rearrangement of antibody genes during development of the immune system. A careful balance between maintenance and variation of genetic information is thus critical to the development of individual organisms as well as to evolution of the species.

DNA Replication

As discussed in Chapter 4, DNA replication is a semiconservative process in which each parental strand serves as a template for the synthesis of a new complementary daughter strand. The central enzyme involved is DNA polymerase, which catalyzes the joining of deoxyribonucleoside 5′-triphosphates (dNTPs) to form the growing DNA chain. However, DNA replication is much more complex than a single enzymatic reaction. Other proteins are involved, and proofreading mechanisms are required to ensure that the accuracy of replication is compatible with the low frequency of errors that is needed for cell reproduction. Additional proteins and specific DNA sequences are also needed both to initiate replication and to copy the ends of eukaryotic chromosomes.

DNA Polymerases

DNA polymerase was first identified in lysates of *E. coli* by Arthur Kornberg in 1956. The ability of this enzyme to accurately copy a DNA template provided a biochemical basis for the mode of DNA replication that was initially proposed by Watson and Crick, so its isolation represented a landmark discovery in molecular biology. Ironically, this first DNA polymerase to be identified (now called DNA polymerase I) is not the major enzyme responsible for *E. coli* DNA replication. Instead, polymerase I is principally involved in repair of damaged DNA, and it is now clear that both prokaryotic and eukaryotic cells contain multiple different DNA polymerases that play distinct roles in DNA replication and repair. In prokaryotic cells, DNA polymerase III is the major polymerase responsible for DNA replication. Eukaryotic cells contain 3 DNA polymerases (α, δ, and ε) that function in replication of nuclear DNA. A distinct DNA polymerase (γ) is localized to mitochondria and is responsible for replication of mitochondrial DNA.

All known DNA polymerases share two fundamental properties that have critical implications for DNA replication (Figure 6.1). First, all polymerases synthesize DNA only in the 5′ to 3′ direction, adding a dNTP to the 3′ hydroxyl group of a growing chain. Second, DNA polymerases can add a new deoxyribonucleotide only to a preformed primer strand that is hydrogen-bonded to the template; they are not able to initiate DNA synthesis *de novo* by catalyzing the polymerization of free dNTPs. In this respect, DNA polymerases differ from RNA polymerases, which can initiate the synthesis of a new strand of RNA in the absence of a primer. As discussed later in this chapter, these properties of DNA polymerases appear critical for maintaining the high fidelity of DNA replication that is required for cell reproduction.

The Replication Fork

DNA molecules in the process of replication were first analyzed by John Cairns in experiments in which *E. coli* were grown in the presence of radioactive thymidine, which allowed subsequent visualization of newly replicated DNA by autoradiography (Figure 6.2). In some cases, complete circular molecules in the process of replicating could be observed. These DNA molecules contained two **replication forks**, representing the regions of active DNA synthesis. At each fork the parental strands of DNA separated and two new daughter strands were synthesized.

The synthesis of new DNA strands complementary to both strands of the parental molecule posed an important problem to understanding the biochemistry of DNA replication. Since the two strands of double-helical DNA run in opposite (antiparallel) directions, continuous synthesis of two new

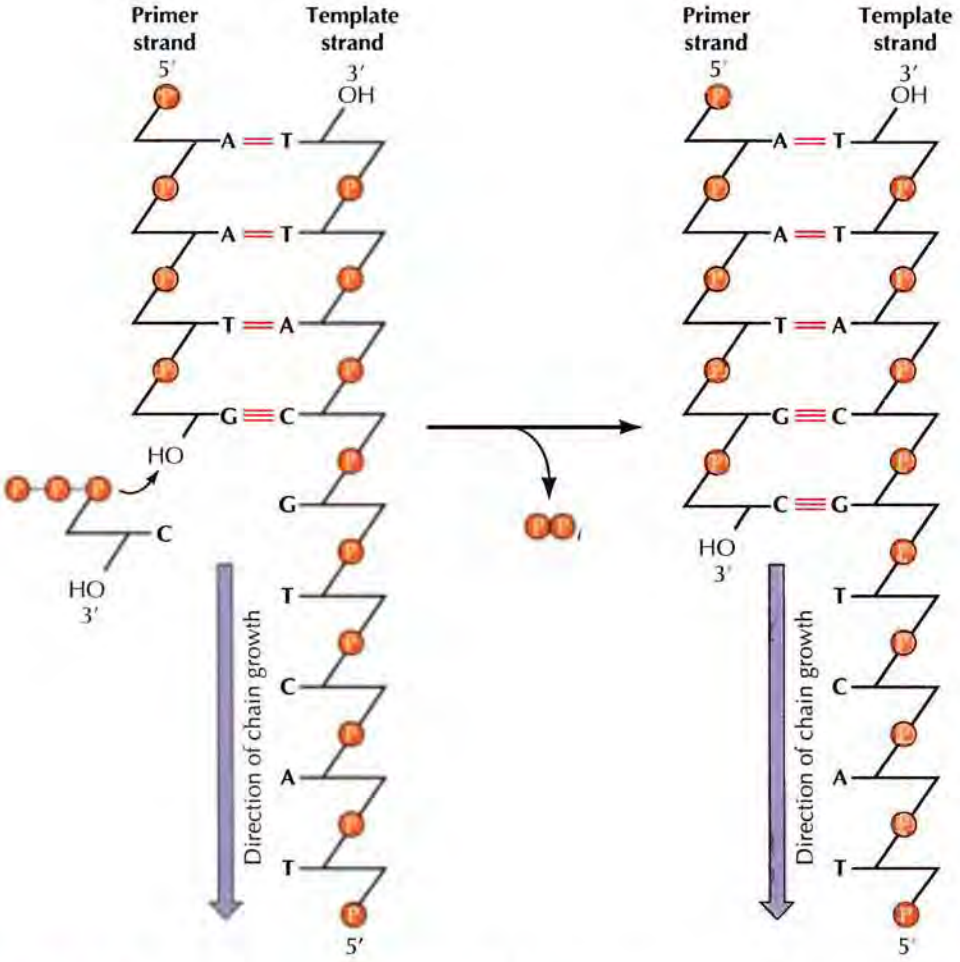


FIGURE 6.1 The reaction catalyzed by DNA polymerase All known DNA polymerases add a deoxyribonucleoside 5′-triphosphate to the 3′ hydroxyl group of a growing DNA chain (the primer strand).

strands at the replication fork would require that one strand be synthesized in the 5′ to 3′ direction while the other is synthesized in the opposite (3′ to 5′) direction. But DNA polymerase catalyzes the polymerization of dNTPs only in the 5′ to 3′ direction. How, then, can the other progeny strand of DNA be synthesized?

This enigma was resolved by experiments showing that only one strand of DNA is synthesized in a continuous manner in the direction of overall DNA replication; the other is formed from short (1–3 kb), discontinuous pieces of DNA that are synthesized backward with respect to the direction of movement of the replication fork (Figure 6.3). These small pieces of newly synthesized DNA (called **Okazaki fragments** after their discoverer, the Japanese biochemist Reiji Okazaki) are joined by the action of **DNA ligase**, forming an intact new DNA strand. The continuously synthesized strand is called the **leading strand**, since its elongation in the direction of replication fork movement exposes the template used for the synthesis of Okazaki fragments (the **lagging strand**).

Although the discovery of discontinuous synthesis of the lagging strand provided a mechanism for the elongation of both strands of DNA at the replication fork, it raised another question: Since DNA polymerase requires a primer and cannot initiate synthesis *de novo*, how is the synthesis of

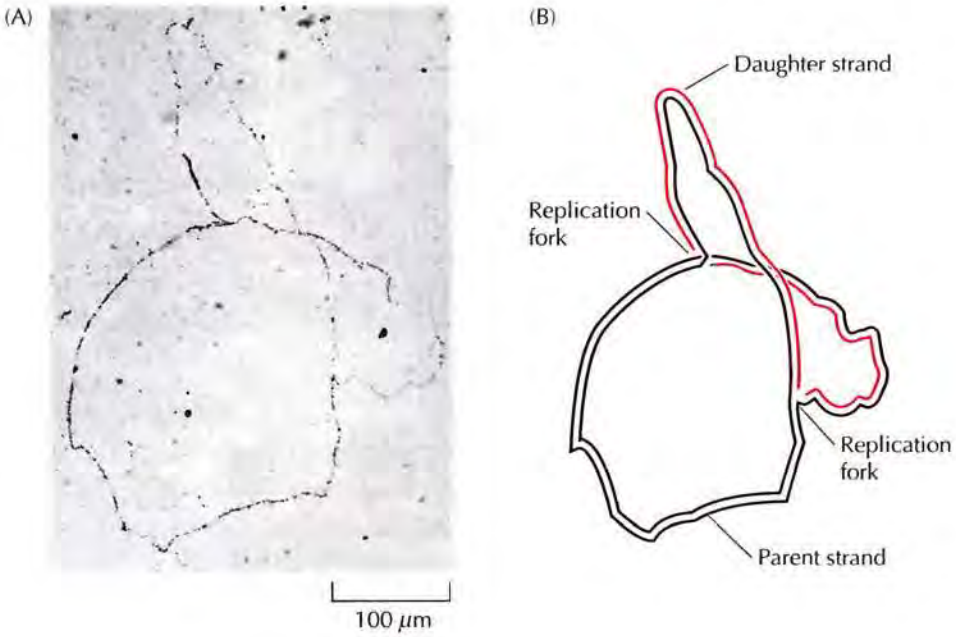


FIGURE 6.2 Replication of *E. coli* DNA (A) An autoradiograph showing bacteria that were grown in [^{3}H]thymidine for two generations to label the DNA, which was then extracted and visualized by exposure to photographic film. (B) This schematic illustrates the two replication forks shown in (A). (From J. Cairns, 1963. *Cold Spring Harbor Symp. Quant. Biol.* 28: 43.)

Okazaki fragments initiated? The answer is that short fragments of RNA serve as primers for DNA replication (Figure 6.4). In contrast to DNA synthesis, the synthesis of RNA can initiate *de novo*, and an enzyme called **primase** synthesizes short fragments of RNA (e.g., three to ten nucleotides long) complementary to the lagging strand template at the replication fork. Okazaki fragments are then synthesized via extension of these RNA primers by DNA polymerase. An important consequence of such RNA priming is that newly synthesized Okazaki fragments contain an RNA-DNA joint, the discovery of which provided critical evidence for the role of RNA primers in DNA replication.

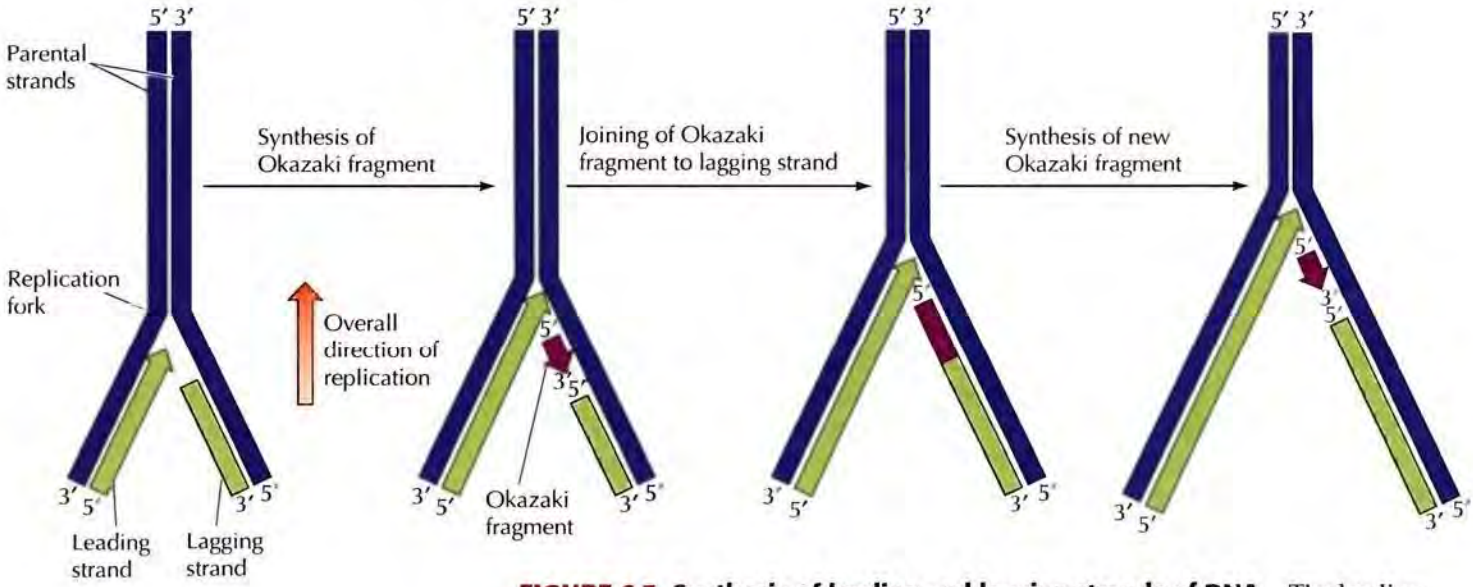


FIGURE 6.3 Synthesis of leading and lagging strands of DNA The leading strand is synthesized continuously in the direction of replication fork movement. The lagging strand is synthesized in small pieces (Okazaki fragments) backward from the overall direction of replication. The Okazaki fragments are then joined by the action of DNA ligase.

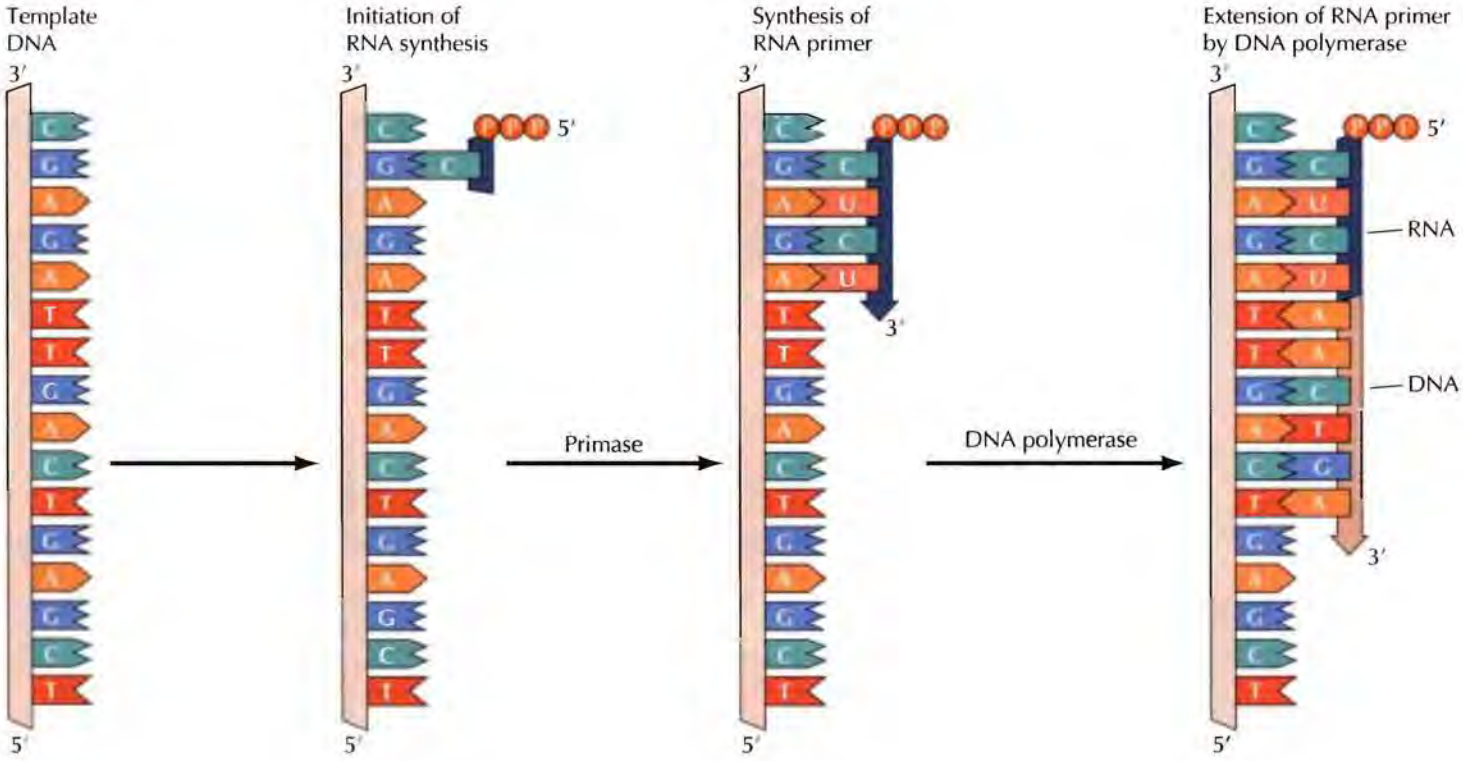


FIGURE 6.4 Initiation of Okazaki fragments with RNA primers
Short fragments of RNA serve as primers that can be extended by DNA polymerase.

To form a continuous lagging strand of DNA, the RNA primers must eventually be removed from the Okazaki fragments and replaced with DNA (Figure 6.5). In prokaryotes, RNA primers are removed by the action of polymerase I. In addition to its DNA polymerase activity, polymerase I acts as an **exonuclease** that can hydrolyze DNA (or RNA) in either the 3′ to 5′ or 5′ to 3′ direction. The action of polymerase I as a 5′ to 3′ exonuclease removes ribonucleotides from the 5′ ends of Okazaki fragments, allowing them to be replaced with deoxyribonucleotides to yield fragments consisting entirely of DNA. In eukaryotic cells, RNA primers are removed by the combined action of **RNase H**, an enzyme that degrades the RNA strand of RNA-DNA hybrids, and 5′ to 3′ exonucleases. The resulting gaps are then filled by polymerase δ and the DNA fragments are joined by DNA ligase, yielding an intact lagging strand.

As noted earlier, different DNA polymerases play distinct roles at the replication fork in both prokaryotic and eukaryotic cells (Figure 6.6). In *E. coli*, polymerase III is the major replicative polymerase, functioning in the synthesis both of the leading strand of DNA and of Okazaki fragments by the extension of RNA primers. In eukaryotic cells, three distinct DNA polymerases (α, δ, and ε) are involved in the replication of nuclear DNA. The roles of these DNA polymerases have been studied in two types of experiments. First, replication of the DNA of some animal viruses, such as SV40, can be studied in cell-free extracts, allowing direct biochemical analysis of the activities of different DNA polymerases as well as other proteins involved in DNA replication. Second, polymerases α, δ, and ε are found in yeasts as well as in mammalian cells, enabling the use of the powerful approaches of yeast genetics (see Chapter 4) to test their biological roles. Biochemical analysis has established that polymerase α is found in a complex with primase, and functions in conjunction with primase to synthesize short RNA-DNA fragments during lagging strand synthesis. Polymerase δ

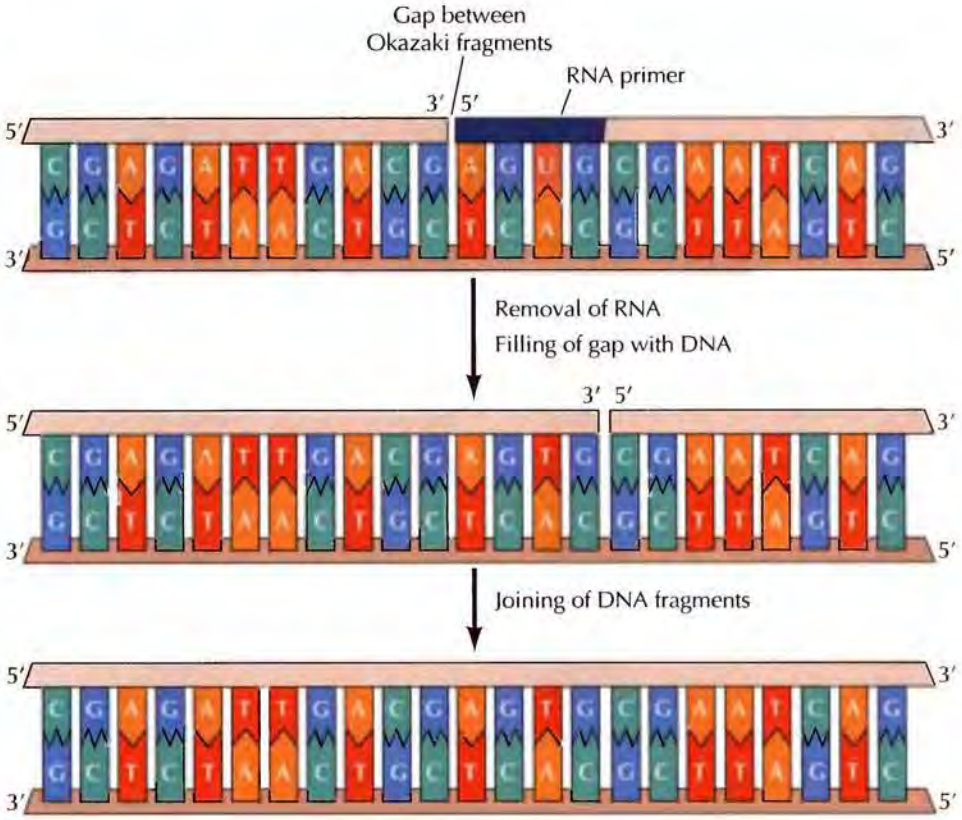


FIGURE 6.5 Removal of RNA primers and joining of Okazaki fragments RNA primers are removed and DNA polymerase fills the gaps between Okazaki fragments with DNA. The resultant DNA fragments can then be joined by DNA ligase.

can then synthesize both the leading and lagging strands, acting to extend the RNA-DNA primers initially synthesized by the polymerase α-primase complex. Polymerase ε is an essential gene in yeast, but it is not required for SV40 DNA replication *in vitro*, so its role remains to be understood. However, its activities seem to be similar to those of polymerase δ, so it probably functions like polymerase δ in the replication process.

Not only polymerases and primase but also a number of other proteins act at the replication fork. These additional proteins have been identified both by the analysis of *E. coli* and yeast mutants defective in DNA replication and by the purification of the mammalian proteins required for *in vitro* replication of SV40 DNA. One class of proteins required for replication

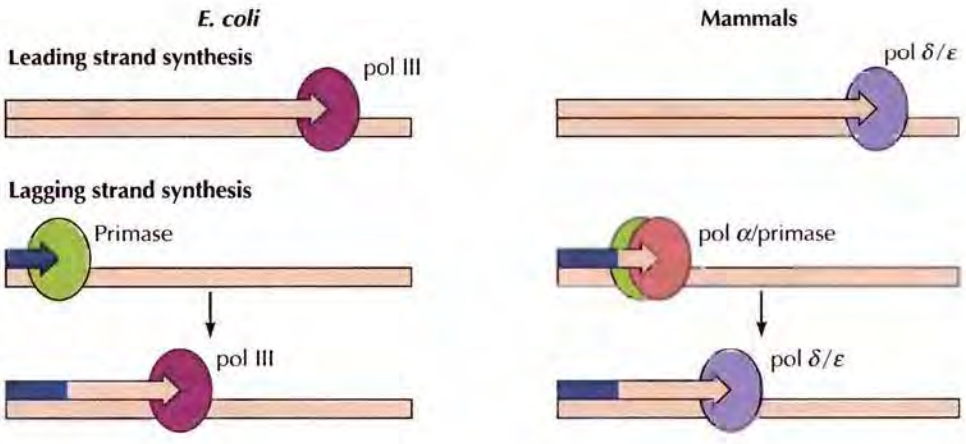
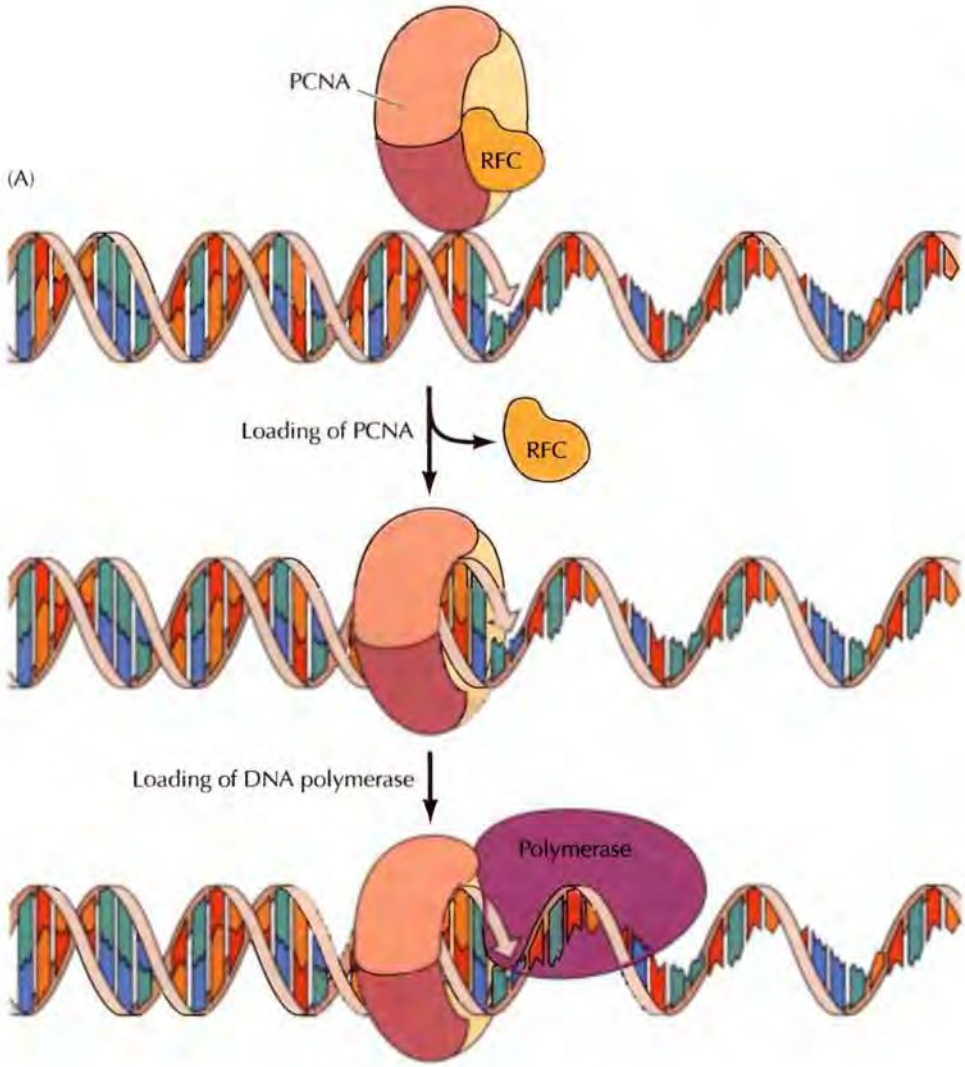


FIGURE 6.6 Roles of DNA polymerases in *E. coli* and mammalian cells The leading strand is synthesized by polymerase III (pol III) in *E. coli* and by polymerases δ and ε (pol δ/ε) in mammalian cells. In *E. coli*, lagging strand synthesis is initiated by primase, and RNA primers are extended by polymerase III. In mammalian cells, lagging strand synthesis is initiated by a complex of primase with polymerase α (pol α). The short RNA-DNA fragments synthesized by this complex are then extended by polymerases δ and ε.



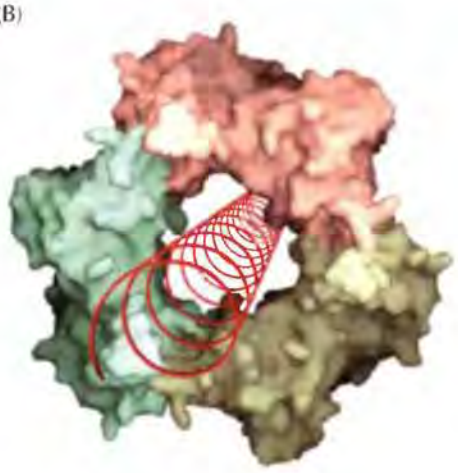


FIGURE 6.7 Polymerase accessory proteins (A) A complex of the clamp-loading and sliding-clamp proteins (RFC and PCNA in mammalian cells, respectively) binds DNA at the junction between primer and template. RFC is then released, loading PCNA onto the DNA. DNA polymerase then binds to PCNA. (B) Model of PCNA bound to DNA. (B, from T. S. Krishna, X. P. Kong, S. Gary, P. M. Burgers and J. Kuriyan, 1994. *Cell* 79: 1233.)

binds to DNA polymerases, increasing the activity of the polymerases and causing them to remain bound to the template DNA so that they continue synthesis of a new DNA strand. Both *E. coli* polymerase III and eukaryotic polymerases δ and ε are associated with two types of accessory proteins (sliding-clamp proteins and clamp-loading proteins) that load the polymerase onto the primer and maintain its stable association with the template (Figure 6.7). The clamp-loading proteins (called replication factor C [RFC] in eukaryotes) and the sliding-clamp proteins (proliferating cell nuclear antigen [PCNA] in eukaryotes) form a complex that specifically recognizes and binds DNA at the junction between the primer and template. The clamp loader then releases the sliding clamp protein, which forms a ring around the template DNA. The sliding clamp protein then loads the DNA polymerase onto DNA at the primer-template junction. The ring formed by the sliding clamp maintains the association of the polymerase with its template as replication proceeds, allowing the uninterrupted synthesis of many thousands of nucleotides of DNA.

Other proteins unwind the template DNA and stabilize single-stranded regions (Figure 6.8). **Helicases** are enzymes that catalyze the unwinding of parental DNA, coupled to the hydrolysis of ATP, ahead of the replication fork. **Single-stranded DNA-binding proteins** (e.g., eukaryotic replication protein A [RPA]) then stabilize the unwound template DNA, keeping it in an extended single-stranded state so that it can be copied by the polymerase.

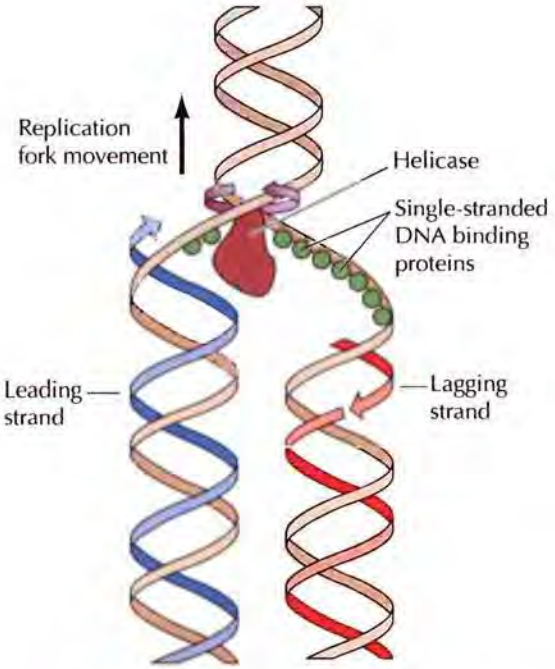


FIGURE 6.8 Action of helicases and single-stranded DNA-binding proteins Helicases unwind the two strands of parental DNA ahead of the replication fork. The unwound DNA strands are then stabilized by single-stranded DNA-binding proteins so that they can serve as templates for new DNA synthesis.

As the strands of parental DNA unwind, the DNA ahead of the replication fork is forced to rotate. Unchecked, this rotation would cause circular DNA molecules (such as SV40 DNA or the *E. coli* chromosome) to become twisted around themselves, eventually blocking replication (Figure 6.9). This problem is solved by **topoisomerases,** enzymes that catalyze the reversible breakage and rejoining of DNA strands. There are two types of these enzymes: Type I topoisomerases break just one strand of DNA; type II topoisomerases introduce simultaneous breaks in both strands. The breaks introduced by type I and type II topoisomerases serve as "swivels" that allow the two strands of template DNA to rotate freely around each other so that replication can proceed without twisting the DNA ahead of the fork (see Figure 6.9). Although eukaryotic chromosomes are composed of linear rather than circular DNA molecules, their replication also requires topoisomerases; otherwise, the complete chromosomes would have to rotate continually during DNA synthesis.

Type II topoisomerase is needed not only to unwind DNA but also to unravel newly replicated circular DNA molecules that become intertwined with each other. In eukaryotic cells, topoisomerase II appears to be involved in mitotic chromosome condensation. In addition, studies of yeast mutants, as well as experiments in *Drosophila* and mammalian cells, indicate that topoisomerase II is required for the separation of daughter chromatids at mitosis, suggesting that it functions to untangle newly replicated loops of DNA in the chromosomes of eukaryotes.

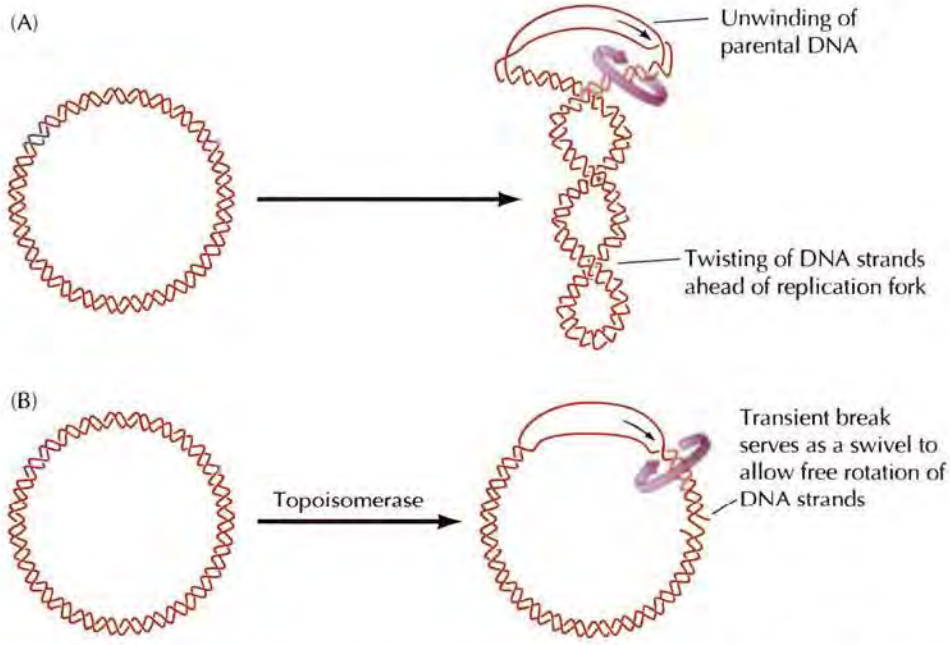


FIGURE 6.9 Action of topoisomerases during DNA replication (A) As the two strands of template DNA unwind, the DNA ahead of the replication fork is forced to rotate in the opposite direction, causing circular molecules to become twisted around themselves. (B) This problem is solved by topoisomerases, which catalyze the reversible breakage and rejoining of DNA strands. The transient breaks introduced by these enzymes serve as swivels that allow the two strands of DNA to rotate freely around each other.

■ **An inhibitor of topoisomerase II, a drug called etoposide, is used as a chemotherapeutic agent for the treatment of several types of cancer.**

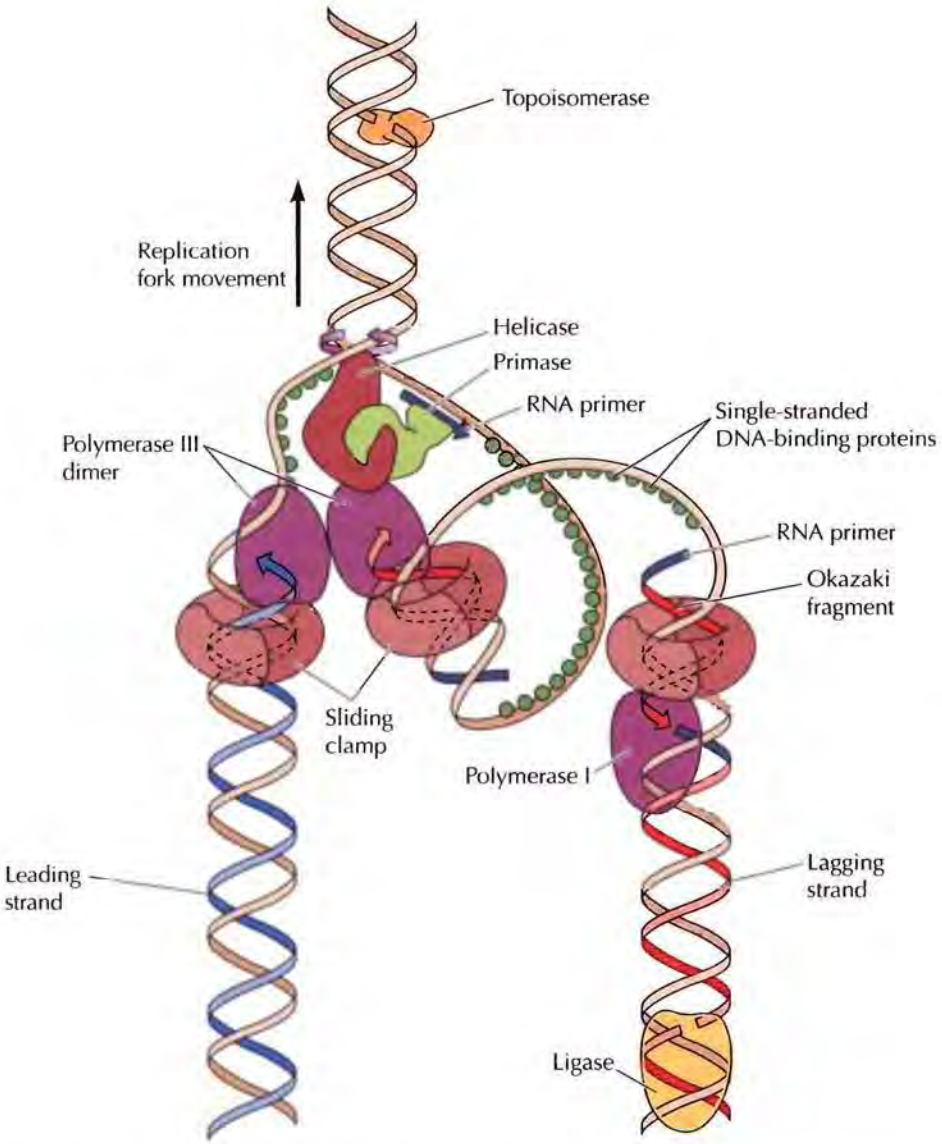


FIGURE 6.10 Model of the *E. coli* replication fork Helicase, primase, and two molecules of DNA polymerases III carry out coordinated synthesis of both the leading and lagging strands of DNA. The lagging strand template is folded so that the polymerase responsible for lagging strand synthesis moves in the same direction as overall movement of the fork. Topoisomerase acts as a swivel ahead of the fork. Behind the fork, RNA primers are removed by DNA polymerase I and Okazaki fragments are joined by DNA ligase.

6.1 WEBSITE ANIMATION

DNA Replication Fork

Helicase, primase, and two molecules of DNA polymerase III carry out the coordinated synthesis of both the leading and lagging strands of DNA at a replication fork.

The enzymes involved in DNA replication act in a coordinated manner to synthesize both leading and lagging strands of DNA simultaneously at the replication fork (Figure 6.10). This task is accomplished by the formation of dimers of the replicative DNA polymerases, each with its appropriate accessory proteins. One molecule of polymerase then acts in synthesis of the leading strand while the other acts in synthesis of the lagging strand. The lagging strand template is thought to form a loop at the replication fork so that the polymerase subunit engaged in lagging strand synthesis moves in the same overall direction as the other subunit, which is synthesizing the leading strand.

In eukaryotic cells, nucleosomes are also disrupted during DNA replication. The histones bound to the parental chromatin are divided between the two daughter strands of DNA, and new histones are then added to reassemble nucleosomes by additional proteins (chromatin assembly factors) that travel along with the replication fork.

The Fidelity of Replication

The accuracy of DNA replication is critical to cell reproduction, and estimates of mutation rates for a variety of genes indicate that the frequency of

errors during replication corresponds to only one incorrect base per 10^8 to 10^9 nucleotides incorporated. This error frequency is much lower than would be predicted simply on the basis of complementary base pairing. In particular, the free energy differences resulting from the changes in hydrogen bonding between correctly matched and mismatched bases are only large enough to favor the formation of correctly matched base pairs by about a hundredfold. Consequently, base selection determined simply by hydrogen bonding between complementary bases would result in an error frequency corresponding to the incorporation of one incorrect base in every 100–1000 nucleotides of newly synthesized DNA. The much higher degree of fidelity actually achieved results largely from the activities of DNA polymerase.

One mechanism by which DNA polymerase increases the fidelity of replication is by helping to select the correct base for insertion into newly synthesized DNA. The polymerase does not simply catalyze incorporation of whatever nucleotide is hydrogen-bonded to the template strand. Instead, it actively discriminates against incorporation of a mismatched base by adapting to the conformation of a correct base pair. In particular, recent structural studies of several DNA polymerases indicate that the binding of correctly matched dNTPs induces conformational changes in DNA polymerase that lead to the incorporation of the nucleotide into DNA. This ability of DNA polymerase to select matched nucleotides for incorporation appears to increase the accuracy of replication about a thousandfold, reducing the expected error frequency from greater than 10^{-3} to between 10^{-5} and 10^{-6}.

The other major mechanism responsible for the accuracy of DNA replication is the **proofreading** activity of DNA polymerase. The replicative DNA polymerases (eukaryotic polymerases δ and ε and *E. coli* polymerase III) have an exonuclease activity that can hydrolyze DNA in the 3′ to 5′ direction. This 3′ to 5′ exonuclease operates in the reverse direction of DNA synthesis, and participates in proofreading newly synthesized DNA (Figure 6.11). Proofreading is effective because DNA polymerase requires a primer and is not able to initiate synthesis *de novo*. Primers that are hydrogen-bonded to the template are preferentially used, so when an incorrect base is incorporated, it is likely to be removed by the 3′ to 5′ exonuclease activity rather than being used to continue synthesis. The 3′ to 5′ exonucleases of the replicative DNA polymerases selectively excise mismatched bases that have been incorporated at the end of a growing DNA chain, thereby increasing the accuracy of replication by a hundred- to a thousandfold.

The importance of proofreading may explain the fact that DNA polymerases require primers and catalyze the growth of DNA strands only in the 5′ to 3′ direction. When DNA is synthesized in the 5′ to 3′ direction, the energy required for polymerization is derived from hydrolysis of the 5′ triphosphate group of a free dNTP as it is added to the 3′ hydroxyl group of the growing chain (see Figure 6.1). If DNA were to be extended in the 3′ to 5′ direction, the energy of polymerization would instead have to be derived from hydrolysis of the 5′ triphosphate group of the terminal nucleotide already incorporated into DNA. This arrangement would eliminate the possibility of proofreading, because removal of a mismatched terminal nucleotide would also remove the 5′ triphosphate group needed as an energy source for further chain elongation. Thus, although the ability of DNA polymerase to extend a primer only in the 5′ to 3′ direction appears to make replication a complicated process, it is necessary for ensuring accurate duplication of the genetic material.

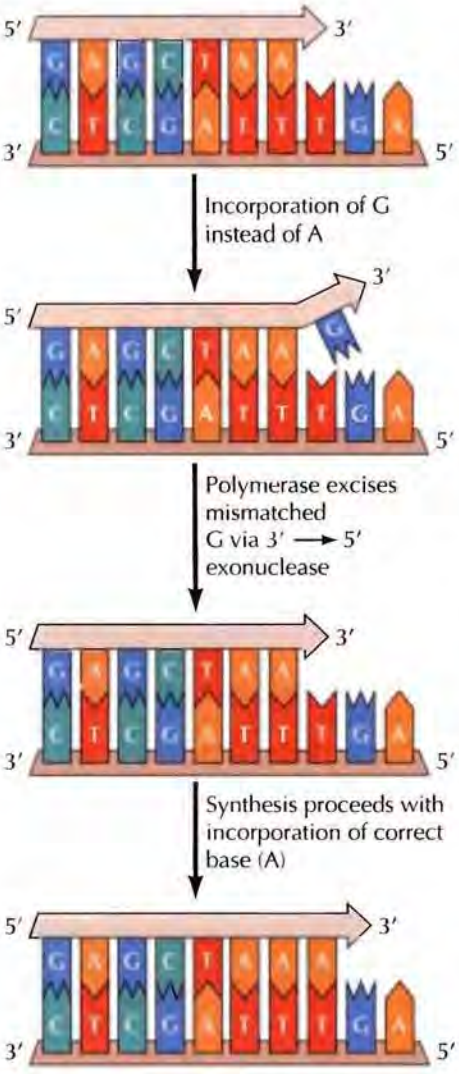


FIGURE 6.11 Proofreading by DNA polymerase G is incorporated in place of A as a result of mispairing with T on the template strand. Because it is mispaired, the 3′ terminal G is not hydrogen-bonded to the template strand. This mismatch at the 3′ terminus of the growing chain is recognized and excised by the 3′ to 5′ exonuclease activity of DNA polymerase, which requires a primer hydrogen-bonded to the template strand in order to continue synthesis. Following excision of the mismatched G, DNA synthesis can proceed with incorporation of the correct nucleotide (A).

Combined with the ability to discriminate against the insertion of mismatched bases, the proofreading activity of DNA polymerases is sufficient to reduce the error frequency of replication to about one mismatched base per 10^8 to 10^9. Additional mechanisms (discussed in the section "DNA Repair") act to remove mismatched bases that have been incorporated into newly synthesized DNA, and further ensure correct replication of the genetic information.

Origins and the Initiation of Replication

The replication of both prokaryotic and eukaryotic DNAs starts at sites called **origins of replication**, which serve as binding sites for proteins that initiate the replication process. The first origin to be defined was that of *E. coli*, in which genetic analysis indicated that replication always begins at a unique site on the bacterial chromosome. The *E. coli* origin has since been studied in detail and found to consist of 245 base pairs of DNA, elements of which serve as binding sites for proteins required to initiate DNA replication (Figure 6.12). The key step is the binding of an initiator protein to specific DNA sequences within the origin. The initiator protein begins to unwind the origin DNA and recruits the other proteins involved in DNA synthesis. Helicase and single-stranded DNA-binding proteins then act to continue unwinding and exposing the template DNA, and primase initiates the synthesis of leading strands. Two replication forks are formed and move in opposite directions along the circular *E. coli* chromosome.

The origins of replication of several animal viruses, such as SV40, have been studied as models for the initiation of DNA synthesis in eukaryotes. SV40 has a single origin of replication (consisting of 64 base pairs) that functions both in infected cells and in cell-free systems. Replication is initiated by a virus-encoded protein (called T antigen) that binds to the origin and also acts as a helicase. A single-stranded DNA-binding protein is required to stabilize the unwound template, and the DNA polymerase α-primase complex then initiates DNA synthesis.

Although single origins are sufficient to direct the replication of bacterial and viral genomes, multiple origins are needed to replicate the much larger genomes of eukaryotic cells within a reasonable period of time. For exam-

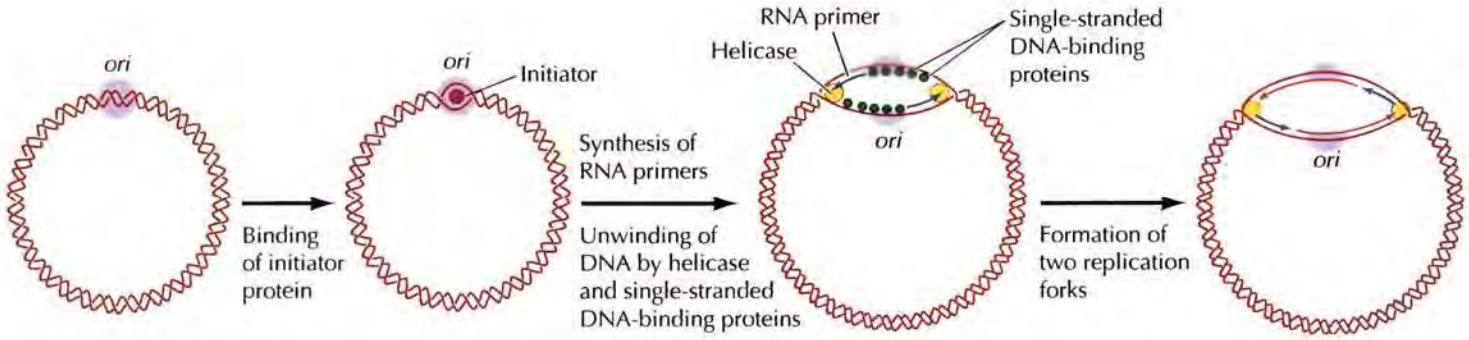


FIGURE 6.12 Origin of replication in *E. coli* Replication initiates at a unique site on the *E. coli* chromosome, designated the origin (*ori*). The first event is the binding of an initiator protein to *ori* DNA, which leads to partial unwinding of the template. The DNA continues to unwind by the actions of helicase and single-stranded DNA-binding proteins, and RNA primers are synthesized by primase. The two replication forks formed at the origin then move in opposite directions along the circular DNA molecule.

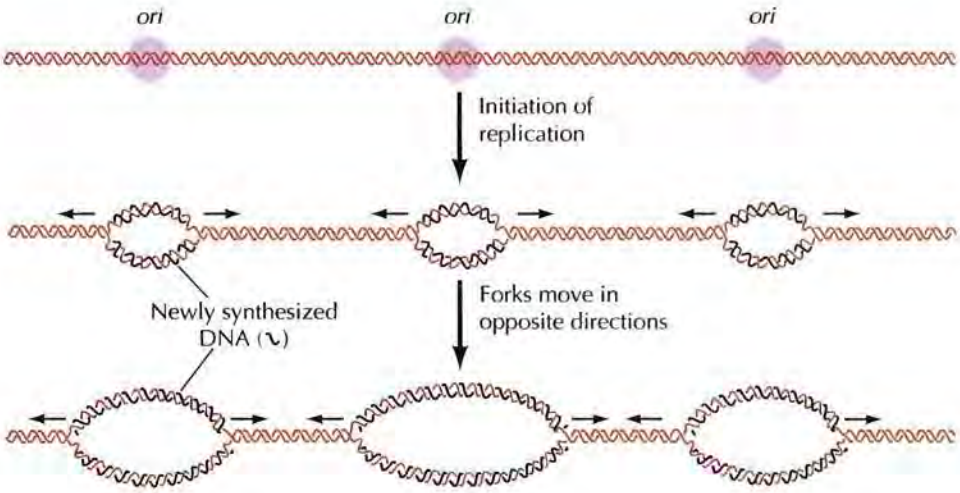


FIGURE 6.13 Replication origins in eukaryotic chromosomes
Replication initiates at multiple origins (*ori*), each of which produces two replication forks.

ple, the entire genome of *E. coli* (4×10^6 base pairs) is replicated from a single origin in approximately 30 minutes. If mammalian genomes (3×10^9 base pairs) were replicated from a single origin at the same rate, DNA replication would require about 3 weeks (30,000 minutes). The problem is further exacerbated by the fact that the rate of DNA replication in mammalian cells is actually about tenfold lower than in *E. coli*, probably as a result of the packaging of eukaryotic DNA in chromatin. Nonetheless, the genomes of mammalian cells are typically replicated within a few hours, necessitating the use of thousands of replication origins.

The presence of multiple replication origins in eukaryotic cells was first demonstrated by the exposure of cultured mammalian cells to radioactive thymidine for different time intervals, followed by autoradiography to detect newly synthesized DNA. The results of such studies indicated that DNA synthesis is initiated at multiple sites, from which it then proceeds in both directions along the chromosome (Figure 6.13). The replication origins in mammalian cells are spaced at intervals of approximately 50 to 300 kb; thus, the human genome has about 30,000 origins of replication. The genomes of simpler eukaryotes also have multiple origins; for example, replication in yeasts initiates at origins separated by intervals of approximately 40 kb.

The origins of replication of eukaryotic chromosomes were first studied in the yeast *S. cerevisiae*, in which they were identified as sequences that can support the replication of plasmids in transformed cells (Figure 6.14). This has provided a functional assay for these sequences, and several such elements (called **autonomously replicating sequences**, or **ARSs**) have been isolated. Their role as origins of replication has been verified by direct biochemical analysis, not only in plasmids but also in yeast chromosomal DNA.

Functional ARS elements span about 100 base pairs, including an 11-base-pair core sequence common to many different ARSs (Figure 6.15). This core sequence is essential for ARS function and has been found to be the binding site of a six-subunit protein complex (called the **origin recognition complex**, or **ORC**) that is required for initiation of DNA replication at *S. cerevisiae* origins. The ORC complex functions to recruit other proteins (including the MCM DNA helicase) to the origin, leading to the initiation of replication. The mechanism of initiation of DNA replication in *S. cerevisiae* thus appears similar to that in bacteria and eukaryotic viruses; that is, an initiator protein specifically binds to specific DNA sequences that define an origin of replication.

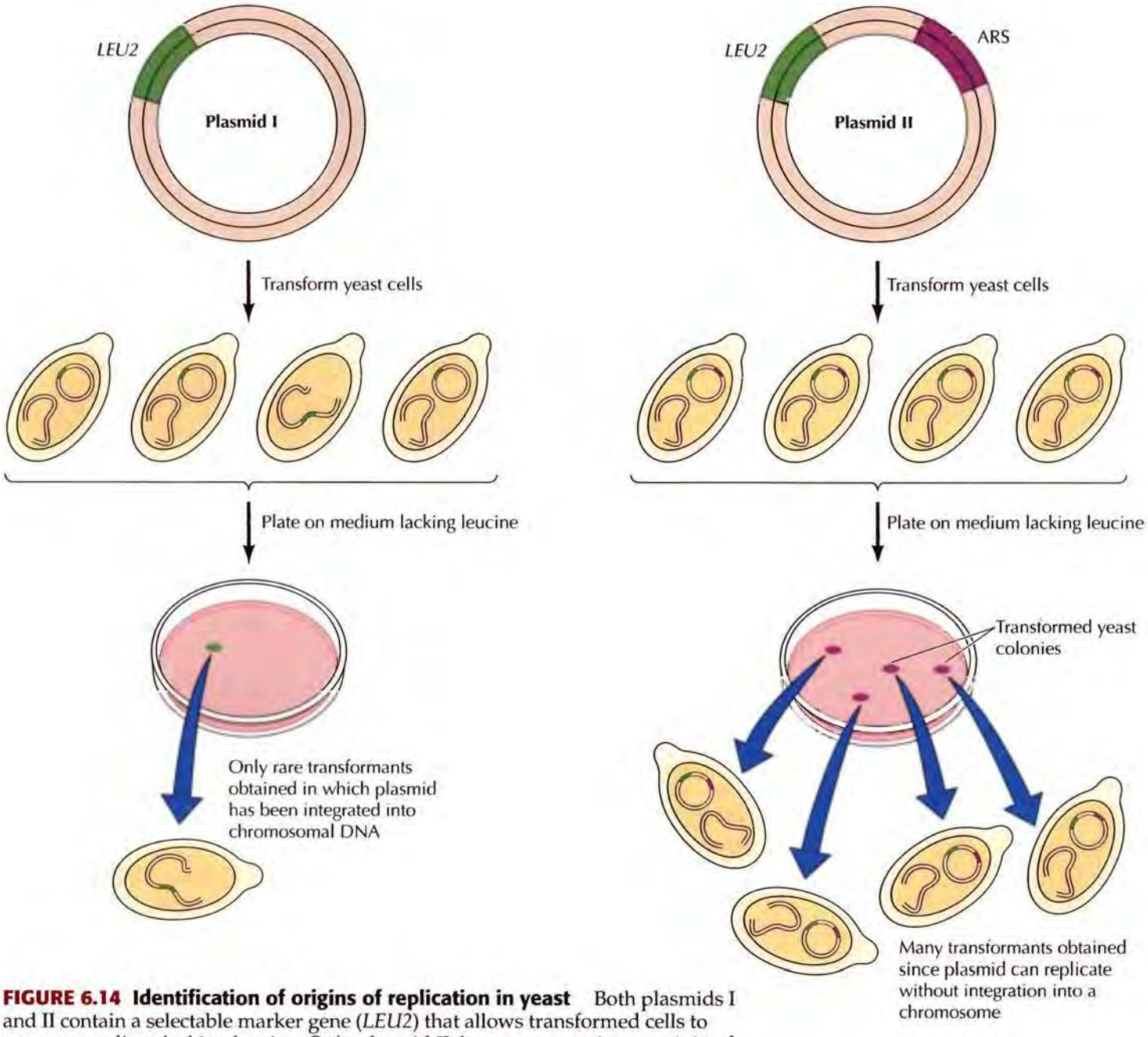


FIGURE 6.14 Identification of origins of replication in yeast Both plasmids I and II contain a selectable marker gene (*LEU2*) that allows transformed cells to grow on medium lacking leucine. Only plasmid II, however, contains an origin of replication (ARS). The transformation of yeasts with plasmid I yields only rare transformants in which the plasmid has integrated into chromosomal DNA. Plasmid II, however, is able to replicate without integration into a yeast chromosome (autonomous replication), so many more transformants result from its introduction into yeast cells.

Subsequent studies have shown that the role of ORC proteins as initiators of replication is conserved in all eukaryotes, from yeasts to mammals. However, replication origins in other eukaryotes are much less well defined than the ARS elements of *S. cerevisiae*. In the fission yeast *S. pombe*, origin sequences are spread over 0.5 to 1 kb of DNA. The *S. pombe* origins lack the clearly defined ORC binding site of the *S. cerevisiae* ARS elements, but they contain repeats of AT-rich sequences that appear to serve as binding sites for the *S. pombe* ORC complex. In *Drosophila* and mammalian cells, however,

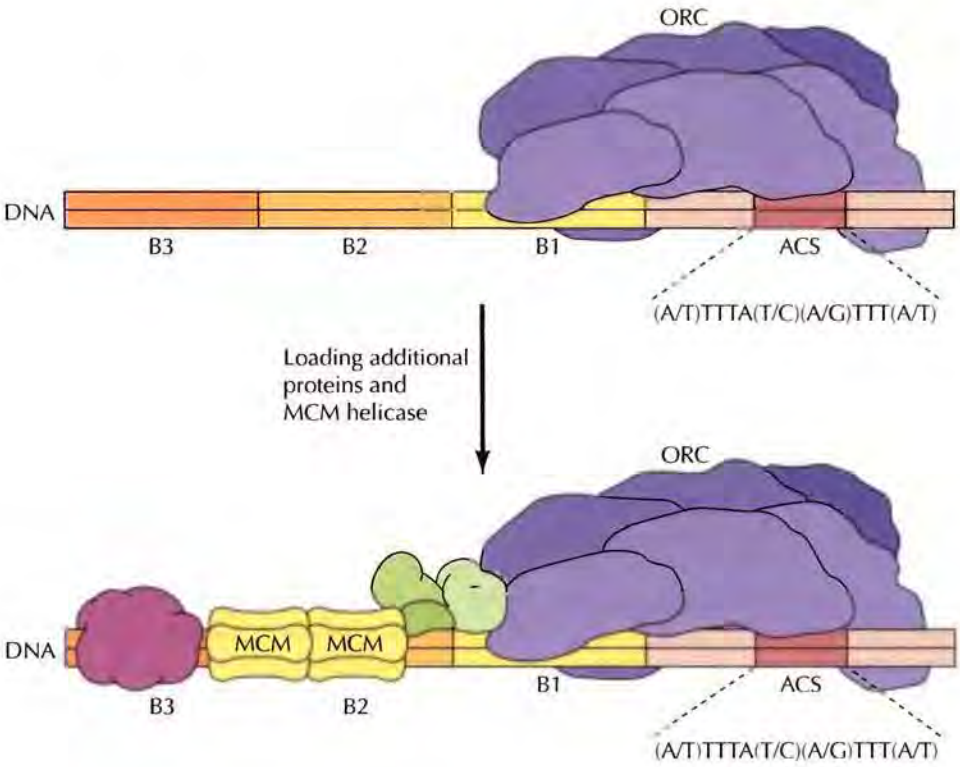


FIGURE 6.15 A yeast ARS element An *S. cerevisiae* ARS element contains an 11-base-pair ARS consensus sequence (ACS) and three additional elements (B1, B2, and B3) that contribute to origin function. The origin recognition complex (ORC) binds to the ACS and B1. ORC then recruits additional proteins, including the MCM DNA helicase, to the origin.

the ORC proteins do not appear to recognize specific DNA sequences. Instead, the sites of ORC binding and initiation of DNA replication in higher eukaryotes may be determined by aspects of chromatin structure that remain to be elucidated by future research.

Telomeres and Telomerase: Maintaining the Ends of Chromosomes

Because DNA polymerases extend primers only in the 5′ to 3′ direction, they are unable to copy the extreme 5′ ends of linear DNA molecules. Consequently, special mechanisms are required to replicate the terminal sequences of the linear chromosomes of eukaryotic cells. These sequences (**telomeres**) consist of tandem repeats of simple-sequence DNA (see Chapter 5). They are maintained by the action of a unique enzyme called **telomerase**, which is able to catalyze the synthesis of telomeres in the absence of a DNA template.

Telomerase is a **reverse transcriptase**, one of a class of DNA polymerases, first discovered in retroviruses (see Chapter 4), that synthesize DNA from an RNA template. Importantly, telomerase carries its own template RNA, which is complementary to the telomere repeat sequences, as part of the enzyme complex. The use of this RNA as a template allows telomerase to generate multiple copies of the telomeric repeat sequences, thereby maintaining telomeres in the absence of a conventional DNA template to direct their synthesis.

The mechanism of telomerase action was initially elucidated in 1985 by Carol Greider and Elizabeth Blackburn during studies of the protozoan *Tetrahymena* (Figure 6.16). The *Tetrahymena* telomerase is complexed to a 159-nucleotide-long RNA that includes the sequence 3′–AACCCCAAC–5′. This sequence is complementary to the *Tetrahymena* telomeric repeat (5′–TTGGGG–3′) and serves as the template for the synthesis of telomeric

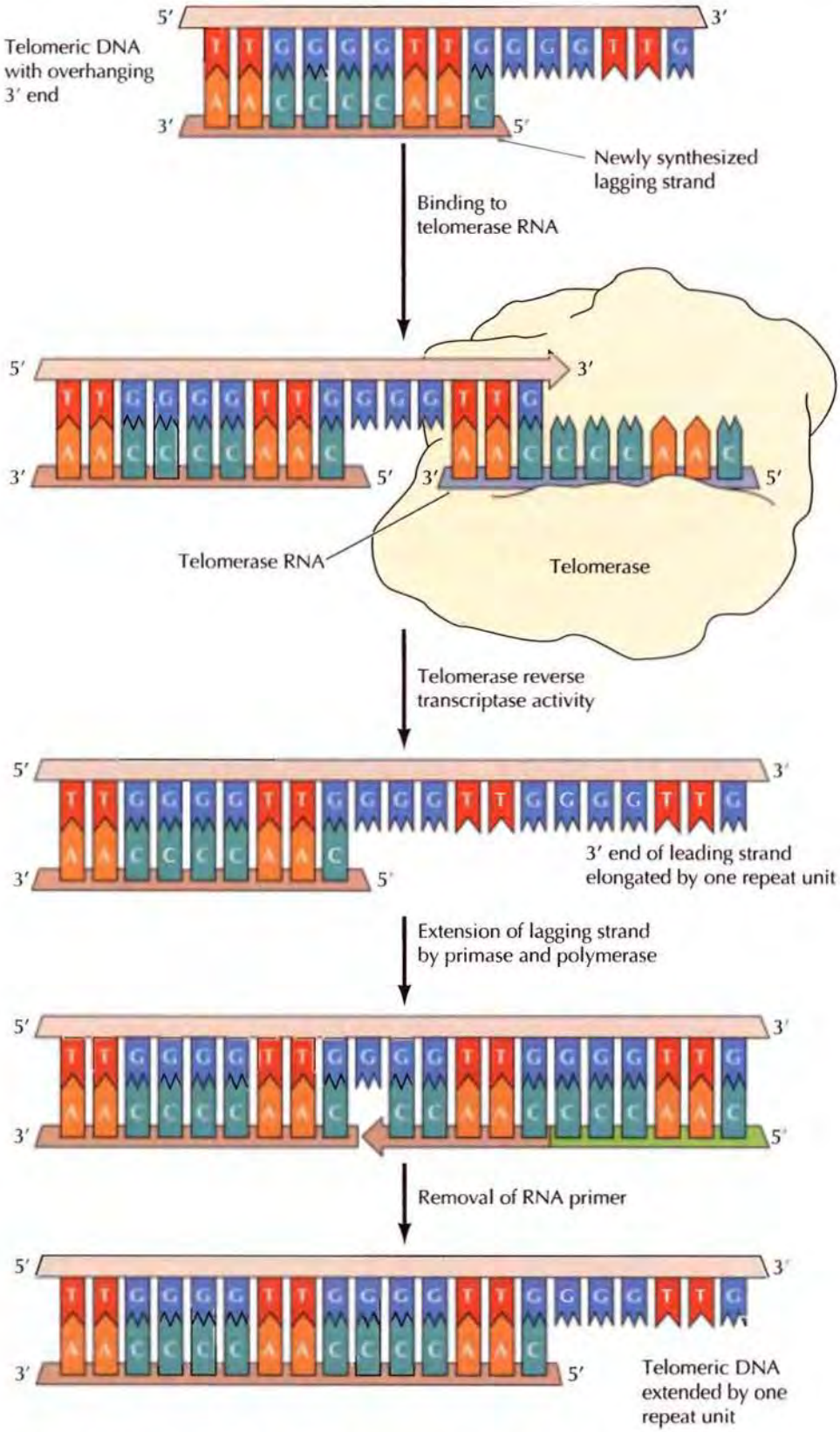


FIGURE 6.16 Action of telomerase Telomeric DNA is a simple repeat sequence with an overhanging 3′ end on the newly synthesized leading strand. Telomerase carries its own RNA molecule, which is complementary to telomeric DNA, as part of the enzyme complex. The overhanging end of telomeric DNA binds to the telomerase RNA, which then serves as a template for extension of the leading strand by one repeat unit. The lagging strand of telomeric DNA can then be elongated by conventional RNA priming and DNA polymerase activity.

DNA. The use of this RNA as a template allows telomerase to extend the 3′ end of chromosomal DNA by one repeat unit beyond its original length. The complementary strand can then be synthesized by the polymerase α-primase complex using conventional RNA priming. Removal of the RNA primer leaves an overhanging 3′ end of chromosomal DNA, which can form loops at the ends of eukaryotic chromosomes (see Figure 5.22).

Telomerase has been identified in a variety of eukaryotes, and genes encoding telomerase RNAs have been cloned from *Tetrahymena*, yeasts, mice, and humans. In each case, the telomerase RNA contains sequences complementary to the telomeric repeat sequence of that organism (see Table 5.5). Moreover, the introduction of mutant telomerase RNA genes into yeasts has been shown to result in corresponding alterations of the chromosomal telomeric repeat sequences, directly demonstrating the function of telomerase in maintaining the termini of eukaryotic chromosomes.

Defects in telomerase and the normal maintenance of telomeres are associated with several human diseases. The activity of telomerase is regulated in dividing cells to maintain telomeres at their normal length. In human cells, for example, telomeres span approximately 10 kb with a 3′ single-strand overhang of 150–200 bases. Although this length is maintained in germ cells, most somatic cells do not have sufficiently high levels of telomerase to maintain the length of their telomeres for an indefinite number of cell divisions. Consequently, telomeres gradually shorten as cells age, and this shortening eventually leads to cell death or senescence. Interestingly, several premature aging syndromes are characterized by an abnormally high rate of telomere loss, and some of these diseases have been found to be caused directly by mutations in telomerase. Conversely, cancer cells express abnormally high levels of telomerase, allowing them to continue dividing indefinitely.

DNA Repair

DNA, like any other molecule, can undergo a variety of chemical reactions. Because DNA uniquely serves as a permanent copy of the cell genome, however, changes in its structure are of much greater consequence than are alterations in other cell components, such as RNAs or proteins. Mutations can result from the incorporation of incorrect bases during DNA replication. In addition, various chemical changes occur in DNA either spontaneously (Figure 6.17) or as a result of exposure to chemicals or radiation (Figure 6.18). Such damage to DNA can block replication or transcription, and can result in a high frequency of mutations—consequences that are unacceptable from the standpoint of cell reproduction. To maintain the integrity of their genomes, cells have therefore had to evolve mechanisms to repair damaged DNA. These mechanisms of DNA repair can be divided into two general classes: (1) direct reversal of the chemical reaction responsible for DNA damage, and (2) removal of the damaged bases followed by their replacement with newly synthesized DNA. Where DNA repair fails, additional mechanisms have evolved to enable cells to cope with the damage.

Direct Reversal of DNA Damage

Most damage to DNA is repaired by removal of the damaged bases followed by resynthesis of the excised region. Some lesions in DNA, however, can be repaired by direct reversal of the damage, which may be a more efficient way of dealing with specific types of DNA damage that occur fre-

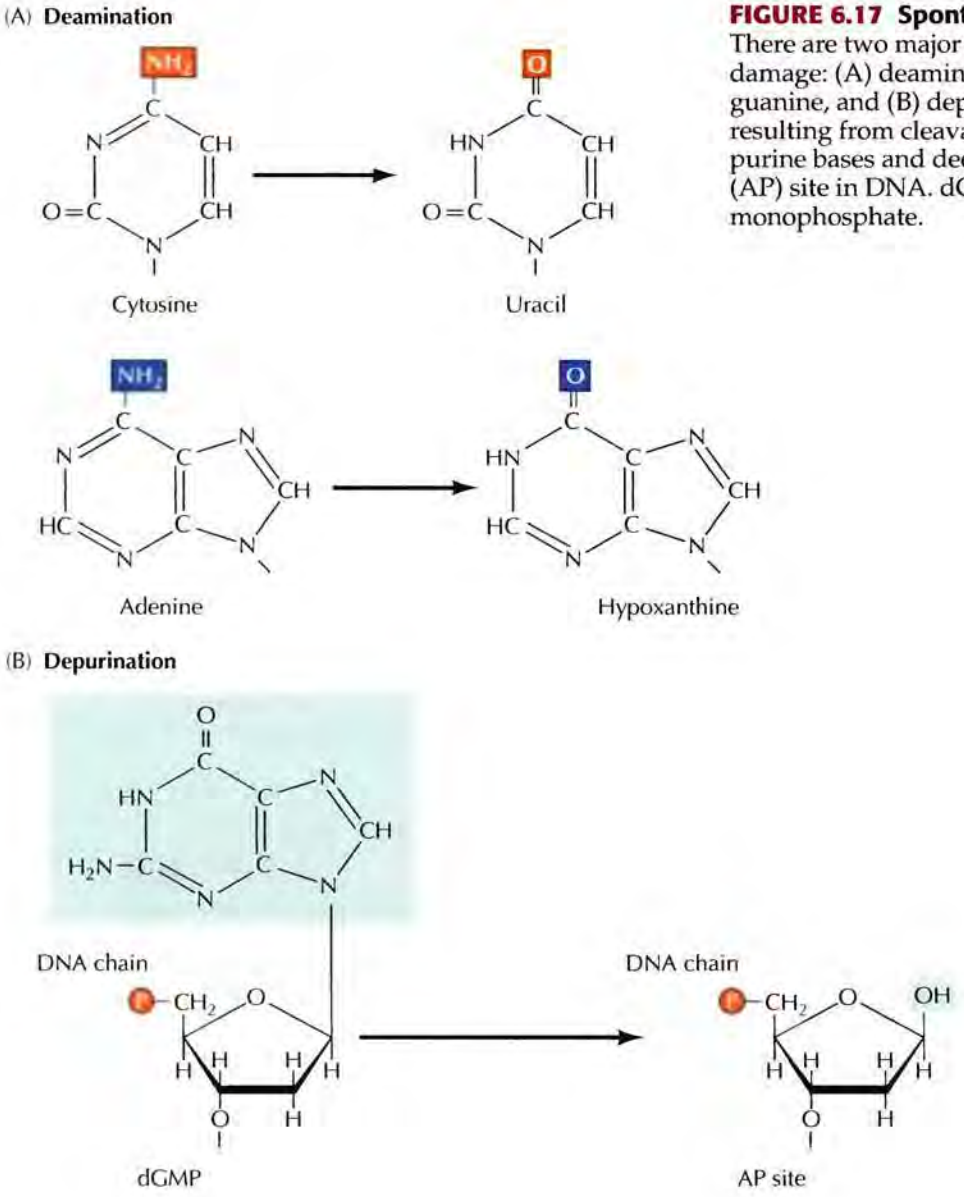


FIGURE 6.17 Spontaneous damage to DNA There are two major forms of spontaneous DNA damage: (A) deamination of adenine, cytosine, and guanine, and (B) depurination (loss of purine bases) resulting from cleavage of the bond between the purine bases and deoxyribose, leaving an apurinic (AP) site in DNA. dGMP = deoxyguanosine monophosphate.

quently. Only a few types of DNA damage are repaired in this way, particularly pyrimidine dimers resulting from exposure to ultraviolet (UV) light and alkylated guanine residues that have been modified by the addition of methyl or ethyl groups at the O^6 position of the purine ring.

UV light is one of the major sources of damage to DNA and is also the most thoroughly studied form of DNA damage in terms of repair mechanisms. Its importance is illustrated by the fact that exposure to solar UV irradiation is the cause of almost all skin cancer in humans. The major type of damage induced by UV light is the formation of **pyrimidine dimers,** in which adjacent pyrimidines on the same strand of DNA are joined by the formation of a cyclobutane ring resulting from saturation of the double bonds between carbons 5 and 6 (see Figure 6.18A). The formation of such dimers distorts the structure of the DNA chain and blocks transcription or replication past the site of damage, so their repair is closely correlated with the ability of cells to survive UV irradiation. One mechanism of repairing UV-induced pyrimidine dimers is direct reversal of the dimerization reac-

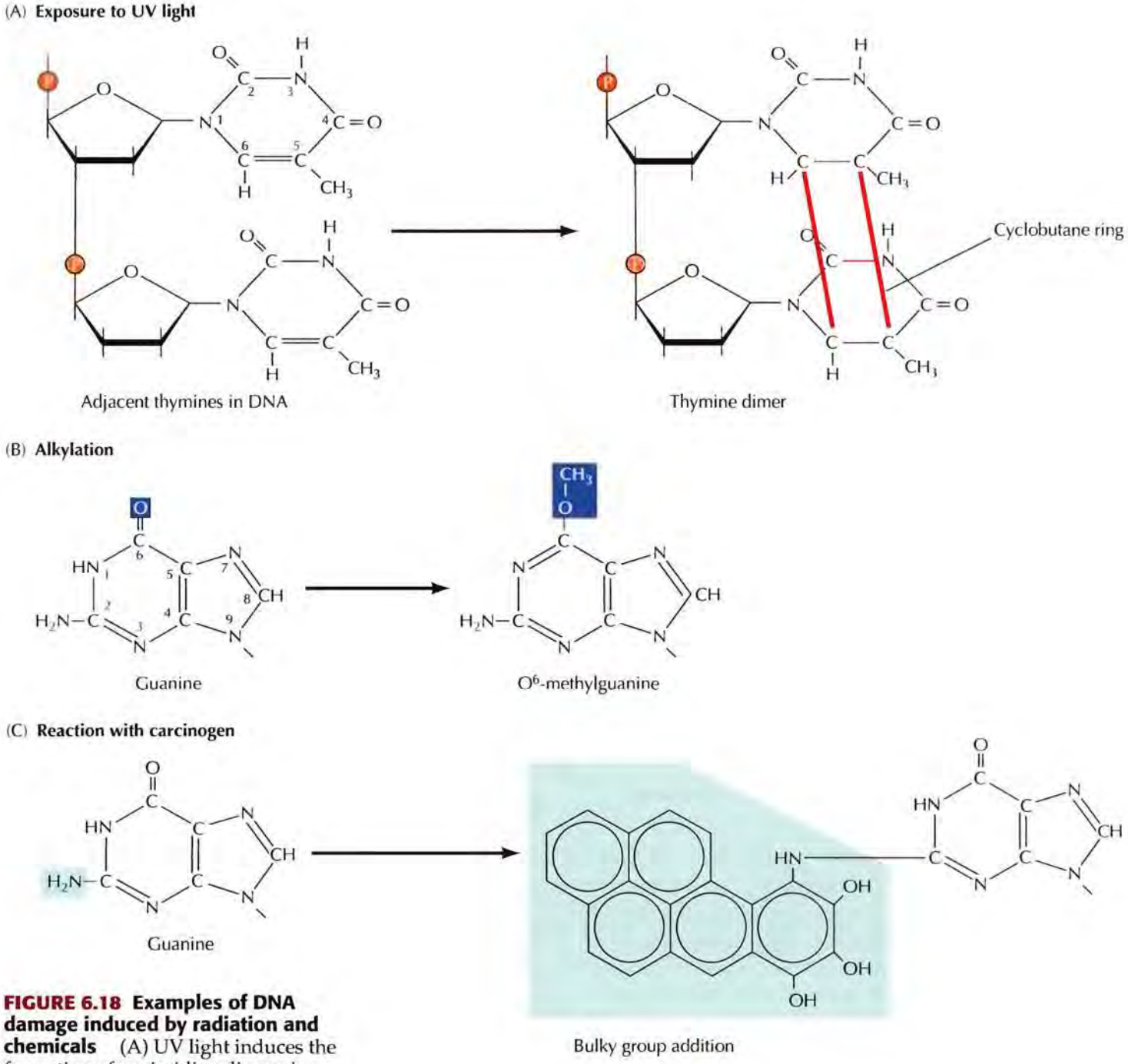


FIGURE 6.18 Examples of DNA damage induced by radiation and chemicals (A) UV light induces the formation of pyrimidine dimers in which two adjacent pyrimidines (e.g., thymines) are joined by a cyclobutane ring structure. (B) Alkylation is the addition of methyl or ethyl groups to various positions on the DNA bases. In this example, alkylation of the O^6 position of guanine results in formation of O^6-methylguanine. (C) Many carcinogens (e.g., benzo[*a*]pyrene) react with DNA bases, resulting in the addition of large bulky chemical groups to the DNA molecule.

tion. The process is called **photoreactivation** because energy derived from visible light is utilized to break the cyclobutane ring structure (Figure 6.19). The original pyrimidine bases remain in DNA, now restored to their normal state. As might be expected from the fact that solar UV irradiation is a major source of DNA damage for diverse cell types, the repair of pyrimidine dimers by photoreactivation is common to a variety of prokaryotic and eukaryotic cells, including *E. coli*, yeasts, and some species of plants and animals. Curiously, however, photoreactivation is not universal; many species (including humans) lack this mechanism of DNA repair.

Another form of direct repair deals with damage resulting from the reaction between alkylating agents and DNA. Alkylating agents are reactive

compounds that can transfer methyl or ethyl groups to a DNA base, thereby chemically modifying the base (see Figure 6.18B). A particularly important type of damage is methylation of the O^6 position of guanine, because the product, O^6-methylguanine, forms complementary base pairs with thymine instead of cytosine. This lesion can be repaired by an enzyme (called O^6-methylguanine methyltransferase) that transfers the methyl group from O^6-methylguanine to a cysteine residue in its active site (Figure 6.20). The potentially mutagenic chemical modification is thus removed, and the original guanine is restored. Enzymes that catalyze this direct repair reaction are widespread in both prokaryotes and eukaryotes, including humans.

Excision Repair

Although direct repair is an efficient way of dealing with particular types of DNA damage, excision repair is a more general means of repairing a wide variety of chemical alterations to DNA. Consequently, the various types of excision repair are the most important DNA repair mechanisms in both prokaryotic and eukaryotic cells. In excision repair, the damaged DNA is recognized and removed, either as free bases or as nucleotides. The resulting gap is then filled in by synthesis of a new DNA strand, using the undamaged complementary strand as a template. Three types of excision repair—base-excision, nucleotide-excision, and mismatch repair—enable cells to cope with a variety of different kinds of DNA damage.

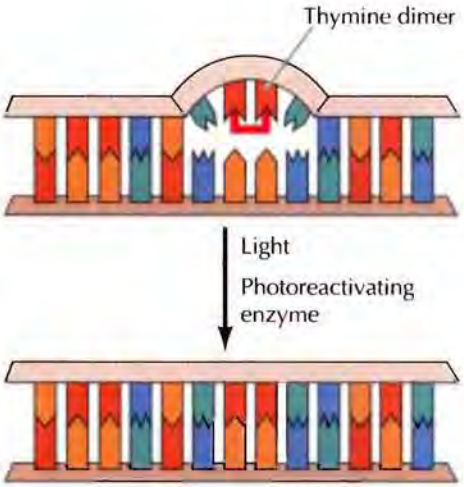


FIGURE 6.19 Direct repair of thymine dimers UV-induced thymine dimers can be repaired by photoreactivation, in which energy from visible light is used to split the bonds forming the cyclobutane ring.

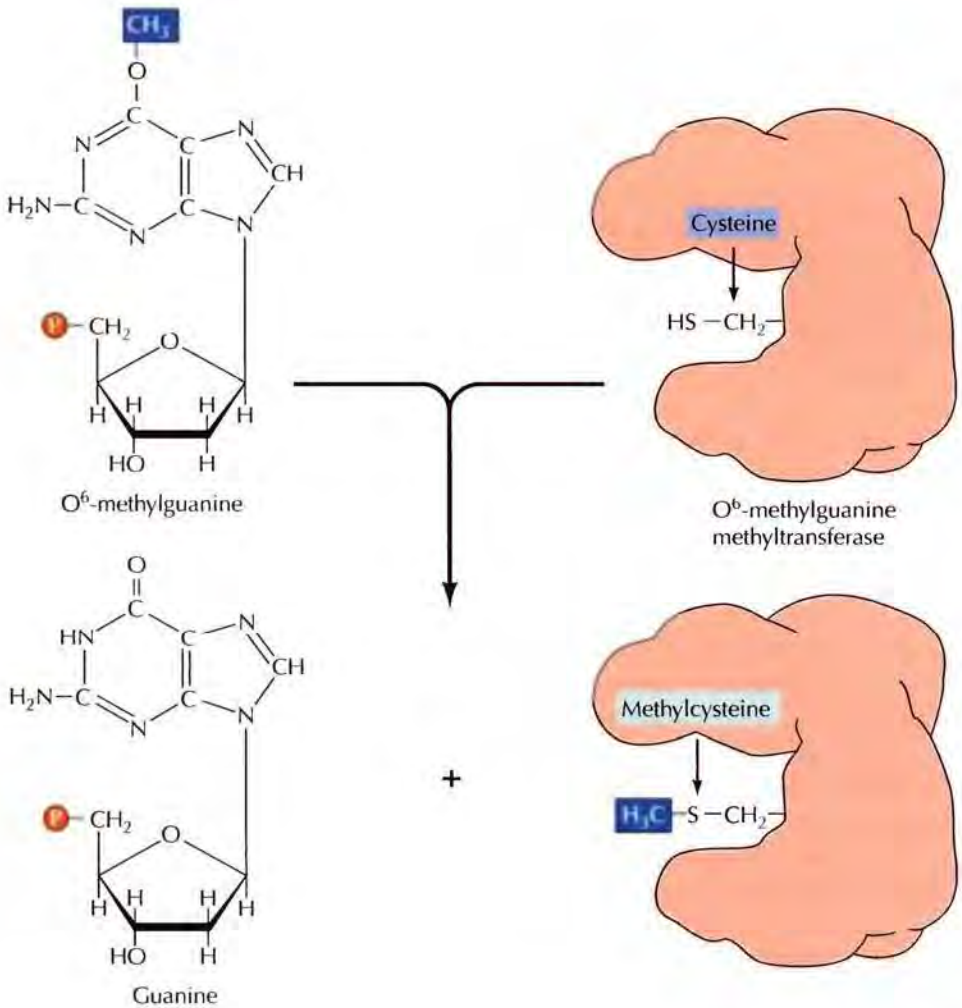


FIGURE 6.20 Repair of O^6-methylguanine O^6-methylguanine methyltransferase transfers the methyl group from O^6-methylguanine to a cysteine residue in the enzyme's active site.

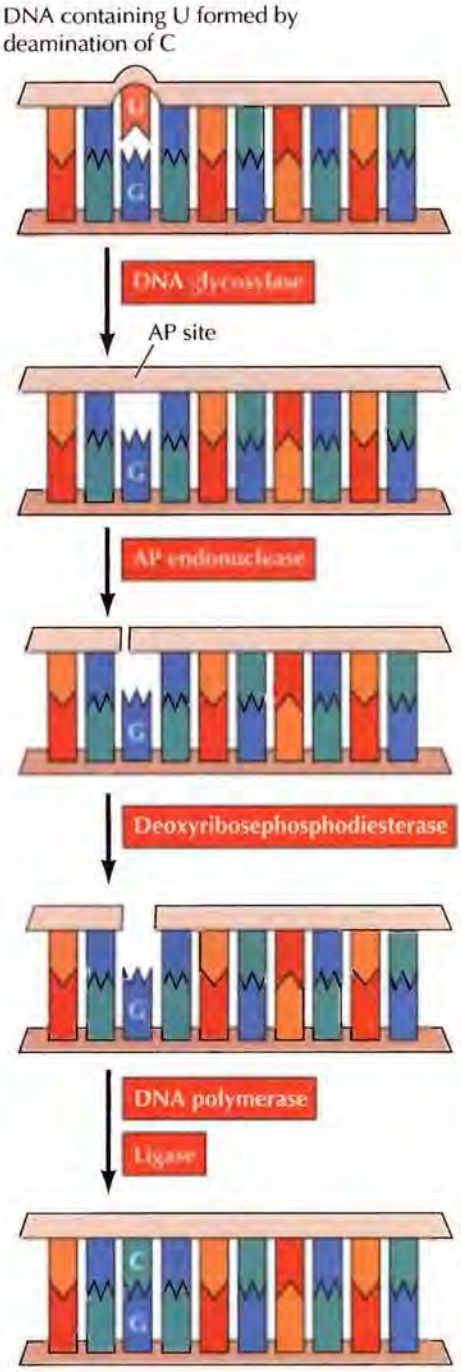


FIGURE 6.21 Base-excision repair In this example, uracil (U) has been formed by deamination of cytosine (C) and is therefore opposite a guanine (G) in the complementary strand of DNA. The bond between uracil and the deoxyribose is cleaved by a DNA glycosylase, leaving a sugar with no base attached in the DNA (an AP site). This site is recognized by AP endonuclease, which cleaves the DNA chain. The remaining deoxyribose is removed by deoxyribosephosphodiesterase. The resulting gap is then filled by DNA polymerase and sealed by ligase, leading to incorporation of the correct base (C) opposite the G.

The repair of uracil-containing DNA is a good example of **base-excision repair**, in which single damaged bases are recognized and removed from the DNA molecule (Figure 6.21). Uracil can arise in DNA by two mechanisms: (1) Uracil (as dUTP [deoxyuridine triphosphate]) is occasionally incorporated in place of thymine during DNA synthesis, and (2) uracil can be formed in DNA by the deamination of cytosine (see Figure 6.17A). The second mechanism is of much greater biological significance because it alters the normal pattern of complementary base pairing and thus represents a mutagenic event. The excision of uracil in DNA is catalyzed by **DNA glycosylase**, an enzyme that cleaves the bond linking the base (uracil) to the deoxyribose of the DNA backbone. This reaction yields free uracil and an apyrimidinic site—a sugar with no base attached. DNA glycosylases also recognize and remove other abnormal bases, including hypoxanthine formed by the deamination of adenine, pyrimidine dimers, alkylated purines other than O^6-alkylguanine, and bases damaged by oxidation or ionizing radiation.

The result of DNA glycosylase action is the formation of an apyrimidinic or apurinic site (generally called an AP site) in DNA. Similar AP sites are formed as a result of the spontaneous loss of purine bases (see Figure 6.17B), which occurs at a significant rate under normal cellular conditions. For example, each cell in the human body is estimated to lose several thousand purine bases daily. These sites are repaired by **AP endonuclease**, which cleaves adjacent to the AP site (see Figure 6.21). The remaining deoxyribose moiety is then removed, and the resulting single-base gap is filled by DNA polymerase and ligase.

Whereas DNA glycosylases recognize only specific forms of damaged bases, other excision repair systems recognize a wide variety of damaged bases that distort the DNA molecule, including UV-induced pyrimidine dimers and bulky groups added to DNA bases as a result of the reaction of many carcinogens with DNA (see Figure 6.18C). This widespread form of DNA repair is known as **nucleotide-excision repair**, because the damaged bases (e.g., a thymine dimer) are removed as part of an oligonucleotide containing the lesion (Figure 6.22).

In *E. coli*, nucleotide-excision repair is catalyzed by the products of three genes (*uvrA*, *uvrB*, and *uvrC*) that were identified because mutations at these loci result in extreme sensitivity to UV light. The protein UvrA recognizes damaged DNA and recruits UvrB and UvrC to the site of the lesion. UvrB and UvrC then cleave on the 3′ and 5′ sides of the damaged site, respectively, thus excising an oligonucleotide consisting of 12 or 13 bases. The UvrABC complex is frequently called an **excinuclease**, a name that reflects its ability to directly *excis*e an oligonucleotide. The action of a helicase is then required to remove the damage-containing oligonucleotide from the double-stranded DNA molecule, and the resulting gap is filled by DNA polymerase and sealed by ligase.

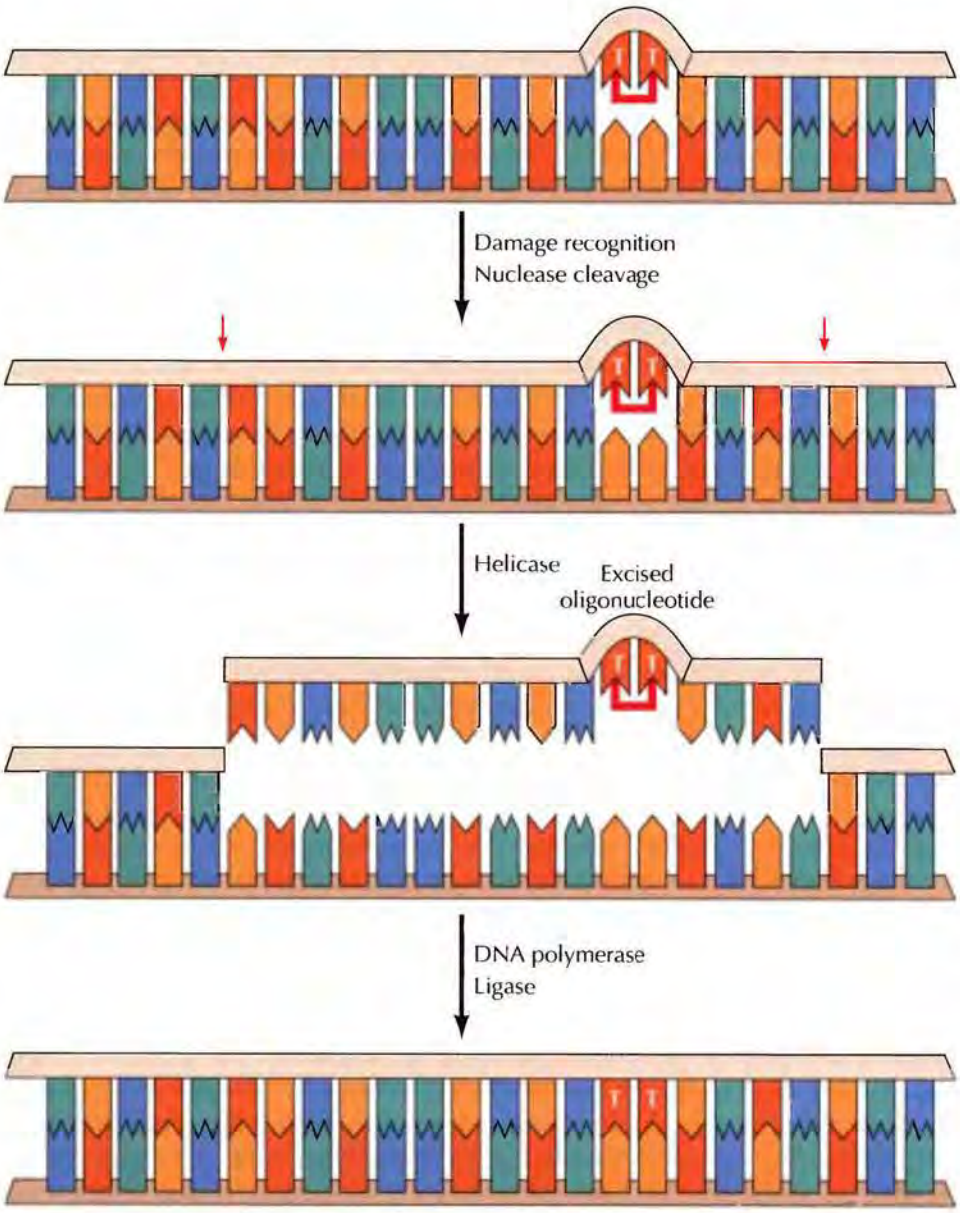
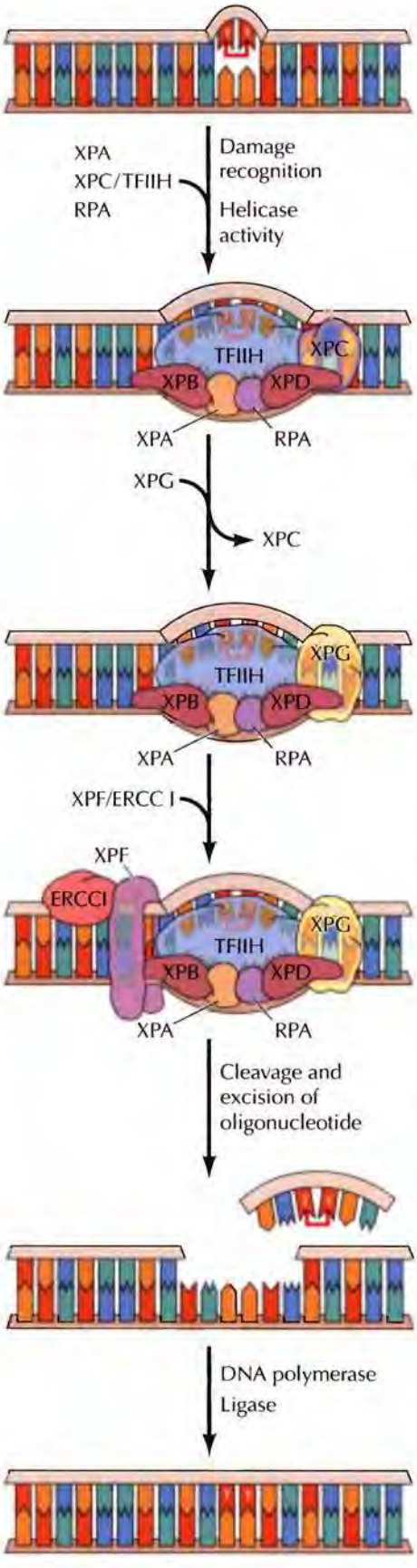


FIGURE 6.22 Nucleotide-excision repair of thymine dimers Damaged DNA is recognized and then cleaved on both sides of a thymine dimer by 3′ and 5′ nucleases. Unwinding by a helicase results in excision of an oligonucleotide containing the damaged bases. The resulting gap is then filled by DNA polymerase and sealed by ligase.

Nucleotide-excision repair systems have also been studied extensively in eukaryotes, including yeasts, rodents, and humans. In yeasts, as in *E. coli*, several genes involved in DNA repair (called *RAD* genes for *rad*iation sensitivity) have been identified by the isolation of mutants with increased sensitivity to UV light. In humans, DNA repair genes have been identified largely by studies of individuals suffering from inherited diseases resulting from deficiencies in the ability to repair DNA damage. The most extensively studied of these diseases is xeroderma pigmentosum (XP), a rare genetic disorder that affects approximately one in 250,000 people. Individuals with this disease are extremely sensitive to UV light and develop multiple skin cancers on the regions of their bodies that are exposed to sunlight. In 1968, James Cleaver made the key discovery that cultured cells from XP patients were deficient in the ability to carry out nucleotide-excision repair. This observation not only provided the first link between DNA repair and cancer, but also suggested the use of XP cells as an experimental system to identify human DNA repair genes. The identification of human DNA repair



genes has been accomplished by studies not only of XP cells, but also of two other human diseases resulting from DNA repair defects (Cockayne's syndrome and trichothiodystrophy) and of UV-sensitive mutants of rodent cell lines. The availability of mammalian cells with defects in DNA repair has allowed the cloning of repair genes based on the ability of wild-type alleles to restore normal UV sensitivity to mutant cells in gene transfer assays, thereby opening the door to experimental analysis of nucleotide-excision repair in mammalian systems.

Molecular cloning has identified seven different repair genes (designated *XPA* through *XPG*) that are mutated in cases of xeroderma pigmentosum, as well as genes that are mutated in Cockayne's syndrome, trichothiodystrophy, and UV-sensitive mutants of rodent cells. The proteins encoded by these mammalian DNA repair genes are closely related to proteins encoded by yeast *RAD* genes, indicating that nucleotide-excision repair is highly conserved throughout eukaryotes. With cloned yeast and mammalian repair genes available, it has been possible to purify their encoded proteins and develop *in vitro* systems to study their roles in the repair process (Figure 6.23). The initial step in excision repair in mammalian cells involves recognition of disrupted base pairing by the cooperative binding of XPA, XPC, and the single-stranded DNA binding protein replication protein A (RPA) discussed earlier in this chapter (see Figure 6.8). XPC is associated with a multisubunit transcription factor called TFIIH, which is required to initiate transcription of eukaryotic genes (see Chapter 7). Two of the subunits of TFIIH are the XPB and XPD proteins, which act as helicases to unwind approximately 25 base pairs of DNA around the site of damage. The XPG protein is then recruited to the complex, displacing XPC, followed by recruitment of XPF as a heterodimer with ERCC1 (a repair protein identified in UV-sensitive rodent cells). XPF/ERCC1 and XPG are endonucleases, which cleave DNA on the 5′ and 3′ sides of the damaged site, respectively. This cleavage excises an oligonucleotide consisting of approximately 30 bases. The resulting gap is then filled by DNA polymerase δ or ε (in association with RFC and PCNA) and sealed by ligase.

Although damaged DNA can be recognized throughout the genome, an alternative form of nucleotide-excision repair, called **transcription-coupled repair**, is specifically dedicated to repairing damage within actively transcribed genes. A connection between transcription and repair was first suggested by experiments showing that transcribed strands of DNA are repaired more rapidly than nontranscribed strands in both *E. coli* and mammalian cells. Since DNA damage blocks transcription, this transcription-repair coupling is thought to be advantageous by allowing the cell to preferentially repair damage to genes that are actively expressed. In both *E. coli* and mammalian cells, the mechanism of transcription-repair coupling involves recognition of RNA polymerase stalled at a lesion in the DNA strand being transcribed. In *E. coli*, the stalled RNA polymerase is recognized by a protein called transcription-repair coupling factor, which dis-

FIGURE 6.23 Nucleotide-excision repair in mammalian cells DNA damage (e.g., a thymine dimer) is recognized by XPA, RPA, and XPC, which is complexed to the transcription factor TFIIH, containing the XPB and XPD helicases. Following unwinding of the DNA by XPB and XPD, XPG is recruited to the complex, replacing XPC. The XPF/ERCC1 complex is then recruited and the DNA is then cleaved by the XPF/ERCC1 and XPG endonucleases, excising the damaged oligonucleotide. The resulting gap is filled by DNA polymerase and sealed by ligase.

places RNA polymerase and recruits the UvrABC excinuclease to the site of damage. In mammalian cells, transcription-coupled repair involves recognition of stalled RNA polymerase by two proteins (CSA and CSB) that are encoded by genes responsible for Cockayne's syndrome. In contrast to patients with xeroderma pigmentosum, patients with Cockayne's syndrome are specifically defective in transcription-coupled repair, consistent with the role of CSA and CSB as transcription-repair coupling factors.

A third excision repair system recognizes mismatched bases that are incorporated during DNA replication. Many such mismatched bases are removed by the proofreading activity of DNA polymerase. The ones that are missed are subject to later correction by the **mismatch repair** system, which scans newly replicated DNA. If a mismatch is found, the enzymes of this repair system are able to identify and excise the mismatched base specifically from the newly replicated DNA strand, allowing the error to be corrected and the original sequence restored.

In *E. coli*, the ability of the mismatch repair system to distinguish between parental DNA and newly synthesized DNA is based on the fact that DNA of this bacterium is modified by the methylation of adenine residues within the sequence GATC to form 6-methyladenine (Figure 6.24). Since methylation occurs after replication, newly synthesized DNA strands are not methylated and thus can be specifically recognized by the mismatch repair enzymes. Mismatch repair is initiated by the protein MutS, which recognizes the mismatch and forms a complex with two other proteins called MutL and MutH. The MutH endonuclease then cleaves the unmethylated DNA strand at a GATC sequence. MutL and MutS then act together with an exonuclease and a helicase to excise the DNA between the strand break and the mismatch, with the resulting gap being filled by DNA polymerase and ligase.

Eukaryotes have a similar mismatch repair system, although the mechanism by which eukaryotic cells identify newly replicated DNA differs from that used by *E. coli*. Eukaryotic cells do not have a homolog of MutH and the strand-specificity of mismatch repair is not determined by DNA methylation. Instead, the presence of single-strand breaks in newly replicated DNA appears to specify the strand to be repaired. Eukaryotic homologs of MutS and MutL bind to the mismatched base, as in *E. coli* and direct excision of the DNA between a strand break and the mismatch. The importance of this repair system is dramatically illustrated by the fact that mutations in the human homologs of *MutS* and *MutL* are responsible for a common type of inherited colon cancer (hereditary nonpolyposis colorectal cancer, or HNPCC). HNPCC is one of the most common inherited diseases; it affects as many as one in 200 people and is responsible for about 15% of all colorectal cancers in this country. The relationship between HNPCC and defects in mismatch repair was discovered in 1993, when two groups of researchers cloned the human homolog of *MutS* and found that mutations in this gene were responsible for about half of all HNPCC cases. Subsequent studies have shown that most of the remaining cases of HNPCC are caused by mutations in one of three human genes that are homologs of *MutL*. Defects in these genes appear to result in a high frequency of mutations in other cell genes, with a correspondingly high likelihood that some of these mutations will eventually lead to the development of cancer.

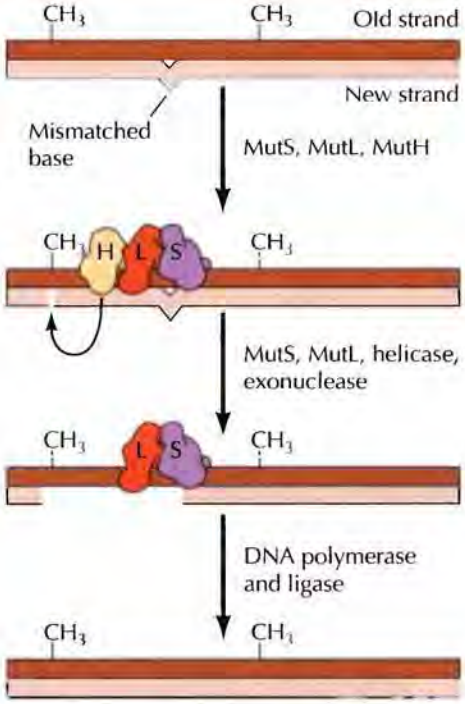


FIGURE 6.24 Mismatch repair in *E. coli* The mismatch repair system detects and excises mismatched bases in newly replicated DNA, which is distinguished from the parental strand because it has not yet been methylated. MutS binds to the mismatched base, followed by MutL. The binding of MutL activates MutH, which cleaves the unmodified strand opposite a site of methylation. MutS and MutL, together with a helicase and an exonuclease, then excise the portion of the unmodified strand that contains the mismatch. The gap is then filled by DNA polymerase and sealed by ligase.

Translesion DNA Synthesis

The direct reversal and excision repair systems act to correct DNA damage before replication, so that replicative DNA synthesis can proceed using an

undamaged DNA strand as a template. Should these systems fail, however, the cell has alternative mechanisms for dealing with damaged DNA at the replication fork. Pyrimidine dimers and many other types of lesions cannot be copied by the normal action of DNA polymerases, so replication is

MOLECULAR MEDICINE

Colon Cancer and DNA Repair

The Disease

Cancers of the colon and rectum (colorectal cancers) are some of the most common types of cancer in Western countries, accounting for about 140,000 cancer cases per year in the United States (approximately 10% of the total cancer incidence). Most colon cancers (like other types of cancer) are not inherited diseases; that is, they are not transmitted directly from parent to offspring. However, two inherited forms of colon cancer have been described. In both of these syndromes, the inheritance of a cancer susceptibility gene results in a very high likelihood of cancer development. One inherited form of colon cancer (familial adenomatous polyposis) is extremely rare, accounting for less than 1% of total colon cancer incidence. The second inherited form of colon cancer (hereditary nonpolyposis colorectal cancer, or HNPCC) is much more common and accounts for up to 15% of all colon cancer cases. Indeed, HNPCC is one of the most common inherited diseases, affecting as many as one in 200 people. Although colon cancers are the most common manifestation of this disease, affected individuals also suffer an increased incidence of other types of cancer, including cancers of the ovaries and endometrium.

Molecular and Cellular Basis

Like other cancers, colorectal cancer results from mutations in genes that regulate cell proliferation, leading to the uncontrolled growth of cancer cells. In most cases these mutations occur sporadically in somatic cells. In hereditary cancers, however, inherited germ-line mutations predispose the individual to cancer development.

A striking advance was made in 1993 with the discovery that a gene responsible for approximately 50% of HNPCC cases encodes an enzyme involved in mismatch repair of DNA; this gene is a human homolog of the *E. coli MutS* gene. Subsequent studies have shown that three other genes, responsible for most remaining cases of HNPCC, are homologs of *MutL* and thus are also involved in the mismatch repair pathway. Defects in these genes appear to result in a high frequency of mutations in other cell genes, with a correspondingly high likelihood that some of these mutations will eventually lead to the development of cancer by affecting genes that regulate cell proliferation.

Prevention and Treatment

As with other inherited diseases, identification of the genes responsible for HNPCC allows individuals at risk for this inherited cancer to be identified by genetic testing. Moreover, prenatal genetic diagnosis may be of great importance to carriers of HNPCC mutations who are planning a family. However, the potential benefits of detecting these mutations are not limited to preventing the transmission of mutant genes to the next generation; their detection may also help prevent the development of cancer in affected individuals.

In terms of disease prevention, a key characteristic of colon cancer is that it develops gradually over several years. Early diagnosis of the disease substantially improves the chances for patient survival. The initial stage of colon cancer development is the outgrowth of small benign polyps, which eventually become malignant and invade the surrounding connective tissue. Prior to the development of malignancy, however, polyps can be easily removed surgically, effectively preventing the outgrowth of a malignant tumor. Polyps and early stages of colon cancer can be detected by examination of the colon with a thin lighted tube (colonoscopy), so frequent colonoscopy of HNPCC patients may allow polyps to be removed before cancer develops. In addition, several drugs are being tested as potential inhibitors of colon cancer development, and these drugs may be of significant benefit to HNPCC patients. By allowing the timely application of such preventive measures, the identification of mutations responsible for HNPCC may make a significant contribution to disease prevention.

A colon polyp visualized by colonoscopy. (David M. Martin, M.D./SPL/Photo Researches, Inc.)

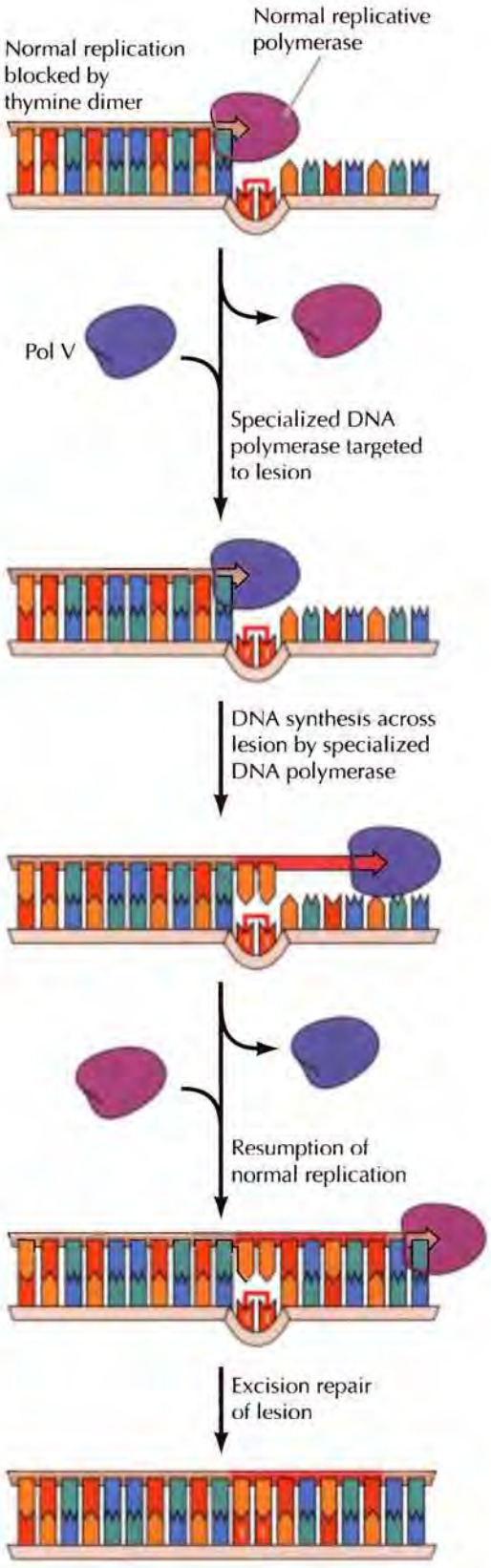


FIGURE 6.25 Translesion DNA synthesis Normal replication is blocked by a thymine dimer, but specialized DNA polymerases such as polymerase V (pol V) recognize and continue DNA synthesis across the lesion. Replication can then be resumed by the normal replicative DNA polymerase, and the thymine dimer subsequently removed by nucleotide-excision repair.

blocked at the sites of such damage. However, cells also possess several specialized DNA polymerases that are capable of replicating across a site of DNA damage. The replication of damaged DNA by these specialized polymerases, called **translesion DNA synthesis**, provides a mechanism by which the cell can bypass DNA damage at the replication fork, which can then be corrected after replication is complete.

The first specialized DNA polymerase responsible for translesion DNA synthesis was discovered in *E. coli* in 1999. This enzyme, called polymerase V, is induced in response to extensive UV irradiation and can synthesize a new DNA strand across from a thymine dimer (Figure 6.25). Two other specialized *E. coli* DNA polymerases, polymerases II and IV, are similarly induced by DNA damage and function in translesion synthesis. Eukaryotic cells also contain multiple specialized DNA polymerases, with ten such enzymes having been identified to date in humans. All of these specialized DNA polymerases exhibit low fidelity when copying undamaged DNA, with error rates ranging from 100 to 10,000 times higher than the error rates of the normal replicative DNA polymerases (e.g., polymerase III in *E. coli* or polymerases δ and ε in eukaryotes). In addition, the specialized polymerases lack the 3′–> 5′ proofreading activity that is characteristic of normal replicative DNA polymerases (see Figure 6.11).

Importantly, however, the specialized polymerases involved in translesion synthesis have evolved to insert the correct base opposite specific lesions in damaged DNA, and are therefore able to accurately synthesize a new strand using some forms of damaged DNA as a template. For example, *E. coli* polymerase V specifically recognizes thymine dimers and correctly inserts AA on the opposite strand. On the other hand, polymerase V makes a high frequency of errors when it synthesizes a new DNA strand opposite other forms of DNA damage. Thus, these enzymes are able to specifically insert correct bases opposite some forms of DNA damage, although they are "error-prone" in inserting bases opposite other forms of damaged DNA or in the synthesis of DNA from a normal undamaged template.

Recombinational Repair

Another means of DNA repair, **recombinational repair**, relies on replacement of the damaged DNA by recombination with an undamaged molecule. This mechanism is frequently used to repair damage encountered during DNA replication, where the presence of thymine dimers or other lesions that cannot be copied by the normal replicative DNA polymerases block the progress of a replication fork. Recombinational repair depends on the fact that one strand of the parental DNA was undamaged and therefore was copied during replication to yield a normal daughter molecule, which can then be used to repair the damaged strand.

The molecular mechanisms of recombinational repair are not entirely understood and may vary between different types of cells, but an illus-

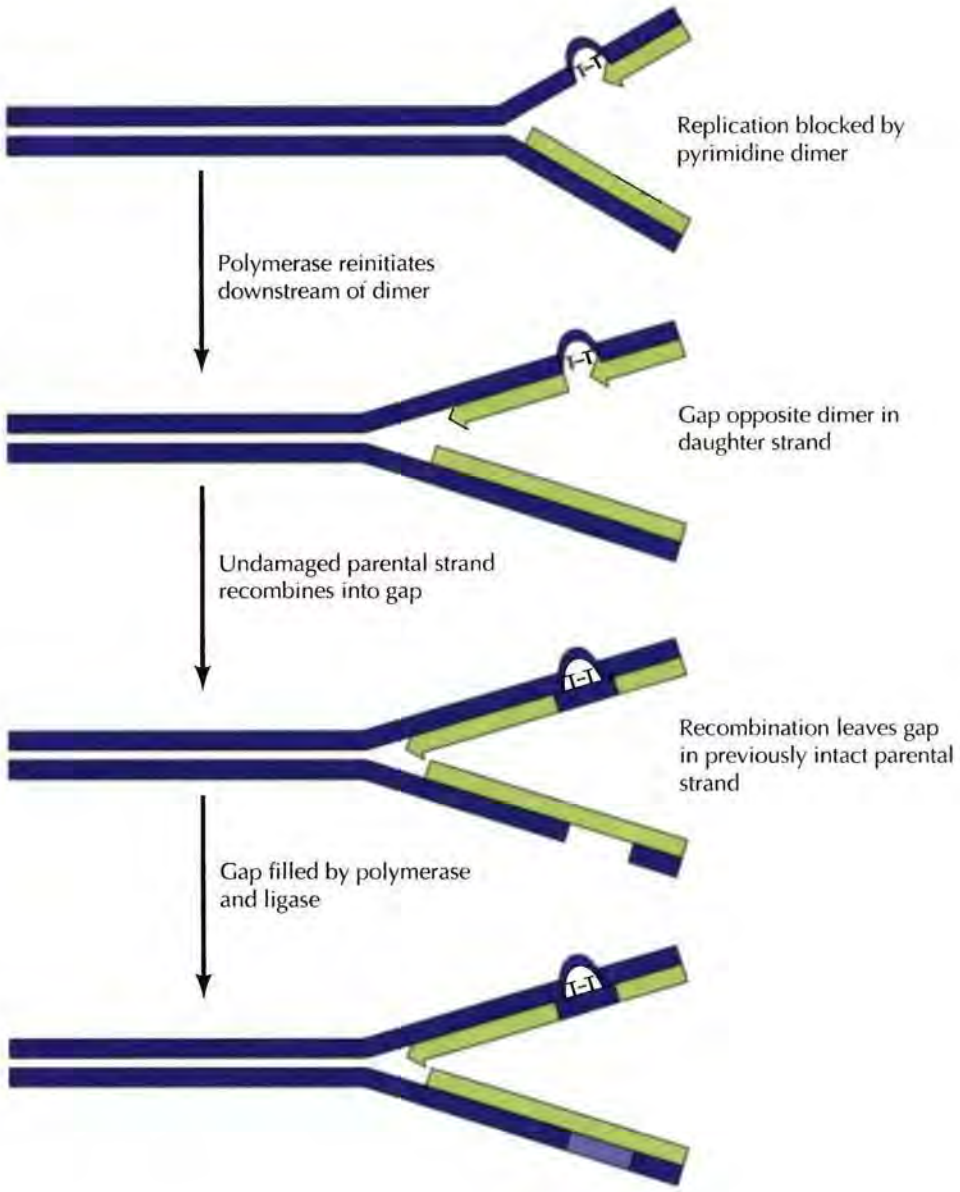


FIGURE 6.26 Recombinational repair The presence of a thymine dimer blocks replication, but DNA polymerase can bypass the lesion and reinitiate replication at a new site downstream of the dimer. The result is a gap opposite the dimer in the newly synthesized DNA strand. This gap is filled by recombination with the undamaged parental strand. Although this leaves a gap in the previously intact parental strand, the gap can be filled by the actions of polymerase and ligase, using the intact daughter strand as a template. Two intact DNA molecules are thus formed, and the remaining thymine dimer eventually can be removed by excision repair.

trative model is presented in Figure 6.26. In this example, normal replication is blocked by the presence of a thymine dimer in one strand of DNA. Downstream of the damaged site, however, replication can be initiated again by the synthesis of an Okazaki fragment and can proceed along the damaged template strand. The result is a daughter strand that has a gap opposite the site of damage to the parental strand. The undamaged parental strand, which has been replicated to yield a normal daughter molecule, can then be used to fill the gap opposite the site of damage by recombination between homologous DNA sequences (see the next section). Because the resulting gap in the previously intact parental strand is opposite an undamaged strand, it can be filled in by DNA polymerase. Although the other parent molecule still retains the original damage (e.g., a thymine dimer), the damage now lies opposite a normal strand and can be dealt with later by excision repair.

Recombinational repair also provides a major mechanism for repair of double strand breaks, which can be introduced into DNA by ionizing radiation (such as X-rays) and some chemicals (Figure 6.27). Since this type of damage affects both strands of DNA, it is particularly difficult to repair.

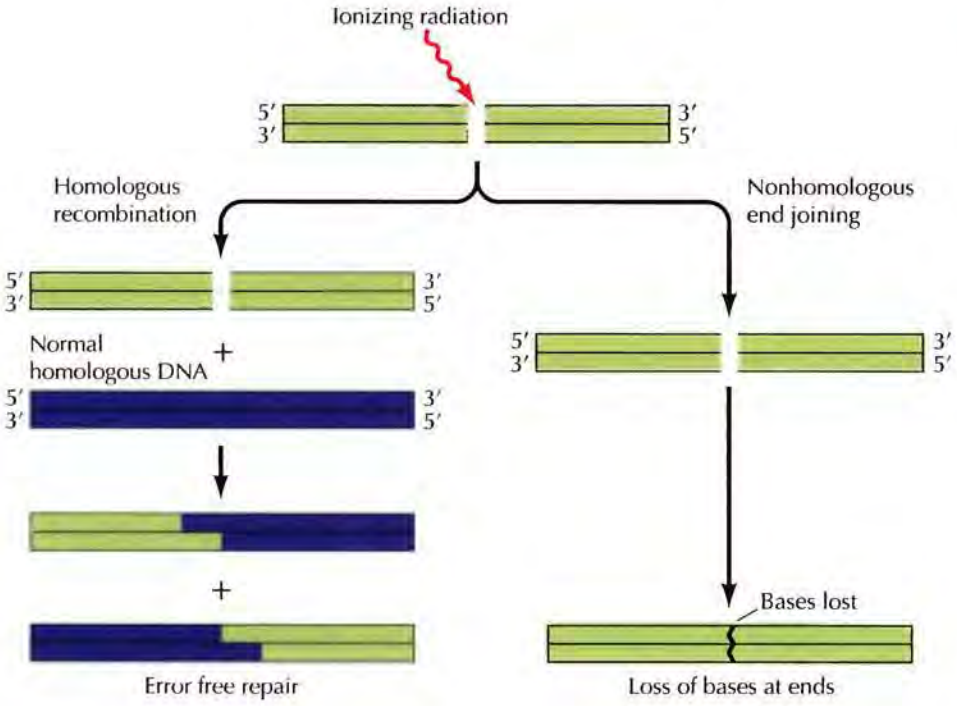


FIGURE 6.27 Repair of double strand breaks Ionizing radiation and some chemicals induce double strand breaks in DNA. These breaks can be repaired by homologous recombination with a normal chromosome, leading to restoration of the original DNA sequence. Alternatively, the ends of the broken molecule can be rejoined by nonhomologous end joining, with the frequent loss of bases around the site of damage.

Recombination with homologous DNA sequences on an undamaged chromosome (discussed in detail in the next section of this chapter) provides a mechanism for repairing such damage and restoring the normal DNA sequence. Alternatively, double strand breaks can be repaired simply by rejoining the broken ends of a single DNA molecule, but this leads to a high frequency of errors resulting from deletion of bases around the site of damage. It is noteworthy that the genes responsible for inherited breast cancer (*BRCA1* and *BRCA2*) encode proteins that are involved in the repair of double strand breaks by homologous recombination, suggesting that defects in this type of DNA repair can lead to the development of one of the most common cancers in women.

■ **DNA repair champion: The bacterium *Deinococcus radiodurans* was isolated in 1956 from cans of meat that were thought to have been sterilized by irradiation. The bacteria survived because of their extreme radiation resistance. *D. radiodurans* is over a hundredfold more resistant to ionizing radiation than *E. coli*.**

Recombination between Homologous DNA Sequences

Accurate DNA replication and repair of DNA damage are essential to maintaining genetic information and ensuring its accurate transmission from parent to offspring. As discussed in the previous section, recombination is an important mechanism for repairing damaged DNA. In addition, recombination is key to the generation of genetic diversity, which is critical from the standpoint of evolution. Genetic differences between individuals provide the essential starting material of natural selection, which allows species to evolve and adapt to changing environmental conditions. Recombination plays a central role in this process by allowing genes to be reassorted into different combinations. For example, genetic recombination results in the exchange of genes between paired homologous chromosomes during meiosis. In addition, recombination is involved in rearrangements of specific DNA sequences that alter the expression and function of some genes during development and differentiation. Thus, recombination plays important roles in the lives of individual cells and organisms, as well as contributing to the genetic diversity of the species.

This section discusses the molecular mechanism of **homologous recombination**, which involves the exchange of information between DNA molecules that share sequence homology over hundreds of bases. Examples include recombination between paired chromosomes during meiosis, as well as homologous recombination during DNA repair. Since this type of recombination involves the exchange of genetic information between two homologous DNA molecules, it does not alter the overall arrangement of the genes on a chromosome. Other types of recombination, however, do not require extensive sequence homology and therefore can occur between unrelated DNAs. Recombination events of this type lead to gene rearrangements, which are discussed later in the chapter.

Models of Homologous Recombination

Recombination results from the breakage and rejoining of two parental DNA molecules, leading to reassortment of the genetic information of the two parental chromosomes. During homologous recombination, this takes place without any other alteration in the genetic information. Thus, a critical question is: How can two parental DNA molecules be broken at precisely the same point, so that they can rejoin without mutations resulting from the gain or loss of nucleotides at the break point? During recombination between homologous DNA molecules, this alignment is provided, not surprisingly, by base pairing between complementary DNA strands (Figure 6.28). Overlapping single strands are exchanged between homologous DNA molecules, leading to the formation of a heteroduplex region, in which the two strands of the recombinant double helix are derived from different parents. If the heteroduplex region contains a genetic difference, the result is a single progeny DNA molecule that contains two genetic markers. In some cases, mispaired bases in a heteroduplex may be recognized and corrected by mismatch repair systems, as discussed in preceding sections of this chapter. Genetic evidence for the formation and repair of such heteroduplex regions, obtained in studies of recombination in fungi as well as in bacteria, led to the development of a molecular model for recombination in 1964. This model, known as the **Holliday model** (after Robin Holliday), has continued to provide the basis for current thinking about recombination mechanisms, although it has been modified as new data have been obtained.

The original version of the Holliday model proposed that recombination is initiated by the introduction of nicks at the same position on the two parental DNA molecules (Figure 6.29). The nicked DNA strands partially unwind, and each invades the other molecule by pairing with the complementary unbroken strand. Ligation of the broken strands then produces a crossed-strand intermediate, known as a **Holliday junction**, which is the central intermediate in recombination. The direct demonstration of Holliday junctions by electron microscopy has provided clear support for this model of recombination (Figure 6.30).

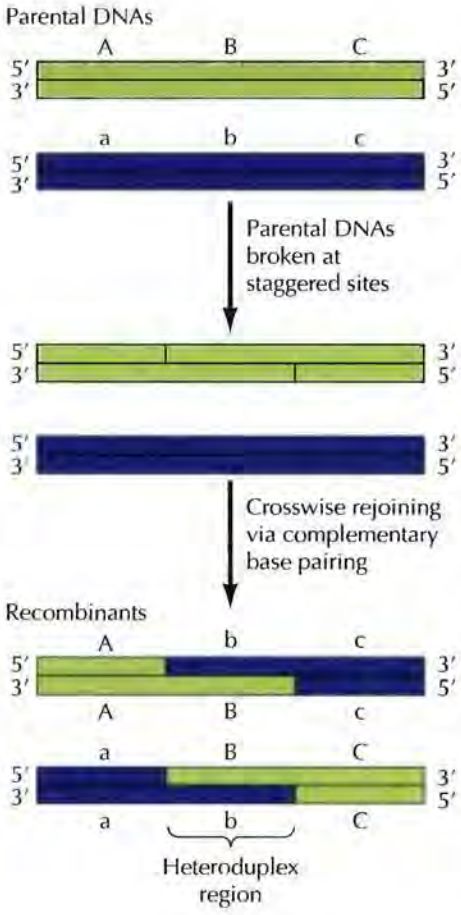


FIGURE 6.28 Homologous recombination by complementary base pairing
Parental DNAs are broken at staggered sites, and overlapping single-stranded regions are exchanged via base pairing with homologous sequences. The result is a heteroduplex region in which the two DNA strands are derived from different parental molecules.

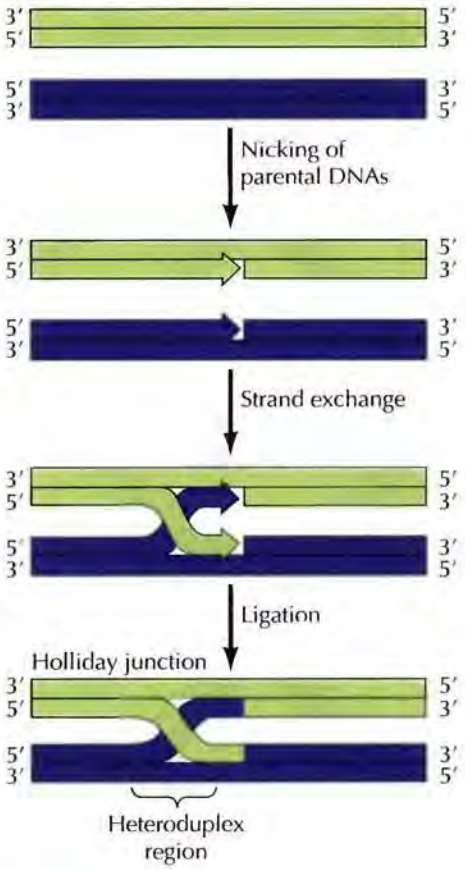


FIGURE 6.29 The Holliday model for homologous recombination Single-strand nicks are introduced at the same position on both parental molecules. The nicked strands then exchange by complementary base pairing, and ligation produces a crossed-strand intermediate called a Holliday junction.

Once a Holliday junction is formed, it can be resolved by cutting and rejoining of the crossed strands to yield recombinant molecules (Figure 6.31). This can occur in two different ways, depending on the orientation of the Holliday junction, which can readily form two different isomers. In the isomer resulting from the initial strand exchange, the crossed strands are those that were nicked at the start of the recombination process. However, simple rotation of this structure yields a different isomer in which the unbroken parental strands are crossed. Resolution of these different isomers has distinct genetic consequences. In the first case, the progeny molecules have heteroduplex regions but are nonrecombinant for DNA that flanks these regions. However, if isomerization occurs, cutting and rejoining of the crossed strands results in progeny molecules that are recombinant for DNA that flanks the heteroduplex regions. The structure of the Holliday junction thus provides the possibility of generating both recombinant and nonrecombinant heteroduplexes, consistent with the genetic data upon which the Holliday model was based.

As initially proposed, the Holliday model failed to explain how recombination was initiated by simultaneously nicking both parental molecules at the same position. It now appears that recombination is generally initiated at double strand breaks, both during DNA repair and during recombination between homologous chromosomes during meiosis (Figure 6.32). Both strands of DNA at the double strand break are first resected by nucleases that digest DNA in the 5′ to 3′ direction, yielding single-stranded ends. These single strands then invade the other parental molecule by homologous base pairing. The gaps are then filled by repair synthesis and the strands are joined by ligation to yield a molecule with a double Holliday junction, which can be resolved to yield either recombinant or nonrecombinant heteroduplex molecules as already described.

Enzymes Involved in Homologous Recombination

Most of the enzymes currently known to be involved in recombination were first identified by analysis of recombination-defective mutants of *E. coli*. Such genetic analysis has established that recombination requires specific enzymes, in addition to proteins (such as DNA polymerase, ligase, and single-stranded DNA-binding proteins) that function in multiple aspects of DNA metabolism. The identification of genes required for efficient recombination in *E. coli* has been followed by considerable progress in eukaryotic cells, leading to the isolation and characterization of the proteins that catalyze the formation and resolution of Holliday junctions.

6.2 WEBSITE ANIMATION

Homologous Recombination In the Holliday model for recombination between two homologous double-stranded DNA molecules, the process begins by nicking one strand of each double-stranded molecule, followed by strand exchange and joining of the nicked strands on the opposite molecules.

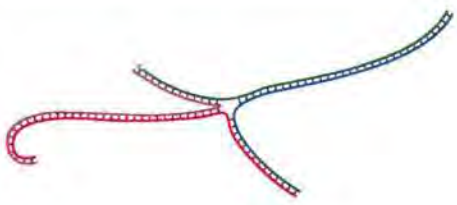


FIGURE 6.30 Identification of Holliday junctions by electron microscopy Electron micrograph of a Holliday junction that was detected during recombination of plasmid DNAs in *E. coli*. An interpretive drawing of the structure is shown below. The molecule illustrates a Holliday junction in the open configuration resulting from rotation of the crossed-strand intermediate (see Figure 6.31). (Courtesy of Huntington Potter, University of South Florida, and David Dressler, University of Oxford.)

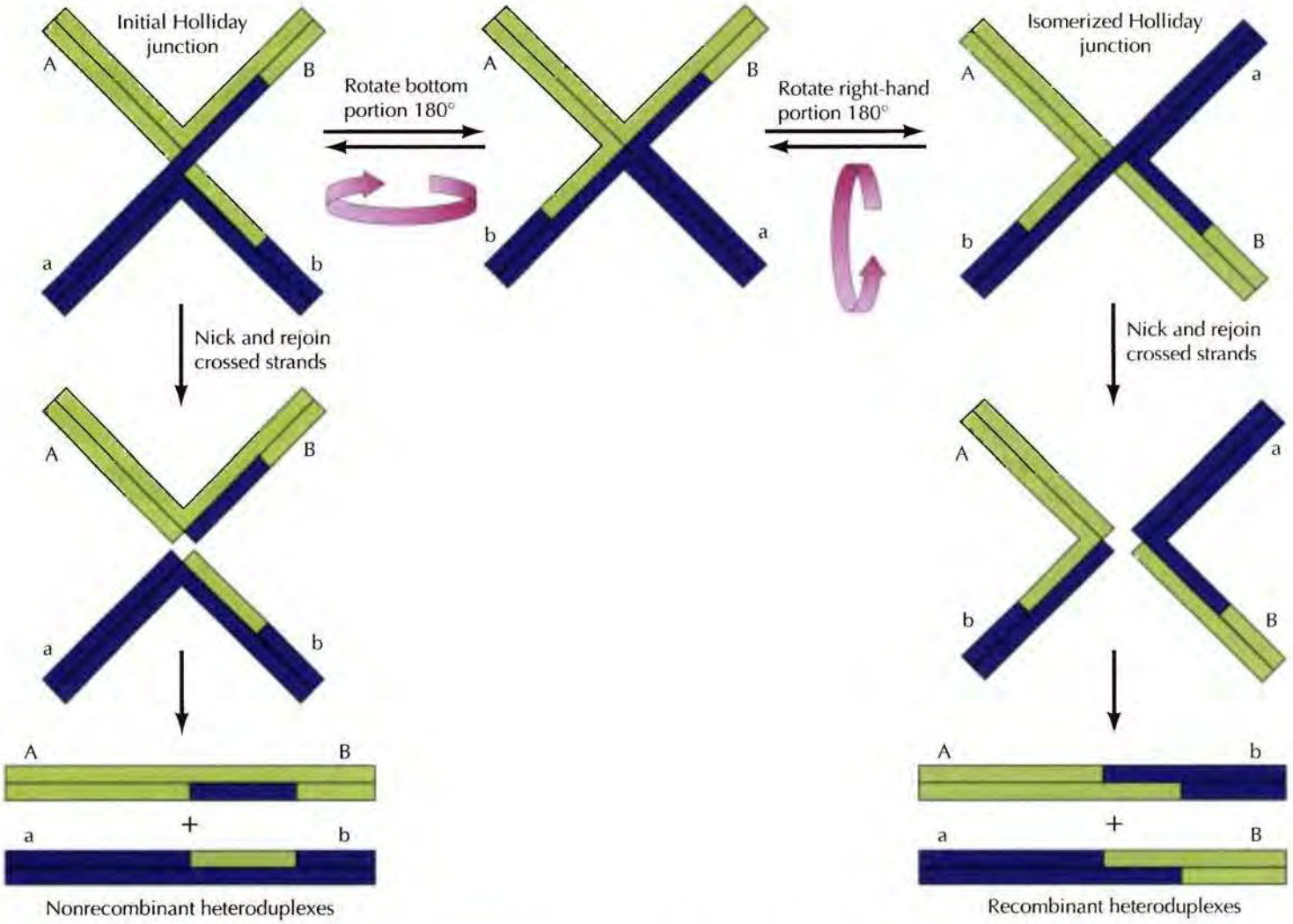


FIGURE 6.31 Isomerization and resolution of Holliday junctions Holliday junctions are resolved by cutting and rejoining of the crossed strands. If the Holliday junction formed by the initial strand exchange is resolved, the resulting progeny are heteroduplexes but are not recombinant for genetic markers outside of the heteroduplex region. Two rotations of the crossed-strand molecule, however, produce an isomer in which the unbroken parental strands, rather than the initially nicked strands, are crossed. Cutting and rejoining of the crossed strands of this isomer yield progeny that are recombinant heteroduplexes.

The key protein involved in the central steps of homologous recombination in *E. coli* is **RecA.** The RecA protein promotes the exchange of strands between homologous DNAs that causes heteroduplexes to form (Figure 6.33). The action of RecA can be considered in three stages. First, the RecA protein binds to single-stranded DNA, coating the DNA to form a protein-DNA filament. Because RecA has 3 distinct DNA binding sites, the RecA protein bound to single-stranded DNA is able to bind a second, double-stranded DNA molecule, forming a complex between the two DNAs. This nonspecific RecA-mediated association is followed by specific base pairing between the single-stranded DNA and its complement. The RecA protein then catalyzes strand exchange, with the single strand originally coated with RecA displacing its homologous strand to form a heteroduplex.

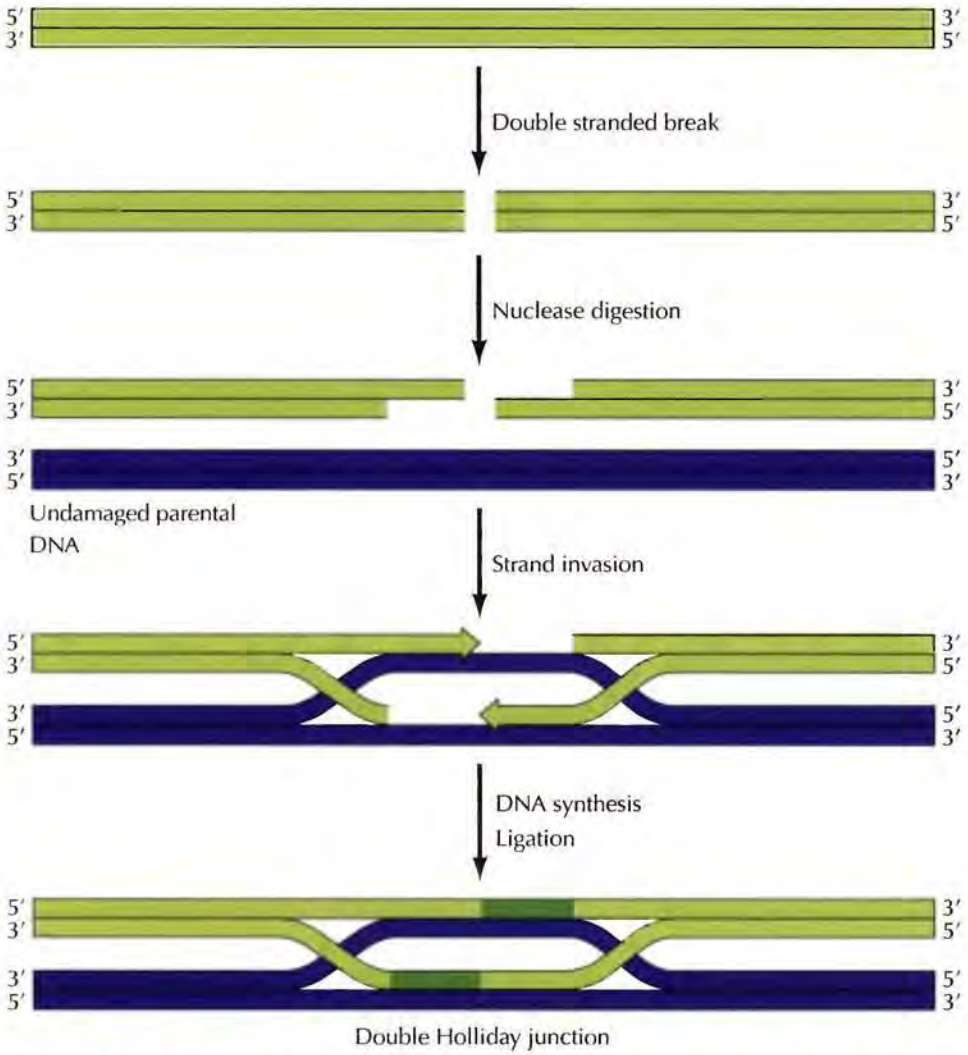


FIGURE 6.32 Initiation of recombination by double strand breaks Both strands of DNA at the double strand break are digested by nucleases in the 5′ to 3′ direction. The single strands then invade the other parental molecule by homologous base pairing. The gaps are then filled by repair synthesis and sealed by ligation, yielding a double Holliday junction.

In eukaryotic cells, a closely-related protein, **Rad51**, functions similarly to RecA. Rad51 was first identified in yeast, as a protein required for genetic recombination as well as for the repair of double strand breaks. Rad51 binds to single-stranded DNA to form protein-DNA filaments similar to those formed by RecA (Figure 6.34) and is also able to catalyze strand exchange reactions *in vitro*. Proteins related to Rad51 have been identified in complex eukaryotes, including humans, indicating that the RecA/Rad51 family of proteins plays key roles in homologous recombination in both prokaryotic and eukaryotic cells.

In *E. coli*, Holliday junctions are resolved by a complex of three proteins, RuvA, RuvB, and RuvC (Figure 6.35). RuvA recognizes the Holliday junction and recruits RuvB. RuvA and RuvB then act as a motor to drive migration of the site at which the DNA strands are crossed, thereby varying the extent of the heteroduplex region and the position at which the crossed strands will be cut and rejoined. RuvC then resolves Holliday junctions by cleaving the

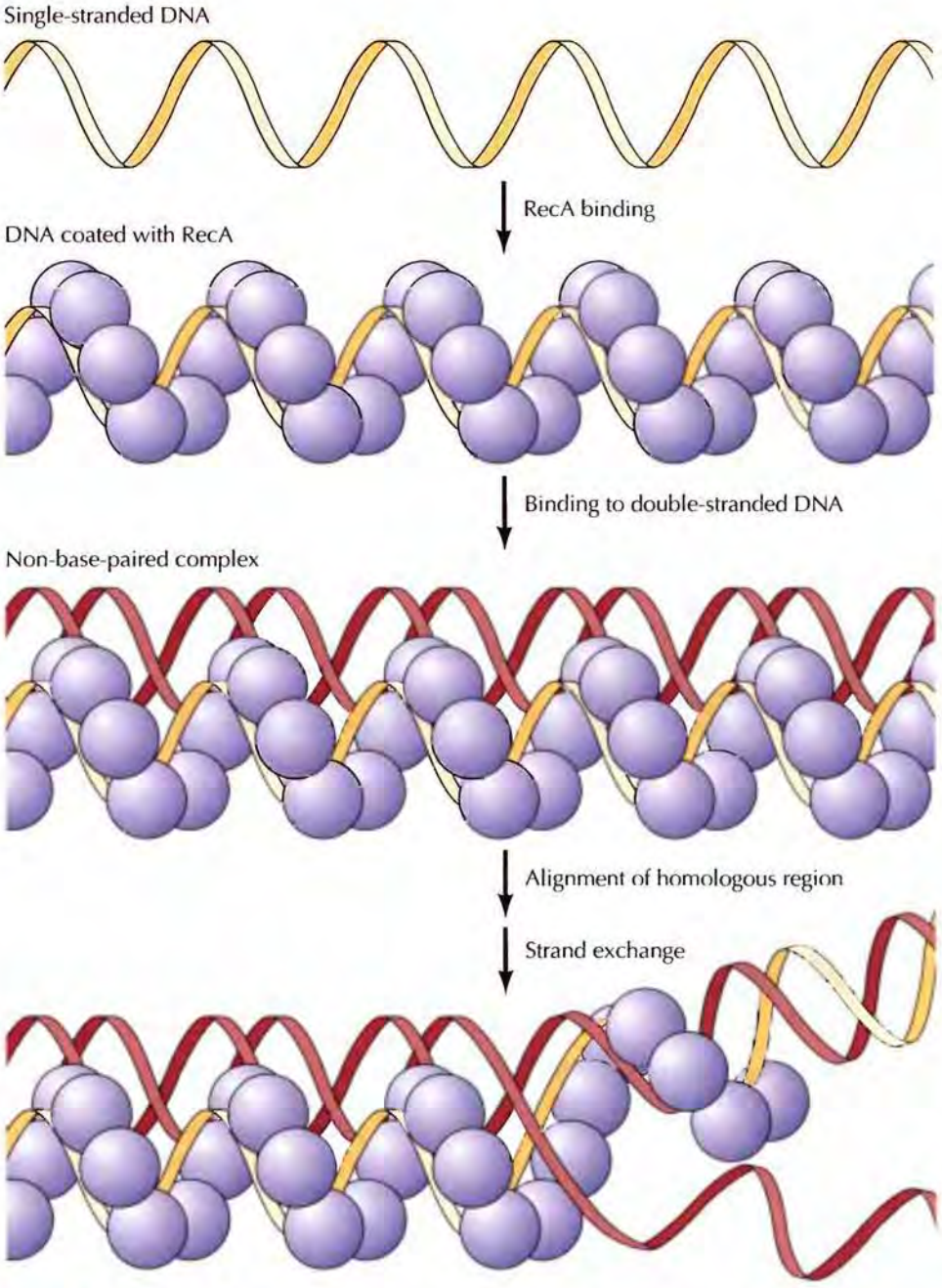


FIGURE 6.33 Function of the RecA protein RecA initially binds to single-stranded DNA to form a protein-DNA filament. The RecA protein that coats the single-stranded DNA then binds to a second, double-stranded DNA molecule to form a non-base-paired complex. Complementary base pairing and strand exchange follow, forming a heteroduplex region.

crossed DNA strands. Rejoining of the cleaved strands by ligation completes the process, yielding two recombinant molecules. Eukaryotic cells do not have homologs of the *E. coli* RuvA, RuvB, and RuvC proteins. Instead, the resolution of Holliday junctions in eukaryotic cells appears to be mediated by other proteins, which remain to be fully characterized. Recently, an endonuclease called Mus81 has been identified as a possible Holliday junction resolvase in yeast and mammalian cells, but the function of Mus81 and other potential resolvase activities still remain to be fully understood.

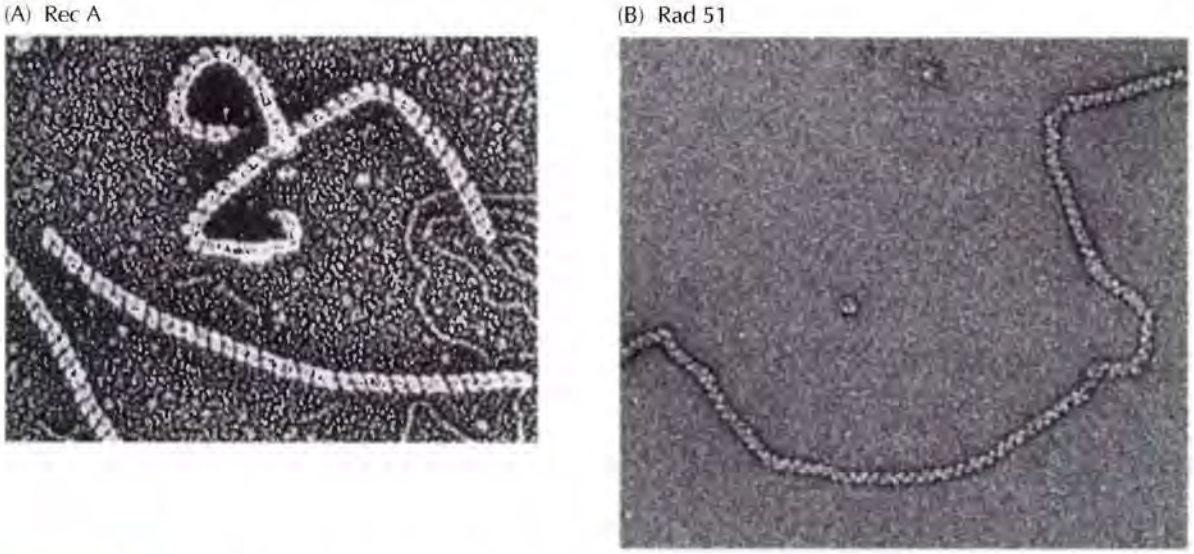


FIGURE 6.34 RecA and Rad51 protein-DNA filaments Electron micrographs of filaments formed by the binding of *E. coli* RecA and human Rad51 proteins to DNA. (From S. C. West 2003. *Nature Rev. Mol. Cell Biol.* 4: 1. Courtesy of A. Stasiak and S. West.)

DNA Rearrangements

Homologous recombination results in the reassortment of genes between chromosome pairs without altering the arrangement of genes within the genome. In contrast, other types of recombinational events lead to rearrangements of genomic DNA. Some of these DNA rearrangements are important in controlling gene expression in specific cell types; others may play an evolutionary role by contributing to genetic diversity.

The discovery that genes can move to different chromosomal locations came from Barbara McClintock's studies of corn in the 1940s. Purely on the basis of genetic analysis, McClintock described novel genetic elements that could move to different locations in the genome and alter the expression of adjacent genes. Nearly three decades elapsed, however, before the physical basis of McClintock's work was elucidated by the discovery of transposable elements in bacteria and the notion of movable genetic elements became widely accepted by scientists. Several types of DNA rearrangements, including the transposition of elements initially described by McClintock, are now recognized in both prokaryotic and eukaryotic cells. Moreover, we now know that transposable elements constitute a large fraction of the genomes of plants and animals, including nearly half of the human genome.

Site-Specific Recombination

In contrast to general homologous recombination, which occurs at any extensive region of sequence homology, **site-specific recombination** occurs between specific DNA sequences, which are usually homologous over only a short stretch of DNA. The principal interaction in this process is mediated by proteins that recognize the specific DNA target sequences rather than by complementary base pairing. Site-specific recombination thus leads to programmed DNA rearrangements that can play important roles in development and the regulation of gene expression.

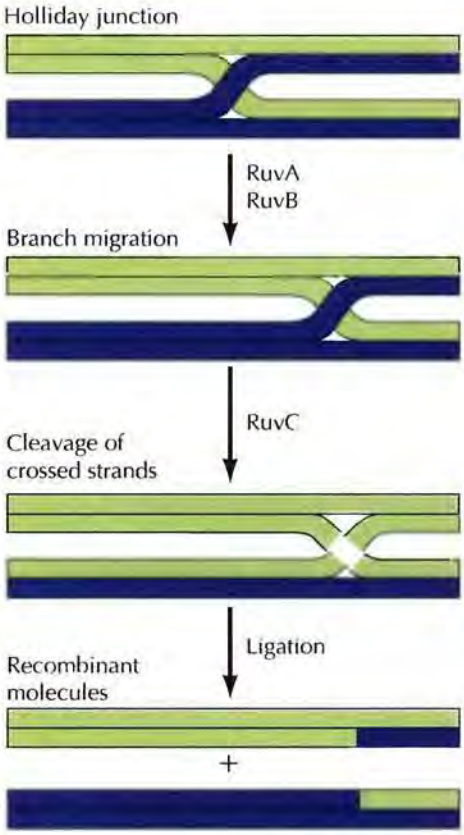


FIGURE 6.35 Branch migration and resolution of Holliday junctions RuvA recognizes the Holliday junction and recruits RuvB, which catalyzes the movement of the crossed-strand site (branch migration). RuvC resolves the Holliday junctions by cleaving the crossed strands, which are then joined by ligase.

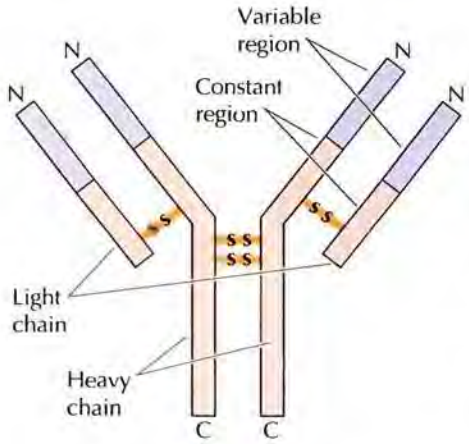


FIGURE 6.36 Structure of an immunoglobulin Immunoglobulins are composed of two heavy chains and two light chains, joined by disulfide bonds. Both the heavy and the light chains consist of variable and constant regions.

Development of the vertebrate immune system, which recognizes foreign substances (**antigens**) and provides protection against infectious agents, is a prime example of site-specific recombination in higher eukaryotes. There are two major classes of immune responses, which are mediated by B and T lymphocytes. B lymphocytes secrete antibodies (**immunoglobulins**) that react with soluble antigens; T lymphocytes express cell surface proteins (called **T cell receptors**) that react with antigens expressed on the surfaces of other cells. The key feature of both immunoglobulins and T cell receptors is their enormous diversity, which enables different antibody or T cell receptor molecules to recognize a vast array of foreign antigens. For example, each individual is capable of producing more than 10^{11} different antibody molecules, which is far in excess of the total number of genes in mammalian genomes (20,000–25,000). Rather than being encoded in germline DNA, these diverse antibodies (and T cell receptors) are encoded by unique lymphocyte genes that are formed during development of the immune system as a result of site-specific recombination between distinct segments of immunoglobulin and T cell receptor genes.

The role of site-specific recombination in the formation of immunoglobulin genes was first demonstrated by Susumu Tonegawa in 1976. Immunoglobulins consist of pairs of identical heavy and light polypeptide chains (Figure 6.36). Both the heavy and light chains are composed of C-terminal constant regions and N-terminal variable regions. The variable regions, which have different amino acid sequences in different immunoglobulin molecules, are responsible for antigen binding, and it is the diversity of variable region amino acid sequences that allows different individual antibodies to recognize unique antigens. Although every individual is capable of producing a vast spectrum of different antibodies, each B lymphocyte produces only a single type of antibody. Tonegawa's key discovery was that each antibody is encoded by unique genes formed by site-specific recombination during B lymphocyte development. These gene rearrangements create different immunoglobulin genes in different individual B lymphocytes, so the population of approximately 10^{12} B lymphocytes in the human body includes cells capable of producing antibodies against a diverse array of foreign antigens.

6.3 WEBSITE ANIMATION

Light-Chain Genes

During B cell development, site-specific recombination joins regions of the immunoglobulin light-chain genes, leading to the production of unique immunoglobulin light chains.

The genes that encode immunoglobulin light chains consist of three regions: a V region that encodes the 95 to 96 N-terminal amino acids of the polypeptide variable region; a joining (J) region that encodes the 12 to 14 C-terminal amino acids of the polypeptide variable region; and a C region that encodes the polypeptide constant region (Figure 6.37). The major class of light-chain genes in the mouse is formed from combinations of approximately 250 V regions and 4 J regions with a single C region. Site-specific recombination during lymphocyte development leads to a gene rearrangement in which a single V region recombines with a single J region to generate a functional light-chain gene. Different V and J regions are rearranged in different B lymphocytes, so the possible combinations of 250 V regions with 4 J regions can generate approximately 1000 (4×250) unique light chains.

6.4 WEBSITE ANIMATION

Heavy-Chain Genes

During B cell development, site-specific recombination joins regions of the immunoglobulin heavy-chain genes, leading to the production of unique immunoglobulin heavy chains.

The heavy-chain genes include a fourth region (known as the diversity, or D, region), which encodes amino acids lying between V and J (Figure 6.38). Assembly of a functional heavy-chain gene requires two recombination events: A D region first recombines with a J region, and a V region then recombines with the rearranged DJ segment. In the mouse, there are about 500 heavy-chain V regions, 12 D regions, and 4 J regions, so the total number of heavy chains that can be generated by the recombination events is 24,000 ($500 \times 12 \times 4$).

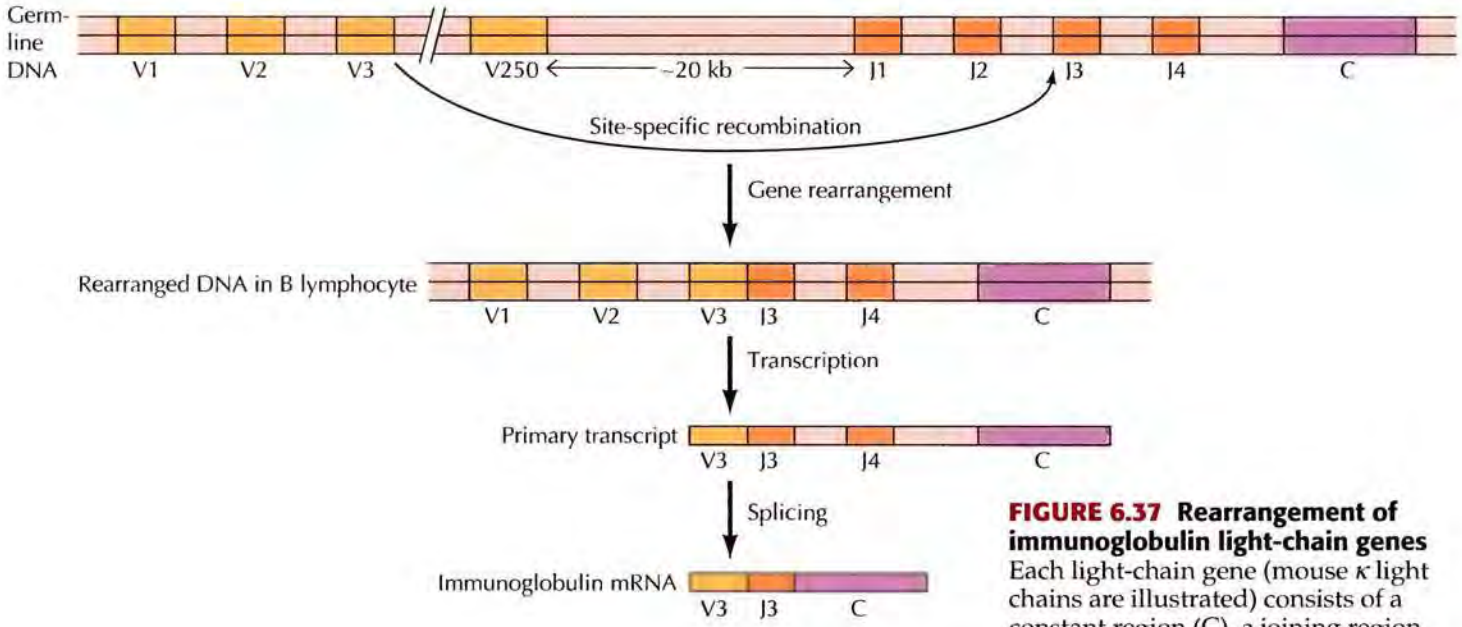


FIGURE 6.37 Rearrangement of immunoglobulin light-chain genes Each light-chain gene (mouse κ light chains are illustrated) consists of a constant region (C), a joining region (J), and a variable region (V). There are approximately 250 different V regions, which are separated from J and C regions by about 20 kb in germ-line DNA. During the development of B lymphocytes, site-specific recombination joins one of the V regions to one of the four J regions. This rearrangement activates transcription, resulting in the formation of a primary transcript containing the rearranged VJ region together with the remaining J regions and C region. The remaining unused J regions and the introns between J and C regions are then removed by splicing, yielding a functional mRNA.

Combinations between the 1000 different light chains and 24,000 different heavy chains formed by site-specific recombination can generate approximately 2×10^7 different immunoglobulin molecules. This diversity is further increased because the joining of immunoglobulin gene segments often involves the loss or gain of one to several nucleotides, as discussed below. The mutations resulting from these deletions and insertions increase the diversity of immunoglobulin variable regions approximately a hundredfold, corresponding to the formation of about 10^5 different light chains and 2×10^6 heavy chains, which can then combine to form more than 10^{11} distinct antibodies.

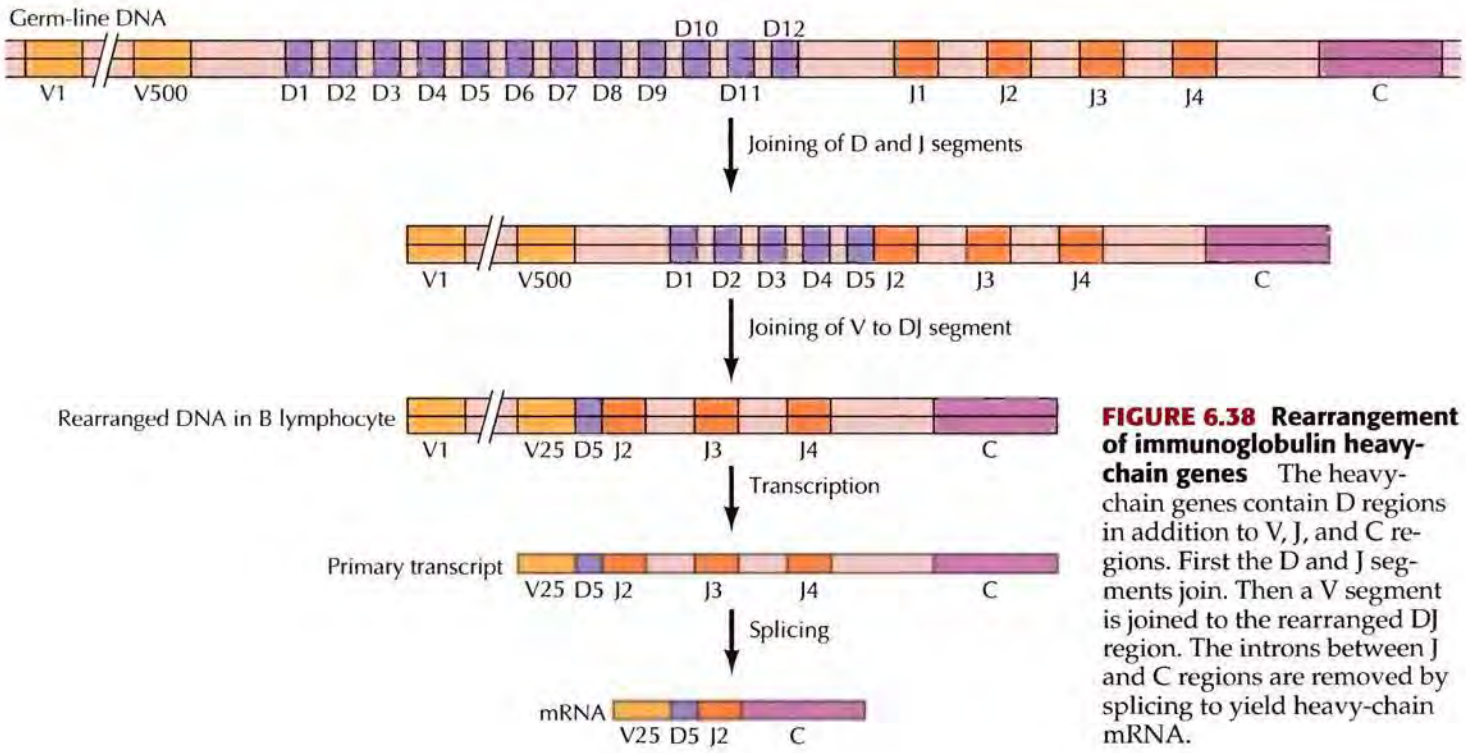


FIGURE 6.38 Rearrangement of immunoglobulin heavy-chain genes The heavy-chain genes contain D regions in addition to V, J, and C regions. First the D and J segments join. Then a V segment is joined to the rearranged DJ region. The introns between J and C regions are removed by splicing to yield heavy-chain mRNA.

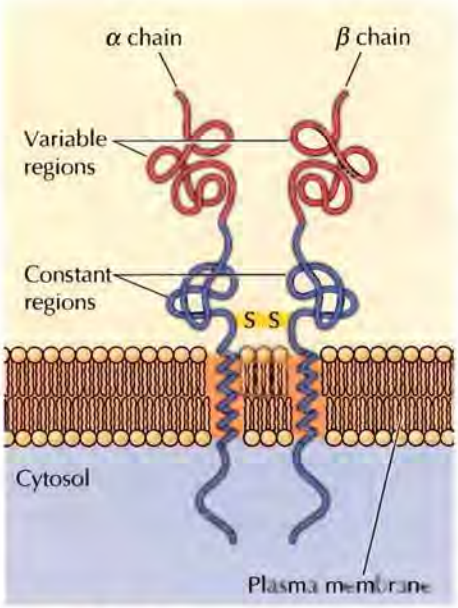


FIGURE 6.39 Structure of a T cell receptor T cell receptors consist of two polypeptide chains (α and β) that span the plasma membrane and are joined by disulfide bonds. Both the α and β chains are composed of variable and constant regions.

T cell receptors similarly consist of two chains (called α and β), each of which contains variable and constant regions (Figure 6.39). The genes encoding these polypeptides are generated by recombination between V and J segments (the α chain) or between V, D, and J segments (the β chain), analogous to the formation of immunoglobulin genes. Site-specific recombination between these distinct segments of DNA, in combination with mutations introduced during recombination, generates a degree of diversity in T cell receptors that is similar to that in immunoglobulins.

V(D)J recombination is mediated by a complex of two proteins, called RAG1 and RAG2, which are specifically expressed in lymphocytes. The RAG proteins recognize recombination signal (RS) sequences adjacent to the coding sequences of each gene segment, and initiate DNA rearrangements by introducing a double strand break between the RS sequences and the coding sequences (Figure 6.40). The coding ends of the gene segments are then joined to yield a rearranged immunoglobulin or T cell receptor gene. Since these double strand breaks are joined by a nonhomologous end joining process (see Figure 6.27), the joining reaction is accompanied by the frequent loss of nucleotides. In addition, lymphocytes contain a specialized enzyme (terminal deoxynucleotide transferase) that adds nucleotides at random to the ends of DNA molecules, so mutations corresponding to both the loss and the gain of nucleotides are introduced during the joining reaction. As noted above, these mutations contribute substantially to the diversity of immunoglobulins and T cell receptors.

Still further antibody diversity is generated after the formation of rearranged immunoglobulin genes by two processes that occur only in B lymphocytes: class switch recombination and somatic hypermutation. **Class switch recombination** results in the association of rearranged V(D)J regions with different heavy chain constant regions, leading to the production of antibodies with distinct functional roles in the immune response. Mammals produce four different classes of immunoglobulins—IgM, IgG, IgE, and IgA—with heavy chains encoded by a variable region joined to the Cμ, Cγ, Cε, and Cα constant regions, respectively. In addition, there are several subclasses of IgG, which are encoded by different Cγ regions. The different classes of immunoglobulins are specialized to remove antigens in different ways. IgM activates complement (a group of serum proteins that destroy invading cells or viruses), so IgM antibodies are an effective first line of defense against bacterial or viral infections. IgG antibodies, the most abundant immunoglobulins in serum, not only activate complement but also bind receptors on phagocytic cells. In addition, IgG antibodies can cross the placenta from the maternal circulation, providing immune protection to the fetus. IgA antibodies are secreted into a variety of bodily fluids, such as nasal mucus and saliva, where they can bind and eliminate invading bacteria or viruses to prevent infection. IgA antibodies are also secreted into the milk of nursing mothers, so they provide immune protection to newborns. IgE antibodies are effective in protecting against parasitic infections, and are also the class of antibodies responsible for allergies.

The initial V(D)J rearrangement produces a variable region joined to Cμ, resulting in the production of IgM antibodies. Class switch recombination then transfers a rearranged variable region to a new downstream constant region, with deletion of the intervening DNA (Figure 6.41). Recombination occurs between highly repetitive sequences in switch (S) regions that are located immediately upstream of each C region. The switch regions are 2–10 kb in length and recombination can take place anywhere within these regions, so class switching is more properly referred to as a region specific

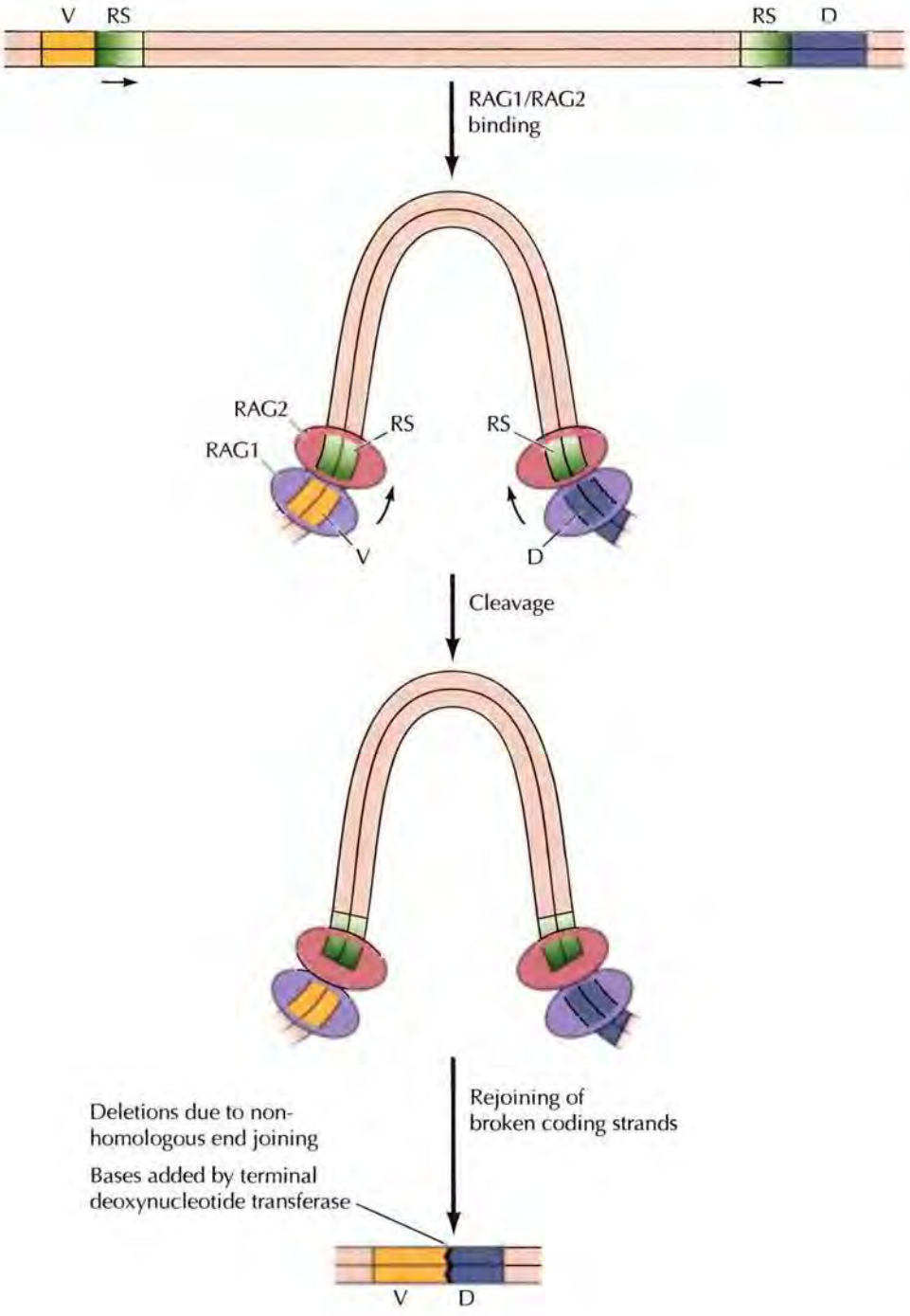


FIGURE 6.40 V(D)J recombination The coding segments of immunoglobulin and T cell receptor genes (e.g., a V and D segment) are flanked by short recombination signal (RS) sequences, which are in opposite orientations at the 5′ and 3′ ends of the coding sequences. The RS sequences are recognized by a complex of the lymphocyte-specific recombination proteins RAG1 and RAG2, which cleave the DNA between the coding sequence and the RS sequence. The broken coding strands are then rejoined to yield a rearranged gene segment. Mutations result from the loss of bases at the ends during nonhomologous end joining, as well as from the addition of bases by terminal deoxynucleotide transferase.

rather than a site specific recombination event. Because the switch regions are within introns, the precise site at which class switch recombination takes place does not affect the immunoglobulin coding sequence.

Somatic hypermutation increases the diversity of immunoglobulins by producing multiple mutations within rearranged variable regions of both heavy and light chains. These mutations, principally single base substitutions, occur with frequencies as high as 10^{-3}, approximately a million times higher than normal rates of spontaneous mutation. They lead to the production of immunoglobulins with a substantially increased affinity for antigen, and therefore are an important contributor to an effective immune response.

■ **Mutations in the genes for RAG1 and RAG2 can lead to severe combined immunodeficiency (SCID). These patients are born without a functional immune system and develop lethal infections if left untreated. Treatments include living in a germ free environment (in plastic "bubbles"), transplantation with stem cells giving rise to a functional immune system, and gene therapy.**

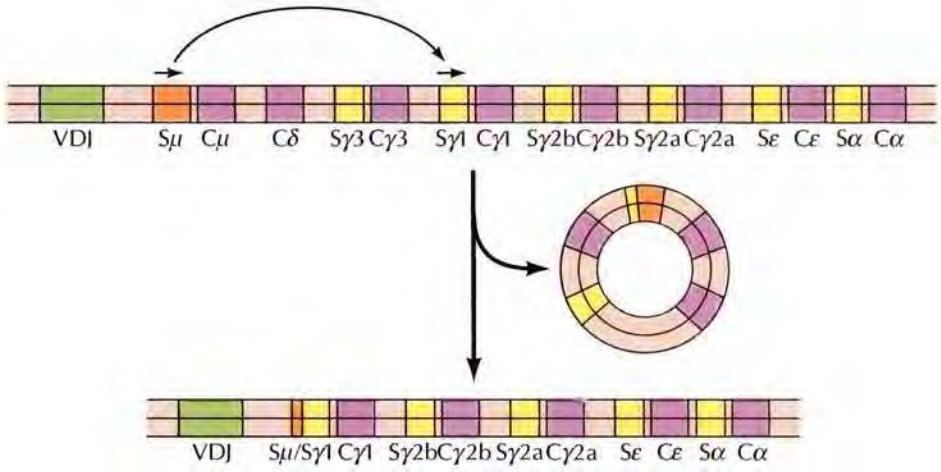


FIGURE 6.41 Class switch recombination Class switching takes place by recombination between repetitive switch (S) regions upstream of a series of constant (C) regions in the heavy chain locus (the mouse locus is shown). In the example shown, a V(D)J region is transferred from Cμ to Cγ1 by recombination between the Sμ and Sγ1 switch regions. The intervening DNA is excised as a circular molecule.

Class switch recombination and somatic hypermutation are novel types of programmed genetic alterations, and the molecular mechanisms involved are very active areas of investigation. A key player in both processes is an enzyme called **activation-induced deaminase (AID)**, which was discovered by Tasuku Honjo and his collaborators in 1999. AID is expressed only in B lymphocytes, and it is required for both class switch recombination and somatic hypermutation. AID catalyzes the deamination of cytosine in DNA to form uracil (Figure 6.42). Its action results in the conversion of C→U in the variable regions and switch regions of immunoglobulin genes, leading to class switch recombination and somatic hypermutation, respectively. The mechanisms by which C→U mutations stimulate these processes are not completely understood, but an important step appears to be the removal of U by base-excision repair (see Figure 6.21). This leads to the formation of single-strand breaks, and it is thought that the formation of multiple breaks in the switch regions (which have a high content of GC base pairs) results in class switch recombination. In the variable regions, somatic hypermutation is thought to result from a high frequency of errors during repair of the C→U mutations, possibly resulting from repair by specialized error-prone DNA polymerases (see Figure 6.25). Although the details of these processes remain to be elucidated, it is clear that AID is an extremely interesting enzyme with the novel role of introducing mutations into DNA at a specific stage of development.

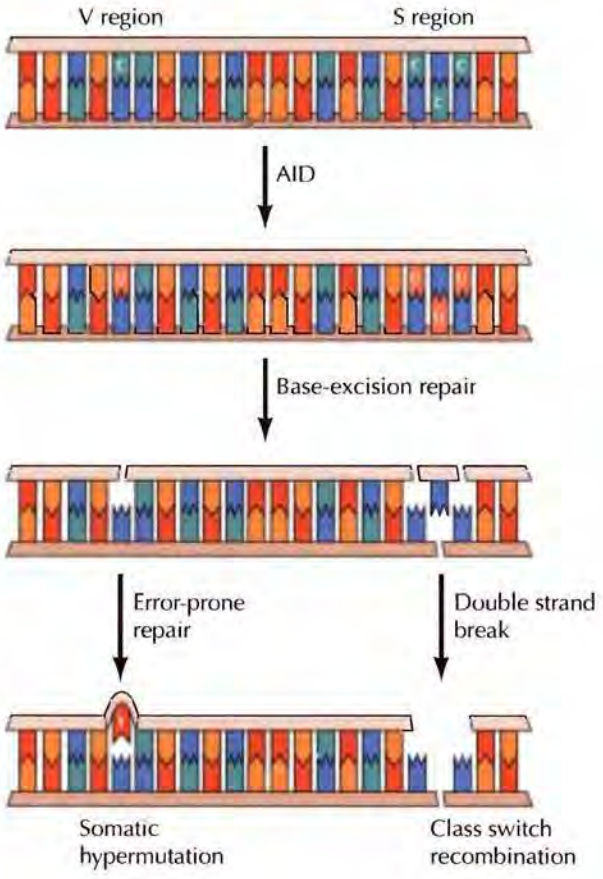


FIGURE 6.42 Model for the role of activation-induced deaminase (AID) in somatic hypermutation and class switch recombination AID deaminates C to U in DNA. Removal of U by base-excision repair (see Figure 6.21) leaves a single-strand gap in the DNA. In the variable regions, errors during repair may lead to somatic hypermutation, possibly resulting from the action of a specialized error-prone DNA polymerase (see Figure 6.25). In the switch regions, the presence of multiple sites of base excision may result in double strand breaks that stimulate class switch recombination.

Transposition via DNA Intermediates

Site-specific recombination occurs between two specific sequences that contain at least a small core of homology. In contrast, transposition involves the movement of sequences throughout the genome and has no requirement for sequence homology. Elements that move by transposition, such as those first described by McClintock, are called **transposable elements,** or **transposons**. They are divided into two general classes, depending on whether they transpose via DNA intermediates or via RNA intermediates. The first class of transposable elements is discussed here; transposition via RNA intermediates is considered in the next section.

The first transposons that were characterized in detail are those of bacteria, which move via DNA intermediates (Figure 6.43). The simplest of these elements are the insertion sequences, ranging in size from about 800 to 2000 nucleotides. Insertion sequences encode a gene for the enzyme involved in transposition (transposase) flanked by short inverted repeats, which are the sites at which transposase acts.

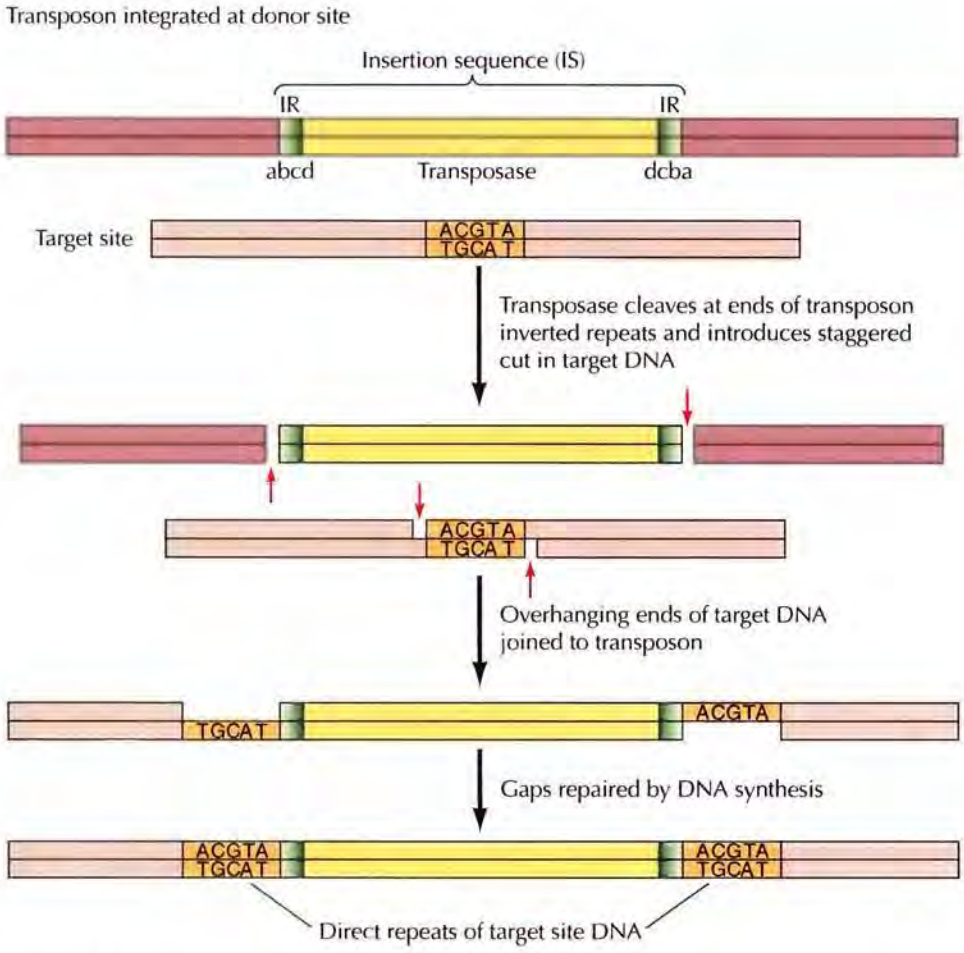


FIGURE 6.43 Bacterial transposons Insertion sequences (IS) range from 800 to 2000 nucleotides and contain a gene for transposase flanked by inverted repeats (IR) of about 20 nucleotides. Transposase cleaves at both ends of the transposon and introduces a staggered cut in the target DNA. The overhanging ends of target DNA are then joined to the transposon, and gaps resulting from the staggered cuts at the target site are repaired. The result is the formation of short direct repeats of target-site DNA (5 to 10 nucleotides long) flanking the integrated transposon.

KEY EXPERIMENT

Rearrangement of Immunoglobulin Genes

Evidence for Somatic Rearrangement of Immunoglobulin Genes Coding for Variable and Constant Regions

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Proceedings of the National Academy of Sciences, USA, Volume 73, 1976, pages 3628–3632

The Context

The ability of the vertebrate immune system to recognize a seemingly infinite variety of foreign molecules implies that lymphocytes can produce a correspondingly vast array of antibodies. Since this antibody diversity is key to immune recognition, understanding the mechanism by which an apparently unlimited number of distinct immunoglobulins are encoded in genomic DNA is a central issue in immunology.

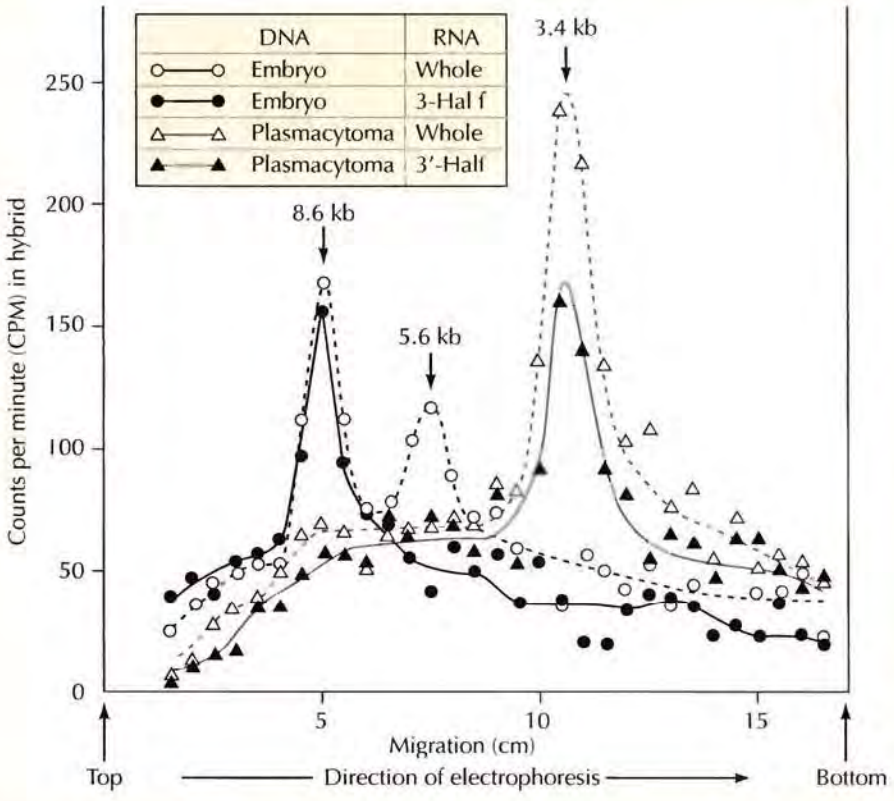
Prior to the experiments of Hozumi and Tonegawa, protein sequencing of multiple immunoglobulins had demonstrated that both heavy and light chains consist of distinct variable and constant regions. Genetic studies further indicated that mice inherit only single copies of the constant-region genes. These observations first led to the proposal that immunoglobulins are encoded by multiple variable-region genes that can associate with a single constant-region gene. The discovery of immunoglobulin gene rearrangements by Hozumi and Tonegawa provided the first direct experimental support for this hypothesis and laid the groundwork for understanding the molecular basis of antibody diversity.

The Experiments

Hozumi and Tonegawa tested the possibility that the genes encoding immunoglobulin variable and constant regions were joined at the DNA level during lymphocyte development. Their experimental approach was to use restriction endonuclease digestion to compare the organization of variable-region and constant-region sequences in DNAs extracted from mouse embryos and from cells of a mouse plasmacytoma (a B lymphocyte tumor that produces a single species of immunoglobulin).

Embryo and plasmacytoma DNAs were digested with the restriction endonuclease *Bam*HI, and DNA fragments of different sizes were separated by electrophoresis in an agarose gel. The gel was then cut into slices, and DNA extracted from each slice was hybridized with radiolabeled probes that had been prepared from immunoglobulin mRNA isolated from the plasmacytoma cells. Two probes were used, corresponding either to the complete immunoglobulin mRNA or to the 3′ half of the mRNA, consisting only of constant-region sequences.

The critical result was that completely different patterns of variable-region and constant-region sequences were detected in embryo versus plasmacytoma DNAs (see figure). In embryo DNA, the complete probe hybridized to two *Bam*HI fragments of approximately 8.6 and 5.6 kb, respectively. Only the 8.6-kb fragment hybridized to the 3′ probe, suggesting that the 8.6-kb fragment contained



Gel electrophoresis of embryo and plasmacytoma DNAs digested with BamHI and hybridized to probes corresponding to either the whole or the 3′ half of the plasmacytoma mRNA. Data are presented as the radioactivity detected in hybrid molecules with DNA from each gel slice.

KEY EXPERIMENT

constant-region sequences and the 5.6-kb fragment contained variable-region sequences. In striking contrast, both probes hybridized to only a single 3.4-kb fragment in plasmacytoma DNA. The interpretation of these results was that the variable- and constant-region sequences were separated in embryo DNA but rearranged to form a single immunoglobulin gene during lymphocyte development.

The Impact

The initial results of Hozumi and Tonegawa, based on the relatively indirect approach of restriction endonuclease mapping, were confirmed and extended by the molecular cloning and sequencing of immunoglobulin genes. Such studies have now unambiguously established that these genes are generated by site-specific recombination between distinct segments of DNA in B lymphocytes. In T lymphocytes, similar DNA rearrangements are responsible for formation of the genes encoding T cell receptors. Thus, site-specific recombination and programmed gene rearrangements are central to the development of the immune system.

Further studies have shown that the variable regions of immunoglobulins and T cell receptors are generated by rearrangements of two or three distinct segments of DNA. The ability of these segments to recombine, together with a high frequency of mutations introduced at the recombination sites, is largely responsible for immunoglobulin and T cell receptor diversity. The discovery of immunoglobulin gene rearrangements thus provided the basis for understanding how the immune system can recognize and respond to a virtually unlimited range of foreign substances.

Susumu Tonegawa

Insertion sequences move from one chromosomal site to another without replicating their DNA. Transposase introduces a staggered break in the target DNA and cleaves at the ends of the transposon inverted-repeat sequences. Although transposase acts specifically at the transposon inverted-repeats, it is usually less specific with respect to the sequence of the target DNA, so it catalyzes the movement of transposons throughout the genome. Following the cleavage of transposon and target site DNAs, transposase joins the overhanging ends of the target DNA to the transposable element. The resulting gap in the target-site DNA is repaired by DNA synthesis, followed by ligation to the other strand of the transposon. The result of this process is a short direct repeat of the target-site DNA on both sides of the transposable element—a hallmark of transposon integration.

This transposition mechanism causes the transposon to move from one chromosomal site to another. Other types of transposons move by a more complex mechanism, in which the transposon is replicated in concert with its integration into a new target site. This mechanism results in the integration of one copy of the transposon into a new position in the genome, while another copy remains at its original location.

Transposons that move via DNA intermediates are present in eukaryotes as well as in bacteria. For example, the human genome contains approximately 300,000 DNA transposons, which account for about 3% of human DNA. The original transposable elements described by McClintock in corn move by a nonreplicative mechanism, as do most transposable elements in other plants and animals. Like bacterial transposons, these elements move to many different target sites throughout the genome. The movement of these transposons to nonspecific sites in the genome is not likely to be useful to the cells in which it occurs, but has undoubtedly played a major role in evolution by promoting DNA rearrangements.

In yeasts and protozoans, however, transposition by a replicative mechanism is responsible for programmed DNA rearrangements that regulate gene expression. In these cases, transposition is initiated by the action of a site-specific nuclease that cleaves a specific target site, at which a copy of

the transposable element is then inserted. Transposable elements are thus capable not only of moving to nonspecific sites throughout the genome, but also of participating in specific gene rearrangements that result in programmed changes in gene expression.

Transposition via RNA Intermediates

Most transposons in eukaryotic cells are **retrotransposons**, which move via reverse transcription of RNA intermediates. In humans, there are almost 3 million copies of retrotransposons, accounting for more than 40% of the genome (see Table 5.2). The mechanism of transposition of these elements is similar to the replication of retroviruses, which have provided the prototype system for studying this class of movable DNA sequences.

Retroviruses contain RNA genomes in their virus particles but replicate via the synthesis of a DNA provirus, which is integrated into the chromosomal DNA of infected cells (see Figure 4.13). A DNA copy of the viral RNA is synthesized by the viral enzyme **reverse transcriptase**. The mechanism by which this occurs results in the synthesis of a DNA molecule that contains direct repeats of several hundred nucleotides at both ends (Figure 6.44). These repeated sequences, called **long terminal repeats**, or **LTRs**, arise from duplication of the sites on viral RNA at which primers bind to initiate DNA synthesis. The LTR sequences thus play central roles in reverse transcription, in addition to being involved in the integration and subsequent transcription of proviral DNA.

Like all DNA polymerases, reverse transcriptase requires a primer, which in the case of retroviruses, is a tRNA molecule bound at a specific site (the primer-binding site) close to the 5′ terminus of the viral RNA (Figure 6.45). Since DNA synthesis proceeds in the 5′ to 3′ direction, only a short piece of DNA is synthesized before reverse transcriptase reaches the end of its template. Continuation of DNA synthesis then depends on the ability of reverse transcriptase to "jump" to the 3′ end of the template RNA molecule. This is accomplished via an RNase H activity of reverse transcriptase, which degrades the RNA strand of DNA-RNA hybrids. As a result, the newly synthesized DNA is converted to a single-stranded molecule, which can hybridize to a short repeated sequence present at both the 5′ and the 3′ ends of the viral RNA. DNA synthesis can then continue, yielding a single-stranded DNA complementary to viral RNA. The viral RNA is then degraded, and synthesis of the opposite strand of DNA is initiated by a fragment of viral RNA that acts as a primer, at a site near the 3′ end of the template DNA strand. Again the result is a short piece of DNA, which includes the primer-binding site copied from the tRNA used as the initial primer for reverse transcription. The primer-binding sequence of the tRNA

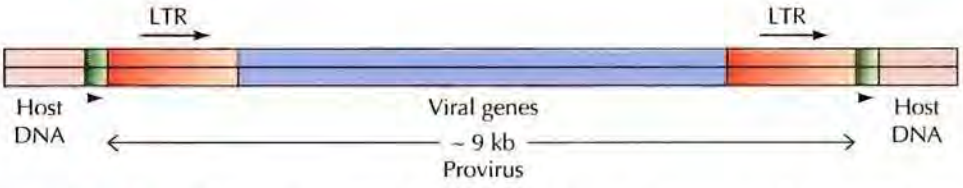


FIGURE 6.44 The organization of retroviral DNA The integrated proviral DNA is flanked by long terminal repeats (LTRs), which are direct repeats of several hundred nucleotides. Viral genes, including genes for reverse transcriptase, integrase, and structural proteins of the virus particle, are located between the LTRs. The integrated provirus is flanked by short direct repeats of host DNA.

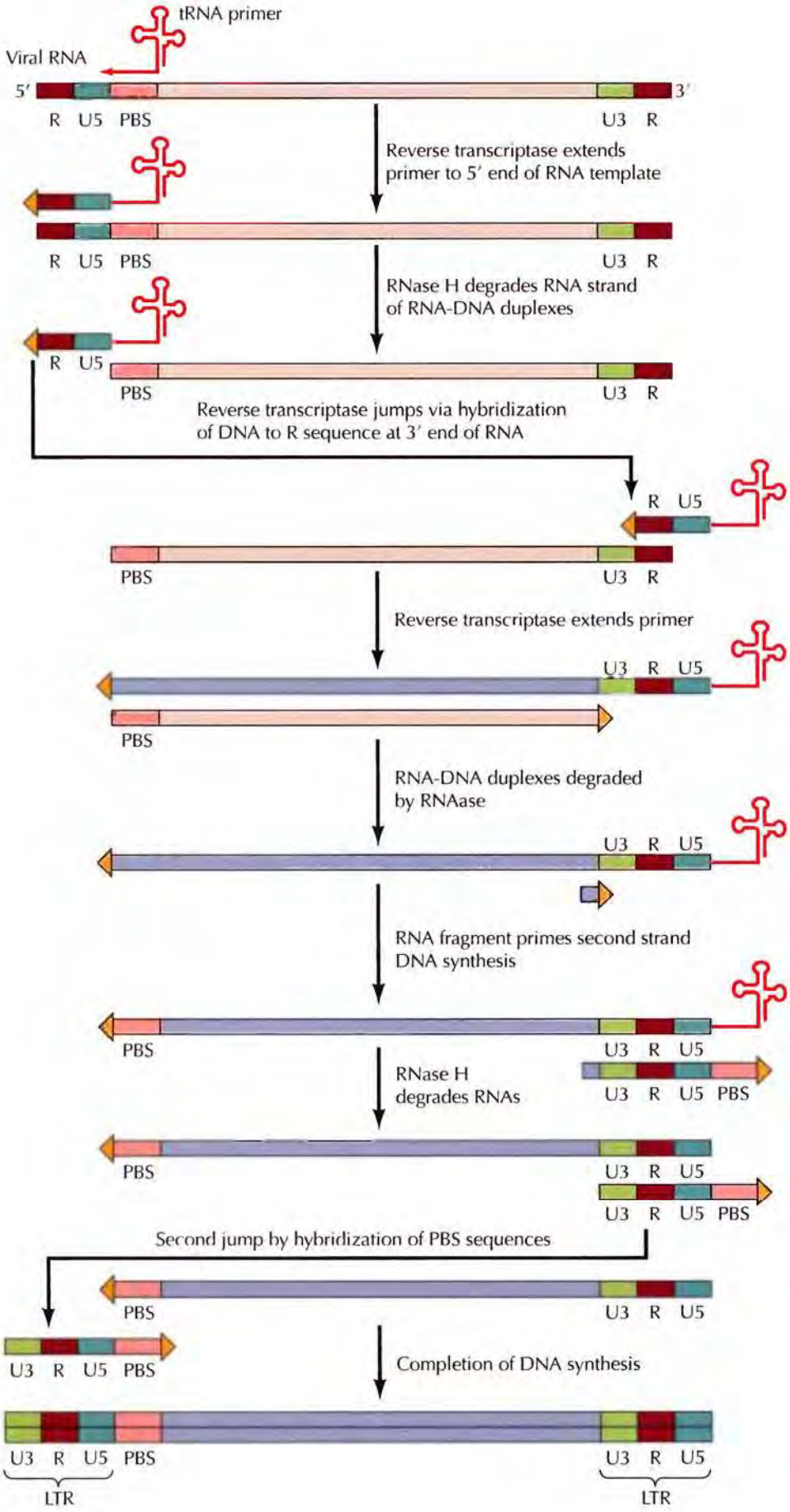


FIGURE 6.45 Generation of LTRs during reverse transcription LTRs consist of three sequence elements: a short repeat sequence (R) of about 20 nucleotides that is present at both ends of the viral RNA; a sequence unique to the 5′ end of viral RNA (U5); and a sequence unique to the 3′ end of viral RNA (U3). Repeats of these sequences are generated during DNA synthesis as reverse transcriptase jumps twice between the ends of its template. Synthesis is initiated using a tRNA primer bound to a primer-binding site (PBS) adjacent to U5 at the 5′ end of the viral RNA. The polymerase copies U5 and R, and the RNA strand of the RNA-DNA hybrid is then degraded by RNase H. The polymerase then jumps to the 3′ end of the viral RNA in order to synthesize a complete DNA strand complementary to the RNA template. The polymerase jumps again during synthesis of the second strand of DNA, which is also initiated by a primer bound close to the 5′ end of its template. The result of these jumps is the formation of LTRs that contain U3-R-U5 sequences.

is then degraded by RNase H, leaving an overhanging DNA strand that again "jumps" to pair with its complementary sequence at the other end of the template. DNA synthesis can then continue once more, finally yielding a linear DNA with LTRs at both ends.

The linear viral DNA integrates into the host cell chromosome by a process that resembles the integration of DNA transposable elements. Integration is catalyzed by a viral integrase and occurs at many different target sequences in cellular DNA. The integrase cleaves two bases from the ends of viral DNA and introduces a staggered cut at the target site in cellular DNA. The overhanging ends of cellular DNA are then joined to the termini of viral DNA, and the gap is filled by DNA synthesis. The integrated provirus is therefore flanked by a direct repeat of cell sequences, similar to the repeats that flank DNA transposons.

The viral life cycle continues with transcription of the integrated provirus, which yields viral genomic RNA as well as mRNAs that direct the synthesis of viral proteins (including reverse transcriptase and integrase). The genomic RNA is then packaged into viral particles, which are released from the host cell. These progeny viruses can infect a new cell, initiating another round of DNA synthesis and integration. The net effect can be viewed as the movement of the provirus from one chromosomal site to another, via the synthesis and reverse transcription of an RNA intermediate.

Other retrotransposons differ from retroviruses in that they are not packaged into infectious particles and therefore cannot spread from one cell to another. However, these retrotransposons can move to new chromosomal sites within the same cell, via mechanisms fundamentally similar to those involved in retrovirus replication.

Some retrotransposons (called retrovirus-like elements or LTR retrotransposons) are structurally similar to retroviruses (Figure 6.46). Retrotransposons of this type account for about 8% of the human genome. They have LTR sequences at both ends; they encode reverse transcriptase and integrase; and they transpose (like retroviruses) via transcription into RNA, synthesis of a new DNA copy by reverse transcriptase, and integration into cellular DNA.

The non-LTR retrotransposons differ from retroviruses in that they do not contain LTR sequences, although they do encode their own reverse transcriptase and an endonuclease involved in transposition. In mammals, the major class of these retrotransposons consists of the highly repetitive long interspersed elements (**LINEs**), which are repeated approximately 850,000 times in the genome and account for about 21% of genomic DNA (see Chapter 5). A full-length LINE element is about 6 kb long, although most members of the family are truncated at their 5′ end (Figure 6.47). At

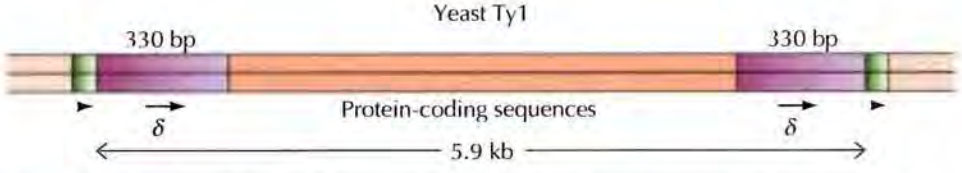


FIGURE 6.46 Structure of a LTR retrotransposon The yeast Ty1 transposable element displays the same organization as a retrovirus. Protein-coding sequences, including genes for reverse transcriptase and integrase, are flanked by LTRs (called δ elements) of about 330 base pairs (bp). The integrated transposon is flanked by short direct repeats of target-site DNA.

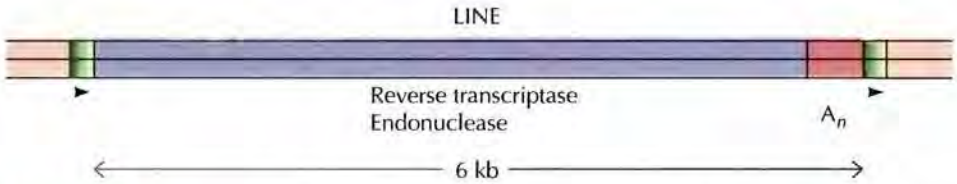


FIGURE 6.47 Structure of human LINEs LINEs lack LTRs, but they do encode reverse transcriptase and an endonuclease. They have tracts of A-rich sequences (designated A_n) at their 3′ ends, which are thought to arise from reverse transcription of poly-A tails added to the 3′ end of mRNAs. Like other transposable elements, LINEs are flanked by short direct repeats of target-site DNA.

their 3′ end, LINEs have tracts of A-rich sequences thought to be derived by reverse transcription of the poly-A tails that are added to mRNAs following transcription (see Chapter 7). Like other transposable elements, LINEs are flanked by short direct repeats of the target-site DNA, indicating that integration involves staggered cuts and repair synthesis.

Since LINEs do not contain LTR sequences, the mechanism of their reverse transcription and subsequent integration into chromosomal DNA must differ from that of retroviruses and LTR-containing retrotransposons. In particular, reverse transcription is primed by a broken end of chromosomal DNA at the integration target site, resulting from cleavage of the target site DNA by the endonuclease encoded by the retrotransposon (Figure 6.48). Reverse transcription then initiates within the poly-A tract at the 3′ end of the transposon RNA and continues along the molecule. The opposite strand

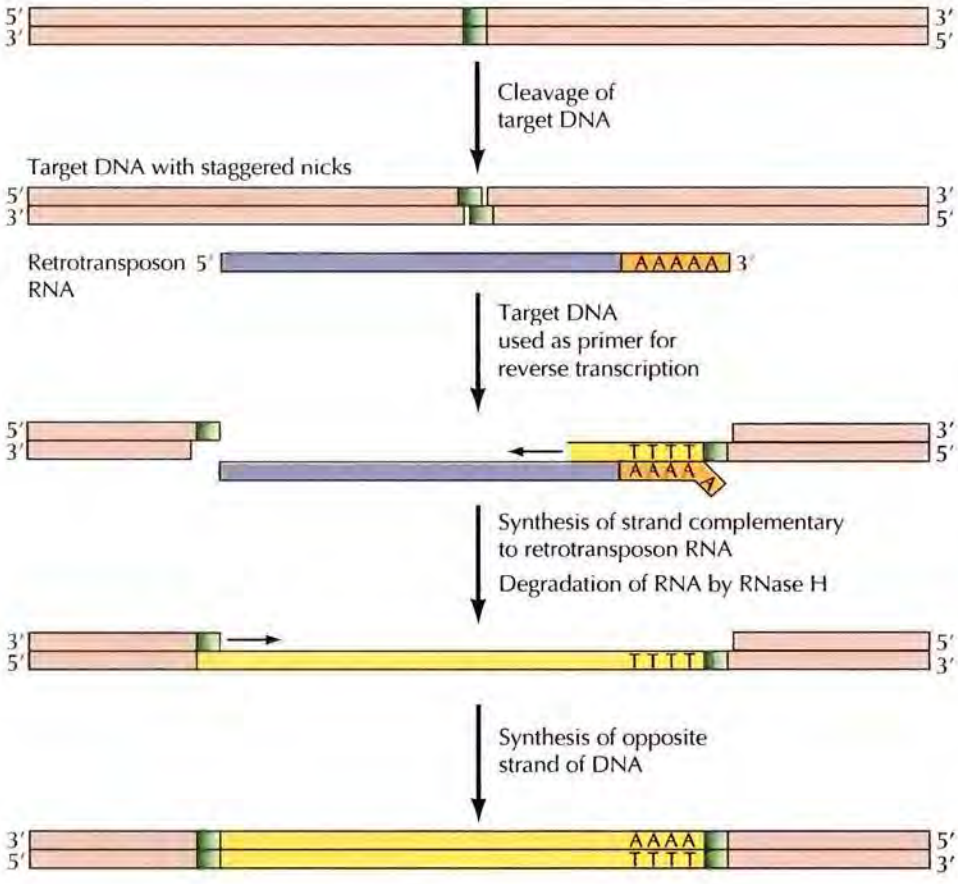


FIGURE 6.48 Model for reverse transcription and integration of LINEs Target site DNA is cleaved by an endonuclease encoded by the retrotransposon. Reverse transcription, primed by a broken end of the target DNA, initiates within the poly-A tail at the 3′ end of retrotransposon RNA. Synthesis of the opposite strand of retrotransposon DNA is similarly primed by the other strand of DNA at the target site.

of DNA is synthesized using the other broken end of the target-site DNA as a primer, resulting in simultaneous synthesis and integration of the retrotransposon DNA.

Other sequence elements, which do not encode their own reverse transcriptase, also transpose via RNA intermediates. These elements include the highly repetitive short interspersed elements (**SINEs**), of which there are approximately 1,500,000 copies in mammalian genomes (see Chapter 5). The major family of these elements in humans consists of the *Alu* sequences, which are about 300 bases long. These sequences have A-rich tracts at their 3′ end and are flanked by short duplications of target-site DNA sequences, a structure similar to that of non-LTR retrotransposons (e.g., LINEs). SINEs arose by reverse transcription of small RNAs, including tRNAs and small cytoplasmic RNAs involved in protein transport. Since SINEs no longer encode functional RNA products, they represent pseudogenes that arose via RNA-mediated transposition. Pseudogenes of many protein-coding genes (called **processed pseudogenes**) have similarly arisen by reverse transcription of mRNAs (Figure 6.49). Such processed pseudogenes are readily recognized not only because they terminate in an A-rich tract but also because the introns present in the corresponding normal gene have been removed during mRNA processing. The transposition of SINEs and of other processed pseudogenes is thought to proceed similarly to the transposition of LINEs. However, since these elements do not include genes for reverse transcriptase or a nuclease, their transposition presumably involves the action of reverse transcriptases and nucleases that are encoded elsewhere in the genome—probably by other retrotransposons, such as LINEs.

Although the highly repetitive SINEs and LINEs account for a significant fraction of genomic DNA, their transpositions to random sites in the genome are not likely to be useful for the cell in which they are located.

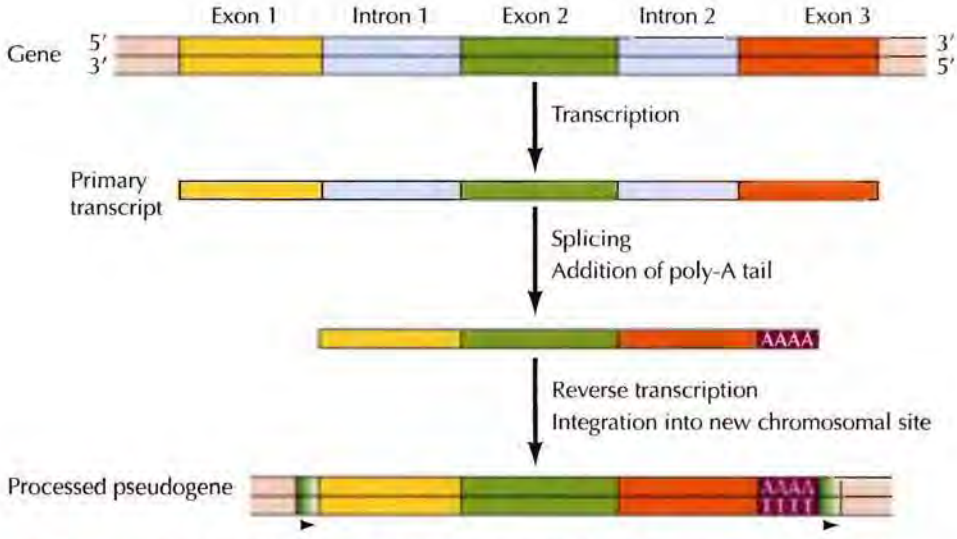


FIGURE 6.49 Formation of a processed pseudogene The gene illustrated contains three exons, separated by two introns. The introns are removed from the primary transcript by splicing, and a poly-A tail is added to the 3′ end of the mRNA. Reverse transcription and integration then yield a processed pseudogene, which does not contain introns and has an A-rich tract at its 3′ end. The processed pseudogene is flanked by short direct repeats of target-site DNA that were generated during its integration.

These transposons induce mutations when they integrate at a new target site, and like mutations induced by other agents, most mutations resulting from transposon integration are expected to be harmful to the cell. For example, mutations resulting from the transposition of both LINEs and SINEs have been associated with some cases of hemophilia, muscular dystrophy, breast cancer, and colon cancer. On the other hand, some mutations resulting from the movement of transposable elements may be beneficial, contributing in a positive way to evolution of the species. For example, some retrotransposons in mammalian genomes have been found to contain regulatory sequences that control the expression of adjacent genes.

In addition to their role as mutagens, retrotransposons have played a major role in shaping the genome by stimulating DNA rearrangements. For example, rearrangements of chromosomal DNA can result from recombination between LINEs integrated at different sites in the genome. Moreover, sequences of cellular DNA adjacent to LINEs are frequently carried along during the process of transposition. Consequently, the transposition of LINEs can result in the movement of cellular DNA sequences to new genomic sites. Since LINEs can integrate into active genes, the associated transposition of cellular DNA sequences can lead to the formation of new combinations of regulatory and/or coding sequences and contribute directly to the evolution of new genes.

The vast majority of the transposable elements in the human genome are inactive, with only about 100 copies of LINEs still retaining the protein-coding sequences required for their transposition. All of the human DNA transposons and most retrotransposons thus represent evolutionary relics rather than currently functional elements. However, this is not the case in other species, including *Arabidopsis*, *C. elegans*, *Drosophila*, and mice, which have a much higher level of ongoing transposon activity. In the mouse, for example, LTR retrotransposons, LINEs, and SINEs are all active. As a consequence, it is estimated that about 10% of all mutations in mice are the result of transposons, compared to only about 1 in 1000 mutations in humans. There is thus a dramatic and intriguing difference in transposon activity between mice and humans, the explanation for which remains to be determined.

Gene Amplification

The DNA rearrangements that have been discussed so far alter the position of a DNA sequence within the genome. **Gene amplification** may be viewed as a different type of alteration in genome structure; it increases the number of copies of a gene within a cell. Gene amplification results from repeated rounds of DNA replication, yielding multiple copies of a particular region (Figure 6.50). The amplified DNA sequences can be found either as free extrachromosomal molecules or as tandem arrays of sequences within a chromosome. In either case, the result is increased expression of the amplified gene, simply because more copies of the gene are available to be transcribed.

In some cases, gene amplification is responsible for developmentally programmed increases in gene expression. The prototypical example is amplification of the ribosomal RNA genes in amphibian oocytes (eggs). Eggs are extremely large cells, with correspondingly high requirements for protein synthesis. Amphibian oocytes in particular are about a million times larger in volume than typical somatic cells and must support large amounts of protein synthesis during early development. This requires increased synthesis of ribosomal RNAs, which is accomplished in part by amplification of the ribosomal RNA genes. As discussed in Chapter 5, there are already several hundred copies of ribosomal RNA genes per genome, so that

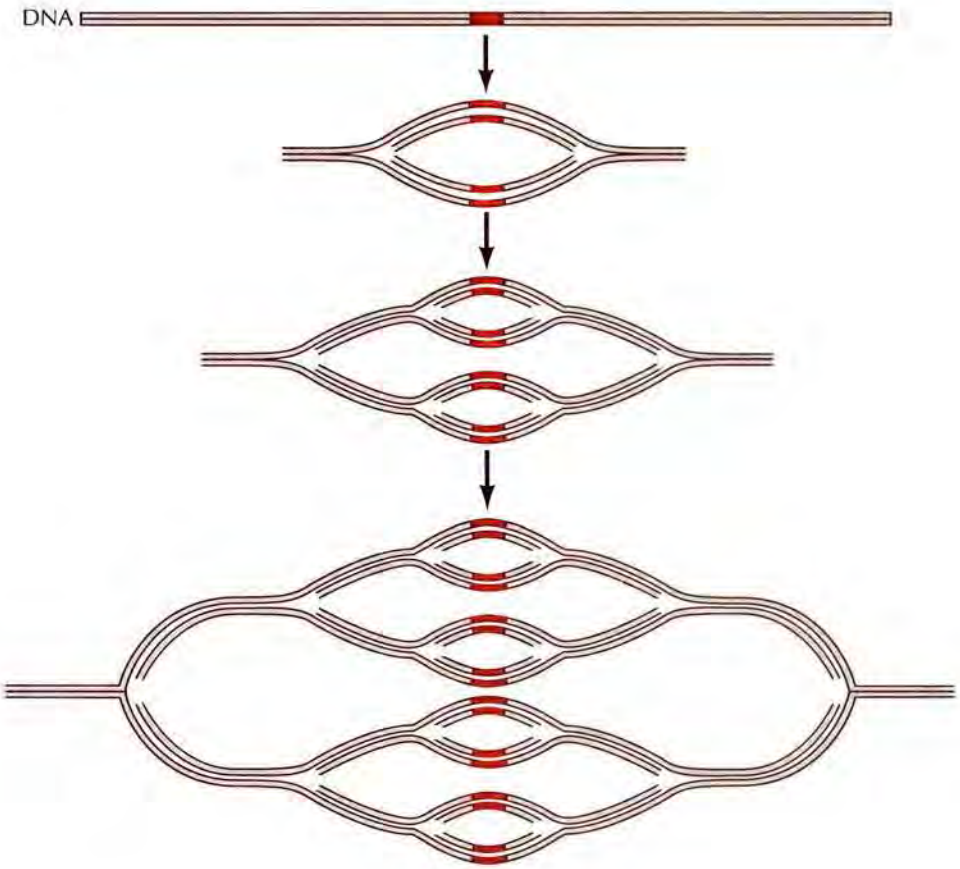


FIGURE 6.50 DNA amplification Repeated rounds of DNA replication yield multiple copies of a particular chromosomal region.

enough ribosomal RNA can be produced to meet the needs of somatic cells. In amphibian eggs, these genes are amplified an additional two thousandfold, to approximately 1 million copies per oocyte. Another example of programmed gene amplification occurs in *Drosophila*, where the genes that encode eggshell proteins (chorion genes) are amplified in ovarian cells to support the requirement for large amounts of these proteins. Like other programmed gene rearrangements, however, gene amplification is a relatively infrequent event that occurs in highly specialized cell types; it is not a common mechanism of gene regulation.

Gene amplification also occurs as an abnormal event in cancer cells, where it results in the increased expression of genes that contribute to uncontrolled cell growth. Such gene amplification was first recognized in cancer cells that had become resistant to methotrexate, a drug commonly used in cancer chemotherapy. Methotrexate inhibits the enzyme dihydrofolate reductase, which is involved in the synthesis of dTTP and is therefore required for DNA synthesis. Resistance to methotrexate frequently develops by amplification of the dihydrofolate reductase gene, leading to increased production of the enzyme and consequently the loss of effective inhibition by methotrexate. In addition, gene amplification in cancer cells frequently results in the increased expression of genes that drive cell proliferation (oncogenes) and thereby directly contributes to tumor development (see Chapter 18). For example, amplification of the oncogene *erb*B-2 is frequently involved in human breast cancers. Thus, as with other types of DNA rearrangements, gene amplification can have either beneficial or deleterious consequences for the cell or organism in which it occurs.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

DNA REPLICATION

DNA Polymerases: Different DNA polymerases play distinct roles in DNA replication and repair in both prokaryotic and eukaryotic cells. All known DNA polymerases synthesize DNA only in the 5′to 3′ direction by the addition of dNTPs to a preformed primer strand of DNA.

DNA polymerase

The Replication Fork: Parental strands of DNA separate and serve as templates for the synthesis of two new strands at the replication fork. One new DNA strand (the leading strand) is synthesized in a continuous manner; the other strand (the lagging strand) is formed by the joining of small fragments of DNA that are synthesized backward with respect to the overall direction of replication. DNA polymerases and various other proteins act in a coordinated manner to synthesize both leading and lagging strands of DNA.

replication fork, Okazaki fragment, DNA ligase, leading strand, lagging strand, primase, exonuclease, RNase H, helicase, single-stranded DNA-binding protein, topoisomerase

The Fidelity of Replication: DNA polymerases increase the accuracy of replication both by selecting the correct base for insertion and by proofreading newly synthesized DNA to eliminate mismatched bases.

proofreading

Origins and the Initiation of Replication: DNA replication starts at specific origins of replication, which contain binding sites for proteins that initiate the process. In higher eukaryotes, origins may be defined by chromatin structure rather than DNA sequence.

origin of replication, autonomously replicating sequence (ARS), origin recognition complex (ORC)

Telomeres and Telomerase: Maintaining the Ends of Chromosomes: Telomeric repeat sequences at the ends of chromosomes are maintained by the action of a reverse transcriptase (telomerase) that carries its own template RNA.

telomere, telomerase, reverse transcriptase

DNA REPAIR

Direct Reversal of DNA Damage: A few types of common DNA lesions, such as pyrimidine dimers and alkylated guanine residues, are repaired by direct reversal of the damage.

pyrimidine dimer, photoreactivation

Excision Repair: Most types of DNA damage are repaired by excision of the damaged DNA. The resulting gap is filled by newly synthesized DNA, using the undamaged complementary strand as a template. In base-excision repair, specific types of single damaged bases are removed from the DNA molecule. In contrast, nucleotide excision repair systems recognize a wide variety of lesions that distort the structure of DNA and remove the damaged bases as part of an oligonucleotide. A third excision repair system specifically removes mismatched bases from newly synthesized DNA strands.

base-excision repair, DNA glycosylase, AP endonuclease, nucleotide-excision repair, excinuclease, transcription-coupled repair, mismatch repair

Translesion DNA Synthesis: Specialized DNA polymerases are capable of replicating DNA across from a site of DNA damage, although the action of these polymerases may result in a high frequency of incorporation of incorrect bases.

translesion DNA synthesis

Recombinational Repair: Damaged DNA can be replaced by recombination with an undamaged molecule. This mechanism plays an important role in repairing damage encountered during DNA replication as well as in the repair of double strand breaks.

recombinational repair

KEY TERMS

KEY TERMS

homologous recombination, Holliday model, Holliday junction

RecA, Rad51

site-specific recombination, antigen, immunoglobulin, T cell receptor, class switch recombination, somatic hypermutation, activation-induced deaminase (AID)

transposable element, transposon

retrotransposon, retrovirus, reverse transcriptase, long terminal repeat (LTR), LINE, SINE, processed pseudogene

gene amplification

SUMMARY

RECOMBINATION BETWEEN HOMOLOGOUS DNA SEQUENCES

Models of Homologous Recombination: Recombination involves the breaking and rejoining of parental DNA molecules. Alignment between homologous DNA molecules is provided by complementary base pairing. Nicked strands of parental DNA invade the other parental molecule, yielding a crossed-strand intermediate known as a Holliday junction. Recombinant molecules are then formed by cleavage and rejoining of the crossed strands.

Enzymes Involved in Homologous Recombination: The central enzyme of homologous recombination is RecA (Rad51 in eukaryotes), which catalyzes the exchange of strands between homologous DNAs. Other enzymes nick and unwind parental DNAs and resolve Holliday junctions.

DNA REARRANGEMENTS

Site-Specific Recombination: Site-specific recombination takes place between specific DNA sequences that are recognized by proteins that mediate the process. In vertebrates, site-specific recombination plays a critical role in generating immunoglobulin and T cell receptor genes during development of the immune system. Additional diversity is provided to immunoglobulin genes by somatic hypermutation and class switch recombination.

Transposition via DNA Intermediates: Most DNA transposons move throughout the genome with no requirement for specific DNA sequences at their sites of insertion. In yeasts and protozoans, however, the transposition of some DNA sequences to specific target sites results in programmed DNA rearrangements that regulate gene expression.

Transposition via RNA Intermediates: Most transposons in eukaryotic cells move by reverse transcription of RNA intermediates, similar to the replication of retroviruses. These retrotransposons include the highly repeated LINE and SINE sequences of mammalian genomes.

Gene Amplification: Gene amplification results from repeated replication of a chromosomal region. In some cases, gene amplification provides a mechanism for increasing gene expression during development. Gene amplification also frequently occurs in cancer cells, where it can result in the elevated expression of genes that contribute to uncontrolled cell proliferation.

Questions

1. You have isolated a temperature sensitive strain of *E. coli* with a mutation in DNA polymerase I. What defects, if any, would you observe in bacteria carrying this mutation at high temperature?

2. Compare and contrast the actions of topoisomerases I and II.

3. What is the approximate number of Okazaki fragments synthesized during replication of the yeast genome?

4. Why is the synthesis of Okazaki fragments initiated by primase instead of a DNA polymerase?

5. What is the function of 3′ to 5′ exonuclease activity in DNA polymerases? What would be the consequence of mutating this activity of DNA polymerase III on the fidelity of replication of *E. coli* DNA?

6. How would you test a sequence of DNA in a yeast cell to determine whether it contains an origin of replication?

7. Yeast cells require telomerase to completely replicate their genome, whereas *E. coli* does not need this special enzyme. Why?

8. What mechanism does a cell use to repair double strand breaks in its DNA? How does this differ from the repair of single strand breaks?

9. Patients with xeroderma pigmentosum suffer an extremely high incidence of skin cancer but have not been found to have correspondingly high incidences of cancers of internal organs (e.g., colon cancer). What might this suggest about the kinds of DNA damage responsible for most internal cancers?

10. Many of the drugs in clinical use for the treatment of AIDS are inhibitors of the HIV reverse transcriptase. What other cellular processes might be affected by these inhibitors?

11. The mismatch repair system in humans includes homologs of *E. coli Mut S* and *Mut L* but not of *Mut H*. Why?

12. What phenotype would you predict for a mutant mouse lacking one of the genes required for site-specific recombination in lymphocytes?

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CHAPTER 7

RNA Synthesis and Processing

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- ***KEY EXPERIMENT:*** The Discovery of snRNPs 294

CHAPTERS 5 AND 6 DISCUSSED THE ORGANIZATION and maintenance of genomic DNA, which can be viewed as the set of genetic instructions governing all cellular activities. These instructions are implemented via the synthesis of RNAs and proteins. Importantly, the behavior of a cell is determined not only by what genes it inherits but also by which of those genes are expressed at any given time. Regulation of gene expression allows cells to adapt to changes in their environments and is responsible for the distinct activities of the multiple differentiated cell types that make up complex plants and animals. Muscle cells and liver cells, for example, contain the same genes; the functions of these cells are determined not by differences in their genomes, but by regulated patterns of gene expression that govern development and differentiation.

The first step in expression of a gene, the transcription of DNA into RNA, is the initial level at which gene expression is regulated in both prokaryotic and eukaryotic cells. RNAs in eukaryotic cells are then modified in various ways—for example, introns are removed by splicing—to convert the primary transcript into its functional form. Different types of RNA play distinct roles in cells: Messenger RNAs (mRNAs) serve as templates for protein synthesis; ribosomal RNAs (rRNAs) and transfer RNAs (tRNAs) function in mRNA translation. Still other small RNAs function in gene regulation, mRNA splicing, rRNA processing, and protein sorting in eukaryotes. In fact, some of the most exciting advances in recent years have pertained to the roles of noncoding RNAs (microRNAs) as regulators of both transcription and translation in eukaryotic cells. Transcription and RNA processing are discussed in this chapter. The final step in gene expression, the translation of mRNA to protein, is the subject of Chapter 8.

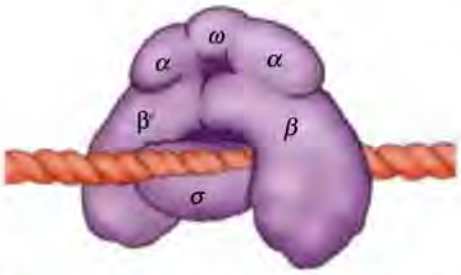


FIGURE 7.1 *E. coli* RNA polymerase The complete enzyme consists of six subunits: two α, one β, one β', one ω and one σ. The σ subunit is relatively weakly bound and can be dissociated from the other five subunits, which constitute the core polymerase.

Transcription in Prokaryotes

As in most areas of molecular biology, studies of *E. coli* have provided the model for subsequent investigations of transcription in eukaryotic cells. As reviewed in Chapter 4, mRNA was discovered first in *E. coli*. *E. coli* was also the first organism from which RNA polymerase was purified and studied. The basic mechanisms by which transcription is regulated were likewise elucidated by pioneering experiments in *E. coli* in which regulated gene expression allows the cell to respond to variations in the environment, such as changes in the availability of nutrients. An understanding of transcription in *E. coli* has thus provided the foundation for studies of the far more complex mechanisms that regulate gene expression in eukaryotic cells.

RNA Polymerase and Transcription

The principal enzyme responsible for RNA synthesis is **RNA polymerase**, which catalyzes the polymerization of ribonucleoside 5′-triphosphates (NTPs) as directed by a DNA template. The synthesis of RNA is similar to that of DNA, and like DNA polymerase, RNA polymerase catalyzes the growth of RNA chains always in the 5′ to 3′ direction. Unlike DNA polymerase, however, RNA polymerase does not require a preformed primer to initiate the synthesis of RNA. Instead, transcription initiates *de novo* at specific sites at the beginning of genes. The initiation process is particularly important because this is a major step at which transcription is regulated.

RNA polymerase, like DNA polymerase, is a complex enzyme made up of multiple polypeptide chains. The intact bacterial enzyme consists of five different types of subunits, called α, β, β', ω and σ (Figure 7.1). The σ subunit is relatively weakly bound and can be separated from the other subunits, yielding a core polymerase consisting of two α, one β, one β' and one ω subunits. The core polymerase is fully capable of catalyzing the polymerization of NTPs into RNA, indicating that σ is not required for the basic catalytic activity of the enzyme. However, the core polymerase does not bind specifically to the DNA sequences that signal the normal initiation of transcription; therefore the σ subunit is required to identify the correct sites for transcription initiation. The selection of these sites is a critical element of transcription because synthesis of a functional RNA must start at the beginning of a gene.

The DNA sequence to which RNA polymerase binds to initiate transcription of a gene is called the **promoter**. The DNA sequences involved in promoter function were first identified by comparisons of the nucleotide sequences of a series of different genes isolated from *E. coli*. These comparisons revealed that the region upstream of the transcription initiation site contains two sets of sequences that are similar in a variety of genes. These common sequences encompass six nucleotides each and are located approximately 10 and 35 base pairs upstream of the transcription start site (Figure 7.2). They are called the –10 and –35 elements, denoting their position relative to the transcription initiation site, which is defined as the +1 position. The sequences at the –10 and –35 positions in different promoters are not

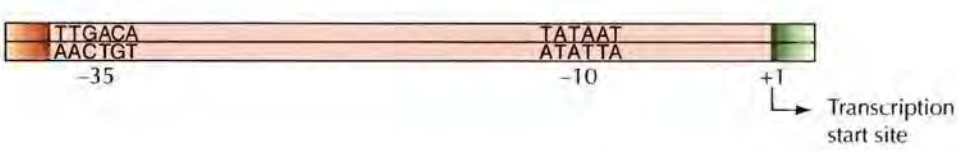


FIGURE 7.2 Sequences of *E. coli* promoters *E. coli* promoters are characterized by two sets of sequences located 10 and 35 base pairs upstream of the transcription start site (+1). The consensus sequences shown correspond to the bases most frequently found in different promoters.

identical, but they are all similar enough to establish consensus sequences—the bases most frequently found at each position.

Several types of experimental evidence support the functional importance of the –10 and –35 promoter elements. First, genes with promoters that differ from the consensus sequences are transcribed less efficiently than genes whose promoters match the consensus sequences more closely. Second, mutations introduced in either the –35 or –10 consensus sequences have strong effects on promoter function. Third, the sites at which RNA polymerase binds to promoters have been directly identified by **footprinting** experiments, which are widely used to determine the sites at which proteins bind to DNA (Figure 7.3). In experiments of this type, a DNA frag-

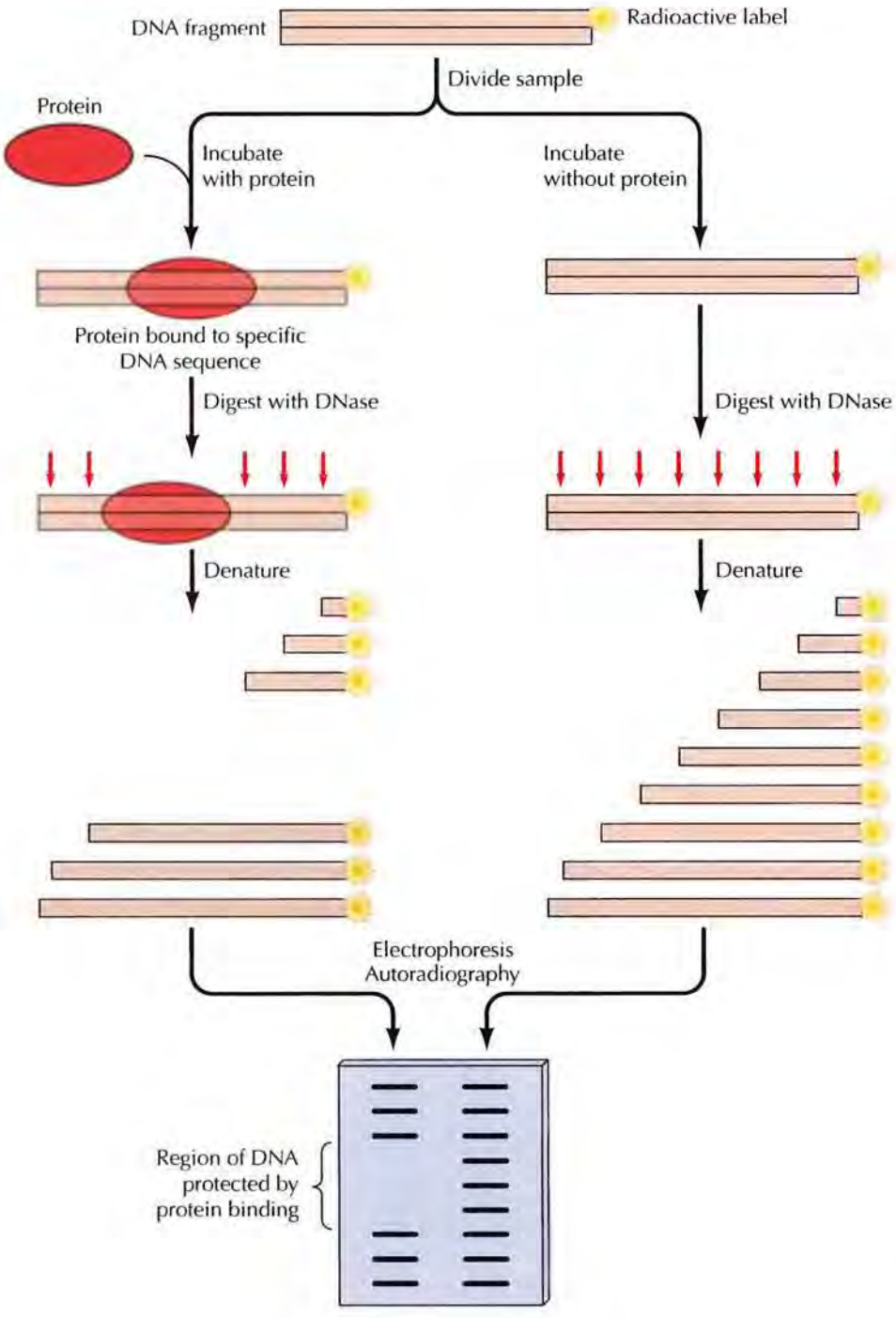


FIGURE 7.3 DNA footprinting A sample containing fragments of DNA radiolabeled at one end is divided in two, and one half of the sample is incubated with a protein that binds to a specific DNA sequence within the fragment. Both samples are then digested with DNase, under conditions such that the DNase introduces an average of one cut per molecule. The region of DNA bound to the protein is protected from DNase digestion. The DNA-protein complexes are then denatured, and the sizes of the radiolabeled DNA fragments produced by DNase digestion are analyzed by electrophoresis. Fragments of DNA resulting from DNase cleavage within the region protected by protein binding are missing from the sample of DNA that was incubated with protein.

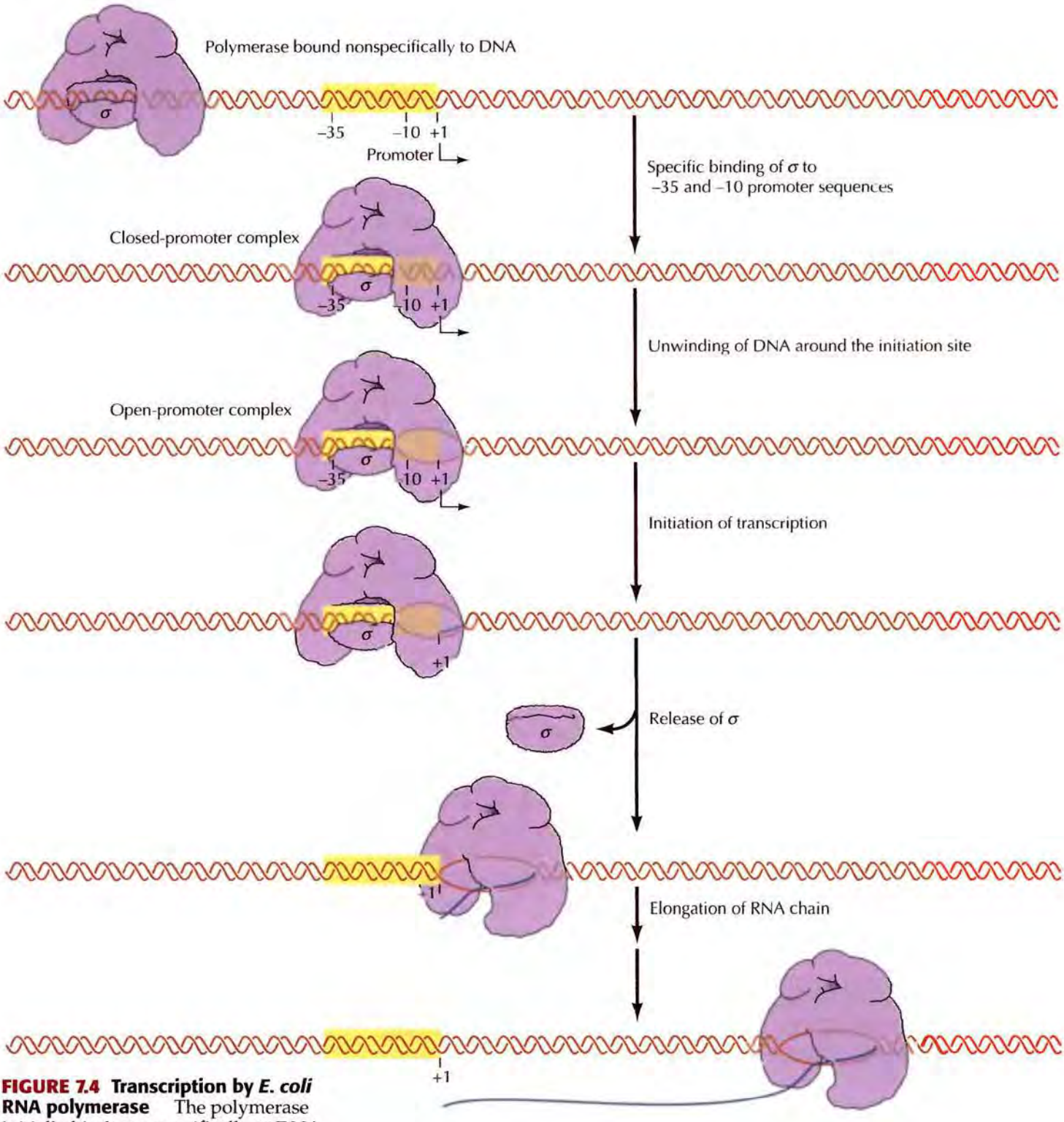


FIGURE 7.4 Transcription by *E. coli* RNA polymerase The polymerase initially binds nonspecifically to DNA and migrates along the molecule until the σ subunit binds to the –35 and –10 promoter elements, forming a closed-promoter complex. The polymerase then unwinds DNA around the initiation site, and transcription is initiated by the polymerization of free NTPs. The σ subunit then dissociates from the core polymerase, which migrates along the DNA and elongates the growing RNA chain.

ment is radiolabeled at one end. The labeled DNA is incubated with the protein of interest (e.g., RNA polymerase) and then subjected to partial digestion with DNase. The principle of the method is that the regions of DNA to which the protein binds are protected from DNase digestion. These regions can therefore be identified by comparison of the digestion products of the protein-bound DNA with those resulting from identical DNase treatment of a parallel sample of DNA that was not incubated with protein. Vari-

ations of this basic method, which employ chemical reagents to modify and cleave DNA at particular nucleotides, can be used to identify the specific DNA bases that are in contact with protein. Such footprinting analysis has shown that RNA polymerase generally binds to promoters over approximately a 60-base-pair region, extending from –40 to +20 (i.e., from 40 nucleotides upstream to 20 nucleotides downstream of the transcription start site). The σ subunit binds specifically to sequences in both the –35 and –10 promoter regions, substantiating the importance of these sequences in promoter function. In addition, some *E. coli* promoters have a third sequence, located upstream of the –35 region, that serves as a specific binding site for the RNA polymerase α subunit.

In the absence of σ, RNA polymerase binds nonspecifically to DNA with low affinity. The role of σ is to direct the polymerase to promoters by binding specifically to both the –35 and –10 sequences, leading to the initiation of transcription at the beginning of a gene (Figure 7.4). The initial binding between the polymerase and a promoter is referred to as a closed-promoter complex because the DNA is not unwound. The polymerase then unwinds 12–14 bases of DNA, from about –12 to +2, to form an open-promoter complex in which single-stranded DNA is available as a template for transcription. Transcription is initiated by the joining of two free NTPs. After addition of about the first 10 nucleotides, σ is released from the polymerase, which then leaves the promoter and moves along the template DNA to continue elongation of the growing RNA chain.

During elongation, the polymerase remains associated with its template while it continues synthesis of mRNAs. As it travels, the polymerase unwinds the template DNA ahead of it and rewinds the DNA behind it, maintaining an unwound region of about 15 base pairs in the region of transcription. Within this unwound portion of DNA, 8–9 bases of the growing RNA chain are bound to the complementary template DNA strand. High resolution structural analysis of bacterial RNA polymerase indicates that the β and β' subunits form a crab claw-like structure that grips the DNA template (Figure 7.5). An internal channel between the β and β' subunits accommodates approximately 20 base pairs of DNA and contains the polymerase active site.

7.1 WEBSITE ANIMATION

Transcription

Transcription is DNA-directed RNA synthesis, catalyzed by the enzyme RNA polymerase.

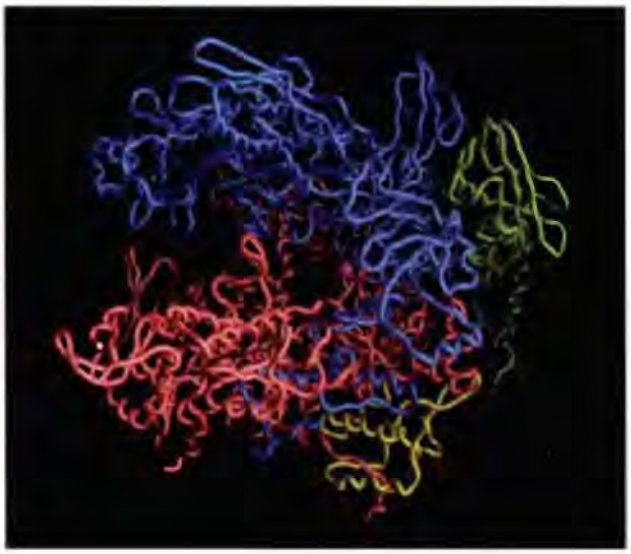


FIGURE 7.5 Structure of bacterial RNA polymerase The α subunits of the polymerase are colored dark green and light green, β blue, β' pink, and ω yellow. (Courtesy of Seth Darst, Rockefeller University.)

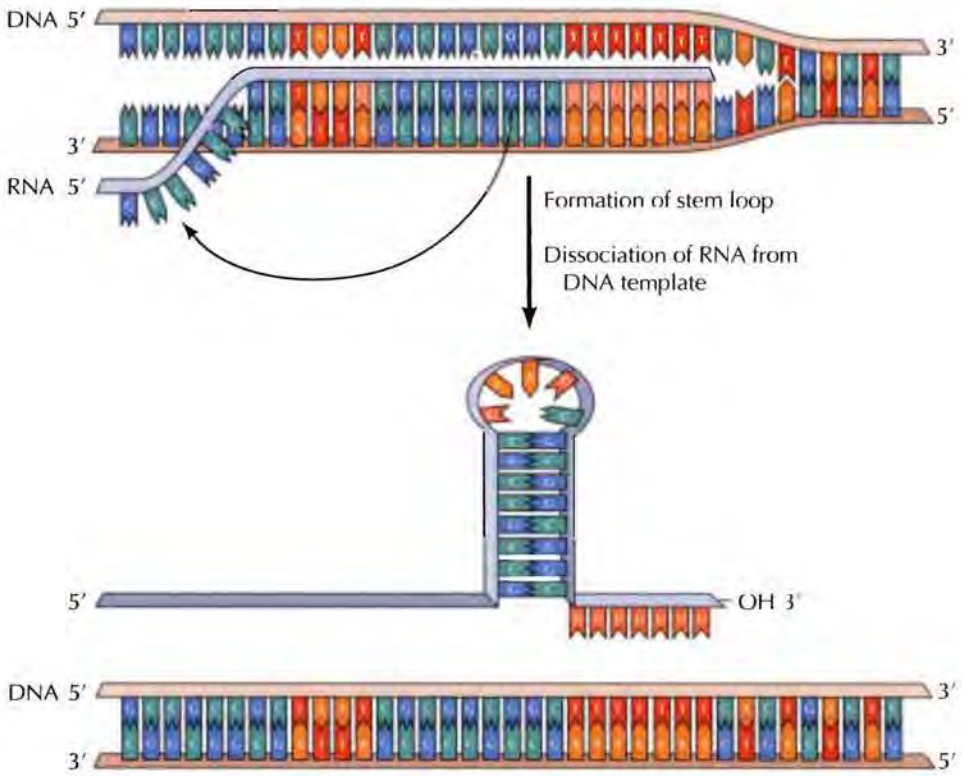


FIGURE 7.6 Transcription termination The termination of transcription is signaled by a GC-rich inverted repeat followed by seven A residues. The inverted repeat forms a stable stem-loop structure in the RNA, causing the RNA to dissociate from the DNA template.

RNA synthesis continues until the polymerase encounters a termination signal, at which point transcription stops, the RNA is released from the polymerase, and the enzyme dissociates from its DNA template. There are two alternative mechanisms for termination of transcription in *E. coli*. The simplest and most common type of termination signal consists of a symmetrical inverted repeat of a GC-rich sequence followed by approximately seven A residues (Figure 7.6). Transcription of the GC-rich inverted repeat results in the formation of a segment of RNA that can form a stable stem-loop structure by complementary base pairing. The formation of such a self-complementary structure in the RNA disrupts its association with the DNA template and terminates transcription. Because hydrogen bonding between A and U is weaker than that between G and C, the presence of A residues downstream of the inverted repeat sequences is thought to facilitate the dissociation of the RNA from its template. Alternatively, the transcription of some genes is terminated by a specific termination protein (called Rho), which binds extended segments (greater than 60 nucleotides) of single-stranded RNA. Since mRNAs in bacteria become associated with ribosomes and are translated while they are being transcribed, such extended regions of single-stranded RNA are exposed only at the end of an mRNA.

Repressors and Negative Control of Transcription

Transcription can be regulated at the stages of both initiation and elongation, but most transcriptional regulation in bacteria operates at the level of initiation. The pioneering studies of gene regulation in *E. coli* were carried out by François Jacob and Jacques Monod in the 1950s. These investigators and their colleagues analyzed the expression of enzymes involved in the metabolism of lactose, which can be used as a source of carbon and energy via cleavage to glucose and galactose (Figure 7.7). The enzyme that catalyzes the cleavage of lactose (β-galactosidase) and other enzymes involved in lac-

tose metabolism are expressed only when lactose is available for use by the bacteria. Otherwise, the cell is able to economize by not investing energy in the synthesis of unnecessary RNAs and proteins. Thus lactose induces the synthesis of enzymes involved in its own metabolism. In addition to requiring β-galactosidase, lactose metabolism involves the products of two other closely linked genes: lactose permease, which transports lactose into the cell, and a transacetylase, which is thought to inactivate toxic thiogalactosides that are transported into the cell along with lactose by the permease. On the basis of purely genetic experiments, Jacob and Monod deduced the mechanism by which the expression of these genes was regulated, thereby formulating a model that remains fundamental to our understanding of transcriptional regulation.

The starting point in this analysis was the isolation of mutants that were defective in regulation of the genes involved in lactose utilization. These mutants were of two types: constitutive mutants, which expressed all three genes even when lactose was not available, and noninducible mutants, which failed to express the genes even in the presence of lactose. Genetic mapping localized these regulatory mutants to two distinct loci, called *o* and *i*, with *o* located immediately upstream of the structural gene for β-galactosidase. Mutations affecting *o* resulted in constitutive expression; mutants of *i* were either constitutive or noninducible.

The function of these regulatory genes was probed by experiments in which two strains of bacteria were mated, resulting in diploid cells containing genes derived from both parents (Figure 7.8). Analysis of gene expression in such diploid bacteria provided critical insights by defining which alleles of these regulatory genes are dominant and which are recessive. For example, when bacteria containing a normal *i* gene (i^+) were mated with bacteria carrying an *i* gene mutation resulting in constitutive expression (an i^- mutation), the resulting diploid bacteria displayed normal inducibility; therefore

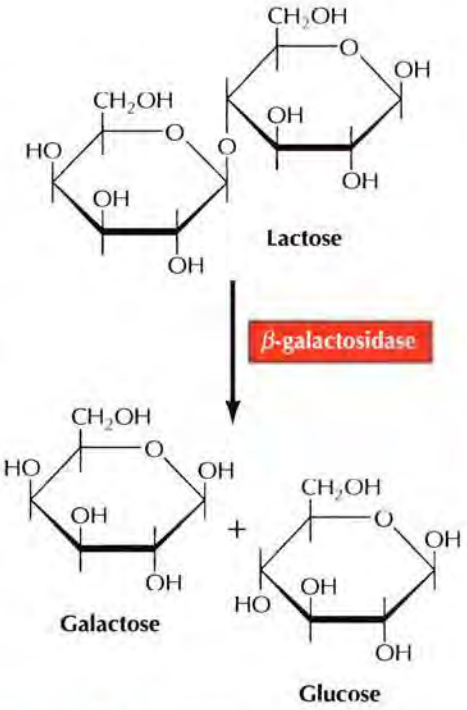


FIGURE 7.7 Metabolism of lactose β-galactosidase catalyzes the hydrolysis of lactose to glucose and galactose.

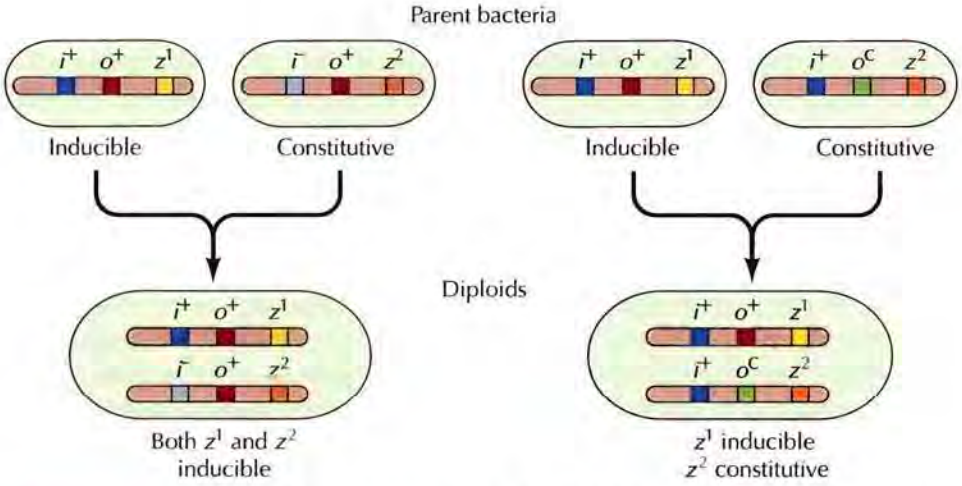


FIGURE 7.8 Regulation of β-galactosidase in diploid *E. coli* The mating of two bacterial strains results in diploid cells that contain genes from both parents. In these examples, it is assumed that the genes encoding β-galactosidase (the z genes) can be distinguished on the basis of structural gene mutations, designated z^1 and z^2. In an i^+/i^- diploid (left), both structural genes are inducible; therefore i^+ is dominant over i^- and affects expression of z genes on both chromosomes. In contrast, in an o^c/o^+ diploid (right), the z gene linked to o^c is constitutively expressed, whereas that linked to o^+ is inducible. Therefore *o* affects expression of only the adjacent z gene on the same chromosome.

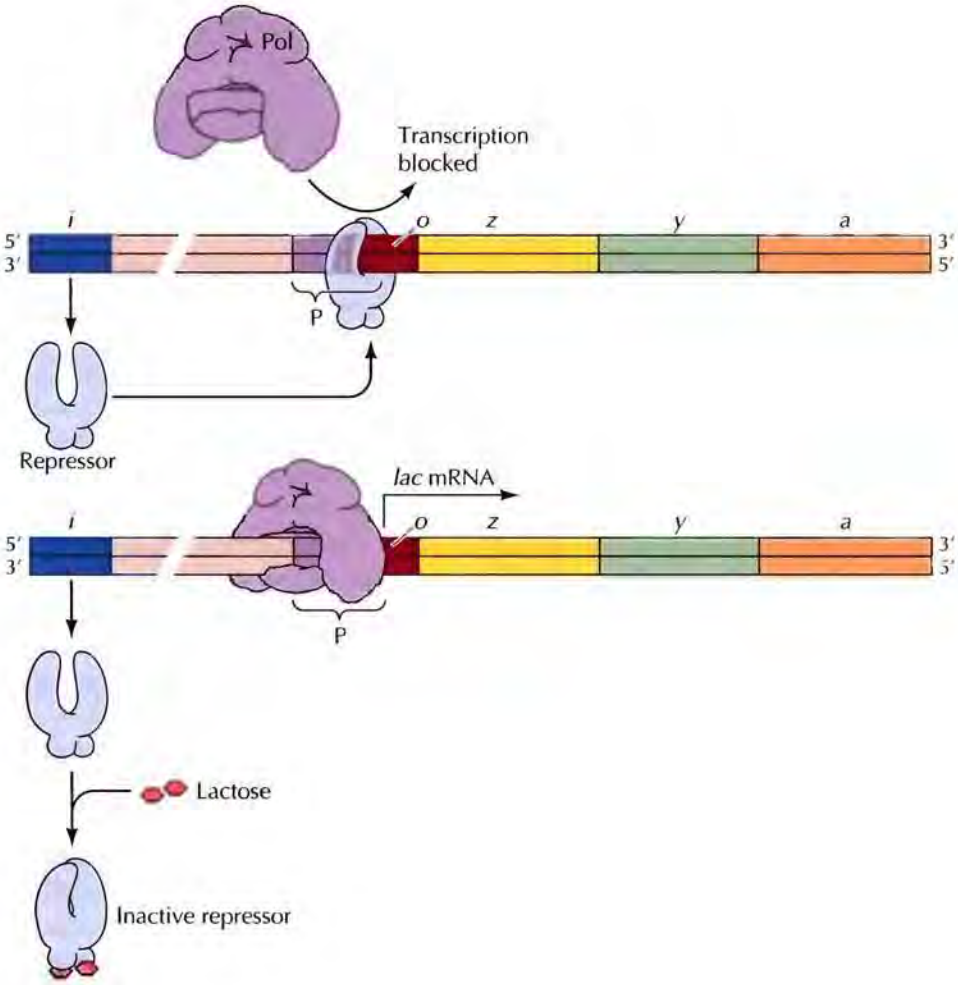


FIGURE 7.9 Negative control of the *lac* operon The *i* gene encodes a repressor which, in the absence of lactose (top), binds to the operator (*o*) and interferes with the binding of RNA polymerase to the promoter, blocking transcription of the three structural genes (*z*, β-galactosidase; *y*, permease; and *a*, transacetylase). Lactose induces expression of the operon by binding to the repressor (bottom), which prevents the repressor from binding to the operator. P = promoter; Pol = polymerase.

the normal i^+ gene was dominant over the i^- mutant. In contrast, matings between normal bacteria and bacteria with an o^c mutation (constitutive expression) yielded diploids with the constitutive expression phenotype, indicating that o^c is dominant over o^+. Additional experiments in which mutations in *o* and *i* were combined with different mutations in the structural genes showed that *o* affects the expression of only the genes to which it is physically linked, whereas *i* affects the expression of genes on both chromosome copies in diploid bacteria. Thus, in an o^c/o^+ cell, only the structural genes that are linked to o^c are constitutively expressed. In contrast, in an i^+/i^- cell, structural genes on both chromosomes are regulated normally. These results led to the conclusion that *o* represents a region of DNA that controls the transcription of adjacent genes, whereas the *i* gene encodes a regulatory factor (e.g., a protein) that can diffuse throughout the cell and control genes on both chromosomes.

The model of gene regulation developed on the basis of these experiments is illustrated in Figure 7.9. The genes encoding β-galactosidase, permease, and transacetylase are expressed as a single unit, called an **operon**. Transcription of the operon is controlled by *o* (the **operator**), which is adjacent to the transcription initiation site. The *i* gene encodes a protein that regulates transcription by binding to the operator. Since i^- mutants (which result in constitutive gene expression) are recessive, it was concluded that these mutants failed to make a functional gene product. This result implies that the normal *i* gene product is a **repressor**, which blocks transcription

when bound to *o*. The addition of lactose leads to induction of the operon because lactose binds to the repressor, thereby preventing it from binding to the operator DNA. In noninducible *i* mutants (which are dominant over i^+), the repressor fails to bind lactose, so expression of the operon cannot be induced.

The model neatly fits the results of the genetic experiments from which it was derived. In i^- cells, the repressor is not made, so the *lac* operon is constitutively expressed. Diploid i^+/i^- cells are normally inducible, since the functional repressor is encoded by the i^+ allele. Finally, in o^c mutants, a functional operator has been lost and the repressor cannot be bound. Consequently, o^c mutants are dominant but affect the expression only of linked structural genes.

Confirmation of this basic model has since come from a variety of experiments, including Walter Gilbert's isolation, in the 1960s, of the *lac* repressor and analysis of its binding to operator DNA. Molecular analysis has defined the operator as approximately 20 base pairs of DNA, starting a few bases before the transcription initiation site. Footprinting analysis has identified this region as the site to which the repressor binds, blocking transcription by interfering with the binding of RNA polymerase to the promoter. As predicted, lactose binds to the repressor, which then no longer binds to operator DNA. Also as predicted, o^c mutations alter sequences within the operator, thereby preventing repressor binding and resulting in constitutive gene expression.

The central principle of gene regulation exemplified by the lactose operon is that control of transcription is mediated by the interaction of regulatory proteins with specific DNA sequences. This general mode of regulation is broadly applicable to both prokaryotic and eukaryotic cells. Regulatory sequences like the operator are called ***cis*-acting control elements**, because they affect the expression of only linked genes on the same DNA molecule. On the other hand, proteins like the repressor are called ***trans*-acting factors** because they can affect the expression of genes located on other chromosomes within the cell. The *lac* operon is an example of negative control because binding of the repressor blocks transcription. This, however, is not always the case; many *trans*-acting factors are activators rather than inhibitors of transcription.

Positive Control of Transcription

The best-studied example of positive control in *E. coli* is the effect of glucose on the expression of genes that encode enzymes involved in the breakdown (catabolism) of other sugars (including lactose) that provide alternative sources of carbon and energy. Glucose is preferentially utilized; so as long as glucose is available, enzymes involved in catabolism of alternative energy sources are not expressed. For example, if *E. coli* are grown in medium containing both glucose and lactose, the *lac* operon is not induced and only glucose is used by the bacteria. Thus glucose represses the *lac* operon even in the presence of the normal inducer (lactose).

Glucose repression (generally called catabolite repression) is now known to be mediated by a positive control system, which is coupled to levels of cyclic AMP (cAMP) (Figure 7.10). In bacteria, the enzyme adenylyl cyclase, which converts ATP to cAMP, is regulated such that levels of cAMP increase when glucose levels drop. cAMP then binds to a transcriptional regulatory protein called catabolite activator protein (CAP). The binding of cAMP stimulates the binding of CAP to its target DNA sequences, which in the *lac* operon are located approximately 60 bases upstream of the transcription start

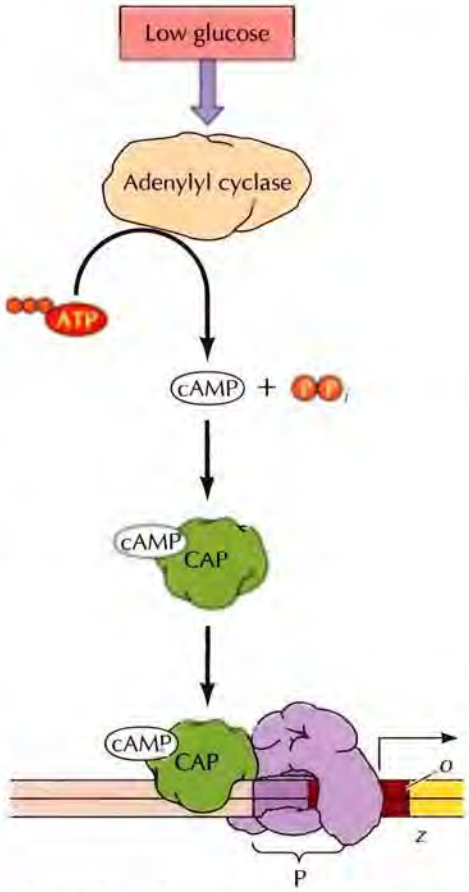


FIGURE 7.10 Positive control of the *lac* operon by glucose Low levels of glucose activate adenylyl cyclase, which converts ATP to cyclic AMP (cAMP). Cyclic AMP then binds to the catabolite activator protein (CAP) and stimulates its binding to regulatory sequences of various operons concerned with the metabolism of alternative sugars, such as lactose. CAP interacts with the α subunit of RNA polymerase to facilitate the binding of polymerase to the promoter.

TABLE 7.1 Classes of Genes Transcribed by Eukaryotic RNA Polymerases

Type of RNA synthesized	RNA polymerase
Nuclear genes	
mRNA	II
tRNA	III
rRNA	
5.8S, 18S, 28S	I
5S	III
snRNA and scRNA	II and III[a]
Mitochondrial genes	Mitochondrial[b]
Chloroplast genes	Chloroplast[b]

[a]Some small nuclear (sn) and small cytoplasmic (sc) RNAs are transcribed by polymerase II and others by polymerase III.

[b]The mitochondrial and chloroplast RNA polymerases are similar to bacterial enzymes.

site. CAP then interacts with the α subunit of RNA polymerase, facilitating the binding of polymerase to the promoter and activating transcription.

Eukaryotic RNA Polymerases and General Transcription Factors

Although transcription proceeds by the same fundamental mechanisms in all cells, it is considerably more complex in eukaryotic cells than in bacteria. This is reflected in three distinct differences between the prokaryotic and eukaryotic systems. First, whereas all genes are transcribed by a single core RNA polymerase in bacteria, eukaryotic cells contain multiple different RNA polymerases that transcribe distinct classes of genes. Second, eukaryotic RNA polymerases need to interact with a variety of additional proteins to specifically initiate and regulate transcription. Finally, transcription in eukaryotes takes place on chromatin rather than on free DNA, and regulation of chromatin structure is an important factor in the transcriptional activity of eukaryotic genes. This increased complexity of eukaryotic transcription presumably facilitates the sophisticated regulation of gene expression needed to direct the activities of the many different cell types of multicellular organisms.

Eukaryotic RNA Polymerases

Eukaryotic cells contain three distinct nuclear RNA polymerases that transcribe different classes of genes (Table 7.1). Protein-coding genes are transcribed by RNA polymerase II to yield mRNAs; ribosomal RNAs (rRNAs) and transfer RNAs (tRNAs) are transcribed by RNA polymerases I and III. RNA polymerase I is specifically devoted to transcription of the three largest species of rRNAs, which are designated 28S, 18S, and 5.8S according to their rates of sedimentation during velocity centrifugation. RNA polymerase III transcribes the genes for tRNAs and for the smallest species of ribosomal RNA (5S rRNA). Some of the small RNAs involved in splicing and protein transport (snRNAs and scRNAs) are also transcribed by RNA polymerase III, while others are polymerase II transcripts. RNA polymerase II also transcribes microRNAs, which are important regulators of both transcription and translation in eukaryotic cells. In addition, separate RNA polymerases (which are similar to bacterial RNA polymerases) are found in chloroplasts and mitochondria, where they specifically transcribe the DNAs of those organelles.

All three of the nuclear RNA polymerases are complex enzymes, consisting of 12 to 17 different subunits each. Although they recognize different promoters and transcribe distinct classes of genes, they share several features in common with each other as well as with bacterial RNA polymerase. In particular, all three eukaryotic RNA polymerases contain nine conserved subunits, five of which are related to the α, β, β', and ω subunits of bacterial RNA polymerase. The structure of yeast RNA polymerase II is strikingly similar to that of the bacterial enzyme (Figure 7.11), suggesting that all RNA polymerases utilize fundamentally conserved mechanisms to transcribe DNA.

General Transcription Factors and Initiation of Transcription by RNA Polymerase II

Because RNA polymerase II is responsible for the synthesis of mRNA from protein-coding genes, it has been the focus of most studies of transcription

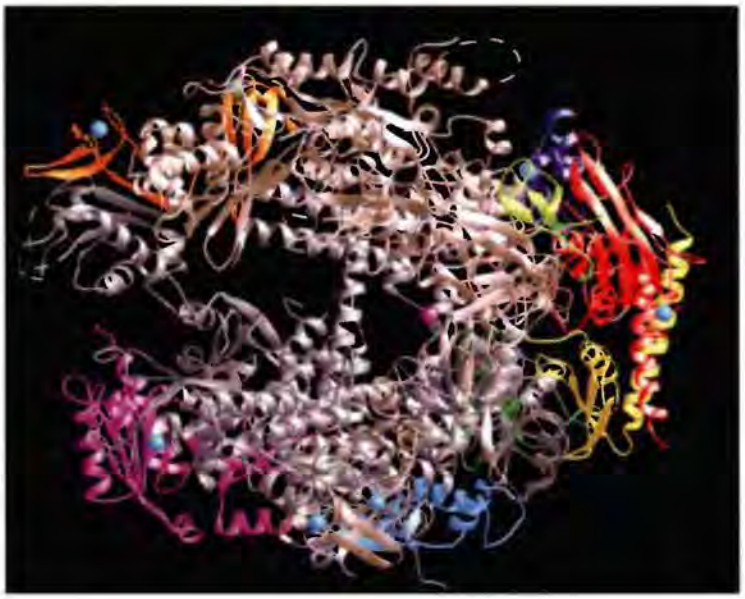


FIGURE 7.11 Structure of yeast RNA polymerase II Individual subunits are distinguished by colors. (From P. D. Kramer et al., 2001. *Science* 292: 1863.)

in eukaryotes. Early attempts at studying this enzyme indicated that its activity is different from that of prokaryotic RNA polymerase. The accurate transcription of bacterial genes that can be accomplished *in vitro* simply by the addition of purified RNA polymerase to DNA containing a promoter is not possible in eukaryotic systems. The basis of this difference was elucidated in 1979, when Robert Roeder and his colleagues discovered that RNA polymerase II is able to initiate transcription only if additional proteins are added to the reaction. Thus transcription in the eukaryotic system appeared to require distinct initiation factors that (in contrast to bacterial σ factors) were not associated with the polymerase.

Biochemical fractionation of nuclear extracts subsequently led to the identification of specific proteins (called **transcription factors**) that are required for RNA polymerase II to initiate transcription. **General transcription factors** are involved in transcription from all polymerase II promoters and therefore constitute part of the basic transcription machinery. Additional gene-specific transcription factors (discussed later in the chapter) bind to DNA sequences that control the expression of individual genes and are thus responsible for regulating gene expression. It is estimated that about 10% of the genes in the human genome encode transcription factors, emphasizing the importance of these proteins.

Five general transcription factors are minimally required for initiation of transcription by RNA polymerase II in reconstituted *in vitro* systems (Figure 7.12). The promoters of many genes transcribed by polymerase II contain a sequence similar to TATAA located 25 to 30 nucleotides upstream of the transcription start site. This sequence (called the **TATA box**) resembles the –10 sequence element of bacterial promoters, and the results of introducing mutations into TATAA sequences have demonstrated their role in the initiation of transcription. The promoters of many genes transcribed by RNA polymerase II also contain a second important sequence element (an initiator, or Inr, sequence) that spans the transcription start site. Although many RNA polymerase II promoters contain both of these elements, some contain only a TATA box and others contain only an Inr element, with no TATA box. Many promoters that lack a TATA box but contain an Inr element also contain an additional downstream promoter element (DPE), located approxi-

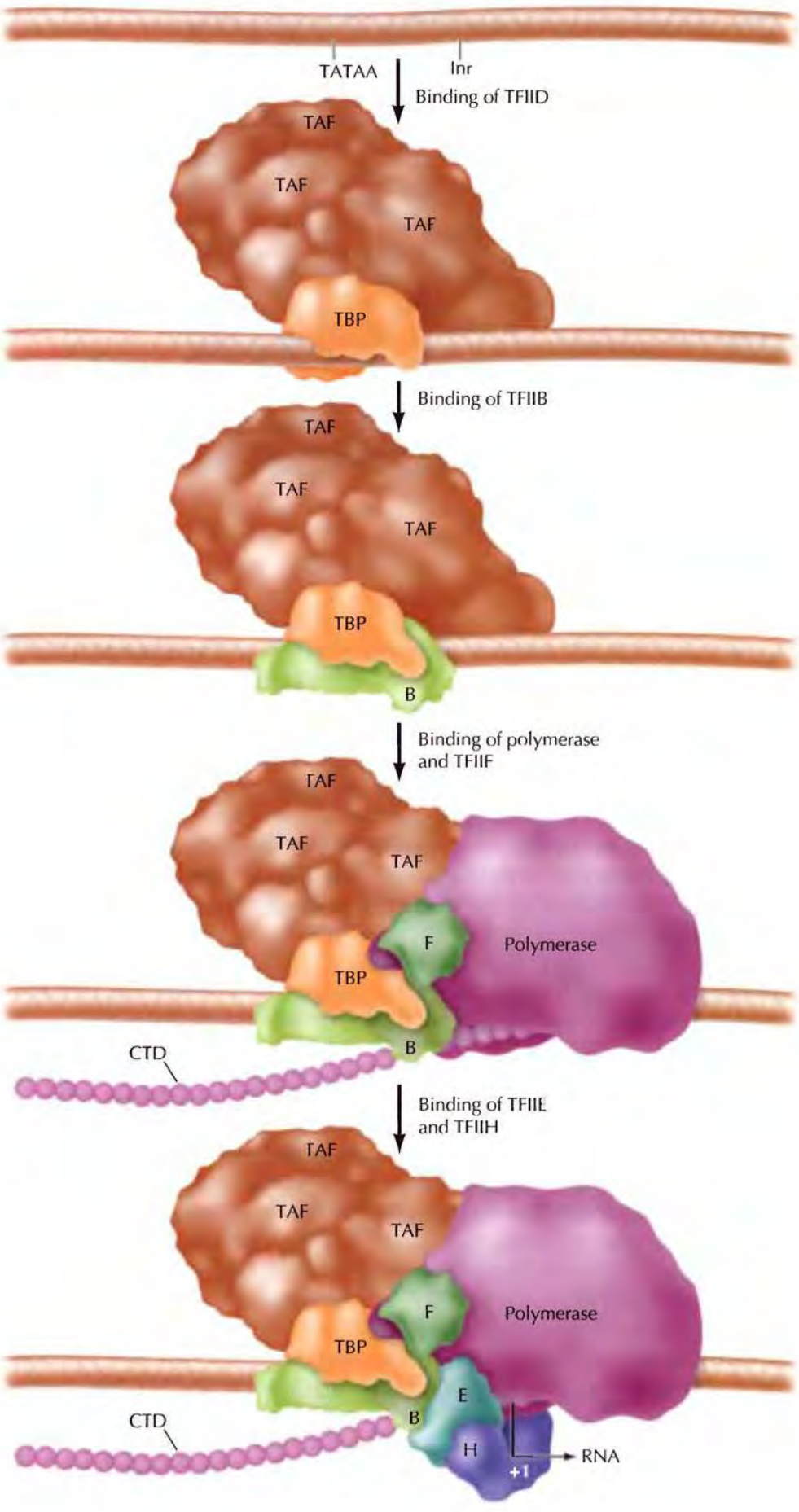


FIGURE 7.12 Formation of a polymerase II transcription initiation complex Many polymerase II promoters have a TATA box (consensus sequence TATAA) 25 to 30 nucleotides upstream of the transcription start site and an initiator (Inr) element that spans the transcription start site. Formation of a transcription complex is initiated by the binding of transcription factor TFIID. One subunit of TFIID, the TATA-binding protein or TBP, binds to the TATA box; other subunits (TBP-associated factors or TAFs) may bind to the Inr. TFIIB(B) then binds to TBP, followed by binding of the polymerase in association with TFIIF(F). Finally, TFIIE(E) and TFIIH(H) associate with the complex.

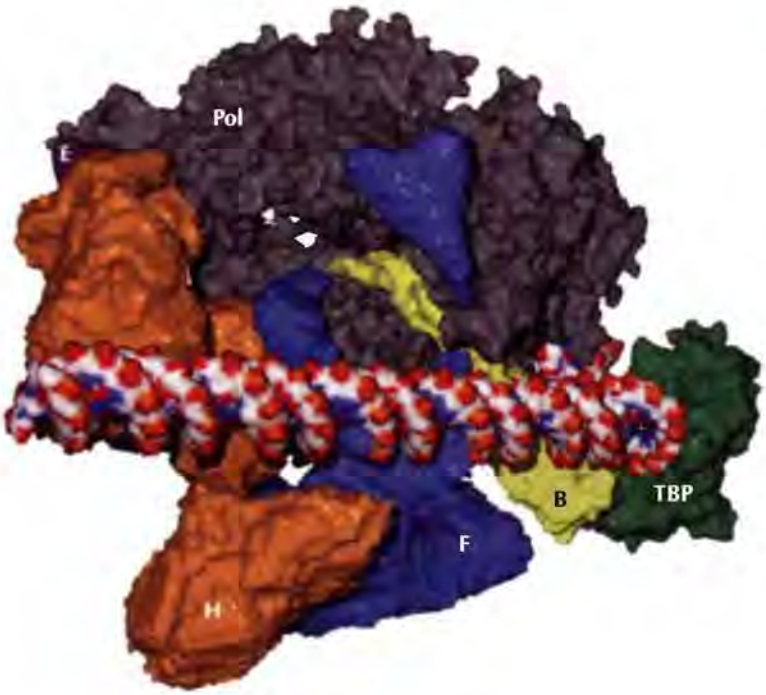


FIGURE 7.13 Model of the polymerase II transcription initiation complex A molecular model of RNA polymerase II (grey), TBP (green), TFIIB (yellow), TFIIE (purple), TFIIF (blue), and TFIIH (beige) assembled on a promoter. The orientation of the initiation complex on the DNA shown here is reversed from that shown in Figure 7.12. (From D. A. Bushnell et al., 2004. *Science* 303: 983.)

mately 30 base pairs downstream of the transcription start site, which functions cooperatively with the Inr sequence.

The first step in formation of a transcription complex is the binding of a general transcription factor called TFIID to the promoter (*TF* indicates *t*ranscription *f*actor; *II* indicates polymerase *II*). TFIID is itself composed of multiple subunits, including the **TATA-binding protein (TBP)** and at least 14 other polypeptides, called **TBP-associated factors (TAFs)**. TBP binds specifically to the TATA box while other subunits of TFIID (TAFs) appear to bind to the Inr and DPE sequences. The binding of TFIID is followed by recruitment of a second general transcription factor (TFIIB), which binds to TBP as well as to DNA sequences that are present upstream of the TATA box in some promoters. TFIIB in turn serves as a bridge to RNA polymerase II, which binds to the TBP-TFIIB complex in association with a third factor, TFIIF.

Following recruitment of RNA polymerase II to the promoter, the binding of two additional factors (TFIIE and TFIIH) completes formation of the initiation complex (Figure 7.13). TFIIH is a multisubunit factor that appears to play at least two important roles. First, two subunits of TFIIH are helicases, which unwind DNA around the initiation site. (These subunits of TFIIH are the XPB and XPD proteins which are also required for nucleotide excision repair, as discussed in Chapter 6.) Another subunit of TFIIH is a protein kinase that phosphorylates repeated sequences present in the C-terminal domain of the largest subunit of RNA polymerase II. The polymerase II C-terminal domain (or CTD) consists of tandem repeats (27 repeats in yeast and 52 in humans) of seven amino acids with the consensus sequence Tyr-Ser-Pro-Thr-Ser-Pro-Ser. Phosphorylation of these amino acids releases the polymerase from its association with the preinitiation complex and leads to the recruitment of other proteins that allow the polymerase to initiate transcription and begin synthesis of a growing mRNA chain.

Although the sequential recruitment of five general transcription factors and RNA polymerase II described here represents the minimal system required for transcription *in vitro*, additional factors are needed to stimulate

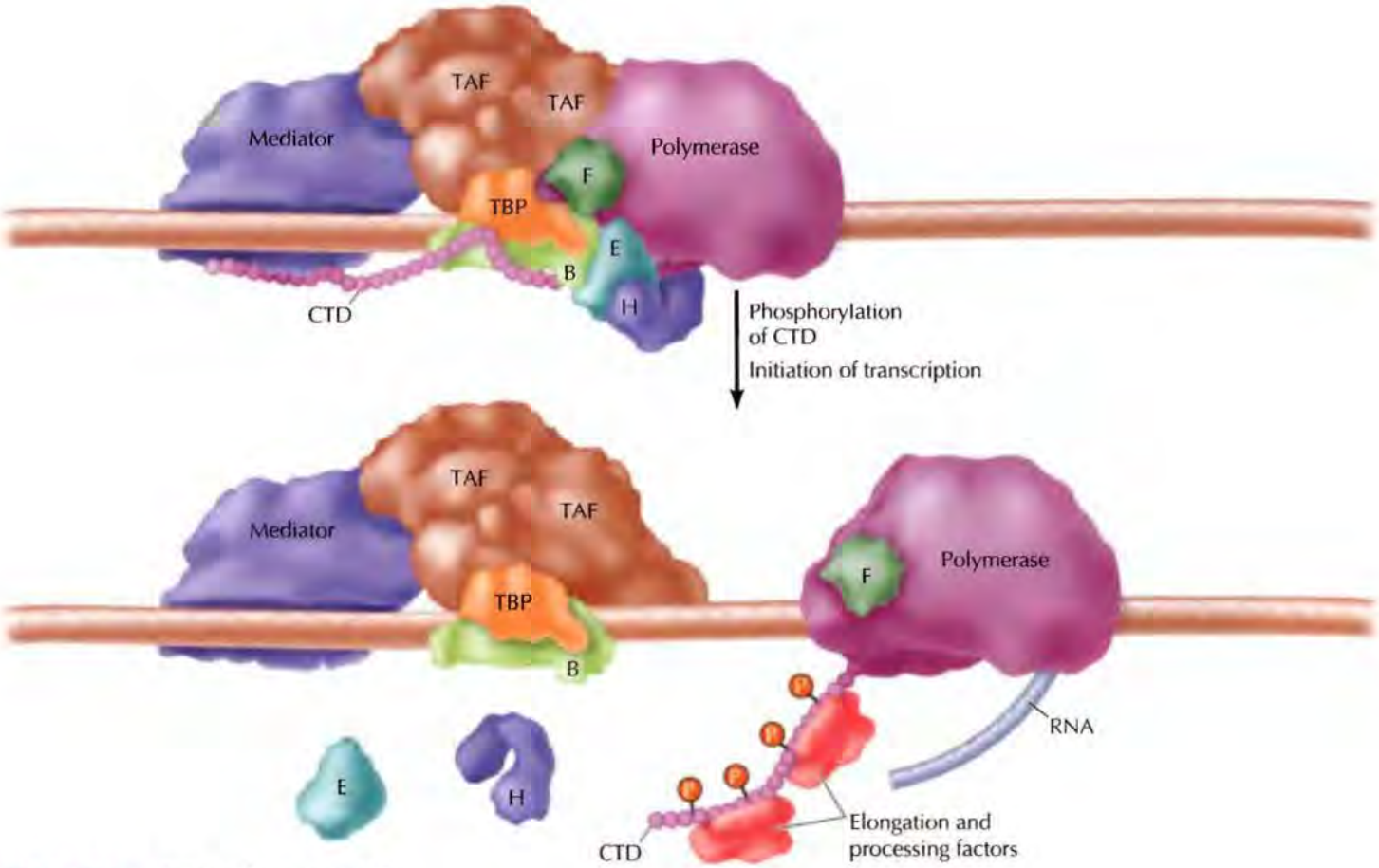


FIGURE 7.14 RNA polymerase II/ Mediator complexes RNA polymerase II is associated with Mediator proteins, as well as with the general transcription factors, at the promoter. The Mediator complex binds to the nonphosphorylated CTD of polymerase II and is released following phosphorylation of the CTD when transcription initiates. The phosphorylated CTD then binds elongation and processing factors that facilitate mRNA synthesis and processing.

transcription within the cell. These factors include a large protein complex, called **Mediator,** which consists of more than 20 distinct subunits. The Mediator complex not only stimulates basal transcription, but also plays a key role in linking the general transcription factors to the gene-specific transcription factors that regulate gene expression. The Mediator proteins are associated with the C-terminal domain of polymerase II, and are released from the polymerase following assembly of the preinitiation complex and phosphorylation of the polymerase C-terminal domain (Figure 7.14). The phosphorylated CTD then binds other proteins that facilitate transcriptional elongation and function in mRNA processing, as discussed later in this chapter.

Transcription by RNA Polymerases I and III

As previously discussed, distinct RNA polymerases are responsible for the transcription of genes encoding ribosomal RNAs, transfer RNAs, and some small noncoding RNAs in eukaryotic cells. All three RNA polymerases, however, require additional transcription factors to associate with appropriate promoter sequences. Furthermore, although the three different polymerases in eukaryotic cells recognize distinct types of promoters, a common transcription factor—the TATA-binding protein (TBP)—appears to be required for initiation of transcription by all three enzymes.

RNA polymerase I is devoted solely to the transcription of ribosomal RNA genes, which are present in tandem repeats. Transcription of these genes yields a large 45S pre-rRNA, which is then processed to yield the 28S, 18S, and 5.8S rRNAs (Figure 7.15). The promoters of ribosomal RNA genes

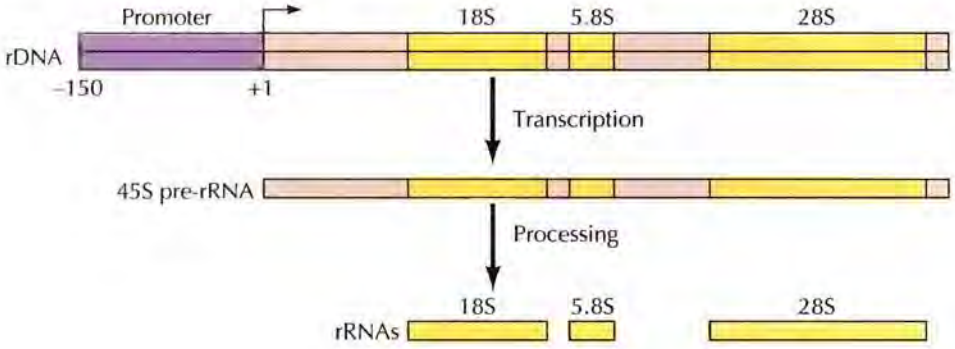


FIGURE 7.15 The ribosomal RNA gene Ribosomal DNA (rDNA) is transcribed to yield a large RNA molecule (45S pre-rRNA), which is then cleaved into 28S, 18S, and 5.8S rRNAs.

span about 150 base pairs just upstream of the transcription initiation site. These promoter sequences are recognized by two transcription factors, UBF (upstream binding factor) and SL1 (selectivity factor 1), which bind cooperatively to the promoter and then recruit polymerase I to form an initiation complex (Figure 7.16). The SL1 transcription factor is composed of four protein subunits, one of which is TBP. Since the promoters for ribosomal RNA genes do not contain a TATA box, TBP does not bind to specific promoter sequences. Instead, the association of TBP with ribosomal RNA genes is mediated by the binding of other proteins in the SL1 complex to the promoter, a situation similar to the association of TBP with the Inr sequences of polymerase II genes that lack TATA boxes.

The genes for tRNAs, 5S rRNA, and some of the small RNAs involved in splicing and protein transport are transcribed by polymerase III. These genes are transcribed from three distinct classes of promoters, two of which lie within, rather than upstream of, the transcribed sequence (Figure 7.17). The most thoroughly studied of the genes transcribed by polymerase III are the 5S rRNA genes of *Xenopus*. TFIIIA (which is the first transcription factor to have been purified) initiates assembly of a transcription complex by binding to specific DNA sequences in the 5S rRNA promoter. This binding is followed by the sequential binding of TFIIIC, TFIIIB, and the polymerase. The promoters for the tRNA genes differ from the 5S rRNA promoter in that they do not contain the DNA sequence recognized by TFIIIA. Instead, TFIIIC binds directly to the promoter of tRNA genes, serving to recruit TFIIIB and polymerase to form a transcription complex. Promoters of the third class of genes transcribed by polymerase III, including genes encoding some of the small nuclear RNAs involved in splicing, are located upstream of the transcription start site. These promoters contain a TATA box (like promoters for polymerase II genes) as well as a binding site for another factor, called SNAP. SNAP and TFIIIB bind cooperatively to these promoters, with

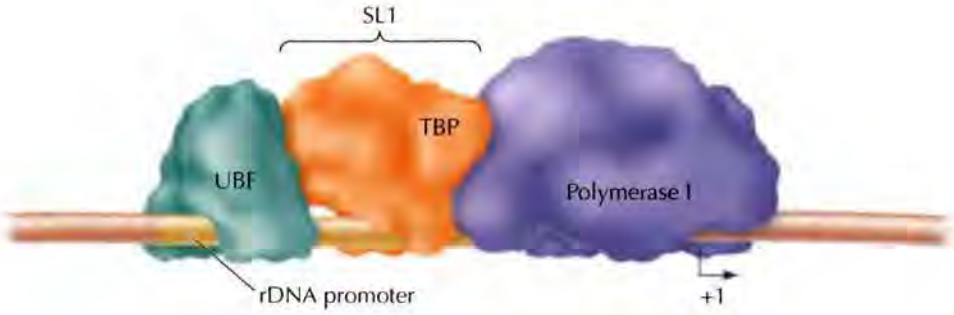


FIGURE 7.16 Initiation of rDNA transcription Two transcription factors, UBF and SL1, bind cooperatively to the rDNA promoters and recruit RNA polymerase I to form an initiation complex. One subunit of SL1 is the TATA-binding protein (TBP).

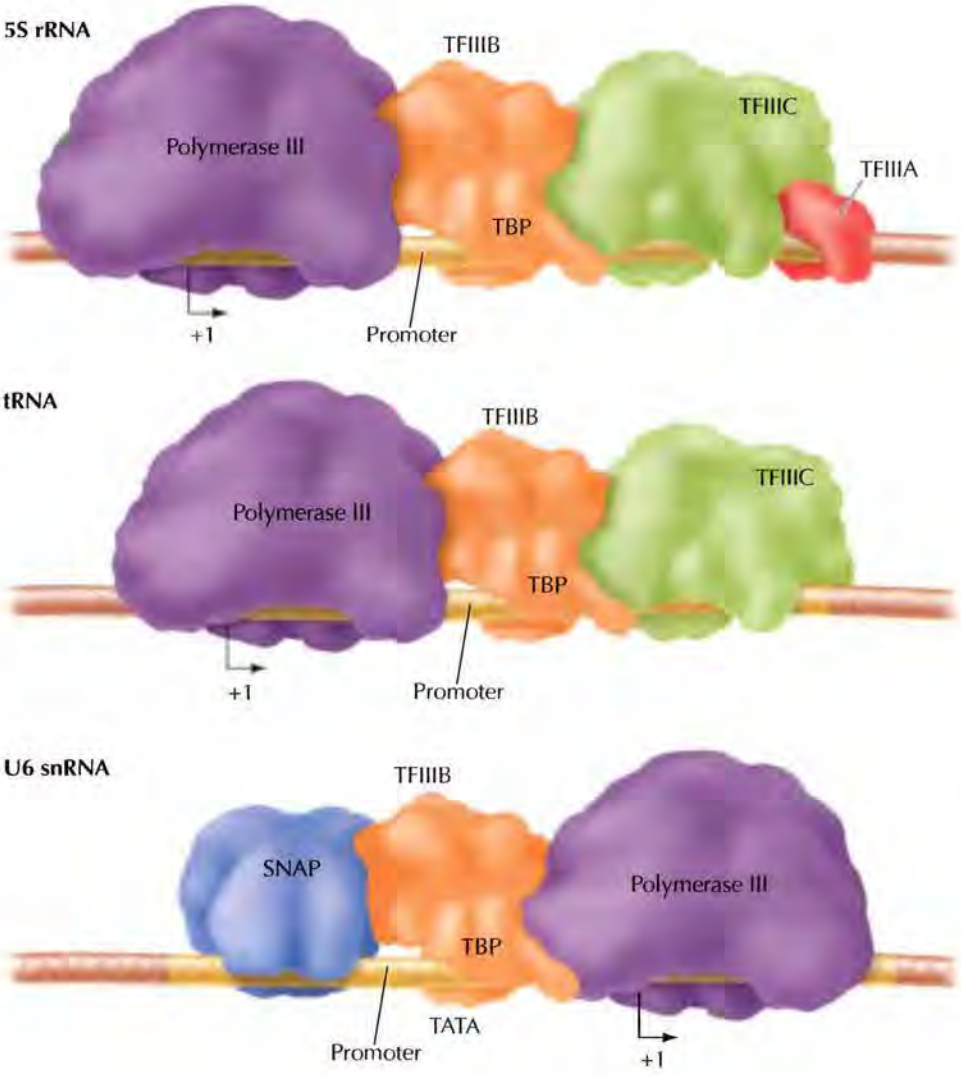


FIGURE 7.17 Transcription of RNA polymerase III genes Genes transcribed by polymerase III are expressed from three types of promoters. The promoters of 5S rRNA and tRNA genes are downstream of the transcription initiation site. Transcription of the 5S rRNA gene is initiated by the binding of TFIIIA, followed by the binding of TFIIIC, TFIIIB, and the polymerase. The tRNA promoters do not contain a binding site for TFIIIA, and TFIIIA is not required for their transcription. Instead, TFIIIC initiates the transcription of tRNA genes by binding to promoter sequences, followed by the association of TFIIIB and polymerase. The promoter of the U6 snRNA gene is upstream of the transcription start site and contains a TATA box, which is recognized by the TATA-binding protein (TBP) subunit of TFIIIB, in cooperation with another factor called SNAP.

TFIIIB binding directly to the TATA box. This is mediated by the TATA-binding protein, TBP, which is one of the subunits of TFIIIB. As in the case of the promoters of other RNA polymerase III genes, TFIIIB then recruits the polymerase to the transcription complex.

Regulation of Transcription in Eukaryotes

The control of gene expression is far more complex in eukaryotes than in bacteria, although some of the same basic principles apply. The expression of eukaryotic genes is controlled primarily at the level of initiation of transcription, although in many cases transcription is also regulated during elongation. As in bacteria, transcription in eukaryotic cells is controlled by proteins that bind to specific regulatory sequences and modulate the activity of RNA polymerase. An important difference between transcriptional regulation in prokaryotes and eukaryotes, however, results from the packaging of eukaryotic DNA into chromatin, which limits its availability as a template for transcription. As a result, modifications of chromatin structure play key roles in the control of transcription in eukaryotic cells. A particularly exciting area of current research is based on the discovery that noncod-

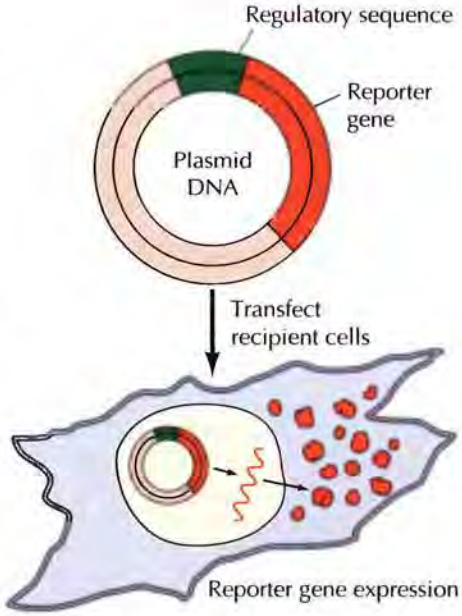


FIGURE 7.18 Identification of eukaryotic regulatory sequences The regulatory sequence of a cloned eukaryotic gene is ligated to a reporter gene that encodes an easily detectable enzyme. The resulting plasmid is then introduced into cultured recipient cells by transfection. An active regulatory sequence directs transcription of the reporter gene, expression of which is then detected in the transfected cells.

ing RNAs, as well as proteins, regulate transcription in eukaryotic cells via modifications in chromatin structure.

cis-Acting Regulatory Sequences: Promoters and Enhancers

As already discussed, transcription in bacteria is regulated by the binding of proteins to *cis*-acting sequences (e.g., the *lac* operator) that control the transcription of adjacent genes. Similar *cis*-acting sequences regulate the expression of eukaryotic genes. These sequences have been identified in mammalian cells largely by the use of gene transfer assays to study the activity of suspected regulatory regions of cloned genes (Figure 7.18). The eukaryotic regulatory sequences are usually ligated to a reporter gene that encodes an easily detectable enzyme. The expression of the reporter gene following its transfer into cultured cells then provides a sensitive assay for the ability of the cloned regulatory sequences to direct transcription. Biologically active regulatory regions can thus be identified, and *in vitro* mutagenesis can be used to determine the roles of specific sequences within the region.

Genes transcribed by RNA polymerase II have core promoter elements, including the TATA box and the Inr sequence, that serve as specific binding sites for general transcription factors. Other *cis*-acting sequences serve as binding sites for a wide variety of regulatory factors that control the expression of individual genes. These *cis*-acting regulatory sequences are frequently, though not always, located upstream of the TATA box. For example, two regulatory sequences that are found in many eukaryotic genes were identified by studies of the promoter of the herpes simplex virus gene that encodes thymidine kinase (Figure 7.19). Both of these sequences are located within 100 base pairs upstream of the TATA box: Their consensus sequences are CCAAT and GGGCGG (called a GC box). Specific proteins that bind to these sequences and stimulate transcription have since been identified.

In contrast to the relatively simple organization of CCAAT and GC boxes in the herpes thymidine kinase promoter, many genes in mammalian cells are controlled by regulatory sequences located farther away (sometimes more than 10 kilobases) from the transcription start site. These sequences,

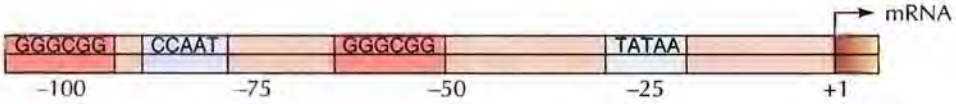


FIGURE 7.19 A eukaryotic promoter The promoter of the thymidine kinase gene of herpes simplex virus contains three sequence elements upstream of the TATA box that are required for efficient transcription: a CCAAT box and two GC boxes (consensus sequence GGGCGG).

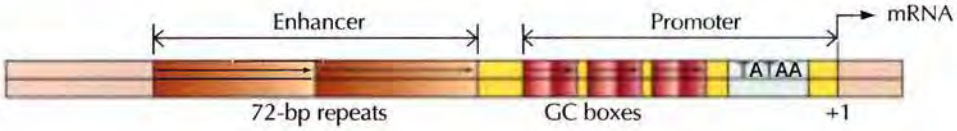


FIGURE 7.20 The SV40 enhancer The SV40 promoter for early gene expression contains a TATA box and six GC boxes arranged in three sets of repeated sequences. In addition, efficient transcription requires an upstream enhancer consisting of two 72-base-pair (bp) repeats.

called **enhancers**, were first identified during studies of the promoter of another virus, SV40 (Figure 7.20). In addition to a TATA box and a set of six GC boxes, two 72-base-pair repeats located farther upstream are required for efficient transcription from this promoter. These sequences were found to stimulate transcription from other promoters as well as from that of SV40, and, surprisingly, their activity depended on neither their distance nor their orientation with respect to the transcription initiation site (Figure 7.21). They could stimulate transcription when placed either upstream or downstream of the promoter, in either a forward or backward orientation.

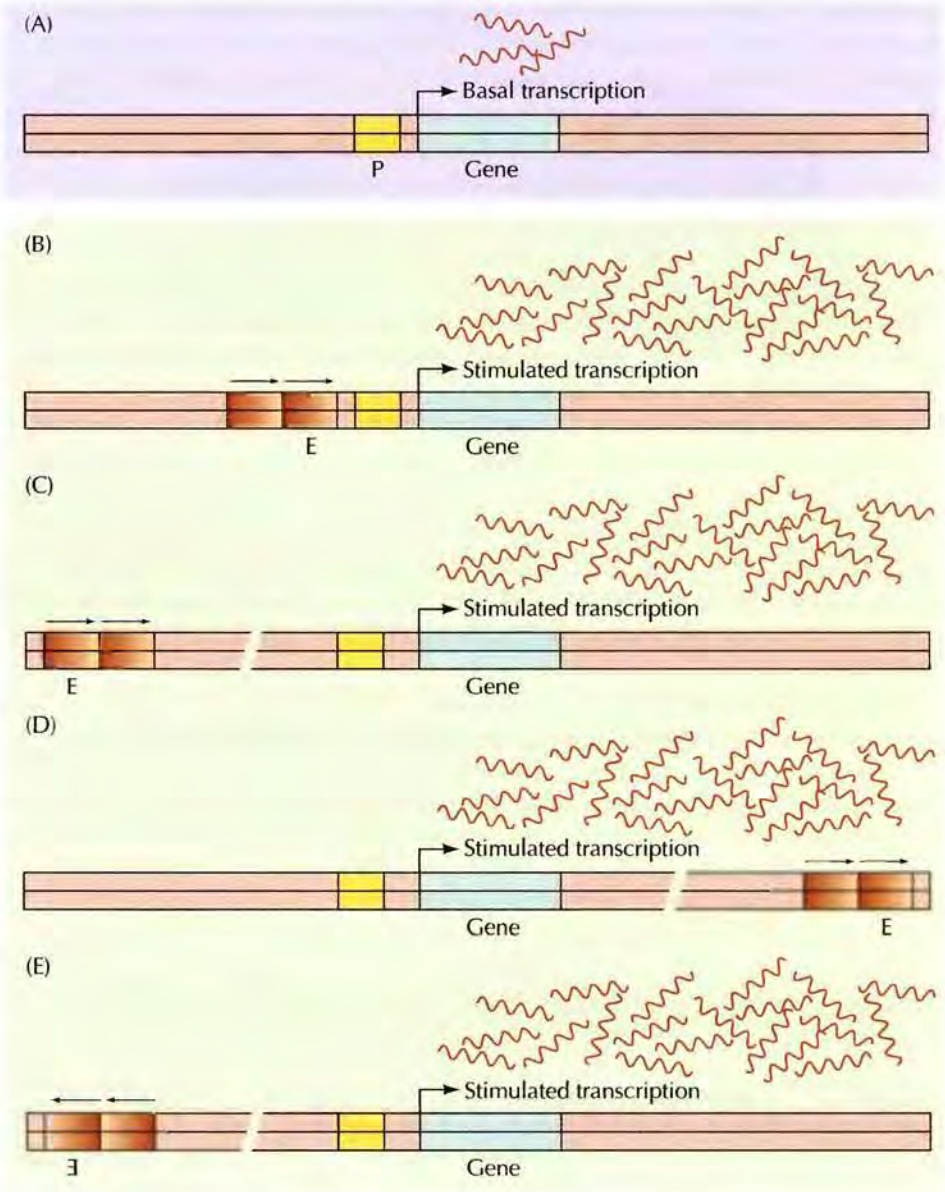


FIGURE 7.21 Action of enhancers Without an enhancer, the gene is transcribed at a low basal level (A). Addition of an enhancer, E—for example, the SV40 72-base-pair repeats—stimulates transcription. The enhancer is active not only when placed just upstream of the promoter (B), but also when inserted up to several kilobases either upstream or downstream from the transcription start site (C and D). In addition, enhancers are active in either the forward or backward orientation (E).

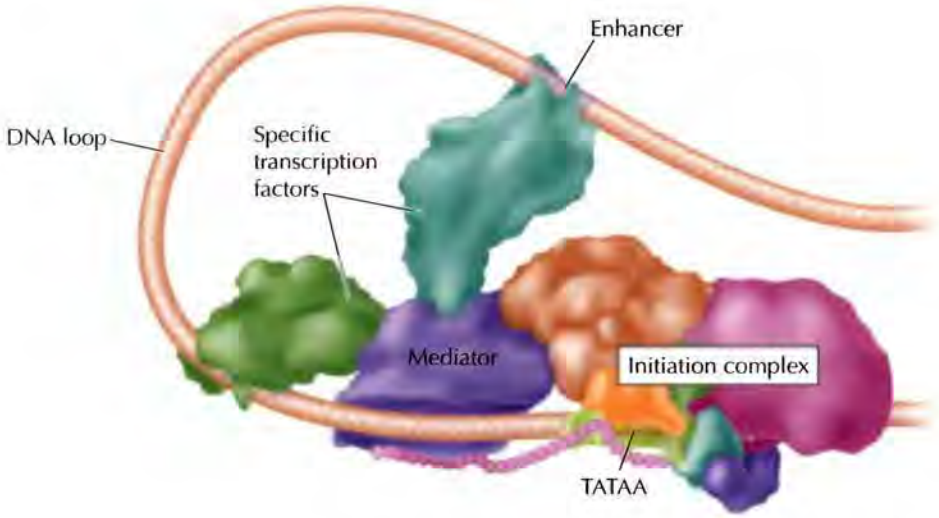


FIGURE 7.22 DNA looping Transcription factors bound at distant enhancers are able to interact with the RNA polymerase II/Mediator complex at the promoter because the intervening DNA can form loops. There is therefore no fundamental difference between the action of transcription factors bound to DNA just upstream of the gene in the promoter and to distant enhancers.

The ability of enhancers to function even when separated by long distances from transcription initiation sites at first suggested that they work by mechanisms different from those of promoters. However, this has turned out not to be the case: Enhancers, like promoters, function by binding transcription factors that then regulate RNA polymerase. This is possible because of DNA looping, which allows a transcription factor bound to a distant enhancer to interact with proteins associated with the RNA polymerase/Mediator complex at the promoter (Figure 7.22). Transcription factors bound to distant enhancers can thus work by the same mechanisms as those bound adjacent to promoters, so there is no fundamental difference between the actions of enhancers and those of *cis*-acting regulatory sequences adjacent to transcription start sites. Interestingly, although enhancers were first identified in mammalian cells, they have subsequently been found in bacteria—an unusual instance in which studies of eukaryotes served as a model for the simpler prokaryotic systems.

The binding of specific transcriptional regulatory proteins to enhancers is responsible for the control of gene expression during development and differentiation, as well as during the response of cells to hormones and growth factors. An example of a thoroughly studied mammalian enhancer controls the transcription of immunoglobulin genes in B lymphocytes. Gene transfer experiments have established that the immunoglobulin enhancer is active in lymphocytes, but not in other types of cells. Thus this regulatory sequence is at least partly responsible for tissue-specific expression of the immunoglobulin genes in the appropriate differentiated cell type.

An important aspect of enhancers is that they usually contain multiple functional sequence elements that bind different transcriptional regulatory proteins. These proteins work together to regulate gene expression. The immunoglobulin heavy-chain enhancer, for example, spans approximately 200 base pairs and contains at least nine distinct sequence elements that serve as protein-binding sites (Figure 7.23). Mutation of any one of these sequences reduces but does not abolish enhancer activity, indicating that the functions of individual proteins that bind to the enhancer are at least partly redundant. Many of the individual sequence elements of the immunoglobulin enhancer by themselves stimulate transcription in nonlymphoid cells. The restricted activity of the intact enhancer in B lymphocytes therefore does not result from the tissue-specific function of each of its components. Instead, tissue-specific expression results from the combina-

■ **Enhancers can function over very long distances, sometimes even from different chromosomes. This process, termed transvection, has been best studied in *Drosophila* and involves *trans*-acting enhancers from one gene regulating the expression of its homolog on a separate chromosome.**

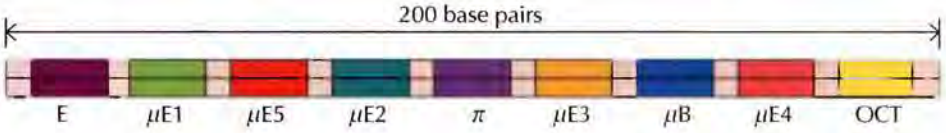


FIGURE 7.23 The immunoglobulin enhancer The immunoglobulin heavy-chain enhancer spans about 200 bases and contains nine functional sequence elements (E, μE1–5, π, μB, and OCT), which together stimulate transcription in B lymphocytes.

tion of the individual sequence elements that make up the complete enhancer. These elements include some *cis*-acting regulatory sequences that bind transcriptional activators that are expressed specifically in B lymphocytes, as well as other regulatory sequences that bind repressors in nonlymphoid cells. Thus the immunoglobulin enhancer contains negative regulatory elements that inhibit transcription in inappropriate cell types, as well as positive regulatory elements that activate transcription in B lymphocytes. The overall activity of the enhancer is greater than the sum of its parts, reflecting the combined action of the proteins associated with each of its individual sequence elements.

Although DNA looping allows enhancers to act at a considerable distance from promoters, the activity of any given enhancer is specific for the promoter of its appropriate target gene. This specificity is maintained in part by **insulators** or **barrier elements**, which divide chromosomes into independent domains and prevent enhancers from acting on promoters located in an adjacent domain. Insulators also prevent the chromatin structure of one domain from spreading to its neighbors, thereby maintaining independently regulated regions of the genome. It is thought that insulators function by organizing independent domains of chromatin within the nucleus, but their mechanism of action remains to be elucidated by further research.

■ **A major hurdle for gene therapy is that introduced genes are often aberrantly regulated or inactivated because of the nearby chromatin structure. The addition of insulator elements is a potential solution to this problem.**

Transcription Factor Binding Sites

The binding sites of transcriptional regulatory proteins in promoter or enhancer sequences have commonly been identified by two types of experiments. The first, footprinting, was described earlier in connection with the binding of RNA polymerase to prokaryotic promoters (see Figure 7.3). The second approach is the **electrophoretic-mobility shift assay** in which a radiolabeled DNA fragment is incubated with a protein preparation and then subjected to electrophoresis through a nondenaturing gel (Figure 7.24). Protein binding is detected as a decrease in the electrophoretic mobility of the DNA fragment, since its migration through the gel is slowed by the bound protein. The combined use of footprinting and electrophoretic-mobility shift assays has led to the correlation of protein-binding sites with the regulatory elements of enhancers and promoters, indicating that these sequences generally constitute the recognition sites of specific DNA-binding proteins.

The binding sites of most transcription factors consist of short DNA sequences, typically spanning 6–10 base pairs. In most cases, these binding sites are degenerate, meaning that the transcription factor will bind not only to the consensus sequence but also to sequences that differ from the consensus at one or more positions. It is thus common to represent transcription factor binding sites as pictograms, representing the frequency of each base at all positions of known binding sites for a given factor (Figure 7.25). Because of their short degenerate nature, sequences matching transcription factor binding sites occur frequently in genomic DNA, so physiologically significant regulatory sequences can not be identified from DNA sequence alone.

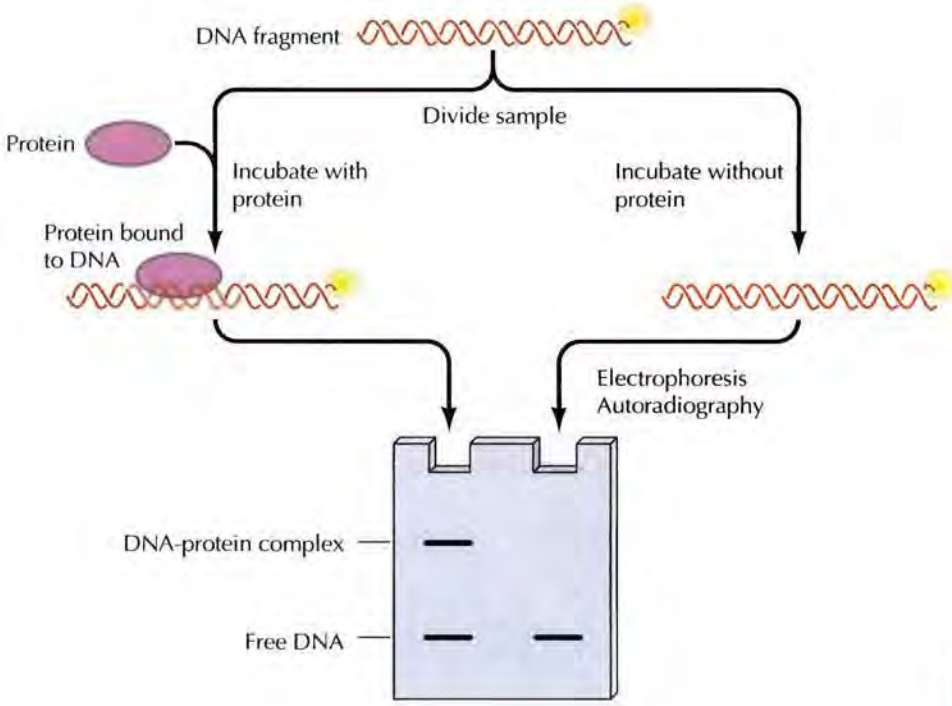


FIGURE 7.24 Electrophoretic-mobility shift assay A sample containing radiolabeled fragments of DNA is divided in two, and one half of the sample is incubated with a protein that binds to a specific DNA sequence. Samples are then analyzed by electrophoresis in a nondenaturing gel so that the protein remains bound to DNA. Protein binding is detected by the slower migration of DNA-protein complexes compared to that of free DNA. Only a fraction of the DNA in the sample is actually bound to protein, so both DNA-protein complexes and free DNA are detected following incubation of the DNA with protein.

As discussed in Chapter 5, identifying functional regulatory sequences in genomic DNA thus remains one of the prime challenges in bioinformatics and systems biology.

An important experimental approach for determining the regions of DNA that bind a transcription factor within the cell has been provided by **chromatin immunoprecipitation** (Figure 7.26). Cells are first treated with formaldehyde, which cross links proteins to DNA. As a result, transcription factors are covalently linked to the DNA sequences to which they were bound within the living cell. Chromatin is then extracted and sheared to fragments of about 500 base pairs. Fragments of DNA linked to a transcription factor of interest can then be isolated by immunoprecipitation with an antibody against the transcription factor (see Figure 4.30). The formaldehyde crosslinks are then reversed, and the immunoprecipitated DNA isolated and analyzed to determine the sites to which the specific transcription factor was bound within the cell. In yeast, chromatin immunoprecipitation has been combined with computational analysis of transcription factor binding sites to produce a global map of transcription factor binding sites throughout the genome.

Transcriptional Regulatory Proteins

A variety of transcriptional regulatory proteins have been isolated based on their binding to specific DNA sequences. One of the proto-

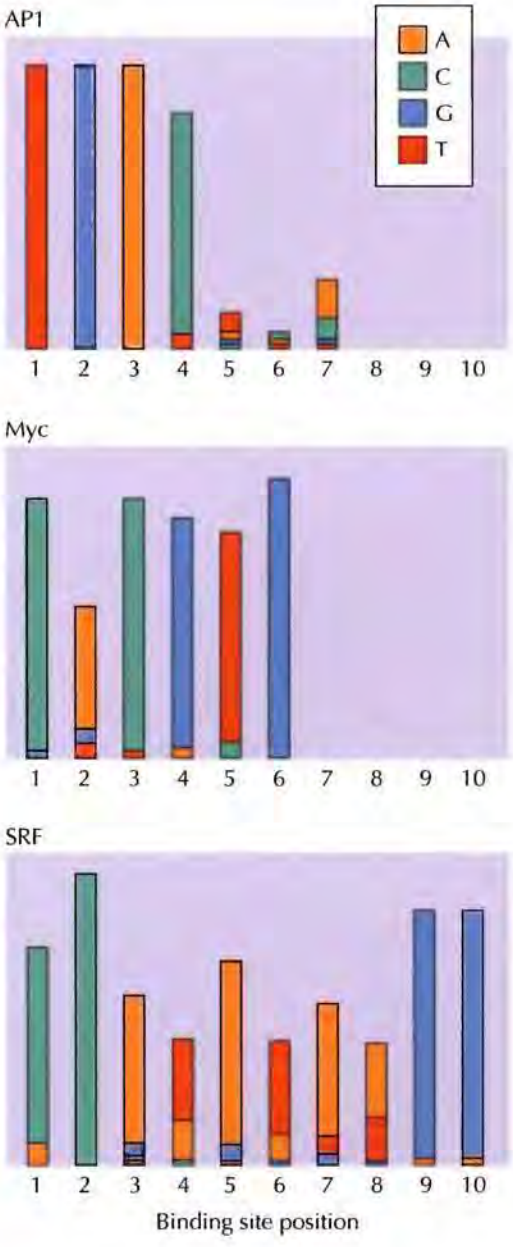


FIGURE 7.25 Representative transcription factor binding sites The binding sites of three mammalian transcription factors (AP1, Myc, and SRF) are shown as pictograms in which the frequency of each nucleotide is represented by the height of the corresponding letter at each position of the binding site.

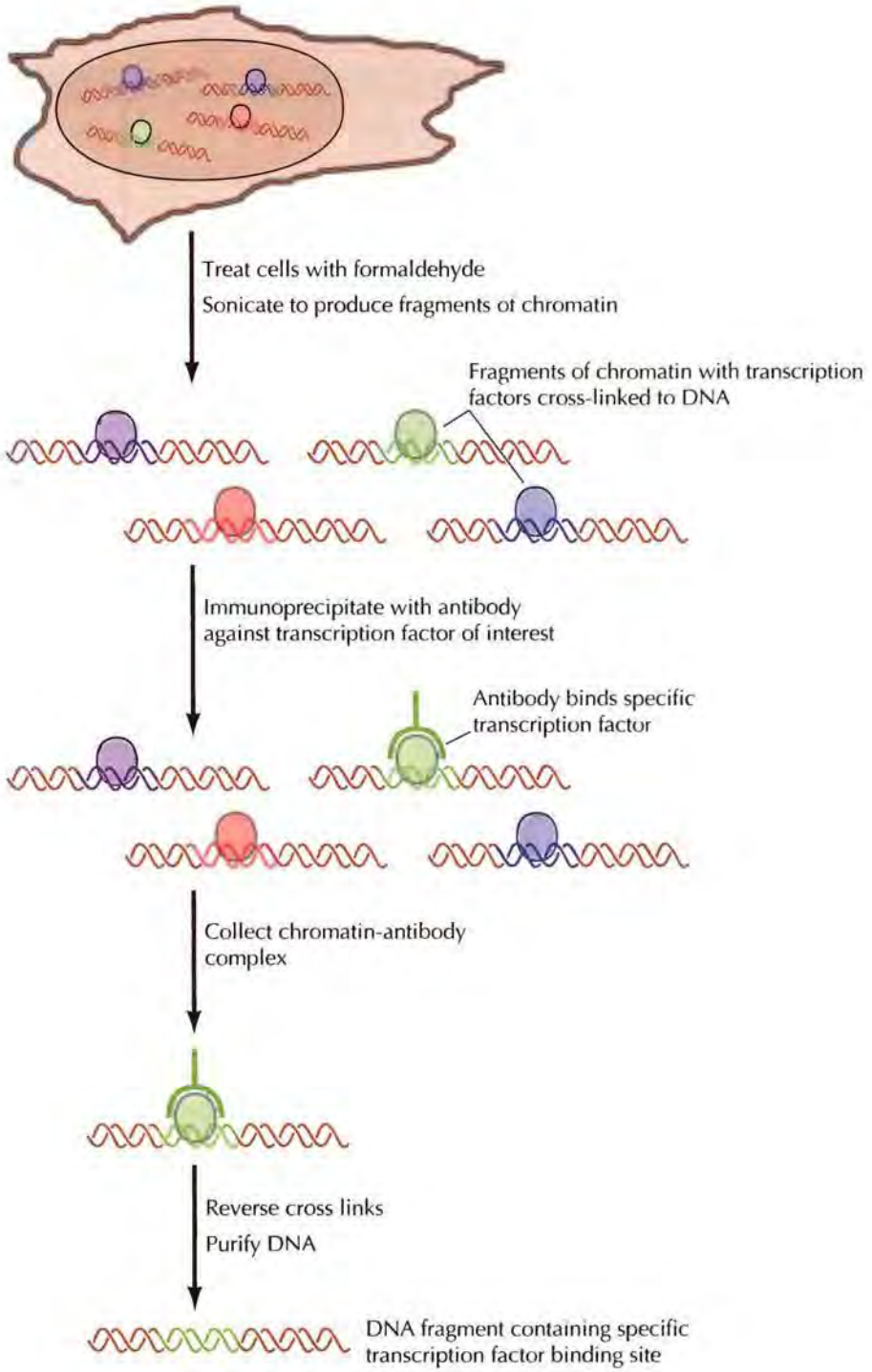


FIGURE 7.26 Chromatin immunoprecipitation Cells are treated with formaldehyde to crosslink DNA and proteins and then sonicated to produce fragments of chromatin. The chromatin fragments are incubated with an antibody against a specific transcription factor, and chromatin fragments bound to the antibody are collected as described in Figure 4.30. The crosslinks are then reversed, and DNA is purified to yield DNA fragments containing binding sites for the specific transcription factor of interest.

types of eukaryotic transcription factors was initially identified by Robert Tjian and his colleagues during studies of the transcription of SV40 DNA. This factor (called Sp1, for specificity protein 1) was found to stimulate transcription via binding to GC boxes in the SV40 promoter. Importantly, the specific binding of Sp1 to the GC box not only established the action of Sp1 as a sequence-specific transcription factor but also suggested a general approach to the purification of transcription factors. The isolation of these proteins initially presented a formidable challenge because they are present in very small quantities (e.g., only 0.001% of total cell protein) that are difficult to purify by conventional biochemical techniques. This problem was

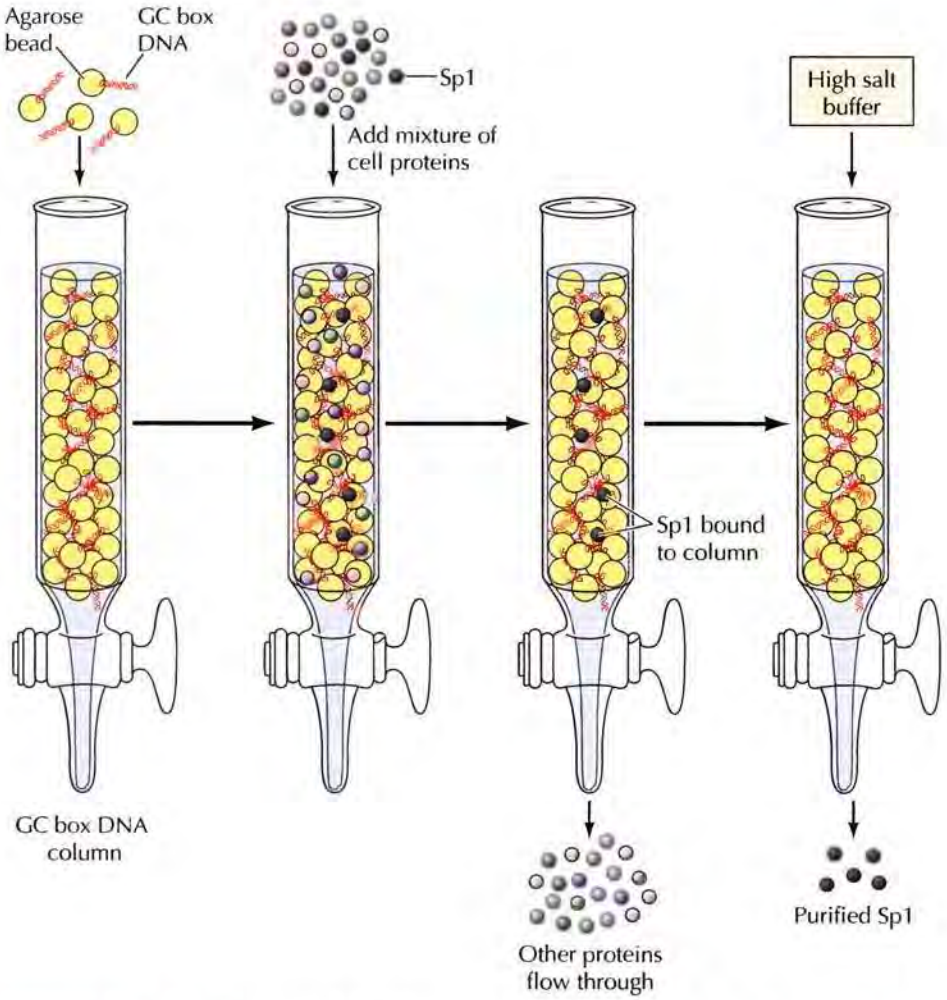


FIGURE 7.27 Purification of Sp1 by DNA-affinity chromatography A double-stranded oligonucleotide containing repeated GC box sequences is bound to agarose beads, which are poured into a column. A mixture of cell proteins containing Sp1 is then applied to the column; because Sp1 specifically binds to the GC box oligonucleotide, it is retained on the column while other proteins flow through. Washing the column with high salt buffer then dissociates Sp1 from the GC box DNA, yielding purified Sp1.

overcome in the purification of Sp1 by **DNA-affinity chromatography** (Figure 7.27). Multiple copies of oligonucleotides corresponding to the GC box sequence were bound to a solid support, and cell extracts were passed through the oligonucleotide column. Because Sp1 bound to the GC box with high affinity, it was specifically retained on the column while other proteins were not. Highly purified Sp1 could thus be obtained and used for further studies.

The general method of DNA-affinity chromatography, first optimized for the purification of Sp1, has been used successfully to isolate a wide variety of sequence-specific DNA-binding proteins from eukaryotic cells. Genes encoding other transcription factors have been isolated by screening cDNA expression libraries to identify recombinant proteins that bind to specific DNA sequences. The cloning and sequencing of transcription factor cDNAs has led to the accumulation of a great deal of information on the structure and function of these critical regulatory proteins.

KEY EXPERIMENT

Isolation of a Eukaryotic Transcription Factor

Affinity Purification of Sequence-Specific DNA-Binding Proteins

James T. Kadonaga and Robert Tjian
University of California, Berkeley

Proceedings of the National Academy of Sciences, USA, 1986, Volume 83, pages 5889–5893

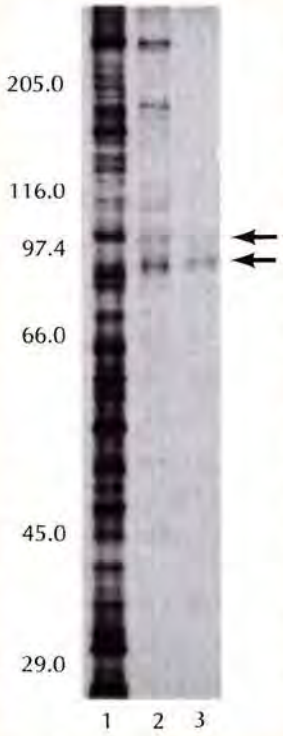
Robert Tjian

The Context

Starting with studies of the *lac* operon by Jacob and Monod, it became clear that transcription is regulated by proteins that bind to specific DNA sequences. One of the prototype systems for studies of gene expression in eukaryotic cells was the monkey virus SV40 in which several regulatory DNA sequences were identified in the early 1980s. In 1983 William Dynan and Robert Tjian first demonstrated that one of these sequence elements (the GC box) is the specific binding site of a protein detectable in nuclear extracts of human cells. This protein (called Sp1, for specificity protein 1) not only binds to the GC box sequence; it also stimulates transcription *in vitro*, demonstrating that it is a sequence-specific transcriptional activator.

To study the mechanism of Sp1 action, it then became necessary to obtain the transcription factor in pure form and eventually to clone the Sp1 gene. The isolation of pure Sp1 thus became a high priority, but it also posed a daunting technical challenge.

Sp1 and other transcription factors appeared to represent only about 0.001% of total cell protein, so they could not readily be purified by conventional biochemical techniques.



Purification of Sp1. Gel electrophoresis of proteins initially present in the crude nuclear extract (lane 1) and of proteins obtained after either one or two sequential cycles of DNA-affinity chromatography (lanes 2 and 3, respectively). The sizes of marker proteins (in kilodaltons) are indicated to the left of the gel, and the Sp1 polypeptides are indicated by arrows.

James Kadonaga and Robert Tjian solved this problem by developing a method of DNA-affinity chromatography that led to the purification not only of Sp1 but also of many other eukaryotic transcription factors, thereby opening the door to molecular analysis of transcriptional regulation in eukaryotic cells.

The Experiments

The DNA-affinity chromatography method developed by Kadonaga and Tjian exploited the specific high-affinity binding of Sp1 to the GC box sequence, GGGCGG. Synthetic oligonucleotides containing multiple copies of this sequence were coupled to solid beads, and a crude nuclear extract was passed through a column consisting of beads linked to GC box DNA. The beads were then washed to remove proteins that had failed to bind specifically to the oligonucleotides. Finally, the beads were washed with a high salt buffer (0.5 *M* KCl), which disrupted the binding of Sp1 to DNA, thereby releasing Sp1 from the column.

Gel electrophoresis demonstrated that the crude nuclear extract initially applied to the column was a complex mixture of proteins (see figure). In contrast, approximately 90% of the protein recovered after two cycles of DNA-affinity chromatography corresponded to only two polypeptides, which were identified as Sp1 by DNA binding and by their activity in *in vitro* transcription assays. Thus Sp1 had been successfully purified by DNA-affinity chromatography.

The Impact

In their 1986 paper, Kadonaga and Tjian stated that the DNA-affinity chromatography technique "should be generally applicable for the purification of other sequence-specific DNA binding proteins." This prediction has been amply verified; many eukaryotic transcription factors have been purified by this method. The genes that encode still other transcription factors have been isolated by an alternative approach (developed independently in 1988 in the laboratories of Phillip Sharp and Steven McKnight) in which cDNA expression libraries are screened with oligonucleotide probes to detect recombinant proteins that bind specifically to the desired DNA sequences. The ability to isolate sequence-specific DNA-binding proteins by these methods has led to detailed characterization of the structure and function of a wide variety of transcriptional regulatory proteins, providing the basis for our current understanding of gene expression in eukaryotic cells.

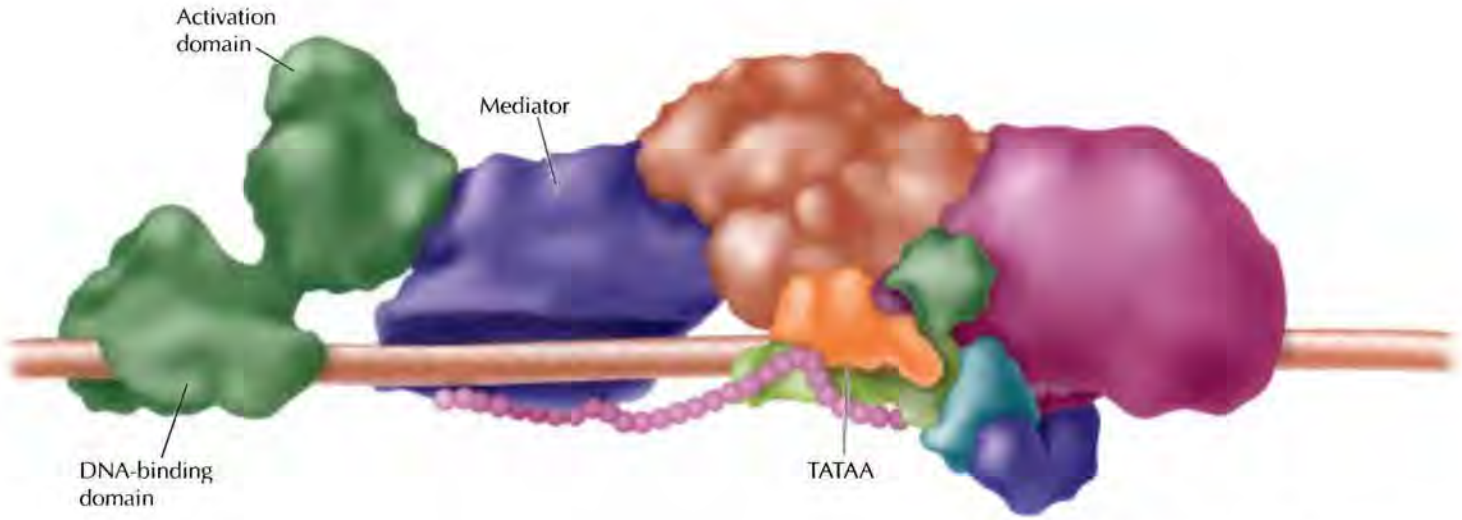


FIGURE 7.28 Structure of transcriptional activators Transcriptional activators consist of two independent domains. The DNA-binding domain recognizes a specific DNA sequence, and the activation domain interacts with Mediator or other components of the transcriptional machinery.

Structure and Function of Transcriptional Activators

Because transcription factors are central to the regulation of gene expression, understanding the mechanisms of their action is a major area of ongoing research in cell and molecular biology. The most thoroughly studied of these proteins are **transcriptional activators**, which, like Sp1, bind to regulatory DNA sequences and stimulate transcription. In general, these factors consist of two independent domains: One region of the protein specifically binds DNA; the other region stimulates transcription by interacting with other proteins, including Mediator or other components of the transcriptional machinery (Figure 7.28). The basic function of the DNA-binding domain is to anchor the transcription factor to the proper site on DNA; the activation domain then independently stimulates transcription through protein-protein interactions.

Many different transcription factors have now been identified in eukaryotic cells, with about 2500 encoded in the human genome. They contain many distinct types of DNA-binding domains, some of which are illustrated in Figure 7.29. The most common is the **zinc finger domain**, which contains repeats of cysteine and histidine residues that bind zinc ions and fold into looped structures ("fingers") that bind DNA. These domains were initially identified in the polymerase III transcription factor TFIIIA but are also common among transcription factors that regulate polymerase II promoters, including Sp1. Other examples of transcription factors that contain zinc finger domains are the **steroid hormone receptors**, which regulate gene transcription in response to hormones such as estrogen and testosterone.

The **helix-turn-helix** motif was first recognized in prokaryotic DNA-binding proteins, including the *E. coli* catabolite activator protein (CAP). In these proteins, one helix makes most of the contacts with DNA, while the other helices lie across the complex to stabilize the interaction. In eukaryotic cells, helix-turn-helix proteins include the **homeodomain** proteins, which play critical roles in the regulation of gene expression during embryonic development. The genes encoding these proteins were first discovered as developmental mutants in *Drosophila*. Some of the earliest recognized *Drosophila* mutants (termed homeotic mutants in 1894) resulted in the development of flies in which one body part was transformed into another. For example, in the homeotic mutant called *Antennapedia*, legs rather than

■ **Artificial transcription factors with zinc-finger domains designed to bind specific sequences within the genome have been developed. By ligating different effector domains to the DNA binding domain, the target gene can be either activated or repressed. This technology can be applied to gene therapy and the development of transgenic plants and animals.**

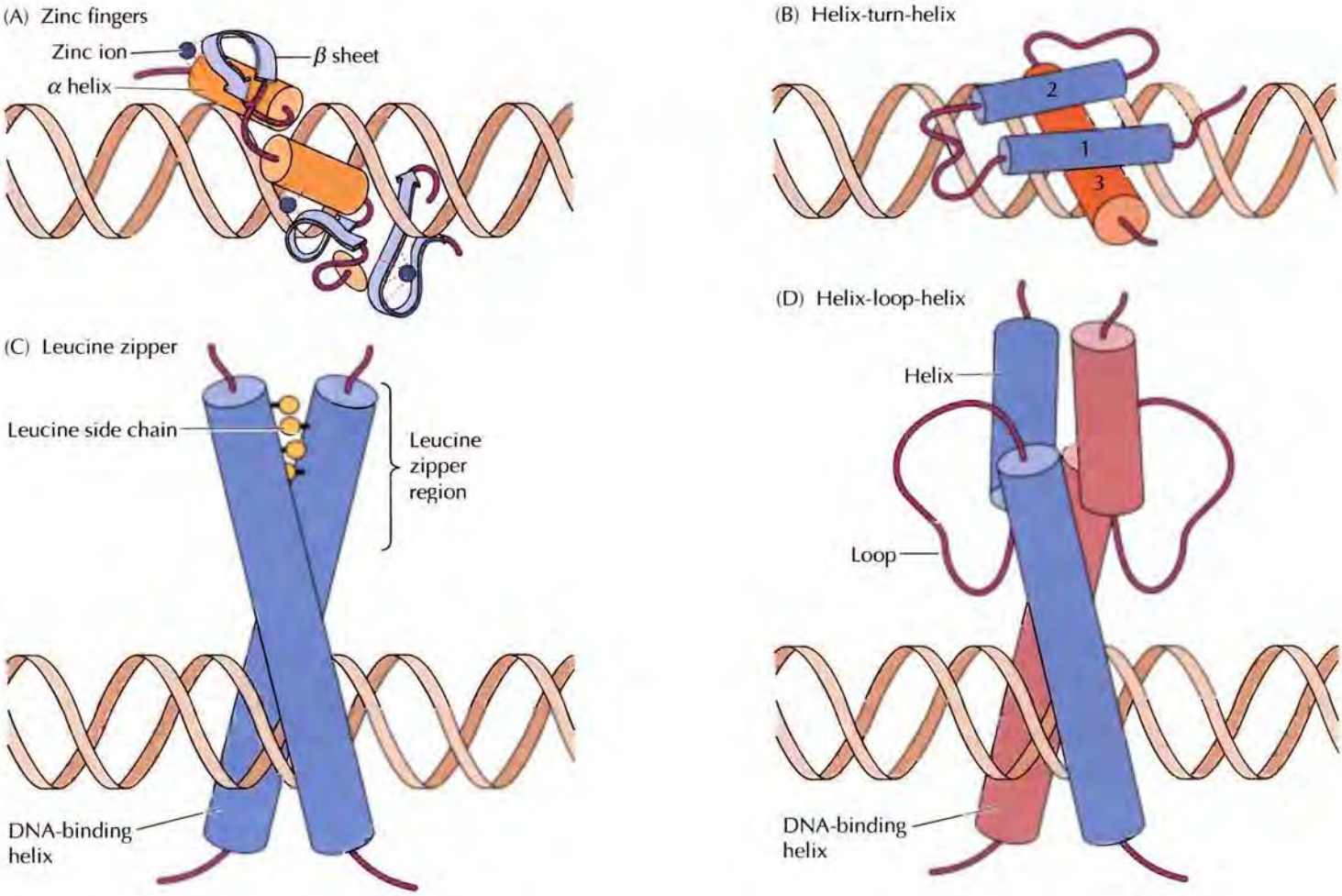


FIGURE 7.29 Examples of DNA-binding domains (A) Zinc finger domains consist of loops in which an α helix and a β sheet coordinately bind a zinc ion. (B) Helix-turn-helix domains consist of three (or in some cases four) helical regions. One helix (helix 3) makes most of the contacts with DNA, while helices 1 and 2 lie on top and stabilize the interaction. (C) The DNA-binding domains of leucine zipper proteins are formed from two distinct polypeptide chains. Interactions between the hydrophobic side chains of leucine residues exposed on one side of a helical region (the leucine zipper) are responsible for dimerization. Immediately following the leucine zipper is a DNA-binding helix, which is rich in basic amino acids. (D) Helix-loop-helix domains are similar to leucine zippers, except that the dimerization domains of these proteins each consist of two helical regions separated by a loop.

antennae grow out of the head of the fly (Figure 7.30). Genetic analysis of these mutants, pioneered by Ed Lewis in the 1940s, has shown that *Drosophila* contains nine homeotic genes, each of which specifies the identity of a different body segment. Molecular cloning and analysis of these genes then indicated that they contain conserved sequences of 180 base pairs (called **homeoboxes**) that encode DNA-binding domains (homeodomains) of transcription factors. A wide variety of similar homeodomain proteins have since been identified in fungi, plants, and other animals, including humans.

Two other families of DNA-binding proteins, **leucine zipper** and **helix-loop-helix** proteins, contain DNA-binding domains formed by dimerization of two polypeptide chains. The leucine zipper contains four or five leucine residues spaced at intervals of seven amino acids, resulting in their hydrophobic side chains being exposed at one side of a helical region. This region serves as the dimerization domain for the two protein subunits, which are held together by hydrophobic interactions between the leucine side chains. Immediately following the leucine zipper is a region rich in positively charged amino acids (lysine and arginine) that binds DNA. The helix-loop-helix proteins are similar in structure, except that their dimerization domains are each formed by two helical regions separated by a loop. An important feature of both leucine zipper and helix-loop-helix transcription factors is that different members of each family can dimerize with one

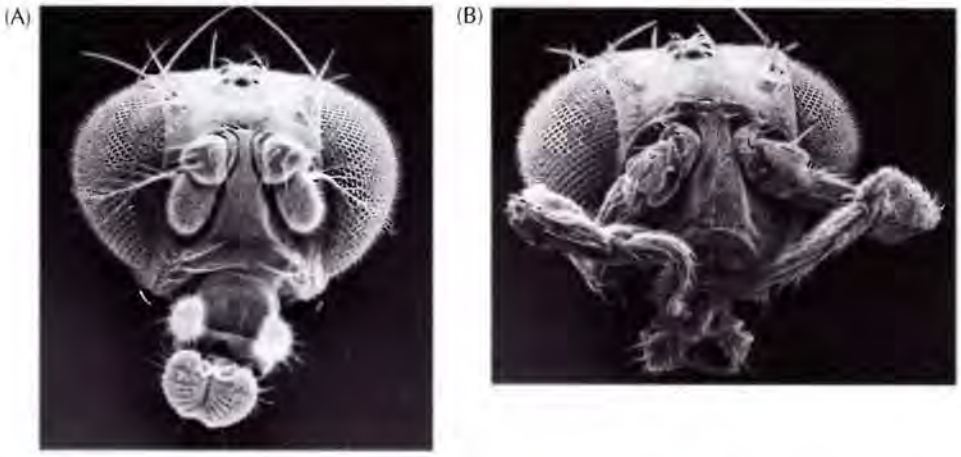


FIGURE 7.30 The Antennapedia mutation *Antennapedia* mutant flies have legs growing out of their heads in place of antennae. (A) Head of a normal fly. (B) Head of an *Antennapedia* mutant. (Courtesy of F. Rudolf Turner, Indiana University.)

another. Thus the combination of distinct protein subunits can form an expanded array of factors that can differ both in DNA sequence recognition and in transcription-stimulating activities. Both leucine zipper and helix-loop-helix proteins play important roles in regulating tissue-specific and inducible gene expression, and the formation of dimers between different members of these families is a critical aspect of the control of their function.

The activation domains of transcription factors are not as well characterized as their DNA-binding domains. Some, called acidic activation domains, are rich in negatively charged residues (aspartate and glutamate); others are rich in proline or glutamine residues. The activation domains of eukaryotic transcription factors stimulate transcription by two distinct mechanisms (Figure 7.31). First, they interact with Mediator proteins and general transcription factors, such as TFIIB or TFIID, to recruit RNA polymerase and facilitate the assembly of a transcription complex on the promoter, similar to transcriptional activators in bacteria (see Figure 7.10). In addition, eukaryotic transcription factors interact with a variety of **coactivators** that stimulate transcription by modifying chromatin structure, as dis-

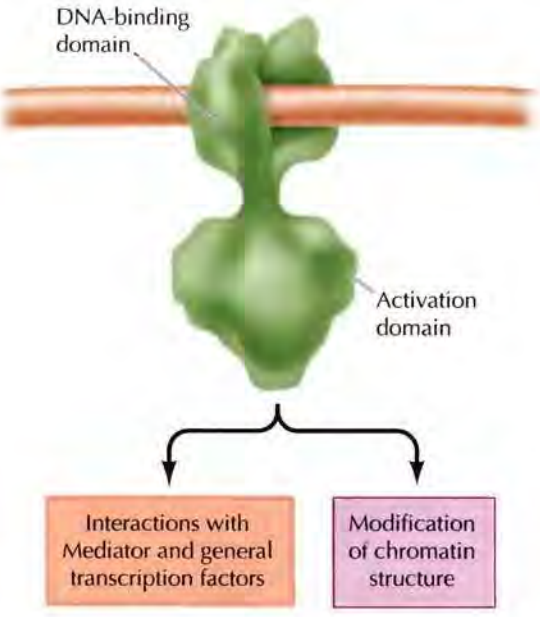


FIGURE 7.31 Action of transcriptional activators Eukaryotic activators stimulate transcription by two mechanisms: 1) they interact with Mediator proteins and general transcription factors to facilitate the assembly of a transcription complex and stimulate transcription, and 2) they interact with coactivators that facilitate transcription by modifying chromatin structure.

cussed later in this chapter. It is also important to note that activators not only regulate the initiation of transcription: Elongation and RNA processing can also be regulated, both by direct modulation of the activity of RNA polymerase and by effects on chromatin structure.

Eukaryotic Repressors

Gene expression in eukaryotic cells is regulated by repressors as well as by transcriptional activators. Like their prokaryotic counterparts, eukaryotic repressors bind to specific DNA sequences and inhibit transcription. In some cases, eukaryotic repressors simply interfere with the binding of other transcription factors to DNA (Figure 7.32A). For example, the binding of a repressor near the transcription start site can block the interaction of RNA polymerase or general transcription factors with the promoter, which is similar to the action of repressors in bacteria. Other repressors compete with activators for binding to specific regulatory sequences. Some such repressors contain the same DNA-binding domain as the activator but lack its activation domain. As a result, their binding to a promoter or enhancer blocks the binding of the activator, thereby inhibiting transcription.

In contrast to repressors that simply interfere with activator binding, many repressors (called active repressors) contain specific functional domains that inhibit transcription via protein-protein interactions (Figure 7.32B). The first such active repressor was described in 1990 during studies

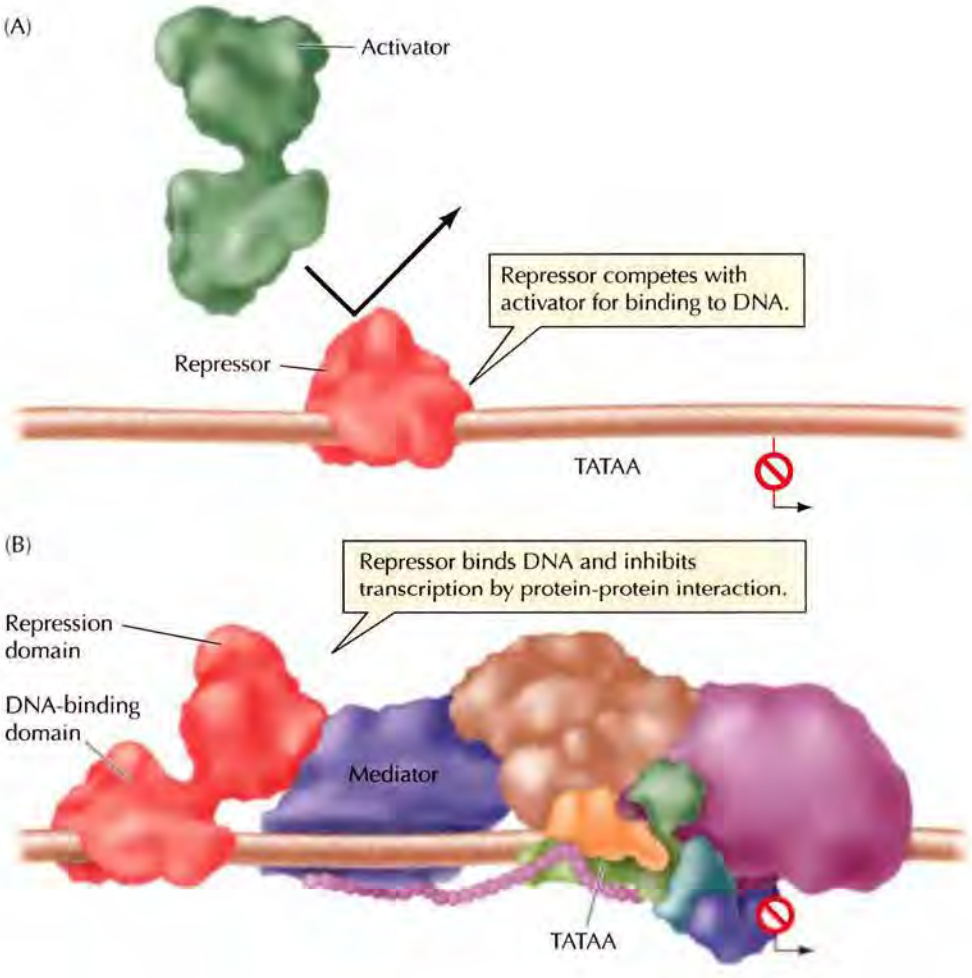


FIGURE 7.32 Action of eukaryotic repressors (A) Some repressors block the binding of activators to regulatory sequences. (B) Other repressors have active repression domains that inhibit transcription by interactions with Mediator proteins or general transcription factors, as well as with corepressors that act to modify chromatin structure.

of a gene called *Krüppel*, which is involved in embryonic development in *Drosophila*. Molecular analysis of the Krüppel protein demonstrated that it contains a discrete repression domain, linked to a zinc finger DNA-binding domain, which inhibits transcription via protein-protein interactions.

Many active repressors have since been found to play key roles in the regulation of transcription in animal cells, in many cases serving as critical regulators of cell growth and differentiation. As with transcriptional activators, several distinct types of repression domains have been identified. For example, the repression domain of Krüppel is rich in alanine residues, whereas other repression domains are rich in proline or acidic residues. The functional targets of repressors are also diverse: Repressors can inhibit transcription by interacting with specific activator proteins, with Mediator proteins or general transcription factors, and with **corepressors** that act by modifying chromatin structure.

The regulation of transcription by repressors as well as by activators considerably extends the range of mechanisms that control the expression of eukaryotic genes. One important role of repressors may be to inhibit the expression of tissue-specific genes in inappropriate cell types. For example, as noted earlier, a repressor-binding site in the immunoglobulin enhancer is thought to contribute to its tissue-specific expression by suppressing transcription in nonlymphoid cell types. Other repressors play key roles in the control of cell proliferation and differentiation in response to hormones and growth factors.

Relationship of Chromatin Structure to Transcription

As noted in the preceding discussion, both activators and repressors regulate transcription in eukaryotes not only by interacting with Mediator and other components of the transcriptional machinery but also by inducing changes in the structure of chromatin. Rather than being present within the nucleus as naked DNA, the DNA of all eukaryotic cells is tightly bound to histones. The basic structural unit of chromatin is the nucleosome, which consists of 147 base pairs of DNA wrapped around two molecules each of histones H2A, H2B, H3, and H4, with one molecule of histone H1 bound to the DNA as it enters the nucleosome core particle (see Figure 5.12). The chromatin is then further condensed by being coiled into higher-order structures organized into large loops of DNA. This packaging of eukaryotic DNA in chromatin clearly has important consequences in terms of its availability as a template for transcription, so chromatin structure is a critical aspect of gene expression in eukaryotic cells.

Actively transcribed genes are found in relatively decondensed chromatin, probably corresponding to the 30-nm chromatin fibers discussed in Chapter 5 (see Figure 5.13). For example, microscopic visualization of the polytene chromosomes of *Drosophila* indicates that regions of the genome that are actively engaged in RNA synthesis correspond to decondensed chromosome regions (Figure 7.33). Nonetheless, actively transcribed genes remain bound to histones and packaged in nucleosomes, so transcription factors and RNA polymerase are still faced with the problem of interacting with chromatin rather than with naked DNA. The tight winding of DNA around the nucleosome core particle is a major obstacle to transcription, affecting both the ability of transcription factors to bind DNA and the ability of RNA polymerase to transcribe through a chromatin template.

Several modifications are characteristic of transcriptionally active chromatin, including modifications of histones, rearrangements of nucleosomes, and the association of two nonhistone chromosomal proteins, called

FIGURE 7.33 Decondensed chromosome regions in *Drosophila*
A light micrograph showing decondensed regions of polytene chromosomes (arrows), which are active in RNA synthesis. (Courtesy of Joseph Gall, Carnegie Institute.)

HMGN proteins, with the nucleosomes of actively transcribed genes. The binding sites of the HMGN proteins on nucleosomes overlap the binding site of histone H1, and it appears that HMGN proteins stimulate transcription by affecting modifications of the core histones and altering the interaction of histone H1 with nucleosomes to maintain a decondensed chromatin structure.

Histone acetylation has been correlated with transcriptionally active chromatin in a wide variety of cell types (Figure 7.34). The core histones (H2A, H2B, H3 and H4) have two domains: a histone fold domain, which is involved in interactions with other histones and in wrapping DNA around the nucleosome core particle, and an amino-terminal tail domain, which extends outside of the nucleosome. The amino-terminal tail domains are rich in lysine and can be modified by acetylation at specific lysine residues.

Studies from two groups of researchers in 1996 provided direct links between histone acetylation and transcriptional regulation by demonstrating that transcriptional activators and repressors are associated with histone acetyltransferases and deacetylases, respectively. This association was first revealed by cloning a gene encoding a histone acetyltransferase from *Tetrahymena*. Unexpectedly, the sequence of this histone acetyltransferase was closely related to a previously known yeast transcriptional coactivator called Gcn5p. Further experiments revealed that Gcn5p has histone acetyltransferase activity, suggesting that transcriptional activation results directly from histone acetylation. These results have been extended by demonstrations that histone acetyltransferases are also associated with a number of mammalian transcriptional coactivators, as well as with the general transcription factor TFIID. Conversely, many transcriptional corepressors in both yeast and mammalian cells function as histone deacetylases, which remove the acetyl groups from histone tails. Histone acetylation is thus targeted directly by both transcriptional activators and repressors, indicating that it plays a key role in regulation of eukaryotic gene expression.

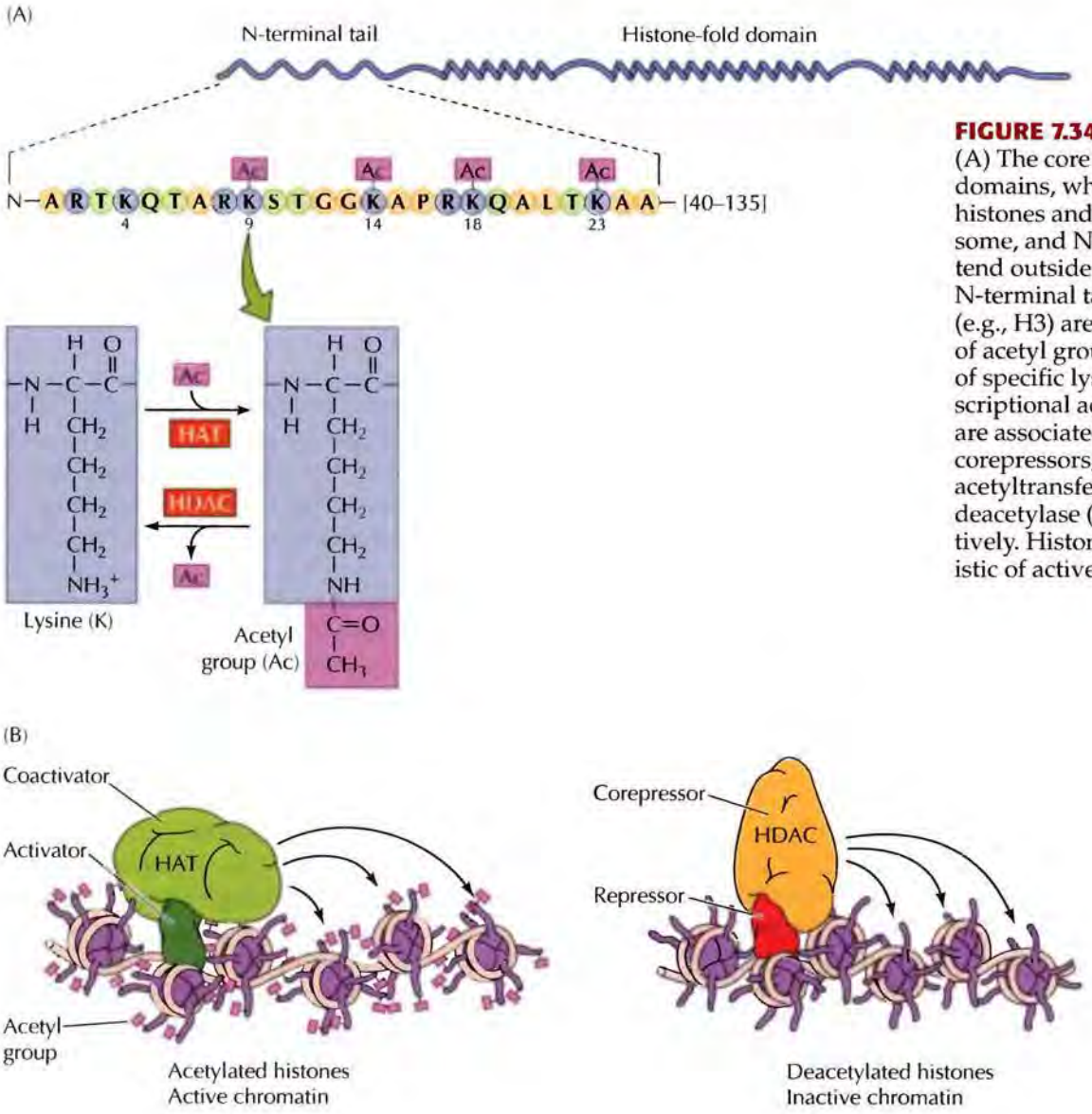


FIGURE 7.34 Histone acetylation
(A) The core histones have histone-fold domains, which interact with other histones and with DNA in the nucleosome, and N-terminal tails, which extend outside of the nucleosome. The N-terminal tails of the core histones (e.g., H3) are modified by the addition of acetyl groups (Ac) to the side chains of specific lysine residues. (B) Transcriptional activators and repressors are associated with coactivators and corepressors, which have histone acetyltransferase (HAT) and histone deacetylase (HDAC) activities, respectively. Histone acetylation is characteristic of actively transcribed chromatin.

Histones are modified not only by acetylation, but also by phosphorylation of serine residues, methylation of lysine and arginine residues, and addition of ubiquitin (a small peptide discussed in Chapter 8) to lysine residues. Like acetylation, these modifications occur at specific amino acid residues in the histone tails and are associated with changes in transcriptional activity (Figure 7.35). It appears that specific histone modifications affect gene expression by providing binding sites for other transcriptional regulatory proteins. According to this hypothesis, combinations of histone modifications constitute a "**histone code**" that regulates gene expression by recruiting other regulatory proteins to the chromatin template. For example, transcriptionally active chromatin is associated with several specific modifications of histone H3, including methylation of lysine-4, phosphorylation of serine-10, acetylation of lysines 9, 14, 18, and 23, and methylation of arginines 17 and 26. In contrast, methylation of lysines 9, 27, and 36 is associated with repression and chromatin condensation. Enzymes that catalyze the methylation of H3 lysine-9 are recruited to target genes by corepressors. The methylated H3 lysine-9 residues have further been shown to

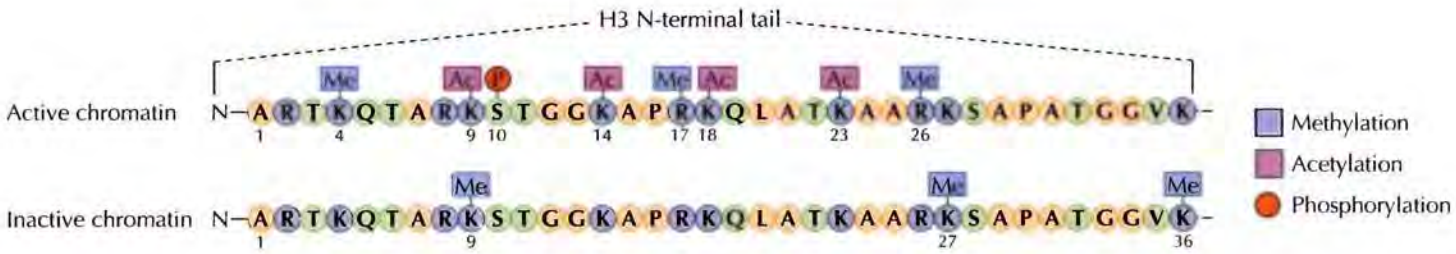


FIGURE 7.35 Patterns of histone modification Transcriptional activity of chromatin is affected by methylation and phosphorylation of specific amino acid residues in histone tails, as well as by their acetylation. Distinct patterns of histone modification are characteristic of transcriptionally active and inactive chromatin.

serve as binding sites for proteins that induce chromatin condensation, directly linking this histone modification to transcriptional repression and the formation of heterochromatin. It is notable that these modifications of histone tails may also regulate one another, leading to the establishment of distinct patterns of histone modification that correlate with stable modifications in transcriptional activity.

In contrast to the enzymes that regulate chromatin structure by modifying histones, **nucleosome remodeling factors** are protein complexes that alter the arrangement or structure of nucleosomes, without removing or covalently modifying the histones (Figure 7.36). One mechanism by which nucleosome remodeling factors act is to catalyze the sliding of histone octamers along the DNA molecule, thereby repositioning nucleosomes to change the accessibility of specific DNA sequences to transcription factors. Alternatively, nucleosome remodeling factors may act by inducing changes in the conformation of nucleosomes, again affecting the ability of specific DNA sequences to interact with transcriptional regulatory proteins. Like histone modifying enzymes, nucleosome remodeling factors can be recruited to DNA in association with either transcriptional activators or repressors, and can alter the arrangement of nucleosomes to either stimulate or inhibit transcription.

The recruitment of histone modifying enzymes and nucleosome remodeling factors by transcriptional activators stimulates the initiation of transcription by altering the chromatin structure of enhancer and promoter regions. However, following the initiation of transcription, RNA polymerase is still faced with the problem of transcriptional elongation through a chromatin template. This is facilitated by **elongation factors** that become associated with the phosphorylated C-terminal domain of RNA polymerase II when transcription is initiated (see Figure 7.14). These elongation factors include histone modifying enzymes (acetyltransferases and methyltransferases), as well as proteins that transiently disrupt the structure of nucleosomes during transcription. Transcriptional elongation is thus a complex process, which involves multiple proteins in addition to RNA polymerase. Although it has not been studied as thoroughly as transcription initiation, the regulation of transcriptional elongation presents an additional level at which gene expression can be controlled in eukaryotic cells.

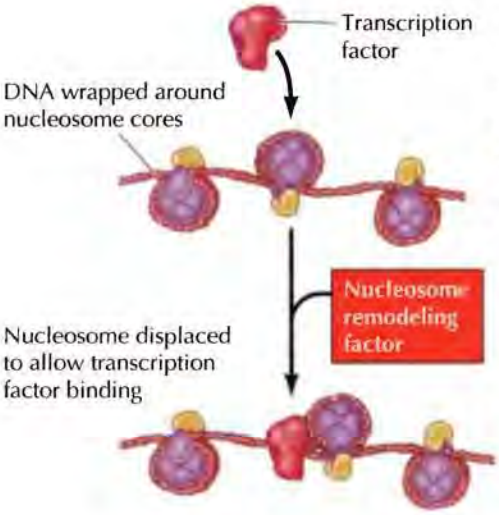


FIGURE 7.36 Nucleosome remodeling factors Nucleosome remodeling factors alter the arrangement or structure of nucleosomes. For example, a nucleosome remodeling factor can facilitate the binding of transcription factors to chromatin by repositioning nucleosomes on the DNA.

Regulation of Transcription by Noncoding RNAs

A series of recent advances indicate that gene expression can be regulated not only by the transcriptional regulatory proteins discussed so far but also by noncoding regulatory RNA molecules. One mode of action of noncoding regulatory RNAs is to inhibit translation by RNA interference–a phenomenon in which short double-stranded RNAs induce degradation of a homologous mRNA (see Figure 4.42). In addition, noncoding RNAs can repress transcription by inducing histone modifications that lead to chromatin condensation and the formation of heterochromatin. **MicroRNAs (miRNAs)** are naturally occurring short noncoding RNAs that function as normal regulators of gene expression. Hundreds of genes encode miRNAs in both plants and animals, so it appears that gene regulation by these noncoding RNAs is a widespread phenomenon, even though the functions of most miRNAs have yet to be determined.

miRNAs are transcribed as precursors containing inverted repeats that form stem-loop structures (Figure 7.37). These precursors are then cleaved to yield mature miRNAs, which are short double-stranded RNAs of approximately 20–25 nucleotides. In RNA interference, miRNAs associate with the RNA-induced silencing complex (RISC), within which the two strands of miRNA separate and target homologous mRNAs for cleavage (see Figure 4.42). In transcriptional repression, the miRNAs associate with a different protein complex called the RITS (RNA-induced transcriptional silencing) complex. The separated miRNA strands then guide the RITS complex to the homologous gene, most likely by base pairing with the mRNA transcript in association with RNA polymerase II. RITS then represses transcription by recruiting a histone methyltransferase that methylates histone H3 lysine-9, leading to the formation of heterochromatin.

The phenomenon of **X chromosome inactivation** provides another example of the role of a noncoding RNA in regulating gene expression in mammals. In many animals, including humans, females have two X chromosomes, and males have one X and one Y chromosome. The X chromosome contains hundreds of genes that are not present on the much smaller Y chromosome (see Figure 5.29). Thus females have twice as many copies of most X chromosome genes as males have. Despite this difference, female and male cells contain equal amounts of the proteins encoded by the majority of X chromosome genes. This results from a dosage compensation mechanism in which most of the genes on one of the two X chromosomes in female cells are inactivated by being converted to heterochromatin early in development. Consequently, only one copy of most genes located on the X chromosome is available for transcription in either female or male cells.

Although the mechanism of X chromosome inactivation is not yet fully understood, the key element appears to be a noncoding RNA transcribed from a regulatory gene, called *Xist*, on the inactive X chromosome. *Xist* RNA remains localized to the inactive X, binding to and coating this chro-

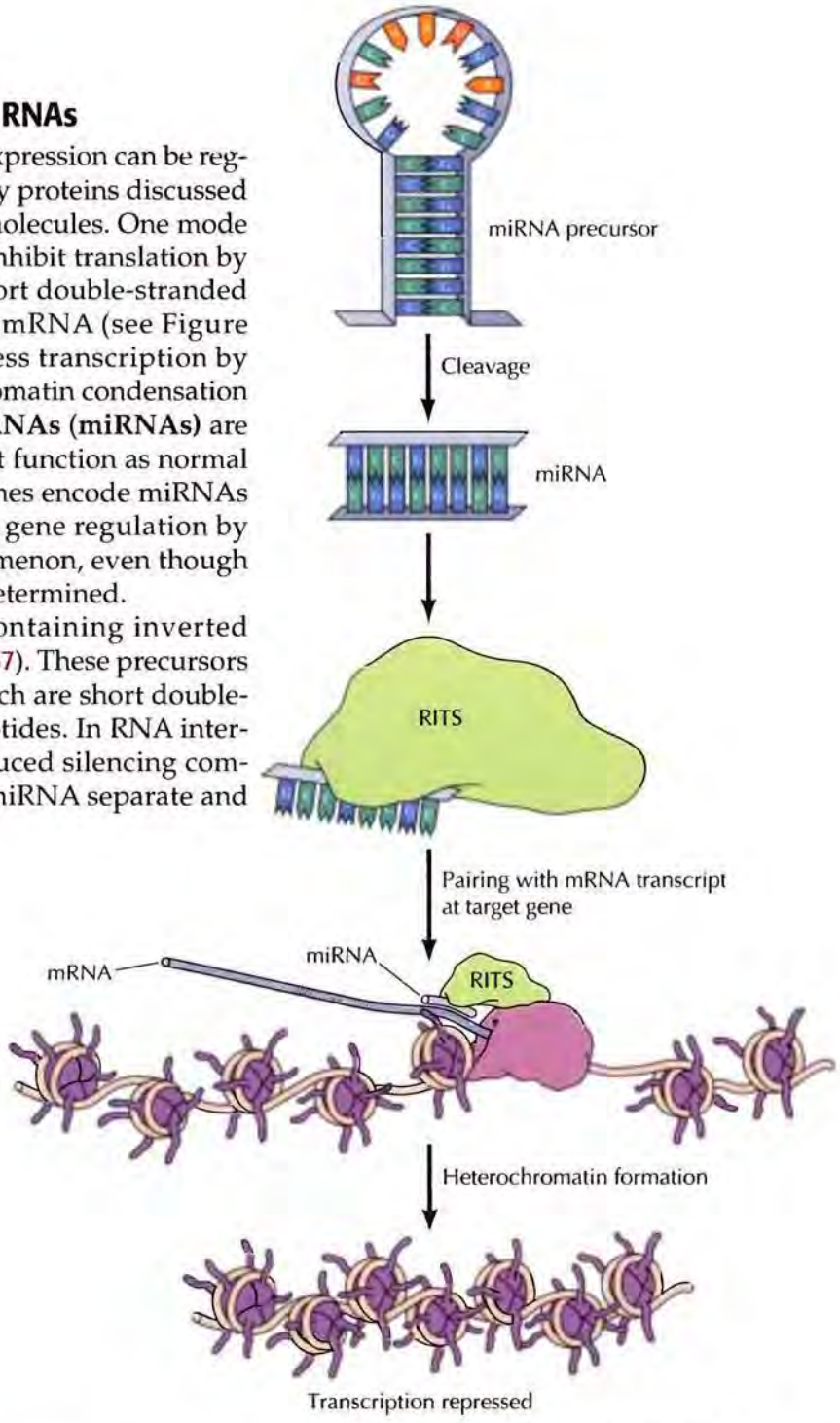


FIGURE 7.37 Regulation of transcription by miRNAs miRNAs are transcribed as precursors containing stem-loop structures, which are cleaved to yield short double-stranded miRNAs. Transcriptional repression is mediated by association of miRNAs with the RITS complex. The miRNA then guides RITS to the homologous target gene, most likely by base pairing with nascent mRNA transcripts associated with RNA polymerase II. RITS then recruits a histone methyltransferase that methylates lysine-9 of histone H3, leading to formation of heterochromatin and repression of transcription.

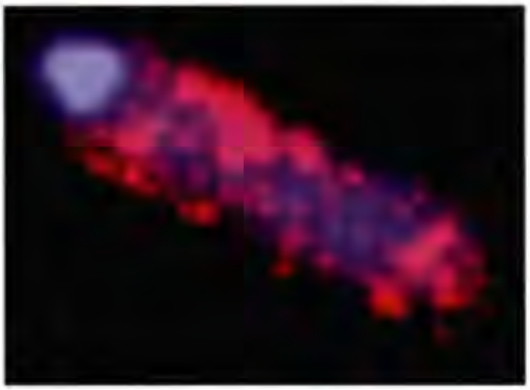


FIGURE 7.38 X chromosome inactivation The inactive X chromosome (blue) is coated by *Xist* RNA (red). (From B. Panning and R. Jaenisch, 1998. *Cell* 93: 305.)

mosome (Figure 7.38). This leads to the recruitment of a protein complex that induces methylation of histone H3 lysine-27 and lysine-9, leading to chromatin condensation and conversion of most of the inactive X to heterochromatin.

DNA Methylation

The methylation of DNA is another general mechanism that controls transcription in eukaryotes. Cytosine residues in DNA of fungi, plants, and animals can be modified by the addition of methyl groups at the 5-carbon position (Figure 7.39). DNA is methylated specifically at the cytosines (C) that precede guanines (G) in the DNA chain (CpG dinucleotides), and this methylation is correlated with transcriptional repression. Methylation commonly occurs within transposable elements, and it appears that methylation plays a key role in suppressing the movement of transposons throughout the genome. In addition, DNA methylation is associated with transcriptional repression of some genes, in concert with alterations in chromatin structure. In plants, miRNAs direct DNA methylation as well as chromatin modifications of repressed genes, although it is not clear whether this also occurs in animals. However, genes on the inactive X chromosome in mammals become methylated following transcriptional repression by *Xist* RNA, so both DNA methylation as well as histone modification appear to play an important role in X inactivation.

One important regulatory role of DNA methylation has been established in the phenomenon known as **genomic imprinting**, which controls the expression of some genes involved in the development of mammalian embryos. In most cases, both the paternal and maternal alleles of a gene are expressed in diploid cells. However, there are some imprinted genes (about 70 have been described in mice and humans) whose expression depends on whether they are inherited from the mother or from the father. In some cases, only the paternal allele of an imprinted gene is expressed, and the maternal allele is transcriptionally inactive. For other imprinted genes, the maternal allele is expressed and the paternal allele is inactive.

DNA methylation appears to play a key role in distinguishing between the paternal and maternal alleles of imprinted genes. A good example is the gene *H19*, which is transcribed only from the maternal copy (Figure 7.40). The *H19* gene is specifically methylated during the development of male, but not female, germ cells. The union of sperm and egg at fertilization therefore yields an embryo containing a methylated paternal allele and an unmethylated maternal allele of the gene. These differences in methylation are maintained following DNA replication by an enzyme that specifically methylates CpG sequences of a daughter strand that is hydrogen-bonded to a methylated parental strand (Figure 7.41). The paternal *H19* allele therefore remains methylated, and transcriptionally inactive, in embryonic cells and somatic tissues. However, the paternal *H19* allele becomes demethylated in the germ line, allowing a new pattern of methylation to be established for transmittal to the next generation.

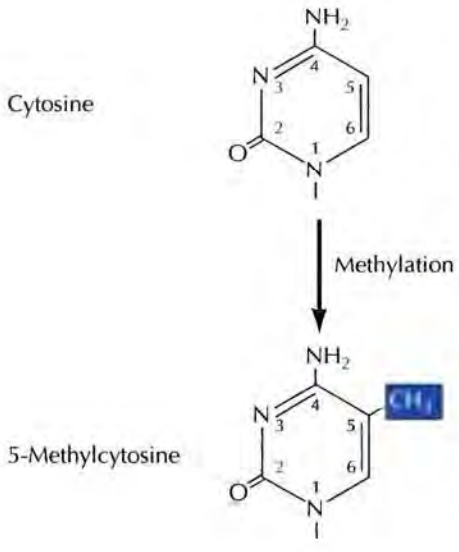


FIGURE 7.39 DNA methylation A methyl group is added to the 5-carbon position of cytosine residues in DNA.

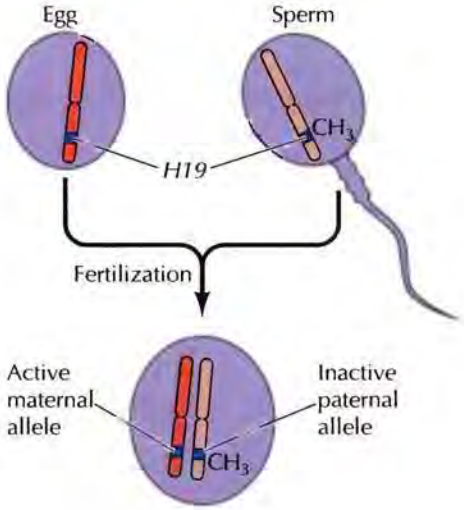


FIGURE 7.40 Genomic imprinting The *H19* gene is specifically methylated during development of male germ cells. Therefore sperm contain a methylated *H19* allele and eggs contain an unmethylated allele. Following fertilization, the methylated paternal allele remains transcriptionally inactive, and only the unmethylated maternal allele is expressed in the embryo.

RNA Processing and Turnover

Although transcription is the first and most highly regulated step in gene expression, it is usually only the beginning of the series of events required to produce a functional RNA. Most newly synthesized RNAs must be modified in various ways to be converted to their functional forms. Bacterial mRNAs are an exception; they are used immediately as templates for protein synthesis while still being transcribed. However, the primary transcripts of both rRNAs and tRNAs must undergo a series of processing steps in prokaryotic as well as eukaryotic cells. Primary transcripts of eukaryotic mRNAs similarly undergo extensive modifications, including the removal of introns by splicing, before they are transported from the nucleus to the cytoplasm to serve as templates for protein synthesis. Regulation of these processing steps provides an additional level of control of gene expression, as does regulation of the rates at which different mRNAs are subsequently degraded within the cell.

Processing of Ribosomal and Transfer RNAs

The basic processing of ribosomal and transfer RNAs in prokaryotic and eukaryotic cells is similar, as might be expected given the fundamental roles of these RNAs in protein synthesis. As discussed previously, eukaryotes

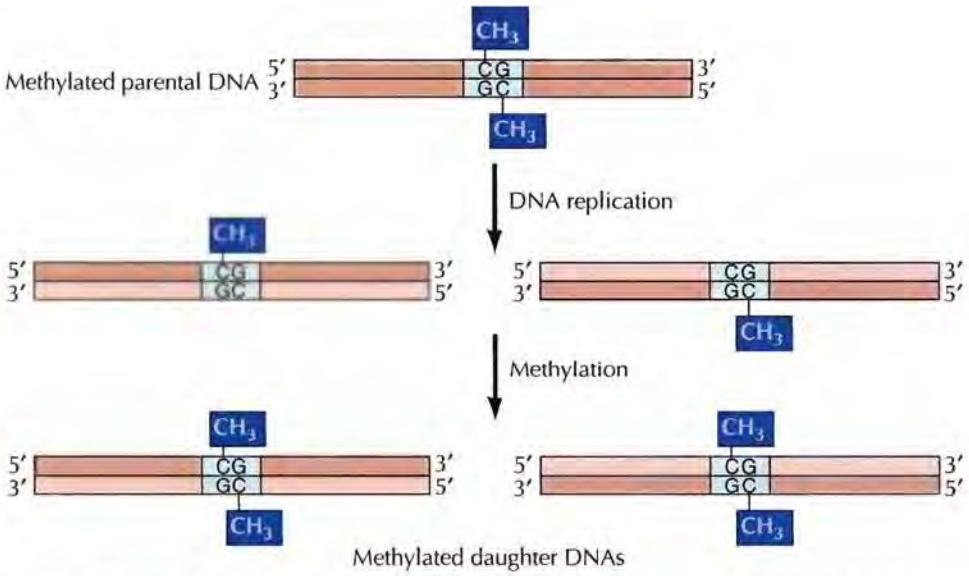


FIGURE 7.41 Maintenance of methylation patterns In parental DNA, both strands are methylated at complementary CpG sequences. Following replication, only the parental strand of each daughter molecule is methylated. The newly synthesized daughter strands are then methylated by an enzyme that specifically recognizes CpG sequences opposite a methylation site.

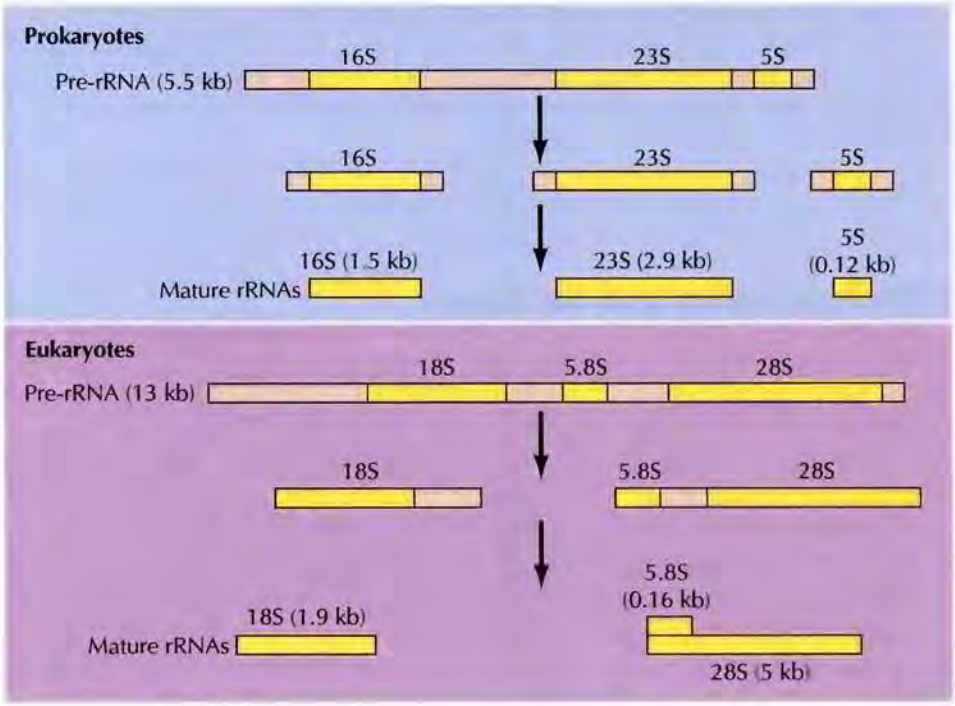


FIGURE 7.42 Processing of ribosomal RNAs Prokaryotic cells contain three rRNAs (16S, 23S, and 5S), which are formed by cleavage of a pre-rRNA transcript. Eukaryotic cells (e.g., human cells) contain four rRNAs. One of these (5S rRNA) is transcribed from a separate gene; the other three (18S, 28S, and 5.8S) are derived from a common pre-rRNA. Following cleavage, the 5.8S rRNA (which is unique to eukaryotes) becomes hydrogen-bonded to 28S rRNA.

have four species of ribosomal RNAs (see Table 7.1), three of which (the 28S, 18S, and 5.8S rRNAs) are derived by cleavage of a single long precursor transcript, called a **pre-rRNA** (Figure 7.42). Prokaryotes have three ribosomal RNAs (23S, 16S, and 5S), which are equivalent to the 28S, 18S, and 5S rRNAs of eukaryotic cells and are also formed by the processing of a single pre-rRNA transcript. The only rRNA that is not processed extensively is the 5S rRNA in eukaryotes, which is transcribed from a separate gene.

Prokaryotic and eukaryotic pre-rRNAs are processed in several steps. Initial cleavages of bacterial pre-rRNA yield separate precursors for the three individual rRNAs; these are then further processed by secondary cleavages to the final products. In eukaryotic cells, pre-rRNA is first cleaved at a site adjacent to the 5.8S rRNA on its 5′ side, yielding two separate precursors that contain the 18S and the 28S + 5.8S rRNAs, respectively. Further cleavages then convert these to their final products, with the 5.8S rRNA becoming hydrogen-bonded to the 28S molecule. In addition to these cleavages, rRNA processing involves the addition of methyl groups to the bases and sugar moieties of specific nucleotides and the conversion of some uridines to pseudouridines. Processing of rRNA takes place within the nucleolus of eukaryotic cells, and will be discussed in detail in Chapter 9.

Like rRNAs, tRNAs in both bacteria and eukaryotes are synthesized as longer precursor molecules (**pre-tRNAs**), some of which contain several individual tRNA sequences (Figure 7.43). In bacteria, some tRNAs are included in the pre-rRNA transcripts. The processing of the 5′ end of pre-tRNAs involves cleavage by an enzyme called **RNase P**, which is of special

interest because it is a prototypical model of a reaction catalyzed by an RNA enzyme. RNase P consists of RNA and protein molecules, both of which are required for maximal activity. In 1983 Sidney Altman and his colleagues demonstrated that the isolated RNA component of RNase P is itself capable of catalyzing pre-tRNA cleavage. These experiments established that RNase P is a **ribozyme**—an enzyme in which RNA rather than protein is responsible for catalytic activity.

The 3′ end of tRNAs is generated by the action of a conventional protein RNase, but the processing of this end of the tRNA molecule also involves an unusual activity: the addition of a CCA terminus. All tRNAs have the sequence CCA at their 3′ ends. This sequence is the site of amino acid attachment, so it is required for tRNA function during protein synthesis. The CCA terminus is encoded in the DNA of some tRNA genes, but in others it is not, instead being added as an RNA processing step by an enzyme that recognizes and adds CCA to the 3′ end of all tRNAs that lack this sequence.

Another unusual aspect of tRNA processing is the extensive modification of bases in tRNA molecules. Approximately 10% of the bases in tRNAs are altered to yield a variety of modified nucleotides at specific positions in tRNA molecules (see Figure 7.43). The functions of most of these modified bases are unknown, but some play important roles in protein synthesis by altering the base-pairing properties of the tRNA molecule (see Chapter 8).

Some pre-tRNAs, as well as pre-rRNAs in a few organisms, contain introns that are removed by splicing. In contrast to other splicing reactions,

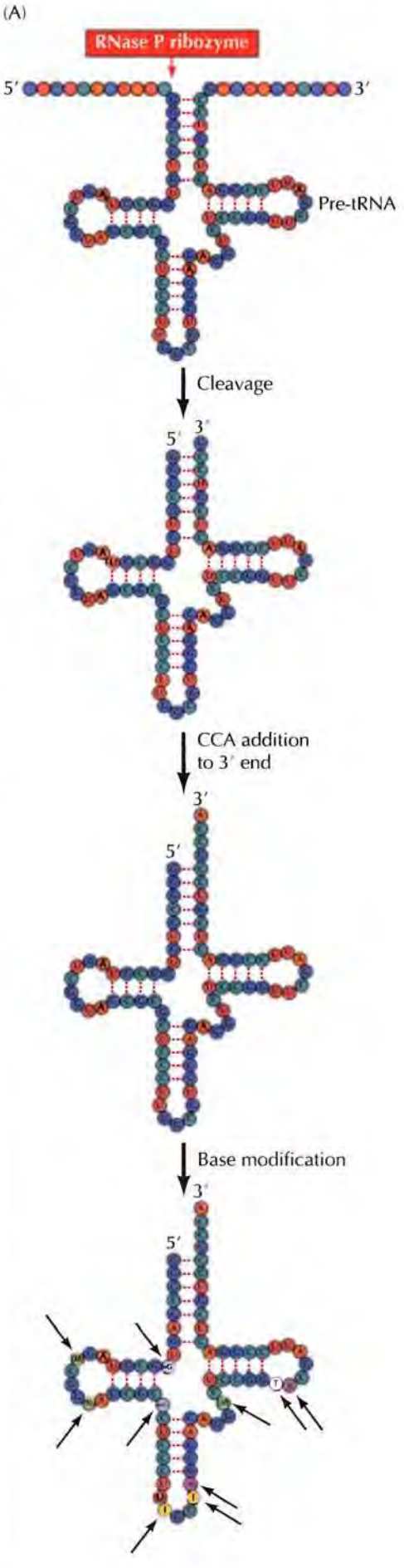
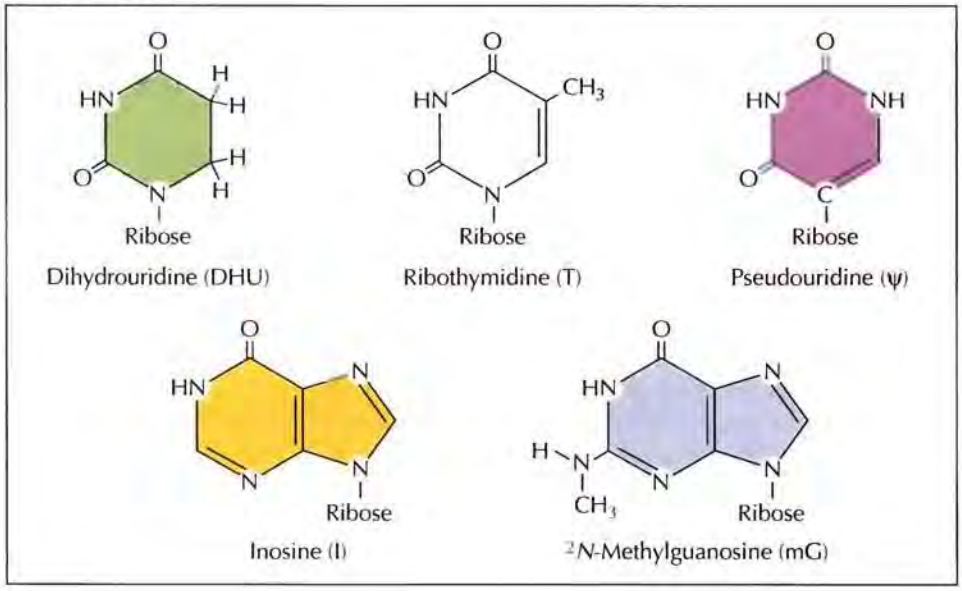


FIGURE 7.43 Processing of transfer RNAs (A) Transfer RNAs are derived from pre-tRNAs, some of which contain several individual tRNA molecules. Cleavage at the 5′ end of the tRNA is catalyzed by the RNase P ribozyme; cleavage at the 3′ end is catalyzed by a conventional protein RNase. A CCA terminus is then added to the 3′ end of many tRNAs in a posttranscriptional processing step. Finally, some bases are modified at characteristic positions in the tRNA molecule. In this example, these modified nucleosides include dihydrouridine (DHU), methylguanosine (mG), inosine (I), ribothymidine (T), and pseudouridine (ψ). (B) Structure of modified bases. Ribothymidine, dihydrouridine, and pseudouridine are formed by modification of uridines in tRNA. Inosine and methylguanosine are formed by the modification of guanosines.

(B) Modified bases



which (as discussed in the next section) involve the activities of catalytic RNAs, tRNA splicing is mediated by conventional protein enzymes. An endonuclease cleaves the pre-tRNA at the splice sites to excise the intron, followed by joining of the exons to form a mature tRNA molecule.

7.2 WEBSITE ANIMATION

RNA Processing

Eukaryotic cells must process the RNA product of transcription to form a mature, translatable messenger RNA.

Processing of mRNA in Eukaryotes

In contrast to the processing of ribosomal and transfer RNAs, the processing of messenger RNAs represents a major difference between prokaryotic and eukaryotic cells. In bacteria, ribosomes have immediate access to mRNA and translation begins on the nascent mRNA chain while transcription is still in progress. In eukaryotes, mRNA synthesized in the nucleus must first be transported to the cytoplasm before it can be used as a template for protein synthesis. Moreover, the initial products of transcription in eukaryotic cells (**pre-mRNAs**) are extensively modified before export from the nucleus. The processing of mRNA includes modification of both ends of the initial transcript, as well as the removal of introns from its middle (Figure 7.44). Rather than occurring as independent events following synthesis of a pre-mRNA, these processing reactions are coupled to transcription so that mRNA synthesis and processing are closely coordinated steps in gene expression. The C-terminal domain (CTD) of RNA polymerase II plays a key role in coordinating these processes by serving as a binding site for the enzyme complexes involved in mRNA processing. The association of these processing enzymes with the CTD of polymerase II accounts for their speci-

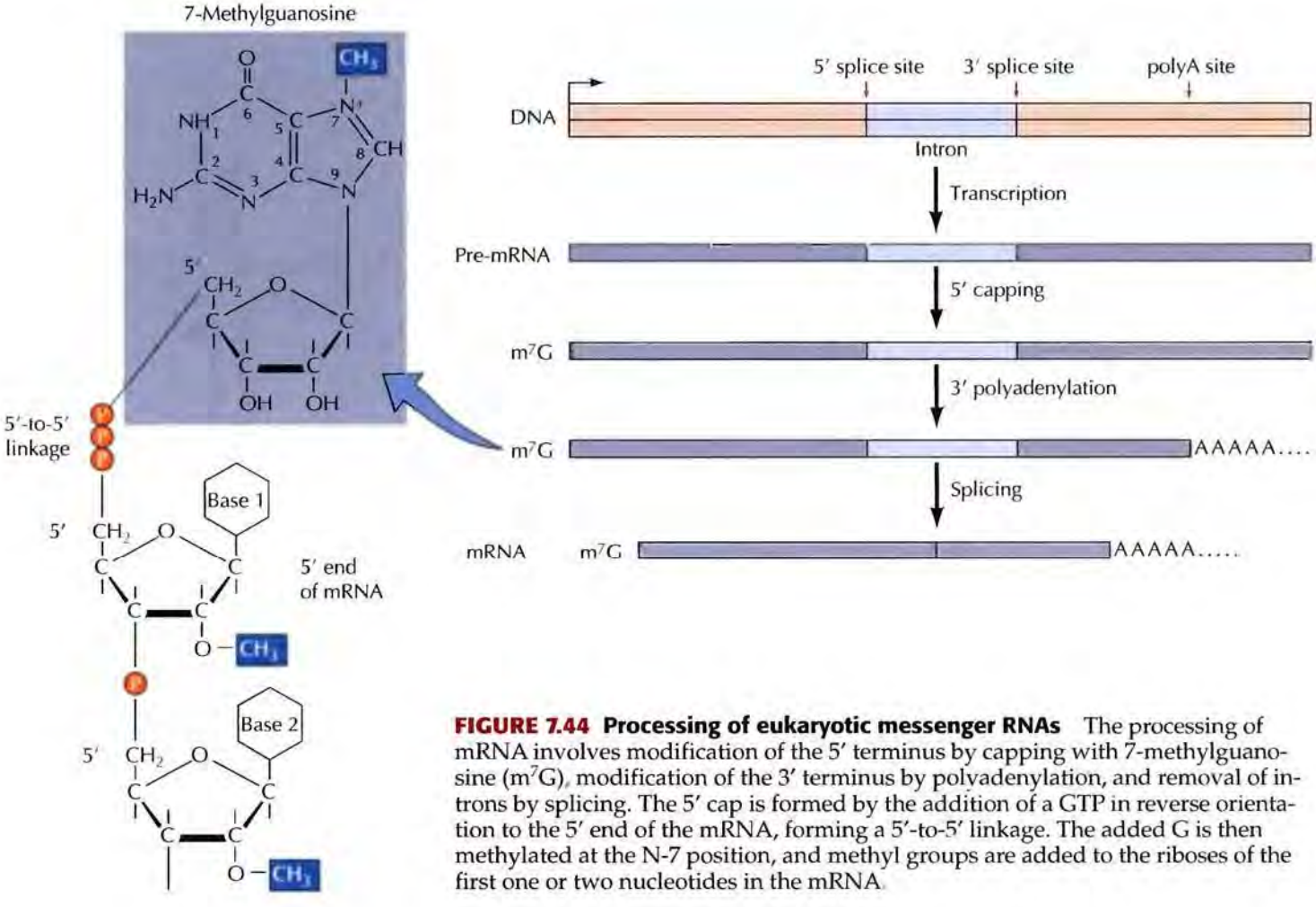


FIGURE 7.44 Processing of eukaryotic messenger RNAs The processing of mRNA involves modification of the 5′ terminus by capping with 7-methylguanosine (m^7G), modification of the 3′ terminus by polyadenylation, and removal of introns by splicing. The 5′ cap is formed by the addition of a GTP in reverse orientation to the 5′ end of the mRNA, forming a 5′-to-5′ linkage. The added G is then methylated at the N-7 position, and methyl groups are added to the riboses of the first one or two nucleotides in the mRNA.

ficity in processing mRNAs; polymerases I and III lack a CTD, so their transcripts are not processed by the same enzyme complexes.

The first step in mRNA processing is the modification of the 5′ end of the transcript by the addition of a structure called a **7-methylguanosine cap**. The enzymes responsible for capping are recruited to the phosphorylated CTD following initiation of transcription, and the cap is added after transcription of the first 20–30 nucleotides of the RNA. Capping is initiated by the addition of a GTP in reverse orientation to the 5′ terminal nucleotide of the RNA. Then methyl groups are added to this G residue and to the ribose moieties of one or two 5′ nucleotides of the RNA chain. The 5′ cap stabilizes the RNA, as well as aligning eukaryotic mRNAs on the ribosome during translation (see Chapter 8).

The 3′ end of most eukaryotic mRNAs is defined not by termination of transcription but by cleavage of the primary transcript and addition of a **poly-A tail**—a processing reaction called **polyadenylation** (Figure 7.45). The signals for polyadenylation include a highly conserved hexanucleotide (AAUAAA in mammalian cells), which is located 10 to 30 nucleotides upstream of the site of polyadenylation, and a G-U rich downstream sequence element. In addition, some genes have a U-rich sequence element upstream of the AAUAAA. These sequences are recognized by a complex of proteins, including an endonuclease that cleaves the RNA chain and a separate poly-A polymerase that adds a poly-A tail of about 200 nucleotides to the transcript. These processing enzymes are associated with the phosphorylated CTD of RNA polymerase II, and may travel with the polymerase all the way from the transcription initiation site. Cleavage and polyadenylation is followed by degradation of the RNA that has been synthesized downstream of the site of poly-A addition, resulting in the termination of transcription.

Almost all mRNAs in eukaryotes are polyadenylated, and poly-A tails have been shown to regulate both translation and mRNA stability. In addition, polyadenylation plays an important regulatory role in early development, where changes in the length of poly-A tails control mRNA translation. For example, many mRNAs are stored in unfertilized eggs in an untranslated form with short poly-A tails (usually 30 to 50 nucleotides long). Fertilization stimulates the lengthening of the poly-A tails of these stored mRNAs, which in turn activates their translation and the synthesis of proteins required for early embryonic development.

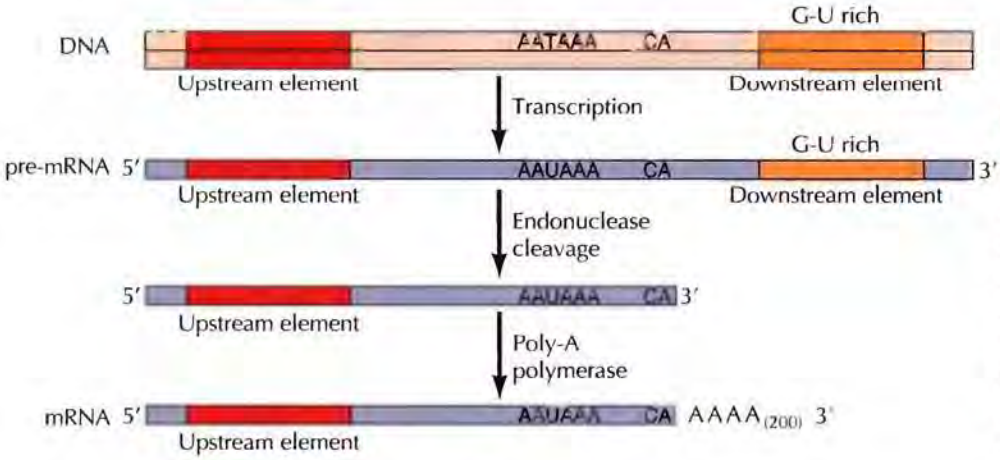


FIGURE 7.45 Formation of the 3′ ends of eukaryotic mRNAs
Polyadenylation signals in mammalian cells consist of the hexanucleotide AAUAAA in addition to upstream and downstream (G-U rich) elements. An endonuclease cleaves the pre-mRNA 10 to 30 nucleotides downstream of the AAUAAA, usually at a CA sequence. Poly-A polymerase then adds a poly-A tail consisting of about 200 adenines (A) to the 3′ end of the RNA.

The most striking modification of pre-mRNAs is the removal of introns by splicing. As discussed in Chapter 5, the coding sequences of most eukaryotic genes are interrupted by noncoding sequences (introns) that are precisely excised from the mature mRNA. In mammals, most genes contain multiple introns, which typically account for about ten times more pre-mRNA sequences than do the exons. The unexpected discovery of introns in 1977 generated an active research effort directed toward understanding the mechanism of splicing, which had to be highly specific to yield functional mRNAs. Further studies of splicing have not only illuminated new mechanisms of gene regulation; they have also revealed novel catalytic activities of RNA molecules.

Splicing Mechanisms

The key to understanding pre-mRNA splicing was the development of *in vitro* systems that efficiently carried out the splicing reaction (Figure 7.46). Pre-mRNAs were synthesized *in vitro* by the cloning of structural genes (with their introns) adjacent to promoters for bacteriophage RNA polymerases, which could readily be isolated in large quantities. Transcription

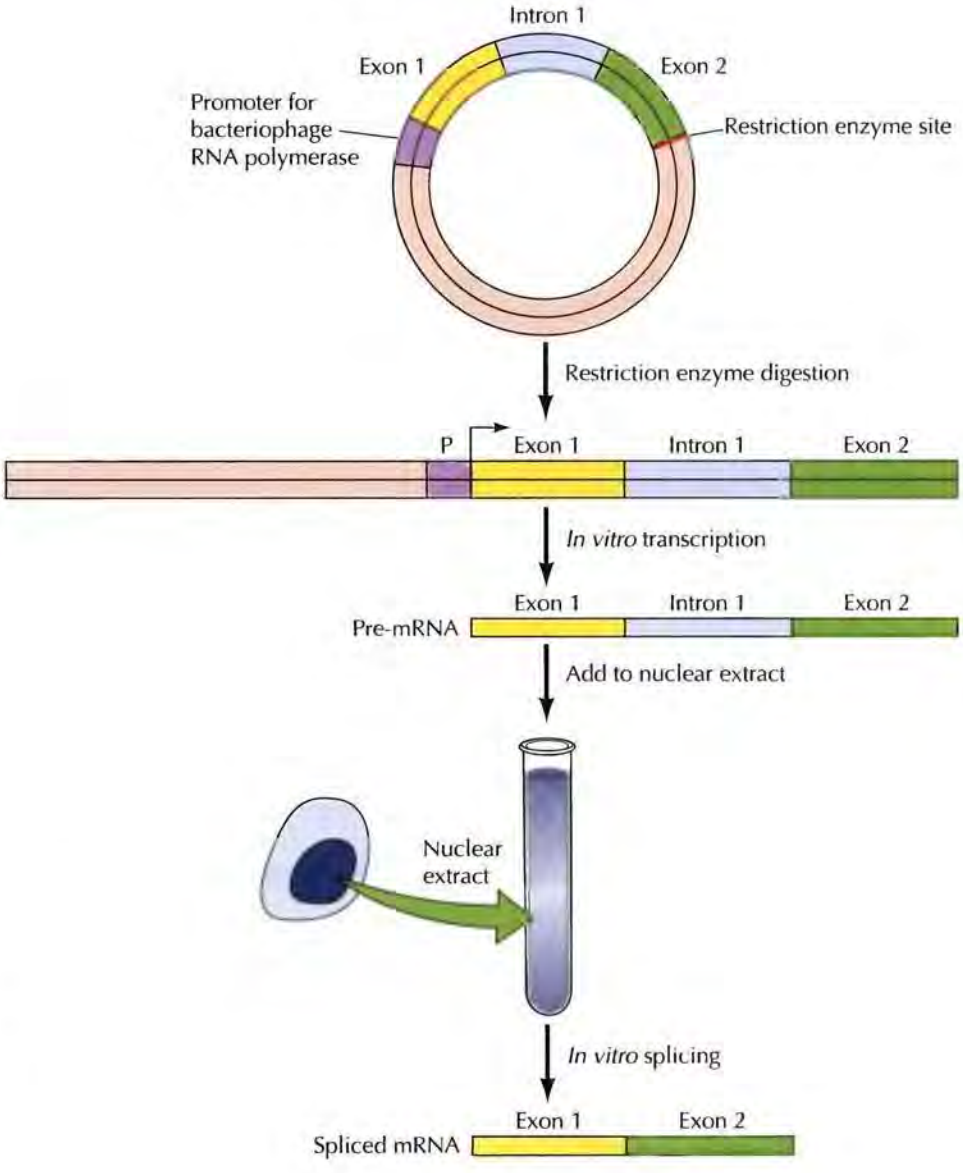


FIGURE 7.46 *In vitro* splicing A gene containing an intron is cloned downstream of a promoter (P) recognized by a bacteriophage RNA polymerase. The plasmid is digested with a restriction enzyme that cleaves at the 3′ end of the inserted gene to yield a linear DNA molecule. This DNA is then transcribed *in vitro* with the bacteriophage polymerase to produce pre-mRNA. Splicing reactions can then be studied *in vitro* by addition of this pre-mRNA to nuclear extracts of mammalian cells.

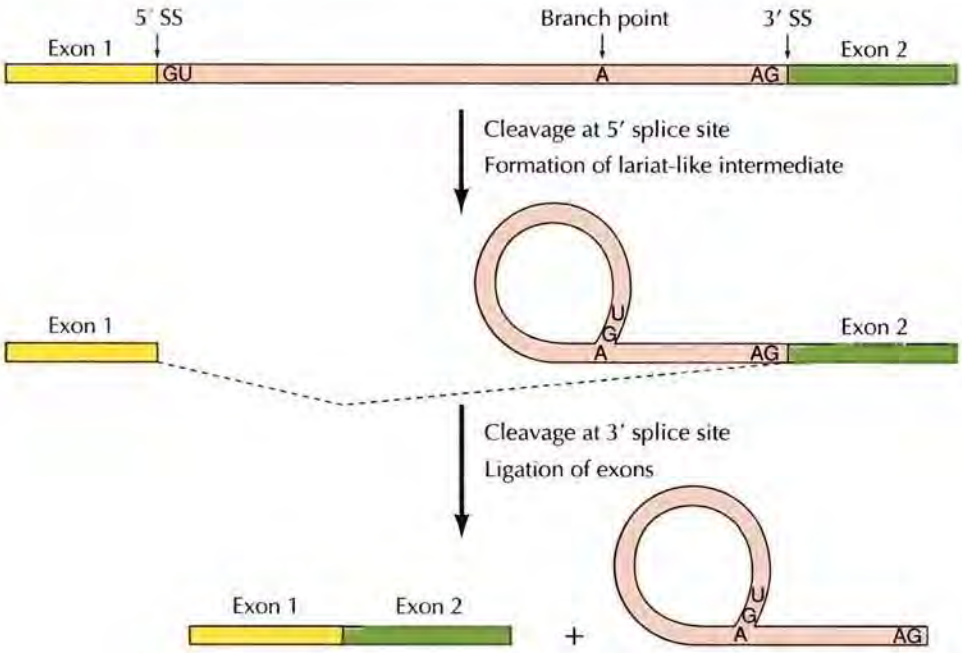


FIGURE 7.47 Splicing of pre-mRNA The splicing reaction proceeds in two steps. The first step involves cleavage at the 5′ splice site (SS) and joining of the 5′ end of the intron to an A within the intron (the branch point). This reaction yields a lariat-like intermediate in which the intron forms a loop. The second step is cleavage at the 3′ splice site and simultaneous ligation of the exons, resulting in excision of the intron as a lariat-like structure.

of these plasmids could then be used to prepare large amounts of pre-mRNAs that, when added to nuclear extracts of mammalian cells, were found to be correctly spliced. As with transcription, the use of such *in vitro* systems has allowed splicing to be analyzed in much greater detail than would have been possible in intact cells.

Analysis of the reaction products and intermediates formed *in vitro* revealed that pre-mRNA splicing proceeds in two steps (Figure 7.47). First, the pre-mRNA is cleaved at the 5′ splice site, and the 5′ end of the intron is joined to an adenine nucleotide within the intron (near its 3′ end). In this step, an unusual bond forms between the 5′ end of the intron and the 2′ hydroxyl group of the adenine nucleotide. The resulting intermediate is a lariat-like structure in which the intron forms a loop. The second step in splicing then proceeds with simultaneous cleavage at the 3′ splice site and ligation of the two exons. The intron is thus excised as a lariat-like structure, which is then linearized and degraded within the nucleus of intact cells.

These reactions define three critical sequence elements of pre-mRNAs: sequences at the 5′ splice site, sequences at the 3′ splice site, and sequences within the intron at the branch point (the point at which the 5′ end of the intron becomes ligated to form the lariat-like structure) (see Figure 7.47). Pre-mRNAs contain similar consensus sequences at each of these positions, allowing the splicing apparatus to recognize pre-mRNAs and carry out the cleavage and ligation reactions involved in the splicing process.

Biochemical analysis of nuclear extracts has revealed that splicing takes place in large complexes, called **spliceosomes**, composed of proteins and RNAs. The RNA components of the spliceosome are five types of **small nuclear RNAs (snRNAs)** called U1, U2, U4, U5, and U6. These snRNAs, which range in size from approximately 50 to nearly 200 nucleotides, are complexed with six to ten protein molecules to form **small nuclear ribonucleoprotein particles (snRNPs)**, which play central roles in the splicing process. The U1, U2, and U5 snRNPs each contain a single snRNA molecule, whereas U4 and U6 snRNAs are complexed to each other in a single snRNP.

KEY EXPERIMENT

The Discovery of snRNPs

Antibodies to small nuclear RNAs complexed with proteins are produced by patients with systemic lupus erythematosus

Michael R. Lerner and Joan A. Steitz
Yale University, New Haven, Connecticut

Proceedings of the National Academy of Sciences, USA, 1979, Volume 76, pages 5495–5499

Joan Steitz

The Context

The discovery of introns in 1977 implied that a totally unanticipated processing reaction was required to produce mRNA in eukaryotic cells. Introns had to be precisely excised from pre-mRNA, followed by the joining of exons to yield a mature mRNA molecule. Given the unexpected nature of pre-mRNA splicing, understanding the mechanism of the splicing reaction captivated the attention of many molecular biologists. One of the major steps in elucidating this mechanism was the discovery of snRNPs and their involvement in pre-mRNA splicing.

Small nuclear RNAs were first identified in eukaryotic cells in the late 1960s. However, the function of snRNAs remained unknown. In this 1979 paper, Michael Lerner and Joan Steitz demonstrated that the most abundant snRNAs were present as RNA-protein complexes called snRNPs. In addition, they provided the first suggestion that these RNA-protein complexes might function in pre-mRNA splicing. This identification of snRNPs led to a variety of experiments that confirmed their roles and elucidated the mechanism by which pre-mRNA splicing takes place.

The Experiments

The identification of snRNPs was based on the use of antisera from patients with systemic lupus erythematosus, an autoimmune disease in which patients produce antibodies against their own normal cell constituents. Many of the antibodies produced by systemic lupus erythematosus patients are directed against components of the nucleus, including DNA, RNA, and histones. The discovery of snRNPs arose from studies in which Lerner and Steitz sought to characterize two antigens, called ribonucleoprotein (RNP) and Sm, which were recognized by antibodies from systemic lupus erythematosus patients. Indirect data suggested that RNP consisted of both protein and RNA, as its name implies, but neither RNP nor Sm had been characterized at the molecular level.

Immunoprecipitation of snRNAs with antisera from systemic lupus erythematosus patients. Lane 1, anti-Sm; lane 2, normal control serum; lane 3, antiserum recognizing primarily the RNP antigen; lane 4, anti-RNP. Note that a nonspecific RNA designated X is present in all immunoprecipitates, including the control.

To identify possible RNA components of the RNP and Sm antigens, nuclear RNAs of mouse cells were radiolabeled with ^{32}P and immunoprecipitated with antisera from different systemic lupus erythematosus patients (see Figure 4.30). Six specific species of snRNAs were found to be selectively immunoprecipitated by antisera from different patients but not by serum from a normal control patient (see figure). Anti-Sm serum immunoprecipitated all six of these snRNAs, which were designated U1a, U1b, U2, U4, U5, and U6. Anti-RNP serum immunoprecipitated only U1a and U1b, and serum from a third patient (which had been characterized as mostly anti-RNP) immunoprecipitated U1a, U1b, and U6. The immunoprecipitated snRNAs were further characterized by sequence analysis, which demonstrated that U1a, U1b, and U2 were identical to the most abundant snRNAs previously reported in mammalian nuclei, with U1a and U1b representing sequence variants of a single species of U1 snRNA present in human cells. In contrast, the U4, U5, and U6 snRNAs were newly identified by Lerner and Steitz in these experiments.

Importantly, the immunoprecipitation of these snRNAs demonstrated that they were components of RNA-protein complexes. The anti-Sm serum,

KEY EXPERIMENT

which immunoprecipitated all six of the snRNAs, had previously been shown to be directed against a protein antigen. Similarly, protein was known to be required for antigen recognition by anti-RNP serum. Moreover, Lerner and Steitz showed that none of the snRNAs could be immunoprecipitated if protein was first removed by extraction of the RNAs with phenol. Further analysis of cells in which proteins had been radiolabeled with ^{35}S-methionine identified seven prominent nuclear proteins that were immunoprecipitated along with the snRNAs by anti-Sm and anti-RNP sera. These data therefore indicated that each of the six snRNAs was present in an snRNP complex with specific nuclear proteins.

The Impact

The finding that snRNAs were components of snRNPs that were recognized by specific antisera opened a new approach to studying snRNA function. Lerner and Steitz noted that a "most intriguing" possible role for snRNAs might be in pre-mRNA splicing, and pointed out that sequences near the 5′ terminus of U1 snRNA were complementary to splice sites.

Steitz and her colleagues then proceeded with a series of experiments that established the critical involvement of snRNPs in splicing. These studies included more extensive sequence analysis that demonstrated the complementarity of conserved 5′ sequences of U1 snRNA to the consensus sequences of 5′ splice sites, suggesting that U1 functioned in 5′ splice site recognition. In addition, antisera against snRNPs were used to demonstrate that U1 was required for pre-mRNA splicing both in isolated nuclei and in *in vitro* splicing extracts. Further studies have gone on to show that the snRNAs themselves play critical roles not only in the identification of splice sites, but also as catalysts of the splicing reaction. The initial discovery that snRNAs were components of snRNPs that could be recognized by specific antisera thus opened the door to understanding the mechanism of pre-mRNA splicing.

The first step in spliceosome assembly is the binding of U1 snRNP to the 5′ splice site of pre-mRNA (Figure 7.48). This recognition of 5′ splice sites involves base pairing between the 5′ splice site consensus sequence and a complementary sequence at the 5′ end of U1 snRNA (Figure 7.49). U2 snRNP then binds to the branch point, by similar complementary base pairing between U2 snRNA and branch point sequences. A preformed complex consisting of U4/U6 and U5 snRNPs is then incorporated into the spliceosome, with U5 binding to sequences upstream of the 5′ splice site. The splicing reaction is then accompanied by rearrangements of the snRNAs. Prior to the first reaction step (formation of the lariat-like intermediate, see Figure 7.47), U6 dissociates from U4 and displaces U1 at the 5′ splice site. U5 then binds to sequences at the 3′ splice site, followed by excision of the intron and ligation of the exons.

Not only do the snRNAs recognize consensus sequences at the branch points and splice sites of pre-mRNAs, but they also catalyze the splicing reaction directly. The catalytic role of RNAs in splicing was demonstrated by the discovery that some RNAs are capable of **self-splicing**; that is, they can catalyze the removal of their own introns in the absence of other protein or RNA factors. Self-splicing was first described by Tom Cech and his colleagues during studies of the 28S rRNA of the protozoan *Tetrahymena*. This RNA contains an intron of approximately 400 bases that is precisely removed following incubation of the pre-rRNA in the absence of added proteins. Further studies have revealed that splicing is catalyzed by the intron, which acts as a ribozyme to direct its own excision from the pre-rRNA molecule. The discovery of self-splicing of *Tetrahymena* rRNA, together with the studies of RNase P already discussed, provided the first demonstrations of the catalytic activity of RNA.

Additional studies have revealed self-splicing RNAs in mitochondria, chloroplasts, and bacteria. These self-splicing RNAs are divided into two

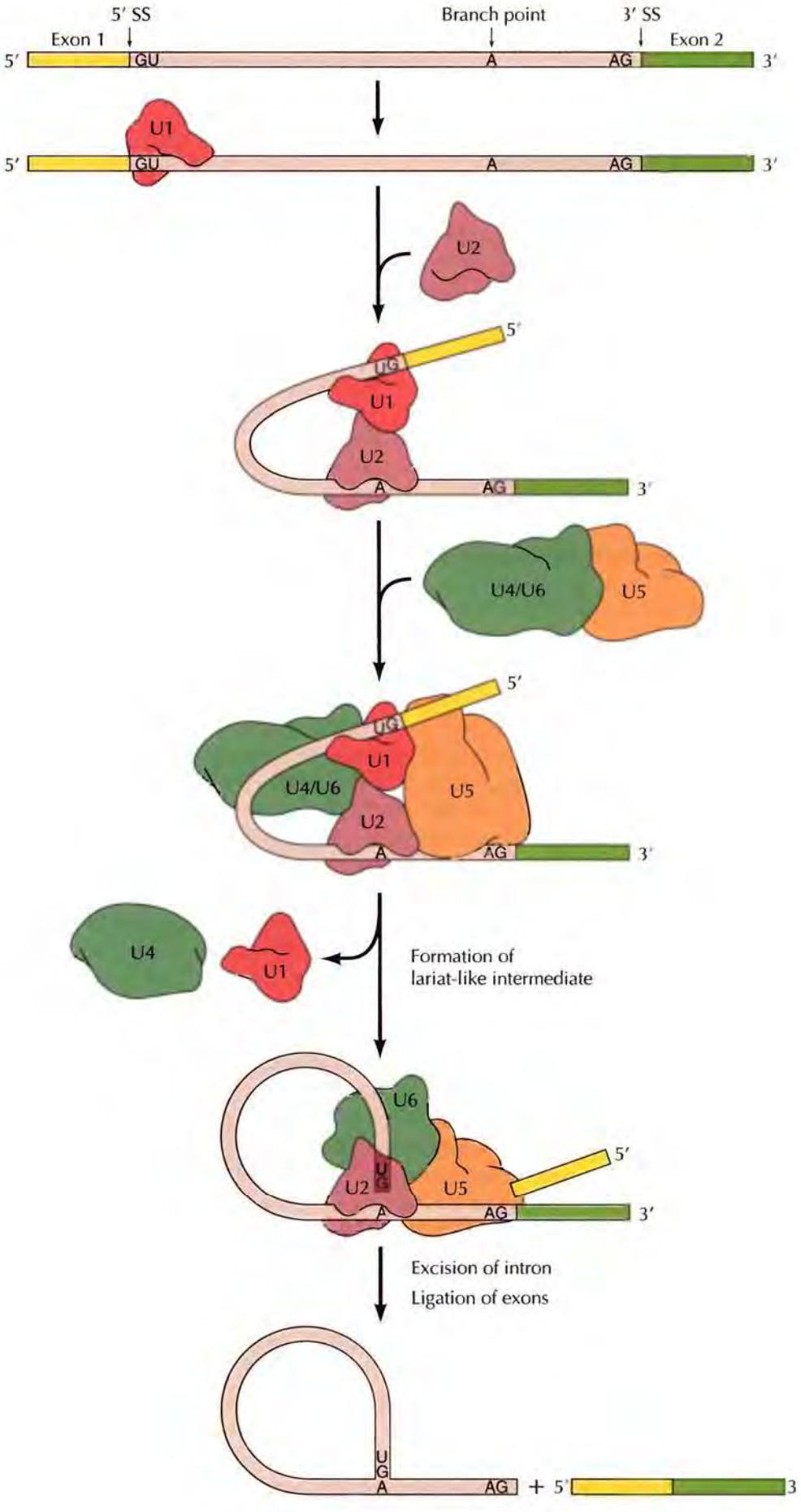


FIGURE 7.48 Assembly of the spliceosome The first step in spliceosome assembly is the binding of U1 snRNP to the 5′ splice site (SS), followed by the binding of U2 snRNP to the branch point. A preformed complex consisting of U4/U6 and U5 snRNPs then enters the spliceosome, with U5 binding to sequences upstream of the 5′ splice site. U6 then dissociates from U4 and displaces U1, leading to formation of the lariat-like intermediate. U5 then binds to the 3′ splice site, followed by excision of the intron and ligation of the exons.

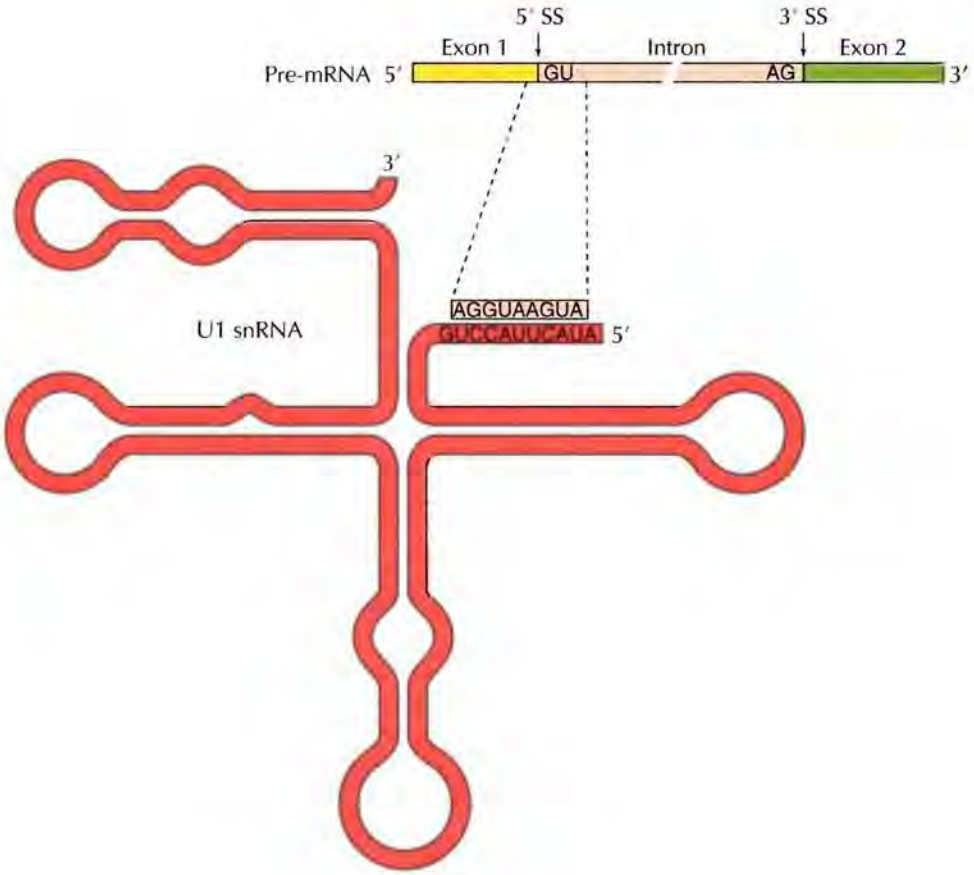


FIGURE 7.49 Binding of U1 snRNA to the 5′ splice site The 5′ terminus of U1 snRNA binds to consensus sequences at 5′ splice sites by complementary base pairing.

classes on the basis of their reaction mechanisms (Figure 7.50). The first step in splicing for group I introns (e.g., *Tetrahymena* pre-rRNA) is cleavage at the 5′ splice site mediated by a guanosine cofactor. The 3′ end of the free exon then reacts with the 3′ splice site to excise the intron as a linear RNA. In contrast, the self-splicing reactions of group II introns (e.g., some mitochondrial pre-mRNAs) closely resemble those characteristic of nuclear pre-mRNA splicing in which cleavage of the 5′ splice site results from attack by an adenosine nucleotide in the intron. As with pre-mRNA splicing, the result is a lariat-like intermediate, which is then excised.

The similarity between spliceosome-mediated pre-mRNA splicing and self-splicing of group II introns strongly suggested that the active catalytic components of the spliceosome were RNAs rather than proteins. In particular, these similarities suggested that pre-mRNA splicing was catalyzed by the snRNAs of the spliceosome. Continuing studies of pre-mRNA splicing have provided clear support for this view, including the demonstration that U2 and U6 snRNAs, in the absence of proteins, can catalyze the first step in pre-mRNA splicing. Pre-mRNA splicing is thus considered to be an RNA-based reaction, catalyzed by spliceosome snRNAs acting analogously to group II self-splicing introns. Within the cell, protein components of the snRNPs are also required, however, and participate in both assembly of the spliceosome and the splicing reaction.

A number of protein splicing factors that are not snRNP components also play critical roles in spliceosome assembly, particularly in identification of the correct splice sites in pre-mRNAs. Mammalian pre-mRNAs typically contain multiple short exons (an average of 150 nucleotides in humans) sep-

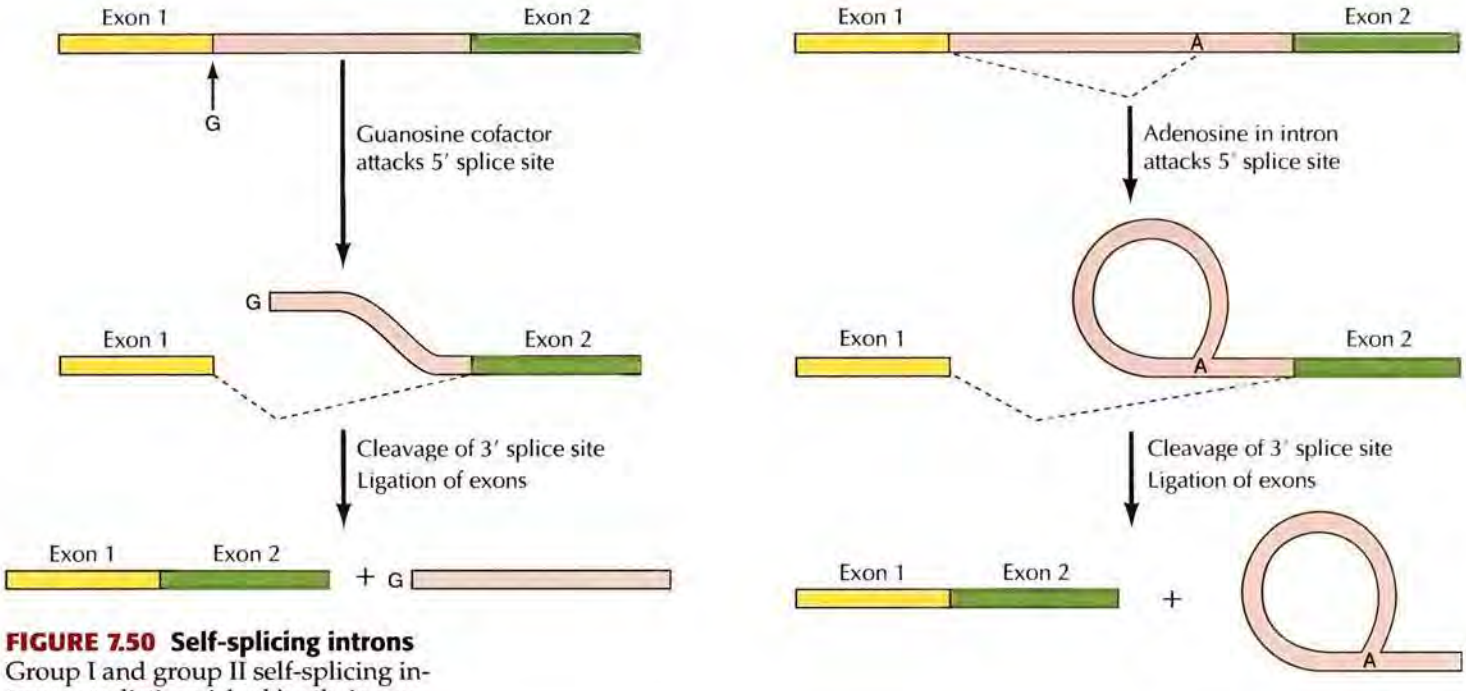


FIGURE 7.50 Self-splicing introns Group I and group II self-splicing introns are distinguished by their reaction mechanisms. In group I introns, the first step in splicing is cleavage of the 5′ splice site by reaction with a guanosine cofactor. The result is a linear intermediate with a G added to the 5′ end of the intron. In group II introns (as in pre-mRNA splicing), the first step is cleavage of the 5′ splice site by reaction with an A within the intron, forming a lariat-like intermediate. In both cases, the second step is simultaneous cleavage of the 3′ splice site and ligation of the exons.

arated by much larger introns (average of 3500 nucleotides). Introns frequently contain many sequences that resemble splice sites, so the splicing machinery must be able to identify the appropriate 5′ and 3′ splice sites at intron/exon boundaries to produce a functional mRNA. Splicing factors serve to direct spliceosomes to the correct splice sites by binding to specific RNA sequences and then recruiting U1 and U2 snRNPs to the appropriate sites on pre-mRNA by protein-protein interactions. For example, the SR splicing factors bind to specific sequences within exons and act to recruit U1 snRNP to the 5′ splice site (Figure 7.51). SR proteins also interact with another splicing factor (U2AF), which binds to pyrimidine-rich sequences at 3′ splice sites and recruits U2 snRNP to the branchpoint. In addition to recruiting the components of the spliceosome to pre-mRNA, splicing factors couple splicing to transcription by associating with the phosphorylated CTD of RNA polymerase II. This anchoring of the splicing machinery to RNA polymerase is thought to be important in ensuring that exons are joined in the correct order as the pre-mRNA is synthesized.

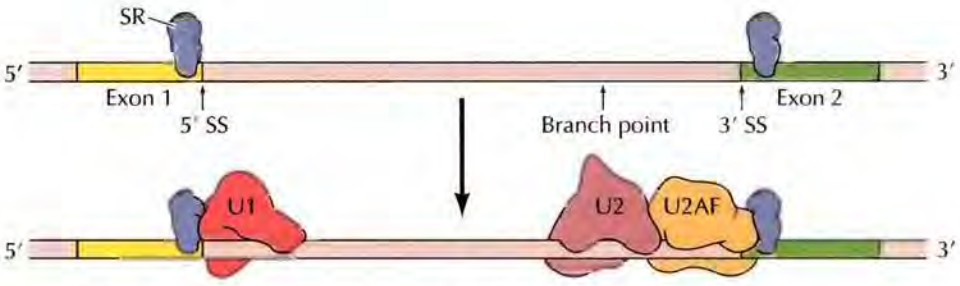


FIGURE 7.51 Role of splicing factors in spliceosome assembly SR splicing factors bind to specific sequences within exons. The SR proteins recruit U1 snRNP to the 5′ splice site and an additional splicing factor (U2AF) to the 3′ splice site. U2AF then recruits U2 snRNP to the branchpoint.

Alternative Splicing

■ **It is estimated that aberrant splicing may cause as many as 15% of all inherited diseases. These include diseases like β-thalassemia and several types of cancer.**

The central role of splicing in the processing of pre-mRNA opens the possibility of regulation of gene expression by control of the splicing machinery. Since most pre-mRNAs contain multiple introns, different mRNAs can be produced from the same gene by different combinations of 5′ and 3′ splice sites. The possibility of joining exons in varied combinations provides a novel means of controlling gene expression by generating multiple mRNAs (and therefore multiple proteins) from the same pre-mRNA. This process, called **alternative splicing**, occurs frequently in genes of complex eukaryotes. For example, it is estimated that about 50% of human genes produce transcripts that are alternatively spliced, considerably increasing the diversity of proteins that can be encoded by the estimated 20,000–25,000 genes in mammalian genomes. Because patterns of alternative splicing can vary in different tissues and in response to extracellular signals, alternative splicing provides an important mechanism for tissue-specific and developmental regulation of gene expression.

One well-studied example of tissue-specific alternative splicing is provided by sex determination in *Drosophila*, where alternative splicing of the same pre-mRNA determines whether a fly is male or female (Figure 7.52). Alternative splicing of the pre-mRNA of a gene called *transformer* is controlled by a protein (SXL) that is only expressed in female flies. The *transformer* pre-mRNA has three exons, but a different second exon is incorporated into the mRNA as a result of using alternate 3′ splice sites in the two different sexes. In males, exon 1 is joined to the most upstream of these 3′ splice sites, which is selected by the binding of the U2AF splicing factor. In females, the SXL protein binds to this 3′ splice site, blocking the binding of U2AF. Consequently, the upstream 3′ splice site is skipped in females, and exon 1 is instead joined to an alternate 3′ splice site that is further downstream. The exon 2 sequences included in the male *transformer* mRNA contain a translation termination codon, so no protein is produced. This termination codon is not included in the female mRNA, so female flies express a functional *transformer* protein, which acts a key regulator of sex determination.

The alternative splicing of *transformer* illustrates the action of a repressor (the SXL protein) that functions by blocking the binding of a splicing factor (U2AF), and a large group of proteins similarly regulate alternative splicing by binding to silencer sequences in pre-mRNAs. In other cases, alternative

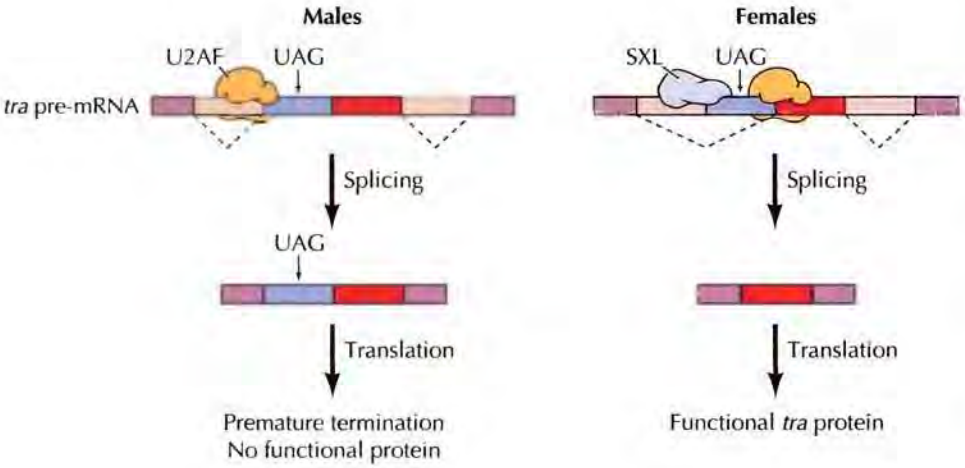


FIGURE 7.52 Alternative splicing in *Drosophila* sex determination
Alternative splicing of *transformer (tra)* mRNA is regulated by the SXL protein, which is only expressed in female flies. In males, the first exon of *tra* mRNA is joined to a 3′ splice site that yields a second exon containing a translation termination codon (UAG), so no *tra* protein is expressed. In females, the binding of SXL protein blocks the binding of U2AF to this 3′ splice site, resulting in the use of an alternative site further downstream in exon 2. This alternative 3′ splice site is downstream of the translation termination codon, so the mRNA expressed in females directs the synthesis of functional *tra* protein.

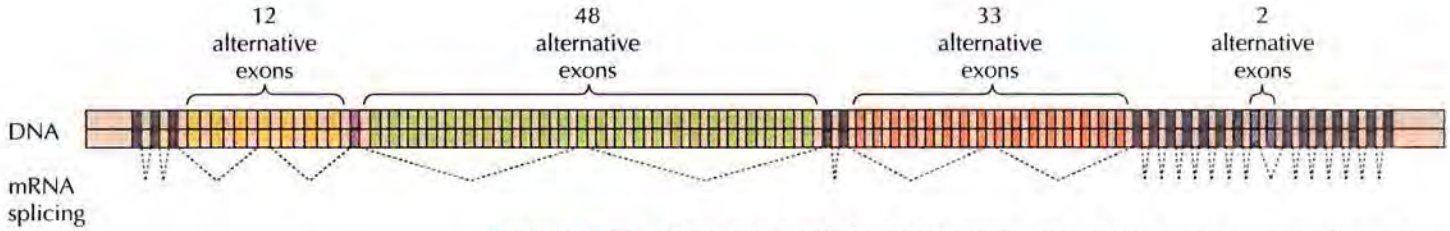


FIGURE 7.53 Alternative splicing of *Dscam* The *Dscam* gene contains four sets of alternative exons: 12 for exon 4, 48 for exon 6, 33 for exon 9, and 2 for exon 17. Any single exon from each of these sets can be incorporated into the mature mRNA, so alternative splicing can produce a total of 38,016 different mRNAs ($12 \times 48 \times 33 \times 2 = 38{,}016$).

splicing is controlled by activators that recruit splicing factors to splice sites that would otherwise not be recognized. The best-studied splicing activators are members of the SR protein family (see Figure 7.51), which bind to specific splicing enhancer sequences.

Multiple mechanisms can thus regulate alternative splicing, and variations in alternative splicing make a major contribution to the diversity of proteins expressed during development and differentiation. One of the most striking examples is the alternative splicing of a *Drosophila* cell surface protein (called Dscam) that is involved in specifying connections between neurons. The *Dscam* gene contains four sets of alternative exons, with a single exon from each set being incorporated into the spliced mRNA (Figure 7.53). These exons can be joined in any combination, so alternative splicing can potentially yield 38,016 different mRNAs and proteins from this single gene: more than twice the total number of genes in the *Drosophila* genome. It is thought that the functional diversity provided by these alternatively spliced forms of Dscam makes an important contribution to specifying the correct connections between neurons during development of the fly brain.

■ **The mammalian ear contains hair cells that are tuned to respond to sounds of different frequency. The tuning of hair cells is thought to be mediated in part by the alternative splicing of a gene encoding a channel protein.**

RNA Editing

RNA editing refers to RNA processing events (other than splicing) that alter the protein-coding sequences of some mRNAs. This unexpected form of RNA processing was first discovered in mitochondrial mRNAs of trypanosomes in which U residues are added and deleted at multiple sites along the pre-mRNA in order to generate the mRNA. More recently, editing has also been described in mitochondrial mRNAs of other organisms, chloroplast mRNAs of higher plants, and nuclear mRNAs of some mammalian genes.

Editing in mammalian nuclear mRNAs, as well as in mitochondrial and chloroplast RNAs of higher plants, involves single base changes as a result of base modification reactions, similar to those involved in tRNA processing. In mammalian cells, RNA editing reactions include the deamination of cytosine to uridine and of adenosine to inosine. One of the best-studied examples is editing of the mRNA for apolipoprotein B, which transports lipids in the blood. In this case, tissue-specific RNA editing results in two different forms of apolipoprotein B (Figure 7.54). In humans, Apo-B100 (4536 amino acids) is synthesized in the liver by translation of the unedited mRNA. However, a shorter protein (Apo-B48, 2152 amino acids) is synthesized in the intestine as a result of translation of an edited mRNA in which a C has been changed to a U by deamination. This alteration changes the

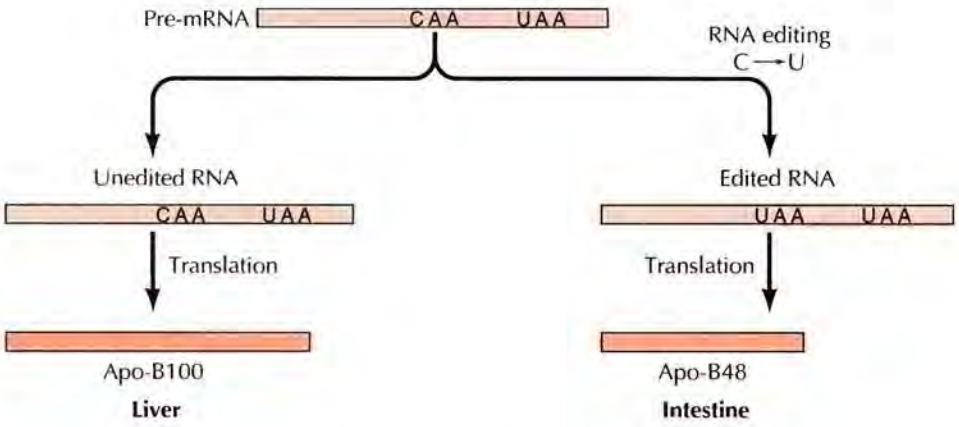


FIGURE 7.54 Editing of apolipoprotein B mRNA In human liver, unedited mRNA is translated to yield a 4536-amino-acid protein called Apo-B100. In human intestine, however, the mRNA is edited by a base modification that changes a specific C to a U. This modification changes the codon for glutamine (CAA) to a termination codon (UAA), resulting in synthesis of a shorter protein (Apo-B48, consisting of only 2152 amino acids).

codon for glutamine (CAA) in the unedited mRNA to a translation termination codon (UAA) in the edited mRNA, resulting in synthesis of the shorter Apo-B protein. Tissue-specific editing of Apo-B mRNA thus results in the expression of structurally and functionally different proteins in liver and intestine. The full-length Apo-B100 produced by the liver transports lipids in the circulation; Apo-B48 functions in the absorption of dietary lipids by the intestine.

RNA editing by the deamination of adenosine to inosine is the most common form of nuclear RNA editing in mammals. This form of editing plays an important role in the nervous system, where A-to-I editing results in single amino acid changes in ion channels and receptors on the surface of neurons. For example, the mRNAs encoding receptors for the neurotransmitter serotonin can be edited at up to five sites, potentially yielding 24 different versions of the receptor with different signaling activities. The importance of A-to-I editing in the nervous system is further demonstrated by the finding that *C. elegans*, *Drosophila*, and mouse mutants lacking the editing enzyme suffer from a variety of neurological defects.

RNA Degradation

The processing steps discussed in the previous section result in the formation of mature mRNAs, which are then transported to the cytoplasm and function to direct protein synthesis. However, most of the sequences transcribed into pre-mRNA are instead degraded within the nucleus. Over 90% of pre-mRNA sequences are introns, which are degraded within the nucleus following their excision by splicing. This is carried out by an enzyme that recognizes the unique 2′-5′ bond formed at the branchpoint, as well as by enzymes that recognize either the 5′ or 3′ ends of RNA molecules and catalyze degradation of the RNA in either direction. The 5′ and 3′ ends of processed mRNAs are protected from this degradation machinery by capping and polyadenylation, respectively, while the unprotected ends of introns are recognized and degraded.

In addition to degrading introns, cells possess a quality-control system (called **nonsense-mediated mRNA decay**) that leads to the degradation of mRNAs that lack complete open-reading frames. This eliminates defective mRNA molecules and prevents the synthesis of abnormal truncated proteins. In yeast, nonsense-mediated mRNA decay takes place in the cytoplasm and is triggered when a premature termination codon is encountered by a ribosome during protein synthesis. In mammals, however, at least some nonsense-mediated mRNA decay may take place while mRNAs are associated with the nucleus. However, it is not clear whether this occurs during nuclear export or within the nucleus, which would require translation of nuclear mRNAs.

What may be considered the final aspect of the processing of an RNA molecule is its eventual degradation in the cytoplasm. Since the intracellular level of any RNA is determined by a balance between synthesis and degradation, the rate at which individual RNAs are degraded is another level at which gene expression can be controlled. Both ribosomal and transfer RNAs are very stable, and this stability largely accounts for the high levels of these RNAs (greater than 90% of all RNA) in both prokaryotic and eukaryotic cells. In contrast, bacterial mRNAs are rapidly degraded, usually having half-lives of only 2 to 3 minutes. This rapid turnover of bacterial mRNAs allows the cell to respond quickly to alterations in its environment, such as changes in the availability of nutrients required for growth. In eukaryotic cells, however, different mRNAs are degraded at different rates, providing an additional parameter to the regulation of eukaryotic gene expression.

The cytoplasmic degradation of most eukaryotic mRNAs is initiated by shortening of their poly-A tails. Then follows removal of the 5′ cap and degradation of the RNA by nucleases acting from both ends. The half-lives of mRNAs in mammalian cells vary from less than 30 minutes to approximately 20 hours. The unstable mRNAs frequently code for regulatory proteins, such as transcription factors, whose levels within the cell vary rapidly in response to environmental stimuli. These mRNAs often contain specific AU-rich sequences near their 3′ ends that appear to signal rapid degradation by promoting deadenylation. In contrast, mRNAs encoding structural proteins or central metabolic enzymes generally have long half-lives.

The degradation of some mRNAs can also be triggered by internal cleavages. In many cases, cleavage is induced by siRNAs or miRNAs acting through the RISC complex (see Figure 4.42). Alternatively, the cleavage of some mRNAs is mediated by proteins and can be regulated in response to extracellular signals. A good example is provided by the mRNA that encodes transferrin receptor—a cell surface protein involved in the uptake of iron by mammalian cells. The amount of transferrin receptor within cells is regulated by the availability of iron, largely as a result of modulation of the stability of its mRNA (Figure 7.55). In the presence of adequate amounts of iron, transferrin receptor mRNA is rapidly degraded as a result of specific nuclease cleavage at a sequence near its 3′ end. If an adequate supply of iron is not available, however, the mRNA is stabilized, resulting in increased synthesis of transferrin receptor and more iron uptake by the cell. This regulation is mediated by an iron regulatory protein (IRP) that binds to specific sequences (called the iron response element, or IRE) near the 3′ end of transferrin receptor mRNA and protects the mRNA from cleavage. The activity of the IRP is in turn controlled by the levels of iron within the cell: If

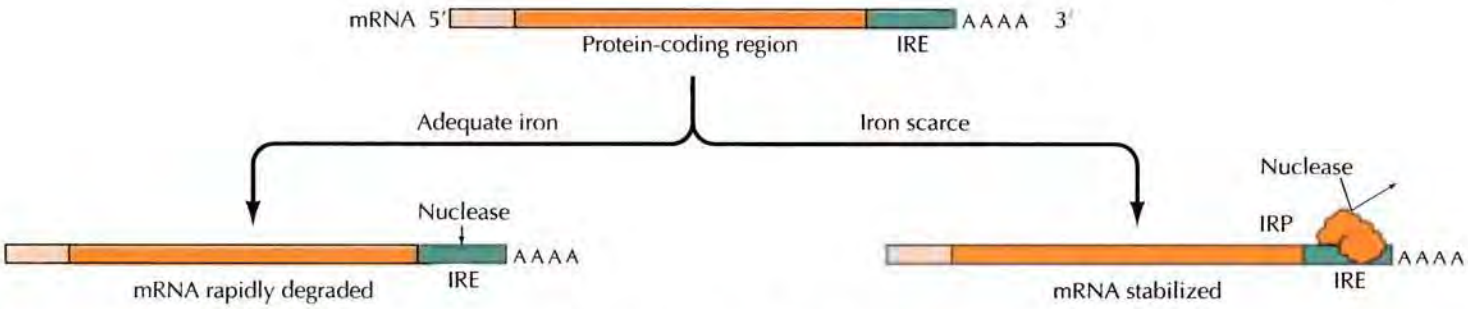


FIGURE 7.55 Regulation of transferrin receptor mRNA stability The levels of transferrin receptor mRNA are regulated by the availability of iron. If the supply of iron is adequate, the mRNA is rapidly degraded as a result of nuclease cleavage near the 3′ end. If iron is scarce, however, a regulatory protein (called the iron regulatory protein, or IRP) binds to a sequence near the 3′ end of the mRNA (the iron response element, or IRE), protecting the mRNA from nuclease cleavage.

iron is scarce, the IRP binds to the IRE and protects transferrin receptor mRNA from degradation. Similar changes in the stability of other mRNAs are involved in the regulation of gene expression by certain hormones. Thus, although transcription remains the primary level at which gene expression is regulated, variations in the rate of mRNA degradation also play an important role in controlling steady-state levels of mRNAs within the cell.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

SUMMARY

TRANSCRIPTION IN PROKARYOTES

RNA Polymerase and Transcription: *E. coli* RNA polymerase consists of α, β, β', ω, and σ subunits. Transcription is initiated by the binding of σ to promoter sequences. After synthesis of about the first ten nucleotides of RNA, the core polymerase dissociates from σ and travels along the template DNA as it elongates the RNA chain. Transcription then continues until the polymerase encounters a termination signal.

Repressors and Negative Control of Transcription: The prototype model for gene regulation in bacteria is the *lac* operon, which is regulated by the binding of a repressor to specific DNA sequences overlapping the promoter.

Positive Control of Transcription: Some bacterial genes are regulated by transcriptional activators rather than repressors.

EUKARYOTIC RNA POLYMERASES AND GENERAL TRANSCRIPTION FACTORS

Eukaryotic RNA Polymerases: Eukaryotic cells contain three distinct nuclear RNA polymerases that transcribe genes encoding mRNAs (polymerase II), rRNAs (polymerases I and III), and tRNAs (polymerase III).

KEY TERMS

RNA polymerase, promoter, footprinting

operon, operator, repressor, *cis*-acting control element, *trans*-acting factor

KEY TERMS

transcription factor, general transcription factor, TATA box, TATA-binding protein (TBP), TBP-associated factor (TAF), Mediator

enhancer, insulator, barrier element

electrophoretic-mobility shift assay, chromatin immunoprecipitation

DNA-affinity chromatography

transcriptional activator, zinc finger domain, steroid hormone receptor, helix-turn-helix, homeodomain, homeobox, leucine zipper, helix-loop-helix, coactivator

corepressor

HMGN proteins, histone acetylation, histone code, nucleosome remodeling factor, elongation factor

SUMMARY

General Transcription Factors and Initiation of Transcription by RNA Polymerase II: Eukaryotic RNA polymerases do not bind directly to promoter sequences; they require additional proteins (general transcription factors) to initiate transcription. The promoter sequences of many polymerase II genes are recognized by the TATA-binding protein, which recruits additional transcription factors and RNA polymerase to the promoter.

Transcription by RNA Polymerases I and III: RNA polymerases I and III also require additional transcription factors to bind to the promoters of rRNA, tRNA, and some snRNA genes.

REGULATION OF TRANSCRIPTION IN EUKARYOTES

cis-Acting Regulatory Sequences: Promoters and Enhancers: Transcription of eukaryotic genes is controlled by proteins that bind to regulatory sequences, which can be located up to several kilobases away from the transcription start site. Enhancers typically contain binding sites for multiple proteins that work together to regulate gene expression.

Transcription Factor Binding sites: Eukaryotic transcription factors bind to short DNA sequences, usually 6–10 base pairs, in promoters or enhancers.

Transcriptional Regulatory Proteins: Many eukaryotic transcription factors have been isolated on the basis of their binding to specific DNA sequences.

Structure and Function of Transcriptional Activators: Transcriptional activators are modular proteins, consisting of distinct DNA-binding and activation domains. DNA-binding domains mediate association with specific regulatory sequences; activation domains stimulate transcription by interacting with Mediator proteins and general transcription factors, as well as with coactivators that modify chromatin structure.

Eukaryotic Repressors: Gene expression in eukaryotic cells is regulated by repressors as well as by activators. Some repressors interfere with the binding of activators or general transcription factors to DNA. Other repressors contain discrete repression domains that inhibit transcription by interacting with Mediator proteins, general transcription factors, transcriptional activators, or corepressors that affect chromatin structure.

Relationship of Chromatin Structure to Transcription: The packaging of DNA in nucleosomes presents an impediment to transcription in eukaryotic cells. Modification of histones by acetylation is tightly linked to transcriptional regulation and enzymes that catalyze histone acetylation are associated with transcriptional activators, whereas histone deacetylases are associated with repressors. Histones are also modified by phosphorylation and methylation, and specific modifications of histones affect gene expression by serving as binding sites for other regulatory proteins. In addition, nucleosome remodeling factors facilitate the binding of transcription factors to DNA by altering the arrangement or structures of nucleosomes.

SUMMARY

Regulation of Transcription by Noncoding RNAs: Transcription can be regulated by noncoding RNAs, as well as by regulatory proteins. MicroRNAs repress transcription of homologous genes by associating with a protein complex (RITS) that induces histone modifications resulting in formation of heterochromatin. X chromosome inactivation in mammals is also mediated by a noncoding RNA.

microRNA (miRNA), X chromosome inactivation

DNA Methylation: Methylation of cytosine residues can inhibit the transcription of eukaryotic genes and is important in silencing transposable elements. Regulation of gene expression by methylation also plays an important role in genomic imprinting, which controls the transcription of some genes involved in mammalian development.

genomic imprinting

RNA PROCESSING AND TURNOVER

Processing of Ribosomal and Transfer RNAs: Ribosomal and transfer RNAs are derived by cleavage of long primary transcripts in both prokaryotic and eukaryotic cells. Methyl groups are added to rRNAs, and various bases are modified in tRNAs.

pre-rRNA, pre-tRNA, RNase P, ribozyme

Processing of mRNA in Eukaryotes: Eukaryotic pre-mRNAs are modified by the addition of 7-methylguanosine caps and 3′ poly-A tails, in addition to the removal of introns by splicing.

pre-mRNA, 7-methylguanosine cap, poly-A tail, polyadenylation

Splicing Mechanisms: Splicing of nuclear pre-mRNAs takes place in large complexes, called spliceosomes, composed of proteins and small nuclear RNAs (snRNAs). The snRNAs recognize sequences at the splice sites of pre-mRNAs and catalyze the splicing reaction. Some mitochondrial, chloroplast, and bacterial RNAs undergo self-splicing in which the splicing reaction is catalyzed by intron sequences.

spliceosome, small nuclear RNA (snRNA), small nuclear-ribonucleoprotein particle (snRNP), self-splicing

Alternative Splicing: Exons can be joined in various combinations as a result of alternative splicing, which provides an important mechanism for tissue-specific control of gene expression in complex eukaryotes.

alternative splicing

RNA Editing: Some mRNAs are modified by processing events that alter their protein-coding sequences. Editing of mitochondrial mRNAs in some protozoans involves the addition and deletion of U residues at multiple sites in the molecule. Other forms of RNA editing in plant and mammalian cells involve the modification of specific bases.

RNA editing

RNA Degradation: Introns are degraded within the nucleus, and abnormal mRNAs lacking complete open-reading frames are eliminated by nonsense-mediated mRNA decay. Functional mRNAs in eukaryotic cells are degraded at different rates, providing an additional mechanism for control of gene expression. In some cases, rates of mRNA degradation are regulated by extracellular signals.

nonsense-mediated mRNA decay

KEY TERMS

Questions

1. How does footprinting identify a protein-binding site on DNA?

2. What is the role of sigma (σ) factors in bacterial RNA synthesis?

3. What is the major mechanism for termination of *E. coli* mRNAs?

4. You are working with two strains of *E. coli*. One contains a wild-type β-galactosidase gene and an i^- mutation; the other contains a temperature-sensitive β-galactosidase gene and an o^c mutation. After mating these strains, you assay for the production of β-galactosidase by the diploid cells at both permissive and nonpermissive temperatures in the absence of lactose. What do you expect to find?

5. You are comparing the requirements for *in vitro* transcription of two RNA polymerase II genes, one containing a TATA box and the other containing only an Inr sequence. Does transcription from these promoters require TBP or TFIID?

6. How do enhancers differ from promoters as *cis*-acting regulatory sequences in eukaryotes?

7. You are studying the enhancer of a gene that normally is expressed only in neurons. Constructs in which this enhancer is linked to a reporter gene are expressed in neuronal cells but not in fibroblasts. However, if you mutate a specific sequence element within the enhancer, you find expression in both fibroblasts and neuronal cells. What type of regulatory protein do you expect binds the sequence element?

8. What are the functions of insulators?

9. Explain the mechanism of X chromosome inactivation in human females.

10. What property of Sp1 was exploited for its purification by DNA affinity chromatography? How would you determine that the purified protein is indeed Sp1?

11. You have developed an *in vitro* splicing reaction in which unspliced pre-mRNA is processed to the mature mRNA. What results would you expect if you added anti-Sm antiserum to the reaction?

12. How can noncoding RNAs regulate the concentration of a specific mRNA within a cell?

13. What is the function of splicing factors that are not components of snRNPs?

14. How are two structurally and functionally different forms of apolipoprotein B synthesized in human liver and intestine?

15. What is nonsense-mediated mRNA decay? What is its significance in the cell?

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CHAPTER 8

Protein Synthesis, Processing, and Regulation

- ***Translation of mRNA 309***
- ***Protein Folding and Processing 329***
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- ***Protein Degradation 344***
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TRANSCRIPTION AND RNA PROCESSING ARE FOLLOWED BY TRANSLATION, the synthesis of proteins as directed by mRNA templates. Proteins are the active players in most cell processes, implementing the myriad tasks that are directed by the information encoded in genomic DNA. Protein synthesis is thus the final stage of gene expression. However, the translation of mRNA is only the first step in the formation of a functional protein. The polypeptide chain must then fold into the appropriate three-dimensional conformation and, frequently, undergo various processing steps before being converted to its active form. These processing steps, particularly in eukaryotes, are intimately related to the sorting and transport of different proteins to their appropriate destinations within the cell.

Although the expression of most genes is regulated primarily at the level of transcription (see Chapter 7), gene expression can also be controlled at the level of translation, and this control is an important element of gene regulation in both prokaryotic and eukaryotic cells. Of even broader significance, however, are the mechanisms that control the activities of proteins within cells. Once synthesized, most proteins can be regulated in response to extracellular signals by either covalent modifications or by association with other molecules. In addition, the levels of proteins within cells can be controlled by differential rates of protein degradation. These multiple controls of both the amounts and activities of intracellular proteins ultimately regulate all aspects of cell behavior.

Translation of mRNA

Proteins are synthesized from mRNA templates by a process that has been highly conserved throughout evolution (reviewed in Chapter 4). All mRNAs are read in the 5′ to 3′ direction, and polypeptide chains are syn-

thesized from the amino to the carboxy terminus. Each amino acid is specified by three bases (a codon) in the mRNA, according to a nearly universal genetic code. The basic mechanics of protein synthesis are also the same in all cells: Translation is carried out on ribosomes, with tRNAs serving as adaptors between the mRNA template and the amino acids being incorporated into protein. Protein synthesis thus involves interactions between three types of RNA molecules (mRNA templates, tRNAs, and rRNAs), as well as various proteins that are required for translation.

Transfer RNAs

During translation, each of the 20 amino acids must be aligned with its corresponding codon on the mRNA template. All cells contain a variety of **tRNAs** that serve as adaptors for this process. As might be expected, given their common function in protein synthesis, different tRNAs share similar overall structures. However, they also possess unique identifying sequences that allow the correct amino acid to be attached and aligned with the appropriate codon in mRNA.

Transfer RNAs are approximately 70 to 80 nucleotides long and have characteristic cloverleaf structures that result from complementary base pairing between different regions of the molecule (Figure 8.1). X-ray crystallography studies have further shown that all tRNAs fold into similar compact L shapes, which are required for the tRNAs to fit onto ribosomes during the translation process. The adaptor function of the tRNAs involves two separated regions of the molecule. All tRNAs have the sequence CCA at their 3′ terminus, and amino acids are covalently attached to the ribose of the terminal adenosine. The mRNA template is then recognized by the **anticodon** loop, located at the other end of the folded tRNA, which binds to the appropriate codon by complementary base pairing.

The incorporation of the correctly encoded amino acids into proteins depends on the attachment of each amino acid to an appropriate tRNA, as well as on the specificity of codon-anticodon base pairing. The attachment of amino acids to specific tRNAs is mediated by a group of enzymes called **aminoacyl tRNA synthetases**, which were discovered by Paul Zamecnik and Mahlon Hoagland in 1957. Each of these 20 enzymes recognizes a sin-

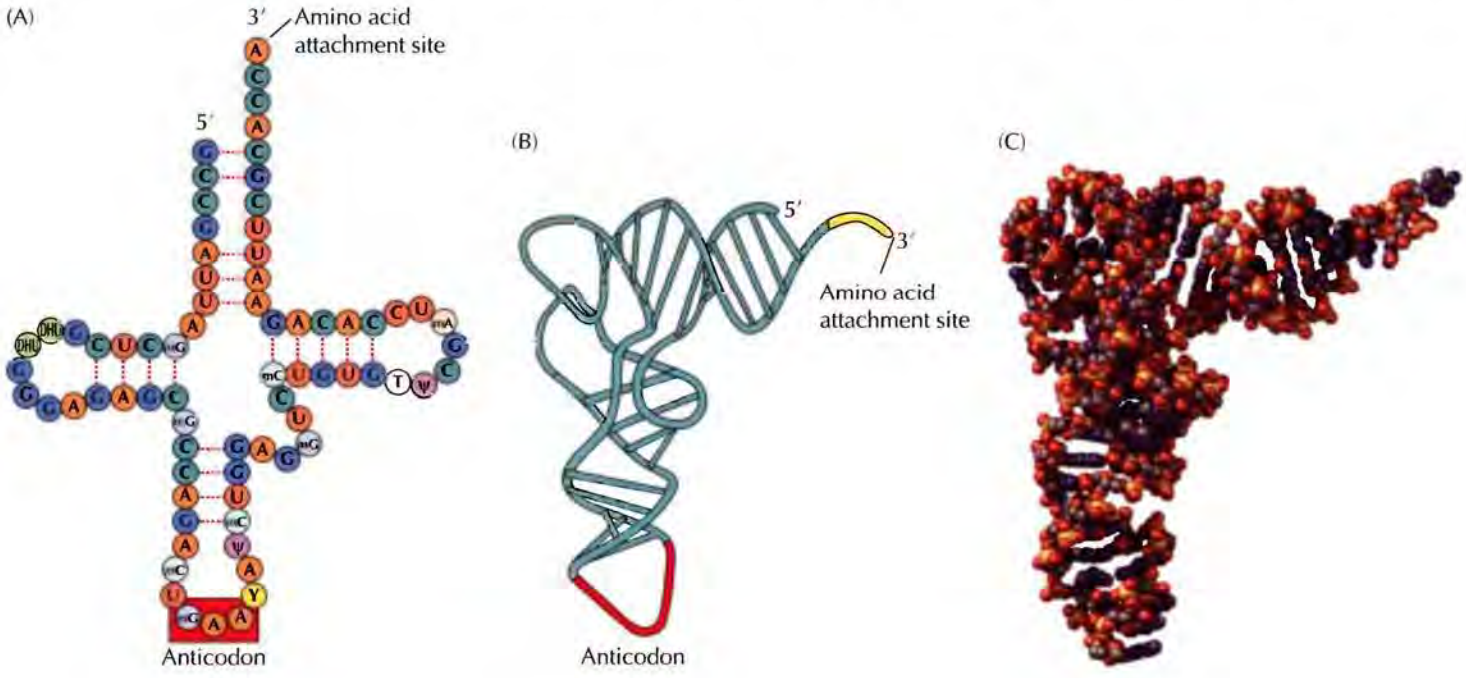


FIGURE 8.1 Structure of tRNAs The structure of yeast phenylalanyl tRNA is illustrated in open "cloverleaf" form (A) to show complementary base pairing. Modified bases are indicated as mG, methylguanosine; mC, methylcytosine; DHU, dihydrouridine; T, ribothymidine; Y, a modified purine (usually adenosine); and ψ, pseudouridine. The folded form of the molecule is shown in (B) and a space-filling model in (C). (C, courtesy of Dan Richardson.)

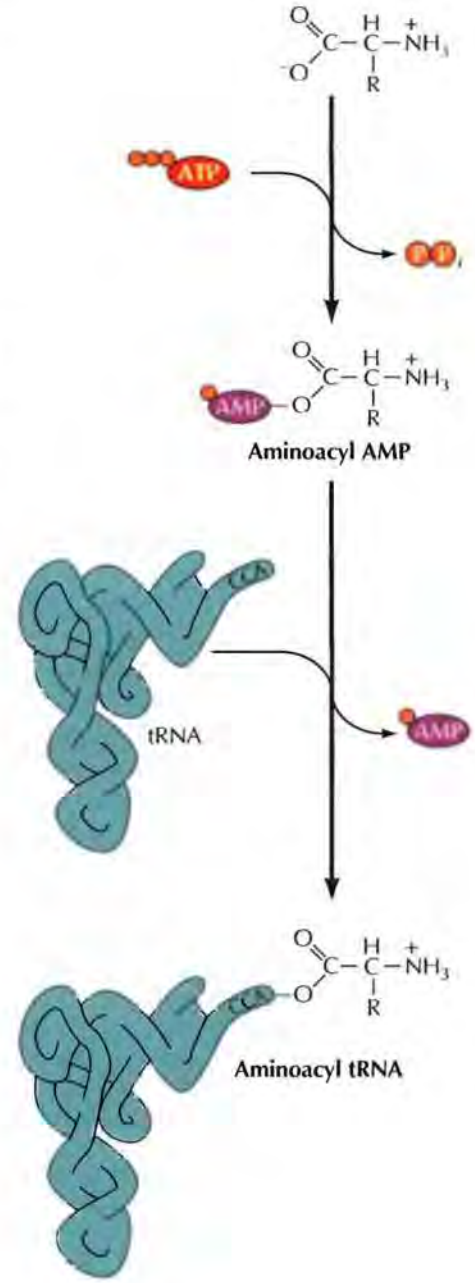


FIGURE 8.2 Attachment of amino acids to tRNAs In the first reaction step, the amino acid is joined to AMP, forming an aminoacyl AMP intermediate. In the second step, the amino acid is transferred to the 3′ CCA terminus of the acceptor tRNA and AMP is released. Both steps of the reaction are catalyzed by aminoacyl tRNA synthetases.

gle amino acid, as well as the correct tRNA (or tRNAs) to which that amino acid should be attached. The reaction proceeds in two steps (Figure 8.2). First, the amino acid is activated by reaction with ATP to form an aminoacyl AMP synthetase intermediate. The activated amino acid is then joined to the 3′ terminus of the tRNA. The aminoacyl tRNA synthetases must be highly selective enzymes that recognize both individual amino acids and specific base sequences that identify the correct acceptor tRNAs. In some cases, the high fidelity of amino acid recognition results in part from a proofreading function by which incorrect aminoacyl AMPs are hydrolyzed rather than being joined to tRNA during the second step of the reaction. Recognition of the correct tRNA by the aminoacyl tRNA synthetase is also highly selective; the synthetase recognizes specific nucleotide sequences (in most cases including the anticodon) that uniquely identify each species of tRNA.

After being attached to tRNA, an amino acid is aligned on the mRNA template by complementary base pairing between the mRNA codon and the anticodon of the tRNA. Codon-anticodon base pairing is somewhat less stringent than the standard A-U and G-C base pairing discussed in preceding chapters. The significance of this unusual base pairing in codon-anticodon recognition relates to the redundancy of the genetic code. Of the 64 possible codons, 3 are stop codons that signal the termination of translation; the other 61 encode amino acids (see Table 4.1). Thus most of the amino acids are specified by more than one codon. In part, this redundancy results from the attachment of many amino acids to more than one species of tRNA. *E. coli*, for example, contain about 40 different tRNAs that serve as acceptors for the 20 different amino acids. In addition, some tRNAs are able to recognize more than one codon in mRNA, as a result of nonstandard base pairing (called wobble) between the tRNA anticodon and the third position of some complementary codons (Figure 8.3). Relaxed base pairing at this position results partly from the formation of G-U base pairs and partly from the modification of guanosine to inosine in the anticodons of several tRNAs during processing (see Figure 7.43). Inosine can base-pair with either C, U, or A in the third position, so its inclusion in the anticodon allows a single tRNA to recognize three different codons in mRNA templates.

The Ribosome

Ribosomes are the sites of protein synthesis in both prokaryotic and eukaryotic cells. First characterized as particles detected by ultracentrifugation of cell lysates, ribosomes are usually designated according to their rates of sedimentation: 70S for bacterial ribosomes and 80S for the somewhat larger ribosomes of eukaryotic cells. Both prokaryotic and eukaryotic ribosomes are composed of two distinct subunits, each containing characteristic proteins and **rRNAs**. The fact that cells typically contain many ribosomes reflects the central importance of protein synthesis in cell metabolism. *E. coli*, for example, contain about 20,000 ribosomes, which account for approximately 25% of the dry weight of the cell, and rapidly growing mammalian cells contain about 10 million ribosomes.

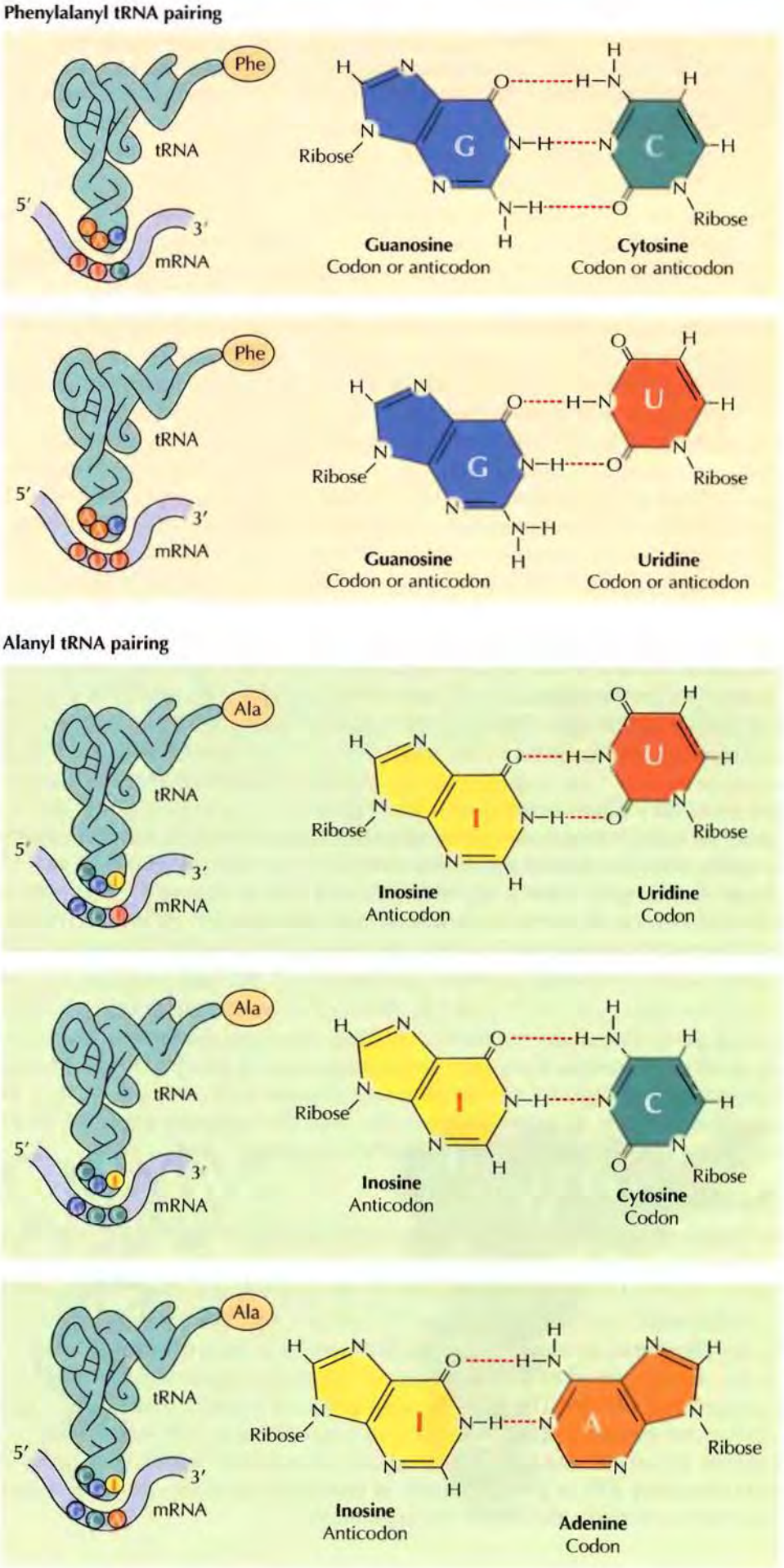


FIGURE 8.3 Nonstandard codon-anticodon base pairing Base pairing at the third codon position is relaxed, allowing G to pair with U, and inosine (I) in the anticodon to pair with U, C, or A. Two examples of abnormal base pairing, allowing phenylalanyl (Phe) tRNA to recognize either UUC or UUU codons and alanyl (Ala) tRNA to recognize GCU, GCC, or GCA, are illustrated.

The general structures of prokaryotic and eukaryotic ribosomes are similar, although they differ in some details (Figure 8.4). The small subunit (designated 30S) of *E. coli* ribosomes consists of the 16S rRNA and 21 proteins; the large subunit (50S) is composed of the 23S and 5S rRNAs and 34 proteins. Each ribosome contains one copy of the rRNAs and one copy of each of the ribosomal proteins, with one exception: One protein of the 50S subunit is present in four copies. The subunits of eukaryotic ribosomes are larger and contain more proteins than their prokaryotic counterparts. The small subunit (40S) of eukaryotic ribosomes is composed of the 18S rRNA and approximately 30 proteins; the large subunit (60S) contains the 28S, 5.8S, and 5S rRNAs and about 45 proteins. Because of their large size and complexity, high-resolution structural analysis of ribosomes by X-ray crystallography was not accomplished until 2000, when structures of both the 50S and 30S subunits were first reported. As discussed below, understand-

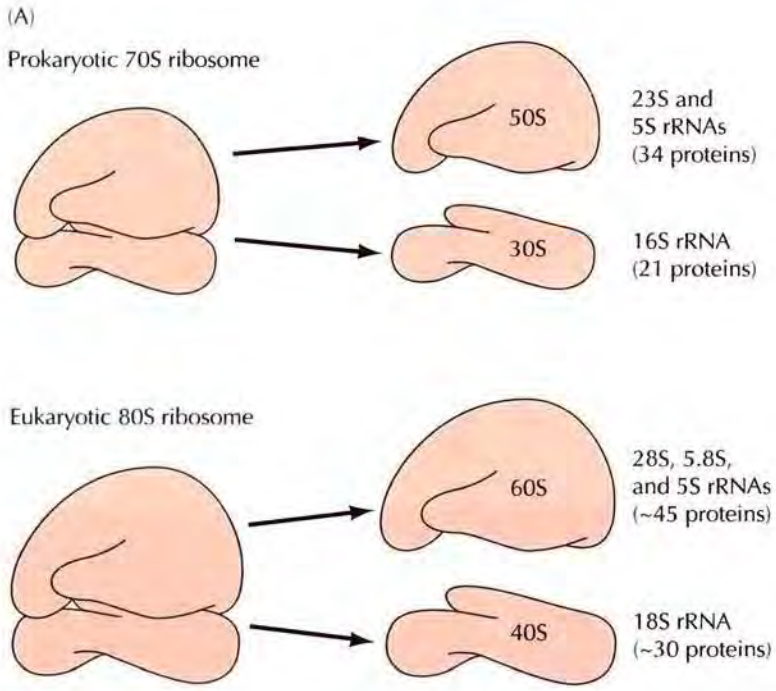


FIGURE 8.4 Ribosome structure (A) Components of prokaryotic and eukaryotic ribosomes. Intact prokaryotic and eukaryotic ribosomes are designated 70S and 80S, respectively, on the basis of their sedimentation rates in ultracentrifugation. They consist of large and small subunits, which contain both ribosomal proteins and rRNAs. (B–C) High resolution X-ray crystal structures of 30S (B) and 50S (C) ribosomal subunits. (B, from B. T. Wimberly, D. E. Brodersen, W. M. Clemons, R. J. Morgan-Warren, A. P. Carter, C. Vonrhein, T. Hartrsch and V. Ramakrishnan, 2000. *Nature* 407: 327. C, from N. Ban, P. Nissen, J. Hansen, P. B. Moore and T. A. Steitz, 2000. *Science* 289: 905.)

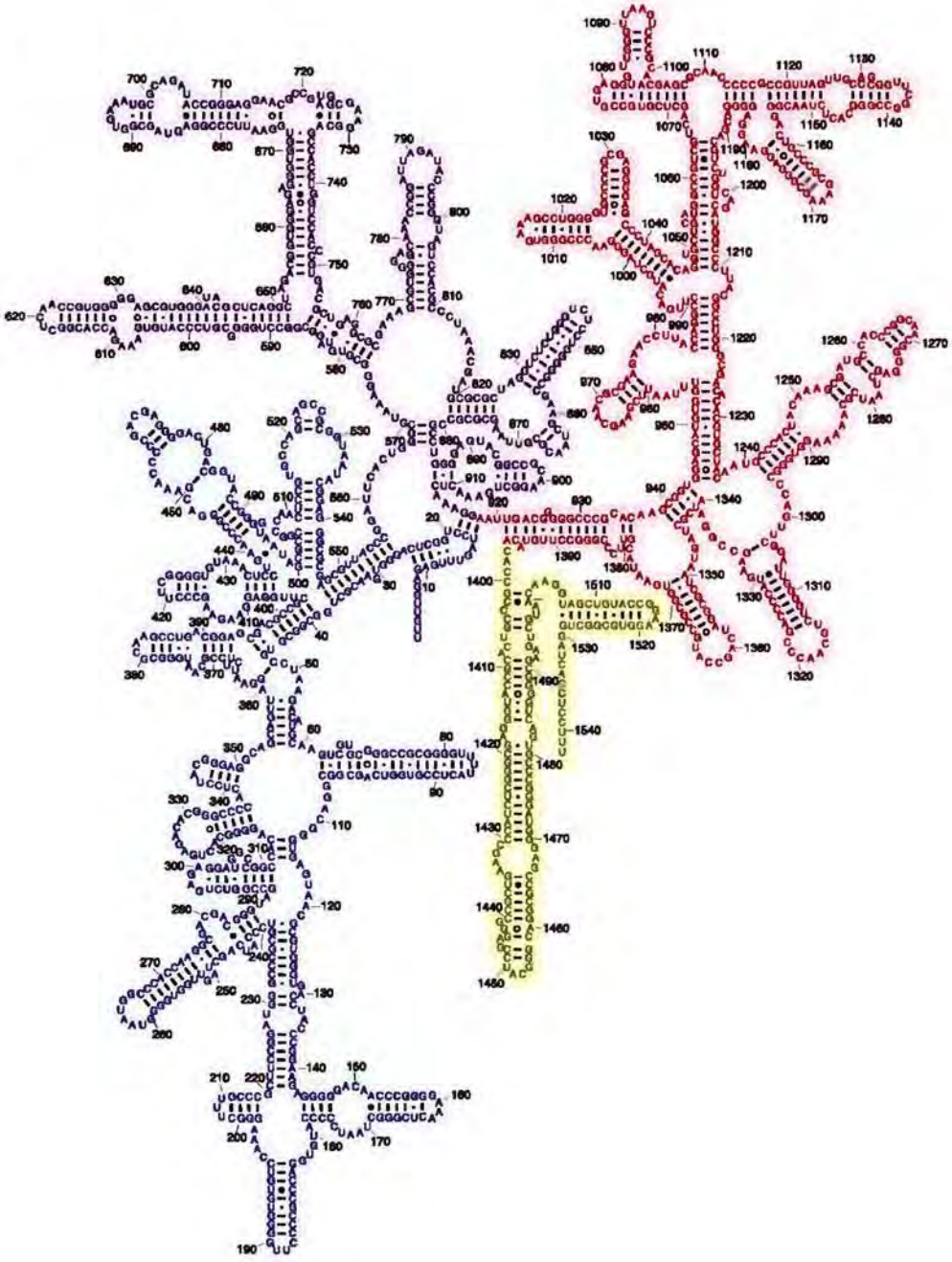


FIGURE 8.5 Structure of 16S rRNA
Complementary base pairing results in the formation of a distinct secondary structure. (From M. M. Yusupov, G. Z. Yusupova, A. Baucom, K. Lieberman, T. N. H. Earnest, J. H. D. Cate and H. F. Noller. 2001. *Science* 292: 883.)

■ **Although the sequence of ribosomal RNAs is highly conserved, substitutions have occurred throughout the course of evolution. The comparison of rRNA sequences is a powerful tool in determining evolutionary relationships between different species.**

ing the structure of ribosomes at the atomic level has had a major impact on our understanding of ribosome function.

A noteworthy feature of ribosomes is that they can be formed *in vitro* by self-assembly of their RNA and protein constituents. As first described in 1968 by Masayasu Nomura, purified ribosomal proteins and rRNAs can be mixed together and, under appropriate conditions, will re-form a functional ribosome. Although ribosome assembly *in vivo* (particularly in eukaryotic cells) is considerably more complicated, the ability of ribosomes to self-assemble *in vitro* has provided an important experimental tool, allowing analysis of the roles of individual proteins and rRNAs.

Like tRNAs, rRNAs form characteristic secondary structures by complementary base pairing (Figure 8.5). In association with ribosomal proteins, the

rRNAs fold further, into distinct three-dimensional structures. Initially, rRNAs were thought to play a structural role, providing a scaffold upon which ribosomal proteins assemble. However, with the discovery of the catalytic activity of other RNA molecules (e.g., RNase P and the self-splicing introns discussed in Chapter 7), the possible catalytic role of rRNA became widely considered. Consistent with this hypothesis, rRNAs were found to be absolutely required for the *in vitro* assembly of functional ribosomes. On the other hand, the omission of many ribosomal proteins resulted in a decrease, but not a complete loss, of ribosomal activity.

Direct evidence for the catalytic activity of rRNA first came from experiments of Harry Noller and his colleagues in 1992. These investigators demonstrated that the large ribosomal subunit is able to catalyze the formation of peptide bonds (the peptidyl transferase reaction) even after approximately 90% of the ribosomal proteins have been removed by standard protein extraction procedures. In contrast, treatment with RNase completely abolishes peptide bond formation, providing strong support for the hypothesis that the formation of a peptide bond is an RNA-catalyzed reaction. However, some of the ribosomal proteins could not be removed under conditions that left the ribosomal RNA intact, so the role of ribosomal proteins as catalysts of peptide bond formation could not be definitively ruled out.

Unambiguous evidence that protein synthesis is catalyzed by rRNA came from the first high-resolution structural analysis of the 50S ribosomal subunit, which was reported by Peter Moore, Thomas Seitz, and their colleagues in 2000 (Figure 8.6). This atomic-level view of ribosome structure revealed that ribosomal proteins were strikingly absent from the site at which the peptidyl transferase reaction occurred, making it evident that rRNA was responsible for catalyzing peptide bond formation. The large

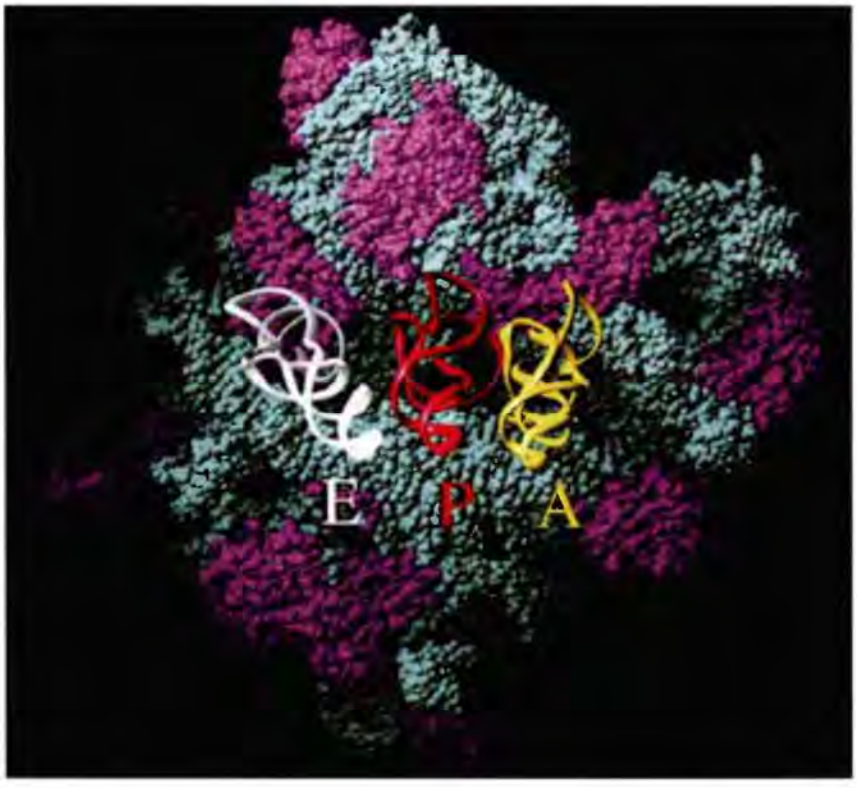


FIGURE 8.6 Structure of the 50S ribosomal subunit A high-resolution model of the 50S ribosomal subunit with three tRNA molecules bound to the A, P, and E sites of the ribosome (see Figure 8.12). Ribosomal proteins are shown in pink and the rRNA in blue. (From P. Nissen, J. Hansen, N. Ban, P. B. Moore and T. A. Steitz. 2000. *Science* 289: 920.)

KEY EXPERIMENT

Catalytic Role of Ribosomal RNA

Unusual Resistance of Peptidyl Transferase to Protein Extraction Procedures

Harry F. Noller, Vernita Hoffarth, and Ludwika Zimniak
University of California at Santa Cruz
Science, Volume 256, 1992, pages 1416–1419

The Context

The role of ribosomes in protein synthesis was elucidated in the 1960s. During this period, ribosomes were characterized as particles consisting of both proteins and RNAs, and the reconstitution of functional ribosomes from purified components was accomplished. At that time peptide bond formation (the peptidyl transferase reaction) was generally assumed to be catalyzed by ribosomal proteins, with the rRNAs delegated to a supporting role in ribosome structure. As early as the 1970s, however, evidence began to suggest that the rRNAs might play a more direct role in protein synthesis. For example, many ribosomal proteins were found to be nonessential for ribosome function. Conversely, the sequences of some parts of rRNA were shown to be extremely well conserved in evolution, suggesting a critical functional role for these portions of the rRNA molecule.

In the early 1980s, the catalytic activity of RNA molecules was established by Tom Cech's studies of the *Tetrahymena* ribozyme and Sidney Altman's studies of RNase P. These discoveries of RNA catalysis provided a precedent for the hypothesis that rRNA is directly involved in catalyzing peptide bond formation. Compelling evidence in support of such a role for rRNA was provided in this 1992 paper by Harry Noller and his colleagues.

The Experiments

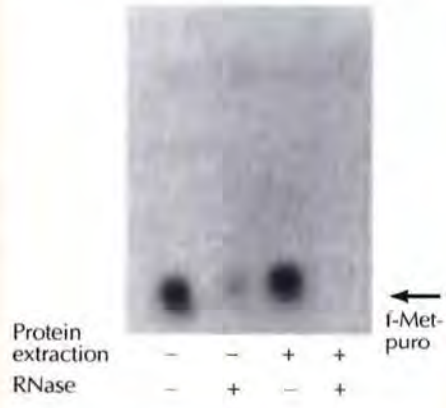
To study the catalytic activity of rRNA, Noller and colleagues used a simplified model reaction to assay peptidyl transferase activity. This reaction measures the transfer of radioactively labeled *N*-formylmethionine from a fragment of tRNA to the amino group of puromycin, an antibiotic that resembles an aminoacyl tRNA and can form peptide bonds with a growing peptide chain. The advantage of this peptidyl transferase model reaction is that it can be carried out by isolated 50S ribosomal subunits; small ribosomal subunits, other protein factors, and mRNA are not required.

The investigators then tested the role of rRNA by assaying the peptidyl transferase activity of 50S subunits from which the ribosomal proteins had been removed by standard protein extraction procedures. An important aspect of the experiments was the use of ribosomes from the bacterium *Thermus aquaticus*. Because these bacteria live at high temperatures, the structure of their rRNA was thought to be more stable than that of *E. coli* rRNA. The critical result was that the peptidyl transferase activity of *T. aquaticus* ribosomes was completely resistant to a vigorous extraction procedure involving treatment with detergents, proteases, and phenol (see figure). Most strikingly, full peptidyl transferase activity was retained even after repeated extractions that removed 90% of the ribosomal protein. In contrast, peptidyl transferase of either intact or extracted ribosomes was extremely sensitive to even a brief treatment with RNase. Although these experiments could not exclude a possible role for the remaining ribosomal proteins, they provided strong support for the direct participation of 23S rRNA in the peptidyl transferase reaction.

Harry F. Noller

The Impact

The results of Noller's experiments were definitively confirmed by the high-resolution structural analysis of the 50S ribosomal subunit reported by Peter Moore, Thomas Steitz and their colleagues in 2000. The ribosomal site of the peptidyl transferase reaction contained only rRNA, with no ribosomal proteins in the vicinity. These results remove any doubts as to the catalytic role of RNA in this reaction and demonstrate that the fundamental reaction of protein synthesis is cat-



The peptidyl transferase reaction is assayed by the formation of radioactive N-formylmethionine-puromycin (f-Met-puro), which is detected by electrophoresis and autoradiography. Ribosomes from *T. aquaticus* were subjected to protein extraction or treatment with RNase, as indicated.

KEY EXPERIMENT

alyzed by ribosomal RNA. These findings not only have a striking impact on our understanding of ribosome function but also profoundly extend the previously described catalytic activities of RNA molecules and provide substantial new support for the hypothesis that an early period of evolution was an RNA world populated by self-replicating RNA molecules. This hypothesis had previously been based on the ability of RNA molecules to catalyze the reactions required for their own replication. The discovery that RNA could catalyze reactions involved in protein synthesis provided a clear link between the RNA world and the flow of genetic information in present-day cells, with rRNA continuing to function in the key reaction of peptide bond formation.

ribosomal subunit thus functions as a ribozyme, with the fundamental reaction of protein synthesis being catalyzed by ribosomal RNA. Rather than being the primary catalytic constituents of ribosomes, ribosomal proteins are now thought to play a largely structural role.

The direct involvement of rRNA in the peptidyl transferase reaction has important evolutionary implications. RNAs are thought to have been the first self-replicating macromolecules (see Chapter 1). This notion is strongly supported by the fact that ribozymes, such as RNase P and self-splicing introns, can catalyze reactions that involve RNA substrates. The role of rRNA in the formation of peptide bonds extends the catalytic activities of RNA beyond self-replication to direct involvement in protein synthesis. Additional studies indicate that the *Tetrahymena* rRNA ribozyme can catalyze the attachment of amino acids to RNA, lending credence to the possibility that the original aminoacyl tRNA synthetases were RNAs rather than proteins. The ability of RNA molecules to catalyze the reactions required for protein synthesis as well as for self-replication may provide an important link for understanding the early evolution of cells.

The Organization of mRNAs and the Initiation of Translation

Although the mechanisms of protein synthesis in prokaryotic and eukaryotic cells are similar, there are also differences, particularly in the signals that determine the positions at which synthesis of a polypeptide chain is initiated on an mRNA template (Figure 8.7). Translation does not simply begin at the 5′ end of the mRNA; it starts at specific initiation sites. The 5′ terminal portions of both prokaryotic and eukaryotic mRNAs are therefore noncoding sequences, referred to as **5′ untranslated regions**. Eukaryotic mRNAs usually encode only a single polypeptide chain, but many prokaryotic mRNAs encode multiple polypeptides that are synthesized independently from distinct initiation sites. For example, the *E. coli lac* operon consists of three genes that are translated from the same mRNA (see Figure 7.9). Messenger RNAs that encode multiple polypeptides are called **polycistronic**, whereas **monocistronic** mRNAs encode a single polypeptide chain. Finally, both prokaryotic and eukaryotic mRNAs end in noncoding **3′ untranslated regions**.

In both prokaryotic and eukaryotic cells, translation always initiates with the amino acid methionine, usually encoded by AUG. Alternative initiation codons, such as GUG, are used occasionally, but when they occur at the beginning of a polypeptide chain, these codons direct the incorporation of methionine rather than the amino acid they normally encode (GUG normally encodes valine). In most bacteria, protein synthesis is initiated with a

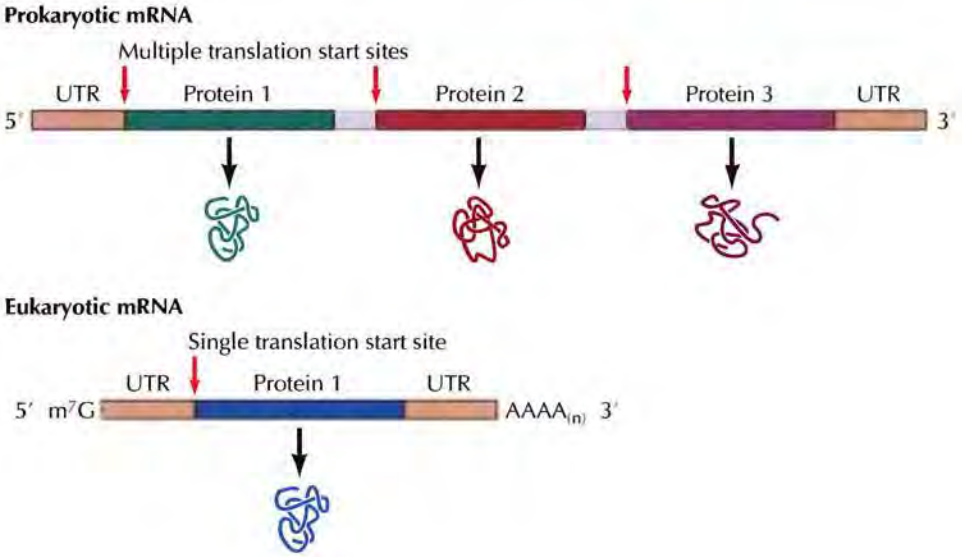


FIGURE 8.7 Prokaryotic and eukaryotic mRNAs Both prokaryotic and eukaryotic mRNAs contain untranslated regions (UTRs) at their 5′ and 3′ ends. Eukaryotic mRNAs also contain 5′ 7-methylguanosine (m^7G) caps and 3′ poly-A tails. Prokaryotic mRNAs are frequently polycistronic: They encode multiple proteins, each of which is translated from an independent start site. Eukaryotic mRNAs are usually monocistronic, encoding only a single protein.

modified methionine residue (*N*-formylmethionine), whereas unmodified methionines initiate protein synthesis in eukaryotes (except in mitochondria and chloroplasts, whose ribosomes resemble those of bacteria).

The signals that identify initiation codons are different in prokaryotic and eukaryotic cells, consistent with the distinct functions of polycistronic and monocistronic mRNAs (Figure 8.8). Initiation codons in bacterial mRNAs are preceded by a specific sequence (called a **Shine-Dalgarno sequence**, after its discoverers, John Shine and Lynn Dalgarno) that aligns the mRNA on the ribosome for translation by base-pairing with a complementary sequence near the 3′ terminus of 16S rRNA. This base-pairing interaction enables bacterial ribosomes to initiate translation not only at the 5′ end of an mRNA but also at the internal initiation sites of polycistronic messages. In contrast, ribosomes recognize most eukaryotic mRNAs by binding to the 7-methylguanosine cap at their 5′ terminus (see Figure 7.44). The ribosomes then scan downstream of the 5′ cap until they encounter the

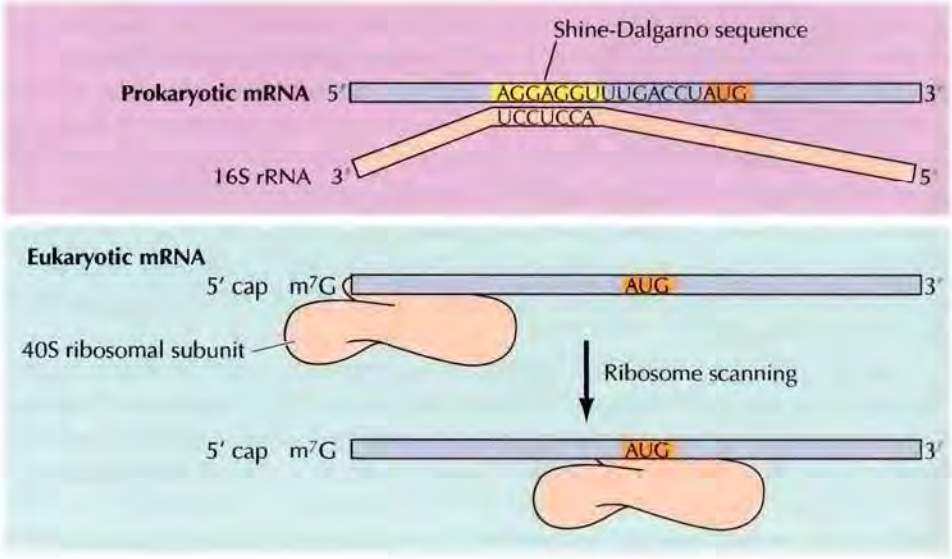


FIGURE 8.8 Signals for translation initiation Initiation sites in prokaryotic mRNAs are characterized by a Shine-Dalgarno sequence that precedes the AUG initiation codon. Base pairing between the Shine-Dalgarno sequence and a complementary sequence near the 3′ terminus of 16S rRNA aligns the mRNA on the ribosome. In contrast, eukaryotic mRNAs are bound to the 40S ribosomal subunit by their 5′ 7-methylguanosine caps. The ribosome then scans along the mRNA until it encounters an AUG initiation codon.

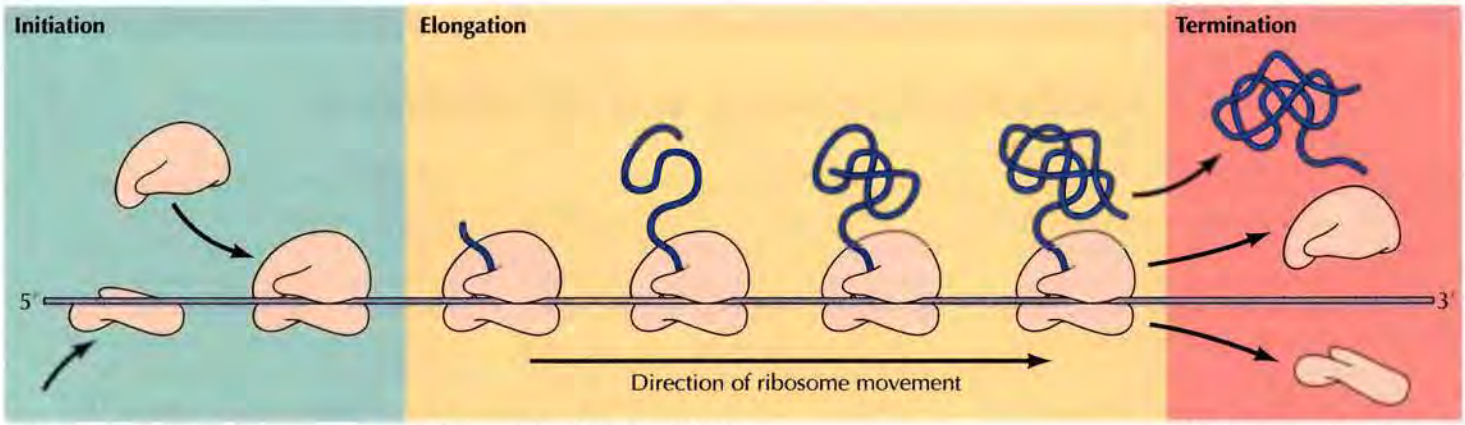


FIGURE 8.9 Overview of translation

initiation codon (usually AUG). Sequences that surround AUGs affect the efficiency of initiation, so in some cases the first AUG in the mRNA is bypassed and translation initiates at an AUG farther downstream. It should also be noted that several viral and cellular mRNAs have internal ribosome entry sites at which translation can initiate by ribosome binding to an internal position on the mRNA. Neither the biological function nor the molecular mechanism by which translation initiates at these internal sites is understood, so the role of internal ribosomal entry sites in eukaryotic mRNAs remains an active area of investigation.

The Process of Translation

Translation is generally divided into three stages: initiation, elongation, and termination (Figure 8.9). In both prokaryotes and eukaryotes the first step of the initiation stage is the binding of a specific initiator methionyl tRNA and the mRNA to the small ribosomal subunit. The large ribosomal subunit then joins the complex, forming a functional ribosome on which elongation of the polypeptide chain proceeds. A number of specific nonribosomal proteins are also required for the various stages of the translation process (Table 8.1).

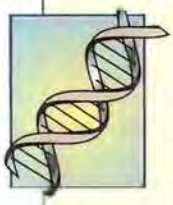
The first translation step in bacteria is the binding of three **initiation factors** (IF1, IF2, and IF3) to the 30S ribosomal subunit (Figure 8.10). The mRNA and initiator *N*-formylmethionyl tRNA then join the complex, with IF2 (which is bound to GTP) specifically recognizing the initiator tRNA. IF3 is then released, allowing a 50S ribosomal subunit to associate with the complex. This association triggers the hydrolysis of GTP bound to IF2, which

TABLE 8.1 Translation Factors

Role	Prokaryotes	Eukaryotes
Initiation	IF1, IF2, IF3	eIF1, eIF1A, eIF2, eIF2B, eIF3, eIF4A, eIF4B, eIF4E, eIF4G, eIF4H, eIF5, eIF5B
Elongation	EF-Tu, EF-Ts, EF-G	eEF1α, eEF1$\beta\gamma$, eEF2
Termination	RF1, RF2, RF3	eRF1, eRF3

MOLECULAR MEDICINE

Antibiotic Resistance and the Ribosome



The Disease

Bacteria are responsible for a wide variety of potentially lethal infectious diseases, including tuberculosis, bacterial pneumonia, childhood meningitis, infections of wounds and burns, syphilis, and gonorrhea. Prior to the 1940s, physicians had no effective treatments for these bacterial infections. At that time, the first antibiotics became available for clinical use. Previously untreatable infections became curable, and a significant increase in average life span has been attributed to the introduction of antibiotics into clinical practice. More than a hundred different antibiotics are now in use, providing effective treatment for bacterial infections. However, bacteria are increasingly becoming resistant to the antibiotics in current use, so the development of new drugs is a continuing challenge.

Molecular and Cellular Basis

To be effective clinically, an antibiotic must kill or inhibit the growth of bacteria without being toxic to humans. Thus most clinically useful antibiotics are directed against targets that are present in bacteria but not in human cells. Penicillin, for example, inhibits synthesis of the bacterial cell wall (see Chapter 14). Many other commonly used antibiotics, however, inhibit protein synthesis by binding to the ribosome. Some of these antibiotics—including streptomycin, tetracycline, chloramphenicol, and erythromycin—are specific for prokaryotic ribosomes and therefore are effective agents for the treatment of bacterial infections. Other antibiotic inhibitors of protein synthesis act against both prokaryotes and eukaryotes (e.g., puromycin) or against eukaryotes only (e.g., cycloheximide). Although these antibiotics are obviously not useful clinically, they have provided important experimental tools for studies of protein synthesis in both prokaryotic and eukaryotic cells.

Prevention and Treatment

The use of antibiotics has had a major impact on modern medicine by enabling physicians to cure otherwise life-threatening bacterial infections. Unfortunately, however, mutations can lead to the development of antibiotic-resistant strains of bacteria. Many strains of bacteria are now resistant to one or more antibiotics, so physicians sometimes have to try several different antibiotics before they find one that is effective. Bacterial resistance to antibiotics is increasing rapidly, particularly in hospitals. It is estimated that about two million patients a year acquire bacterial infections in hospitals, and more than half of these bacteria are resistant to at least one antibiotic. Most seriously, strains of bacteria that are resistant to multiple antibiotics have emerged, and a few strains of bacteria, frequently found in hospitals, are now resistant to all but one or a few known antibiotics. The emergence of these strains with multiple resistance raises the specter of untreatable infections caused by the spread of antibiotic-resistant bacteria—a scenario that would herald a return to the pre-antibiotic era of infectious diseases. The continuing development of new antibiotics thus represents an important concern in drug development. Our increasing understanding of the structures of several antibiotics bound to bacterial ribosomes may provide a rational approach to this problem and allow the targeted development of new antibiotics to replace those against which bacterial resistance has emerged.

References

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Steitz, T. A. 2005. On the structural basis of peptide-bond formation and antibiotic resistance from atomic structures of the large ribosomal subunit. *FEBS Lett.* 579: 955–958.

Antibiotic Inhibitors of Protein Synthesis

Antibiotic	Target cells	Effect
Streptomycin	Prokaryotic	Inhibits initiation and causes misreading
Tetracycline	Prokaryotic	Inhibits binding of aminoacyl tRNAs
Chloramphenicol	Prokaryotic	Inhibits peptidyl transferase activity
Erythromycin	Prokaryotic	Inhibits translocation
Puromycin	Prokaryotic and eukaryotic	Causes premature chain termination
Cycloheximide	Eukaryotic	Inhibits peptidyl transferase activity

The structures of seven different antibiotics bound to a bacterial ribosome. (From T. A. Steitz. 2005. *FEBS Lett.* 579: 955.)

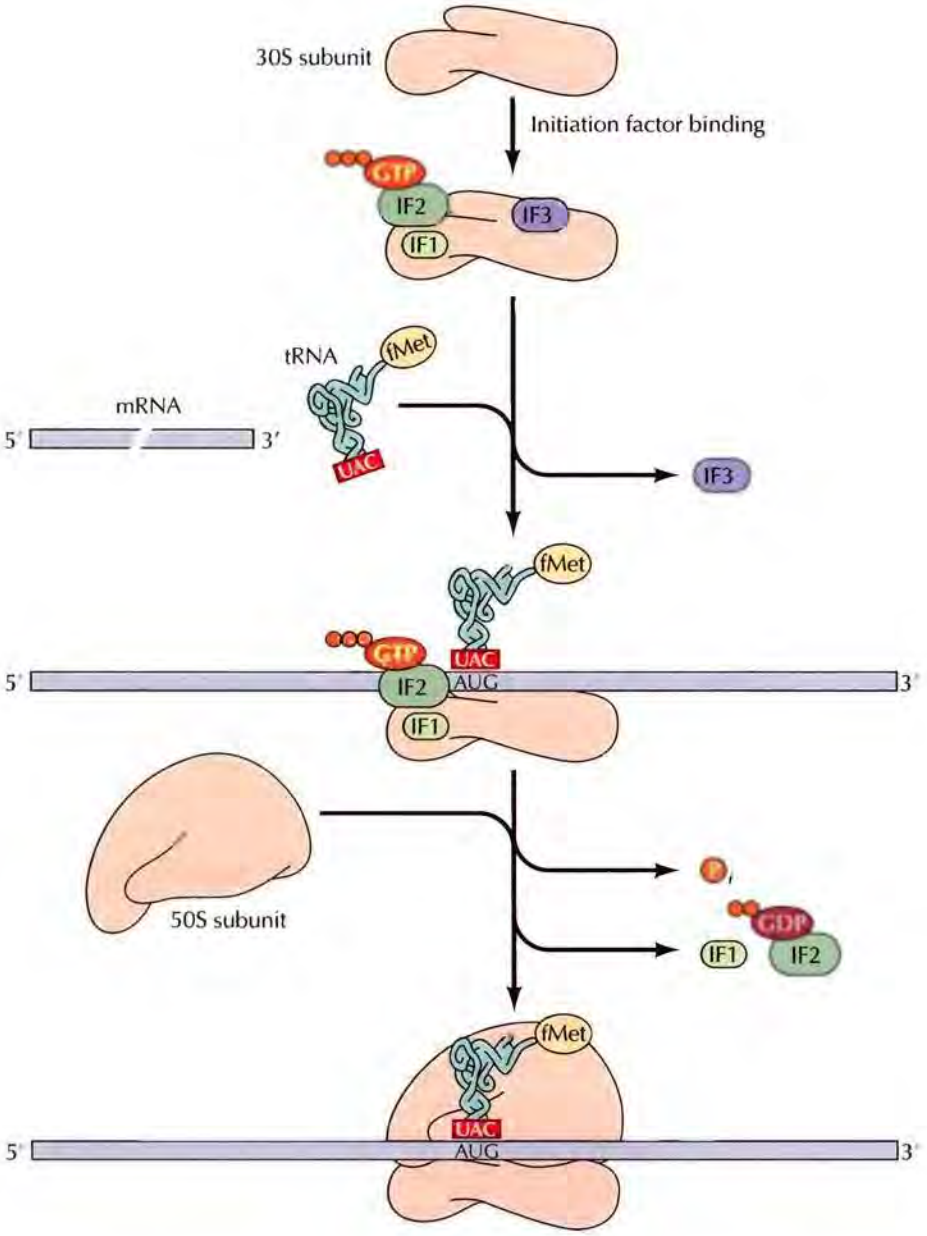


FIGURE 8.10 Initiation of translation in bacteria Three initiation factors (IF1, IF2, and IF3) first bind to the 30S ribosomal subunit. This step is followed by binding of the mRNA and the initiator *N*-formylmethionyl (fMet) tRNA, which is recognized by IF2 bound to GTP. IF3 is then released, and a 50S subunit binds to the complex, triggering the hydrolysis of bound GTP, followed by the release of IF1 and IF2 bound to GDP.

leads to the release of IF1 and IF2 (bound to GDP). The result is the formation of a 70S initiation complex (with mRNA and initiator tRNA bound to the ribosome) that is ready to begin peptide bond formation during the elongation stage of translation.

Initiation in eukaryotes is more complicated and requires at least twelve proteins (each consisting of multiple polypeptide chains), which are designated eIFs (*e*ukaryotic *i*nitiation *f*actors; see Table 8.1). The factors eIF1A and eIF3 bind to the 40S ribosomal subunit, and eIF2 (in a complex with GTP) binds to the initiator methionyl tRNA (Figure 8.11). A preinitiation complex is then formed by association of the 40S subunit, the initiator tRNA, and eIF1. The mRNA is recognized and brought to the ribosome by the eIF4 group of factors. The 5′ cap of the mRNA is recognized by eIF4E, which forms a complex with eIF4A and eIF4G. eIF4G also binds to poly-A binding protein (PABP), which is associated with the poly-A tail at the 3′ end of the mRNA. Eukaryotic initiation factors thus recognize both the 5′ and 3′ ends of mRNAs, accounting for the stimulatory effect of polyadeny-

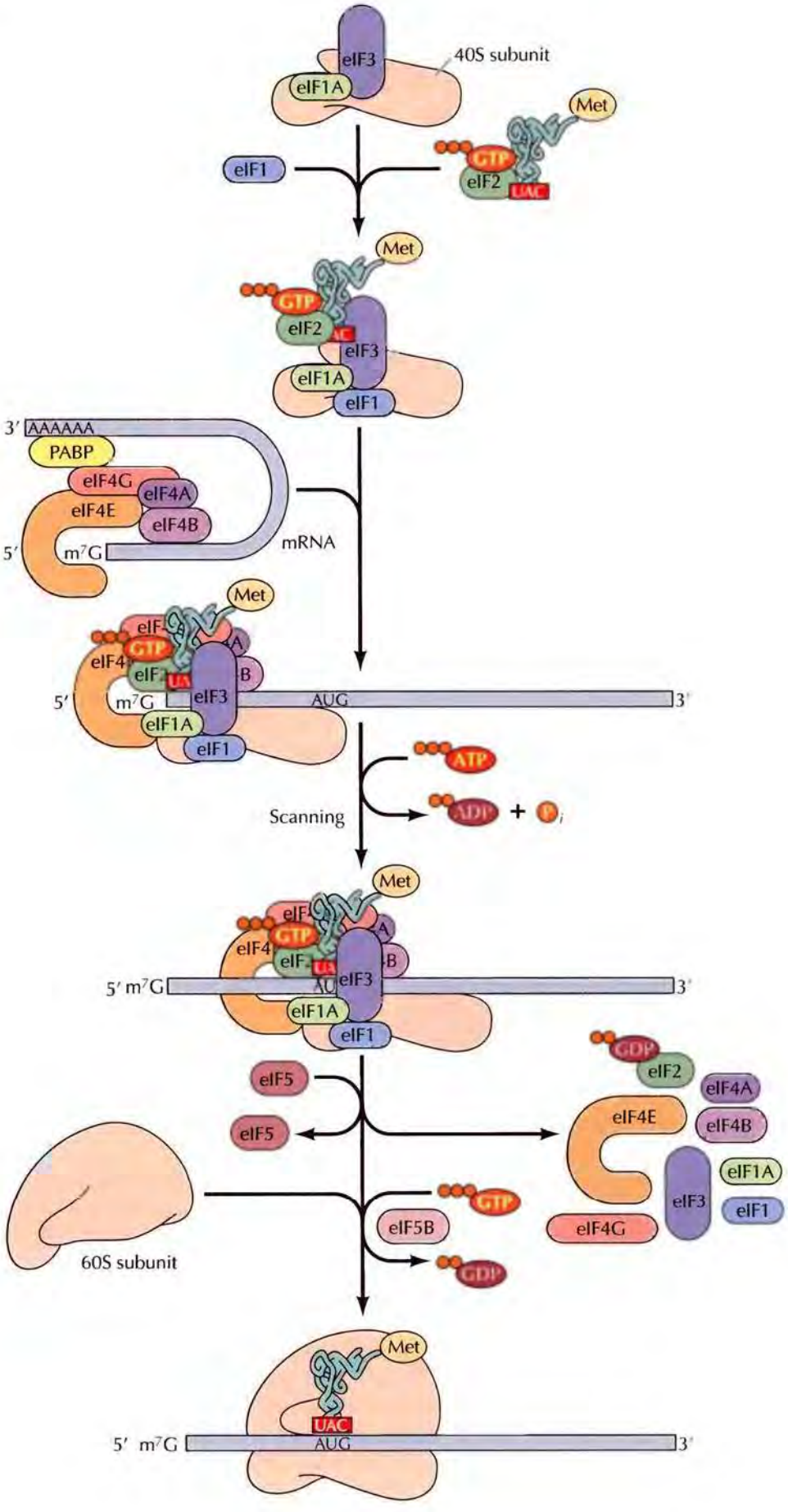


FIGURE 8.11 Initiation of translation in eukaryotic cells Initiation factors eIF1A and eIF3 bind to the 40S ribosomal subunit. The initiator methionyl tRNA is bound by eIF2 (complexed to GTP), and forms a complex with the 40S subunit and eIF1. The mRNA is brought to the 40S subunit by eIF4E (which binds to the 5′ cap), eIF4G (which binds to both eIF4E at the 5′ cap and PABP at the 3′ poly-A tail), eIF4A, and eIF4B. The ribosome then scans down the mRNA to identify the first AUG initiation codon. Scanning requires energy and is accompanied by ATP hydrolysis. When the initiating AUG is identified, eIF5 triggers the hydrolysis of GTP bound to eIF2, followed by the release of eIF2 (complexed to GDP) and other initiation factors. The 60S ribosomal subunit then joins the 40S complex, facilitated by eIF5B.

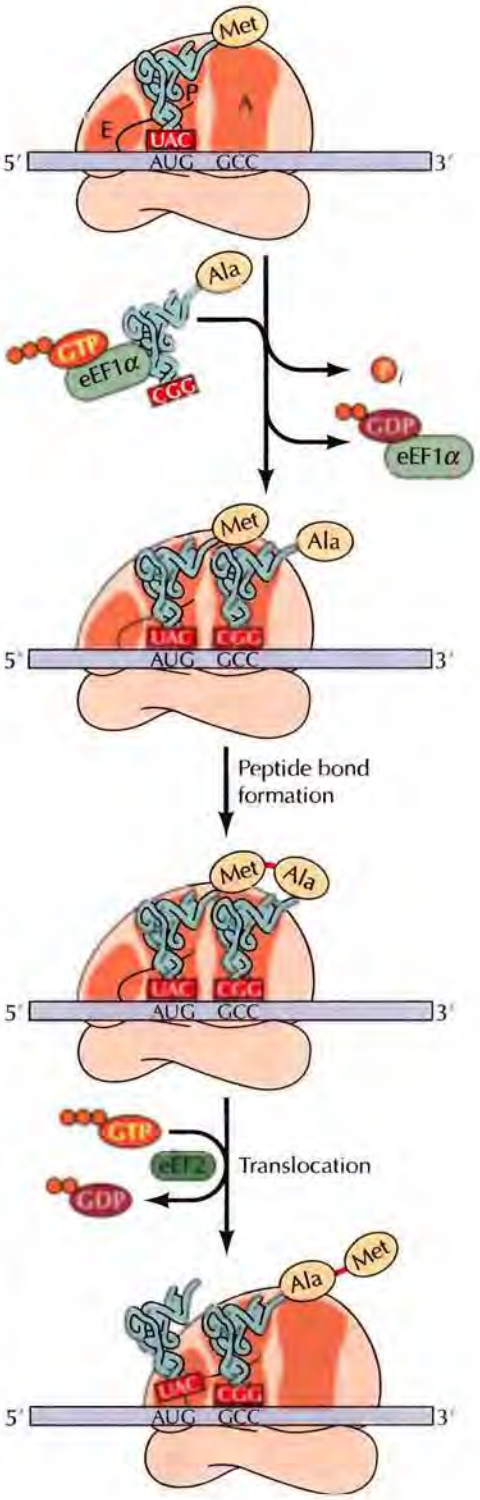
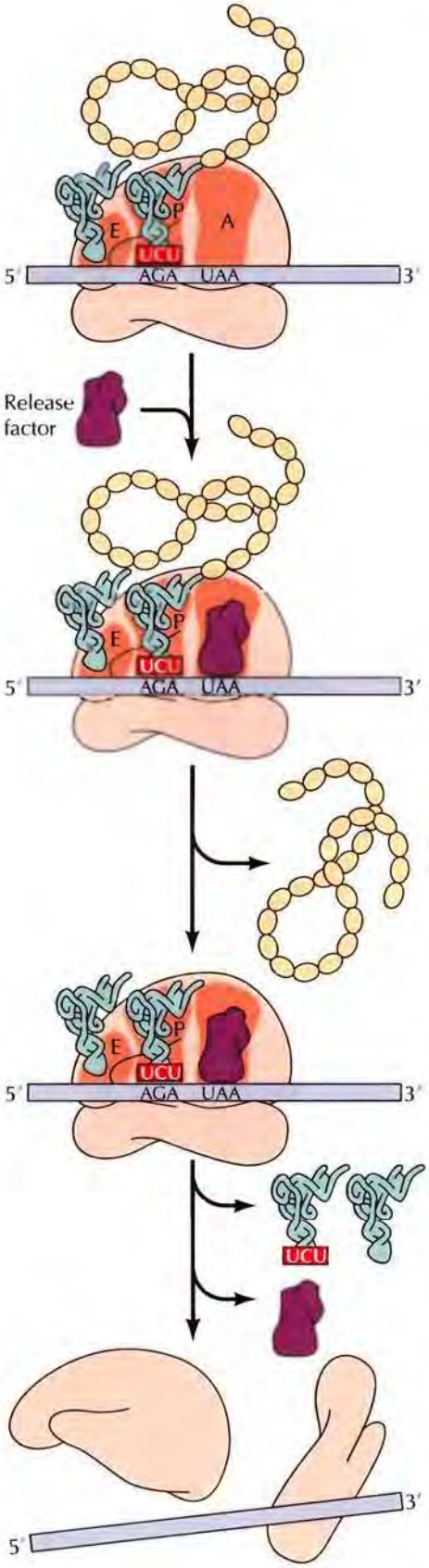


FIGURE 8.12 Elongation stage of translation The ribosome has three tRNA-binding sites, designated P (peptidyl), A (aminoacyl), and E (exit). The initiating methionyl tRNA is positioned in the P site, leaving an empty A site. The second aminoacyl tRNA (e.g., alanyl tRNA) is then brought to the A site by eEF1α (complexed with GTP). Following GTP hydrolysis, eEF1α (complexed with GDP) leaves the ribosome, with alanyl tRNA inserted into the A site. A peptide bond is then formed, resulting in the transfer of methionine to the aminoacyl tRNA at the A site. The ribosome then moves three nucleotides along the mRNA. This movement translocates the peptidyl (Met-Ala) tRNA to the P site and the uncharged tRNA to the E site, leaving an empty A site ready for addition of the next amino acid. Translocation is mediated by eEF2, coupled to GTP hydrolysis. The process, illustrated here for eukaryotic cells, is very similar in prokaryotes. (Table 8.1 gives the names of the prokaryotic elongation factors.)

lation on translation. The initiation factors eIF4E and eIF4G, in association with eIF4A and eIF4B, then bring the mRNA to the 40S ribosomal subunit, with eIF4G interacting with eIF3. The 40S ribosomal subunit, in association with the bound methionyl tRNA and eIFs, then scans the mRNA to identify the AUG initiation codon. When the AUG codon is reached, eIF5 triggers the hydrolysis of GTP bound to eIF2. Initiation factors (including eIF2 bound to GDP) are then released, and a 60S subunit binds to the 40S subunit to form the 80S initiation complex of eukaryotic cells.

After the initiation complex has formed, translation proceeds by elongation of the polypeptide chain. The mechanism of elongation in prokaryotic and eukaryotic cells is very similar (Figure 8.12). The ribosome has three sites for tRNA binding, designated the P (peptidyl), A (aminoacyl), and E (exit) sites. The initiator methionyl tRNA is bound at the P site. The first step in elongation is the binding of the next aminoacyl tRNA to the A site by pairing with the second codon of the mRNA. The aminoacyl tRNA is escorted to the ribosome by an **elongation factor** (EF-Tu in prokaryotes, eEF1α in eukaryotes), which is complexed to GTP. The selection of the correct aminoacyl tRNA for incorporation into the growing polypeptide chain is the critical step that determines the accuracy of protein synthesis. Although this selection is based on base pairing between the codon on mRNA and the anticodon on tRNA, base pairing alone is not sufficient to account for the accuracy of protein synthesis, which has an error rate of less than 10^{-3}. This accuracy is provided by a "decoding center" in the small ribosomal subunit, which recognizes correct codon-anticodon base pairs and discriminates against mismatches. Insertion of a correct aminoacyl tRNA into the A site triggers a conformational change that induces the hydrolysis of GTP bound to eEF1α and release of the elongation factor bound to GDP. A notable result of the recent structural studies of the ribosome is that recognition of correct codon-anticodon pairing in the decoding center, like peptidyl transferase activity, is principally based on the activity of ribosomal RNA rather than proteins.

Once eEF1α has left the ribosome, a peptide bond can be formed between the initiator methionyl tRNA at the P site and the second aminoacyl tRNA at the A site. This reaction is catalyzed by the large ribosomal subunit, with the rRNA playing a critical role (as already discussed). The result is the transfer of methionine to the aminoacyl tRNA at the A site of the ribosome, forming a peptidyl tRNA at this position and leaving the uncharged initiator tRNA at the P site. The next step in elongation is translocation, which requires another elongation factor (EF-G in prokaryotes, eEF2 in



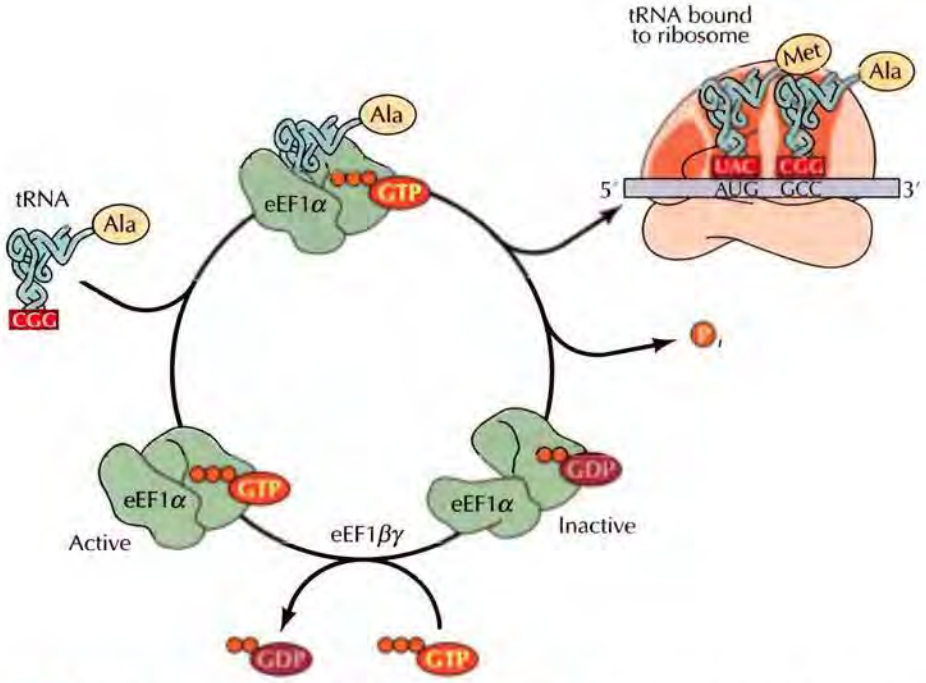


FIGURE 8.13 Regeneration of eEF1α/GTP eEF1α complexed to GTP escorts the aminoacyl tRNA to the ribosome. The bound GTP is hydrolyzed as the correct tRNA is inserted, so eEF1α complexed to GDP is released. The eEF1α/GDP complex is inactive and unable to bind another tRNA. In order for translation to continue, the active eEF1α/GTP complex must be regenerated by another factor, eEF1βγ, which stimulates the exchange of the bound GDP for free GTP.

eukaryotes) and is again coupled to GTP hydrolysis. During translocation, the ribosome moves three nucleotides along the mRNA, positioning the next codon in an empty A site. This step translocates the peptidyl tRNA from the A site to the P site, and the uncharged tRNA from the P site to the E site. The ribosome is then left with a peptidyl tRNA bound at the P site, and an empty A site. The binding of a new aminoacyl tRNA to the A site then induces the release of the uncharged tRNA from the E site, leaving the ribosome ready for insertion of the next amino acid in the growing polypeptide chain.

As elongation continues, the eEF1α (or EF-Tu) that is released from the ribosome bound to GDP must be reconverted to its GTP form (Figure 8.13). This conversion requires a third elongation factor, eEF1βγ (EF-Ts in prokaryotes), which binds to the eEF1α/GDP complex and promotes the exchange of bound GDP for GTP. This exchange results in the regeneration of eEF1α /GTP, which is now ready to escort a new aminoacyl tRNA to the A site of the ribosome, beginning a new cycle of elongation. The regulation of eEF1α by GTP binding and hydrolysis illustrates a common means of the regulation of protein activities. As will be discussed in later chapters, similar mechanisms control the activities of a wide variety of proteins involved in the regulation of cell growth and differentiation, as well as in protein transport and secretion.

FIGURE 8.14 Termination of translation A termination codon (e.g., UAA) at the A site is recognized by a release factor rather than by a tRNA. The result is the release of the completed polypeptide chain, followed by the dissociation of tRNA and mRNA from the ribosome.

Elongation of the polypeptide chain continues until a stop codon (UAA, UAG, or UGA) is translocated into the A site of the ribosome. Cells do not contain tRNAs with anticodons complementary to these termination signals; instead, they have **release factors** that recognize the signals and terminate protein synthesis (Figure 8.14). Prokaryotic cells contain two release factors that recognize termination codons: RF1 recognizes UAA or UAG, and RF2 recognizes UAA or UGA (see Table 8.1). In eukaryotic cells a single release factor (eRF1) recognizes all three termination codons. Both prokaryotic and eukaryotic cells also contain release factors (RF3 and eRF3, respectively) that do not recognize specific termination codons but act together with RF1 (or eRF1) and RF2. The release factors bind to a termination codon at the A site and stimulate hydrolysis of the bond between the tRNA and the polypeptide chain at the P site, resulting in release of the completed polypeptide from the ribosome. The tRNA is then released, and the ribosomal subunits and the mRNA template dissociate.

Messenger RNAs can be translated simultaneously by several ribosomes in both prokaryotic and eukaryotic cells. Once one ribosome has moved away from the initiation site, another can bind to the mRNA and begin synthesis of a new polypeptide chain. Thus mRNAs are usually translated by a series of ribosomes, spaced at intervals of about 100 to 200 nucleotides (Figure 8.15). The group of ribosomes bound to an mRNA molecule is called a polyribosome, or **polysome**. Each ribosome within the group functions independently to synthesize a separate polypeptide chain.

■ **In some cases, selenocysteine, which has been termed the 21st amino acid, is inserted at stop codons. Selenocysteine is a selenium-containing amino acid that is found in proteins from diverse organisms, including humans. Another modified amino acid, pyrrolysine, can also be inserted at stop codons. Pyrrolysine is a modified lysine found in proteins from some archaeabacteria and is important in the production of methane.**

Regulation of Translation

Although transcription is the initial level at which gene expression is controlled, the translation of mRNAs is also regulated in both prokaryotic and eukaryotic cells. The translation of particular mRNAs can be regulated by translational repressor proteins, noncoding microRNAs, and controlled polyadenylation. In addition, the global translational activity of cells is modulated in response to cell stress, nutrient availability, and growth factor stimulation.

One mechanism of translational regulation is the binding of repressor proteins (which block translation) to specific mRNA sequences. A well understood example of this mechanism in eukaryotic cells is regulation of the synthesis of ferritin, a protein that stores iron within the cell. The trans-

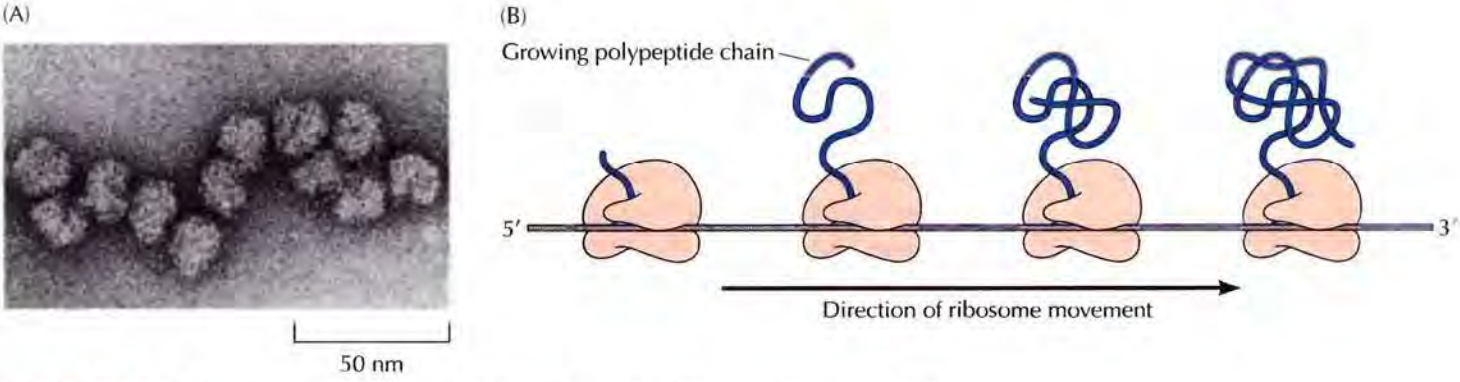


FIGURE 8.15 Polysomes Messenger RNAs are translated by a series of multiple ribosomes (a polysome). (A) Electron micrograph of a eukaryotic polysome. (B) Schematic of a generalized polysome. Note that the ribosomes closer to the 3′ end of the mRNA have longer polypeptide chains. (A, from M. Boublik et al., 1990. *The Ribosome*, p. 117. Courtesy of American Society for Microbiology.)

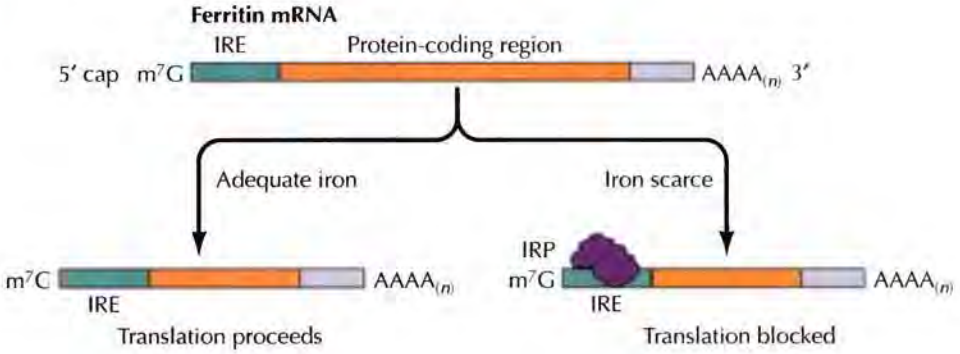


FIGURE 8.16 Translational regulation of ferritin The mRNA contains an iron response element (IRE) near its 5′ cap. In the presence of adequate supplies of iron, translation of the mRNA proceeds normally. If iron is scarce, however, a protein (called the iron regulatory protein, or IRP) binds to the IRE, blocking translation of the mRNA.

lation of ferritin mRNA is regulated by the supply of iron: More ferritin is synthesized if iron is abundant (Figure 8.16). This regulation is mediated by a protein, which (in the absence of iron) binds to a sequence (the iron response element, or IRE) in the 5′ untranslated region of ferritin mRNA, blocking its translation. In the presence of iron, the repressor no longer binds to the IRE and ferritin translation is able to proceed.

It is interesting to note that the regulation of translation of ferritin mRNA by iron is similar to the regulation of transferrin receptor mRNA stability, which was discussed in the previous chapter (see Figure 7.55). Namely, the stability of transferrin receptor mRNA is regulated by protein binding to an IRE in its 3′ untranslated region. The same protein binds to the IREs of both ferritin and transferrin receptor mRNAs. However, the consequences of protein binding to the two IREs are quite different. Protein bound to the transferrin receptor IRE protects the mRNA from degradation rather than inhibiting its translation. These distinct effects presumably result from the different locations of the IRE in the two mRNAs. To function as a repressor-binding site, the IRE must be located within 70 nucleotides of the 5′ cap of ferritin mRNA, suggesting that protein binding to the IRE blocks translation by interfering with cap recognition and binding of the 40S ribosomal subunit. Rather than inhibiting translation, protein binding to the same sequence in the 3′ untranslated region of transferrin receptor mRNA protects the mRNA from nuclease degradation. Binding of the same regulatory protein to different sites on mRNA molecules can thus have distinct effects on gene expression, in one case inhibiting translation and in the other case stabilizing the mRNA to increase protein synthesis.

Translation can also be regulated by proteins that bind to specific sequences in the 3′ untranslated regions of some mRNAs. In some cases, these translational repressors function by interacting with the initiation factor eIF4E to inhibit the initiation of translation (Figure 8.17). Proteins that bind to the 3′ untranslated regions of mRNAs are also responsible for localizing mRNAs to specific regions of cells, allowing proteins to be produced in specific subcellular locations. Localization of mRNAs is an important part of translational regulation in a variety of cell types, including eggs, embryos, nerve cells, and moving fibroblasts. For example, localization of mRNAs to specific regions of eggs or embryos plays an important role in development by allowing the encoded proteins to be synthesized at the appropriate sites in the developing embryo (Figure 8.18). The localization of mRNAs is coupled to regulation of their translation, so that their encoded proteins are synthesized when the mRNA becomes properly localized at the appropriate developmental stage.

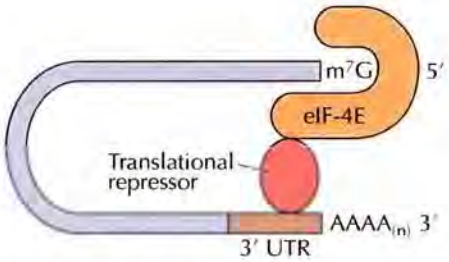


FIGURE 8.17 Translational repressor binding to 3′ untranslated sequences Translational repressors can bind to regulatory sequences in the 3′ untranslated region (UTR) and inhibit translation by binding to the initiation factor eIF4E, bound to the 5′ cap. This interferes with translation by blocking the binding of eIF4E to eIF4G in a normal initiation complex (see Figure 8.11).

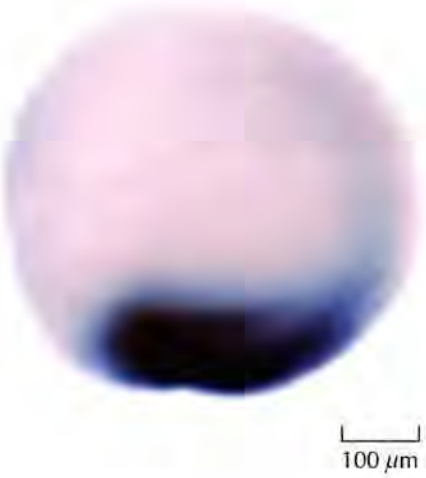


FIGURE 8.18 Localization of mRNA in *Xenopus* oocytes *In situ* hybridization illustrating the localization of Xlerk mRNA to the vegetal cortex of *Xenopus* oocytes. (Courtesy of James Deshler, Boston University.)

Translational regulation is particularly important during early development. As discussed in Chapter 7, a variety of mRNAs are stored in oocytes in an untranslated form; the translation of these stored mRNAs is activated at fertilization or later stages of development. One mechanism of such translational regulation is the controlled polyadenylation of oocyte mRNAs. Many untranslated mRNAs are stored in oocytes with short poly-A tails (approximately 30–50 nucleotides). These stored mRNAs are subsequently recruited for translation at the appropriate stage of development by the lengthening of their poly-A tails to several hundred nucleotides. This allows the binding of poly-A binding protein (PABP), which stimulates translation by interacting with eIF4G (see Figure 8.11).

■ **Specific mRNAs are localized at synapses between neurons. The regulated translation of these localized mRNAs appears to be required for the formation and maintenance of synapses and may play a role in the process of learning.**

As discussed in Chapter 7, gene regulation by noncoding RNAs is a recently discovered phenomenon, the full significance of which is just beginning to be appreciated by cell biologists. **MicroRNAs (miRNAs)** are short (approximately 20–25 bases) double-stranded RNAs that can regulate gene expression at both the transcriptional and translational levels. Transcriptional regulation is mediated by the association of miRNAs with a protein complex called RITS, which represses transcription by chromatin modification of target genes (see Figure 7.37). Translational regulation is mediated by association of miRNAs with a distinct protein complex (called RISC), which targets homologous mRNAs (Figure 8.19). In some cases, the miRNAs direct cleavage of the targeted mRNA by RISC, as discussed in chapter 4 with respect to RNA interference. In other cases, however, the miRNA/RISC complex represses translation without inducing cleavage of the mRNA. It appears that cleavage is generally induced by miRNAs that pair perfectly with their mRNA targets, whereas translational repression in the absence of cleavage results from miRNAs that are mismatched with their targets. The mechanism of translational repression is not yet understood and remains an area of active investigation. It is noteworthy that sequence analysis of human mRNAs has predicted that approximately one-third of human genes are targets for regulation by miRNAs, implying that translational regulation by miRNAs is an extremely widespread mechanism of gene regulation.

Another mechanism of translational regulation in eukaryotic cells, resulting in global effects on overall translational activity rather than on the translation of specific mRNAs, involves modulation of the activity of initiation factors, particularly eIF2 and eIF4E. As already discussed, eIF2 (complexed

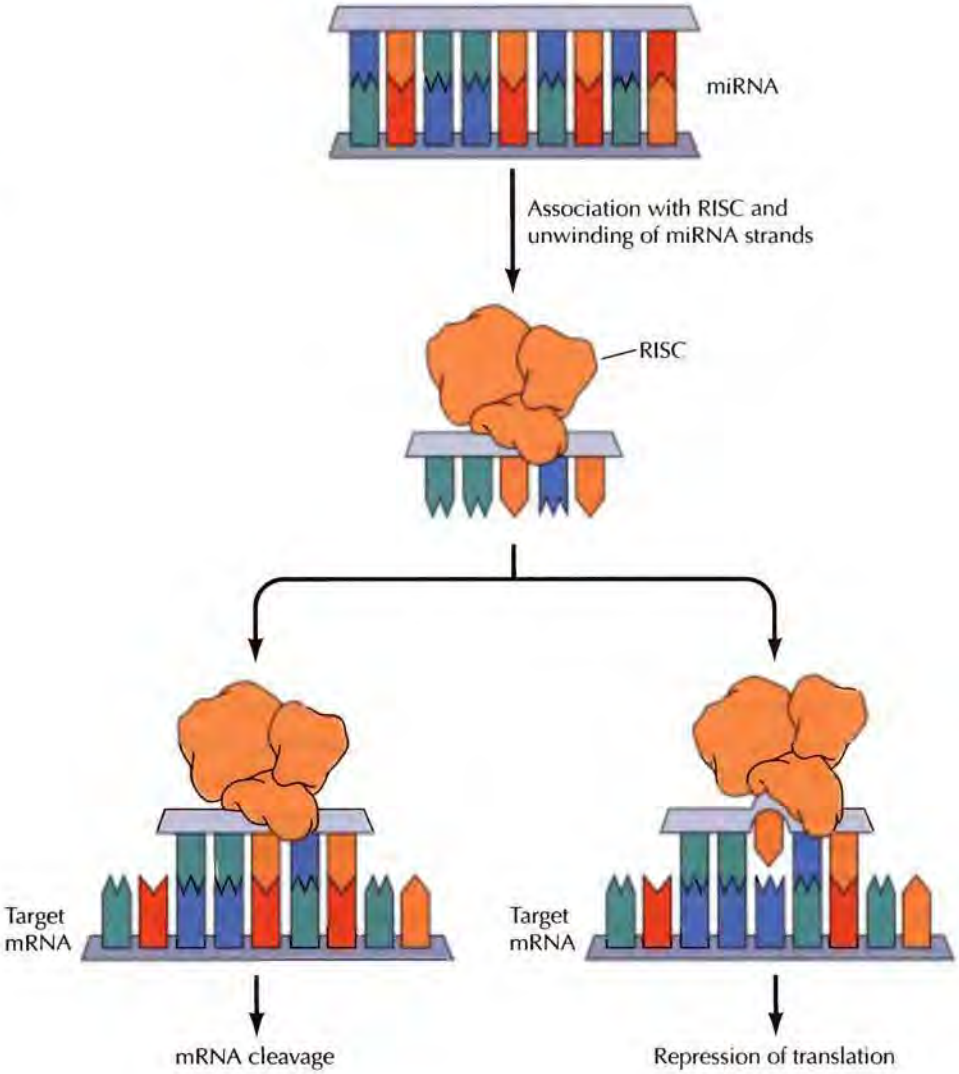


FIGURE 8.19 Regulation of translation by miRNAs miRNAs associate with the RISC complex in which the two strands of the miRNA are unwound. The miRNA then targets RISC to a homologous mRNA, leading either to mRNA cleavage or repression of translation. Cleavage generally results from miRNAs that pair perfectly with their targets, whereas translational repression results from miRNAs that are mismatched with their targets.

with GTP) binds to the initiator methionyl tRNA, bringing it to the ribosome. The subsequent release of eIF2 is accompanied by GTP hydrolysis, leaving eIF2 as an inactive GDP complex. To participate in another cycle of initiation, the eIF2/GTP complex must be regenerated by the exchange of bound GDP for GTP (Figure 8.20). This exchange is mediated by another factor, eIF2B. The control of eIF2 activity by GTP binding and hydrolysis is thus similar to that of eEF1α (see Figure 8.13). However, the regulation of eIF2 provides a critical control point in a variety of eukaryotic cells. In particular, both eIF2 and eIF2B can be phosphorylated by regulatory protein kinases. These phosphorylations inhibit the exchange of bound GDP for GTP, thereby inhibiting initiation of translation. For example, if mammalian cells are subjected to stress or starved of growth factors, protein kinases that phosphorylate eIF2 and eIF2B become activated, inhibiting further protein synthesis.

Regulation of the activity of eIF4E, which binds to the 5′ cap of mRNAs, is another critical point at which growth factors act to control protein synthesis. For example, growth factors that stimulate protein synthesis in mammalian cells activate protein kinases that phosphorylate regulatory proteins (called eIF4E binding proteins or 4E-BPs) that bind to eIF4E. In the absence of the appropriate growth factors, the nonphosphorylated 4E-BPs

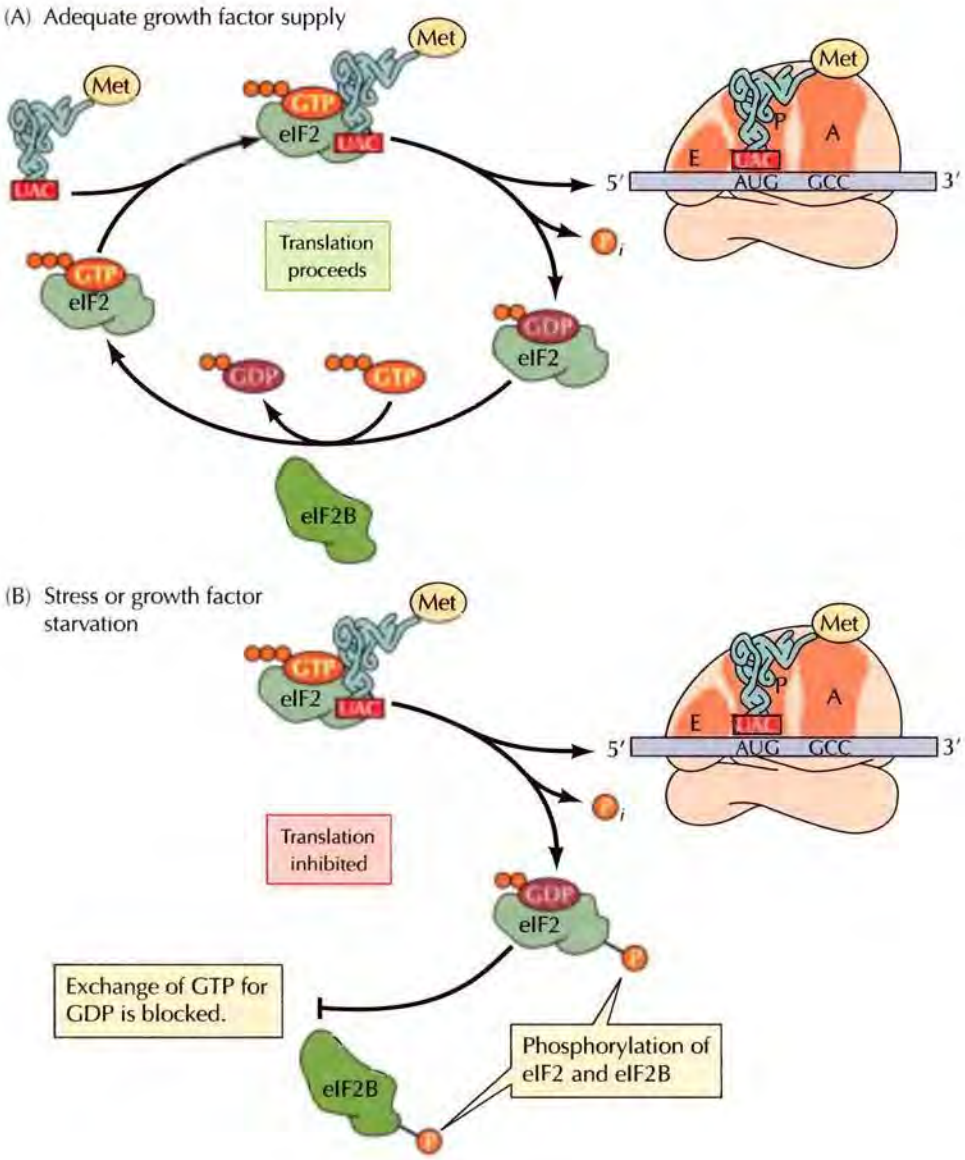


FIGURE 8.20 Regulation of translation by phosphorylation of eIF2 and eIF2B The active form of eIF2 (complexed with GTP) escorts initiator methionyl tRNA to the ribosome (see Figure 8.11). The eIF2 is then released from the ribosome in an inactive GDP-bound form. In order to continue translation, eIF2 must be reactivated by eIF2B, which stimulates the exchange of GTP for the bound GDP. Translation can be inhibited (for example, if cells are stressed or starved of growth factors) by regulatory protein kinases that phosphorylate either eIF2 or eIF2B. These phosphorylations block the exchange of GTP for GDP, so eIF2/GTP cannot be regenerated.

bind to eIF4E and inhibit translation by interfering with the interaction of eIF4E with eIF4G (see Figure 8.11). When growth factors are present in adequate supply, phosphorylation of the 4E-BPs prevents their interaction with eIF4E, leading to increased rates of translation initiation.

Protein Folding and Processing

Translation completes the flow of genetic information within the cell. The sequence of nucleotides in DNA has now been converted to the sequence of amino acids in a polypeptide chain. The synthesis of a polypeptide, however, is not equivalent to the production of a functional protein. To be useful, polypeptides must fold into distinct three-dimensional conformations, and in many cases multiple polypeptide chains must assemble into a functional complex. In addition, many proteins undergo further modifications, including cleavage and the covalent attachment of carbohydrates and lipids that are critical for the function and correct localization of proteins within the cell.

Chaperones and Protein Folding

The three-dimensional conformations of proteins result from interactions between the side chains of their constituent amino acids, as reviewed in Chapter 2. The classic principle of protein folding is that all the information required for a protein to adopt the correct three-dimensional conformation is provided by its amino acid sequence. This was initially established by Christian Anfinsen's experiments demonstrating that denatured RNase can spontaneously refold *in vitro* to its active conformation (see Figure 2.17). Protein folding thus appeared to be a self-assembly process that did not require additional cellular factors. More recent studies, however, have shown that this is not an adequate description of protein folding within the cell. The proper folding of proteins within cells is mediated by the activities of other proteins.

Proteins that facilitate the folding of other proteins are called molecular **chaperones**. The term "chaperone" was first used by Ron Laskey and his colleagues to describe a protein (nucleoplasmin) that is required for the assembly of nucleosomes from histones and DNA. Nucleoplasmin binds to histones and mediates their assembly into nucleosomes, but nucleoplasmin itself is not incorporated into the final nucleosome structure. Chaperones thus act as catalysts that facilitate assembly without being part of the assembled complex. Subsequent studies have extended the concept to include proteins that mediate a variety of other assembly processes, particularly protein folding.

It is important to note that chaperones do not convey additional information required for the folding of polypeptides into their correct three-dimensional conformations; the folded conformation of a protein is determined solely by its amino acid sequence. Rather, chaperones catalyze protein folding by assisting the self-assembly process. They appear to function by binding to and stabilizing unfolded or partially folded polypeptides that are intermediates along the pathway leading to the final correctly folded state. In the absence of chaperones, unfolded or partially folded polypeptide chains would be unstable within the cell, frequently folding incorrectly or aggregating into insoluble complexes. The binding of chaperones stabilizes these unfolded polypeptides, thereby preventing incorrect folding or aggregation and allowing the polypeptide chain to fold into its correct conformation.

A good example is provided by chaperones that bind to nascent polypeptide chains that are still being translated on ribosomes, thereby preventing incorrect folding or aggregation of the amino-terminal portion of the polypeptide before synthesis of the chain is finished (Figure 8.21). Proteins fold into domains of approximately 100–300 amino acids, so it is necessary to protect the nascent chain from aberrant folding or aggregation with other proteins until synthesis of the entire domain is complete and the protein can fold into its correct conformation. Chaperone binding stabilizes the amino-terminal portion in an unfolded conformation until the rest of the polypeptide chain is synthesized and the completed protein can fold correctly. Chaperones also stabilize unfolded polypeptide chains during their transport into subcellular organelles—for example, during the transfer of proteins into mitochondria from the cytosol (Figure 8.22). Proteins are transported across the mitochondrial membrane in partially unfolded conformations that are stabilized by chaperones in the cytosol. Chaperones within the mitochondrion then facilitate transfer of the polypeptide chain across the membrane and its subsequent folding within the organelle. In addition, chaperones are involved in the assembly of proteins that consist of

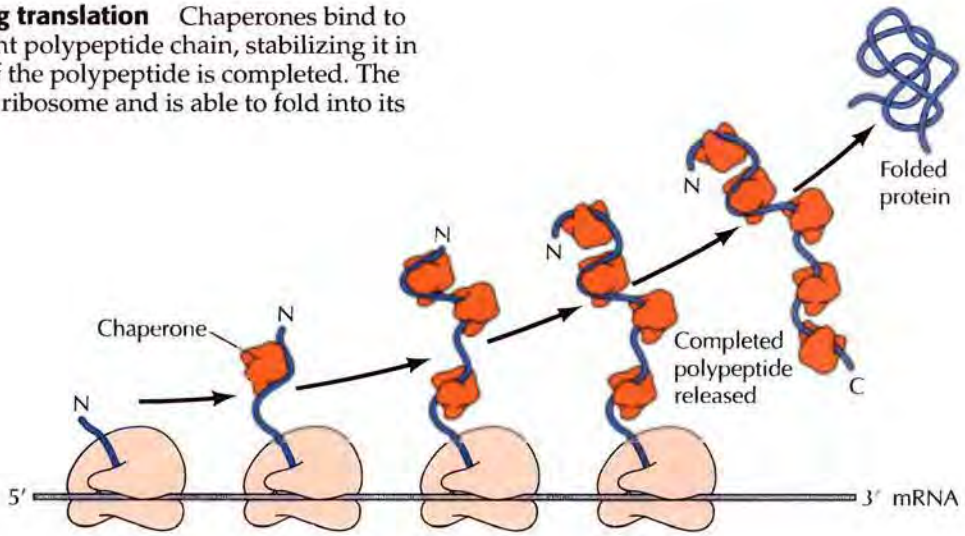


FIGURE 8.21 Action of chaperones during translation Chaperones bind to the amino (N) terminal portion of the nascent polypeptide chain, stabilizing it in an unfolded configuration until synthesis of the polypeptide is completed. The completed protein is then released from the ribosome and is able to fold into its correct three-dimensional conformation.

multiple polypeptide chains and in the assembly of macromolecular structures (e.g., nucleoplasmin).

Many of the proteins now known to function as chaperones were initially identified as heat-shock proteins, a group of proteins expressed in cells that have been subjected to elevated temperatures. The heat-shock proteins (abbreviated Hsp) are thought to stabilize and facilitate the refolding of proteins that have been partially denatured as a result of exposure to elevated temperatures. Two families of chaperone proteins, the Hsp70 chaperones and the chaperonins, act in a general pathway of protein folding in both prokaryotic and eukaryotic cells (Figure 8.23). Members of the Hsp70 and chaperonin families are found in the cytosol and in subcellular

■ **Protein misfolding can have dire consequences for the cell. Some misfolded proteins can aggregate and form insoluble fibers, called amyloid fibers, that accumulate in extracellular spaces and within cells. Amyloid fibers are hallmarks of several human neurodegenerative diseases, such as Alzheimer's and Parkinson's disease.**

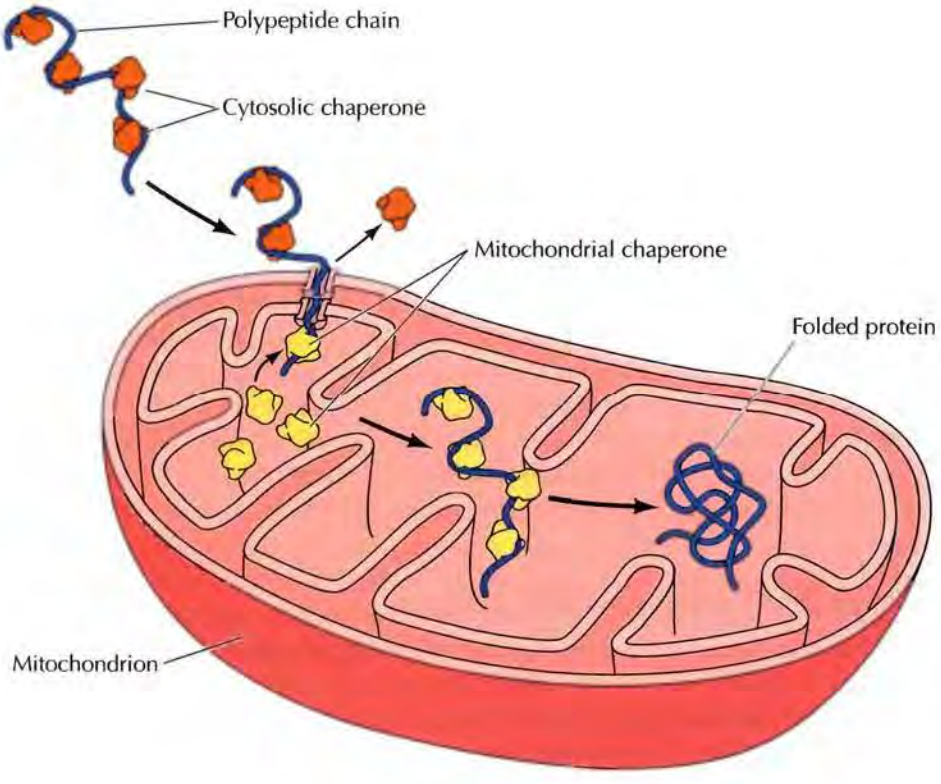


FIGURE 8.22 Action of chaperones during protein transport A partially unfolded polypeptide is transported from the cytosol to a mitochondrion. Cytosolic chaperones stabilize the unfolded configuration. Mitochondrial chaperones facilitate transport and subsequent folding of the polypeptide chain within the organelle.

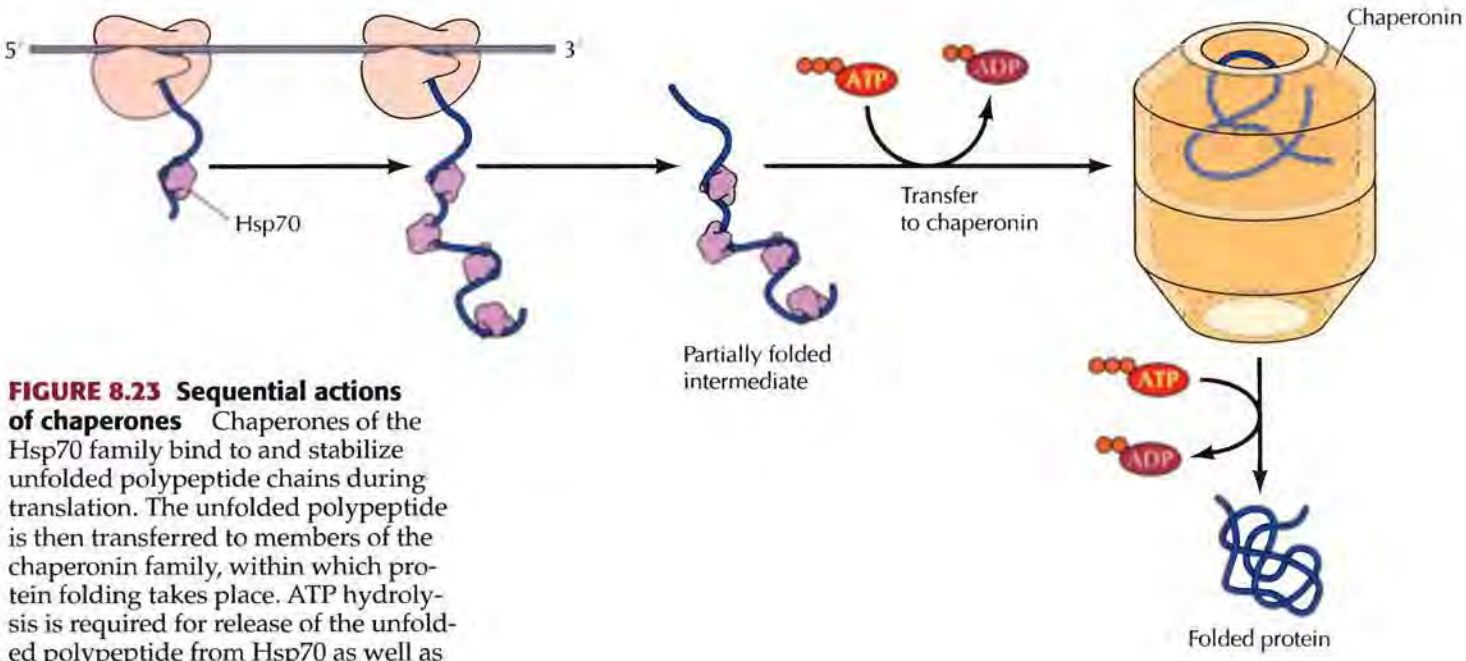


FIGURE 8.23 Sequential actions of chaperones Chaperones of the Hsp70 family bind to and stabilize unfolded polypeptide chains during translation. The unfolded polypeptide is then transferred to members of the chaperonin family, within which protein folding takes place. ATP hydrolysis is required for release of the unfolded polypeptide from Hsp70 as well as for folding within the chaperonin.

organelles (endoplasmic reticulum, mitochondria, and chloroplasts) of eukaryotic cells, as well as in bacteria. Members of the Hsp70 family stabilize unfolded polypeptide chains during translation (see, for example, Figure 8.21) as well as during the transport of polypeptides into a variety of subcellular compartments, such as mitochondria and the endoplasmic reticulum. These proteins bind to short hydrophobic segments (approximately seven amino acid residues) of unfolded polypeptides, maintaining the polypeptide chain in an unfolded configuration and preventing aggregation.

The unfolded polypeptide chain is then transferred from an Hsp70 chaperone to a chaperonin, within which protein folding takes place, yielding a protein correctly folded into its functional three-dimensional conformation. The chaperonins consist of multiple protein subunits arranged in two stacked rings to form a double-chambered structure. Unfolded polypeptide chains are shielded from the cytosol within the chamber of the chaperonin. In this isolated environment, protein folding can proceed while aggregation of unfolded segments of the polypeptide chain with other unfolded polypeptides is prevented.

Both bacteria and eukaryotic cells also contain additional families of chaperones, and the number of chaperones is considerably larger in eukaryotes. For example, an alternative pathway for the folding of some proteins in the cytosol and endoplasmic reticulum of eukaryotic cells involves the sequential actions of Hsp70 and Hsp90 family members. The majority of substrates for folding by Hsp90 are proteins that are involved in cell signaling, including receptors for steroid hormones and a variety of protein kinases.

Enzymes that Catalyze Protein Folding

In addition to chaperones that facilitate protein folding by binding to and stabilizing partially folded intermediates, cells contain at least two types of enzymes that act as chaperones by catalyzing protein folding. The formation of disulfide bonds between cysteine residues is important in stabilizing

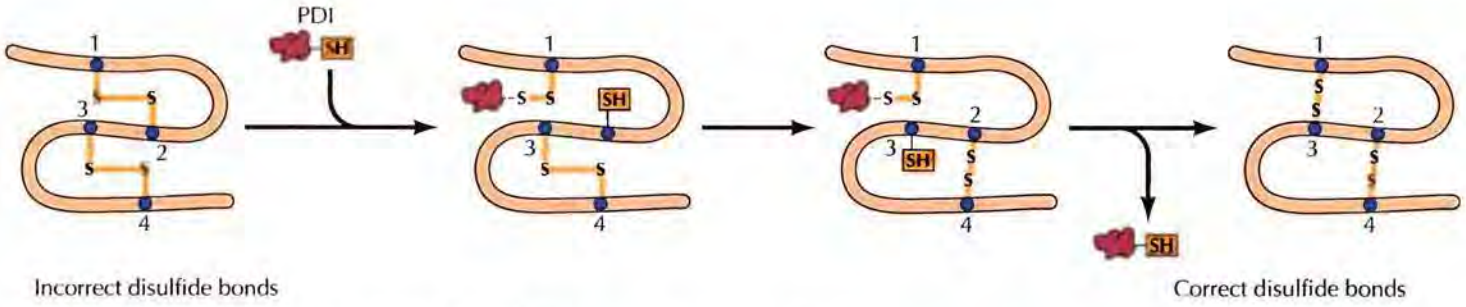


FIGURE 8.24 The action of protein disulfide isomerase Protein disulfide isomerase (PDI) catalyzes the breakage and rejoining of disulfide bonds, resulting in exchanges between paired disulfides in a polypeptide chain. The enzyme forms a disulfide bond with a cysteine residue of the polypeptide and then exchanges its paired disulfide with another cysteine residue. In this example, PDI catalyzes the conversion of two incorrect disulfide bonds (1-2 and 3-4) to the correct pairing (1-3 and 2-4).

the folded structures of many proteins (see Figure 2.16). **Protein disulfide isomerase (PDI)**, which was discovered by Christian Anfinsen in 1963, catalyzes disulfide bond formation. For proteins that contain multiple cysteine residues, PDI plays an important role by promoting rapid exchanges between paired disulfides, thereby allowing the protein to attain the pattern of disulfide bonds that is compatible with its stably folded conformation (Figure 8.24). Disulfide bonds are generally restricted to secreted proteins and some membrane proteins because the cytosol contains reducing agents that maintain cysteine residues in their reduced (—SH) form, thereby preventing the formation of disulfide (S—S) linkages. In eukaryotic cells, disulfide bonds form in the endoplasmic reticulum in which an oxidizing environment is maintained. PDI is a critical chaperone and catalyst of protein folding in the endoplasmic reticulum and is one of the most abundant proteins in that organelle.

The second enzyme that plays a role in protein folding catalyzes the isomerization of peptide bonds that involve proline residues (Figure 8.25). Proline is an unusual amino acid in that the equilibrium between the *cis* and *trans* conformations of peptide bonds that precede proline residues is only slightly in favor of the *trans* form. In contrast, peptide bonds between other amino acids are almost always in the *trans* form. Isomerization between the *cis* and *trans* configurations of prolyl-peptide bonds, which could otherwise represent a rate-limiting step in protein folding, is catalyzed by the enzyme **peptidyl prolyl isomerase**. This enzyme is widely distributed in both prokaryotic and eukaryotic cells and plays an important role in the folding of some proteins.

Protein Cleavage

Cleavage of the polypeptide chain (**proteolysis**) is an important step in the maturation of many proteins. A simple example is removal of the initiator methionine from the amino terminus of many polypeptides, which occurs soon after the amino terminus of the growing polypeptide chain emerges from the ribosome. Additional chemical groups, such as acetyl groups or fatty acid chains (discussed shortly), are then frequently added to the amino-terminal residues.

Proteolytic modifications of the amino terminus also play a part in the translocation of many proteins across membranes, including secreted pro-

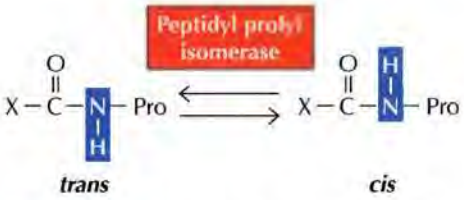


FIGURE 8.25 The action of peptidyl prolyl isomerase Peptidyl prolyl isomerase catalyzes the isomerization of peptide bonds that involve proline between the *cis* and *trans* conformations.

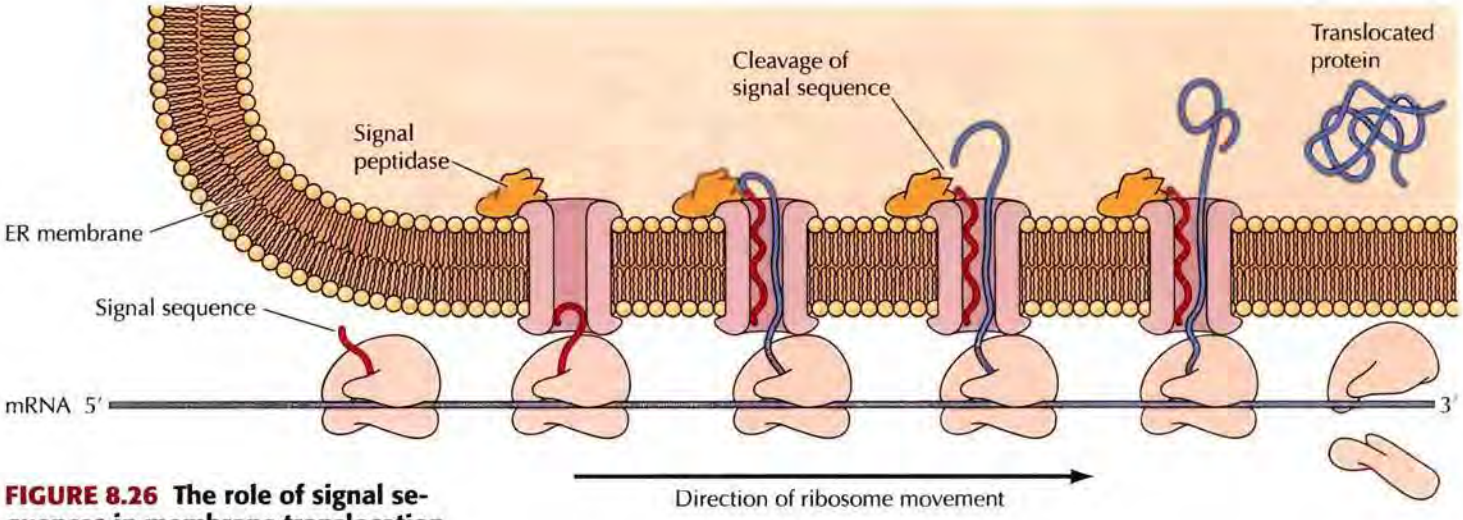


FIGURE 8.26 The role of signal sequences in membrane translocation Signal sequences target the translocation of polypeptide chains across the plasma membrane of bacteria or into the endoplasmic reticulum of eukaryotic cells (shown here). The signal sequence, a stretch of hydrophobic amino acids at the amino terminus of the polypeptide chain, inserts into a membrane channel as it emerges from the ribosome. The rest of the polypeptide is then translocated through the channel and the signal sequence is cleaved by the action of signal peptidase, releasing the mature translocated protein.

teins in both bacteria and eukaryotes as well as proteins destined for incorporation into the plasma membrane, lysosomes, mitochondria, and chloroplasts of eukaryotic cells. These proteins are targeted for transport to their destinations by amino-terminal sequences that are removed by proteolytic cleavage as the protein crosses the membrane. For example, amino-terminal **signal sequences**, usually about 20 amino acids long, target many secreted proteins to the plasma membrane of bacteria or to the endoplasmic reticulum of eukaryotic cells while translation is still in progress (Figure 8.26). The signal sequence, which consists predominantly of hydrophobic amino acids, is inserted into a membrane channel as it emerges from the ribosome. The remainder of the polypeptide chain passes through the channel membrane as translation proceeds. The signal sequence is then cleaved by a specific membrane protease (**signal peptidase**), and the mature protein is released. In eukaryotic cells, the translocation of growing polypeptide chains into the endoplasmic reticulum is the first step in targeting proteins for secretion, incorporation into the plasma membrane, or incorporation into lysosomes. The mechanisms that direct the transport of proteins to these destinations, as well as the role of other targeting sequences in directing the import of proteins into mitochondria and chloroplasts, will be discussed in detail in Chapters 10 and 11.

In other important instances of proteolytic processing, active enzymes or hormones form via cleavage of larger precursors. Insulin, which is synthesized as a longer precursor polypeptide, is a good example. Insulin forms by two cleavages. The initial precursor (preproinsulin) contains an amino-terminal signal sequence that targets the polypeptide chain to the endoplasmic reticulum (Figure 8.27). Removal of the signal sequence during transfer to the endoplasmic reticulum yields a second precursor, called proinsulin. This precursor is then converted to insulin (which consists of two chains held together by disulfide bonds) by proteolytic removal of an internal peptide. Other proteins activated by similar cleavage processes include digestive enzymes, proteins involved in blood clotting, and a cascade of proteases that regulate programmed cell death in animals.

It is interesting to note that the proteins of many animal viruses are derived from the cleavage of larger precursors. One particularly important example of the role of proteolysis in virus replication is provided by HIV. In the replication of HIV, a virus-encoded protease cleaves precursor polypep-

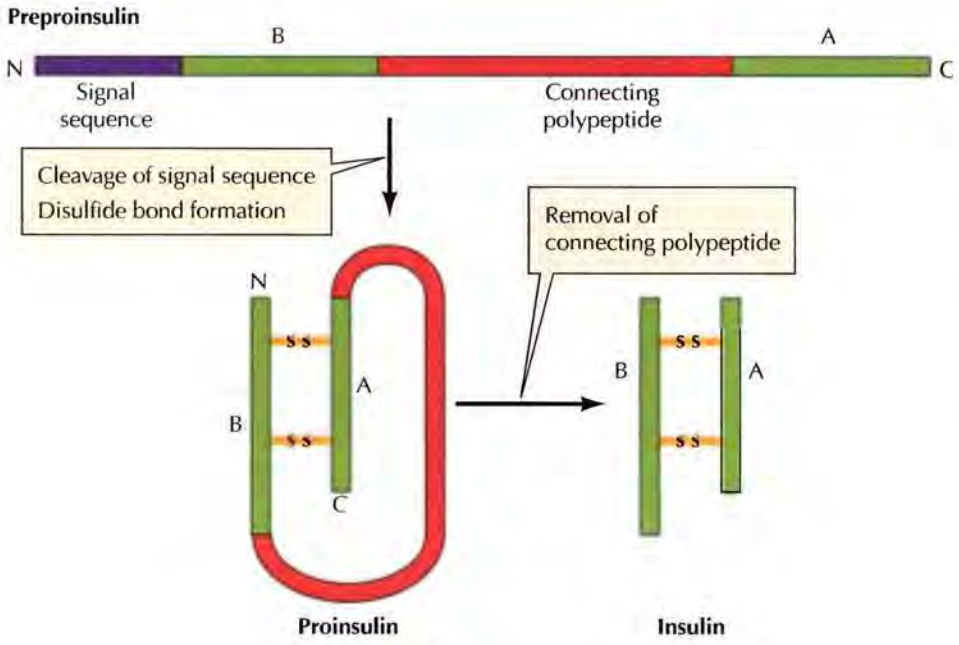


FIGURE 8.27 Proteolytic processing of insulin The mature insulin molecule consists of two polypeptide chains (A and B) joined by disulfide bonds. It is synthesized as a precursor polypeptide (preproinsulin) containing an amino-terminal signal sequence that is cleaved during transfer of the growing polypeptide chain to the endoplasmic reticulum. This cleavage yields a second precursor (proinsulin), which is converted to insulin by further proteolysis, removing the internal connecting polypeptide.

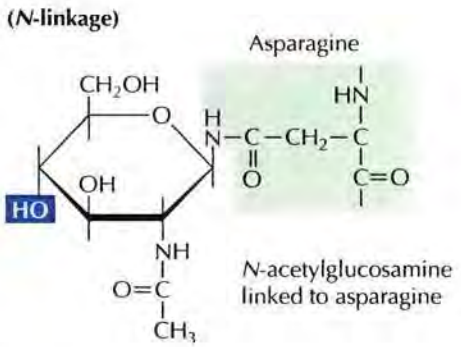
tides to form the viral structural proteins. Because of its central role in virus replication, the HIV protease (in addition to reverse transcriptase) is an important target for the development of drugs used for treating AIDS. Indeed, such protease inhibitors are now among the most effective agents available for combating this disease.

Glycosylation

Many proteins, particularly in eukaryotic cells, are modified by the addition of carbohydrates, a process called **glycosylation**. The proteins to which carbohydrate chains have been added (called **glycoproteins**) are usually secreted or localized to the cell surface, although many nuclear and cytosolic proteins are also glycosylated. The carbohydrate moieties of glycoproteins play important roles in protein folding in the endoplasmic reticulum, in the targeting of proteins for delivery to the appropriate intracellular compartments, and as recognition sites in cell-cell interactions.

Glycoproteins are classified as either *N*-linked or *O*-linked, depending on the site of attachment of the carbohydrate side chain (Figure 8.28). In *N*-linked glycoproteins, the carbohydrate is attached to the nitrogen atom in the side chain of asparagine. In *O*-linked glycoproteins, the oxygen atom in the side chain of serine or threonine is the site of carbohydrate attachment. The sugars directly attached to these positions are usually either *N*-acetylglucosamine or *N*-acetylgalactosamine, respectively.

Most glycoproteins in eukaryotic cells are destined either for secretion or for incorporation into the plasma membrane. These proteins are usually transferred into the endoplasmic reticulum while their translation is still in progress. Glycosylation is also initiated in the endoplasmic reticulum before



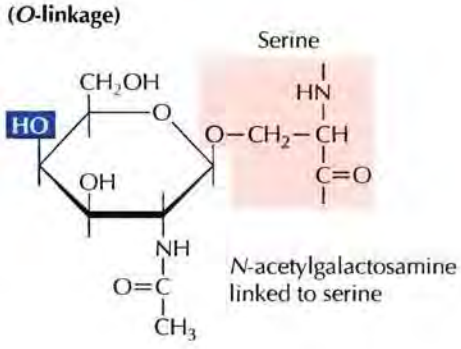


FIGURE 8.28 Linkage of carbohydrate side chains to glycoproteins The carbohydrate chains of *N*-linked glycoproteins are attached to asparagine; those of *O*-linked glycoproteins are attached to either serine (shown) or threonine. The sugars joined to the amino acids are usually either *N*-acetylglucosamine (*N*-linked) or *N*-acetylgalactosamine (*O*-linked).

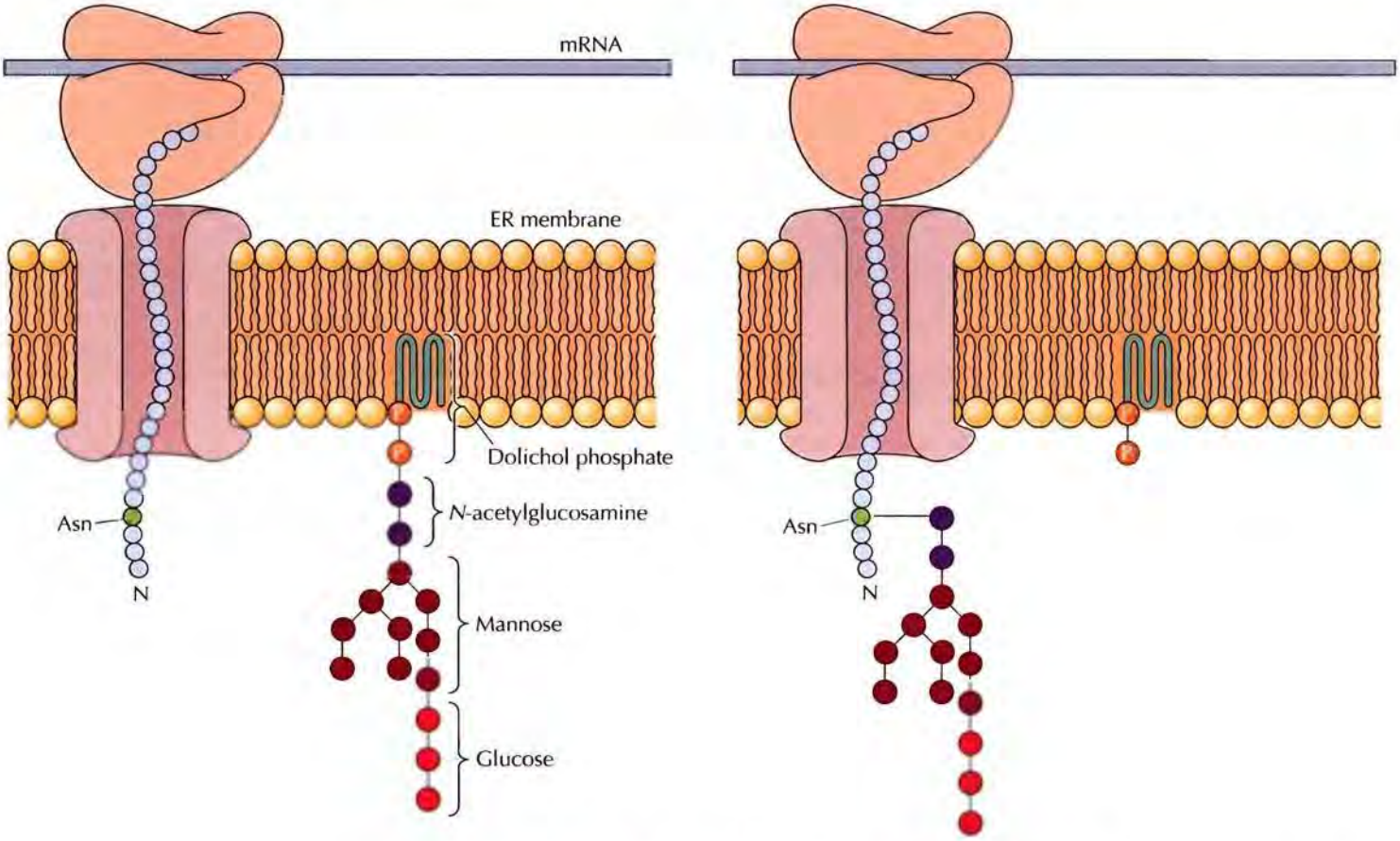


FIGURE 8.29 Synthesis of *N*-linked glycoproteins The first step in glycosylation is the addition of an oligosaccharide consisting of 14 sugar residues to a growing polypeptide chain in the endoplasmic reticulum (ER). The oligosaccharide (consisting of two *N*-acetylglucosamine, nine mannose, and three glucose residues) is assembled on a lipid carrier (dolichol phosphate) in the ER membrane. It is then transferred as a unit to an acceptor asparagine residue of the polypeptide.

translation is complete. The first step is the transfer of a common oligosaccharide consisting of 14 sugar residues (two *N*-acetylglucosamine, three glucose, and nine mannose) to an asparagine residue of the growing polypeptide chain (Figure 8.29). The oligosaccharide is assembled within the endoplasmic reticulum on a lipid carrier (**dolichol phosphate**). It is then transferred as an intact unit to an acceptor asparagine (Asn) residue within the sequence Asn-X-Ser or Asn-X-Thr (where X is any amino acid other than proline).

In further processing, the common *N*-linked oligosaccharide is modified (Figure 8.30). Three glucose residues are removed while the glycoprotein is in the endoplasmic reticulum. As discussed in Chapter 10, these modifications play a critical role in protein folding. The oligosaccharide is then further modified in the Golgi apparatus, to which glycoproteins are transferred from the endoplasmic reticulum. These modifications (which will be discussed in Chapter 10) include both the removal and addition of carbohydrate residues as the glycoprotein is transported through the compartments of the Golgi. The *N*-linked oligosaccharides of different glycoproteins are processed to different extents, depending on both the enzymes present in different cells and on the accessibility of the oligosaccharide to the enzymes

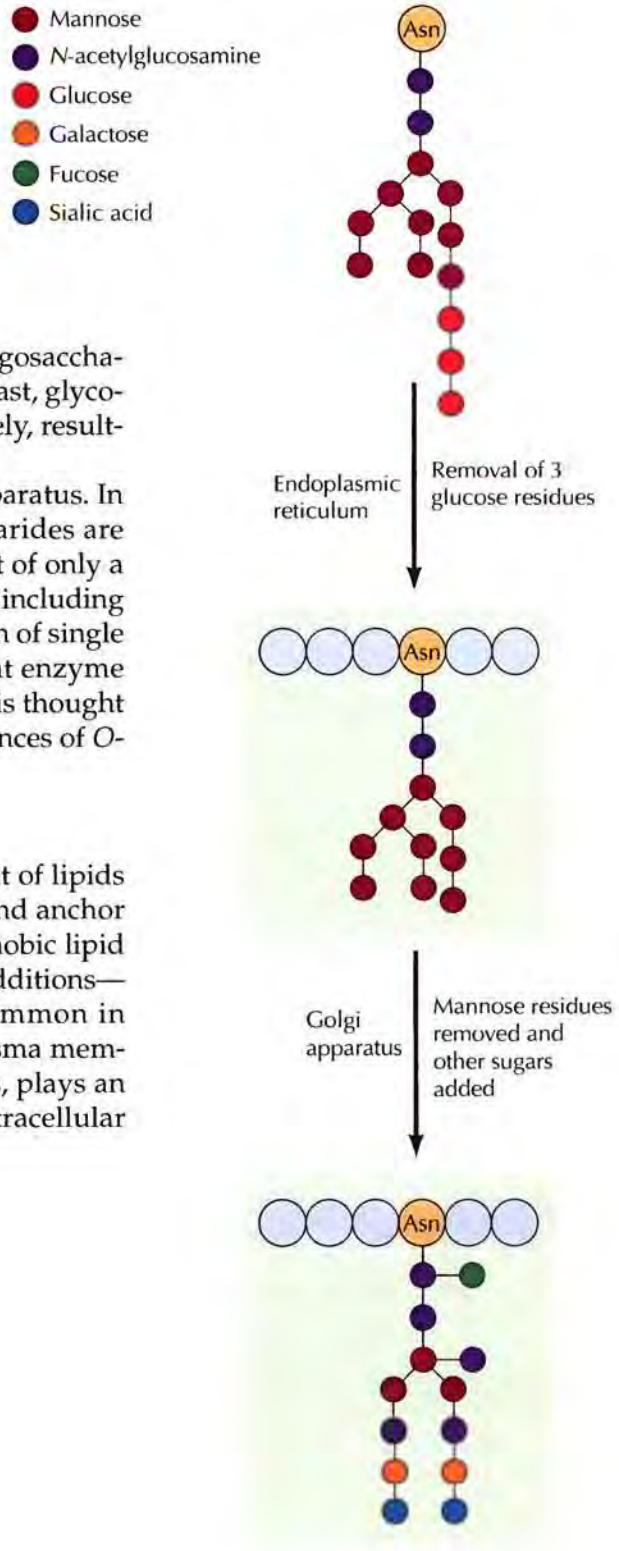


FIGURE 8.30 Processing of *N*-linked oligosaccharides Various oligosaccharides form from further modifications of the common 14-sugar unit initially added in the endoplasmic reticulum. Three glucose residues are removed in the endoplasmic reticulum. The glycoprotein is then transferred to the Golgi apparatus in which mannose residues are removed and other sugars are added. The structure shown is a representative example.

that catalyze its modification. Glycoproteins with inaccessible oligosaccharides do not have new sugars added to them in the Golgi. In contrast, glycoproteins with accessible oligosaccharides are processed extensively, resulting in the formation of a variety of complex oligosaccharides.

O-linked oligosaccharides are also added within the Golgi apparatus. In contrast to the *N*-linked oligosaccharides, *O*-linked oligosaccharides are formed by the addition of one sugar at a time and usually consist of only a few residues (Figure 8.31). Many cytoplasmic and nuclear proteins, including a variety of transcription factors, are also modified by the addition of single *O*-linked *N*-acetylglucosamine residues, catalyzed by a different enzyme system. Glycosylation of these cytoplasmic and nuclear proteins is thought to play a role in regulating their activities, although the consequences of *O*-glycosylation remain to be fully understood.

Attachment of Lipids

Some proteins in eukaryotic cells are modified by the attachment of lipids to the polypeptide chain. Such modifications frequently target and anchor these proteins to the plasma membrane, with which the hydrophobic lipid is able to interact (see Figure 2.25). Three general types of lipid additions—*N*-myristoylation, prenylation, and palmitoylation—are common in eukaryotic proteins associated with the cytosolic face of the plasma membrane. A fourth type of modification, the addition of glycolipids, plays an important role in anchoring some cell surface proteins to the extracellular face of the plasma membrane.

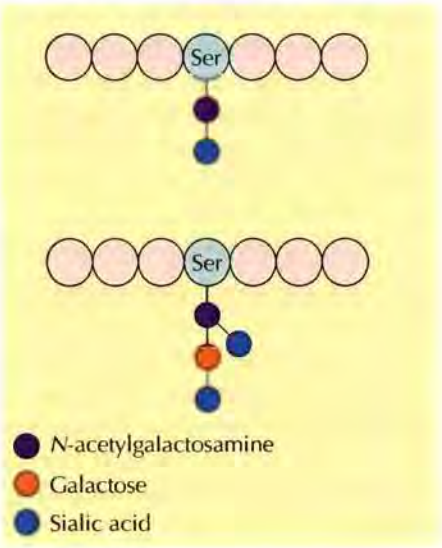


FIGURE 8.31 Examples of *O*-linked oligosaccharides *O*-linked oligosaccharides usually consist of only a few carbohydrate residues, which are added one sugar at a time.

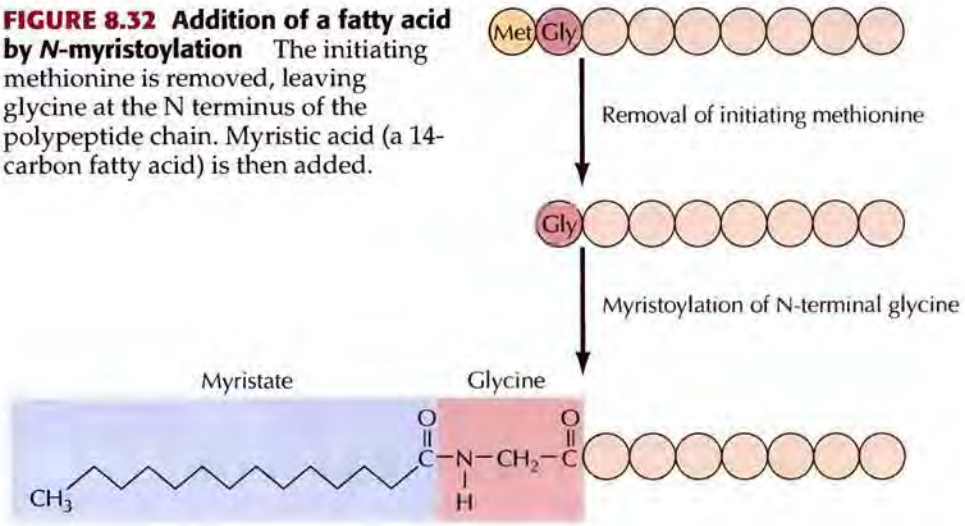


FIGURE 8.32 Addition of a fatty acid by *N*-myristoylation The initiating methionine is removed, leaving glycine at the N terminus of the polypeptide chain. Myristic acid (a 14-carbon fatty acid) is then added.

In some proteins, a fatty acid is attached to the amino terminus of the growing polypeptide chain during translation. In this process, called ***N*-myristoylation**, myristic acid (a 14-carbon fatty acid) is attached to an N-terminal glycine residue (Figure 8.32). The glycine is usually the second amino acid incorporated into the polypeptide chain; the initiator methionine is removed by proteolysis before fatty acid addition. Many proteins that are modified by *N*-myristoylation are associated with the inner face of the plasma membrane, and the role of the fatty acid in this association has been clearly demonstrated by analysis of mutant proteins in which the N-terminal glycine is changed to an alanine. This substitution prevents myristoylation and blocks the function of the mutant proteins by inhibiting their membrane association.

Lipids can also be attached to the side chains of cysteine, serine, and threonine residues. One important example of this type of modification is **prenylation** in which specific types of lipids (prenyl groups) are attached to the sulfur atoms in the side chains of cysteine residues located near the C terminus of the polypeptide chain (Figure 8.33). Many plasma membrane-associated proteins involved in the control of cell growth and differentiation are modified in this way, including the Ras oncogene proteins, which are responsible for the uncontrolled growth of many human cancers (see Chapter 18). Prenylation of these proteins proceeds by three steps. First, the prenyl group is added to a cysteine located three amino acids from the carboxy terminus of the polypeptide chain. The prenyl groups added in this reaction are either farnesyl (15 carbons, as shown in Figure 8.33) or geranylgeranyl (20 carbons). The amino acids following the cysteine residue are then removed, leaving cysteine at the carboxy terminus. Finally, a methyl group is added to the carboxyl group of the C-terminal cysteine residue.

The biological significance of prenylation is indicated by the fact that mutations of the critical cysteine block the membrane association and function of Ras proteins. Because farnesylation is a relatively rare modification of cellular proteins, interest in this reaction has been stimulated by the possibility that inhibitors of the key enzyme (farnesyl transferase) might prove useful as drugs for the treatment of cancers that involve Ras proteins. Such inhibitors of farnesylation have been found to interfere with the growth of cancer cells in experimental models and are undergoing evaluation of their efficacy against human tumors in clinical trials.

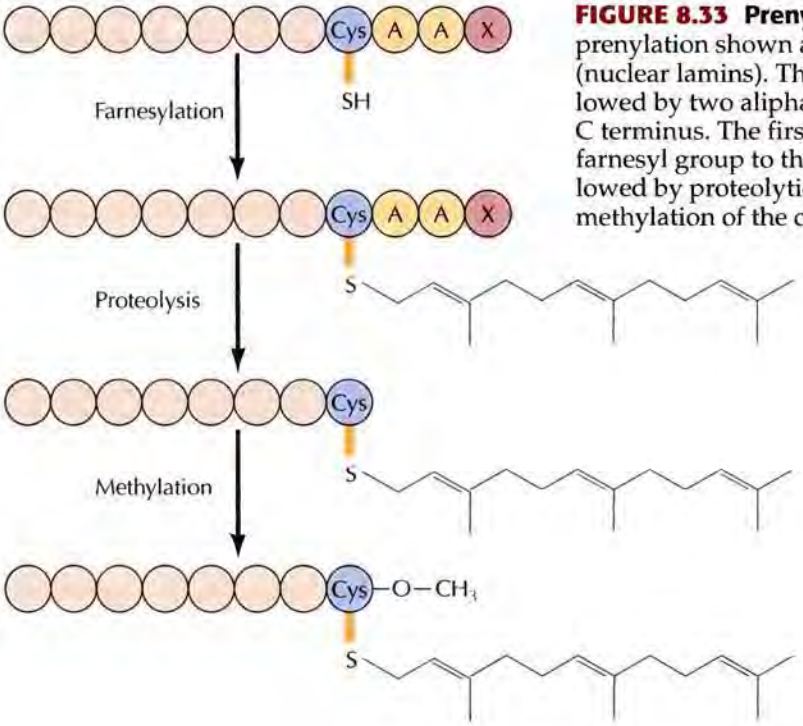


FIGURE 8.33 Prenylation of a C-terminal cysteine residue The type of prenylation shown affects Ras proteins and proteins of the nuclear envelope (nuclear lamins). These proteins terminate with a cysteine residue (Cys) followed by two aliphatic amino acids (A) and any other amino acid (X) at the C terminus. The first step in their modification is addition of the 15-carbon farnesyl group to the side chain of cysteine (farnesylation). This step is followed by proteolytic removal of the three C-terminal amino acids and methylation of the cysteine, which is now at the C terminus.

In the third type of fatty acid modification, **palmitoylation**, palmitic acid (a 16-carbon fatty acid) is added to sulfur atoms of the side chains of internal cysteine residues (Figure 8.34). Like *N*-myristoylation and prenylation, palmitoylation plays an important role in the association of some proteins with the cytosolic face of the plasma membrane.

Finally, lipids linked to oligosaccharides (**glycolipids**) are added to the C-terminal carboxyl groups of some proteins, where they serve as anchors that attach the proteins to the external face of the plasma membrane. Because the glycolipids attached to these proteins contain phosphatidylinositol, they are usually called **glycosylphosphatidylinositol**, or **GPI anchors** (Figure 8.35). The oligosaccharide portions of GPI anchors are attached to the terminal carboxyl group of polypeptide chains. The inositol head group of phosphatidylinositol is in turn attached to the oligosaccharide, so the carbohydrate serves as a bridge between the protein and the fatty acid chains of the phospholipid. The GPI anchors are synthesized and added to proteins as a preassembled unit within the endoplasmic reticulum. Their addition is accompanied by cleavage of a peptide consisting of about 20 amino acids from the C terminus of the polypeptide chain. The modified protein is then transported to the cell surface, where the fatty acid chains of the GPI anchor mediate its attachment to the plasma membrane.

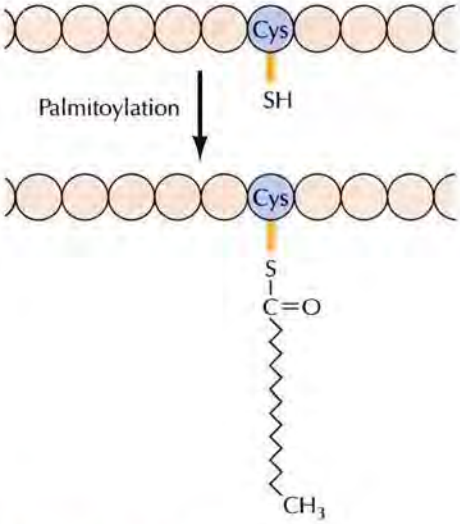


FIGURE 8.34 Palmitoylation Palmitate (a 16-carbon fatty acid) is added to the side chain of an internal cysteine residue.

Regulation of Protein Function

A critical function of proteins is their activity as enzymes, which are needed to catalyze almost all biological reactions. Regulation of enzyme activity thus plays a key role in governing cell behavior. This is accomplished in part at the level of gene expression, which determines the amount of any enzyme (protein) synthesized by the cell. A further level of control is then obtained by regulation of protein function, which allows the cell to regulate

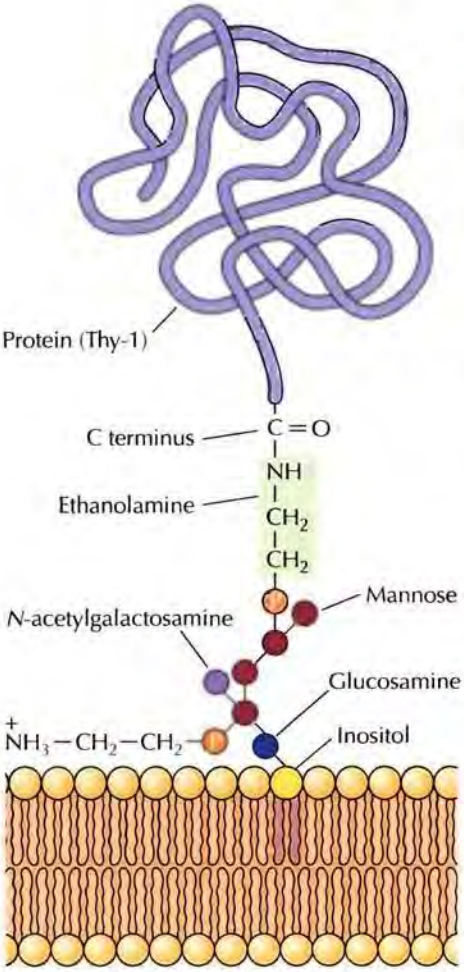


FIGURE 8.35 Structure of a GPI anchor The GPI anchor, attached to the C terminus, anchors the protein in the plasma membrane. The anchor is joined to the C-terminal amino acid by an ethanolamine, which is linked to an oligosaccharide that consists of mannose, *N*-acetylgalactosamine, and glucosamine residues. The oligosaccharide is in turn joined to the inositol head group of phosphatidylinositol. The two fatty acid chains of the lipid are embedded in the plasma membrane. The GPI anchor shown here is that of a rat protein, Thy-1.

not only the amounts but also the activities of its protein constituents. Regulation of the activities of some of the proteins involved in transcription and translation has already been discussed in this and the preceding chapter, and many further examples of regulated protein function in the control of cell behavior will be evident throughout the remainder of this book. This section discusses three general mechanisms by which the activities of cellular proteins are controlled.

Regulation by Small Molecules

Most enzymes are controlled by changes in their conformation, which in turn alter catalytic activity. In many cases such conformational changes result from the binding of small molecules, such as amino acids or nucleotides, that regulate enzyme activity. This type of regulation commonly is responsible for controlling metabolic pathways by feedback inhibition. For example, the end products of many biosynthetic pathways (e.g., amino acids) inhibit the enzymes that catalyze the first step in their synthesis, thus ensuring an adequate supply of the product while preventing the synthesis of excess amounts (Figure 8.36). Feedback inhibition is an example of **allosteric regulation** in which a regulatory molecule binds to a site on an enzyme that is distinct from the catalytic site (*allo* = "other"; *steric* = "site"). The binding of such a regulatory molecule alters the conformation of the protein, thereby changing the shape of the catalytic site and affecting catalytic activity (see Figure 3.8). Many transcription factors (discussed in Chapter 7) are also regulated by the binding of small molecules. For

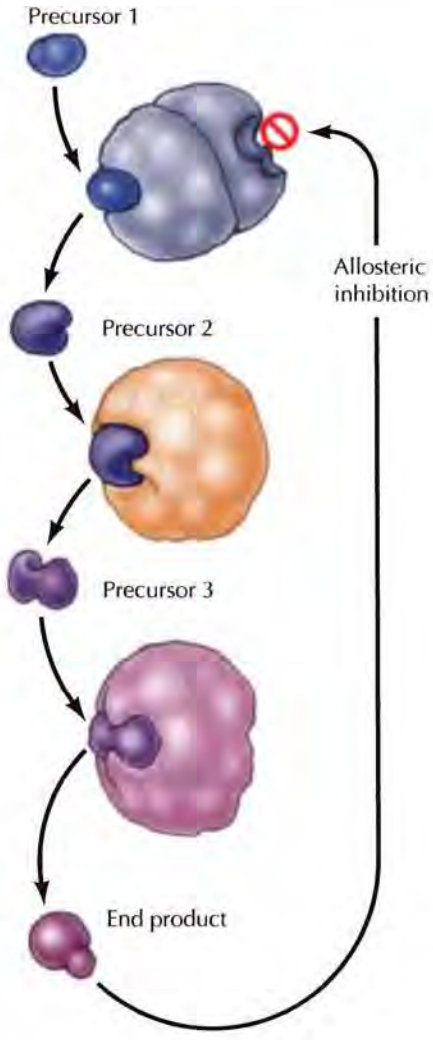


FIGURE 8.36 Feedback inhibition The end product of a biochemical pathway acts as an allosteric inhibitor of the enzyme that catalyzes the first step in its synthesis.

FIGURE 8.37 Conformational differences between active and inactive Ras proteins The Ras proteins alternate between active GTP-bound and inactive GDP-bound forms. The major effect of GTP binding versus GDP binding is alteration of the conformation of two regions of the molecule, designated the switch I and switch II regions. The backbone of the GTP complex is shown here in white; the backbone of the GDP complex of switch I and switch II are in blue and yellow, respectively. The guanine nucleotide is in red and Mg^{2+} is the yellow sphere. (Courtesy of Sung-Hou Kim, University of California, Berkeley.)

example, the binding of lactose to the *E. coli lac* repressor induces a conformational change that prevents the repressor from binding DNA (see Figure 7.9). In eukaryotic cells, steroid hormones similarly control gene expression by binding to transcriptional regulatory proteins.

The regulation of translation factors such as eEF1α by GTP binding (see Figure 8.13) illustrates another common mechanism by which the activities of intracellular proteins are controlled. In this case, the GTP-bound form of the protein is its active conformation, while the GDP-bound form is inactive. Many cellular proteins are similarly regulated by GTP or GDP binding. These proteins include the Ras oncogene proteins, which have been studied intensively because of their roles in the control of cell proliferation and in human cancers. X-ray crystallography of these proteins has been particularly interesting, revealing subtle but functionally very important conformational differences between the inactive GDP-bound and active GTP-bound forms (Figure 8.37). This small difference in protein conformation determines whether Ras (in the active GTP-bound form) can interact with its target molecule, which signals the cell to divide. The importance of such subtle differences in protein conformation is dramatically illustrated by the fact that mutations in *ras* genes contribute to the development of about 20% of human cancers. Such mutations alter the structure of the Ras proteins so that they are locked in the active GTP-bound conformation and continually signal cell division, thereby driving the uncontrolled growth of cancer cells. In contrast, normal Ras proteins alternate between the GTP- and GDP-bound conformations, such that they are active only following stimulation by the hormones and growth factors that normally control cell proliferation in multicellular organisms.

Protein Phosphorylation

The examples discussed in the previous section involve noncovalent associations of proteins with small-molecule inhibitors or activators. Since no covalent bonds form, the binding of these regulatory molecules to the protein is readily reversible, allowing the cell to respond rapidly to environmental changes. The activity of many proteins, however, is also regulated by covalent modifications. One example of this type of regulation is the activation of some enzymes by proteolytic cleavage of inactive precursors. As noted previously in this chapter, digestive enzymes and proteins involved in blood clotting are regulated by this mechanism. Since proteolysis is irreversible, however, it provides a means of controlling enzyme activation rather than of turning proteins on and off in response to changes in the environment. In contrast, other covalent modifications—particularly phosphorylation—are readily reversible within the cell and function, as allosteric regulation does, to reversibly activate or inhibit a wide variety of cellular proteins in response to environmental signals.

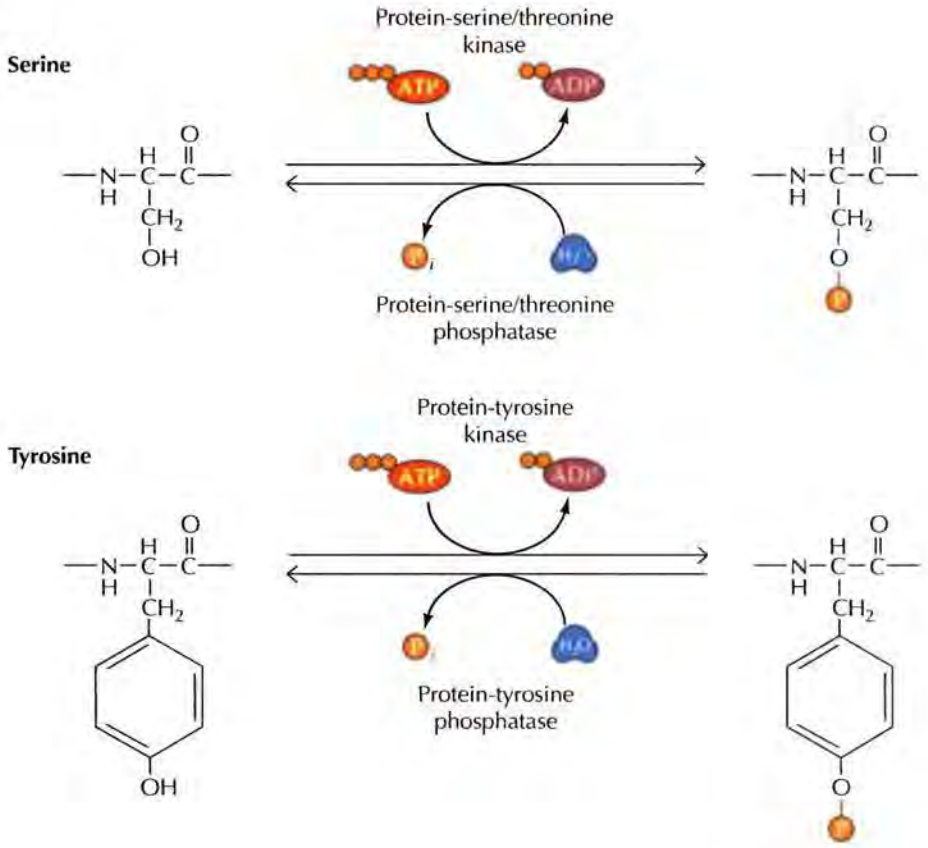


FIGURE 8.38 Protein kinases and phosphatases Protein kinases catalyze the transfer of a phosphate group from ATP to the side chains of serine and threonine (protein-serine/threonine kinases) or tyrosine (protein-tyrosine kinases) residues. Protein phosphatases catalyze the removal of phosphate groups from the same amino acids by hydrolysis.

Protein phosphorylation is catalyzed by **protein kinases**, most of which transfer phosphate groups from ATP to the hydroxyl groups of the side chains of serine, threonine, or tyrosine residues (Figure 8.38). The protein kinases are one of the largest protein families in eukaryotes, accounting for approximately 2% of eukaryotic genes. Most protein kinases phosphorylate either serine and threonine or tyrosine residues: These enzymes are called **protein-serine/threonine kinases** or **protein-tyrosine kinases**, respectively. Protein phosphorylation is reversed by **protein phosphatases**, which catalyze the hydrolysis of phosphorylated amino acid residues. Like protein kinases, most protein phosphatases are specific either for serine and threonine or for tyrosine residues, although some protein phosphatases recognize all three phosphoamino acids.

The combined action of protein kinases and protein phosphatases mediates the reversible phosphorylation of many cellular proteins. Frequently, protein kinases function as components of signal transduction pathways in which one kinase activates a second kinase, which may act on yet another kinase. The sequential action of a series of protein kinases can transmit a signal received at the cell surface to target proteins within the cell, resulting in changes in cell behavior in response to environmental stimuli.

The prototype of the action of protein kinases came from studies of glycogen metabolism by Ed Fischer and Ed Krebs in 1955. In muscle cells the hormone epinephrine (adrenaline) signals the breakdown of glycogen to glucose-1-phosphate, providing an available source of energy for increased muscular activity. Glycogen breakdown is catalyzed by the enzyme glycogen phosphorylase, which is regulated by a protein kinase (Figure 8.39). Epinephrine binds to a cell surface receptor that triggers the conversion of ATP to cyclic AMP (cAMP), which then binds to and activates

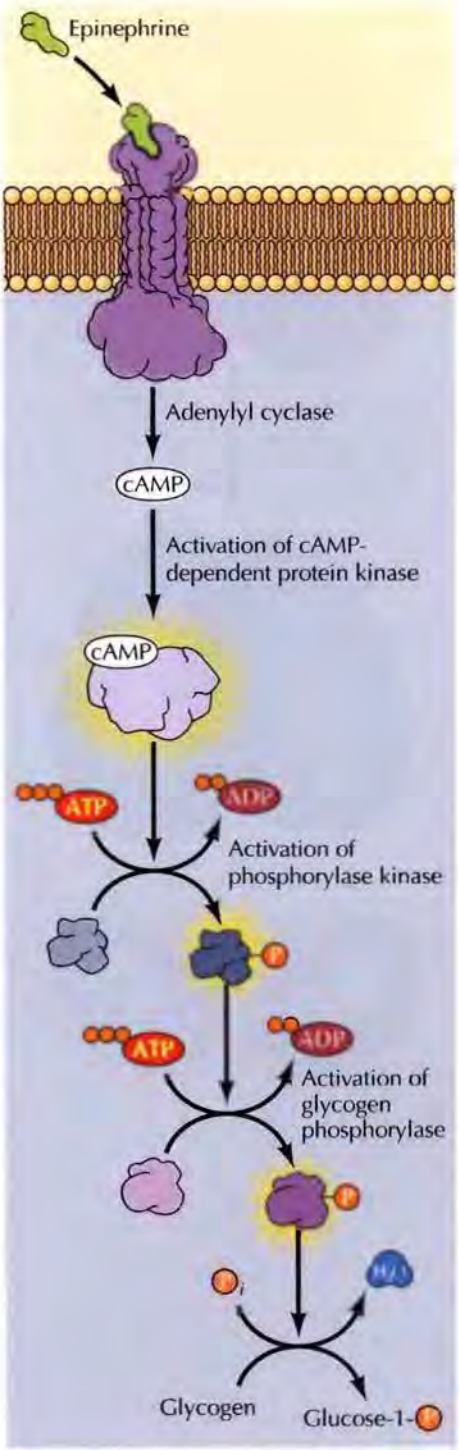


FIGURE 8.39 Regulation of glycogen breakdown by protein phosphorylation
The binding of epinephrine (adrenaline) to its cell surface receptor triggers the production of cyclic AMP (cAMP), which activates cAMP-dependent protein kinase. cAMP-dependent protein kinase phosphorylates and activates phosphorylase kinase, which in turn phosphorylates and activates glycogen phosphorylase. Glycogen phosphorylase then catalyzes the breakdown of glycogen to glucose-1-phosphate.

a protein kinase, called cAMP-dependent protein kinase. This kinase phosphorylates and activates a second protein kinase, called phosphorylase kinase. Phosphorylase kinase in turn phosphorylates and activates glycogen phosphorylase, leading to glucose production. The activating phosphorylations of both phosphorylase kinase and glycogen phosphorylase can be reversed by specific phosphatases, so removal of the initial stimulus (epinephrine) inhibits further glycogen breakdown.

The signaling pathway that leads to activation of glycogen phosphorylase is initiated by the binding of small molecules at the cell surface—epinephrine binding to its receptor and cAMP binding to cAMP-dependent protein kinase. The signal is then transmitted to its intracellular target by the sequential action of protein kinases. Similar signaling pathways, in which protein kinases and phosphatases play central roles, are involved in regulating almost all aspects of the behavior of eukaryotic cells (see Chapters 15 and 16). Aberrations in these pathways, frequently involving abnormalities of protein kinases, are also responsible for many diseases associated with improper regulation of cell growth and differentiation, particularly the development of cancer.

Although phosphorylation is the most common and best studied type of covalent modification that regulates protein activity, several other types of protein modifications also play important roles. These include methylation and acetylation of lysine residues (discussed in Chapter 7), as well as **nitrosylation**—the addition of NO groups to the side chains of cysteine residues (Figure 8.40). As noted earlier in this chapter, *O*-linked glycosylation of nuclear and cytosolic proteins may also play a regulatory role. In addition, some proteins are regulated by the covalent attachment of small polypeptides, such as ubiquitin and SUMO, discussed in the next section of this chapter.

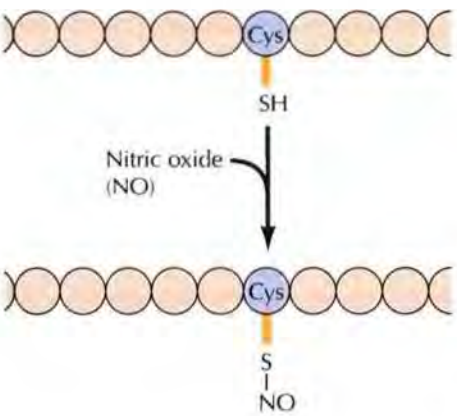


FIGURE 8.40 Nitrosylation
Ntiric oxide (NO) can react with the side chain of cysteine residues.

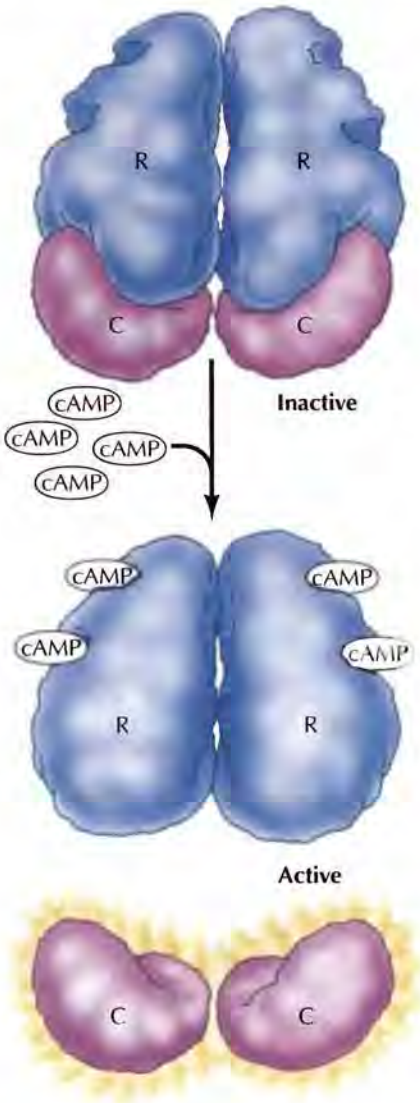


FIGURE 8.41 Regulation of cAMP-dependent protein kinase In the inactive state, the enzyme consists of two regulatory (R) and two catalytic (C) subunits. Cyclic AMP binds to the regulatory subunits, inducing a conformational change that leads to their dissociation from the catalytic subunits. The free catalytic subunits are enzymatically active.

Protein-Protein Interactions

Many proteins consist of multiple subunits, each of which is an independent polypeptide chain. In some proteins the subunits are identical; other proteins are composed of two or more distinct polypeptides. In either case, interactions between the polypeptide chains are important in regulation of protein activity. The importance of these interactions is evident in many allosteric enzymes in which the binding of a regulatory molecule alters protein conformation by changing the interactions between subunits.

Many other enzymes are similarly regulated by protein-protein interactions. A good example is cAMP-dependent protein kinase, which is composed of two regulatory and two catalytic subunits (Figure 8.41). In this state, the enzyme is inactive; the regulatory subunits inhibit the enzymatic activity of the catalytic subunits. The enzyme is activated by cAMP, which binds to the regulatory subunits and induces a conformational change leading to dissociation of the complex; the free catalytic subunits are then enzymatically active protein kinases. Cyclic AMP thus acts as an allosteric regulator by altering protein-protein interactions. As discussed in later chapters, similar protein-protein interactions, which can themselves be regulated by the binding of small molecules and phosphorylation, play critical roles in the control of many different aspects of cell behavior.

Protein Degradation

The levels of proteins within cells are determined not only by rates of synthesis but also by rates of degradation. The half-lives of proteins within cells vary widely, from minutes to several days, and differential rates of protein degradation are an important aspect of cell regulation. Many rapidly degraded proteins function as regulatory molecules, such as transcription factors. The rapid turnover of these proteins is necessary to allow their levels to change quickly in response to external stimuli. Other proteins are rapidly degraded in response to specific signals, providing another mechanism for the regulation of intracellular enzyme activity. In addition, faulty or damaged proteins are recognized and rapidly degraded within cells, thereby eliminating the consequences of mistakes made during protein synthesis. In eukaryotic cells, two major pathways—the ubiquitin-proteasome pathway and lysosomal proteolysis—mediate protein degradation.

The Ubiquitin-Proteasome Pathway

The major pathway of selective protein degradation in eukaryotic cells uses **ubiquitin** as a marker that targets cytosolic and nuclear proteins for rapid proteolysis (Figure 8.42). Ubiquitin is a 76-amino-acid polypeptide that is highly conserved in all eukaryotes (yeasts, animals, and plants). Proteins are marked for degradation by the attachment of ubiquitin to the amino group of the side chain of a lysine residue. Additional ubiquitins are then added to form a multiubiquitin chain. Such polyubiquinated proteins are

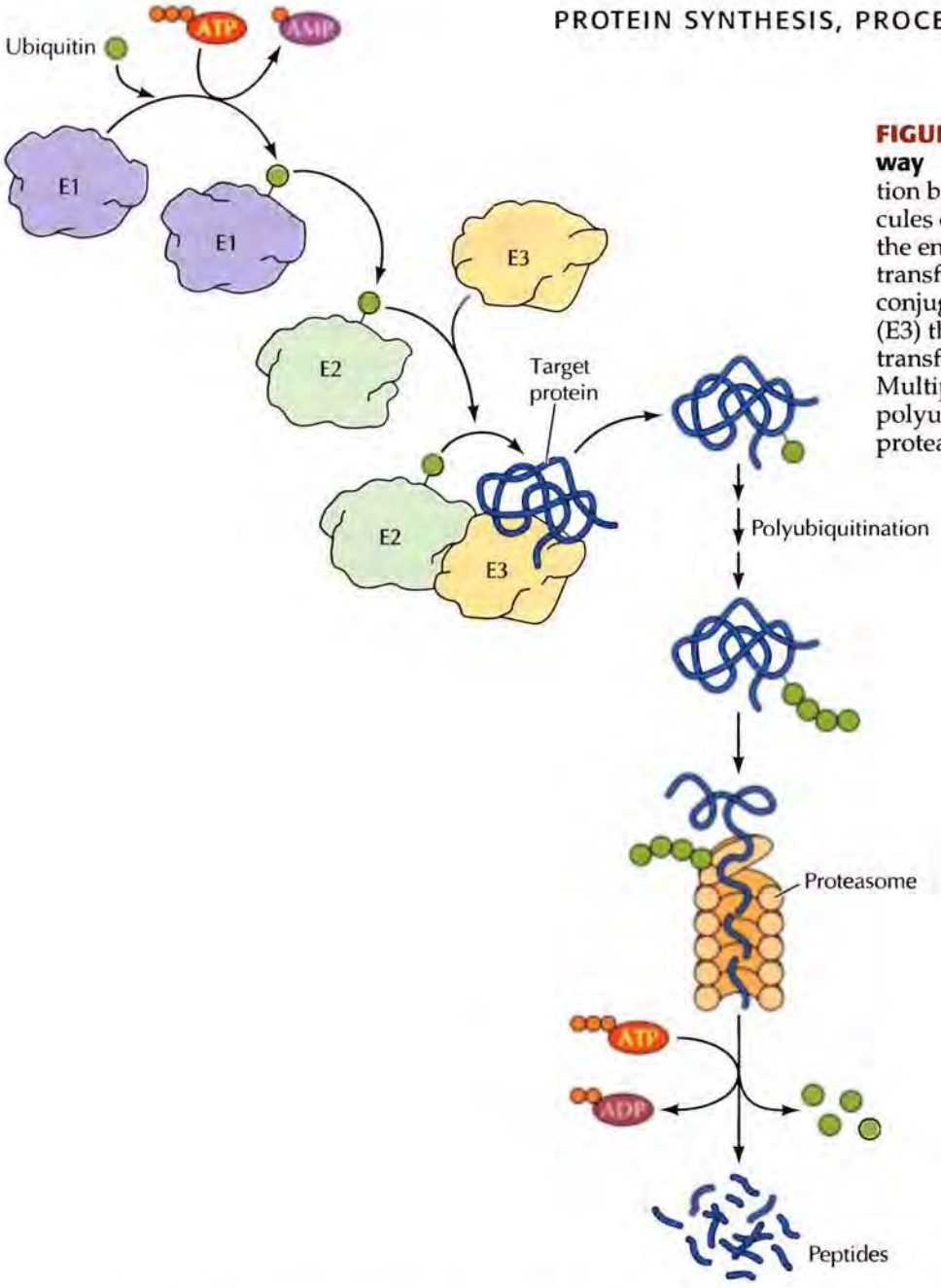


FIGURE 8.42 The ubiquitin-proteasome pathway Proteins are marked for rapid degradation by the covalent attachment of several molecules of ubiquitin. Ubiquitin is first activated by the enzyme E1. Activated ubiquitin is then transferred to one of several different ubiquitin-conjugating enzymes (E2). A ubiquitin ligase (E3) then associates with E2 and directs the transfer of ubiquitin to a specific target protein. Multiple ubiquitins are then added, and the polyubiquinated proteins are degraded by a protease complex (the proteasome).

recognized and degraded by a large, multisubunit protease complex, called the **proteasome**. Ubiquitin is released in the process, so it can be reused in another cycle. Both the attachment of ubiquitin and the degradation of marked proteins require energy in the form of ATP.

Since the attachment of ubiquitin marks proteins for rapid degradation, the stability of many proteins is determined by whether they become ubiquitinated. Ubiquitination is a multistep process. First, ubiquitin is activated by being attached to a ubiquitin-activating enzyme, E1. The ubiquitin is then transferred to a second enzyme, called ubiquitin-conjugating enzyme (E2). The ubiquitin is then transferred to the target protein in association with a third enzyme, called ubiquitin ligase or E3, which is responsible for the selective recognition of appropriate substrate proteins. Most cells contain a single E1 but have several E2s and a large number of E3 enzymes. Different E3s recognize different substrate proteins, and the specificity of these enzymes is what selectively targets cellular proteins for degradation by the ubiquitin-proteasome pathway.

A number of proteins that control fundamental cellular processes, such as gene expression and cell proliferation, are targets for regulated ubiquiti-

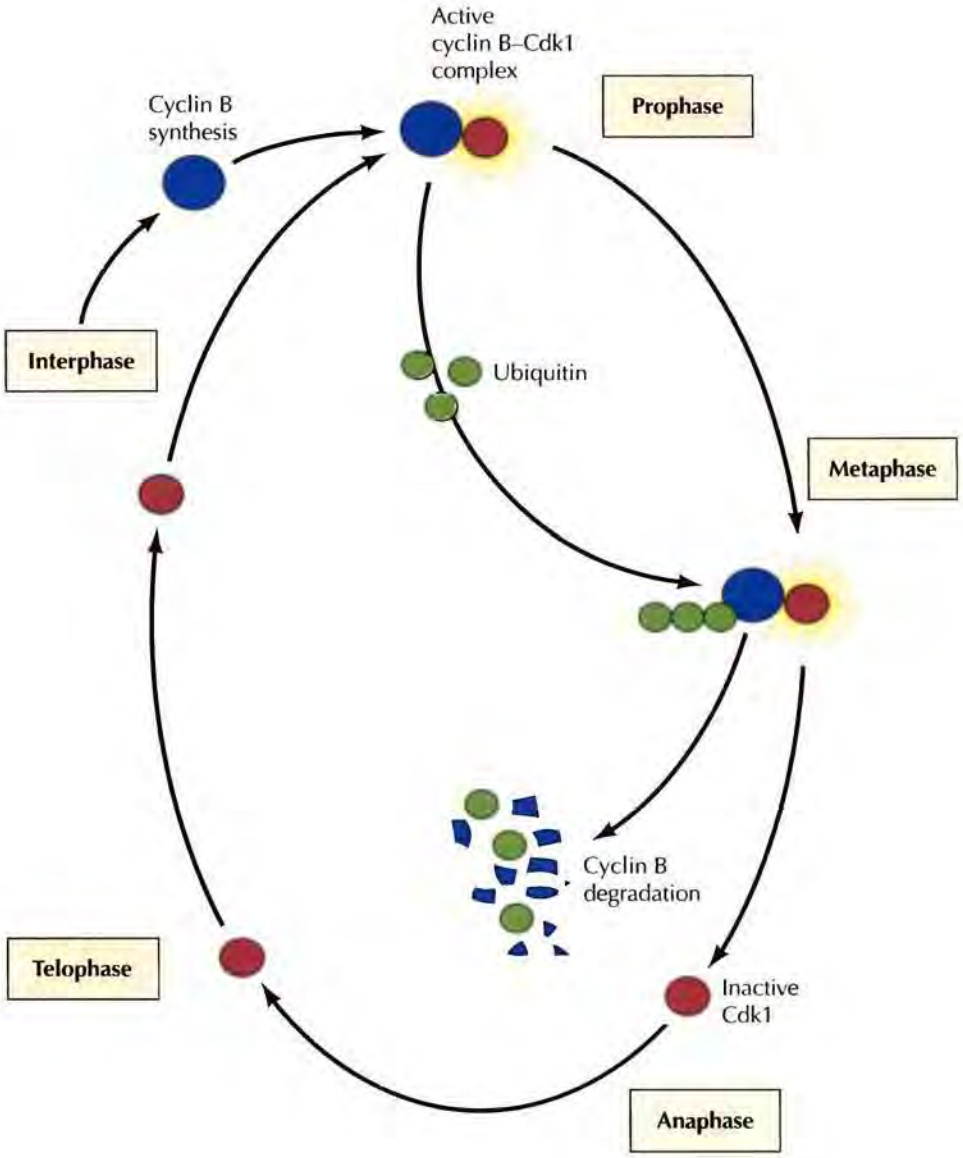


FIGURE 8.43 Cyclin degradation during the cell cycle The progression of eukaryotic cells through the division cycle is controlled in part by the synthesis and degradation of cyclin B, which is a regulatory subunit of the Cdk1 protein kinase. Synthesis of cyclin B during interphase leads to the formation of an active cyclin B–Cdk1 complex, which induces entry into mitosis. Rapid degradation of cyclin B then leads to inactivation of the Cdk1 kinase, allowing the cell to exit mitosis and return to interphase of the next cell cycle.

nation and proteolysis. An interesting example of such controlled degradation is provided by proteins (known as cyclins) that regulate progression through the division cycle of eukaryotic cells (Figure 8.43). The entry of all eukaryotic cells into mitosis is controlled in part by cyclin B, which is a regulatory subunit of a protein kinase called Cdk1 (see Chapter 16). The association of cyclin B with Cdk1 is required for activation of the Cdk1 kinase, which initiates the events of mitosis (including chromosome condensation and nuclear envelope breakdown) by phosphorylating various cellular proteins. Cdk1 also activates a ubiquitin ligase that targets cyclin B for degradation toward the end of mitosis. This degradation of cyclin B inactivates Cdk1, allowing the cell to exit mitosis and progress to interphase of the next cell cycle. The ubiquitination of cyclin B is a highly selective reaction, targeted by a 9-amino-acid cyclin B sequence called the destruction box. Mutations of this sequence prevent cyclin B proteolysis and lead to the arrest of dividing cells in mitosis, demonstrating the importance of regulated protein degradation in controlling the fundamental process of cell division.

Although ubiquitination usually targets proteins for degradation, the addition of ubiquitin to some proteins serves other functions. For example,

ubiquitination of some proteins serves as a marker for endocytosis, and ubiquitination of histones may constitute an element of the "histone code" discussed in Chapter 7. In addition, proteins can be modified by the attachment of other ubiquitin-related polypeptides such as SUMO (small *u*biquitin-related *mo*difier). SUMO does not target proteins for degradation but instead serves as a marker for protein localization and a regulator of protein activity. Many of the proteins modified by SUMO are transcription factors and other nuclear proteins with roles in maintenance of chromatin structure and DNA repair.

■ **The ubiquitin-proteasome pathway is responsible for the degradation of several important regulatory proteins, including proteins that control cell proliferation and cell survival. Since the growth of cancer cells depends on the destruction of these regulatory proteins, the proteasome has emerged as a target for anti-cancer drugs. Bortezomib was the first proteasome inhibitor approved for the treatment of a human cancer, multiple myeloma.**

Lysosomal Proteolysis

The other major pathway of protein degradation in eukaryotic cells involves the uptake of proteins by **lysosomes**. Lysosomes are membrane-enclosed organelles that contain an array of digestive enzymes, including several proteases (see Chapter 10). They have several roles in cell metabolism, including the digestion of extracellular proteins taken up by endocytosis as well as the turnover of cytoplasmic organelles and cytosolic proteins.

The containment of proteases and other digestive enzymes within lysosomes prevents uncontrolled degradation of the contents of the cell. Therefore, in order to be degraded by lysosomal proteolysis, cellular proteins must first be taken up by lysosomes. The principal pathway for this uptake of cellular proteins, **autophagy**, involves the formation of vesicles (autophagosomes) in which small areas of cytoplasm or cytoplasmic organelles are enclosed in membranes derived from the endoplasmic reticulum (Figure 8.44). These vesicles then fuse with lysosomes, and the degradative lysosomal enzymes digest their contents. The uptake of pro-

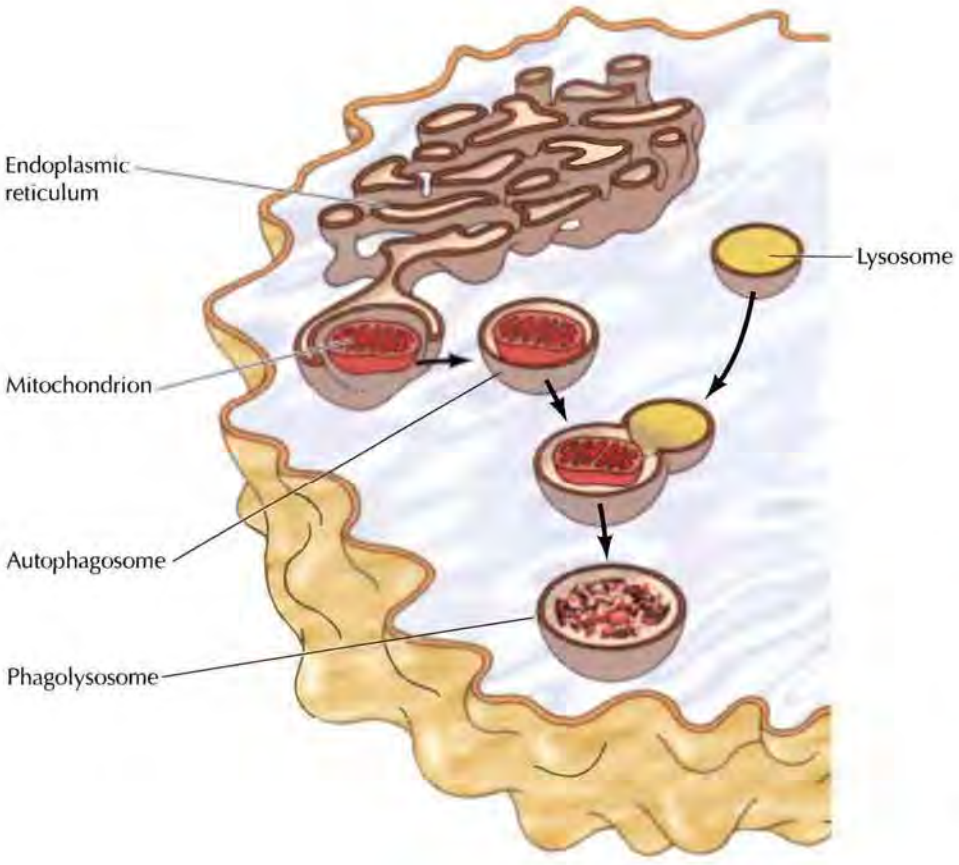


FIGURE 8.44 Autophagy
Lysosomes contain various digestive enzymes, including proteases. Lysosomes take up cellular proteins by fusion with autophagosomes, which are formed by the enclosure of areas of cytoplasm or organelles (e.g., a mitochondrion) in fragments of the endoplasmic reticulum. This fusion yields a phagolysosome, which digests the contents of the autophagosome.

teins into autophagosomes appears to be nonselective, so it results in the eventual slow degradation of long-lived cytoplasmic proteins.

Autophagy is regulated both in response to the availability of nutrients and during development of multicellular organisms. Autophagy is generally activated under conditions of nutrient starvation, allowing cells to degrade nonessential proteins and organelles so that their components can be reutilized. In addition, autophagy plays an important role in many developmental processes, such as insect metamorphosis, which involve extensive tissue remodeling and degradation of cellular components. Recent studies have also linked defects in autophagy to several human diseases, including neurodegenerative diseases and cancer.

COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

KEY TERMS

tRNA, anticodon, aminoacyl tRNA synthetase

ribosome, rRNA

5′ untranslated region, polycistronic, monocistronic, 3′ untranslated region, Shine-Dalgarno sequence

initiation factor, elongation factor, release factor, polysome,

microRNA (miRNA)

SUMMARY

TRANSLATION OF mRNA

Transfer RNAs: Transfer RNAs serve as adaptors that align amino acids on the mRNA template. Aminoacyl tRNA synthetases attach amino acids to the appropriate tRNAs, which then bind to mRNA codons by complementary base pairing.

The Ribosome: Ribosomes consist of two subunits, which are composed of proteins and ribosomal RNAs. The 23S rRNA is the primary catalyst of peptide bond formation.

The Organization of mRNAs and the Initiation of Translation: Translation of both prokaryotic and eukaryotic mRNAs initiates with a methionine residue. In bacteria, initiation codons are preceded by a sequence that aligns the mRNA on the ribosome by base pairing with 16S rRNA. In eukaryotes, most initiation codons are identified by scanning from the 5′ end of the mRNA, which is recognized by its 7-methylguanosine cap.

The Process of Translation: Translation is initiated by the binding of methionyl tRNA and mRNA to the small ribosomal subunit. The large ribosomal subunit then joins the complex, and the polypeptide chain elongates until the ribosome reaches a termination codon in the mRNA. A variety of nonribosomal factors are required for initiation, elongation, and termination of translation in both prokaryotic and eukaryotic cells.

Regulation of Translation: Translation of specific mRNAs can be regulated by the binding of repressor proteins and by proteins that localize mRNAs to specific regions of cells. Controlled polyadenylation of mRNA is also an important mechanism for the regulation of translation during early development. In addition, the translation of many mRNAs may be controlled by noncoding microRNAs that either repress translation or target homologous mRNAs for degradation. Finally, the general translational activity of cells can be regulated by modification of initiation factors.

SUMMARY

PROTEIN FOLDING AND PROCESSING

Chaperones and Protein Folding: Molecular chaperones facilitate protein folding by binding to and stabilizing unfolded or partially folded polypeptide chains.

chaperone

Enzymes that Catalyze Protein Folding: At least two types of enzymes, protein disulfide isomerase and peptidyl prolyl isomerase, catalyze protein folding.

protein disulfide isomerase (PDI), peptidyl prolyl isomerase

Protein Cleavage: Proteolysis is an important step in the processing of many proteins. For example, secreted proteins and proteins incorporated into most eukaryotic organelles are targeted to their destinations by amino-terminal sequences that are removed by proteolytic cleavage as the polypeptide chain crosses the membrane.

proteolysis, signal sequence, signal peptidase

Glycosylation: Many eukaryotic proteins, particularly secreted and plasma membrane proteins, are modified by the addition of carbohydrates in the endoplasmic reticulum and Golgi apparatus.

glycosylation, glycoprotein, dolichol phosphate

Attachment of Lipids: Covalently attached lipids frequently target and anchor proteins to the plasma membrane.

***N*-myristoylation, prenylation, pamitoylation, glycolipid, glycosylphosphatidylinositol (GPI) anchor**

REGULATION OF PROTEIN FUNCTION

Regulation by Small Molecules: Many proteins are regulated by the binding of small molecules, such as amino acids and nucleotides, which induce changes in protein conformation and activity.

allosteric regulation

Protein Phosphorylation: Reversible phosphorylation, which controls the activities of a wide variety of cellular proteins, results from the action of protein kinases and phosphatases. Other modifications, such as nitrosylation, also regulate the activities of some proteins.

protein kinase, protein-serine/threonine kinase, protein-tyrosine kinase, protein phosphatase, nitrosylation

Protein-Protein Interactions: Interactions between polypeptide chains are important in the regulation of allosteric enzymes and other cellular proteins.

PROTEIN DEGRADATION

The Ubiquitin-Proteasome Pathway: The major pathway of selective protein degradation in eukaryotic cells uses ubiquitin as a marker that targets proteins for rapid proteolysis by the proteasome.

ubiquitin, proteasome

Lysosomal Proteolysis: Lysosomal proteases degrade extracellular proteins taken up by endocytosis and are responsible for the degradation of cytoplasmic organelles and long-lived cytosolic proteins. Autophagy is activated as a response to cell starvation.

lysosome, autophagy

KEY TERMS

Questions

1. *E. coli* contain 64 different codons in their mRNAs, 61 of which code for amino acids. How can they synthesize proteins when they have only 40 different tRNAs?

2. You wish to express a cloned eukaryotic cDNA in bacteria. What type of sequence must you add in order for the mRNA to be translated on prokaryotic ribosomes?

3. Discuss the evidence that ribosomal RNA is the major catalytic component of the ribosome.

4. What effect would an inhibitor of polyadenylation have on protein synthesis in fertilized eggs?

5. What are chaperones? Why is it beneficial for the synthesis of heat-shock proteins to be induced by exposure of cells to elevated temperatures?

6. You are interested in studying a protein expressed on the surface of liver cells. How could treatment of these cells with a phospholipase (an enzyme that cleaves phospholipids) enable you to determine whether your protein is a transmembrane protein or one that is attached to the cell surface by a GPI anchor?

7. What was the first evidence that ubiquitination and degradation of specific proteins by proteasomes requires a specific target sequence on the protein?

8. Does ubiquitination of a protein always signal its destruction by the proteasome?

9. How do miRNAs regulate the translation of specific mRNAs?

10. What is the function of 3′ UTRs in mRNAs?

11. Why did Noller and colleagues use ribosomes of *T. aquaticus* for their studies?

12. Why is regulated proteolytic cleavage important for the activity of certain proteins?

13. How does the ribosome ensure that the correct aminoacyl tRNA is inserted opposite a codon?

14. You are studying the pathway responsible for the secretion of ribonuclease (RNase) in cultured pancreatic cells, by assaying the activity of secreted RNase in the culture medium. How would the expression of siRNA targeted against protein disulfide isomerase (PDI) affect the amount of active RNase you detect in your experiments?

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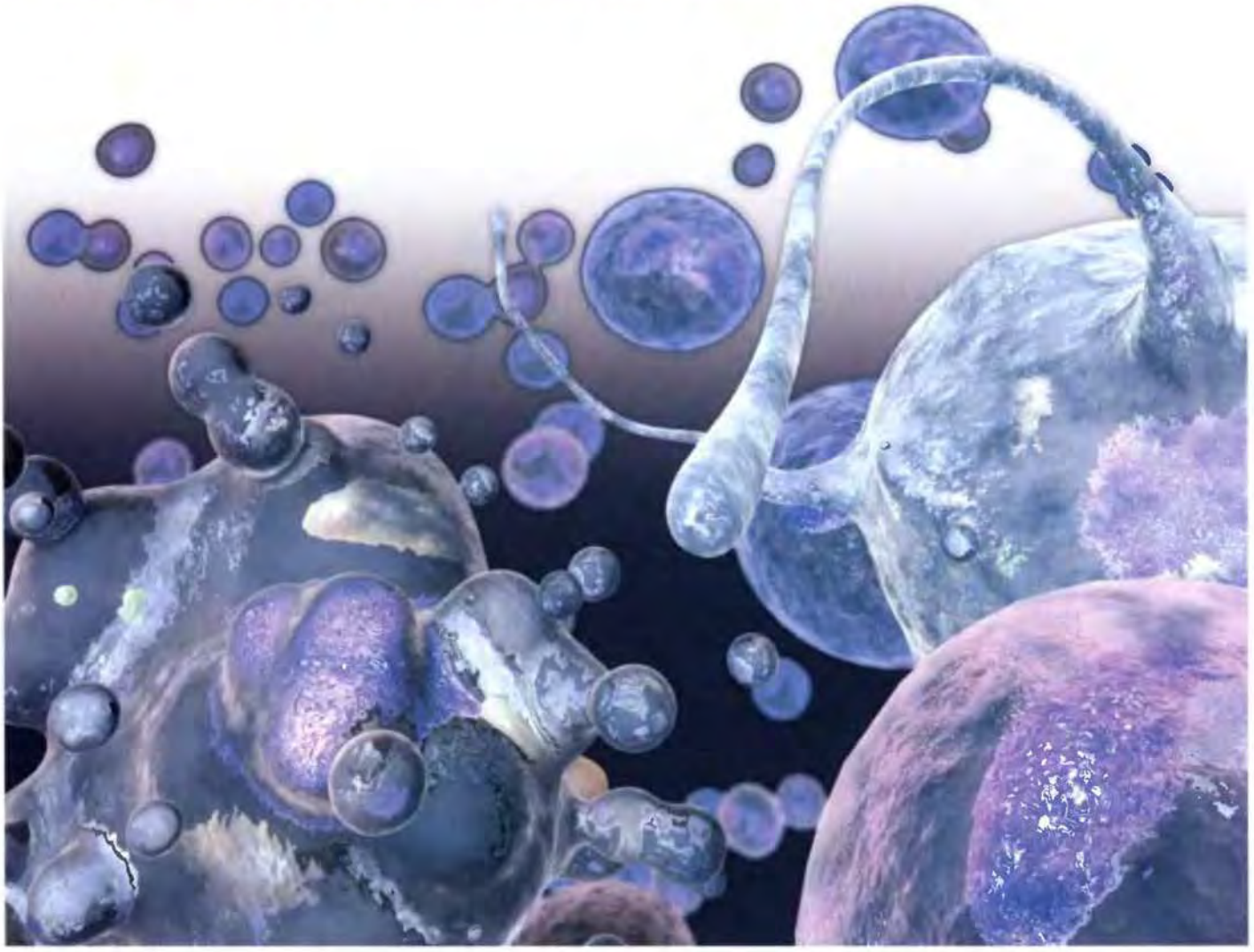
PART III

Cell Structure and Function



CHAPTER 9 ■ The Nucleus

CHAPTER 10 ■ Protein Sorting and Transport

CHAPTER 11 ■ Bioenergetics and Metabolism

CHAPTER 12 ■ The Cytoskeleton and Cell Movement

CHAPTER 13 ■ The Plasma Membrane

CHAPTER 14 ■ Cell Walls, the Extracellular Matrix, and Cell Interactions

CHAPTER 9

The Nucleus

- ***The Nuclear Envelope and Traffic between the Nucleus and the Cytoplasm 355***
- ***Internal Organization of the Nucleus 371***
- ***The Nucleolus and rRNA Processing 375***
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THE PRESENCE OF A NUCLEUS IS THE PRINCIPAL FEATURE that distinguishes eukaryotic from prokaryotic cells. By housing the cell's genome, the nucleus serves both as the repository of genetic information and as the cell's control center. DNA replication, transcription, and RNA processing all take place within the nucleus, with only the final stage of gene expression (translation) localized to the cytoplasm.

By separating the genome from the cytoplasm, the nuclear envelope allows gene expression to be regulated by mechanisms that are unique to eukaryotes. Whereas prokaryotic mRNAs are translated while their transcription is still in process, eukaryotic mRNAs undergo several forms of posttranscriptional processing before being transported from the nucleus to the cytoplasm. The presence of a nucleus thus allows gene expression to be regulated by posttranscriptional mechanisms, such as alternative splicing. By limiting the access of selected proteins to the genetic material, the nuclear envelope also provides novel opportunities for the control of gene expression at the level of transcription. For example, the expression of some eukaryotic genes is controlled by the regulated transport of transcription factors from the cytoplasm to the nucleus —a form of transcriptional regulation unavailable to prokaryotes. Separation of the genome from the site of mRNA translation thus plays a central role in eukaryotic gene expression.

The Nuclear Envelope and Traffic between the Nucleus and the Cytoplasm

The nuclear envelope separates the contents of the nucleus from the cytoplasm and provides the structural framework of the nucleus. The two envelope membranes, acting as barriers that prevent the free passage of molecules between the nucleus and the cytoplasm, maintain the nucleus as a distinct biochemical compartment. The only channels through the

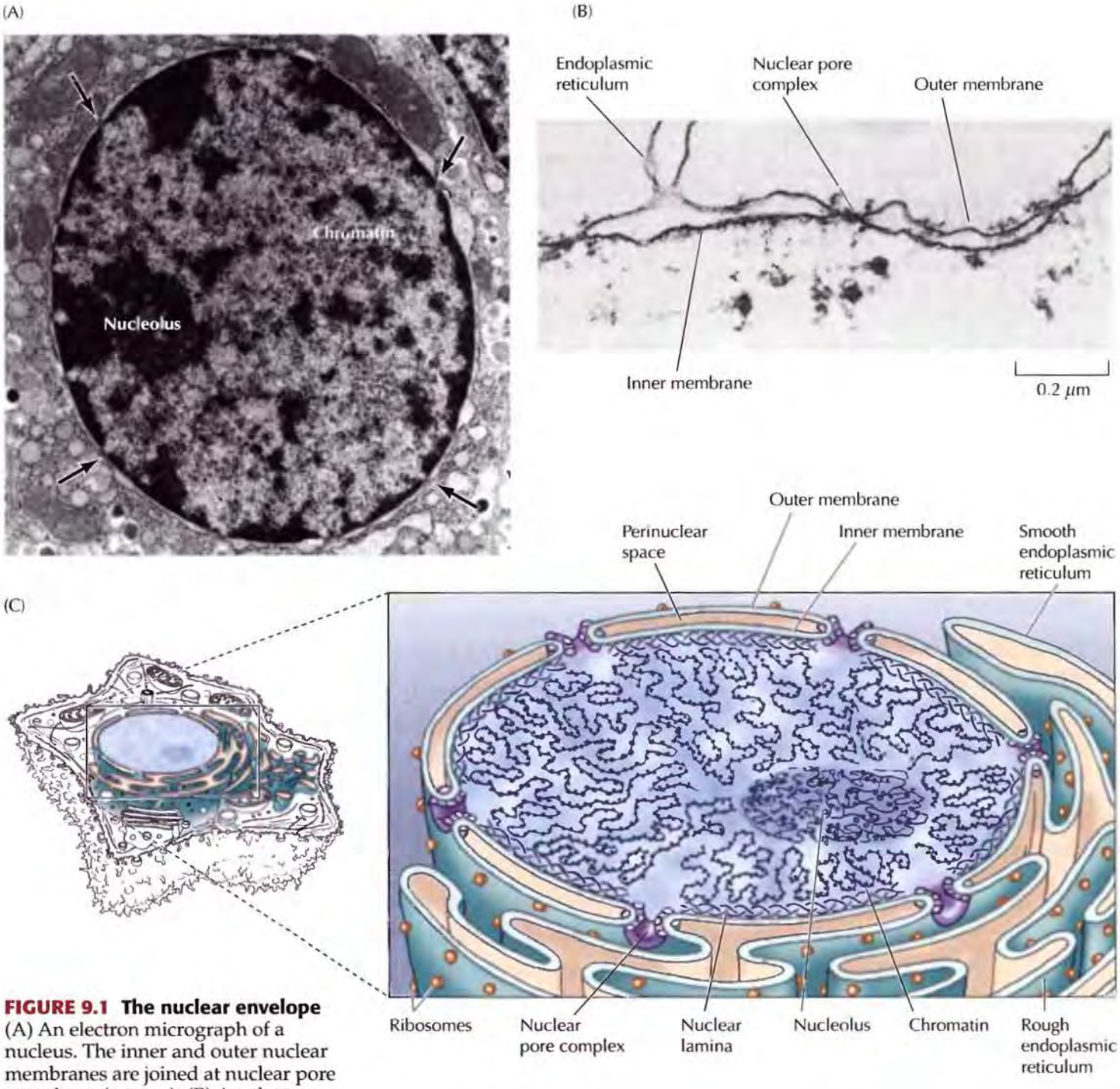


FIGURE 9.1 The nuclear envelope (A) An electron micrograph of a nucleus. The inner and outer nuclear membranes are joined at nuclear pore complexes (arrows). (B) An electron micrograph illustrating the continuity of the outer nuclear membrane with the endoplasmic reticulum. (C) Schematic of the nuclear envelope. The inner nuclear membrane is lined by the nuclear lamina, which serves as an attachment site for chromatin. (A, David M. Phillips/Photo Researchers, Inc.; B, courtesy of Dr. Werner W. Franke, German Cancer Research Center, Heidelberg.)

nuclear envelope are provided by the nuclear pore complexes, which allow the regulated exchange of molecules between the nucleus and the cytoplasm. The selective traffic of proteins and RNAs through the nuclear pore complexes not only establishes the internal composition of the nucleus but also plays a critical role in regulating eukaryotic gene expression.

Structure of the Nuclear Envelope

The **nuclear envelope** has a complex structure consisting of two nuclear membranes, an underlying nuclear lamina, and nuclear pore complexes (Figure 9.1). The nucleus is surrounded by a system of two concentric membranes, called the inner and outer **nuclear membranes**. The outer nuclear

FIGURE 9.2 Electron micrograph showing nuclear pores Many nuclear pores (arrows) are visible in this freeze-fracture preparation of the nuclear envelope. (Don Wo Fawcett/ Photo Researchers, Inc.)

membrane is continuous with the endoplasmic reticulum, so the space between the inner and outer nuclear membranes is directly connected with the lumen of the endoplasmic reticulum. In addition, the outer nuclear membrane is functionally similar to the membranes of the endoplasmic reticulum (see Chapter 10) and has ribosomes bound to its cytoplasmic surface. In contrast, the inner nuclear membrane carries proteins that are specific to the nucleus, such as those that bind the nuclear lamina (discussed below).

The critical function of the nuclear membranes is to act as a barrier that separates the contents of the nucleus from the cytoplasm. Like other cell membranes, each nuclear membrane is a phospholipid bilayer permeable only to small nonpolar molecules (see Figure 2.27). Other molecules are unable to diffuse through the bilayer. The inner and outer nuclear membranes are joined at nuclear pore complexes—the sole channels through which small polar molecules and macromolecules pass through the nuclear envelope (Figure 9.2). As discussed in the next section, the nuclear pore complex is a complicated structure that is responsible for the selective traffic of proteins and RNAs between the nucleus and the cytoplasm.

■ **Erythrocytes (or red blood cells) in mammals are devoid of nuclei. As the erythrocytes develop from precursor cells, the nucleus is extruded from the cell.**

Underlying the inner nuclear membrane is the **nuclear lamina**, a fibrous meshwork that provides structural support to the nucleus (Figure 9.3). The nuclear lamina is composed of 60- to 80-kilodalton (kd) fibrous proteins called **lamins**. Lamins are a class of intermediate filament proteins; the other classes are found in the cytoskeleton (see Chapter 12). Mammalian cells have three lamin genes, designated A, B, and C, which code for at least seven distinct proteins. Like other intermediate filament proteins, the lamins associate with each other to form higher order structures (Figure 9.4), although the extent and polarity of this association is thought to differ from that of other intermediate filaments. The first stage of this association is the interaction of two lamins to form a dimer in which the α-helical regions of two polypeptide chains are wound around each other in a structure called a coiled coil. These lamin dimers then associate with each other to form the nuclear lamina.

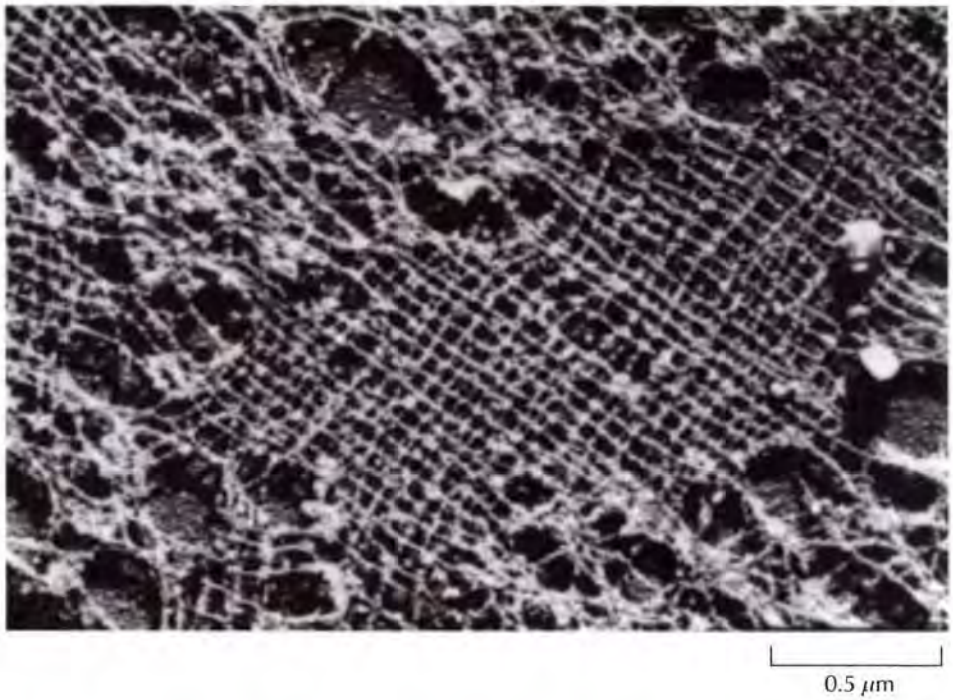


FIGURE 9.3 Electron micrograph of the nuclear lamina The lamina is a meshwork of filaments underlying the inner nuclear membrane and extending into the interior of the nucleus. (From U. Aebi, J. Cohn, L. Buhle and L. Gerace, 1986. *Nature* 323: 560.)

The association of lamins with the inner nuclear membrane is facilitated by the posttranslational addition of lipid—in particular, prenylation of C-terminal cysteine residues (see Figure 8.33). In addition, the lamins bind to specific inner nuclear membrane proteins such as emerin and the lamin B receptor, mediating their attachment to the nuclear envelope and localizing

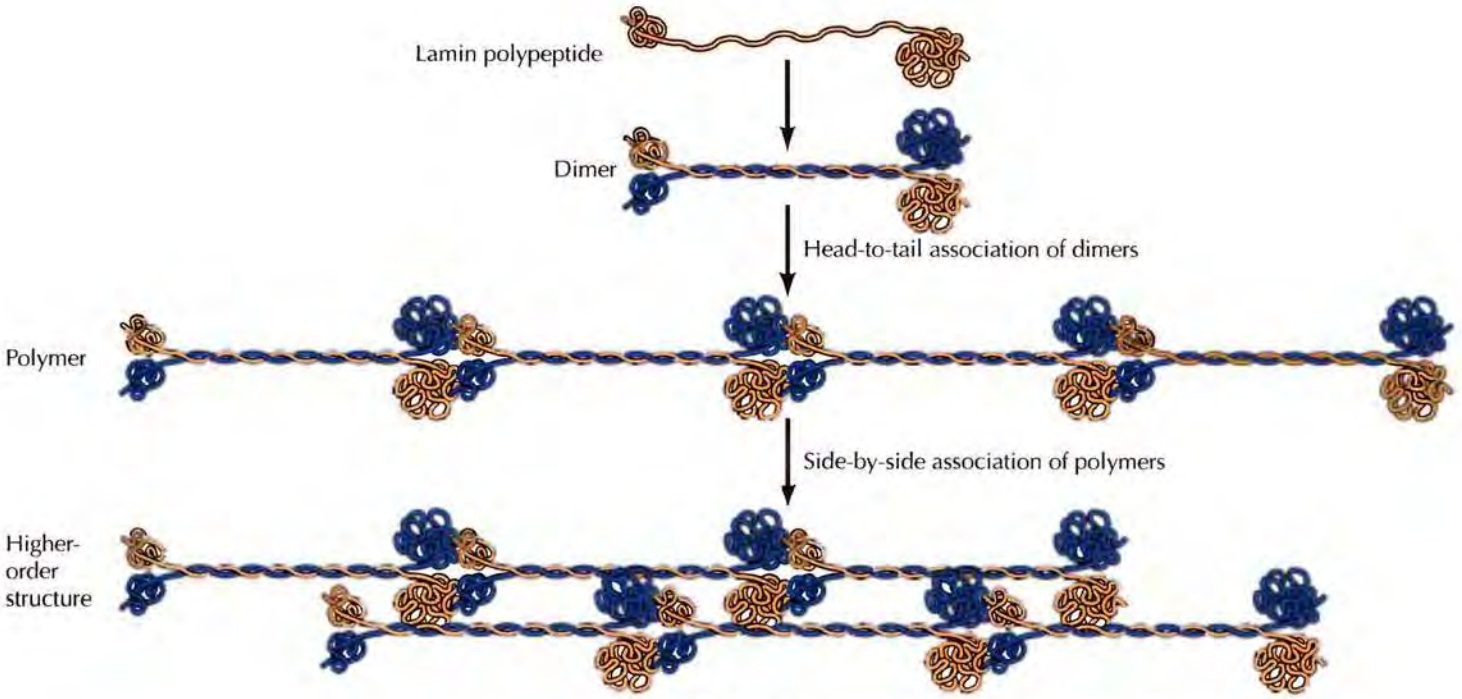
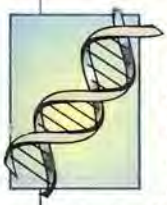


FIGURE 9.4 Model of lamin assembly The lamin polypeptides form dimers in which the central α-helical regions of two polypeptide chains are wound around each other. Further assembly may involve the head-to-tail association of dimers to form linear polymers and the side-by-side association of polymers to form higher order structures.

MOLECULAR MEDICINE

Nuclear Lamina Diseases



The Diseases

In 1966 Emery and Dreifuss described a new muscular dystrophy linked to the X-chromosome. Early in the disease the elbows, neck, and heels of affected individuals become stiff, and often there is a conduction block in the heart. These symptoms are seen by age 10 and include "toe-walking" because of stiff Achilles' tendons in the heels and difficulty bending the elbows. Heart problems develop by age 20 and may require a pacemaker. There is a gradual wasting and weakness of the shoulder and upper arm muscles and the calf muscles of the legs, but this occurs slowly and is often not a problem until late in life.

Nearly 30 years later researchers showed that mutations in a novel transmembrane protein were responsible for this X-linked Emery–Dreifuss muscular dystrophy. They named the protein emerin, after Alan Emery. Soon several groups found that emerin was a protein localized to the inner nuclear membrane and absent in patients with the X-linked Emery–Dreifuss muscular dystrophy. This was unexpected; mutations in a nuclear envelope protein expressed in all cells apparently caused a tissue-specific disease. While all cells in the body were missing the protein, the pathology occurred only in muscle. Subsequent investigators found that the same dystrophy could also be inherited in a non-sex-linked manner. Families with this non-sex-linked Emery–Dreifuss muscular dystrophy had mutations in *LMNA*, the single gene encoding nuclear lamins A and C. So mutations in two genes—one coding for an inner nuclear membrane protein and one coding for a major nuclear lamin—caused clinically identical muscular dystrophy.

More surprising was that parallel investigations on different diseases, Dunnigan-type partial lipodystrophy, Charcot–Marie–Tooth disorder type 2B1, and a disease that causes premature ageing—Hutchinson-Gilford progeria syndrome—traced them to different mutations in the *LMNA* gene. Previously, physicians classified these as distinct diseases based on their clinical features and modes of inheritance. Recent work shows that mutations in another protein in the inner nuclear membrane, the lamin B receptor, are the basis for Pelger-Huët anomaly.

Molecular and Cellular Basis

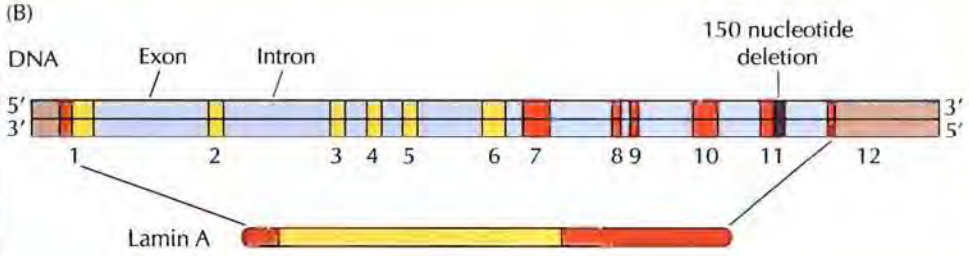
Most biologists thought that mutations in lamins would cause generalized defects in nuclear architecture and serious problems in rapidly dividing cells. However, only minor aberrations of nuclear structure occur in the patients. Thus the puzzle is how mutations in nuclear lamins or lamin-binding proteins cause different tissue-specific diseases. The answer is not yet known but there are two major hypotheses. The first is the "gene expression" hypothesis. This posits that the correct interaction of the two lamin proteins, A and C, with the nuclear envelope is essential for normal tissue-specific expression of certain genes. Transcriptionally inactive genes are located preferentially at the nuclear periphery, whereas expressed genes are concentrated in the center of the nucleus with a cell-type specificity. Thus the basis of these diseases would be a change in gene expression caused by defective protein interactions.

In the "mechanical stress" hypothesis the mutations in the nuclear lamins-emerin complex are thought to weaken the structural integrity of an integrated cytoskeletal network. In all cells the lamina, inner nuclear membrane, and nuclear pore complexes are tightly connected. This hypothesis, which works best for the muscular dystrophies, suggests that through filaments attached to the nuclear pore complex, the lamina could be connected indirectly with the muscle cell cytoskeleton.

(Continued on next page)

(A)

(B)



(A) A child with Hutchinson-Gilford progeria. (B) Diagram of the intron-exon structure of the *LMNA* gene and lamin A protein, with the globular domains indicated in red and the rod domains in yellow. In the mutant gene shown, the gene has a 150-bp deletion (black) in exon 11. (A, courtesy of Maggie Bartlett, NHGRI.)

MOLECULAR MEDICINE

Prevention and Treatment

The discovery that mutations in commonly expressed proteins of the nuclear lamina complex cause different inherited tissue-specific diseases has come as a surprise and has changed the way scientists look at nuclear lamins. Further research is needed to learn whether the bases of the pathologies in each of these diseases is mechanical regulation or gene expression. However, the known molecular nature of the diseases greatly simplifies their diagnosis and makes eventual treatment more likely.

The recent development of a mouse model where the *LMNA* gene is knocked out represents a first step. As the embryos develop they show symptoms of Emery–Dreifuss muscular dystrophy. Finally, researchers are now aware that several slow-developing congenital diseases might be new members of the nuclear "laminopathies."

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and organizing them within the nucleus (Figure 9.5). The nuclear lamina also binds to chromatin through histones H2A and H2B as well as other chromatin proteins. While it binds DNA directly, it is not clear if this interaction is important in the cell. Lamins also extend in a loose meshwork throughout the interior of the nucleus. Many nuclear proteins that function in DNA synthesis, transcription, or chromatin modification are known to bind lamins, although the significance of these interactions is only beginning to be understood.

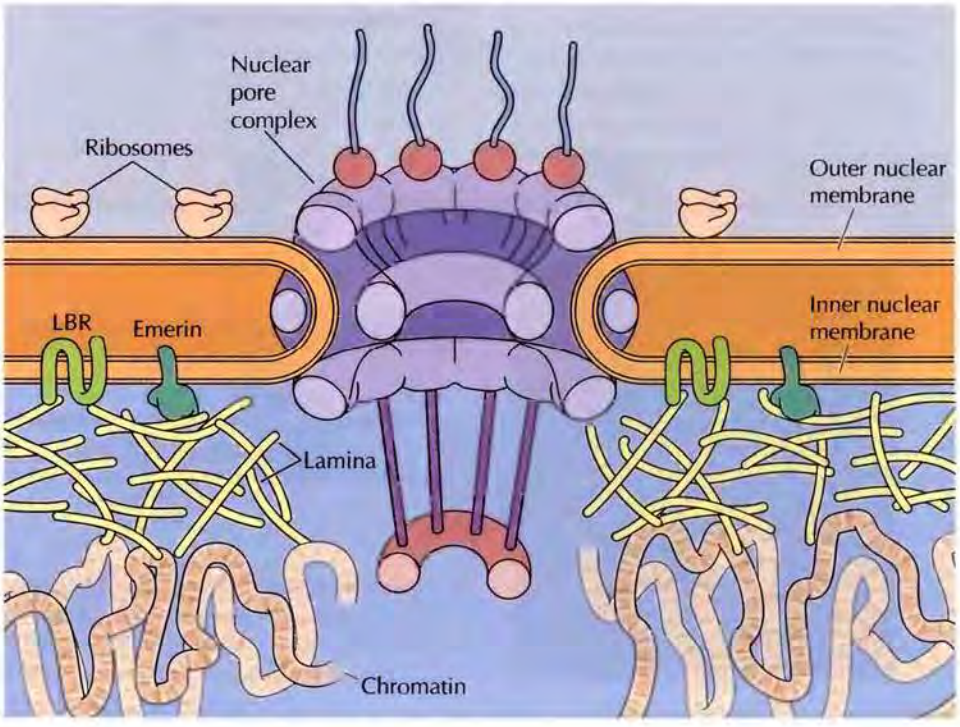


FIGURE 9.5 The nuclear lamina
The inner nuclear membrane contains several integral proteins, such as emerin and the lamin B receptor (LBR) that interact with nuclear lamins. The lamins also interact with chromatin.

The Nuclear Pore Complex

The **nuclear pore complexes** are the only channels through which small polar molecules, ions, and macromolecules (proteins and RNAs) can travel between the nucleus and the cytoplasm. The nuclear pore complex is an extremely large structure with a diameter of about 120 nm and an estimated molecular mass of approximately 125 million daltons—about 30 times the size of a ribosome. In vertebrates, the nuclear pore complex is composed of 30–50 different pore proteins (called nucleoporins), most of which are present in multiple copies. By controlling the traffic of molecules between the nucleus and the cytoplasm, the nuclear pore complex plays a fundamental role in the physiology of all eukaryotic cells. RNAs synthesized in the nucleus must be efficiently exported to the cytoplasm where they function in protein synthesis. Conversely, proteins required for nuclear functions (e.g., transcription factors) must be transported to the nucleus from their sites of synthesis in the cytoplasm. In addition, many proteins shuttle continuously between the nucleus and the cytoplasm.

Depending on their size, molecules can travel through the nuclear pore complex by one of two different mechanisms (Figure 9.6). Small molecules and some proteins with molecular mass less than approximately 20–40 kd pass freely through the pore in either direction: cytoplasm to nucleus or nucleus to cytoplasm. These molecules diffuse passively through open aqueous channels, estimated to have diameters of approximately 9 nm, in the nuclear pore complex. Most proteins and RNAs, however, are unable to pass through these open channels. Instead, these macromolecules pass through the nuclear pore complex by an active process in which appropriate proteins and RNAs are recognized and selectively transported in a specific direction (nucleus to cytoplasm or cytoplasm to nucleus).

Visualization of nuclear pore complexes by electron microscopy reveals a structure with eightfold symmetry organized around a large central channel (Figure 9.7), which is the route through which proteins and RNAs cross the nuclear envelope. Detailed structural studies, including computer-

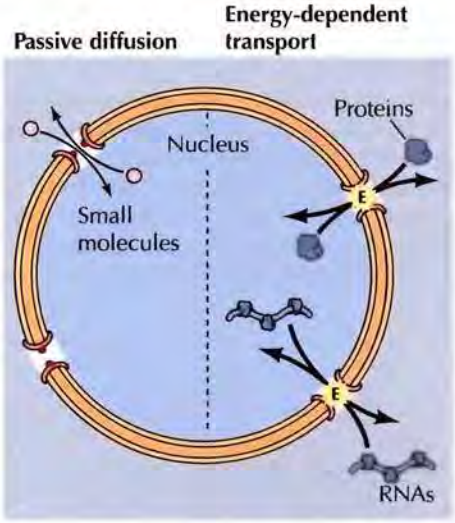


FIGURE 9.6 Molecular traffic through nuclear pore complexes Small molecules are able to pass rapidly through open channels in the nuclear pore complex by passive diffusion. In contrast, macromolecules (proteins and RNAs) are transported by a selective, energy-dependent mechanism.

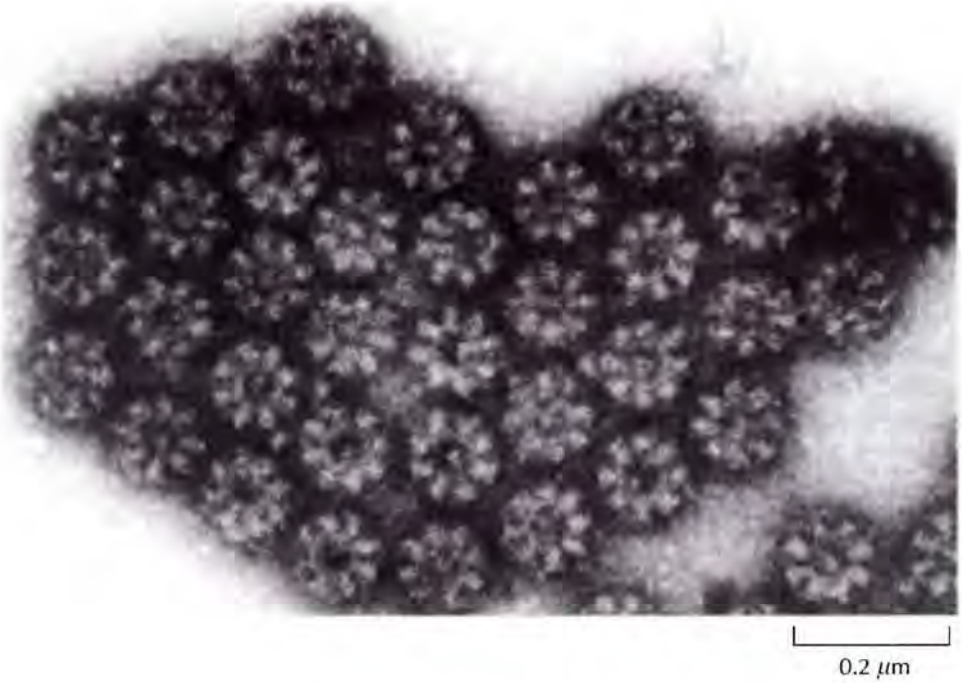


FIGURE 9.7 Electron micrograph of nuclear pore complexes In this face-on view, isolated nuclear pore complexes appear to consist of eight structural subunits surrounding a central channel. (Courtesy of Dr. Ron Milligan, The Scripps Research Institute.)

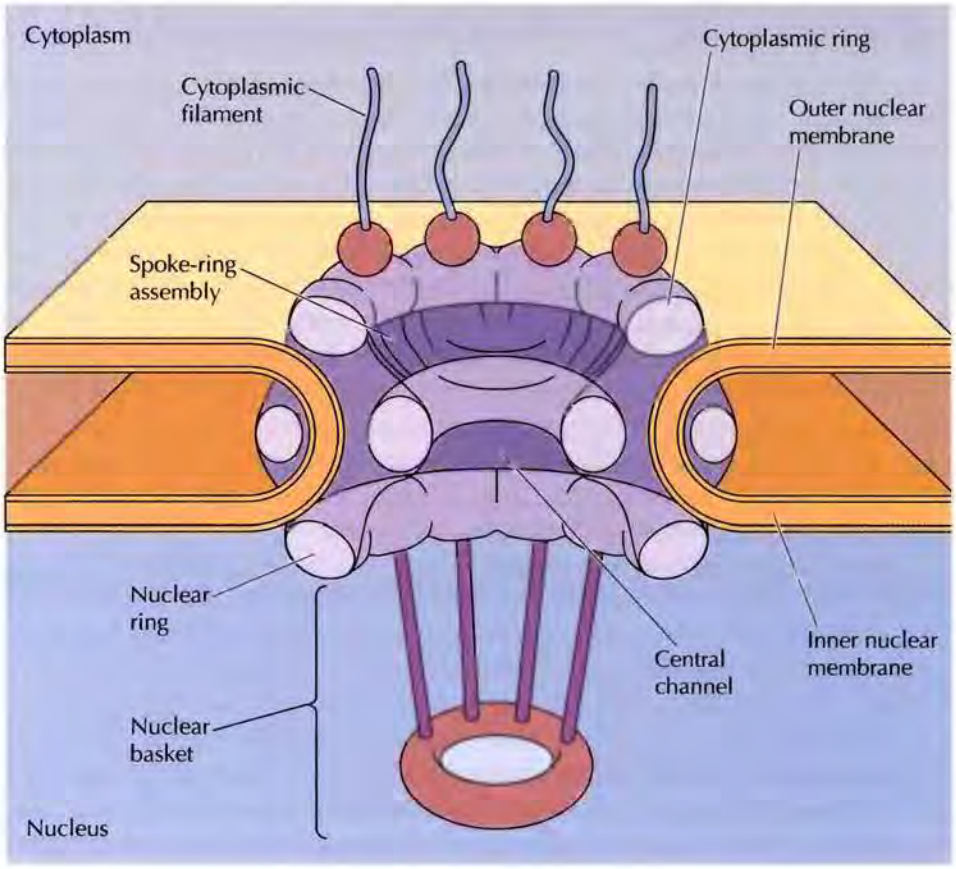


FIGURE 9.8 Model of the nuclear pore complex The complex consists of an assembly of eight spokes attached to rings on the cytoplasmic and nuclear sides of the nuclear envelope. The spoke-ring assembly surrounds a central channel. Cytoplasmic filaments extend from the cytoplasmic ring, and filaments forming the nuclear basket extend from the nuclear ring.

based image analysis, have led to the development of three-dimensional models of the nuclear pore complex (Figure 9.8). These studies show that the nuclear pore complex consists of an assembly of eight spokes arranged around a central channel. The spokes are connected to rings at the nuclear and cytoplasmic surfaces, and the spoke-ring assembly is anchored within the nuclear envelope at sites of fusion between the inner and outer nuclear membranes. Protein filaments extend from both the cytoplasmic and nuclear rings, forming a distinct basketlike structure on the nuclear side.

Selective Transport of Proteins to and from the Nucleus

Several million macromolecules selectively pass between the nucleus and the cytoplasm every minute. The basis for selective traffic across the nuclear envelope is best understood for proteins imported from the cytoplasm to the nucleus. Such proteins are responsible for all aspects of genome structure and function; they include histones, DNA polymerases, RNA polymerases, transcription factors, splicing factors, and many others. These proteins are targeted to the nucleus by specific amino acid sequences called **nuclear localization signals** that are recognized by transport receptors and direct the transport of the proteins through the nuclear pore complex.

The first nuclear localization signal to be mapped in detail was characterized by Alan Smith and colleagues in 1984. These investigators studied simian virus 40 (SV40) T antigen, a virus-encoded protein that initiates viral DNA replication in infected cells (see Chapter 6). As expected for a replication protein, T antigen is normally localized to the nucleus. The signal responsible for its nuclear localization was first identified by the finding

that mutation of a single lysine residue prevents nuclear import, resulting instead in the accumulation of T antigen in the cytoplasm. Subsequent studies defined the T antigen nuclear localization signal as the seven-amino-acid sequence Pro-Lys-Lys-Lys-Arg-Lys-Val. Not only was this sequence necessary for the nuclear transport of T antigen but its addition to other, normally cytoplasmic, proteins was sufficient to direct their accumulation in the nucleus.

Nuclear localization signals have since been identified in many other proteins. Some of these sequences, like that of T antigen, are short stretches rich in basic amino acid residues (lysine and arginine). Often, however, the amino acids that form the nuclear localization signal are close together but not immediately adjacent to each other. For example, the nuclear localization signal of nucleoplasmin (a protein involved in chromatin assembly) consists of two parts: a Lys-Arg pair followed by four lysines located ten amino acids farther downstream (Figure 9.9). Both the Lys-Arg and Lys-Lys-Lys-Lys sequences are required for nuclear targeting, but the ten amino acids between these sequences can be mutated without affecting nuclear localization. Because this nuclear localization sequence is composed of two separated elements, it is called bipartite. Similar bipartite motifs appear to function as the localization signals of many nuclear proteins; thus they may be more common than the simpler nuclear localization signal of T antigen. While many nuclear localization signals consist of these basic amino acid residues—often termed the basic or "classical" nuclear localization signal—the amino acid sequences and structures of other nuclear localization signals vary considerably. Some are far apart in the amino acid sequence and depend on normal folding of the protein for their activity.

Nuclear localization signals are recognized by proteins that function as **nuclear transport receptors**. Most are members of the **karyopherin** protein family and function either as **importins**, which transport macromolecules to the nucleus from the cytoplasm, or **exportins**, which transport macromolecules from the nucleus to the cytoplasm (Table 9.1). Some importins (Kapβ1) act with an adapter karyopherin (Kapα) in a heterodimer to import proteins containing the basic nuclear localization signal, and some karyopherins act as importins for one protein and exportins for another.

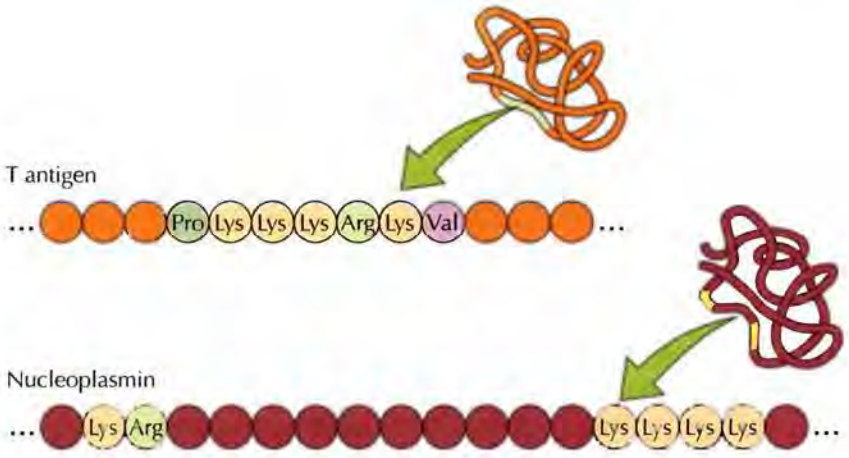


FIGURE 9.9 Nuclear localization signals The T antigen nuclear localization signal is a single stretch of amino acids. In contrast, the nuclear localization signal of nucleoplasmin is bipartite, consisting of a Lys-Arg sequence, followed by a Lys-Lys-Lys-Lys sequence located ten amino acids farther downstream.

KEY EXPERIMENT

Identification of Nuclear Localization Signals

A Short Amino Acid Sequence Able to Specify Nuclear Location

Daniel Kalderon, Bruce L. Roberts, William D. Richardson, and Alan E. Smith
National Institute for Medical Research, Mill Hill, London

Cell, Volume 39, 1984, pages 499–509

Alan Smith

The Context

Maintaining the nucleus as a distinct biochemical compartment requires a mechanism by which proteins are segregated between the nucleus and the cytoplasm. Studies in the 1970s established that small molecules diffuse rapidly across the nuclear envelope but that most proteins are unable to do so. It therefore appeared likely that nuclear proteins are specifically recognized and selectively imported to the nucleus from their sites of synthesis on cytoplasmic ribosomes.

Earlier experiments of Günter Blobel and his colleagues had established that proteins are targeted to the endoplasmic reticulum by signal sequences consisting of short stretches of amino acids (see Chapter 10). In this 1984 paper, Alan Smith and his colleagues extended this principle to the targeting of nuclear proteins by identifying a short amino acid sequence that serves as a nuclear localization signal.

The Experiments

The viral protein SV40 T antigen was used as a model for studies of nuclear localization in animal cells. T antigen is a 94-kd protein that is required for SV40 DNA replication and is normally localized to the nucleus of SV40-infected cells. Previous studies in both Alan Smith's laboratory and in the laboratory of Janet Butel (Lanford and Butel, 1984, *Cell* 37: 801–813) had shown that mutation of Lys-128 to either Thr or Asn prevented the normal nuclear accumulation of T antigen in both rodent and monkey cells. Rather than being transported to the nucleus these mutant T antigens remained in the cytoplasm, suggesting that Lys-128 was part of a nuclear localization signal. Kalderon and colleagues tested this hypothesis using two distinct experimental approaches.

First, they determined the effects of different deletions on the subcellular localization of T antigen. Mutant T antigens bearing deletions that eliminated amino acids either between residues 1 and 126 or between residue

TABLE 9.1 Karyopherins with Known Substrates

Karyopherin	Substrates
Import	
Kapα/Kapβ1 dimer	Proteins with a basic amino acid nuclear localization signal (e.g., nucleoplasmin)
Snurportin/Kapβ1	snRNPs (U1, U2, U4, U5)
Kapβ1 alone	Cdk/cyclin complexes
Kapβ2 (transportin)	mRNA binding proteins, ribosomal proteins
Importin7/Kapβ1 dimer	Histone H1, ribosomal proteins
Export	
Crm1	Proteins with a leucine-rich nuclear export signal, snurportin
CAS	Kapα
Exportin-t	tRNAs
Exportin4	Elongation factor 5A

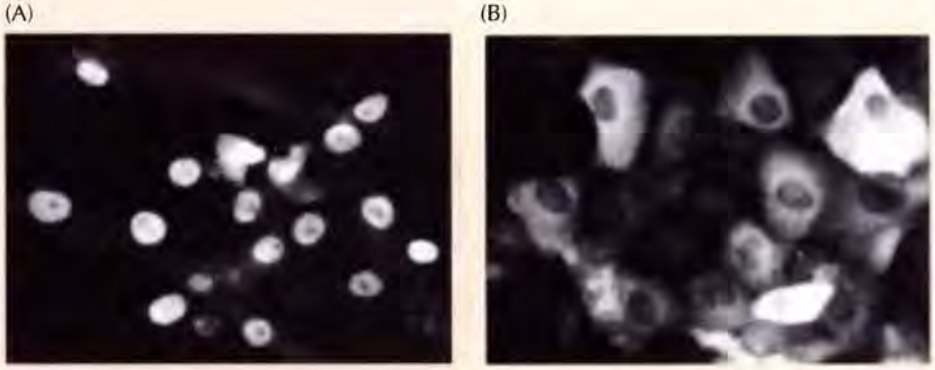
KEY EXPERIMENT

136 and the C terminus were found to accumulate normally in the nucleus. In contrast, a mutant with a deletion of amino acids 127 to 132 remained in the cytoplasm. Thus the amino acid sequence extending from residue 127 to 132 appeared to be responsible for nuclear localization of T antigen.

To determine whether this amino acid sequence was able to target other proteins to the nucleus the investigators constructed chimeras in which the T antigen amino acid sequence was fused to proteins that were normally cytoplasmic. These experiments established that the addition of T antigen amino acids 126 to 132 to either β-galactosidase or pyruvate kinase is sufficient to specify the nuclear accumulation of these otherwise cytoplasmic proteins (see figure). This short amino acid sequence of SV40 T antigen thus functions as a nuclear localization signal, which is both necessary and sufficient to target proteins for nuclear import.

The Impact

As Kalderon and colleagues suggested in their 1984 paper, the nuclear localization signal of SV40 T antigen has proved to "represent a prototype of similar sequences in other nuclear proteins." By targeting proteins for nuclear import, these signals are key to establishing the biochemical identity of the nucleus and maintaining the fundamental division of eukaryotic cells into nuclear and cytoplasmic compartments. Nuclear localization signals are now known to be recognized by cytoplasmic receptors that transport their substrate proteins through the nuclear pore complex. The identification of nuclear localization signals was thus a key advance in understanding nuclear protein import.



Cells were microinjected with plasmid DNAs encoding chimeric proteins in which SV40 amino acids were fused to pyruvate kinase. Cellular localization of the fusion proteins was then determined by immunofluorescence microscopy. (A) The fusion protein contains an intact SV40 nuclear localization signal (amino acids 126 to 132). (B) The nuclear localization signal has been inactivated by deletion of amino acids 131 and 132.

Movement of macromolecules through the nuclear pore is regulated by a protein called **Ran**. This is one of several types of small GTP-binding proteins whose conformation and activity are regulated by GTP binding and hydrolysis. Ras (see Figure 8.37), several of the translation factors involved in protein synthesis (see Figure 8.13), Arf and Rab (discussed in Chapter 10), and Rac, Rho and Cdc42 (discussed in Chapter 15) are other examples of small GTP-binding proteins. For Ran, enzymes that stimulate GTP hydrolysis to GDP are localized to the cytoplasmic side of the nuclear envelope, whereas enzymes that stimulate the exchange of GDP for GTP are localized to the nuclear side (Figure 9.10). Consequently, there is an unequal distribution of Ran/GTP across the nuclear pore, with a high concentration of Ran/GTP in the nucleus. This high concentration of Ran/GTP in the nucleus determines the directionality of nuclear transport of cargo proteins.

Ran regulates movement through the nuclear pore by controlling the activity of the nuclear transport receptors. Protein import through the nuclear pore complex begins when a specific importin binds to the nuclear

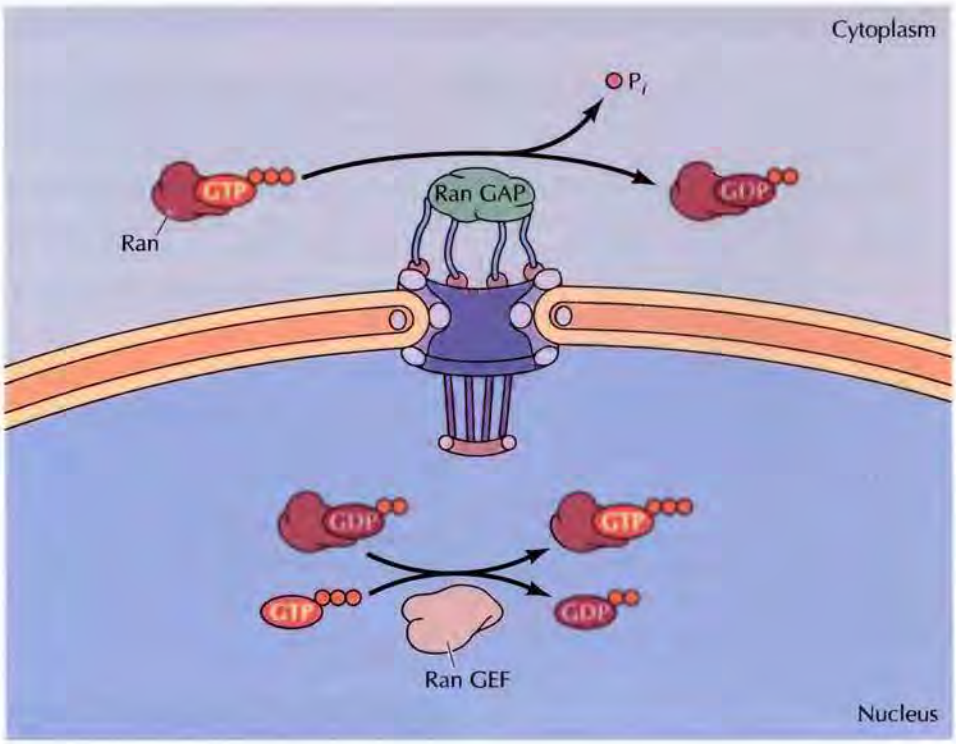


FIGURE 9.10 Distribution of Ran/GTP across the nuclear envelope An unequal distribution of Ran/GTP across the nuclear envelope is maintained by the localization of Ran GTPase-activating protein (Ran GAP) in the cytoplasm and Ran guanine nucleotide exchange factor (Ran GEF) in the nucleus. In the cytoplasm, Ran GAP (which is bound to the cytoplasmic filaments of the nuclear pore complex) stimulates the hydrolysis of GTP bound to Ran, leading to the conversion of Ran/GTP to Ran/GDP. In the nucleus, Ran GEF stimulates the exchange of GDP bound to Ran for GTP, leading to the conversion of Ran/GDP to Ran/GTP. Consequently, a high concentration of Ran/GTP is maintained within the nucleus.

localization signal of a cargo protein in the cytoplasm (Figure 9.11). This importin/cargo complex then binds to proteins in the cytoplasmic filaments of the nuclear pore complex, and transport proceeds by sequential binding to specific nuclear pore proteins located further and further toward the nuclear side of the pore complex. At the nuclear side of the pore complex the importin/cargo complex is disrupted by the binding of Ran/GTP. This causes a change in the conformation of the importin, which displaces the cargo protein and releases it into the nucleus.

The importin-Ran/GTP complex is then exported back through the nuclear pore complex. In the cytoplasm the GTP is hydrolyzed to GDP. This releases the importin so that it can bind to a new cargo protein in the cytoplasm and participate in another round of transport. The Ran/GDP formed in the cytoplasm is then transported back to the nucleus by its own import receptor (a protein called NTF2), where Ran/GTP is regenerated.

Some proteins remain within the nucleus following their import from the cytoplasm, but many others shuttle back and forth between the nucleus and the cytoplasm. Some of these proteins act as carriers in the transport of other molecules, such as RNAs; others coordinate nuclear and cytoplasmic functions (e.g., by regulating the activities of transcription factors). Proteins

■ **Many viruses must gain entry to the nucleus in order to replicate. Following infection of a cell, retroviruses, such as HIV, reverse transcribe their genomic RNA to synthesize a DNA provirus in the cytoplasm. HIV has evolved special mechanisms for the transport of proviral DNA to the nucleus where it can be transcribed.**

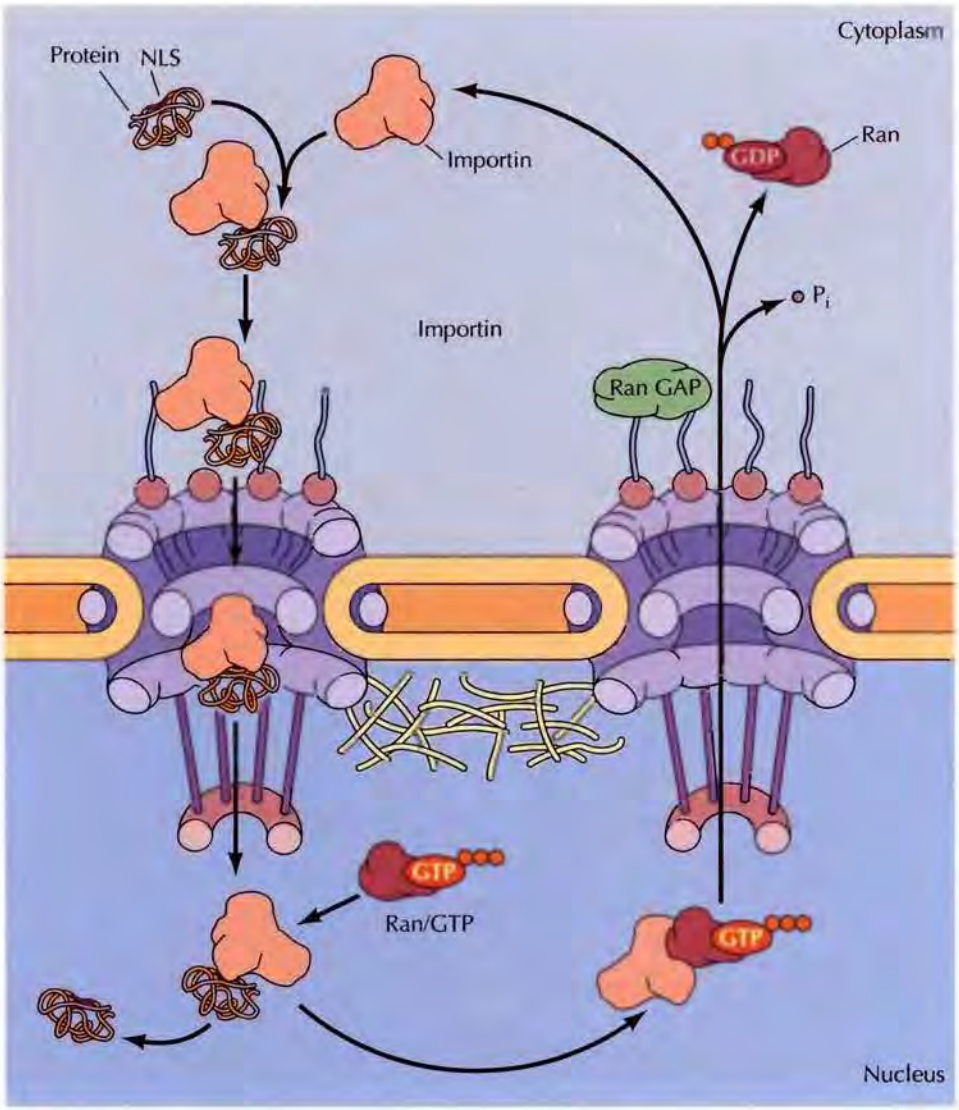


FIGURE 9.11 Protein import through the nuclear pore complex
Transport of a protein through the nuclear pore complex begins when its nuclear localization sequence (NLS) is recognized by an importin nuclear transport receptor. The cargo (the protein with the nuclear localization sequence)/importin complex binds to specific nuclear pore proteins in the cytoplasmic filaments. By sequential binding to more interior nuclear pore proteins, the complex is translocated through the nuclear pore. At the nuclear side of the pore, the cargo/importin complex is disrupted by the binding of Ran/GTP to the importin. The change in conformation of the importin displaces the cargo protein and releases it into the nucleus. The importin-Ran/GTP complex is re-exported through the nuclear pore and the GTPase-activating protein (Ran GAP) in the cytoplasm hydrolyzes the GTP on Ran to GDP, releasing the importin.

are targeted for export from the nucleus by specific amino acid sequences, called **nuclear export signals**. Like nuclear localization signals, nuclear export signals are recognized by receptors within the nucleus—exportins that direct protein transport through the nuclear pore complex to the cytoplasm. Many exportins are also members of the karyopherin protein family (see Table 9.1). Like importins these exportins bind to Ran, which is required for nuclear export as well as for nuclear import (Figure 9.12). However, Ran/GTP promotes the formation of stable complexes between exportins and their cargo proteins, whereas it dissociates the complexes between importins and their cargos. This effect of Ran/GTP binding on exportins dictates the movement of proteins containing nuclear export signals from the nucleus to the cytoplasm. Thus exportins form stable complexes with their cargo proteins in association with Ran/GTP within the nucleus. Following transport to the cytosolic side of the nuclear envelope, GTP hydrolysis and release of Ran/GDP leads to dissociation of the cargo protein, which is released into the cytoplasm.

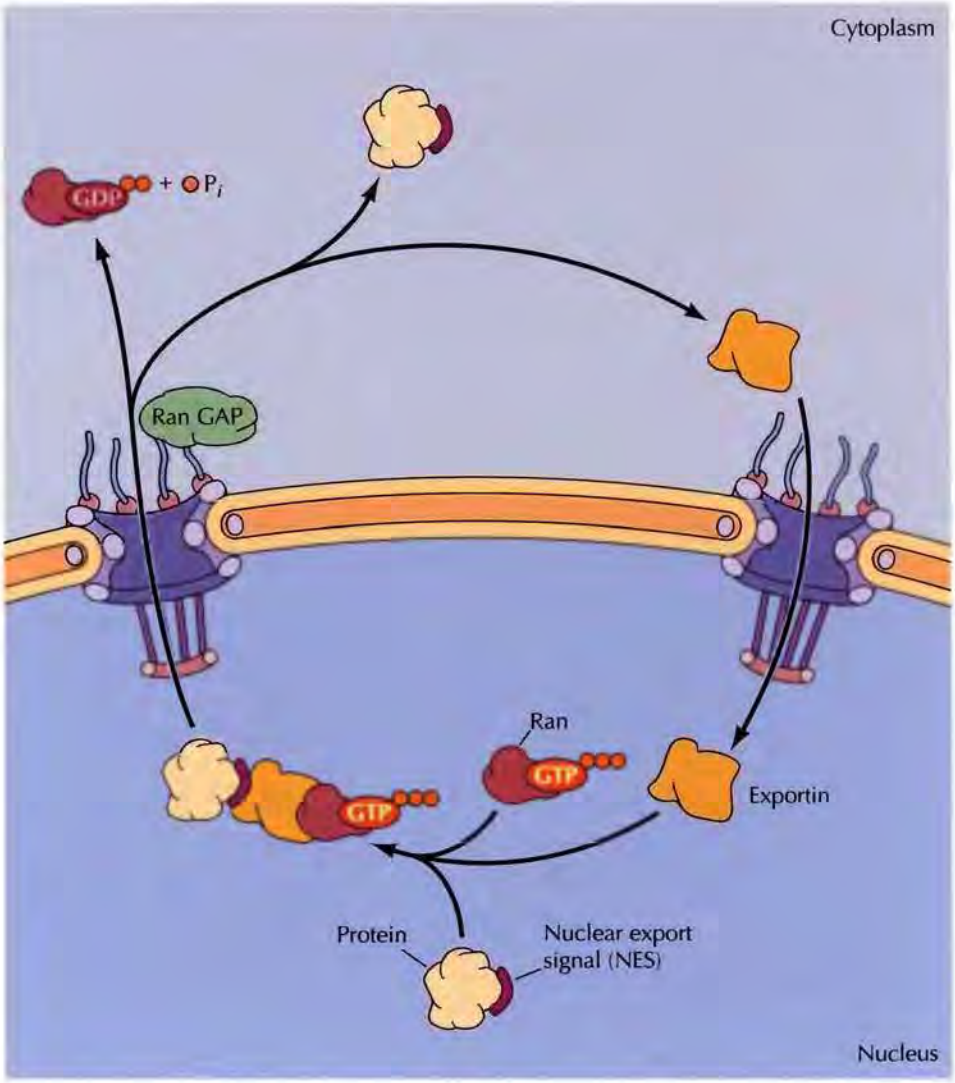


FIGURE 9.12 Nuclear export Complexes between cargo proteins bearing nuclear export signals (NES), exportins, and Ran/GTP form in the nucleus. Following transport through the nuclear pore complex, Ran GAP stimulates the hydrolysis of bound GTP, leading to formation of Ran/GDP and release of the cargo protein and exportin into the cytoplasm. Exportin is then transported back to the nucleus.

Regulation of Nuclear Protein Import

The transport of proteins to the nucleus is yet another level at which the activities of nuclear proteins can be controlled. Transcription factors, for example, are functional only when they are present in the nucleus, so regulation of their import to, and export from, the nucleus is a novel means of controlling gene expression. As will be discussed in Chapter 15, the regulated nuclear import of both transcription factors and protein kinases plays an important role in controlling the behavior of cells in response to changes in the environment, because it provides a mechanism by which signals received at the cell surface can be transmitted to the nucleus. The importance of regulated nuclear import is demonstrated by the finding that changes in nuclear transport receptor affinity of only two nuclear pore complex proteins apparently contributed to the evolutionary split between *Drosophila melanogaster* and *Drosophila simulans*.

In one mechanism of regulation, transcription factors (or other proteins) associate with cytoplasmic proteins that mask their nuclear localization signals; because their signals are no longer recognizable, these proteins remain in the cytoplasm. A good example is provided by the transcription factor NF-κB, which is activated in response to a variety of extracellular signals in

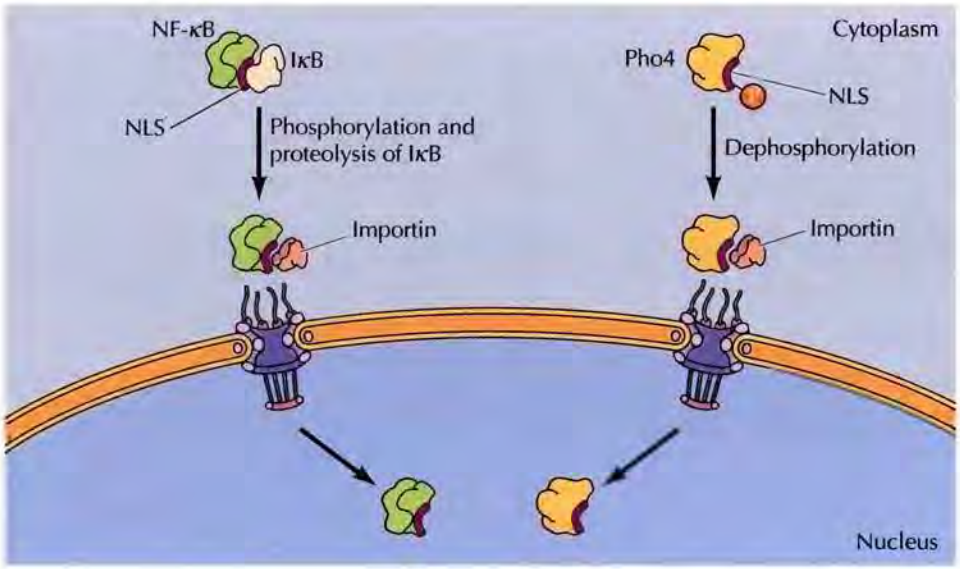


FIGURE 9.13 Regulation of nuclear import of transcription factors The transcription factor NF-κB is maintained as an inactive complex with IκB, which masks its nuclear localization sequence (NLS) in the cytoplasm. In response to appropriate extracellular signals, IκB is phosphorylated and degraded by proteolysis, allowing the import of NF-κB to the nucleus. The yeast transcription factor Pho4 is maintained in the cytoplasm by phosphorylation in the vicinity of its nuclear localization sequence. Regulated dephosphorylation exposes the NLS and allows Pho4 to be transported to the nucleus.

mammalian cells (Figure 9.13). In unstimulated cells, NF-κB is found as an inactive complex with an inhibitory protein (IκB) in the cytoplasm. Binding to IκB masks the NF-κB nuclear localization signal, thus preventing NF-κB from being transported to the nucleus. In stimulated cells, IκB is phosphorylated and degraded by ubiquitin-mediated proteolysis, allowing NF-κB to enter the nucleus and activate transcription of its target genes.

The nuclear import of other transcription factors is regulated directly by their phosphorylation rather than by association with inhibitory proteins (see Figure 9.13). For example, the yeast transcription factor Pho4 is phosphorylated at a serine residue adjacent to its nuclear localization signal. Phosphorylation at this site inhibits Pho4 by interfering with its nuclear import. Under appropriate conditions, regulated dephosphorylation of this site activates Pho4 by permitting its translocation to the nucleus.

Transport of RNAs

Whereas many proteins are selectively transported from the cytoplasm to the nucleus, most RNAs are exported from the nucleus to the cytoplasm. Since proteins are synthesized in the cytoplasm, the export of mRNAs, rRNAs, and tRNAs is a critical step in gene expression in eukaryotic cells. Like protein import, the export of all RNAs through the nuclear pore complex is an active, energy-dependent process requiring the transport receptors to interact with the nuclear pore complex. Karyopherin importins and exportins (see Table 9.1) transport most tRNAs, rRNAs, and small nuclear RNAs in a Ran/GTP–dependent manner. However, mRNAs are exported by a complex of two proteins (the "mRNA exporter"), one of which is related to the Ran/GDP transporter, NTF2. This transport of mRNAs appears to be independent of Ran.

RNAs are transported across the nuclear envelope as ribonucleoprotein complexes (RNPs) (Figure 9.14). Ribosomal RNAs are first associated with both ribosomal proteins and specific RNA processing proteins in the nucleolus, and nascent 60S and 40S ribosomal subunits are then transported to the cytoplasm (see Figure 9.30). Their export from the nucleus is mediated by nuclear export signals present on proteins within the subunit complex. Pre-mRNAs and mRNAs are associated with a set of at least 20 proteins

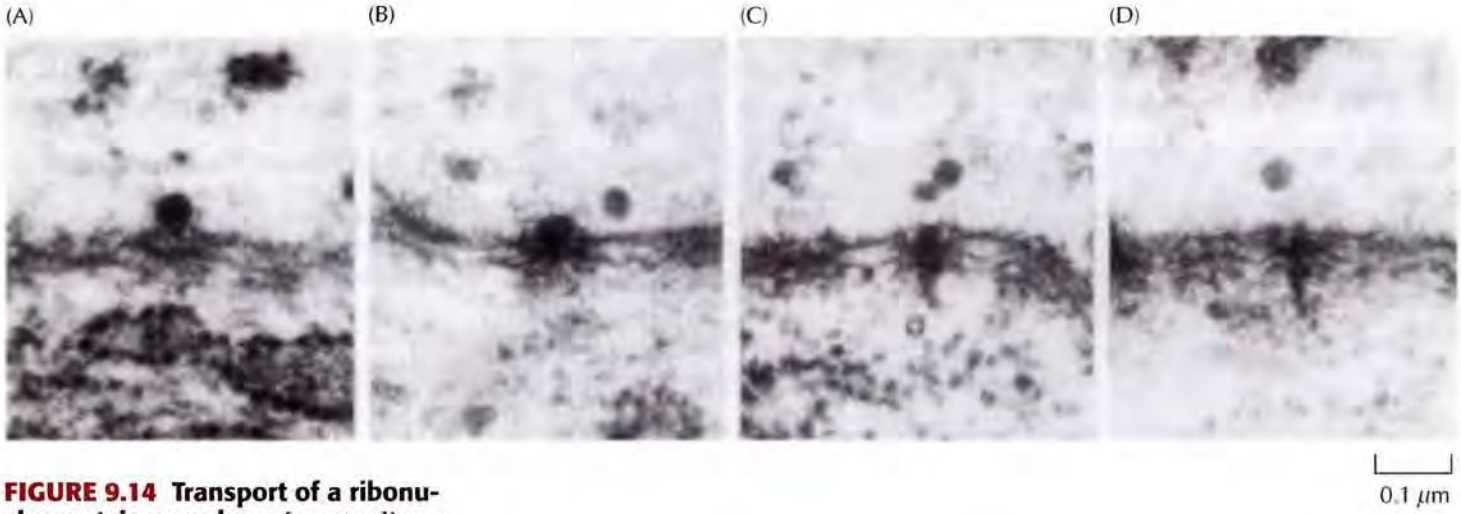


FIGURE 9.14 Transport of a ribonucleoprotein complex Insect salivary gland cells produce large ribonucleoprotein complexes (RNPs), which contain 35 to 40 kb of RNA and have a total mass of approximately 30 million daltons. This series of electron micrographs shows the attachment of such an RNP to a nuclear pore complex (A) and the unfolding of the RNA during its translocation to the cytoplasm (B–D). (From H. Mehlin et al., 1992. *Cell* 69: 605.)

throughout their processing in the nucleus and eventual transport to the cytoplasm, which is mediated by the mRNA exporter complex after its recruitment to the processed mRNA. tRNAs are exported from the nucleus by exportin-t, which binds directly to the tRNAs.

In contrast to mRNAs, tRNAs, and rRNAs, which function in the cytoplasm, many small RNAs (snRNAs and snoRNAs) function within the nucleus as components of the RNA processing machinery. Perhaps surprisingly, snRNAs are initially transported from the nucleus to the cytoplasm, where they associate with proteins to form functional snRNPs and then return to the nucleus (Figure 9.15). Transport receptor proteins that bind to the 5′ caps of snRNAs appear to be involved in the export of the snRNAs to the cytoplasm. In contrast, sequences present on the snRNP proteins are responsible for the transport of snRNPs from the cytoplasm to the nucleus.

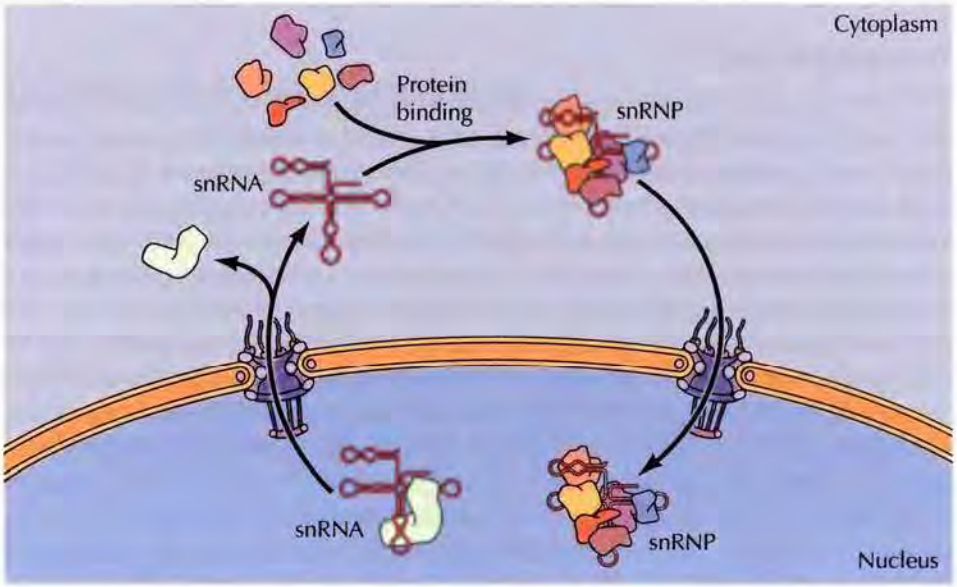


FIGURE 9.15 Transport of snRNAs between nucleus and cytoplasm Small nuclear RNAs (snRNAs) are initially exported from the nucleus to the cytoplasm, where they associate with proteins to form snRNPs. The assembled snRNPs are then transported back to the nucleus.

Internal Organization of the Nucleus

The nucleus is more than a container in which chromatin, RNAs, and nuclear proteins move freely in aqueous solution. Instead, the nucleus has an internal structure that organizes the genetic material and localizes nuclear functions. A loosely organized matrix of nuclear lamins extends from the nuclear lamina into the interior of the nucleus. These lamins serve as sites of chromatin attachment and organize other proteins into functional nuclear bodies. Chromatin within the nucleus is organized into large loops of DNA, and specific regions of these loops are bound to the lamin matrix by lamin-binding proteins in the chromatin. Many other nuclear proteins form lamin-dependent complexes, and these complexes form nuclear bodies that have roles in DNA repair, chromatin organization, gene regulation, and signal transduction. It is thought that this role of the nuclear lamina and lamin proteins in localizing DNA repair and gene transcription is the basis for the variety of lamin-related genetic diseases.

Chromosomes and Higher-Order Chromatin Structure

Chromatin becomes highly condensed during mitosis to form the compact metaphase chromosomes that are distributed to daughter nuclei (see Figure 5.15). During interphase, some of the chromatin (**heterochromatin**) remains highly condensed and is transcriptionally inactive; the remainder of the chromatin (**euchromatin**) is decondensed and distributed throughout the nucleus (Figure 9.16). Cells contain two types of heterochromatin: Constitutive heterochromatin, which contains DNA sequences that are generally not transcribed, such as the satellite sequences present at centromeres; and facultative heterochromatin, which contains sequences that are not transcribed in the cell being examined but are transcribed in other cell types. Conse-

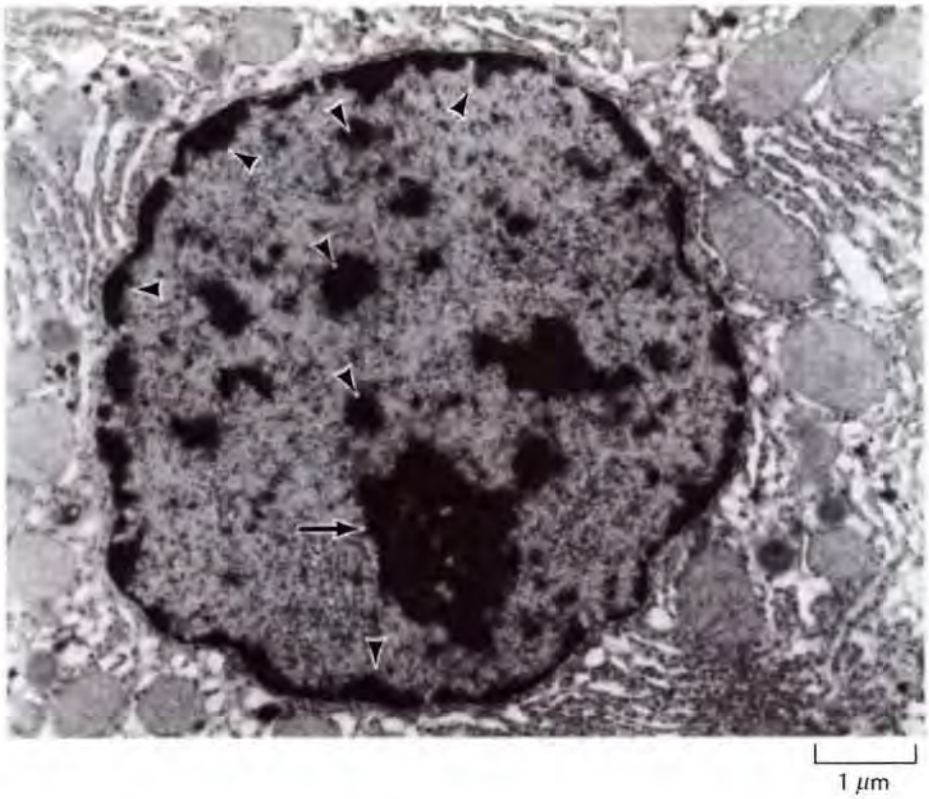


FIGURE 9.16 Heterochromatin in interphase nuclei The euchromatin is distributed throughout the nucleus. The heterochromatin is indicated by arrowheads and the nucleolus by an arrow. (Courtesy of Ada L. Olins and Donald E. Olins, Oak Ridge National Laboratory.)

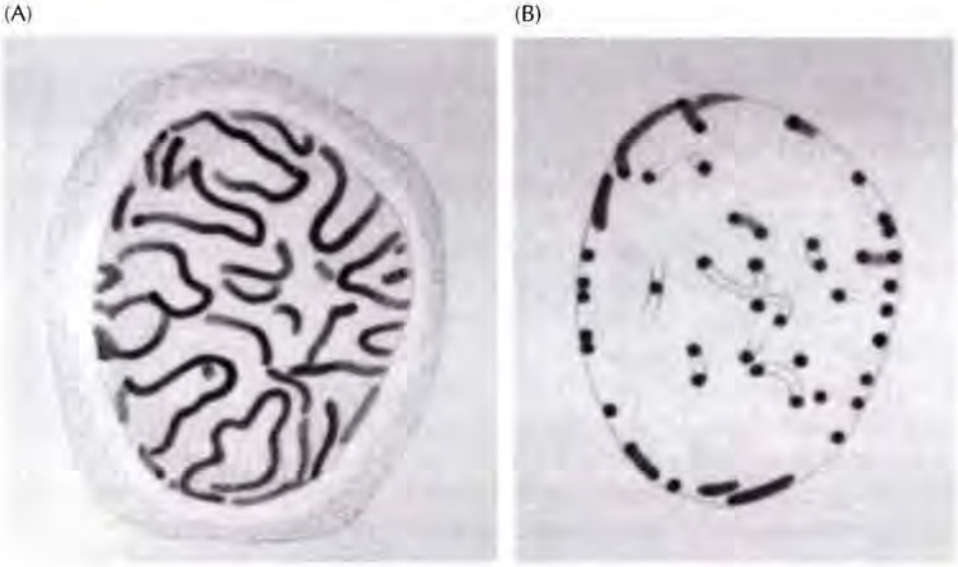


FIGURE 9.17 Chromosome organization Reproduction of hand-drawn sketches of chromosomes in salamander cells. (A) Complete chromosomes. (B) Telomeres only (located at the nuclear membrane). (From C. Rabl, 1885. *Morphologisches Jahrbuch* 10: 214.)

■ **Ciliated protozoa contain two types of nuclei: a polyploid macronucleus that contains transcriptionally active genes, and one or more diploid transcriptionally inactive micronuclei that participate in sexual reproduction.**

quently, the amount of facultative heterochromatin varies depending on the transcriptional activity of the cell.

Although interphase chromatin appears to be uniformly distributed, the chromosomes are actually arranged in an organized fashion and divided into discrete functional domains that play an important role in regulating gene expression. The nonrandom distribution of chromatin within the interphase nucleus was first suggested in 1885 by Carl Rabl who proposed that each chromosome occupies a distinct territory, with centromeres and telomeres attached to opposite sides of the nuclear envelope (Figure 9.17). This basic model of chromosome organization was confirmed nearly a hundred years later (in 1984) by detailed studies of polytene chromosomes in *Drosophila* salivary glands. Rather than randomly winding around one another, each chromosome was found to occupy a discrete region of the nucleus (Figure 9.18). The chromosomes are closely associated with the nuclear envelope at many sites.

Individual chromosomes also occupy distinct territories within the nuclei of mammalian cells (Figure 9.19). Actively transcribed genes are

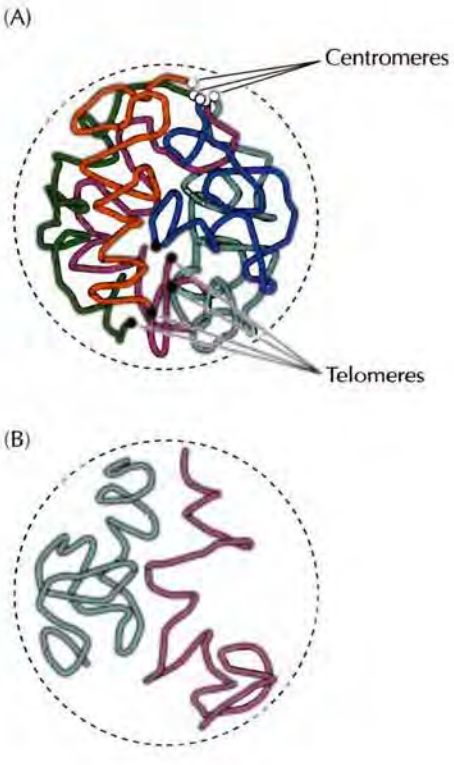


FIGURE 9.18 Organization of *Drosophila* chromosomes (A) A model of the nucleus, showing the five chromosome arms in different colors. The positions of telomeres and centromeres are indicated. (B) The two arms of chromosome 3 are shown to illustrate the topological separation between chromosomes. (From D. Mathog et al., 1984. *Nature* 308: 414.)

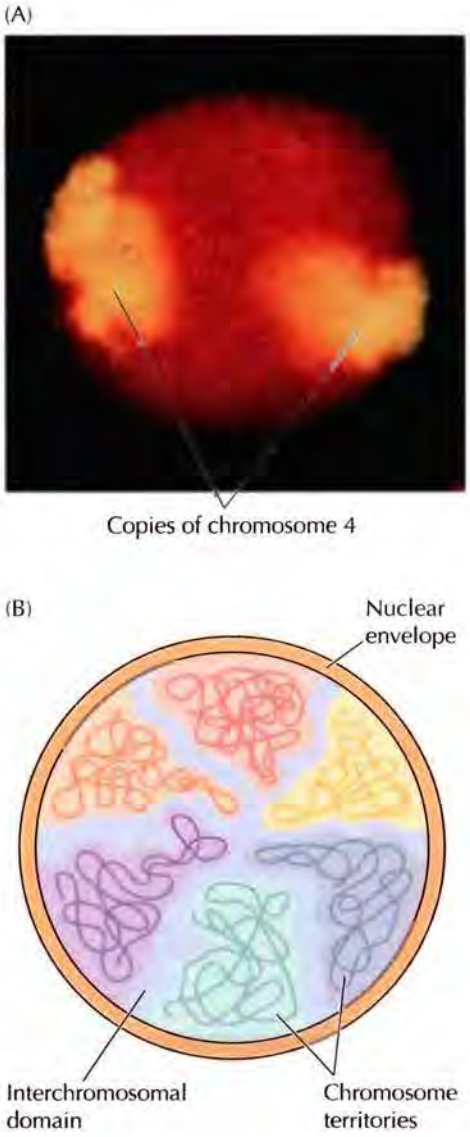


FIGURE 9.19 Organization of chromosomes in the mammalian nucleus (A) Probes to repeated sequences on chromosome 4 were hybridized to a human cell. The two copies of chromosome 4, identified by yellow fluorescence, occupy distinct territories in the nucleus. (B) A model of chromosome organization. The chromosomes occupy discrete territories, separated by intrachromosomal domains in which RNA processing and transport are thought to occur. (A, courtesy of Thomas Cremer, Ludwig Maximilians University, from A. I. Lamond and W. C. Earnshaw, 1999. *Science* 280: 547.)

localized to the periphery of these territories, adjacent to channels separating the chromosomes. Newly transcribed RNAs are thought to be released into these channels between chromosomes where RNA processing takes place. Much of the heterochromatin is localized to the periphery of the nucleus because proteins associated with heterochromatin bind to the matrix of the nuclear lamina. Since different cell types express different genes their facultative heterochromatin is different, and varying regions of the chromosomes interact with the nuclear lamina in different cells and tissues. Some cells have their centromeres and telomeres clustered at opposite poles while others have their chromosomes arranged radially. While the locations of the chromosomes within the nucleus are not random, they differ across different tissues, different organisms, and during cell differentiation.

Like the DNA in metaphase chromosomes (see Figure 5.16), the chromatin in interphase nuclei is organized into looped domains containing approximately 50 to 100 kb of DNA. A good example of this looped-domain organization is provided by the highly transcribed chromosomes of amphibian oocytes in which actively transcribed regions of DNA can be visualized as large loops of extended chromatin (Figure 9.20). These chromatin domains appear to represent discrete functional units, which independently regulate gene expression (see Chapter 7).

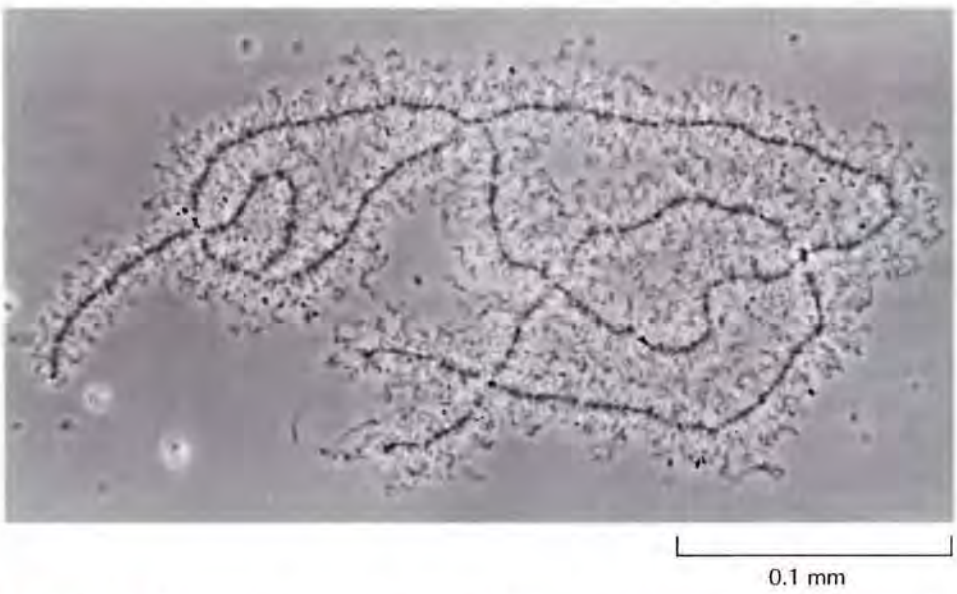


FIGURE 9.20 Looped chromatin domains Light micrograph of a chromosome of amphibian oocytes showing decondensed loops of actively transcribed chromatin extending from an axis of highly condensed nontranscribed chromatin. (Courtesy of Joseph Gall, Carnegie Institute.)

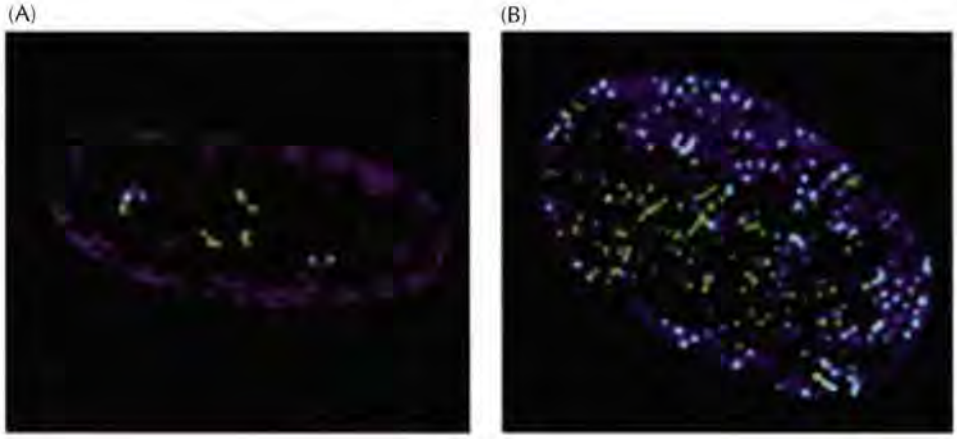


FIGURE 9.21 Clustered sites of DNA replication Newly replicated DNA was labeled by a brief exposure of cells to bromodeoxyuridine, which is incorporated into DNA in place of thymidine. This substitution allows detection of newly synthesized DNA by immunofluorescence following staining with an antibody against bromodeoxyuridine. Note that the newly replicated DNA is present in discrete clusters distributed throughout the nucleus. The two panels show the distribution early and late in DNA synthesis, respectively. (From B. K. Kennedy et al. 2000. *Genes Dev.* 14: 2855)

Sub-Compartments within the Nucleus

The internal organization of the nucleus is further demonstrated by the localization of most nuclear processes to distinct regions of the nucleus. Many important enzymes and other proteins of the nucleus are localized to discrete subnuclear bodies that have a low-density, sponge-like structure that allows macromolecules from the rest of the nucleus to move in and out. Targeting or retention signals for some of these structures have been identified but not yet characterized. The nature and function of these nuclear substructures are not yet clear, and understanding the organization of specific biochemical processes within the nucleus is an incompletely explored area of cell biology.

The nuclei of mammalian cells contain clustered sites of DNA replication within which the replication of multiple DNA molecules takes place. These discrete sites of DNA replication have been defined by experiments in which newly synthesized DNA was visualized within cell nuclei by labeling cells with bromodeoxyuridine, an analog of thymidine that is incorporated into DNA and then detected by staining with fluorescent antibodies (Figure 9.21). At the beginning of DNA synthesis, the newly replicated DNA was detected in approximately 20 discrete clusters distributed around the nucleolus. These perinucleolar sites were associated with nuclear lamins. Later in DNA synthesis the process spread to several hundred sites that were distributed over the nucleus. Since approximately 4000 origins of replication can be active in a diploid mammalian cell at any given time, each of these clustered sites of DNA replication must contain multiple replication forks. Thus DNA replication appears to take place in large structures that contain multiple replication complexes organized into distinct functional bodies, which have been called replication factories.

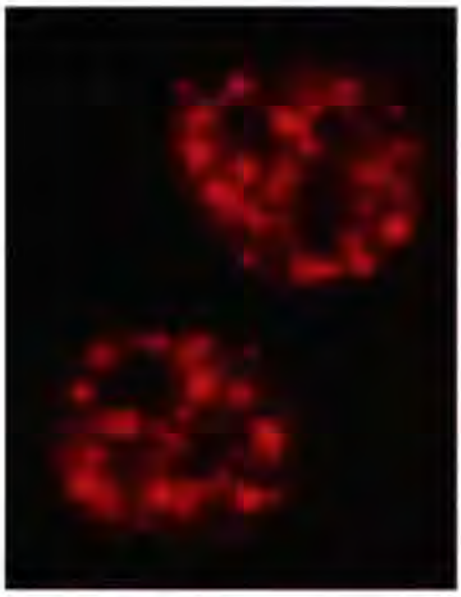


FIGURE 9.22 Localization of splicing components Staining with immunofluorescent antibodies indicates that splicing factors are concentrated in discrete bodies within the nucleus. (Courtesy of David L. Spector, Cold Spring Harbor Laboratory.)

While actively transcribed genes appear to be distributed throughout the nucleus, components of the mRNA splicing machinery are concentrated in discrete nuclear bodies termed nuclear speckles. Immunofluorescent staining with antibodies against snRNPs and splicing factors showed that, rather than being distributed uniformly throughout the nucleus, these components of the RNA splicing apparatus are concentrated in these 20 to 50 discrete structures (Figure 9.22). It is thought that the speckles are storage sites of splicing components, which are then recruited to actively transcribed genes where pre-mRNA processing occurs.

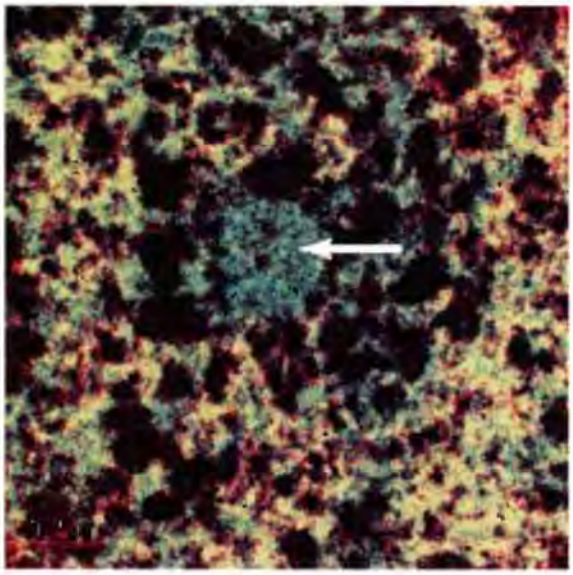


FIGURE 9.23 A PML body The PML body (arrow) is surrounded by chromatin. (From G. Dellaire, R. Nisman and D. P. Bazett-Jones. 2004. *Met. Enzymol.* 375: 456; courtesy of D. Bazett-Jones.)

In addition to speckles, nuclei contain several other types of distinct structures. Besides nucleoli (discussed below), these include PML bodies and Cajal bodies. PML bodies (typically 5–20 within a nucleus) were first identified as distinct sites of localization of a transcriptional regulatory protein involved in acute *prom*yelocytic *l*eukemia. PML bodies are known to interact with chromatin (Figure 9.23) and are the sites of accumulation of transcription factors and chromatin-modifying proteins (such as histone deacetylases), which may be targeted to PML bodies by the small polypeptide SUMO (discussed in Chapter 8). However, the function of PML bodies remains largely unknown. Cajal bodies contain the characteristic protein coilin (Figure 9.24) and are enriched in small RNPs. They are believed to function as sites of RNP assembly, especially in the final steps of snRNP processing.

The Nucleolus and rRNA Processing

The most prominent nuclear body is the **nucleolus** (see Figure 9.1), which is the site of rRNA transcription and processing as well as aspects of ribosome assembly. As discussed in the preceding chapter, cells require large num-

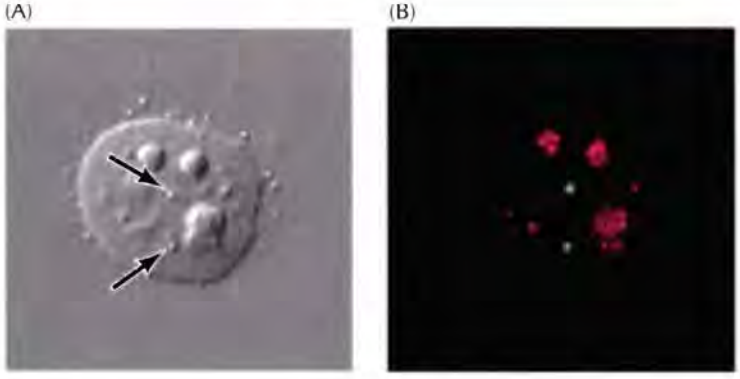


FIGURE 9.24 Cajal bodies in the nucleus (A) Differential interference contrast microscope image of the nucleus of a HeLa cell. The arrows indicate the two Cajal bodies. (B) Immunofluorescent staining of the same nucleus with antibodies to the proteins Coilin (green) and Fibrillarin (red). Fibrillarin is present in both the dense fibrillar zones of the nucleoli and in the Cajal bodies. Coilin is detectable only in the Cajal bodies. (From J. G. Gall, 2000. *Ann. Rev. Cell Dev. Biol.* 16: 273.)

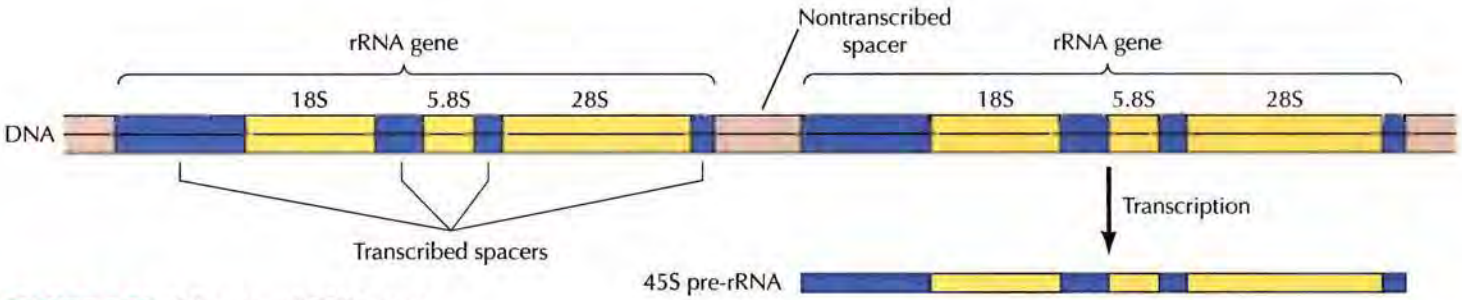


FIGURE 9.25 Ribosomal RNA genes Each rRNA gene is a single transcription unit containing the 18S, 5.8S, and 28S rRNAs as well as transcribed spacer sequences. The rRNA genes are organized in tandem arrays, separated by nontranscribed spacer DNA.

bers of ribosomes at specific times to meet their needs for protein synthesis. Actively growing mammalian cells, for example, contain 5 million to 10 million ribosomes that must be synthesized each time the cell divides. The nucleolus is a ribosome production factory, designed to fulfill the need for regulated and efficient production of rRNAs and assembly of the ribosomal subunits. Recent evidence suggests that nucleoli also have a more general role in RNA modification and that several types of RNA move in and out of the nucleolus at specific stages during their processing.

Ribosomal RNA Genes and the Organization of the Nucleolus

The nucleolus, which is not surrounded by a membrane, is associated with the chromosomal regions that contain the genes for the 5.8S, 18S, and 28S rRNAs. Ribosomes of higher eukaryotes contain four types of RNA designated the 5S, 5.8S, 18S, and 28S rRNAs (see Figure 8.4). The 5.8S, 18S, and 28S rRNAs are transcribed as a single unit within the nucleolus by RNA polymerase I, yielding a 45S ribosomal precursor RNA (Figure 9.25). The 45S pre-rRNA is processed to the 18S rRNA of the 40S (small) ribosomal subunit and to the 5.8S and 28S rRNAs of the 60S (large) ribosomal subunit. Transcription of the 5S rRNA, which is also found in the 60S ribosomal subunit, takes place outside the nucleolus in higher eukaryotes and is catalyzed by RNA polymerase III.

To meet the need for transcription of large numbers of rRNA molecules, all cells contain multiple copies of the rRNA genes. The human genome, for example, contains about 200 copies of the gene that encodes the 5.8S, 18S, and 28S rRNAs and approximately 2000 copies of the gene that encodes 5S rRNA. The genes for 5.8S, 18S, and 28S rRNAs are clustered in tandem arrays on five different human chromosomes (chromosomes 13, 14, 15, 21, and 22); the 5S rRNA genes are present in a single tandem array on chromosome 1.

The importance of ribosome production is particularly evident in oocytes in which the rRNA genes are amplified to support the synthesis of the large numbers of ribosomes required for early embryonic development. In *Xenopus* oocytes, the rRNA genes are amplified approximately two thousandfold, resulting in about one million copies per cell. These amplified rRNA genes are distributed to thousands of nucleoli (Figure 9.26), which support the accumulation of nearly 10^{12} ribosomes per oocyte.

Morphologically, nucleoli consist of three distinguishable regions: the fibrillar center, dense fibrillar component, and granular component (Figure 9.27).

FIGURE 9.26 Nucleoli in amphibian oocytes The amplified rRNA genes of *Xenopus* oocytes are clustered in multiple nucleoli (darkly stained spots). (From D. D. Brown and I. B. Dawid, 1969. *Science* 160: 272.)

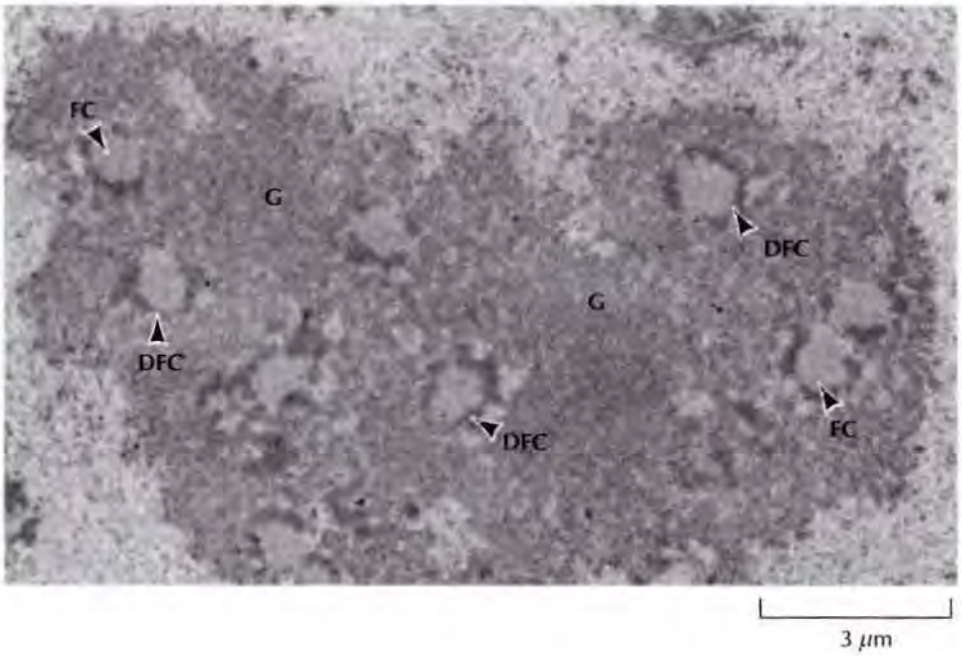


FIGURE 9.27 Structure of the nucleolus An electron micrograph illustrating the fibrillar center (FC), dense fibrillar component (DFC), and granular component (G) of a nucleolus. (Courtesy of David L. Spector, Cold Spring Harbor Laboratory.)

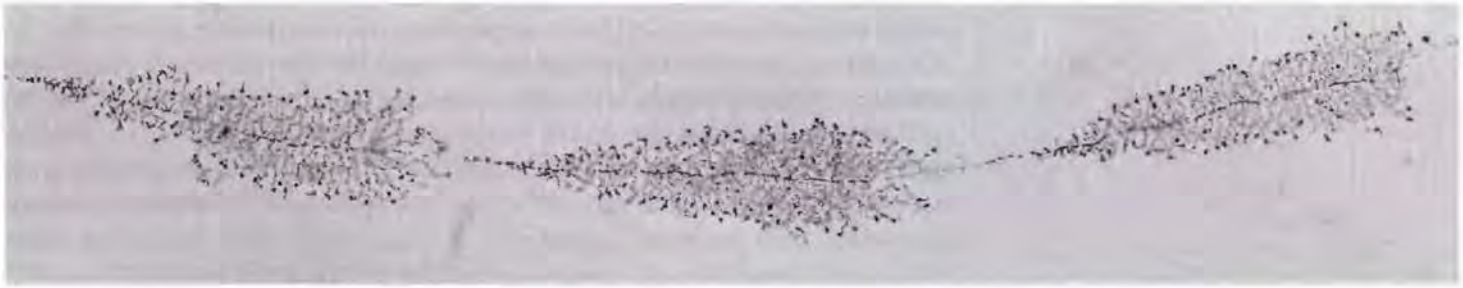
These different regions are thought to represent the sites of progressive stages of rRNA transcription, processing, and ribosome assembly. Modification of other small RNAs, such as that of the signal recognition particle (see Chapter 10), occurs elsewhere in the nucleolus.

Following each cell division, nucleoli become associated with the chromosomal regions that contain the 5.8S, 18S, and 28S rRNA genes, which are therefore called **nucleolar organizing regions**. The formation of nucleoli requires the transcription of 45S pre-rRNA, which appears to lead to the fusion of small prenucleolar bodies that contain processing factors and other components of the nucleolus. In most cells, the initially separate nucleoli then fuse to form a single nucleolus. The size of the nucleolus depends on the metabolic activity of the cell, with large nucleoli found in cells that are actively engaged in protein synthesis. This variation is due primarily to differences in the size of the granular component, reflecting the levels of ribosome assembly.

Transcription and Processing of rRNA

Each nucleolar organizing region contains a cluster of tandemly repeated rRNA genes separated from each other by nontranscribed spacer DNA. These genes are very actively transcribed by RNA polymerase I, allowing their transcription to be readily visualized by electron microscopy (Figure 9.28). In such electron micrographs, each of the tandemly arrayed rRNA

FIGURE 9.28 Transcription of rRNA genes An electron micrograph of nucleolar chromatin showing three rRNA genes separated by nontranscribed spacer DNA. Each rRNA gene is surrounded by an array of growing RNA chains, resulting in a Christmas tree appearance. (Courtesy of O. L. Miller, Jr.)



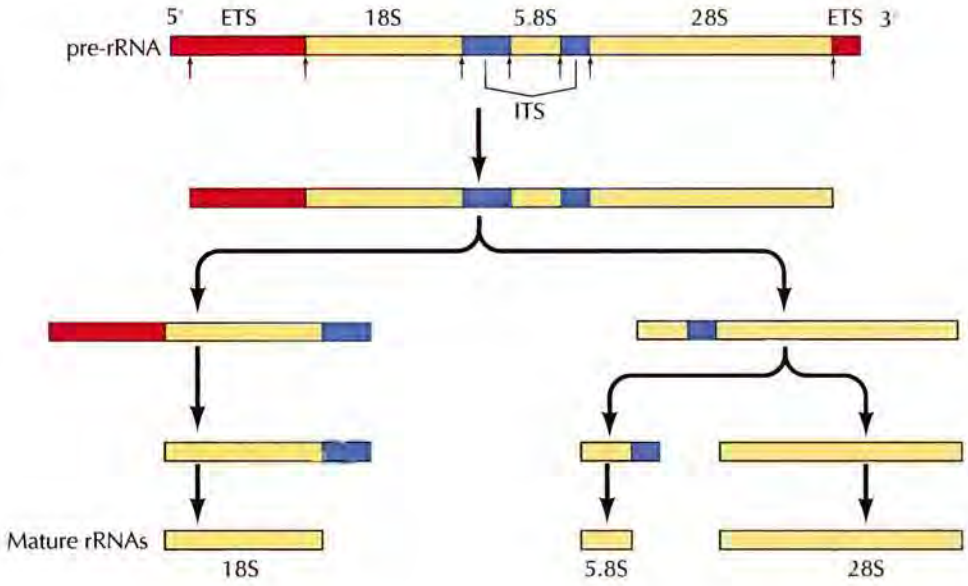


FIGURE 9.29 Processing of pre-rRNA The higher eukaryote 45S pre-rRNA transcript contains external transcribed spacers (ETS) at both ends and internal transcribed spacers (ITS) between the sequences of 18S, 5.8S, and 28S rRNAs. The pre-rRNA is processed via a series of cleavages to yield the mature rRNA species.

genes is surrounded by densely packed growing RNA chains forming a structure that looks like a Christmas tree. The high density of growing RNA chains reflects that of RNA polymerase molecules, which are present at a maximal density of approximately one polymerase per hundred base pairs of template DNA.

In higher eukaryotes the primary transcript of the rRNA genes is the large 45S pre-rRNA, which contains the 18S, 5.8S, and 28S rRNAs as well as transcribed spacer regions (Figure 9.29). External transcribed spacers are present at both the 5′ and 3′ ends of the pre-rRNAs, and two internal transcribed spacers lie between the 18S, 5.8S, and 28S rRNA sequences. Early steps in the processing of this transcript are cleavages within the external transcribed spacer near the 5′ end of the pre-rRNA and removal of the external transcribed spacer at the 3′ end of the molecule. Additional cleavages then result in formation of the mature rRNAs. Processing follows a similar pattern in all eukaryotes, although there may be differences in the order or the number of cleavages.

In addition to cleavage, the processing of pre-rRNA involves a substantial amount of base modification resulting both from the addition of methyl groups to specific bases and ribose residues and from the conversion of uridine to pseudouridine (see Figure 7.43). In animal cells, pre-rRNA processing involves the methylation of approximately a hundred ribose residues and ten bases, in addition to the formation of about a hundred pseudouridines. Most of these modifications occur during or shortly after synthesis of the pre-rRNA, although a few take place at later stages of pre-rRNA processing.

The processing of pre-rRNA requires the action of both proteins and RNAs that are localized to the nucleolus. The involvement of small nuclear RNAs (snRNAs) in pre-mRNA splicing was discussed in Chapter 7. Nucleoli contain more than 300 proteins and a large number (about 200) of **small nucleolar RNAs (snoRNAs)** that function in pre-rRNA processing. Like the spliceosomal snRNAs, the snoRNAs are complexed with proteins, forming snoRNPs. Individual snoRNPs consist of single snoRNAs associated with eight to ten proteins. The snoRNPs then assemble on the pre-rRNA to form processing complexes in a manner analogous to the formation of spliceosomes on pre-mRNA.

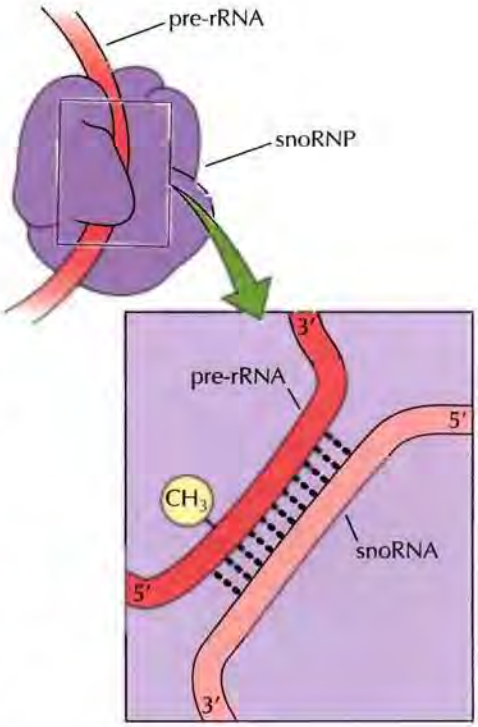


FIGURE 9.30 Role of snoRNAs in base modification of pre-rRNA The snoRNAs contain short sequences complementary to rRNA. Base pairing between snoRNAs and pre-rRNA targets the enzymes that catalyze base modification (e.g., methylation) to the appropriate sites on pre-rRNA.

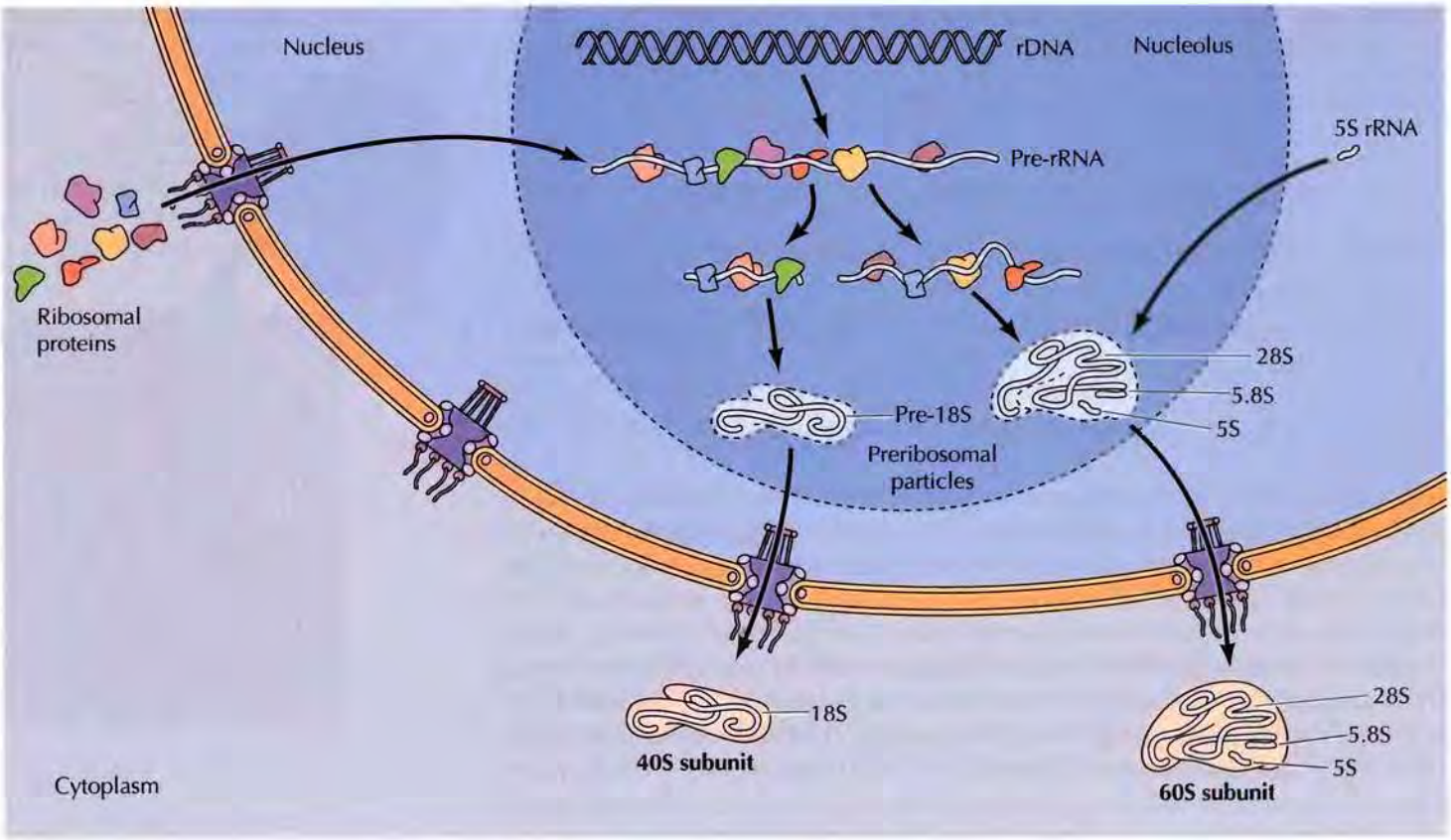
Some snoRNAs are responsible for the cleavages of pre-rRNA into 18S, 5.8S, and 28S products. For example, the most abundant nucleolar snoRNA is U3, which is present in about 200,000 copies per cell and is required for cleavage of pre-rRNA within the 5′ external transcribed spacer sequences. Similarly, U8 snoRNA is required for cleavage of pre-rRNA to 5.8S and 28S rRNAs, and U22 snoRNA is required for cleavage of pre-rRNA to 18S rRNA.

Most snoRNAs, however, function in rRNA synthesis as guide RNAs to direct the specific base modifications of pre-rRNA, including the methylation of specific ribose residues and the formation of pseudouridines (Figure 9.30). Most of the snoRNAs contain short sequences of approximately 15 nucleotides that are complementary to 18S or 28S rRNA. Importantly, these regions of complementarity include the sites of base modification in the rRNA. By base pairing with specific regions of the pre-rRNA, the snoRNAs act as guide RNAs that target the enzymes responsible for ribose methylation or pseudouridylation to the correct site on the pre-rRNA molecule. Other RNAs besides rRNA require modified bases, and it is the localization of snoRNPs in the nucleolus that is thought to be the basis for its more general role in RNA modification. One example is the signal recognition particle RNA (see Chapter 10).

Ribosome Assembly

The formation of ribosomes involves the assembly of the ribosomal precursor RNA with both ribosomal proteins and 5S rRNA (Figure 9.31). The genes that encode ribosomal proteins are transcribed outside the nucleolus by RNA polymerase II yielding mRNAs that are translated on cytoplasmic ribosomes. The ribosomal proteins are then transported from the cytoplasm to the nucleolus where they are assembled with rRNAs to form preribosomal particles. Although the genes for 5S rRNA are also transcribed outside the nucleolus—in this case by RNA polymerase III—5S rRNAs similarly are assembled into preribosomal particles within the nucleolus.

The association of ribosomal proteins with rRNA begins while the pre-rRNA is still being synthesized, and more than half of the ribosomal proteins are complexed with the pre-rRNA before its cleavage. The remaining ribosomal proteins and the 5S rRNA are incorporated into preribosomal particles as cleavage of the pre-rRNA proceeds. Early in ribosome assembly, the processing of the two nascent ribosomal subunits separates. Processing of the smaller subunit, which contains only the 18S rRNA, is simpler and involves only four endonuclease cleavages. In higher eukaryotes this is completed within the nucleus but in yeast the final cleavage to the mature 18S rRNA actually occurs after export of the 40S subunit to the cytosol. Processing of the larger subunit, which contains the 28S, 5.8S and 5S rRNAs, involves extensive nuclease cleavages and is completed within the nucleolus. Consequently, most of the preribosomal particles in the nucleolus represent precursors to the large (60S) subunit. The final stages of ribosome maturation follow the export of preribosomal particles to the cytoplasm, forming the active 40S and 60S subunits of eukaryotic ribosomes.



COMPANION WEBSITE

Visit the website that accompanies **The Cell** (www.sinauer.com/cooper) for animations, videos, quizzes, problems, and other review material.

FIGURE 9.31 Ribosome assembly Ribosomal proteins are imported to the nucleolus from the cytoplasm and begin to assemble on pre-rRNA prior to its cleavage. As the pre-rRNA is processed, additional ribosomal proteins and the 5S rRNA (which is synthesized elsewhere in the nucleus) assemble to form preribosomal particles. The final steps of maturation follow the export of preribosomal particles to the cytoplasm, yielding the 40S and 60S ribosomal subunits.

KEY TERMS

nuclear envelope, nuclear membrane, nuclear lamina, lamin

nuclear pore complex

SUMMARY

THE NUCLEAR ENVELOPE AND TRAFFIC BETWEEN THE NUCLEUS AND THE CYTOPLASM

Structure of the Nuclear Envelope: The nuclear envelope separates the contents of the nucleus from the cytoplasm, maintaining the nucleus as a distinct biochemical compartment that houses the genetic material and serves as the site of transcription and RNA processing in eukaryotic cells. The nuclear envelope consists of the inner and outer nuclear membranes (which are joined at nuclear pore complexes) and an underlying nuclear lamina.

The Nuclear Pore Complex: Nuclear pore complexes are large structures that provide the only routes through which molecules can travel between the nucleus and the cytoplasm. Small molecules can diffuse freely through open channels in the nuclear pore complex. Macromolecules are selectively transported in an energy-dependent process.

SUMMARY

Selective Transport of Proteins to and from the Nucleus: Proteins destined for import to the nucleus contain nuclear localization signals that are recognized by receptors that direct transport through the nuclear pore complex. Proteins that shuttle back and forth between the nucleus and the cytoplasm contain nuclear export signals that target them for transport from the nucleus to the cytoplasm. In most cases, the small GTP-binding protein Ran is required for translocation through the nuclear pore complex and determines the directionality of transport.

Regulation of Nuclear Protein Import: The activity of some proteins, such as transcription factors, is controlled by regulation of both their import to, and export from, the nucleus.

Transport of RNAs: RNAs are transported through the nuclear pore complex as ribonucleoprotein complexes. Messenger RNAs, ribosomal RNAs, and transfer RNAs are exported from the nucleus to function in protein synthesis. Several classes of small nuclear RNAs are initially transported from the nucleus to the cytoplasm where they associate with proteins to form RNPs; they then return to the nucleus.

KEY TERMS

nuclear localization signal, nuclear transport receptor, karyopherin, importin, exportin, Ran, nuclear export signal

INTERNAL ORGANIZATION OF THE NUCLEUS

Chromosomes and Higher-Order Chromatin Structure: The interphase nucleus contains transcriptionally inactive, highly condensed heterochromatin as well as decondensed euchromatin. Interphase chromosomes are organized within the nucleus and divided into large looped domains that function as independent units.

heterochromatin, euchromatin

Sub-Compartments within the Nucleus: Some nuclear processes, such as DNA replication and mRNA metabolism, may be localized to discrete subnuclear structures.

THE NUCLEOLUS AND rRNA PROCESSING

Ribosomal RNA Genes and the Organization of the Nucleolus: The nucleolus is associated with the genes for ribosomal RNAs. It is the site of rRNA transcription, rRNA processing, ribosome assembly, and the modification of several small RNAs.

nucleolus, nucleolar organizing region

Transcription and Processing of rRNA: The primary transcript of the rRNA genes is 45S pre-rRNA, which is processed to yield 18S, 5.8S, and 28S rRNAs. Processing of pre-rRNA and other small RNAs is mediated by small nucleolar RNAs (snoRNAs).

small nucleolar RNA (snoRNA)

Ribosome Assembly: Ribosomal subunits are assembled within the nucleolus from rRNAs and ribosomal proteins.

Questions

1. By separating transcription from translation, the nuclear envelope allows eukaryotes to regulate gene expression by processes that are not found in prokaryotes. What are these regulatory mechanisms that are unique to eukaryotes?

2. What roles do lamins play in nuclear structure and function?

3. You inject a frog egg with two globular proteins, one 15 kd and the other 100 kd—both of which lack nuclear localization signals. Will either protein enter the nucleus?

4. What determines the directionality of nuclear import?

5. Describe how the activity of a transcription factor can be regulated by nuclear import.

6. You are studying a transcription factor that is regulated by phosphorylation of serine residues that inactivate its nuclear localization signal. How would mutating these serines to alanines affect subcellular localization of the transcription factor and expression of its target gene?

7. How would mutational inactivation of the nuclear export signal of a protein that normally shuttles back and forth between the nucleus and cytoplasm affect its subcellular distribution?

8. DNA replication appears to occur in specific sites or replication factories. How would you locate these sites in mammalian cells in culture?

9. How did Kalderon and colleagues demonstrate that the T antigen amino acid sequence 126 to 132 was sufficient for nuclear accumulation of the protein?

10. What is the significance of nuclear speckles?

11. What is the function of snoRNAs?

12. How would RNAi against human exportin-t affect human fibroblasts in culture?

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CHAPTER 10

Protein Sorting and Transport

The Endoplasmic Reticulum, Golgi Apparatus, and Lysosomes

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IN ADDITION TO THE PRESENCE OF A NUCLEUS, eukaryotic cells are distinguished from prokaryotic cells by the presence of membrane-enclosed organelles within their cytoplasm. These organelles provide discrete compartments in which specific cellular activities take place, and the resulting subdivision of the cytoplasm allows eukaryotic cells to function efficiently in spite of their large size—at least a thousand times the volume of bacteria.

Because of the complex internal organization of eukaryotic cells, the sorting and targeting of proteins to their appropriate destinations are considerable tasks. The first step of protein sorting takes place while translation is still in progress. Many proteins destined for the endoplasmic reticulum, the Golgi apparatus, lysosomes, the plasma membrane, and secretion from the cell are synthesized on ribosomes that are bound to the membrane of the endoplasmic reticulum. As translation proceeds, the polypeptide chains are transported into the endoplasmic reticulum where protein folding and processing take place. From the endoplasmic reticulum, proteins are transported in vesicles to the Golgi apparatus where they are further processed and sorted for transport to endosomes, lysosomes, the plasma membrane, or secretion from the cell. Some of these organelles also participate in the sorting and transport of proteins being taken up from outside the cell (see Chapter 13). The endoplasmic reticulum, Golgi apparatus, endosomes, and lysosomes are thus distinguished from other cytoplasmic organelles by their common involvement in protein processing and connection by vesicular transport.

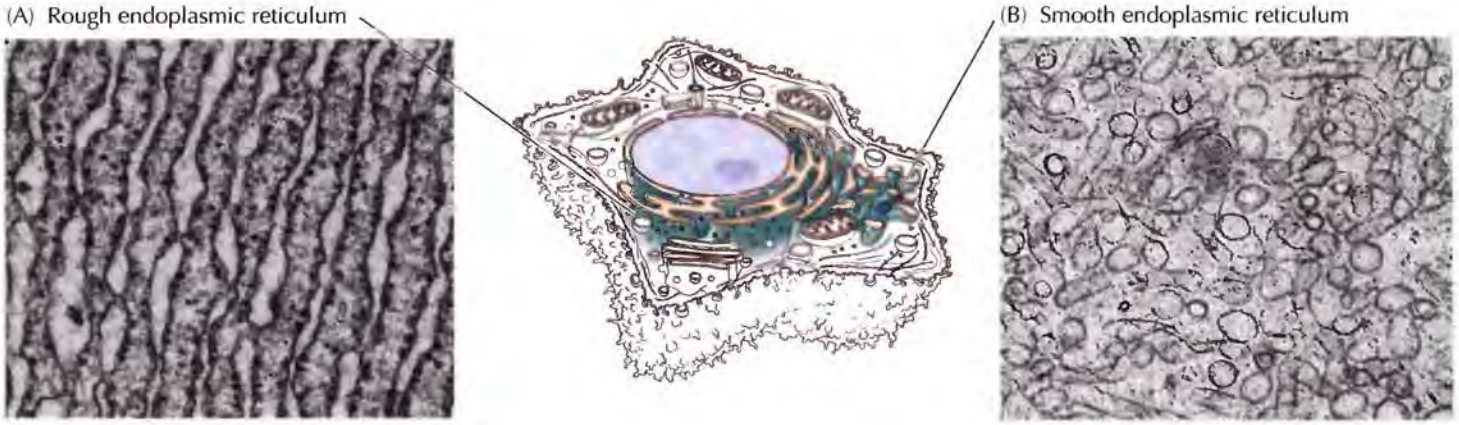


FIGURE 10.1 The endoplasmic reticulum (ER) (A) Electron micrograph of rough ER in rat liver cells. Ribosomes are attached to the cytosolic face of the ER membrane. (B) Electron micrograph of smooth ER in Leydig cells of the testis, which are active in steroid hormone synthesis. (A, Richard Rodewald, University of Virginia/Biological Photo Service; B, Don Fawcett/Photo Researchers, Inc.)

The Endoplasmic Reticulum

The **endoplasmic reticulum (ER)** is a network of membrane enclosed tubules and sacs (cisternae) that extends from the nuclear membrane throughout the cytoplasm (Figure 10.1). The entire endoplasmic reticulum is enclosed by a continuous membrane and is the largest organelle of most eukaryotic cells. Its membrane may account for about half of all cell membranes, and the space enclosed by the ER (the lumen, or cisternal space) may represent about 10% of the total cell volume. As discussed below, there are three contiguous membrane domains within the ER that perform different functions within the cell. The **rough ER**, which is covered by ribosomes on its outer surface, and the **transitional ER,** where vesicles exit to the Golgi apparatus, both function in protein processing. The **smooth ER** is not associated with ribosomes and is involved in lipid, rather than protein, metabolism.

The Endoplasmic Reticulum and Protein Secretion

The role of the endoplasmic reticulum in protein processing and sorting was first demonstrated by George Palade and his colleagues in the 1960s (Figure 10.2). These investigators studied the fate of newly synthesized proteins in specialized cells of the pancreas (pancreatic acinar cells) that secrete digestive enzymes into the small intestine. Because most proteins synthesized by these cells are secreted, Palade and coworkers were able to study the pathway taken by secreted proteins simply by labeling newly synthesized proteins with radioactive amino acids. The location of the radiolabeled proteins within the cell was then determined by autoradiography, revealing the cellular sites involved in the events leading to protein secretion. After a brief exposure of pancreatic acinar cells to radioactive amino acids, newly synthesized proteins were detected in the rough ER, which was therefore identified as the site of synthesis of proteins destined for secretion. If the cells were then incubated for a short time in media containing nonradioactive amino acids (a process known as a "chase"), the radiolabeled proteins were detected in the Golgi apparatus. Following longer chase periods, the radiolabeled proteins traveled from the Golgi apparatus to the cell surface in **secretory vesicles**, which then fused with the plasma membrane to release their contents outside of the cell.

These experiments defined a pathway taken by secreted proteins—the **secretory pathway**: rough ER → Golgi → secretory vesicles → cell exterior. Further studies extended these results and demonstrated that this pathway

■ **The endoplasmic reticulum also plays a key role in signal transduction by acting as a major repository of intracellular calcium. The release of calcium from the ER in response to appropriate signals alters the activity of key cytosolic proteins and plays a very important role in muscle contraction (see Chapters 12 and 15).**

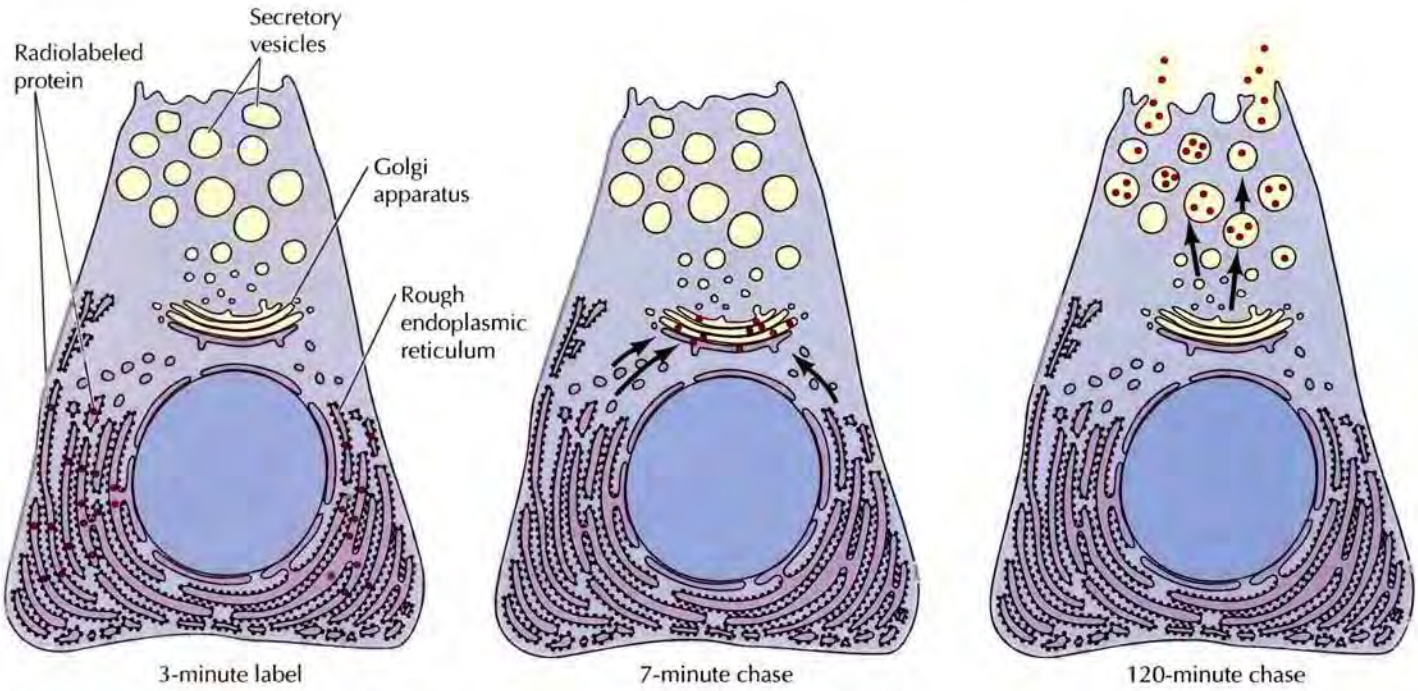


FIGURE 10.2 The secretory pathway Pancreatic acinar cells, which secrete most of their newly synthesized proteins into the digestive tract, were labeled with radioactive amino acids to study the intracellular pathway taken by secreted proteins. After a short incubation with radioactive amino acids (3-minute label), autoradiography revealed that newly synthesized proteins were localized to the rough ER. Following further incubation with nonradioactive amino acids (a "chase"), proteins were found to move from the ER to the Golgi apparatus and then, within secretory vesicles, to the plasma membrane and cell exterior.

is not restricted to proteins destined for secretion from the cell. Portions of it are shared by proteins destined for other compartments. Plasma membrane and lysosomal proteins also travel from the rough ER to the Golgi and then to their final destinations. Still other proteins travel through the initial steps of the secretory pathway but are then retained and function within either the ER or the Golgi apparatus.

The entrance of proteins into the ER thus represents a major branch point for the traffic of proteins within eukaryotic cells. Proteins destined for secretion or incorporation into the ER, Golgi apparatus, lysosomes, or plasma membrane are initially targeted to the ER. In mammalian cells most proteins are transferred into the ER while they are being translated on membrane bound ribosomes (Figure 10.3). In contrast, proteins destined to remain in the cytosol or to be incorporated into the nucleus, mitochondria, chloroplasts, or peroxisomes are synthesized on free ribosomes and released into the cytosol when their translation is complete.

Targeting Proteins to the Endoplasmic Reticulum

Proteins can be translocated into the ER either during their synthesis on membrane-bound ribosomes (cotranslational translocation) or after their translation has been completed on free ribosomes in the cytosol (posttranslational translocation). In mammalian cells, most proteins enter the ER cotranslationally, whereas both cotranslational and posttranslational pathways are used in yeast. The first step in the cotranslational pathway is the association of ribosomes with the ER. Ribosomes are targeted for binding to

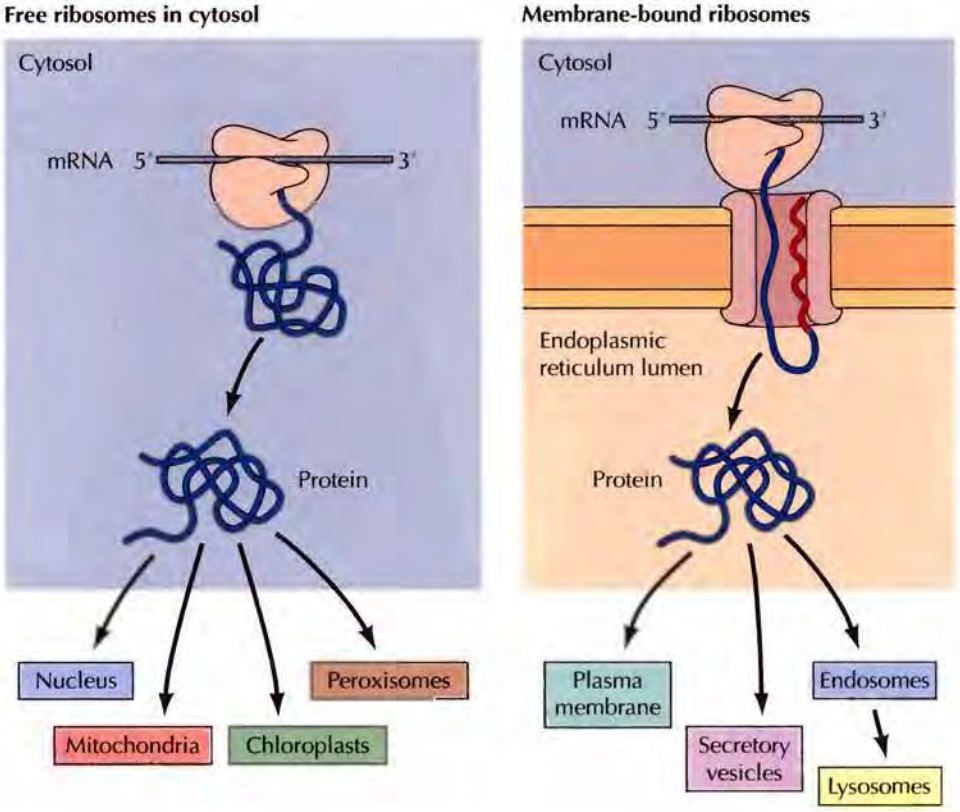


FIGURE 10.3 Overview of protein sorting In higher eukaryotic cells, the initial sorting of proteins to the ER takes place while translation is in progress. Proteins synthesized on free ribosomes either remain in the cytosol or are transported to the nucleus, mitochondria, chloroplasts, or peroxisomes. In contrast, proteins synthesized on membrane-bound ribosomes are translocated into the ER while their translation is in progress. They may be either retained within the ER or transported to the Golgi apparatus and, from there, to lysosomes, the plasma membrane, or the cell exterior via secretory vesicles.

the ER membrane by the amino-acid sequence of the polypeptide chain being synthesized, rather than by intrinsic properties of the ribosome itself. Free and membrane bound ribosomes are functionally indistinguishable, and all protein synthesis initiates on ribosomes that are free in the cytosol. Ribosomes engaged in the synthesis of proteins that are destined for secretion are then targeted to the endoplasmic reticulum by a **signal sequence** at the amino terminus of the growing polypeptide chain. These signal sequences are short stretches of hydrophobic amino acids that are usually cleaved from the polypeptide chain during its transfer into the ER lumen.

The general role of signal sequences in targeting proteins to their appropriate locations within the cell was first elucidated by studies of the import of secretory proteins into the ER. These experiments used *in vitro* preparations of rough ER, which were isolated from cell extracts by density-gradient centrifugation (Figure 10.4). When cells are disrupted and the nuclei cen-

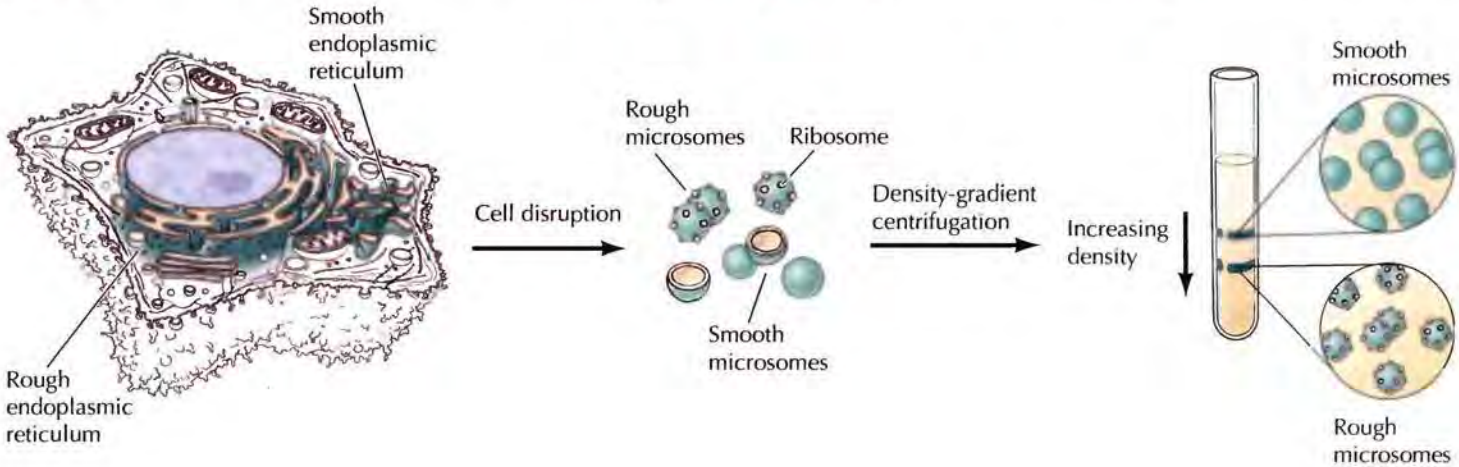


FIGURE 10.4 Isolation of rough ER When cells are disrupted, the ER fragments into small vesicles called microsomes. The microsomes derived from the rough ER (rough microsomes) are lined with ribosomes on their outer surface. Because ribosomes contain large amounts of RNA, the rough microsomes are denser than smooth microsomes and can be isolated by equilibrium density-gradient centrifugation.

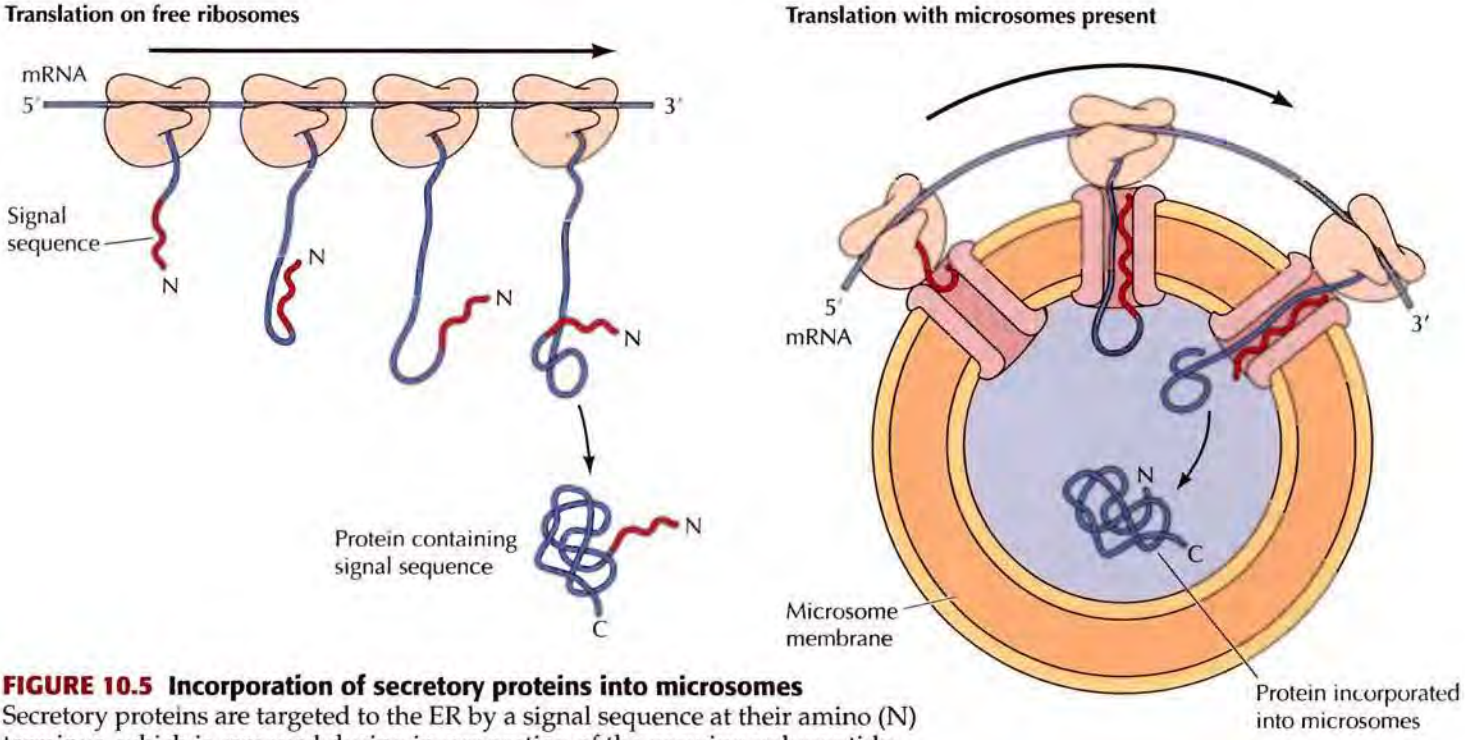


FIGURE 10.5 Incorporation of secretory proteins into microsomes
Secretory proteins are targeted to the ER by a signal sequence at their amino (N) terminus, which is removed during incorporation of the growing polypeptide chain into the ER. This was demonstrated by experiments showing that translation of secretory protein mRNAs on free ribosomes yielded proteins that retained their signal sequences and were therefore slightly larger than the normal secreted proteins. However, when microsomes were added to the system, the growing polypeptide chains were incorporated into the microsomes and the signal sequences were removed by proteolytic cleavage.

trifuged out, the ER breaks up into small vesicles called **microsomes**. Because the vesicles derived from the rough ER are covered with ribosomes, they can be separated from similar vesicles derived from the smooth ER or from other membranes (e.g., the plasma membrane). In particular, the large amount of RNA within ribosomes increases the density of the membrane vesicles to which they are attached, allowing purification of vesicles derived from the rough ER (rough microsomes) by equilibrium centrifugation in density gradients.

David Sabatini and Günter Blobel first proposed in 1971 that the signal for ribosome attachment to the ER might be an amino acid sequence near the amino terminus of the growing polypeptide chain. This hypothesis was supported by the results of *in vitro* translation of mRNAs encoding secreted proteins, such as immunoglobulins (Figure 10.5). If an mRNA encoding a secreted protein was translated on free ribosomes *in vitro*, it was found that the protein produced was slightly larger than the normal secreted protein. If microsomes were added to the system, however, the *in vitro* translated protein was incorporated into the microsomes and cleaved to the correct size. These experiments led to a more detailed formulation of the signal hypothesis, which proposed that an amino terminal leader sequence targets the polypeptide chain to the microsomes and is then cleaved by a microsomal protease. Many subsequent findings have substantiated this model, including recombinant DNA experiments demonstrating that addition of a signal sequence to a normally nonsecreted protein is sufficient to direct the incorporation of the recombinant protein into the rough ER.

KEY EXPERIMENT

The Signal Hypothesis

Transfer of Proteins across Membranes. I. Presence of Proteolytically Processed and Unprocessed Nascent Immunoglobulin Light Chains on Membrane-Bound Ribosomes of Murine Myeloma

Günter Blobel and Bernhard Dobberstein
Rockefeller University, New York
Journal of Cell Biology, 1975, Volume 67, pages 835–851

Günter Blobel

The Context

How are specific polypeptide chains transferred across the appropriate membranes? Studies in the 1950s and 1960s indicated that secreted proteins are synthesized on membrane-bound ribosomes and transferred across the membrane during their synthesis. However, this did not explain why ribosomes—engaged in the synthesis of secreted proteins—attach to membranes, while ribosomes synthesizing cytosolic proteins do not. A hypothesis to explain this difference was first suggested by Günter Blobel and David Sabatini in 1971. At that time, they proposed that (1) mRNAs to be translated on membrane-bound ribosomes contain a unique set of codons just 3′ of the initiation site, (2) translation of these codons yields a unique sequence at the amino terminus of the growing polypeptide chain (the signal sequence), and (3) the signal sequence triggers attachment of the ribosome to the membrane. In 1975 Blobel and Dobberstein reported a series of experiments that provided critical support for this notion. In addition, they proposed "a somewhat more detailed version of this hypothesis, henceforth referred to as the signal hypothesis."

The Experiments

Myelomas are cancers of B lymphocytes that actively secrete immunoglobulins, so they provide a good model for studies of secreted proteins. Previous studies in Cesar Milstein's laboratory had shown that the proteins produced by *in vitro* translation of immunoglobulin light-chain mRNA contain about 20 amino acids at their amino terminus that are not present in the secreted light chains. This result led to the suggestion that these amino acids direct binding of the ribosome to the membrane. To test this idea, Blobel and Dobberstein investigated the synthesis of light chains by membrane-bound ribosomes from myeloma cells.

As expected from earlier work, *in vitro* translation of light-chain mRNA on free ribosomes yielded a protein that was larger than the secreted light chain (see figure). In contrast, *in vitro* translation of mRNA associated with membrane-bound ribosomes from myeloma cells yielded a protein that was the same size as the normally secreted light chain. Moreover, the light chains synthesized by ribosomes that remained bound to microsomes were resistant to digestion by added proteases, indicating that the light chains had been transferred into the microsomes.

These results indicated that an amino-terminal signal sequence is removed by a microsomal protease as growing polypeptide chains are transferred across the membrane. The results were interpreted in terms of a more detailed version of the signal

The mechanism by which secretory proteins are targeted to the ER during their translation (the cotranslational pathway) is now well understood. The signal sequences span about 20 amino acids, including a stretch of hydrophobic residues usually located at the amino terminus of the polypeptide chain (Figure 10.6). As they emerge from the ribosome, signal

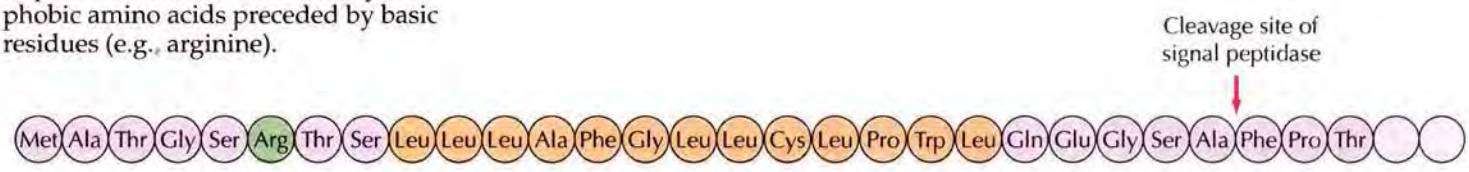
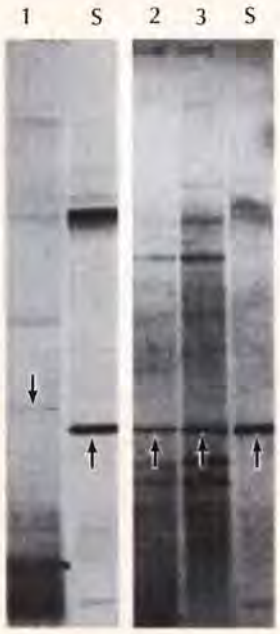


FIGURE 10.6 The signal sequence of growth hormone Most signal sequences contain a stretch of hydrophobic amino acids preceded by basic residues (e.g., arginine).

KEY EXPERIMENT

hypothesis. As stated by Blobel and Dobberstein, "the essential feature of the signal hypothesis is the occurrence of a unique sequence of codons, located immediately to the right of the initiation codon, which is present only in those mRNAs whose translation products are to be transferred across a membrane."

In vitro translation of immunoglobulin light-chain mRNA on free ribosomes (lane 1) yields a product that migrates slower than secreted light chains (lane S) in gel electrophoresis. In contrast, light chains synthesized by *in vitro* translation on membrane-bound ribosomes (lane 2) are the same size as secreted light chains. In addition, the products of *in vitro* translation on membrane-bound ribosomes were unaffected by subsequent digestion with proteases (lane 3), indicating that they were protected from the proteases by insertion into microsomes.



The Impact

The selective transfer of proteins across membranes is critical to the maintenance of the membrane-enclosed organelles of eukaryotic cells. To maintain the identity of these organelles, proteins must be translocated specifically across the appropriate membranes. The signal hypothesis provided the conceptual basis for understanding this phenomenon. Not only has this basic model been firmly substantiated for the transfer of secreted proteins into the endoplasmic reticulum, but it also has provided the framework for understanding the targeting of proteins to all the other membrane-enclosed compartments of the cell, thereby impacting virtually all areas of cell biology.

sequences are recognized and bound by a **signal recognition particle (SRP)** consisting of six polypeptides and a small cytoplasmic RNA (**srpRNA**) (Figure 10.7). The SRP binds the ribosome as well as the signal sequence, inhibiting further translation and targeting the entire complex (the SRP, ribosome, and growing polypeptide chain) to the rough ER by binding to the **SRP receptor** on the ER membrane (Figure 10.8). Recent structural studies have shown that both the SRP proteins and the srpRNA are involved in interac-

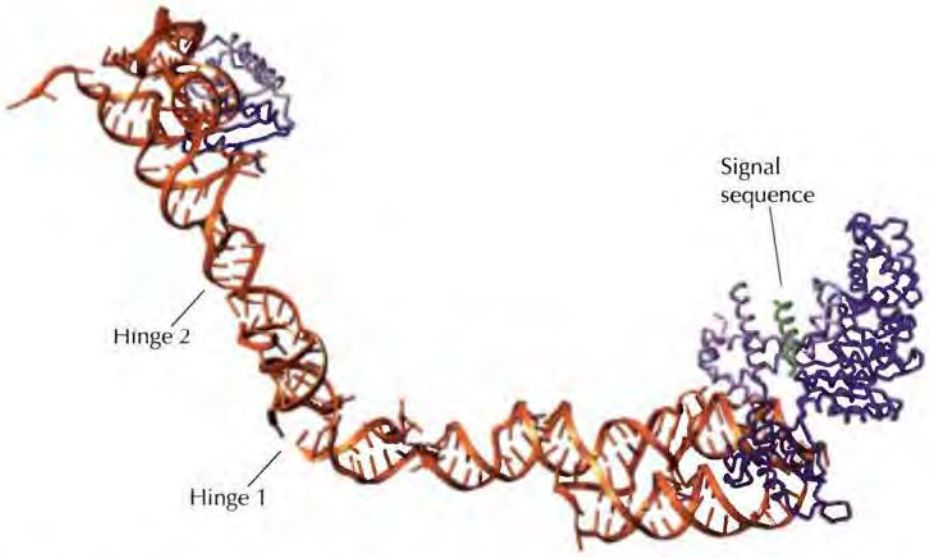


FIGURE 10.7 Structure of the SRP The SRP proteins (blue) are associated with the SRP RNA (orange), which has two flexible hinge regions. The signal sequence on the nascent protein (green) binds to a pocket in the SRP proteins. (After Halic et al., 2004. *Nature* 427: 808.)

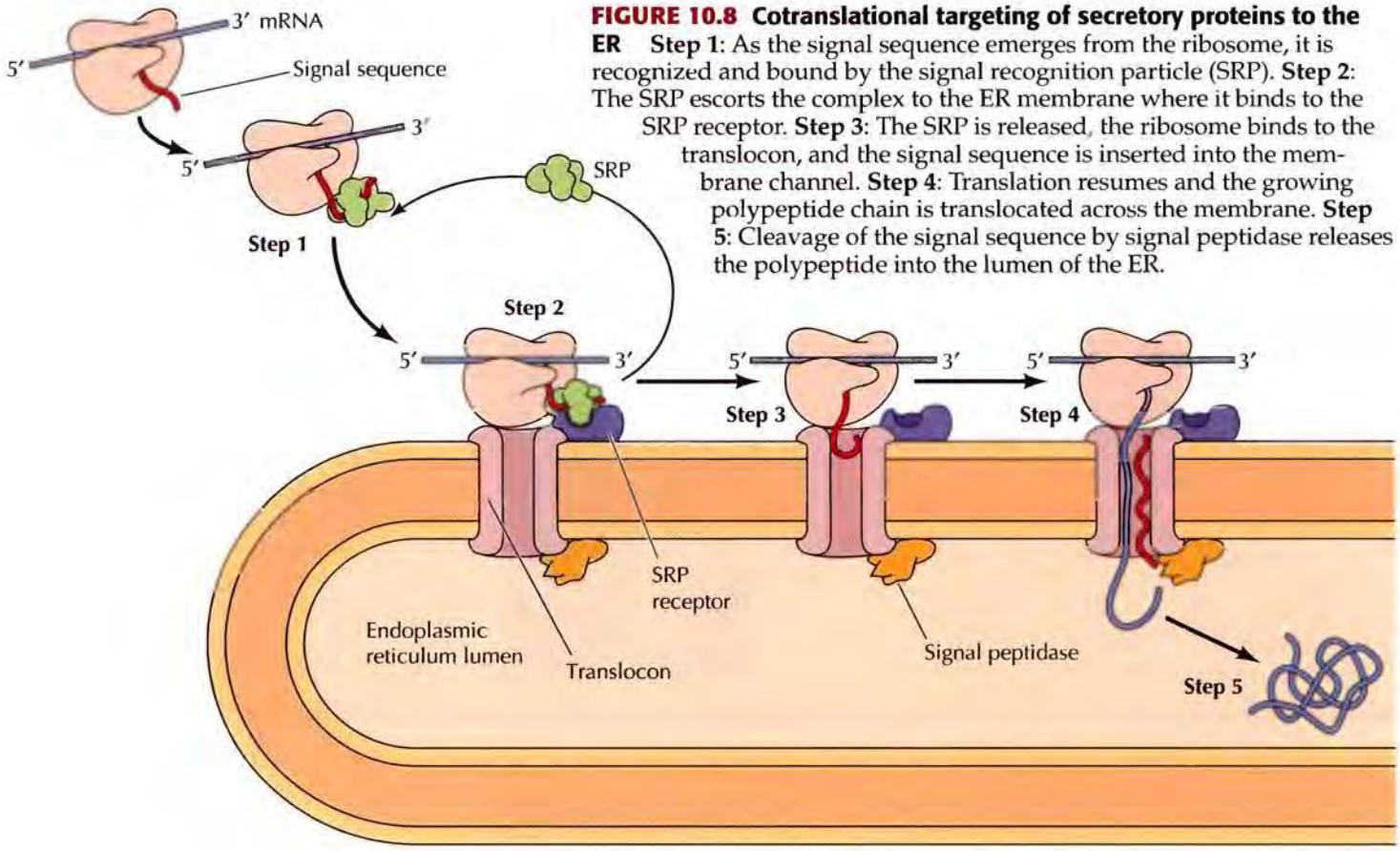


FIGURE 10.8 Cotranslational targeting of secretory proteins to the ER **Step 1**: As the signal sequence emerges from the ribosome, it is recognized and bound by the signal recognition particle (SRP). **Step 2**: The SRP escorts the complex to the ER membrane where it binds to the SRP receptor. **Step 3**: The SRP is released, the ribosome binds to the translocon, and the signal sequence is inserted into the membrane channel. **Step 4**: Translation resumes and the growing polypeptide chain is translocated across the membrane. **Step 5**: Cleavage of the signal sequence by signal peptidase releases the polypeptide into the lumen of the ER.

tion with the ribosome, with srpRNA binding to both proteins and rRNA of the large ribosomal subunit. Binding to the receptor releases the SRP from both the ribosome and the signal sequence of the growing polypeptide chain. The ribosome then binds to a protein translocation complex in the ER membrane, and the signal sequence is inserted into a membrane channel or **translocon**. This entire process is coordinated by GTP binding to both the SRP and the SRP receptor, with hydrolysis of GTP to GDP leading to dissociation of SRP from both the receptor and the ribosome-mRNA complex.

In both yeast and mammalian cells, the translocons through the ER membrane are complexes of three transmembrane proteins called the Sec61 proteins. The yeast and mammalian translocon proteins are closely related to the plasma membrane proteins that translocate secreted polypeptides in bacteria, demonstrating a striking conservation of the protein secretion machinery in prokaryotic and eukaryotic cells. Transfer of the ribosome-mRNA complex from the SRP to the translocon opens the gate on the translocon and allows translation to resume, and the growing polypeptide chain is transferred directly into the translocon channel and across the ER membrane as translation proceeds. Thus the process of protein synthesis directly drives the transfer of growing polypeptide chains through the translocon and into the ER. As translocation proceeds, the signal sequence is cleaved by **signal peptidase** and the polypeptide is released into the lumen of the ER.

Many proteins in yeast, as well as a few proteins in mammalian cells, are targeted to the ER after their translation is complete (posttranslational translocation) rather than being transferred into the ER during synthesis on

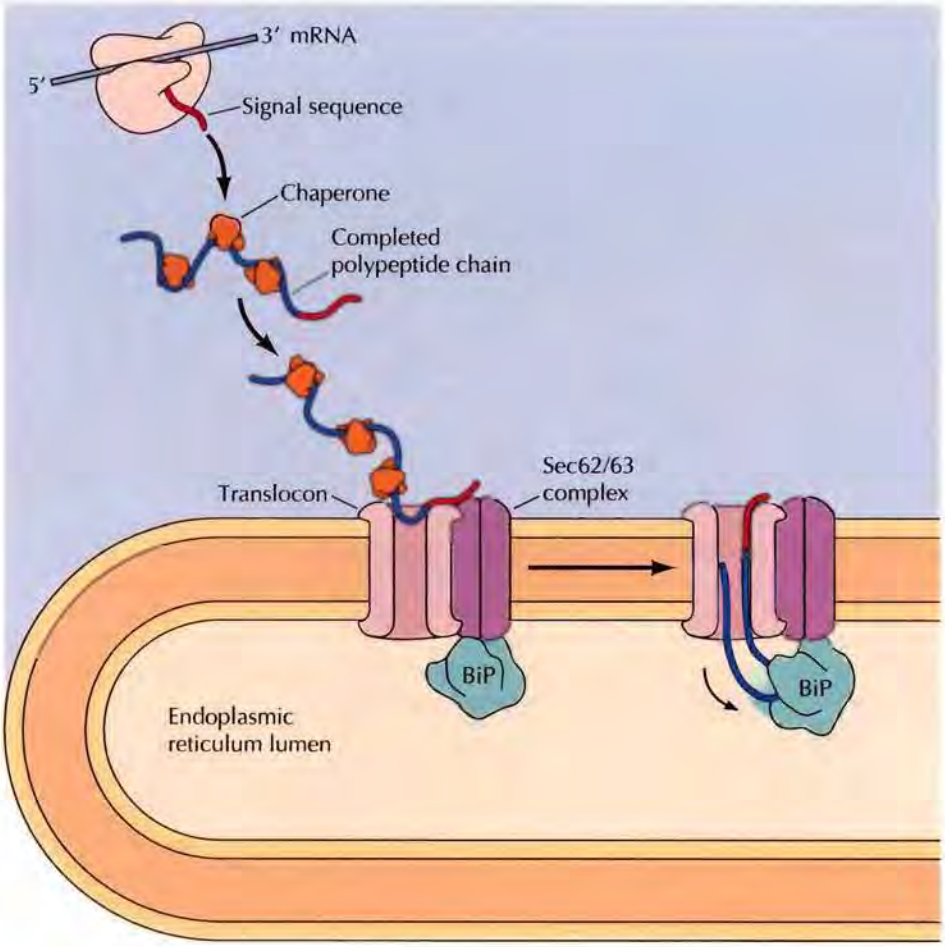


FIGURE 10.9 Posttranslational translocation of proteins into the ER Proteins destined for posttranslational import to the ER are synthesized on free ribosomes and maintained in an unfolded conformation by cytosolic chaperones. Their signal sequences are recognized by the Sec62/63 complex, which is associated with the translocon in the ER membrane. The Sec63 protein is also associated with a chaperone protein (BiP), which acts as a molecular ratchet to drive protein translocation into the ER.

membrane bound ribosomes. These proteins are synthesized on free cytosolic ribosomes, and their posttranslational incorporation into the ER does not require SRP. Instead, their signal sequences are recognized by distinct receptor proteins (the Sec62/63 complex) associated with the translocon in the ER membrane (Figure 10.9). Cytosolic Hsp70 chaperones are required to maintain the polypeptide chains in an unfolded conformation so they can enter the translocon, and another Hsp70 chaperone within the ER (called BiP) is required to pull the polypeptide chain through the channel and into the ER. BiP acts as a ratchet to drive the posttranslational translocation of proteins into the ER, whereas the cotranslational translocation of growing polypeptide chains is driven directly by the process of protein synthesis.

Insertion of Proteins into the ER Membrane

Proteins destined for secretion from the cell or residence within the lumen of the ER, Golgi apparatus, endosomes, or lysosomes are translocated across the ER membrane and released into the lumen of the ER as already described. However, proteins destined for incorporation into the plasma membrane or the membranes of these compartments are initially inserted into the ER membrane instead of being released into the lumen. From the ER membrane, they proceed to their final destination along the same pathway as that of secretory proteins: ER → Golgi → plasma membrane or endosomes → lysosomes. However, these proteins are transported along this pathway as membrane components rather than as soluble proteins.

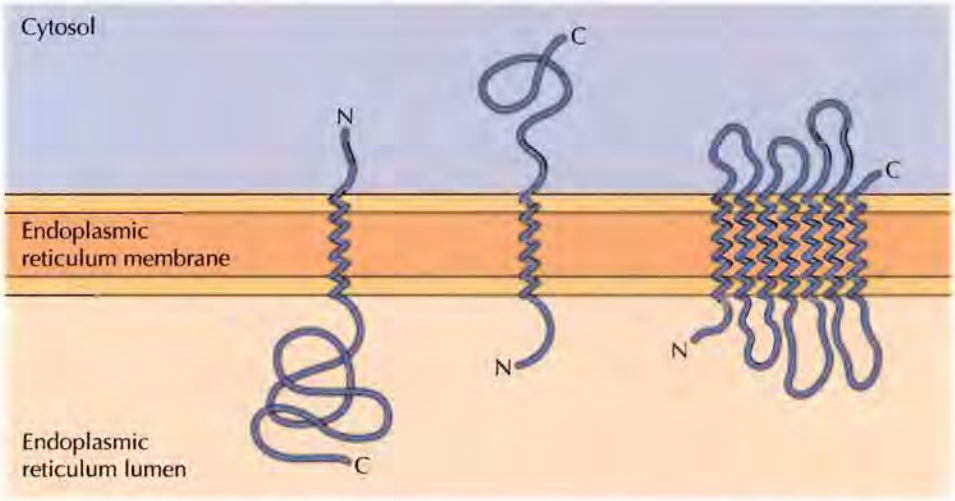


FIGURE 10.10 Orientations of membrane proteins Integral membrane proteins span the membrane via α-helical regions of 20 to 25 hydrophobic amino acids, which can be inserted in a variety of orientations. The two proteins at left and center each span the membrane once, but they differ in whether the amino (N) or carboxy (C) terminus is on the cytosolic side. On the right is an example of a protein that has multiple membrane-spanning regions.

Integral membrane proteins are embedded in the membrane by hydrophobic sequences that span the phospholipid bilayer (see Figure 2.25). The membrane spanning portions of these proteins are usually α helical regions consisting of 20 to 25 hydrophobic amino acids. The formation of an α helix maximizes hydrogen bonding between the peptide bonds, and the hydrophobic amino acid side chains interact with the fatty acid tails of the phospholipids in the bilayer. However, different integral membrane proteins vary in how they are inserted (Figure 10.10). For example, whereas some integral membrane proteins span the membrane only once, others have multiple membrane spanning regions. In addition, some proteins are oriented in the membrane with their amino terminus on the cytosolic side; others have their carboxy terminus exposed to the cytosol. These orientations of proteins inserted into the ER, Golgi, lysosomal, and plasma membranes are generally established as the growing polypeptide chains are translocated into the ER. The lumen of the ER is topologically equivalent to the exterior of the cell, so the domains of plasma membrane proteins that are exposed on the cell surface correspond to the regions of polypeptide chains that are translocated into the ER lumen (Figure 10.11).

The most straightforward mode of insertion into the ER membrane results in the synthesis of transmembrane proteins oriented with their carboxy termini exposed to the cytosol (Figure 10.12). These proteins have a normal amino terminal signal sequence, which is cleaved by signal peptidase during translocation of the polypeptide chain across the ER membrane through the translocon. They are then anchored in the membrane by a second membrane spanning α helix in the middle of the protein. This transmembrane sequence, called a stop-transfer sequence, signals a change in the translocon channel. Further translocation of the polypeptide chain across the ER membrane is thus blocked, so the carboxy terminal portion of

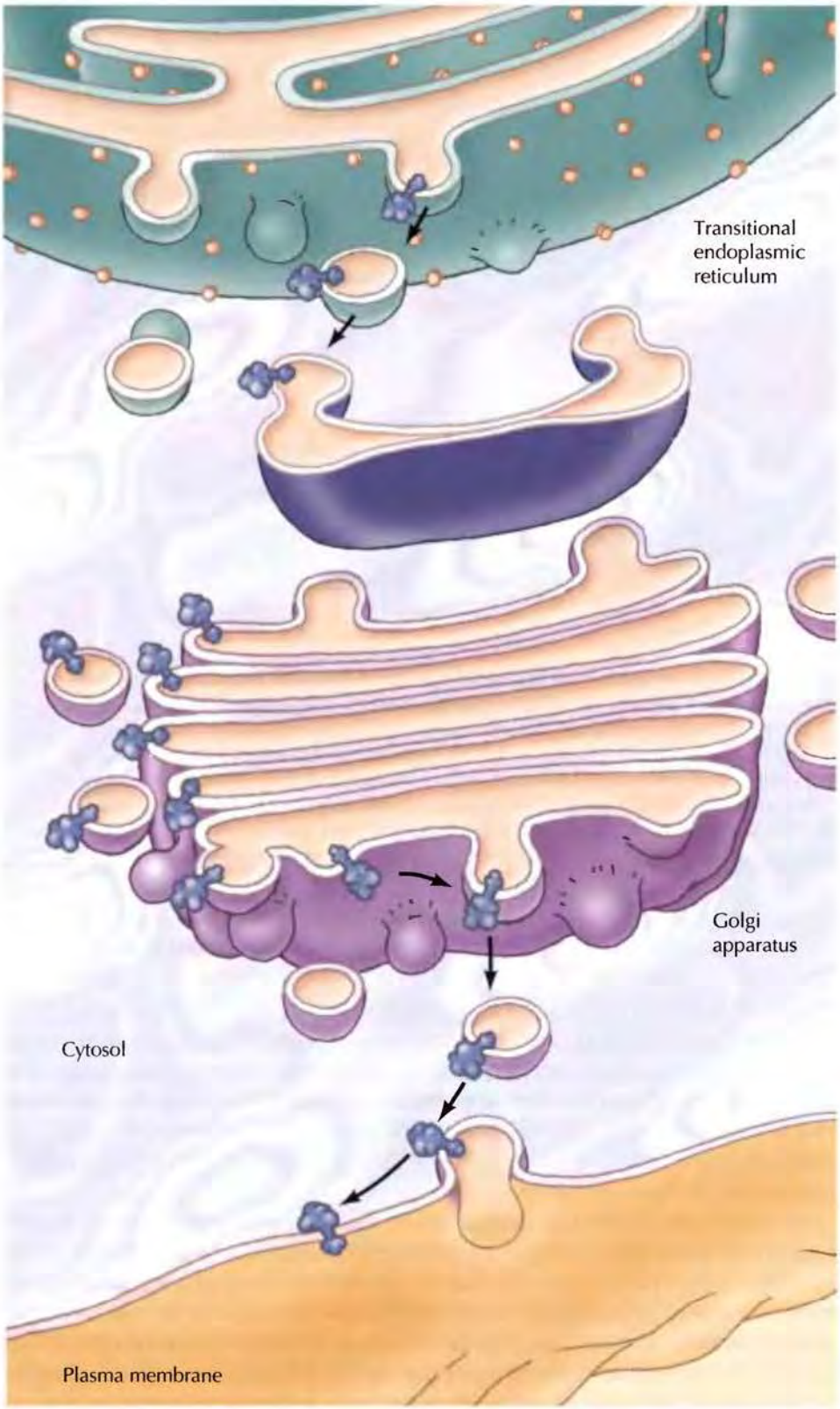


FIGURE 10.11 Topology of the secretory pathway The lumens of the endoplasmic reticulum and Golgi apparatus are topologically equivalent to the exterior of the cell. Consequently, those portions of polypeptide chains that are translocated into the ER are exposed on the cell surface following transport to the plasma membrane.

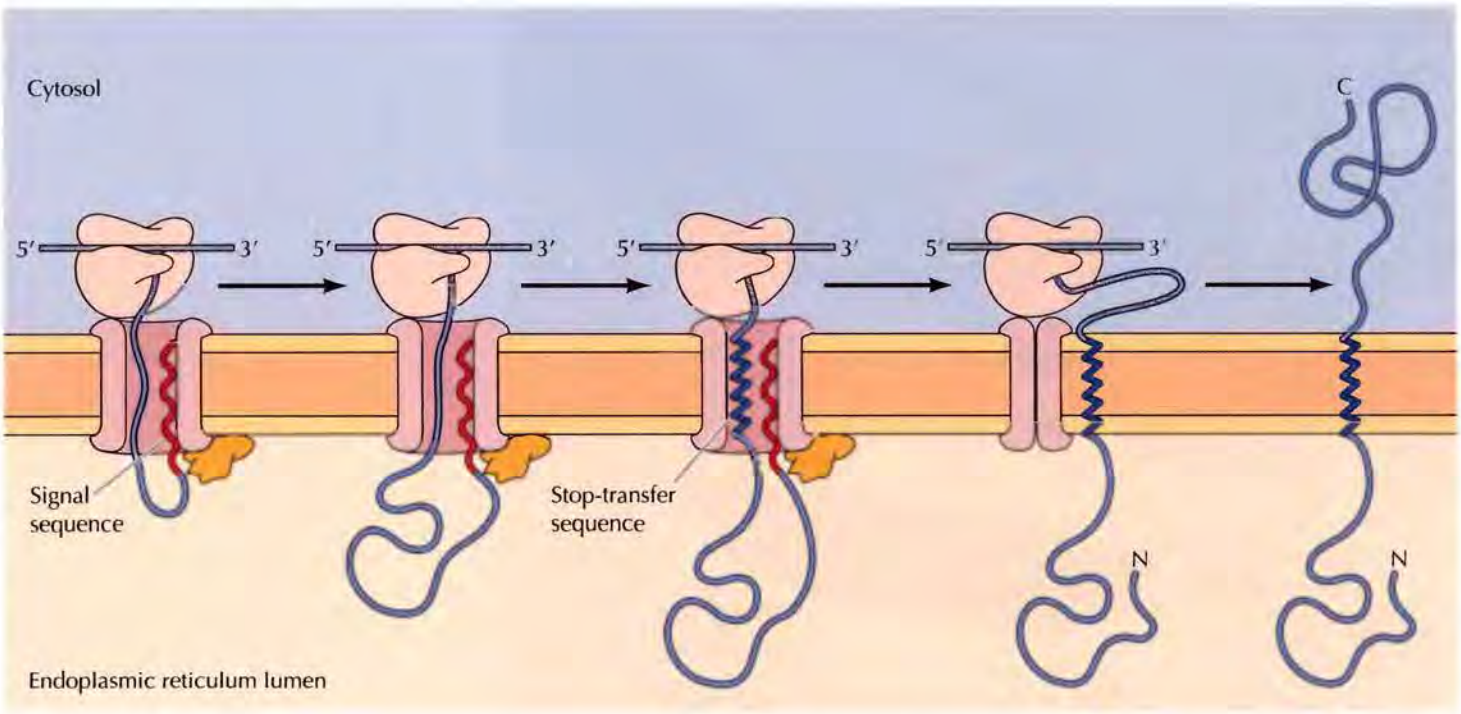


FIGURE 10.12 Insertion of a membrane protein with a cleavable signal sequence and a single stop-transfer sequence The signal sequence is cleaved as the polypeptide chain crosses the membrane, so the amino terminus of the polypeptide chain is exposed in the ER lumen. However, translocation of the polypeptide chain across the membrane is halted by a transmembrane stop-transfer sequence that closes the translocon and exits the channel laterally to anchor the protein in the ER membrane. Continued translation results in a membrane-spanning protein with its carboxy terminus on the cytosolic side.

the growing polypeptide chain remains in the cytosol. As the subunits of the translocon separate, the transmembrane domain of the protein enters the lipid bilayer. The insertion of these transmembrane proteins into the membrane thus involves the sequential action of two distinct elements: a cleavable amino terminal signal sequence that initiates translocation across the membrane, and a transmembrane stop-transfer sequence that anchors the protein in the membrane.

Proteins can also be anchored in the ER membrane by internal signal sequences that are not cleaved by signal peptidase (Figure 10.13). These internal signal sequences are recognized by the SRP and brought to the ER membrane as already discussed. Because they are not cleaved by signal peptidase, however, these signal sequences act as transmembrane α helices that exit the translocon and anchor proteins in the ER membrane. Importantly, internal signal sequences can be oriented so as to direct the translocation of either the amino or carboxy terminus of the polypeptide chain across the membrane. Therefore, depending on the orientation of the signal sequence, proteins inserted into the membrane by this mechanism can have either their amino or carboxy terminus exposed to the cytosol.

Proteins that span the membrane multiple times are thought to be inserted as a result of an alternating series of internal signal sequences and transmembrane stop-transfer sequences. For example, an internal signal

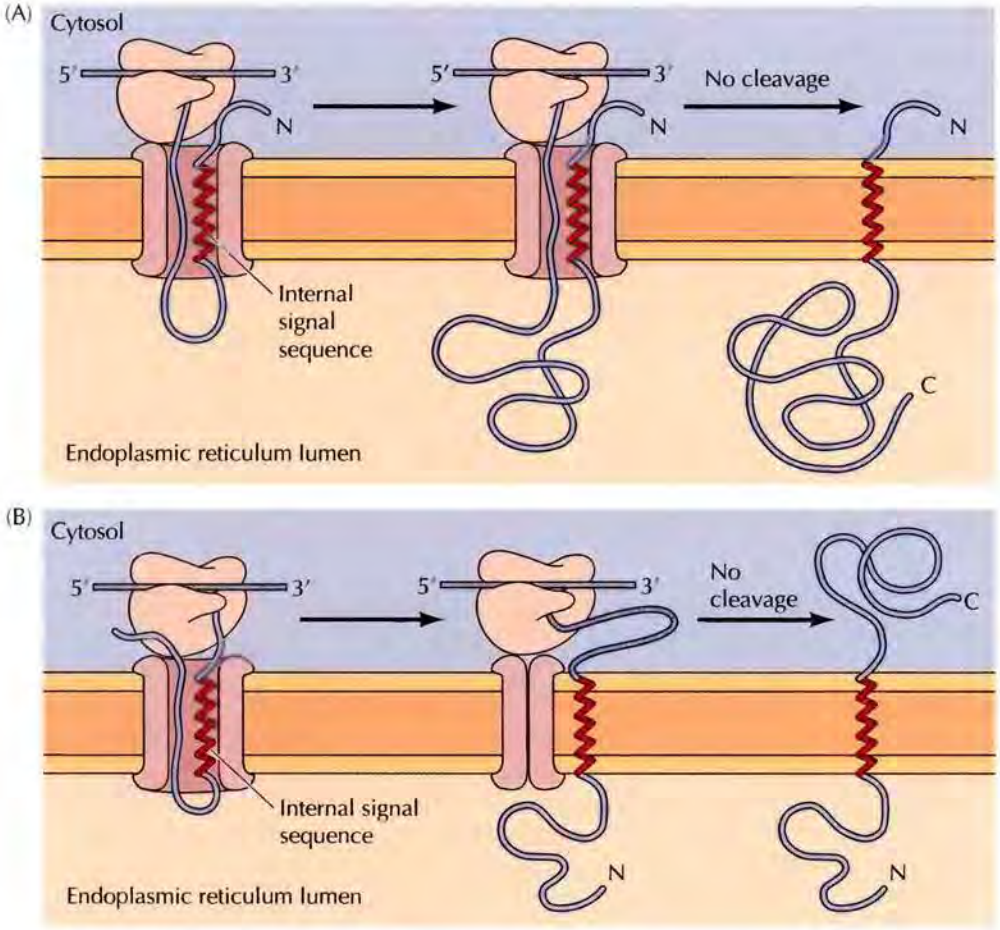


FIGURE 10.13 Insertion of membrane proteins with internal noncleavable signal sequences Internal noncleavable signal sequences can lead to the insertion of polypeptide chains in either orientation in the ER membrane. (A) The signal sequence directs insertion of the polypeptide such that its amino terminus is exposed on the cytosolic side. The remainder of the polypeptide chain is translocated into the ER as translation proceeds. The signal sequence is not cleaved, so it acts as a membrane-spanning sequence that anchors the protein in the membrane with its carboxy terminus in the lumen of the ER. (B) Other internal signal sequences are oriented to direct the transfer of the amino-terminal portion of the polypeptide across the membrane. Continued translation results in a protein that spans the ER membrane with its amino terminus in the lumen and its carboxy terminus in the cytosol. Note that this orientation is the same as that resulting from insertion of a protein that contains a cleavable signal sequence followed by a stop-transfer sequence (see Figure 10.12).

sequence can result in membrane insertion of a polypeptide chain with its amino terminus on the cytosolic side (Figure 10.14). If a stop-transfer sequence is then encountered, the polypeptide will form a loop in the ER lumen, and protein synthesis will continue on the cytosolic side of the membrane. If a second signal sequence is encountered, the growing polypeptide chain will again be inserted into the ER, forming another looped domain on the cytosolic side of the membrane. This can be followed by yet another stop-transfer sequence and so forth, so that an alternating series of signal and stop-transfer sequences can result in the insertion of proteins that span the membrane multiple times, with looped domains exposed on both the lumenal and cytosolic sides.

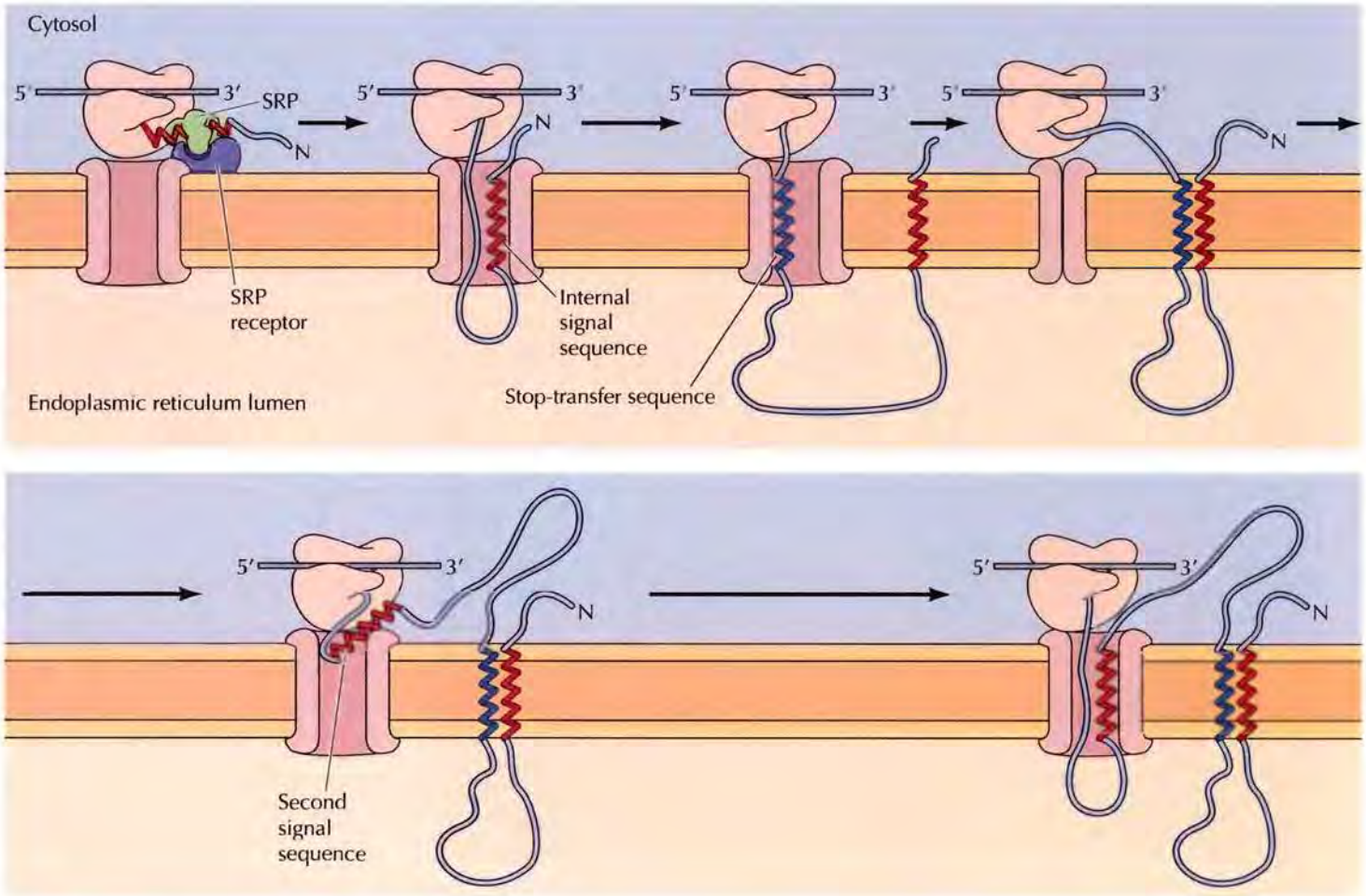


FIGURE 10.14 Insertion of a protein that spans the membrane multiple times In this example, an internal signal sequence results in insertion of the polypeptide chain with its amino terminus on the cytosolic side of the membrane. A stop-transfer sequence then signals closure of the translocation channel causing the polypeptide chain to form a loop within the lumen of the ER; translation continues in the cytosol. A second internal signal sequence reopens the channel, triggering reinsertion of the polypeptide chain into the ER membrane and forming a loop in the cytosol. The process can be repeated many times, resulting in the insertion of proteins with multiple membrane-spanning regions.

As discussed below, most transmembrane proteins destined for other compartments in the secretory pathway are delivered to them in transport vesicles. However, the inner nuclear membrane is continuous with the ER and it is thought that proteins destined for the inner nuclear membrane (such as emerin or LBR; see Chapter 9) diffuse in the plane of the membrane and then are retained in the inner nuclear membrane by interactions with nuclear components, such as lamins or chromatin. In addition, recent studies suggest that inner nuclear membrane proteins contain specific transmembrane sequences that alter their interaction with the translocon and signal their transport to the inner nuclear membrane.

Protein Folding and Processing in the ER

The folding of polypeptide chains into their correct three-dimensional conformations, the assembly of polypeptides into multisubunit proteins, and the covalent modifications involved in protein processing were discussed in Chapter 8. For proteins that enter the secretory pathway, many of these events occur either during translocation across the ER membrane or within the ER lumen. One such processing event is the proteolytic cleavage of the signal peptide as the polypeptide chain is translocated across the ER membrane. The ER is also the site of protein folding, assembly of multisubunit proteins, disulfide bond formation, the initial stages of glycosylation, and the addition of glycolipid anchors to some plasma membrane proteins. In

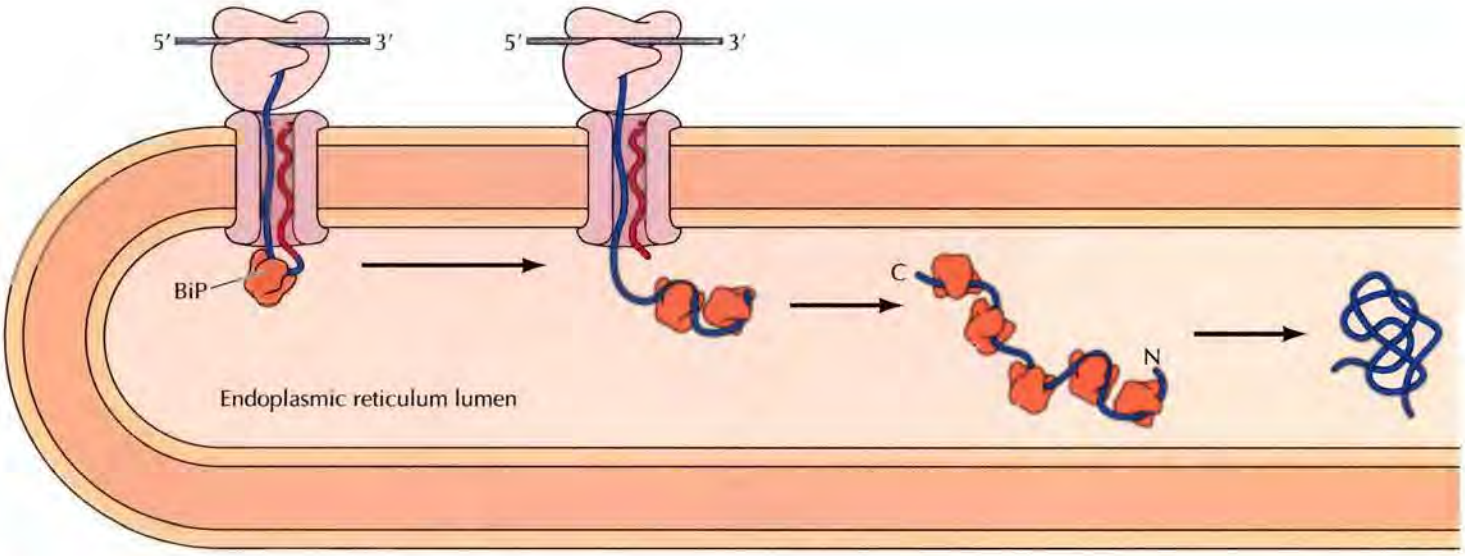


FIGURE 10.15 Protein folding in the ER The molecular chaperone BiP binds to polypeptide chains as they cross the ER membrane and facilitates protein folding and assembly within the ER.

fact, the primary role of lumenal ER proteins is to assist the folding and assembly of newly translocated polypeptides.

As already discussed, proteins are translocated across the ER membrane as unfolded polypeptide chains while their translation is still in progress. These polypeptides, therefore, fold into their three-dimensional conformations within the ER, assisted by molecular chaperones that facilitate the folding of polypeptide chains (see Chapter 8). The Hsp70 chaperone, BiP, is thought to bind to the unfolded polypeptide chain as it crosses the membrane and then mediate protein folding and the assembly of multisubunit proteins within the ER (Figure 10.15). Correctly assembled proteins are released from BiP (and other chaperones) and are available for transport to the Golgi apparatus. Abnormally folded or improperly assembled proteins are targets for degradation, as will be discussed later.

The formation of disulfide bonds between the side chains of cysteine residues is an important aspect of protein folding and assembly within the ER. These bonds do not form in the cytosol, which is characterized by a reducing environment that maintains cysteine residues in their reduced (—SH) state. In the ER, however, an oxidizing environment promotes disulfide (S—S) bond formation, and disulfide bonds formed in the ER play important roles in the structure of secreted and cell surface proteins. Disulfide bond formation is facilitated by the enzyme **protein disulfide isomerase** (see Figure 8.24), which is located in the ER lumen.

Proteins are also glycosylated on specific asparagine residues (*N*-linked glycosylation) within the ER while their translation is still in process (Figure 10.16). As discussed in Chapter 8 (see Figures 8.29 and 8.30), oligosaccharide units consisting of 14 sugar residues are added to acceptor asparagine residues of growing polypeptide chains as they are translocated into the ER. The oligosaccharide is synthesized on a lipid (dolichol) carrier anchored in the ER membrane. It is then transferred as a unit to acceptor asparagine residues in the consensus sequence Asn-X-Ser/Thr by a membrane-bound enzyme called oligosaccharyl transferase. Three glucose residues are

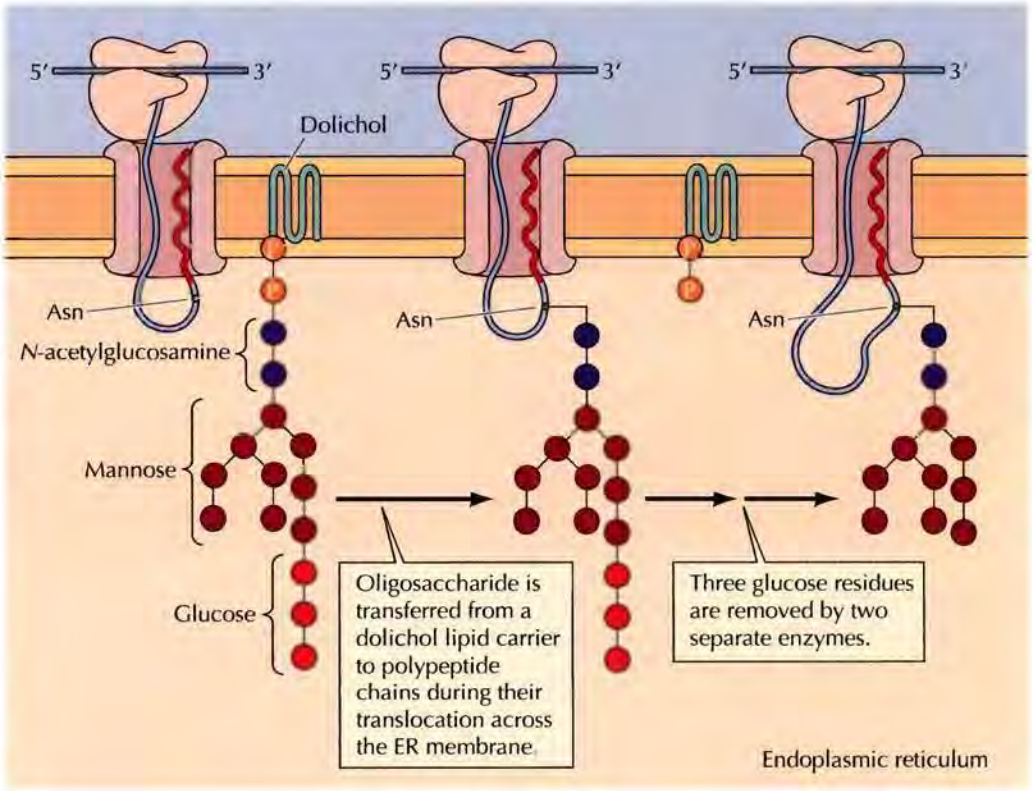


FIGURE 10.16 Protein glycosylation in the ER

removed while the protein is still within the ER, and the protein is modified further after being transported to the Golgi apparatus (discussed later in this chapter).

Some proteins are attached to the plasma membrane by glycolipids rather than by membrane-spanning regions of the polypeptide chain. Because these membrane-anchoring glycolipids contain phosphatidylinositol, they are called **glycosylphosphatidylinositol (GPI) anchors**, the structure of which was illustrated in Figure 8.35. The GPI anchors are assembled in the ER membrane. They are then added immediately after completion of protein synthesis to the carboxy terminus of some proteins that are retained in the membrane by a C-terminal hydrophobic sequence (Figure 10.17). The C-terminal sequence of the protein is cleaved and exchanged for the GPI anchor, so these proteins remain attached to the membrane only by their associated glycolipid. Like transmembrane proteins, they are transported to the cell surface as membrane components via the secretory pathway. Their orientation within the ER dictates that GPI-anchored proteins are exposed on the outside of the cell, with the GPI anchor mediating their attachment to the plasma membrane.

Quality Control in the ER

Many proteins synthesized in the ER are rapidly degraded, primarily because they fail to fold correctly; others reside in the ER for several hours while they are properly folded. Thus an important role of the ER is to identify misfolded proteins, mark them, and divert them to a degradation pathway. Because they assist proteins in correct folding, chaperones and protein processing enzymes in the ER lumen often act as sensors of misfolded pro-

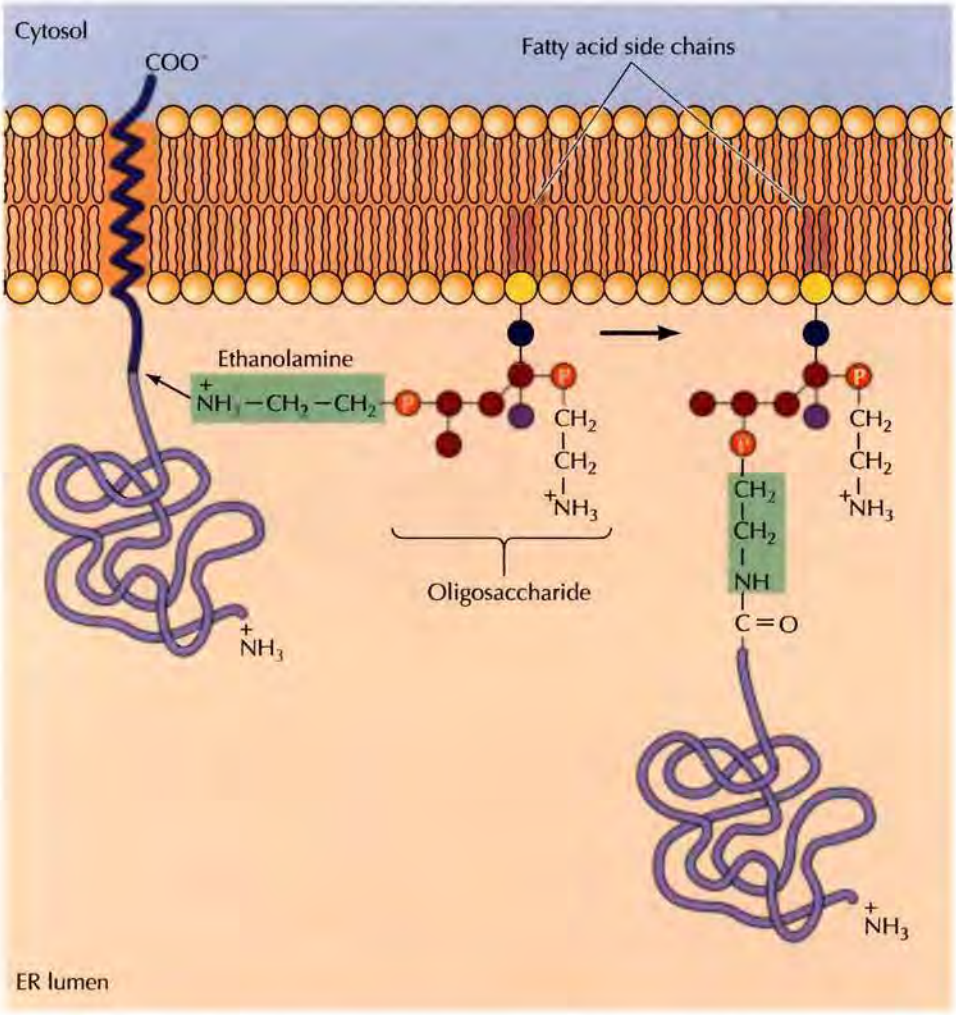


FIGURE 10.17 Addition of GPI anchors Glycosylphosphatidylinositol (GPI) anchors contain two fatty acid chains, an oligosaccharide portion consisting of inositol and other sugars, and ethanolamine (see Figure 8.35 for a more detailed structure). The GPI anchors are assembled in the ER and added to polypeptides anchored in the membrane by a carboxy-terminal membrane-spanning region. The membrane-spanning region is cleaved, and the new carboxy terminus is joined to the NH_2 group of ethanolamine immediately after translation is completed, leaving the protein attached to the membrane by the GPI anchor.

teins. The process of ER protein quality control is complex and involves at least four chaperones, protein disulfide isomerase, and many supporting proteins. While the process is not fully understood, much is known about the roles of the glycoprotein chaperones, calnexin and calreticulin (Figure 10.18). These proteins bind sugar residues on the partially folded glycoproteins before translocation is complete and assist the glycoprotein in folding correctly. If the glycoprotein fails to fold after multiple cycles, the chaperone complex will send it into a degradation pathway that involves retro-translocation of the protein back through the translocon channel. In the cytosol it is marked by ubiquitination and degraded in the proteasome as described in Chapter 8.

In addition to acting as a chaperone, BiP plays a pivotal role as a sensor of the general state of protein folding within the cell. If an excess of